

NATIONAL BUILDING CONSTRUCTION STANDARDS

PART D BUILDING SERVICES

SECTION 4 ACOUSTICS, SOUND INSULATION AND NOISE CONTROL

BUREAU OF INDIAN STANDARDS

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FOREWORD

This NBCS (Part D/Section 4) covers the acoustical, sound insulation and noise control requirements in buildings. Emphasis is laid on planning of buildings *vis-a-vis* its surroundings to reduce noise, and in addition, sound insulation aspects of different occupancies are covered for achieving acceptable noise levels.

This Section was first published in 1970 as part of the National Building Code of India and was subsequently revised in 1983, 2005 and 2016. Some of the important changes in the 1983 version included: specifying of approximate measured noise levels due to various types of traffic (air, rail and road) conditions; elaboration of planning and design features of buildings against outdoor noise; modification of impact sound insulation in residential buildings and hearing damage risk criteria in industrial buildings; recommendations regarding planning of open plan schools against noise; planning and design aspects of hotels and hostels, laboratories and test houses, and other miscellaneous buildings, and planning of office buildings with light weight partitions; and elaboration of the public address system to cover public address system at passenger terminals.

Some of the important changes in the 2005 version included: addition of large numbers of important definitions in line with the existing international practice of usage of terms in the field of acoustics, sound insulation and noise control; inclusion of a new clause on highway noise barrier under provisions on planning and design against outdoor noise; deletion of the clause on public address system; addition of a new clause on cinema; and deletion of existing appendices on 'Constructional measures for sound insulation of buildings' and 'Sound Insulation values for various types of materials and construction'; and addition of eight new informative annexes on noise calculations, specification of sound insulation, noise rating, outdoor noise regulations in India, special problems requiring expert advice, airborne and impact sound insulation, basic design techniques for noise control in HVAC and suggested equipment noise data sheet.

The significant changes that were incorporated in the 2016 revision included the review and update of definitions of various existing terms, along with the introduction of new terms such as noise criteria, sound level difference, ground-borne noise, and structure-borne noise. A new clause on construction noise was introduced, while the provision relating to noise control in open-plan schools was deleted. For auditoria and theatres, a provision for the effective isolation of mechanical equipment, such as lifts, from the building structure was included to prevent noise transmission. Additionally, in auditoria and theatres, a requirement was introduced for adequate damping under lightweight metal roofs, along with an additional lightweight under-deck noise barrier to reduce noise generated by rainfall. The reverberation time of assembly halls in schools, corresponding to 'Maximum for noise control (empty)', was modified, and the reverberation times of classrooms were also revised. The requirement for insulation (R_w) for walls or partitions between rooms in hospitals was modified, as also the recommended maximum reverberation times for very large offices and canteens. The recommended sound isolation value (D_w) between one room and another room in office was modified, and a similar modification was made for clerical offices where noise does not constitute a major nuisance. The provisions for assembly for partitions in hotels, including those between guest rooms as well as between rooms, corridors, and floors, were also modified. Examples of common types of wall and floor construction with sound insulation were reviewed and updated as required. Additionally, the provisions were updated to address the requirements of persons with disabilities.

In this revision, this publication has been renamed as National Building Construction Standards (NBCS).

This revision has been brought out to incorporate changes based on the experience gained during usage of the last version of this Section. Following significant changes have been made in this revision:

- a) Provisions for open plan offices have been updated to address concerns regarding data theft and protection of sensitive information;
- b) Provisions have been added for noise control in specific industrial occupancies;
- c) A new clause has been included for neighbourhood acoustic shielding during infrastructure development to reduce noise impact on nearby areas;
- d) Provisions on traffic noise barriers has been introduced;

- e) A new [Annex F](#) has been introduced detailing the design requirements and performance evaluation of noise barriers to ensure effective noise mitigation;
- f) Provisions on the integration of noise barriers and noise abatement measures in zonal and urban planning have been included in [Annex E](#); and
- g) Room criteria (RC) has been added in [Annex J](#) for rating the heating ventilation and air-conditioning system-related noise in buildings.

There are two types of noises, that is, air-borne and structure-borne noise. To reduce the intensity of air-borne noise, sound absorbent materials may be used. An absorbent material is one which reduces the intensity of sound reflected from its surface. It may be applied to walls, floors, ceilings or used as furnishings to reduce the sound level by absorption. However, the materials selected for sound absorption shall be consistent with fire safety requirements of the buildings.

To reduce the transmission of air-borne noise, sound insulating materials may be used. Sound insulating materials block the passage of noise through them by virtue of their mass and physical properties. The extent of noise reduction provided by a single homogeneous panel is proportional to the logarithm of mass per unit area. For high values of sound insulation, normally heavy panels are required. Thin sheets of materials do not have adequate mass for providing any appreciable sound transmission loss by themselves. However, when thin sheet materials are used in a double panel construction with an intervening air cavity, this special construction can give extremely high sound transmission loss values considering the mass of the partition, if designed properly. Porous materials lack the mass required to provide any appreciable sound transmission loss, and readily allow sound at most frequencies to be transmitted through them.

To reduce the transmission of structure-borne noise (such as noise generated by impacts) special construction methods and elastic discontinuity in the structure may be used. Structure-borne noise reduction is effected by corner joints, changes in cross-section, changes in materials, etc, in construction. The reduction by these construction methods is, however, not appreciable especially when a large amount of noise reduction is required over a short distance. In such cases, introduction of an elastic discontinuity in the structure can result in a very large amount of noise reduction. The noise transmission is affected only above a certain lower frequency which depends on the material thickness and the elastic properties of the material. Bonded fibrous materials, rubber elastomers, cork, etc, are suitable for curtailing structure-borne noise transmission.

This Section is largely based on the following standards:

IS 1950 : 1962	Code of practice for sound insulation of non-industrial buildings
IS 3483 : 1965	Code of practice for noise reduction in industrial buildings
IS 4954 : 1968	Recommendations for noise abatement in town planning
IS 11050 (Part 1) : 2023	Rating of sound insulation in buildings and of building elements: Part 1 Airborne sound insulation (<i>first revision</i>)
IS 11050 (Part 2) : 2023	Rating of sound insulation in buildings and of building elements: Part 2 Impact sound insulation (<i>first revision</i>)

Assistance has also been drawn from the following publication, in the formulation of this Section:

BS 8233 : 2014	Code of practice for sound insulation and noise reduction for buildings
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All standards, whether given herein above or cross-referred to in the main text of this Section, are subject to revision. The parties to agreement based on this Section are encouraged to investigate the possibility of applying the most recent editions of the standards.

As the regulation of land development and buildings is a State subject and is dealt by the States and local bodies such as urban local bodies within their jurisdiction through building regulatory documents like building regulations, building byelaws, and fire regulations, this document is voluntary in nature and is non-binding. The implementation depends on adoption by concerned parties or stipulations in a contract or by appropriate adoption by the concerned authorities.

For the purpose of deciding whether a particular requirement of this Section is complied with, the final value, observed or calculated, expressing the result of a test or analysis, shall be rounded off in accordance with IS 2 : 2022 'Rules for rounding off numerical values (*second revision*)'. The number of significant places retained in the rounded off value should be the same as that of the specified value in this Section.

Important Explanatory Note for Users of this NBCS

In any Part/Section of this NBCS, where reference is made to ‘**good practice**’ in relation to **design, constructional procedures or other related information**, and where reference is made to “**accepted standard**” in relation to **material specification, testing, or other related information**, the Indian Standards listed at the end of the Part/Section shall be used as a guide to the interpretation.

At the time of publication, the editions indicated in the standards were valid. All standards are subject to revision and parties to agreements based on any Part/ Section are encouraged to investigate the possibility of applying the most recent editions of the standards.

In the list of standards given at the end of a Part/Section, the number appearing within parentheses in the first column indicates the number of the reference of the standard in the Part/Section. For example:

a) Good practice [D-4(1)] refers to the Indian Standard(s) given at serial number (1) of the list of standards given at the end of this Part D/Section 4, that is, IS 11050 (Part 1) : 2023 ‘Rating of sound insulation in buildings and of building elements: Part 1 Airborne sound insulation (*first revision*)’.

NATIONAL BUILDING CONSTRUCTION STANDARDS

PART D BUILDING SERVICES

SECTION 4 ACOUSTICS, SOUND INSULATION AND NOISE CONTROL

1 SCOPE

This NBCS (Part D/Section 4) covers requirements and guidelines regarding planning against noise, acceptable noise levels and the requirements for sound insulation in buildings with different occupancies.

2 TERMINOLOGY

For the purpose of this Section, the definitions given below shall apply.

2.1 Ambient Noise — The sound pressure levels associated with a given environment. Ambient noise is usually a composite of sounds from near and far sources, none of which are particularly dominant.

2.2 Audible Frequency Range — The range of sound frequencies normally heard by the human ear. The audible range spans from 20 Hz to 20 000 Hz.

2.3 A-Weighted Sound Pressure, p_A — Value of overall sound pressure, measured in pascal (Pa), after the electrical signal derived from a microphone has been passed through an A-weighting network.

NOTE — The A-weighting network modifies the electrical response of a sound level meter with frequency in approximately the same way as the sensitivity of the human hearing system.

2.4 A-Weighted Sound Pressure Level, L_{pA} — The quantity of A-weighted sound pressure, in decibels (dB), is given by the following formula:

$$L_{pA} = 10 \log_{10} (p_A / p_0)^2$$

where

p_A = A-weighted sound pressure, in pascal (Pa); and

p_0 = reference sound pressure (20 μ Pa).

NOTE — Measurements of A-weighted sound pressure level can be made with a meter and correlate roughly with subjective assessments of loudness, and are usually made to assist in judging the effects of noise on people. The size of A-weighting in 1/3 octave bands, is shown in Annex A (see [A-5](#)). An increase or decrease in level of 10 dBA corresponds roughly to a doubling or halving of loudness.

2.5 Background Noise — The sound pressure levels in a given environment from all sources excluding a specific sound source being investigated or measured.

2.6 Break-in — Unwanted sound transmission into a duct or a quiet environment, from outside.

2.7 Break-out — Unwanted sound transmission from the inside of a duct or a noisy enclosure, to the outside.

2.8 Broad Band Noise — Spectrum consisting of a large number of frequency components, none of which is individually dominant.

2.9 Cross-talk — Unwanted sound transmission between one room and another room or space, *via* a duct.

2.10 C_{tr} — Correction term applied against the sound insulation single-number values (R_w , D_w , and $D_{nT,w}$) to provide A-weighting against low frequency performance.

NOTE — The reference values used within the C_{tr} calculation are based on urban traffic noise.

2.11 Decibel — Ten times the logarithm (to the base 10) of the ratio of two mean square values of sound pressure, sound power or sound intensity. The abbreviation for decibel is dB.

2.12 Effective Perceived Noise Level, in Decibel (EPN dB) — The number for rating the noise of an individual aircraft flying overhead is the effective perceived noise level in decibels (EPN dB). The effective perceived noise decibel value takes into account the subjectively annoying effects of the noise including pure tones and duration. In principle, it is a kind of time-integrated loudness level.

2.13 Equivalent Continuous A-Weighted Sound Pressure Level, $L_{Aeq,T}$ — Value of the A-weighted sound pressure level in decibels (dB) of a continuous, steady sound, that within a specified time interval, T , has the same mean squared sound pressure as the sound under consideration that varies with time, and is given by the formula:

$$L_{Aeq,T} = 10 \log_{10} \left[\frac{1}{T} \int_0^T \frac{p_A^2(t)}{p_0^2} dt \right]$$

where

$p_A(t)$ = instantaneous A-weighted sound pressure, in pascals (Pa); and
 p_0 = reference sound pressure (20 μ Pa).

NOTE — Equivalent continuous A-weighted sound pressure level is mainly used for the assessment of environmental noise and occupational noise exposure.

2.14 Equivalent Sound Absorption Area of a Room, A — Hypothetical area of a totally absorbing surface without diffraction effects, expressed in square metres (m^2) which, if it were the only absorbing element in the room, would give the same reverberation time as the room under consideration.

2.15 Facade Level — Sound pressure level measured 1 m to 2 m in front of the facade.

NOTE — Facade level measurements of L_{pA} are usually 2 dB to 3 dB higher than corresponding free-field measurements.

2.16 Free-Field Level — Sound pressure level measured outside, far away from reflecting surfaces.

NOTE — Measurements made 1.2 m to 1.5 m above the ground and at least 3.5 m away from other reflecting surfaces are usually regarded as being free-field measurements. To minimise the effect of reflections the measuring position should be at least 3.5 m to the side of the reflecting surface (that is, not 3.5 m from the reflecting surface in the direction of the source). Estimates of noise from aircraft overhead usually include a correction of 2 dB to allow for reflections from the ground.

2.17 Frequency — The number of cyclical variations per unit time. Frequency is generally expressed in cycles per second (cps) and is also denoted as Hertz (Hz).

2.18 Impact Sound Pressure Level, L_i — Average sound pressure level in a specific frequency band in a room below a floor, when it is excited by a standard tapping machine.

2.19 Indoor Ambient Noise — Pervasive noise in a given situation at a given time, usually composed of noise from many sources, inside and outside the building, but excluding noise from activities of the occupants.

2.20 Insertion Loss (L_{IL}) — Insertion loss is generally defined as the difference, in decibels, between two sound pressure levels (or power levels or intensity levels) which are measured at the same point in space before and after a muffler or any other noise control device is inserted between the measurement point and the noise source.

2.21 Noise — Unwanted sound which may be hazardous to health, interferes with communications or is disturbing.

2.22 Noise Exposure Forecast (NEF) — The noise exposure forecast at any location is the summation of the noise levels in EPN dB from all aircraft types, on all runways, suitably weighted for the number of operations during day time and night time.

2.23 Noise Criteria — Numerical indices used to define design goals for the maximum allowable noise in a given space.

2.24 Noise Rating (NR) — Graphical method for rating a noise by comparing the noise spectrum with a family of noise rating curves.

NOTE — Noise rating is described in [Annex B](#).

2.25 Noise Reduction Coefficient (NRC) — A single figure descriptor of the sound absorption property of a material. It is the arithmetic mean of the sound absorption coefficients at 250 Hz, 500 Hz, 1 000 Hz and 2 000 Hz rounded off to the nearest multiple of 0.05.

2.26 Normalized Impact Sound Pressure Level, L_n — Impact sound pressure level normalized for a standard absorption area in the receiving room.

NOTE — Normalized impact sound pressure level is usually used to characterize the insulation of a floor in a laboratory against impact sound in a stated frequency band (see [Annex C](#)).

2.27 Octave Band — Band of frequencies in which the upper limit of the band is twice the frequency of the lower limit.

2.28 Percentile Level, $L_{AN,T}$ — A-weighted sound pressure level obtained using time-weighting ‘F’, which is exceeded for N percent of a specified time interval.

Example:

$L_{A90,1h}$ is the A-weighted level exceeded for 90 percent of 1 h.

NOTE — Percentile levels, determined over a certain time interval cannot accurately be extrapolated to other time intervals. Time-weighting ‘F’ or ‘S’ can be selected on most modern measuring instruments and used to determine the speed at which the instrument responds to changes in the amplitude of the signal. Time-weighting ‘F’ is faster than ‘S’ and so its use can lead to higher values when rapidly changing signals are measured.

2.29 Pink Noise — Sound with an uninterrupted frequency spectrum and a power which is steady within frequency band and proportional to centre frequency. An example is constant power level per octave band.

2.30 Pure Tone — A sound emitted at a single frequency.

2.31 Rating Level, L_{Ar,T_r} — Equivalent continuous A-weighted sound pressure level of the noise, plus any adjustment for the characteristic features of the noise.

NOTE — This definition is used for rating industrial noise, where the noise is the specific noise from the source under investigation.

2.32 Reverberation Time, T — Time that would be required for the sound pressure level to decrease by 60 dB after the sound source has stopped.

NOTE — Reverberation time is usually measured in octave or third octave bands. It is not necessary to measure the decay over the full 60 dB range. The decay measured over the range 5 dB to 35 dB below the initial level is denoted by T_{30} , and over the range 5 dB to 25 dB below the initial level by T_{20} .

2.33 Sound — A vibrational disturbance, exciting hearing mechanisms, transmitted in a predictable manner determined by the medium through which it propagates. To be audible the disturbance shall have to fall within the frequency range of 20 Hz to 20 000 Hz.

2.34 Sound Exposure Level, L_{AE} — Level of a sound, of 1 s duration, that has the same sound energy as the actual noise event considered.

NOTES

1 The L_{AE} of a discrete noise event is given by the formula:

$$L_{AE} = 10 \log_{10} \left(\frac{1}{t_0} \int_{t_1}^{t_2} \frac{p_A^2(t)}{p_0^2} dt \right)$$

where

$p_A(t)$ = instantaneous A-weighted sound pressure in pascals (Pa);

$t_2 - t_1$	=	stated time interval in seconds (s) long enough to encompass all significant sound energy of the event;
p_0	=	reference sound pressure level ($20 \mu\text{ Pa}$); and
t_0	=	reference time interval (1 s).

2 L_{AE} is also known as L_{AX} (single-event noise exposure level).

2.35 Sound Level Difference, D — Difference between the sound pressure level in the source room and the sound pressure level in the receiving room.

NOTE — D is given by the following formula:

$$D = L_1 - L_2$$

where

L_1	=	average sound pressure level in the source room; and
L_2	=	average sound pressure level in the receiving room.

2.36 Sound Power — The acoustic power of a sound source, expressed in Watts.

2.37 Sound Power Level (L_w) — The acoustic power radiated from a given sound source as related to a reference power level (typically 10-12 watt) and expressed in decibel as:

$$L_p = 10 \log_{10} \left\{ \frac{W}{10^{-12}} \right\}$$

or

$$L_w = 10 \log W + 120$$

where

W = acoustic power, in watts.

By definition, 1 W therefore corresponds to 120 dB for L_w .

2.38 Sound Pressure, p — Root-mean-square value of the variation in air pressure measured in pascal (Pa), above and below atmospheric pressure, caused by the sound.

2.39 Sound Pressure Level, L_p — Quantity of sound pressure, in decibel (dB), given by the formula:

$$L_p = 10 \log_{10} (p/p_0)^2$$

where

p	=	root mean square sound pressure in pascals (Pa); and
p_0	=	reference sound pressure ($20 \mu\text{ Pa}$).

NOTE — The range of sound pressures for ordinary sounds is very wide. The use of decibels gives a smaller, more convenient range of numbers. For example, sound pressure levels ranging from 40 dB to 94 dB correspond to sound pressures ranging from 0.002 Pa to 1 Pa. A doubling of sound energy corresponds to an increase in level of 3 dB.

2.40 Sound Receiver — One or more observation points at which sound is evaluated or measured. The effect of sound on an individual receiver is usually evaluated by measurements near the ear or close to the body.

2.41 Sound Reduction Index, R — Laboratory measure of the sound insulating properties of a material or building element in a stated frequency band.

NOTE — For further information see [Annex C](#).

2.42 Sound Source — Equipment or phenomena which generate sound. Source room is the room containing sound source.

2.43 Spectrum — A quantity expressed as a function of frequency, such as sound pressure versus frequency curve.

2.44 Standardized Impact Sound Pressure Level, L'_{nT} — Impact sound pressure level normalized to that in a receiving room having a reverberation time of 0.5 s.

NOTE — Standardized impact sound pressure level is used to characterize the insulation of floors in buildings against impact sound in a stated frequency band (see [Annex C](#)).

2.45 Speech Interference Level (SIL) — A descriptor for rating steady noise according to its ability to interfere with conversation between two people. SIL is the arithmetic average of the sound pressure levels in the three octave bands with centre frequencies at 500 Hz, 1 000 Hz and 2 000 Hz.

2.46 Standardized Level Difference, D_{nT} — Difference in sound level between a pair of rooms, in a stated frequency band, normalized to a reverberation time of 0.5 s.

NOTE — Standardized level difference takes account of all sound transmission paths between the rooms (see [Annex C](#)).

2.47 Structure-borne Noise and Ground-borne

NOTE — When elements of a structure vibrate they radiate noise and, if the vibration is high enough, this noise can be audible. Ground-borne and structure-borne noises are rarely an issue outside buildings or structures.

2.47.1 Ground-borne Noise — Audible noise caused by the vibration of elements of a structure, for which the vibration propagation path from the source is partially or wholly through the ground.

NOTE — Common sources of ground-borne noise include railways and heavy construction work on adjacent construction sites.

2.47.2 Structure-borne Noise — Audible noise caused by the vibration of elements of a structure, the source of which is within a building or structure with common elements.

NOTE — Common sources of structure-borne noise include building services plant, manufacturing machinery and construction or demolition of the structure.

2.48 Transient Sound — Sound which is audible for a limited period of time, for example sound from over flight of an airplane.

2.49 Third Octave Band — Band of frequencies in which the upper limit of the band is $2\frac{1}{3}$ times the frequency of the lower limit.

2.50 Threshold of Hearing — The lowest continuous sound pressure level which will create an auditory sensation for the average human ear. Any sound below these levels will be inaudible and any sound above the threshold will vary in loudness dependent on intensity.

2.51 Vibration Isolation — Reduction of force or displacement transmitted by a vibratory source, often attained by use of a resilient mount.

2.52 Wavelength — The length in space of one complete cycle of a sound wave given by.

$$\lambda = \frac{\text{Speed of Sound}}{\text{frequency}} = \frac{C}{f}$$

2.53 Weighted Level Difference, D_w — Single-number quantity that characterizes airborne sound insulation between rooms but which is not adjusted to reference conditions.

NOTE — Weighted level difference is used to characterize the insulation between rooms in a building as they are; values cannot normally be compared with measurements made under other conditions {see good practice [D-4(1)]}.

2.54 Weighted Sound Reduction Index, R_w — Single number quantity which characterizes the airborne sound insulating properties of a material or building element over a range of frequencies.

NOTE — The weighted sound reduction index is used to characterize the insulation of a material or product that has been measured in a laboratory (see [Annex C](#)).

2.55 Weighted Standardized Impact Sound Pressure Level, $L'_{nT,w}$ — Single number quantity used to characterize the impact sound insulation of floors over a range of frequencies.

NOTE — Weighted standardized impact sound pressure level is used to characterize the insulation of floors in buildings (see [Annex C](#)).

2.56 Weighted Standardized Level Difference, $D_{nT,w}$ — Single-number quantity, which characterizes the airborne sound insulation between rooms.

NOTE — Weighted standardized level difference is used to characterize the insulation between rooms in a building (see [Annex C](#)).

2.57 Weighted Normalized Impact Sound Pressure Level, $L'_{n,w}$ — Single number quantity used to characterize the impact sound insulation of floors over a range of frequencies.

NOTE — Weighted normalized impact sound pressure level is usually used to characterize the insulation of floors tested in a laboratory (see [Annex C](#)).

2.58 White Noise — A noise whose spectrum (level) density is substantially independent of frequency over a specified range and has equal power for any range of frequencies of constant band width.

3 PLANNING AND DESIGN AGAINST OUTDOOR NOISE

3.1 General

Planning against noise should be an integral part of town and country planning proposals, ranging from regional proposals to detailed zoning, and three-dimensional layouts and road design within built-up areas. Noise nuisance should be fully recognized in zoning regulations.

3.1.1 Noise is either generated by traffic (road, air and surface and underground railway) or it arises from zones and buildings within built-up areas (industry, commerce, offices and public buildings), and from public gatherings and social activities. For planning, the noise survey should examine all the possible causes of noise and consider the various factors causing actual nuisance.

3.1.2 Noise by night, causing disturbance of sleep, is more of a nuisance than noise by day. For this reason, housing colonies that adjoin areas with heavy traffic movement during the night are liable to cause serious complaints. Also, the factories that work by night are liable to cause serious complaints if housing estates adjoin them. While planning, care should be taken that housing colonies are adequately set back from busy airports, state and national highways, factories, main railway lines and marshalling yards.

3.1.3 There are two aspects of defence by planning. The first is to plan so as to keep the noise at a distance. Under this aspect comes the separation of housing from traffic noise by interposing buffer zones, and the protection of schools and hospitals by green belts, public gardens, etc. The second is the principle of shading or screening. This consists of deliberately interposing a less vulnerable building to screen a more vulnerable one or by providing a solid barrier, such as a wall, between the source and the location to be protected.

3.2 Traffic Noise Levels

3.2.1 For Air Traffic

For guidance, approximate noise levels due to various types of aircrafts, measured on ground, when the aircrafts fly overhead at a height of 450 m, are given in [Table 1](#).

3.2.2 For Rail Traffic

Noise levels of some typical railway traffic are given in [Table 2](#).

3.2.3 For Road Traffic

The level of noise generated by road traffic depends upon such factors as the number of vehicles passing per hour, the type of traffic, the preponderance of heavy vehicles, average speed, gradient and smoothness of traffic flow. The smoothness of traffic flow also affects variability of the noise and is governed by such features as roundabouts and traffic lights, and the volume of traffic and pedestrian movement with their effects on stopping, starting, overtaking and honking. The level of traffic noise fluctuates continuously and the way it does, has a considerable effect on the nuisance caused. For assessing traffic noise, noise is measured in dBA. Because of the fluctuating nature of traffic, noise levels due to different volumes of traffic flow with a varying mix of vehicles are given in [Table 3](#).

Table 1 Typical Noise Levels of Some Aircraft Types (Clause 3.2.1)		
Sl No.	Type of Aircraft	Flyover Noise Levels at 450 m with Take-off Thrust EPN dB
(1)	(2)	(3)
i)	Boeing 737	107
ii)	Boeing 747 to 200	103
iii)	Airbus A300	101

Table 2 Typical Noise Levels of Railway Trains (Clause 3.2.2)		
Sl No.	Type of Train	Noise Level at 30 m, Measured on the Side or in the Direction of Train dBA
(1)	(2)	(3)
i)	Steam train, 60 km/h	85
ii)	Diesel train, 60 km/h	83
iii)	Electric train, 60 km/h	77

Table 3 Typical Noise Levels Due to Free-Flowing Road Traffic (Clause 3.2.3)		
Sl No.	Type of Traffic	L_{10} 30 m from Edge of Road dBA
(1)	(2)	(3)
i)	5 000 vehicles per 18 hour day (10 percent heavy vehicles), 50 kmph	65
ii)	10 000 vehicles per 18 hour day (20 percent heavy vehicles), 60 kmph	70
iii)	10 000 vehicles per 18 hour day (40 percent heavy vehicles), 80 kmph	75
iv)	20 000 vehicles per 18 hour day (40 percent heavy vehicles), 80 kmph	77
NOTE — The values are applicable to free flowing traffic without honking.		

3.3 Outdoor Noise Regulations

The outdoor noise regulations in force from time to time shall be complied with (see [Annex D](#)).

3.4 Planning and Design

3.4.1 For Air Traffic

Near airports two sources of aircraft noise should be considered.

- a) *Flyover Noise* — Flyover noise is that which occurs under flight paths close to airports and is the most serious and common problem. As the aircraft passes overhead the noise level at any particular location rises to a peak and then decreases; and
- b) *Ground Noise* — The noise emitted by an aircraft during ground operations is less variable in direction than flyover noise, but is usually of a longer duration.

3.4.1.1 Aircraft noise may disturb sleep, rest and communication, and as such may be considered potentially harmful to health. It is important that no new development is carried out within areas where the expected noise levels will cause mental and physical fatigue or permanent loss of hearing. In case development in such areas is essential, adequate sound insulation shall be provided for the building.

3.4.1.2 As the problems caused by aircraft noise have become more acute, a number of methods have been devised for evaluating noise exposure in the vicinity of airports. They all combine many factors into a single number evaluation. A commonly used criterion is the noise exposure forecast (NEF). The NEF is used primarily to develop noise contours for areas around airports. It has been accepted generally that noise exposure forecast levels greater than NEF 40 are unacceptable to people while levels less than NEF 25 are normally acceptable. Levels between NEF 25 and 40 may lead to subjective complaints.

3.4.1.3 While it is theoretically possible to provide sufficient insulation to achieve an acceptable indoor noise environment in the area of very high outdoor noise, there is a level above which aircraft noise seriously affects living conditions no matter how much sound insulation has been applied to the dwelling unit. For this reason, it is recommended that no residential development be allowed beyond the NEF 35 level.

3.4.1.4 During summer months, the windows are normally kept open for adequate ventilation. In view of this, no matter how much sound insulation is provided for the building structure, the noise level inside the room can never be less than 10 dB below the outdoor noise level. For very critical buildings, such as buildings necessary for maintaining and supplementing the airport services, and for commercial development, such as hotels, it is possible to provide sealed windows and to centrally air-condition the entire building. However, it is not feasible for most of the residential developments in the country. In such cases proper zoning regulations and siting of vulnerable buildings away from aircraft noise are of vital importance.

3.4.2 Rail Traffic

This is a very serious source of noise in built-up areas, both by day and by night. Railway cuttings reduce the spread of noise, whereas embankments extend it. The elevated railway on viaducts or embankment is very common in built-up areas. The elevation increases exposure to noise but in addition the construction of the viaduct may affect the propagation of noise. In this respect solid embankments are preferable to built-up arches, which tend to act as sound boxes. Worst of all are the steel bridges, which greatly magnify the noise due to vibration. Uphill gradients are another feature tending to increase noise, especially of heavy goods trains.

3.4.2.1 Wherever possible, no residential or public building zone should abut onto railway lines, especially on the marshalling yards which is particularly objectionable because of the shrill, clanging and intermittent noise they generate, often at night. The appropriate zones alongside railway lines are industrial and commercial buildings other than office buildings. Where these precautions are not practicable and housing has to abut on to railway lines, every attempt may be made to house as few people as possible in the vicinity of the railway lines.

3.4.2.2 Underground transportation system can be a major cause of disturbance for the neighbouring community. Very high noise levels are propagated to long distances by the underground high speed railway, as a result of

wheel-rail interaction. Both airborne noise and ground or structure-borne vibrations are potential sources of complaints. Noise control measures, therefore, need to be considered for the following:

- a) In stations, where high noise levels are produced at the arrival and departure of trains;
- b) In tunnels, during high speed train movement;
- c) Where an underground rail transit system passes close to existing structures or high rise buildings, adequate attention should also be paid to the problem of ground vibration transmitted to the building, and proper isolation should be provided for critical areas;
- d) Wherever elevated railway tracks are provided, adequate measures should be taken to avoid the spread of noise in the surrounding built up areas; and
- e) In transit cars, where sound insulation is of vital importance to provide comfortable conditions for the commuters.

3.4.3 Road Traffic

3.4.3.1 Convoys of long-distance heavy trucks at night moving past through built-up areas cause serious noise complaints. On busy roads, the noise of continuous traffic may be a worse nuisance than that of railways. At least the same precautions may, therefore, be taken in the planning of dwellings in relation to arterial and trunk roads as with railways. Care may be taken that local housing roads do not provide short cuts for heavy traffic through residential areas. Hilly roads present the additional noise of gear changing. Trees with heavy foliage planted on both sides of carriageway help slightly to muffle the noise, provided the foliage extends for a considerable distance (30 m or above).

3.4.3.2 Road traffic may give rise to serious nuisance particularly on busy thoroughfares, between continuous high buildings in main streets, at the traffic lights, near bus stops, on steep slopes and in parking spaces and enclosed yards.

3.4.3.3 For zoning and planning of new buildings in urban areas, it is recommended that external L_{A10} be limited to a maximum of 70 dBA when the dwellings are proposed to have sealed windows and 60 dBA when the dwellings are proposed to have open windows. Indeed, it is desirable to confine major new residential development to locations subject to L_{A10} levels substantially lower than those given above.

It is recognized, however, that within the large urban areas, the use of sites where the external L_{A10} is greater than 60 dBA to 70 dBA cannot always be avoided. In that case it is suggested to utilize such design solutions as barrier blocks in order to reduce external L_{A10} noise levels to at least 60 dBA to 70 dBA at any point 1.0 m from any inward looking facade. When the orientation of site and the density of development are such that this cannot be fully achieved some form of dwelling insulation will have to be provided. It should be appreciated that where open windows are necessary, the occupants would have to put up with discomfort if the above conditions are not met.

3.4.3.4 Certain other methods can often be utilized to provide economical and effective protection from noise, as follows:

- a) Methods should be adopted to improve the smoothness of flow and reduce number of stopping and starting. This leads to an improvement even, if it leads to increased flows. Flow linking of traffic lights, for example, may reduce noise nuisance;
- b) Use of roads passing through residential areas should be prohibited to heavy commercial vehicles. An alternative would be to limit use by commercial vehicles to certain times of the day;
- c) Use of honking shall be prohibited near sensitive buildings, such as hospitals and the like; and
- d) Barriers shall be provided to shield sites from noise.

3.4.4 Construction Noise

3.4.4.1 Construction activities relating to building and infrastructure facilities cause noise disturbance to occupants in existing built facilities located around the site.

Construction noise involves both stationary and mobile sources. Common activities and sources of concern in case of construction noise include excavation and site clearing activities, blasting, piling, operation of batching

plants, concrete mixers, pumping, running of diesel generators, drilling and chipping, loading/unloading activities, and truck and other vehicular movement.

The source noise intensity and spectrum is primarily characterized by the type of construction activity, equipment used, ground conditions and associated vehicular movement. Apart from the source characteristics, community noise impact is influenced by the time and sequence of construction related activities.

3.4.4.2 Noise control can be done at the source or in the noise propagation path. While it is preferable to adopt noise control at the source, it may not be practically possible to implement it for some of the construction equipment. Given the diversity of noise sources, mitigation strategies in case of construction noise have to be dealt with appropriately depending on the spectral characteristics of sources.

Source noise control strategies include providing acoustic enclosures for sources like diesel generators, choosing less noisy construction machinery and selecting alternative construction practices which involve lesser noise generation.

Noise control in the propagation paths primarily involves provision of noise barriers around the construction sites.

Positioning of major fixed construction equipment/installations such as batching plants and diesel generators have a significant impact on the noise spectrum and its impact. Also, time of operation and duration of noisy construction activities have significant impact on the resultant annoyance.

3.4.4.3 Due consideration for environmental noise regulation should be given while scheduling and sequencing noisy construction activities.

3.5 Zoning

The zoning of the different cities shall be done by the town planning authorities, taking into account besides other aspects, the noise levels from different occupancies. Wherever necessary, experts in the field may be consulted. For detailed information on noise reduction for town planning schemes, reference may be made to good practice [D-4(2)]. Details regarding the integration of noise barriers and noise abatement measures in zonal and urban planning are provided in [Annex E](#).

3.6 Green Belts and Landscaping

Where relief from noise is to be provided by means of green belts these may be of considerable width and be landscaped. In case of railway tracks, a minimum distance of 50 m to 70 m may be provided between the buildings and the tracks. The extent of relief that may be derived from the above may be estimated only after considering other environmental factors. Only thick belts of planting (greater than 30 m) are of real value. Strong leafy trees may be planted to act as noise baffles. Shrubs or creepers may also be planted for additional protection between tree trunks; artificial mounds and banks should be formed where practicable. As little hard paving and as much grass as possible may be used. The creation of green belt is particularly advisable on the perimeter of aerodromes, along railway lines and arterial roads, through or past built-up areas and adjoining noisy industrial zones.

3.7 Highway Noise Barriers

Barriers are often the most effective means of reducing traffic noise around residential areas. They have the great advantage that they generally protect most or all of the site. In nearly all situations, a well-designed barrier of even a modest height (say 3 m) can at least ensure that all areas of open space are free from excessive noise levels.

There are two types of barriers that can be built to protect sites; one which are built solely for the purpose of reducing noise and the other which form part of the building complex (barrier blocks). Free standing walls and artificial mounds are typical examples of the first type while single and multi-storeyed utility buildings and garages are the most common form of the second.

Of the two types, laying out barrier blocks of a complex in an appropriate fashion is a better option because they are cheaper and also tend to form a more effective barrier overall because of their greater height and width. Barrier walls or mounds are more limited in their effect than barrier blocks for they protect little more than the area of the site close to ground level essentially because of the lack of height, as continuous walls much higher than 3m are often difficult to construct.

3.8 Use of Acoustic Barriers for Residential, Educational and Hospital Buildings along Highways on City Outskirts (Approximately 4 m to 5 m in Height)

3.8.1 Noise barriers can be constructed from a wide variety of materials and designs, but the basic purpose of all the noise barriers is to reduce noise levels at noise sensitive receptors.

3.8.2 A reduction in noise levels up to 10 dBA can be obtained at ground level in an area that is close behind a barrier of considerable height with sufficient insulation value and absorption value. Noise barriers are relatively ineffective at more than 250 m distance from the road.

3.8.3 Innovative combinations, including perforated metal sheeting on one side with the inclusion of absorptive material and a corrugated profile, should be utilized to enhance the insertion loss provided by the barrier.

3.8.4 Noise barriers may be constructed from a variety of materials, including but not limited to earth, concrete, masonry, metal and wood. Conventional concrete or masonry structures should be considered the best practicable option for highway noise barriers due to their long-term stability and maintenance-free service characteristics. Laminated glass requires frequent cleaning due to accumulation of dirt, while polycarbonate sheets become opalescent over time as it can absorb water. Plastics are prone to damage from fire and vandalism, and some, such as polyethylene, become brittle after prolonged exposure to sunlight.

3.8.5 The application of multiple-edge noise barriers, also known as profiled noise barriers, should be implemented in specific applications where enhanced acoustical performance is required to effectively screen a single noise source.

3.8.6 The noise barriers, although highly effective in reducing highway noise levels by significant extent, are subject to certain limitations. To ensure effectiveness, noise barriers shall be designed to be of sufficient height and length to block the view of the road. Barriers are less effective for homes on hillsides or buildings that rise above the barrier.

3.8.7 The design of noise barriers (see [Annex F](#)) should take into account the following factors:

- a) Visual impact;
- b) Maintenance requirements; and
- c) Cost implications.

3.8.8 The noise barriers may also have negative impacts on the surrounding environment, such as blocking views or creating shadows, which should be taken into account during the design process.

3.8.9 Transparent barriers are associated with lower perceived loudness and noise annoyance compared to opaque barriers, contributing to improved user satisfaction.

3.8.10 The utilization of waste materials, such as plastic, rubber, and bottom ash, is recommended for achieving high noise attenuation with improved cost efficiency.

3.9 Special Problems Requiring Expert Advice

The purpose of noise control is to ensure that people are neither harmed nor disturbed by noise. In addition to provisions given in this Section, special advice may be required for more complex situations, such as those listed in [Annex G](#).

4 PLANNING AND DESIGN AGAINST INDOOR NOISE

4.1 Acceptable Indoor Noise Levels in Buildings

The generally acceptable noise levels inside buildings are given in [Table 4](#).

Table 4 Acceptable Indoor Noise Levels for Various Buildings

(Clause [4.1](#))

Sl No.	Location	Noise Level dBA
(1)	(2)	(3)
i)	Auditoria and concert halls	20 to 25
ii)	Radio and TV studios	20 to 25
iii)	Cinemas	25 to 30
iv)	Music rooms	25 to 30
v)	Hospitals	35 to 40
vi)	Apartments, hotels and homes	35 to 40
vii)	Conference rooms, small offices and libraries	35 to 40
viii)	Court rooms and class rooms	40 to 45
ix)	Large public offices, banks and stores	45 to 50
x)	Restaurants	50 to 55

4.2 Vulnerable Buildings

Some buildings or parts of buildings are especially vulnerable to noise, for example, recording and radio studios, hospitals and research laboratories. These should not be sited near loud noise sources. Most vulnerable buildings contain some areas which are themselves noisy and in such buildings, the less vulnerable elements should be planned to act as noise buffers. Most noisy buildings also contain quiet accommodation, which equally may be planned to act as a buffer between the noisy part of the building and adjoining vulnerable buildings.

4.3 The details of site and internal planning and insulation requirements are covered under individual occupancies (see [5](#) to [12](#)) as applicable to the respective character and sources of noise in different buildings.

4.3.1 Equitable Inclusion of Hearing Impaired Persons in Public Places

People with hearing impairments have particular difficulty in making out sounds and words in noisy environments. Therefore, adequate sound insulation would minimise noise from both outside and inside the building. Also, low reverberation times are more suitable for hearing impaired persons and should thus be planned while designing the size and shape of the room.

Persons using hearing aids may require quiet areas with induction loops in very noisy information counters or where announcements are made. Induction loops may also be provided in all areas where there are verbal inputs provided, such as conference halls, auditoria, class rooms and cinema halls.

4.4 Sound Insulation of Non-Industrial Buildings by Constructional Measures

The desired (acceptable) noise levels and the recommended insulation values for the various areas may be achieved by providing sound insulation treatments by constructional measures. The details of the same are given in [Annex H](#). The recommendations given in [Annex H](#) are applicable to non-industrial buildings like residences, educational buildings, hospitals and office buildings.

4.5 Special Problems Requiring Expert Advice — See [3.9](#) and [Annex G](#).

5 RESIDENTIAL BUILDINGS

5.1 Sources of Noise Nuisance

5.1.1 Outdoor Noise

The main sources of outdoor noise in residential areas are traffic (aeroplane, railways, roadways), children playing, hawkers, services deliveries, road repairs, blaring loud-speakers, various types of moving machinery in the neighbourhood and building operations, and captive power generation machinery.

5.1.2 Indoor Noise

5.1.2.1 As far as indoor noises are concerned, conversation of the occupants, footsteps, banging of doors, shifting of the furniture, operation of the cistern and water closet, playing of radio, television, music system, cooling and ventilation machinery, etc, contribute most of the noise emanating from an adjacent room or an adjacent building. Noise conditions vary from time to time and noise which may not be objectionable during the day may assume annoying proportions in the silence of the night when quiet conditions are essential.

5.1.2.2 In the case of flats the main sources of noise are from other flats and from stairs, lifts and access balconies. Plumbing noise is another cause. In semi-detached buildings, outdoor noises from streets are noticed more than indoor noises from neighbours.

5.2 Recommendations

5.2.1 Site Planning

The most desirable method is to locate the residential buildings in a quiet area away from the noisy sources like the industrial areas, rail tracks, aerodromes, roads carrying heavy traffic, etc.

5.2.1.1 To minimize ground reflection, the dwellings should be surrounded by the maximum amount of planting and grassed areas and the minimum amount of hard surfacing. This applies particularly to high density areas. Where for maintenance reasons a large amount of hard paving is necessary, it should be broken up by areas of planting and grassing. Narrow hard paved courts should be avoided between adjacent tall buildings.

5.2.1.2 Roads within a residential area should be kept to a minimum both in width and length, and should be designed to discourage speeding. Area-wise planning, with zones from which vehicular traffic is altogether excluded will greatly help to reduce noise. Roads with through traffic should be excluded from residential areas, but where sites have to be developed adjacent to existing major roads the same principles should be observed in the siting of blocks as with railway lines as covered under [3.4.2.1](#).

5.2.1.3 Play areas for older children should be sited as far away from dwellings as possible. Special care should be taken with old peoples' dwellings. They should not be placed immediately adjacent to service entries, play spaces, or to any entrances where children may tend to congregate.

5.2.2 Internal Planning

The orientation of buildings in a locality should be planned in such a way as to reduce the noise disturbance from neighbourhood areas. The non-critical areas, such as corridors, kitchens, bathrooms, elevators and service spaces may be located on the noisy side and the critical areas, such as bedrooms and living space, on the quiet side.

5.2.2.1 Windows and doors

Windows and doors should be kept away from the noisy side of the building as given below, wherever possible:

- a) When windows of a building, particularly those of bedrooms in apartments or flats, face roads carrying heavy traffic or other noises where the external noise is of the order of 80 dBA to 90 dBA, the building should be located at a distance of about 30 m from the road, but a distance of 45 m or more, where possible, should be aimed at for greater relief from noise;
- b) When the windows are at right angles to the direction of the above type of noise, the distance from the road should be arranged to be about 15 m to 25 m; and

- c) In case another building, boundary wall or trees and plantations intervene between the road traffic and the house/flat further noise reduction is achieved and in such cases the above distances may be reduced suitably.

5.2.2.2 Layout plans

It is desirable that rooms adjoining party walls and above/below party floors should be of similar use. By this means, bedrooms are not exposed to noise from adjoining living rooms, and there is less risk of disturbance of sleep.

In semi-detached houses, the staircase, hall and kitchen should adjoin each other on each side of the party wall, thus providing a sound baffle between rooms requiring quiet conditions.

Bedrooms should not be planned alongside access balconies, and preferably not underneath them. Where the approach is by an internal corridor, a sound baffle may usefully be provided by arranging internal passages and bathrooms between the corridor and the living room or bedrooms.

Kitchens and water-closets should not be planned over living rooms and bedrooms, whether within the same dwelling or over other dwellings. Soil pipes should not be carried in ducts which adjoin living rooms or bedrooms unless the side of the duct next to these rooms is a solid wall containing no inspection openings. Refuse chutes should not be planned next to living rooms or bedrooms.

5.2.3 Sound Insulation

5.2.3.1 Reduction of airborne noise

The weighted sound reduction index, R_w , of partitions between individual rooms or apartments of a building unit shall be as given in [Table 5](#). These values may, however, be suitably increased, where required, for critical areas.

Table 5 Sound Insulation between Individual Rooms (Airborne)		
(Clause 5.2.3.1)		
SI No.	Situation	R_w dB
(1)	(2)	(3)
i)	Between the living room in one house or flat and the living room and bed-rooms in another	50
ii)	Elsewhere between houses or flats	45
iii)	Between one room and another in the same house or flat	35
NOTES		
1 Where communicating doors are provided, all doors should be so designed as to provide recommended insulation between the rooms.		
2 There are cases when a set of houses or flats have to be built for the people who work at night and sleep during the day. It is desirable to consider the design of at least one such room in each of the houses or flats which will provide an insulation of about 45 dB in that room.		
3 The insulation values referred to are applicable with doors and windows shut.		

5.2.3.2 Suppression of noise at the source itself

All items of equipment that are potentially noisy should be selected with care. Water-closet cisterns should not be fixed on partitions next to bedrooms or living rooms. Plumbing pipes should be isolated from the structures. Lift motors should be mounted on resilient supports. Access doors from machine rooms to internal staircases should be well fitting and of solid construction. Special noise control measures may be required for electrical and mechanical services such as diesel generators, outdoor air conditioning units, cooling towers, etc.

5.2.3.3 Reduction of airborne noise transmitted through the structure

Reduction of airborne noise requires the use of rigid and massive walls, or acoustically designed dry walls, without any openings. Openings are the major cause of penetration of noise through a barrier. While designing it should be borne in mind that all components should provide a sound transmission compatible with that of the rest of the barrier so that an equivalent amount of sound energy is transmitted through each portion of the barrier.

Ventilating ducts or air transfer openings where provided should be designed to minimise transmission of noise. For this purpose, sound attenuating devices having necessary insertion loss may be installed in these openings.

All partitions should be sealed effectively where they butt against rest of the structure. All doors and windows should be properly gasketed where a high degree of sound insulation is desired.

5.2.3.4 Reduction of structure-borne noise

This requires the use of discontinuous or non-homogeneous materials in the construction of the structure.

5.2.3.5 Reduction of impact noise

The floor of a room immediately above the bed room or living room shall result in an impact sound pressure level ($L'_{nT,w}$) not greater than 60 dB. Typically, a 150 mm thick concrete floor with thick carpet (12 mm) covering would satisfy this requirement.

5.2.3.6 Main staircases in blocks of flats are often highly reverberant. Some of the surfaces at least (for example, the soffits of stairs and landings) should be finished with sound absorbent materials, wherever required.

6 EDUCATIONAL BUILDINGS

6.1 Sources of Noise Nuisance

6.1.1 Outdoor Noise

The outdoor sources of noise produced on school premises, which cause disturbance within the school, include the noise arising from playgrounds, playing fields and open-air swimming pools. Though playgrounds are used mainly during break periods, they are also used for games and physical education at times when teaching is in progress in the adjoining class rooms.

6.1.2 Indoor Noise

Indoor sources of noise are as follows:

- a) Singing, instrumental and reproduced music which may take place in class rooms and in dining and assembly halls particularly in primary schools. In secondary schools, specialized music rooms are generally provided;
- b) Movement of chairs, desks and tables at the end of one period may disturb a class engaged in a lesson in a room below;
- c) Shutting and openings of doors and windows which may occur at any time during teaching periods;
- d) Audio-visual presentations in class rooms;
- e) Wood and metal workshops, machine shops (engineering laboratories), typing rooms, etc that produce continuous or intermittent sound of considerable loudness;
- f) Practical work carried out in general teaching areas;
- g) Gymnasia and swimming pools;
- h) School kitchens and dining spaces where food preparation and the handling of crockery and utensils persist for the greater part of the school day;
- j) Corridors and other circulation spaces; and
- k) Plumbing and mechanical services.

6.2 Recommendations

6.2.1 Site Planning

Where outdoor noise nuisance exists from local industry, busy roads, railway, airfields, sport grounds or other sources beyond the control of the school authority, school buildings should be sited as far away from the sources of noise, as possible.

6.2.1.1 Rooms should be planned in a manner so that the minimum amount of glazing is placed on the side facing the external noise.

6.2.1.2 Noises arising from the activities of a school and from the use of the buildings after school hours may constitute a nuisance to occupants of surrounding property; therefore, it is desirable to place playgrounds, workshops, swimming pools, music rooms, assembly halls and gymnasias as far away as possible from buildings which require a quiet environment.

6.2.2 Internal Planning

The following principles should be observed in the detailed planning of educational buildings:

- a) *Grouping* — Noisy rooms should be separated from quiet ones, if possible. In general, it is desirable that rooms should be grouped together in accordance with the classification given in [6.2.4.1](#);
- b) *Windows and ventilators* — Windows of noisy and quiet rooms should not open on to the same courtyard or be near to one another. Skylights and ventilators over noisy rooms should be avoided, if they are likely to be a source of nuisance to adjacent upper floors;
- c) *Doors* — Swing doors into rooms should only be used where no problem of sound transmission exists. Reduction of insulation between rooms and corridors due to doors shall be borne in mind. The type and method of fitting of doors is important and necessary care should be paid in this respect;
- d) Sliding partitions, if used, should be acoustic operable partitions. Pass-through doors, if provided, should be acoustic doors with drop seals;
- e) *Open planning and circulation areas* — Where open planning is used to permit spaces, such as assembly halls, dining rooms or entrance halls to be used in association with each other or for circulation, the degree of disturbance caused by interfering noise to teaching areas needs careful consideration; traffic through such areas should be strictly controlled; full use should be made of sound absorbent treatments to reduce the spread of noise from one space to another (see [6.2.3](#));

If rooms have large glazed panels or ventilation openings facing directly on the circulation areas, human traffic passing by the rooms should be controlled. Preferably baffled ventilation system or double windows should be used. Fan-lights over doors should be fixed and glazed; and

- f) *Furniture* — In all educational buildings, regardless of the character of the floor finish, rubber buffers should be fitted to the legs of chairs and tables.

6.2.3 Noise Reduction within Rooms

Sound absorbent materials play a useful part in reducing the built-up or air-borne noise at source. In rooms, such as class-rooms, assembly halls and music rooms, a fairly short reverberation time under occupied conditions is one of the requirements of the acoustic design. The maximum reverberation times permissible for this purpose are usually short enough to give adequate noise control but in addition, the reverberation time should not be excessive under empty conditions, because noise may occur in these rooms with very few occupants. [Table 6](#) gives the reverberation times often arranged in occupied rooms for acoustic reasons and the maximum times recommended in the empty rooms for noise reduction; the times given are for a frequency of 500 Hz, but they should not be greatly exceeded at any frequency. When rooms are used for a variety of purposes, the reverberation period appropriate to the major use should be adopted.

6.2.3.1 Special attention should be given to noise reduction in schools for the hearing impaired and schools for the visually impaired. Hearing impaired children are taught by means of hearing aids which cannot be used satisfactorily in high noise levels or in reverberant conditions. Visually impaired children depend on good hearing for understanding speech and for detecting changes in environment. In both these types of schools, noise levels should be kept low and reverberation times short. As an example, the reverberation times in empty class-rooms should not exceed 1.0 s in schools for the visually impaired or 0.5 s in schools for the hearing impaired.

Table 6 Reverberation Times in Schools

(Clause 6.2.3)

Sl No.	Room	Reverberation Time ^s	
		Usual for Acoustic Reasons (Full)	Maximum ¹⁾ for Noise Control (Empty)
(1)	(2)	(3)	(4)
i)	Assembly halls	1.0 to 1.25 according to size	1.5 to 2.0 according to volume of hall
ii)	Music teaching rooms	0.75 to 1.25	1.5
iii)	Gymnasia and indoor swimming pools	—	1.5
iv)	Dining rooms	—	1.25
v)	Classrooms	0.6	1.1
vi)	Headmasters room and staff rooms	0.5 to 1.00	1.0

¹⁾ Shorter reverberation times are desirable for noise control, whenever possible.

6.2.4 Sound Insulation

6.2.4.1 Airborne noise

For purposes of sound insulation, rooms in educational buildings may be classified as follows:

Class A	Noise Producing	a) Workshops b) Kitchens c) Dining Rooms d) Gymnasiums e) Indoor Swimming Pools
Class B	Noise producing but needing quiet at times	a) Assembly halls b) Lecture halls c) Music rooms d) Typing rooms
Class C	Average	a) General class rooms b) Practical rooms c) Laboratories d) Offices
Class D	Rooms needing quiet	a) Libraries b) Studies
Class E	Rooms needing privacy	a) Medical rooms b) Staff rooms

6.2.4.2 The recommended minimum sound reduction (D_w) between rooms of the same class is as follows:

Class A	—	25 dB
Class C or D	—	35 dB
Class B or E	—	45 dB

6.2.4.3 Where a room is likely to have a dual use, for example, a dining room to be used as a class-room, the higher sound insulation value should be used.

6.2.4.4 The recommended minimum sound reduction (D_w) between rooms in different classes is 45 dB subject to the following:

- In schools or institutes with a technical bias where noisy activities, such as sheet metal work, plumbing and woodwork, are likely to be practised extensively in normal hours, workshops should be regarded as a special category requiring more than 45 dB isolation (D_w) from rooms of any other class;
- Assembly halls and music rooms are special cases in that, as well as producing noise, they also require protection from it and may need more than 45 dB isolation (D_w) from rooms in Class A, if the latter are very noisy;
- Circulation spaces may vary from a long and frequented corridor to a small private lobby and it is therefore difficult to give precise recommendations to cover them. For partitions between rooms in Class C and most corridors, an R_w of 35 dB for the partition itself is adequate. For partitions between rooms in other classes and corridors, more or less insulation may be necessary, depending upon the specific usage; and
- The problem of noise in circulation areas is as a rule greatly mitigated in schools by the fact that classes usually change rooms together at regular times. In colleges and evening institutes, however, this is much less true and in such buildings particular attention should be paid to insulation between rooms and corridors.

6.2.4.5 Impact noise

In the case of schools, the concrete floor of the room immediately above the teaching rooms shall provide an impact sound pressure level, $L'_{nT,w}$ not greater than 70 dB. For example, a covering of 6 mm linoleum or cork tiles on concrete floor (hollow or solid) weighing not less than 220 kg/m² will usually meet the above requirement.

7 HOSPITAL BUILDINGS

7.1 General

Problems of noise control vary from hospital to hospital but the principles outlined below apply to all types. A quiet environment in hospitals is desirable for patients who are acutely ill. Staffs require quiet conditions for consultations and examinations and also in their living and sleeping quarters. There have been rapid rises in noise levels in hospitals due to the higher levels of outdoor noise, increasing use of mechanical and mobile equipment (some of which is now brought much nearer to the patient in order to facilitate nursing procedure) and the introduction of loudspeaker, radio, television and call systems. Noise control in the hospital is made much more difficult by the extensive use of hard washable surfaces which reflect and intensify the noise. In most hospitals, windows to the open air and fanlights to corridors are usually open for the purpose of ventilation, admitting noise from outside and allowing it to spread through the building.

7.2 Sources of Noise Nuisance

7.2.1 Outdoor Noise

This may be classified into two main categories:

- Noise from sources outside the hospital premises, for example, traffic and industrial noises; and
- Noise from sources outside the building but usually within the control of the hospital authority, for example, ambulances, motor-cars and service vehicles, fuel and stores deliveries, laundries, refuse collection, trucks and trolleys.

7.2.2 Indoor Noise

A hospital is a complex building with many services and the numerous internal sources of structure-borne and airborne noises are grouped into three main categories:

- a) Noise consequent upon hospital routines. This category includes sources which transmit noise through both structure-borne and air-borne paths, many of which may be quite near to patients particularly those in wards, such as the following:
 - 1) Wheeled trolleys of various kinds, for food and medical supplies;
 - 2) Sterilizing equipment;
 - 3) Sluice room equipment including bedpan washers;
 - 4) Ward kitchen equipment;
 - 5) Footsteps;
 - 6) Doors banging;
 - 7) Handling of metal or glass equipment;
 - 8) Noises caused during maintenance and overhaul of engineering services; and
 - 9) Vacuum cleaners, mechanical polishers, etc.
- b) Loudspeaker, radio or television, audible call system, telephone bells and buzzers, and other air-borne noises, such as loud conversation; and
- c) Noises from fixed or mobile equipment and services not directly concerned with hospital routines. These include all the fixed services as given below:
 - 1) Plumbing and sanitary fittings;
 - 2) Steam, hot and cold water and central heating pipes;
 - 3) Ventilation shafts and ducts;
 - 4) Fans
 - 5) Boilers;
 - 6) Pumps;
 - 7) Air compressors;
 - 8) Pneumatic tubes;
 - 9) Electrical and mechanical motors and equipment;
 - 10) Lifts;
 - 11) Laundry equipment; and
 - 12) Main kitchen equipment (refrigerators, mixers, steam boilers, etc).

7.3 Recommendations

7.3.1 Site Planning

Hospital sites with their high degree of sensitivity to outside noise should be as far away from outside sources as may be compatible with other considerations, such as accessibility and availability of services. The building should be so arranged on the site that sensitive areas like wards, consulting and treatment rooms, operating theatres and staff bedrooms are placed away from outdoor sources of noise, if possible, with their windows overlooking areas of acoustic shadow.

7.3.2 Detailed Planning

There is a very large number of unit and room classification in hospital design and in planning the units in relation to each other and to the common services, (such as X-ray departments, operating theatre suits and main kitchens), noise reduction in the sensitive areas should be weighed carefully against other design factors. Special care in

overall planning and internal planning against noise is required in the planning within the building of units which are themselves potential noise sources, for example, children's wards and outpatients' departments, parts of which require protection against noise.

7.3.2.1 Unloading bays, refuse disposal areas, boiler houses, workshops and laundries are examples of service units which should be as far from sensitive areas as possible.

7.3.2.2 The kitchen is a constant source of both air-borne and structure-borne noise and should preferably be in a separate building away from or screened from the sensitive areas. If this is not possible and the main kitchens shall form part of a multi-storey building, noise control is easier if they are placed below and not above the wards and other sensitive rooms so as to facilitate the insulation of the equipment and machinery in order to reduce the transmission of structure-borne noise to a minimum.

7.3.2.3 In ward units, the kitchens, sluice rooms, utility rooms, sterilizing rooms and other ancillary rooms, need to be placed quite near to the beds if they are to fulfil their purposes, which are all sources of noise. Some form of noise baffling between open wards and rooms of this kind will be needed.

7.3.3 Reduction of Noise at Source

In view of the difficulty of suppressing noise in hospital buildings, it is important to eliminate noise at its source, wherever possible.

7.3.3.1 Use of resilient material

Mats of rubber or other resilient material on draining boards and rubber-shod equipment will greatly reduce noise from utility rooms, sluice rooms and ward kitchens. The use of plastics or other resilient materials for sinks, draining boards, utensils and bowls would also reduce the noise. Many items of equipment especially mobile equipment, such as trolleys, beds and movable furniture, may be silenced by means of rubber-tyred wheels and rubber bumper and the provision of resilient floor finishes (see [7.3.4.1](#)). The latter also reduces footstep noise. Silent type curtain rails, rings and runners should be used. Lift gates and doors should be fitted with buffers and silent closing gear. Fans and other machinery should be mounted on suitable resilient mountings to prevent the spread of noise through the structure.

7.3.3.2 Other measures

Noise from water or heating pipes may be reduced by installing systems which operate at comparatively low pressure and velocities. Silencing pipes and specially designed flushing action reduce water closet noise at source and make structural measures easier to apply. The ventilation system should be designed so as not to create a noise problem. Silent closers should be fitted to doors.

7.3.4 Reduction of Noise by Structural Means

7.3.4.1 Insulation

Since the various departments or units may be planned in many ways, only general guidance on the insulation values for walls and partitions is as given below:

- a) It is recommended that walls or partitions between rooms should normally have an R_w of at least 45 dB. Higher values of R_w of at least 50 dB are necessary where a noisy room is adjacent to one requiring quiet conditions. Doors should be solid with close fitting in the frames;
- b) There is little insulation value in double swing doors and where these are fitted to a noisy room the opening should be planned so that it is screened from areas requiring quiet by a baffle lobby lined with absorbent material. Very high insulation values may be necessary in special cases and exceptional measures may be required;
- c) Solid floors with floating finishes and resilient surfaces are necessary particularly between wards and other parts of the building. Ordinary timber board on joist floors should never be used; and
- d) Conduits, ventilation ducts, chases etc, should be constructed so as not to form easy by-pass for disseminating noise about the building, and should be provided with sufficient sound insulation. Pipe

ducts should be completely sealed around the pipes where they pass through walls or floors. Ducts carrying waste or water pipes should be lined with sound insulating material to prevent noise from the pipes passing through duct walls into the rooms through which they pass.

7.3.4.2 Absorption

Most surfaces in hospitals should be easily cleanable, so as to prevent the build-up of bacteria which may cause cross-infection. Many sound absorbent materials of a soft nature and difficult to clean are unsuitable for use in some hospital areas and lose much of their effectiveness, if painted for hygienic reasons.

Some porous materials with very thin non-porous coverings (like mineral wool covered with thin plastic sheets) have good sound absorption and when covered with a perforated sheet metal facing can be used in most areas requiring a washable acoustical treatment. In noisy areas, such as corridors and waiting rooms, however, a wider choice of absorbents is available.

In the ward, bed curtains, window curtain, etc, add to the absorbent properties of the room and help reduce reverberation in otherwise hard surfaced surroundings.

7.3.5 Sensitive areas such as operation theatres, doctors' consultation rooms, intensive care units (ICU) require special consideration against noise control. Apart from outdoor noise, a common problem is the transmission of sound between the consulting room and the waiting room. To ensure silence, a sound isolation D_w of 45 dBA, between the rooms shall be provided. If the doors are directly connected by a single communicating door it will not be possible to achieve these values of isolation D_w . To obtain 40 dBA to 45 dBA insulation between communicating rooms, it is necessary to provide two doors separated by an air gap, such as a lobby or corridor.

8 OFFICE BUILDINGS

8.1 General

Modern office buildings are often noisier than older buildings due to the use of thinner and more rigid forms of construction, harder finishes, more austere furnishings and use of business machines.

8.2 Sources of Noise Nuisance

8.2.1 Indoor Noise

Main sources of indoor noise include the following:

- a) Office machines, such as keyboards, typewriters, and printers;
- b) Telephonic conversation;
- c) Noise from the public admitted to the building;
- d) Footsteps, voices and slamming of doors in circulation spaces, lift doors and gates;
- e) Sound reproduction in staff training rooms, conference rooms and recreation rooms, etc;
- f) Handling of crockery and utensils in canteens and kitchens; and
- g) HVAC and lift machinery.

8.3 Recommendations

8.3.1 Site Planning

Rooms demanding quiet conditions should be placed on the quiet side of the site. Even on quiet thoroughfares, these rooms should also not be planned at street level. They should also not be planned on enclosed yards used for parking of cars, scooters, etc. Where, however, the problems cannot be resolved by planning, the provision of double windows may be necessary.

8.3.2 Detailed Planning

8.3.2.1 Noise reduction within rooms

The reverberation time should not exceed 1.0 s in all general offices of the types listed in [8.3.2.2](#) to [8.3.2.6](#). In small private offices, the reverberation time should not exceed 0.75 s, in very large offices the reverberation time may be increased to 1.0 s. For canteens, the recommended maximum reverberation time is 1.0 s.

NOTE — All reverberation times are specified for a frequency of 500 Hz.

8.3.2.2 Large general offices

The grouping of departments and machines together in one room should be avoided, wherever possible. Where supervision is necessary the provision of glazed screens carried up to the ceiling should be considered. If it is essential to the work of an office for machine operators and clerks to work side by side in the same room, the machines should be enclosed by panels or low screens lined with absorbent material and the ceiling should be sound absorbent. In addition, the machines should be as quiet as possible in operation and mounted on suitable resilient mountings.

NOTE — A quiet area should be planned for prolonged telephonic conversation.

8.3.2.3 Light weight construction

Modern construction methods and economy dictate the use of light weight construction for many office buildings. While the lightweight materials lead to fast fabrication and erection and also effect considerable economy in the building structure, they may lead to tremendous sound insulation problems between adjacent offices and areas. Light weight construction is also frequently employed for the subdivision of large space into executive cabins and secretarial areas. Where such construction is considered desirable, efforts should be made to provide a double-skin panel. The panels should be isolated from each other as far as possible either by the use of separate framing or by the use of elastic discontinuities in the construction, and a sound absorbing material may be introduced in the air cavity between the panel. The partitions should be full height up to the bottom of the roof above and any openings required for air movement should be provided with sound attenuators compatible with the rest of the partition.

When light weight floors are provided in multi-use buildings, adequate attention should be paid to the question of air-borne and structure-borne noise transmission from the upper floors to the floors below. For effective reduction of air-borne noise, a double panel hollow floor construction may be employed with some heavy sound damping material introduced between the panels and the panel isolated from each other. The sound damping material could be sand, mineral wool, etc. In case impact noise isolation is also required, the upper panel should be effectively isolated from the rest of the floors and building structure. The choice of the isolation layer would of course depend upon the lowest frequency of interest.

Another point to be kept in mind when going in for light weight construction is to ensure that the light weight panels are not in resonance with the natural frequencies of any mechanical equipment installed inside the building. Lightweight materials have high natural frequencies well within the audio range and may resonate or vibrate due to an applied vibratory force. This vibratory force is caused by mechanical equipment, road traffic, rail traffic, etc. Special measures also need to be taken to isolate either the source or the building so as to reduce the amount of vibration transmitted to the building structure.

8.3.2.4 Open plan offices

A new concept in office planning is the use of open plan offices. Large open floor spaces are converted into an office area with senior executives, junior executives and secretarial staff all seated within the same area without the use of any partitions or walls. While this method of planning is appreciated, it leads to a problem of inadequate acoustical privacy between adjacent work spaces. Speech privacy in open plan offices is defined by the speech interference level of intruding noise. Speech privacy between two adjacent rooms or spaces is, therefore, a function of two key parameters; noise reduction of the intervening partition and background noise levels.

Special design measures are, therefore, required to reduce the level of intruding sounds at work places to acceptable low value so that people are not disturbed and adequate privacy is maintained. Some special measures which might be considered for such open plan offices are the use of an acoustical ceiling together with partial

height barriers between work spaces, all designed to provide adequate privacy between adjacent work spaces. In addition, use may have to be made of an electronic background masking noise system which provides a constant level of a generally acceptable background noise in the entire office area. The masking noise system is a very useful concept in open plan office design because by raising the background level at every workplace, intruding noises are made less disturbing. A background music system cannot serve as a noise masking system because the music does not have a constant spectrum or sound level. In fact, the background noise masking system shall be introduced gradually without the knowledge of employees. The air-conditioning system can also be used to generate background masking noise if the noise level from the fans, ducts and grills is suitably tailored to generate the desired frequency spectrum. However, it is not simple to predict the noise level of air-conditioning components accurately. On the other hand, the electronic system enables both the level and the spectrum of the background noise to be accurately adjusted to suit individual job requirements.

8.3.2.4.1 In addition to comfort, speech privacy and sound insulation play a critical role in preventing data theft or leakage of sensitive information. Improper acoustical design can lead to unauthorized access to confidential conversations, especially in organizations handling financial data, legal matters or classified projects. Speech leakage across open-plan offices occurs when sound waves are transmitted either directly or indirectly through reflective surfaces, openings, or poorly isolated barriers. The sound control measures given below should be employed to mitigate this risk and ensure information security.

a) *Control of direct and indirect sound paths*

Partial-height barriers alone are often insufficient to block sound transmission, as sound can travel over the barriers and reflect off ceilings, windows, or walls. To prevent sound leakage, barriers between workstations should be designed with sound-absorptive materials to reduce reflections. In high-security zones, barriers should ideally reach ceiling height to limit the direct path of sound waves. Additional sealing measures, such as acoustic baffles or sound-blocking panels near ceilings, can help to further contain sound within specific zones;

b) *Masking noise for speech concealment*

Even with physical barriers in place, speech masking systems play a crucial role in enhancing speech privacy by rendering conversations unintelligible beyond a specific range. These systems should generate uniform background noise across the entire office, tuned to a frequency spectrum that aligns with typical human speech, thereby reducing the clarity of overheard conversations. Care should be taken to avoid variations in masking noise across different zones, as uneven noise profiles can inadvertently create 'quiet spots' where speech becomes more easily discernible;

c) *Acoustic zoning for sensitive areas*

Workspaces dealing with confidential or sensitive data should be zoned acoustically to limit sound propagation beyond their intended boundaries. These zones can be created through sound-dampening partitions, low-noise areas, and placement of workstations away from high-traffic zones. Additionally, team discussions or meetings involving classified topics should occur in acoustically treated rooms with sealed doors and walls to avoid inadvertent sound leakage into adjacent spaces;

d) *Control of reflective surfaces*

Reflective surfaces such as glass partitions, metal ceilings, and large windows can amplify sound reflections, making speech intelligible at greater distances. Such surfaces shall be treated with sound-absorbing films, acoustic panels, or curtains to minimize reflection. Flooring materials such as carpets or rugs may also be used to reduce the transmission of footfall noise, which could otherwise interfere with the effectiveness of masking systems; and

e) *Air-conditioning systems and noise control*

Although air-conditioning systems can generate a degree of background noise, it is critical to ensure that the noise they produce remains predictable and stable. Inconsistent air-conditioning noise can undermine masking systems, creating intermittent quiet periods that increase the risk of speech leakage. Ducts, grilles, and vents shall be designed and placed carefully to maintain a uniform noise level throughout the office area, ensuring no unintended listening opportunities arise.

8.3.2.5 Office equipment rooms

It is important that machines like keyboards, typewriters, printers, etc, should be quiet in themselves and also be fitted with resilient pads, to prevent the floors or tables on which they stand from acting as large radiating panels. It is desirable to locate machines further apart and to apply sound absorbent treatment to the ceiling.

8.3.2.6 Banking halls

If banking halls are large and lofty, noise nuisance tends to be aggravated. It is advisable to avoid high reflective ceilings. The worst effects may be reduced by segregating the noise from the quiet operations and screening one from the other and by applying sound absorbent materials to the surfaces of the ceilings, screens and nearby walls. Resilient flooring is also recommended.

8.3.2.7 Public offices and waiting spaces

Noise nuisance may be minimised by the provision of resilient flooring, sound absorbent ceilings and heavy full height screens between the public space and the clerical office.

8.3.2.8 Canteens

The provision of a sound absorbent ceiling, resilient flooring and the use of plastics/wooden trays and tables with quiet tops are recommended.

8.3.2.9 Circulation spaces

The effective length of long corridors should be limited by providing swing doors at intervals. Hard floor finishes and board and batten floors in corridors should be avoided. The provision of a sound absorbent ceiling in corridors is recommended. Floor ducts should be planned on one side of corridors.

The noise from slamming of doors may be reduced by fitting automatic quiet action type door closers. Door buffers are useful but may reduce insulation of air-borne sound due to the inevitable gaps between buffers. Continuous soft, resilient strip let into the door frames is preferable. The use of quiet action door latches is recommended.

Staircases and lifts should be isolated from quiet rooms and should have silent type doors.

8.3.3 Requirement of Sound Insulation

With open window (single or double) the sound reduction (D_w) will be 5 dB to 10 dB, and with sealed double windows it will be 40 dB to 45 dB. Intermediate values are obtainable with closed openable windows (single or double) but only, of course, at such times as ventilation may be dispensed with. Having to choose between ventilation and noise exclusion is a serious handicap to efficient working in offices. In large office blocks on noisy sites, consideration should be given to the provision of sealed double windows and mechanical ventilation at least in the offices on the sides of the building exposed to noise.

8.3.3.1 The insulation necessary between adjoining rooms, both horizontally and vertically, depends upon the amount of noise created within the rooms, the amount of intruding noise and whether it is important that conversation should not be overheard between rooms. Generally, a sound isolation value (D_w) of 40 dB between one room and another room in office is recommended.

8.3.3.2 The following list may be considered as broad classification of noise producing rooms and rooms requiring quiet though many offices fall into both categories. Where rooms in opposing categories are planned adjacent to each other, a sound reduction (D_w) of at least 45 dB should be provided between them.

<i>Sl No.</i>	<i>Noise Producing Rooms</i>
(1)	(2)
i)	Entrance and reception halls, staircases and corridors used by the public
ii)	Lifts and lift halls

<i>Sl No.</i>	<i>Noise Producing Rooms</i>
(1)	(2)
iii)	Motor and plant rooms
iv)	Lavatories
v)	Public offices
vi)	Canteen and kitchens
vii)	Office machine rooms and typing pools
viii)	Recreation rooms
ix)	Large general offices
x)	Cinemas and projection rooms

<i>Sl No.</i>	<i>Rooms Requiring Quiet Conditions</i>
(1)	(2)
i)	Executive's rooms, Conference rooms and Board rooms
ii)	Interview rooms
iii)	Offices for one or two persons
iv)	Medical officer's rooms
v)	Sick rooms
vi)	Rest rooms
vii)	Libraries
viii)	Telephoning rooms

The recommended sound reductions (D_w) are as follows:

a)	Rooms requiring quiet (as listed above) on a quiet site where privacy is required	45 dB
b)	Rooms requiring quiet (as listed above) but on a noisy site or where a lower degree of privacy is tolerable	40 dB
c)	Clerical offices in which noise does not constitute a major nuisance	30 dB

8.3.3.3 It is recommended that the minimum sound reduction index, R_w for floors should be 45 dB, and the floors should have a resilient finish.

9 HOTELS AND HOSTELS

9.1 General

Hotels and hostels are primarily used as dwelling units, and hotels also provide for public entertainment. The most serious risk, of course, is disturbance to sleep, and adequate care, therefore, needs to be taken to protect the occupants from being disturbed by outdoor and indoor noise.

9.1.1 Outdoor Noise

Hotels near railway stations, airports, highways and those situated in highly urbanized areas are especially vulnerable to outdoor noise. The outdoor noise in many of the areas is of a high level even late at night and in the early morning. The noise could also be due to other types of activities such as building construction activity

(pile driving, concrete mixing, etc) and various types of portable utility equipment, such as compressors or generators.

9.1.2 Indoor Noise

Insofar as indoor noise is concerned, the noise could be due to the occupants themselves, which is transmitted from one room to the other. It could also be due to public functions and late night use of restaurants located in the hotel, as also due to miscellaneous utility equipment installed for providing and maintaining the services in the hotel, such as air-conditioning equipment, pumping equipment, power laundry and kitchen. Sometimes hotels equipped with standby generators are a potential source of noise. Another source which could lead to disturbance to the occupants is the plumbing system.

9.2 Recommendations

9.2.1 Site Planning

While it is desirable to locate the hotel, or hostel away from an area where there is a high ambient noise level, many a time these have to be located in noisy areas for public convenience. Hotels near airports and railway stations are becoming popular because they are convenient for passengers in transit. Hotels located in the commercial areas of a city are also a commercially viable proposition and many a time this factor outweighs the other problems associated with such a location. When a reasonably quiet location is not possible, it is desirable that adequate measures be considered to provide a comfortable acoustical environment for the occupants.

9.2.2 Internal Planning

Where a hotel is located in a noisy environment, the provision of sealed windows (single or double) and provision of an air conditioning system is desirable for rooms exposed to noise. The requirements for the windows would of course depend upon the level and character of noise in the area.

The general recommendations for satisfactory acoustical design of hotels and hostels are given in [9.2.2.1](#) to [9.2.2.7](#).

9.2.2.1 Hotels of all classes should by necessity provide good protection against indoor noise. Since hotels can be considered as flats, the standards of protection recommended for flats are also applicable to hotels. Partition between guest rooms and between rooms, corridors and floors should not be less than 230 mm brick wall plastered or equivalent ($R_w = 50$ dB). The floors should have proper impact insulation. Special attention should be paid to built-in wall cupboards as these are potential areas of sound leakage. These will not serve as sound insulating partitions and may not be relied upon to increase the insulation value of partitions against which they may be built. In fact, partitions between adjoining rooms should be continuous behind the cupboards. Use of silent type door gear and cupboard catches is also highly desirable.

9.2.2.2 Door openings on opposite sides of corridors should be staggered and doors shall be provided with gaskets on head, sides and threshold. As an alternative to thresholds, drop seals may be provided. Inter-communicating doors should be double doors, fully gasketed. Doors should also have quiet action latches. Whenever possible, rooms should be entered through a baffle lobby. Wherever possible, corridor walls should not have ventilators unless they are double glazed and non-openable.

9.2.2.3 Corridors and staircases may have resilient floor coverings and sound absorbent ceilings are desirable unless the corridor is fully carpeted. Staircases and lift wells may be cut off from corridors by means of swing doors and, if possible, isolated from guest rooms by linen stores or similar rooms. Room service pantries on floors can also be a source of noise and may be separated from corridors by baffle lobbies, unless the rooms themselves have baffle lobbies.

9.2.2.4 Except within the same suite, bathrooms should not be planned next to bedrooms. Where this is unavoidable, internal pipe shafts with heavy walls, unpierced on bedrooms' side may be used as means of separation. It is important to choose quiet type of sanitary fittings and to design the plumbing system so as not to create noise, that is, by avoiding sharp bends, restrictions of flow, quick-action valves that might cause water hammer, etc.

9.2.2.5 Air-conditioning system should be quiet in operation. Care should also be taken that the air-conditioning ducts do not lead to a cross-talk problem between rooms. Suitable acoustical lining should be provided in the ducts consistent with the fire safety requirements of the buildings.

9.2.2.6 Large hotels often have banquet halls and conference halls which are separately hired out for public and private functions. Late night restaurants and night clubs are also popular and functions in all these areas may go on well into the night. It is therefore essential that these rooms be effectively isolated from bedrooms and effective insulation from all possible noise sources is considered. Here it is not only necessary to consider the air-borne sound insulation but it is also necessary to consider the question of structure-borne and impact noise transmitted from areas where there might be dancing late into the night. Floating floors may be considered for structure borne sound isolation for dance floors and loudspeakers.

9.2.2.7 While most of the noise problems encountered in hotels are applicable to hostels, the latter are normally of more economical construction and, therefore, cannot cater for special sound insulation provisions. However, as far as possible, precautions should be taken to provide comfortable conditions in hostel rooms. This is especially true for student hostels where each room is also a living room. Students might play music or have loud discussions late into the night.

This may disturb sleep or study of other students. Proper precautions should, therefore, be taken to provide satisfactory conditions.

10 INDUSTRIAL BUILDINGS

10.1 General

Industrial buildings are primarily producers rather than receivers of noise. The level of industrial noise commonly exceeds that from any other source with the exception of aircraft. As compared with traffic noise, its effects are less widespread but it is often more annoying in character.

10.1.1 Many industrial noises contain very strong high frequency whines, screeches and clatter - these components are relatively more attenuated by passage through the air and by the insulation of light structure than are lower frequencies.

10.1.2 Intermittent noises are either isolated explosions or reports, or noises of a periodic nature, such as those of pressure relief valves of blow off, or the noises of work occurring at random intervals, for example, hammering, grinding and sawing operations; the latter class may be especially irritating because of high frequency components.

10.2 Sources of Industrial Noise

10.2.1 Noises in industrial buildings are mainly of indoor origin. Noise in factories and workshops is generally caused by machine tools and by operations involved in making and handling the product and they are classified into the following groups, depending upon how the noise energy is generated.

10.2.1.1 Impact

Noise caused by impact is the most intense and widespread of all industrial noises. It is normally coupled with resonant response of the structural members connected to the impacting surface. Common sources of this type of noise are forging, riveting, chipping, pressing, tumbling, cutting, weaving, etc. Intense impact noise may also be produced during handling of materials as in the case of sheared steel plates falling one over another in collecting trays in a steel factory. Impact noise is usually intermittent and impulsive in character, but it may also be continuous as in the case of tumbling.

10.2.1.2 Friction

Most of the noise due to friction is produced in such processes as sawing, grinding and sanding. Friction also occurs at the cutting edge on lathes and other machine tools and in brakes and from bearings. The spectrum of frictional noise often predominates in high frequency and is very unpleasant in character.

10.2.1.3 Rotation and reciprocation

A rotating or reciprocating machine generates noise due to unbalanced forces and/or pressure fluctuations in the fluids inside the machines. In many cases, the moving surfaces radiate noise directly and in other cases, the pressure fluctuations are transmitted to the outer casings of the machine from where they are radiated as noise. Interaction of rotating component with the fluid stream can also give rise to pure tone components, such as the whine in a turbine. Since most machine casings have radiation efficiencies of unity in the higher frequency range, the amount of sound radiated is often substantial.

10.2.1.4 Air turbulence

Noise may be generated by rapid variation in air pressure caused by turbulence from high velocity air, steam or gases. Common examples are the exhaust noise from pneumatic tools and air jets. The noise is intense, and broad based in character and the frequency criteria depends on the size of the jet. The intensity increases rapidly with the velocity of the air stream.

10.2.1.5 Noises with pure tone components

Whining noise from turbines and humming noise from transformers come under this group.

10.3 Noise Criteria

10.3.1 Hearing Damage – Risk Criteria

Continuous exposure to high noise levels may result in permanent noise induced hearing loss in the course of time. Damage-risk criteria specify the maximum levels and duration of noise exposure that may be considered safe. Generally accepted damage-risk criteria for exposure to continuous, steady broad band noise are shown in [Table 7](#). Whenever the sound levels at the worker's position in a factory exceed the levels and the duration suggested, feasible engineering controls shall be utilized to reduce the sound to the limits shown. If such controls fail to reduce sound levels within the levels of [Table 7](#), personal hearing protection equipment shall be provided and used to reduce sound levels within the level shown.

Table 7 Permissible Exposure Limits for Steady-State Noise (Clause 10.3.1)		
SI No.	Sound Level (Slow Response) dBA	Time Permitted, T h : min
(1)	(2)	(3)
i)	85	16 : 00
ii)	86	13 : 56
iii)	87	12 : 08
iv)	88	10 : 34
v)	89	9 : 11
vi)	90	8 : 00
vii)	91	6 : 58
viii)	92	6 : 04
ix)	93	5 : 17
x)	94	4 : 36
xi)	95	4 : 00
xii)	96	3 : 29
xiii)	97	3 : 02

Table 7 (Concluded)

SI No.	Sound Level (Slow Response) dBA	Time Permitted, <i>T</i> h : min
(1)	(2)	(3)
xiv)	98	2 : 50
xv)	99	2 : 15
xvi)	100	2 : 00
xvii)	101	1 : 44
xviii)	102	1 : 31
xix)	103	1 : 19
xx)	104	1 : 09
xxi)	105	1 : 00
xxii)	106	0 : 52
xxiii)	107	0 : 46
xxiv)	108	0 : 40
xxv)	109	0 : 34
xxvi)	110	0 : 30
xxvii)	111	0 : 26
xxviii)	112	0 : 23
xxix)	113	0 : 20
xxx)	114	0 : 17
xxxi)	115	0 : 15

NOTES

1 Where the table does not reflect the actual exposure times and levels, the permissible exposure to continuous noise at a single level shall not exceed the time *T* (in h) computed from the formula:

$$T = \frac{16}{2^{[0.2(L-85)]}}$$

where

L = work place sound level measured in dBA.

2 When the daily noise exposure is composed of two or more periods of different levels, their combined effect should be considered rather than the individual effect of each. The combined levels may not exceed a daily noise dose, *D* of unity where *D* is computed from the formula:

$$D = \frac{C_1}{T_1} + \frac{C_2}{T_2} + \dots + \frac{C_n}{T_n}$$

where, *C*₁, *C*₂ *C*_{*n*} indicate the total duration of exposure (in hour) at a given steady-state noise level; and *T*₁, *T*₂.....*T*_{*n*} are the noise exposure limits (in hour for the respective levels given in the table or computed by the equation in Note 1). Exposure to continuous noise shall not exceed 115 dBA regardless of any value computed by the formula for the daily noise dose, *D* or by the equation in Note 2.

10.3.2 Interference with Communication

In factories where audible warning signals are used, or where an operator follows the operation of his machine by ear, the background noise should not be so loud as to mask the signal or desired sound (the information sound) to be heard. Noise may be the cause of accidents by hindering communication or by masking warning signals.

10.4 Methods of Reducing Noise

10.4.1 Noise Control by Location

Machines, processes and work areas which are approximately equally noisy should be located together as far as possible. Areas that are particularly noisy should be segregated from quiet areas by buffer zones that produce and may tolerate intermediate noise levels.

10.4.2 Noise Reduction by Layout

The office space in a factory should be as far as possible located preferably in a separate building. This building should not have a wall common with the production area. Where a common wall is unavoidable, it should be heavy with few connecting doors and no permanent openings.

10.4.3 Noise Reduction at Source

10.4.3.1 Selection of machinery

Noise should be reduced as near the source as possible. While the operational processes in a factory may be fixed and may have no quieter alternative, careful selection of the machine tools and equipment to be used may considerably help attaining lower noise levels in the machine shop.

10.4.3.2 Reducing noise from potential sources

Impact that is not essential to a process should be quietened. Noise from handling and dropping of materials on hard surface may be reduced by using soft resilient materials on containers, fixing rubber tyres on trucks, trolleys, etc. Machine noise may be kept to a minimum by proper maintenance. Proper lubrication reduces noise by friction conveyors, rollers, etc.

10.4.3.3 The noise from the radiating surfaces may be reduced by reducing the radiating area. For example, if the area is halved, the noise intensity will be reduced by 3 dB and at low frequencies the reduction will be much greater.

10.4.3.4 Supporting structures for vibrating machines and other equipment should be frames rather than cabinets or sheeted enclosures. If an enclosure is used, precaution should be taken to isolate it and line it on the inside with sound-absorbent material. Penetration through the enclosure should be adequately sealed. The noise radiated by machinery guards can be minimised by making them of perforated sheet or of wire mesh.

10.4.3.5 Reducing transmission of mechanical vibration

A vibrating source does not usually contain a large radiating surface but the vibration is conducted along mechanically rigid paths to surfaces that can act as effective radiator. If the rigid connecting paths are interrupted by resilient materials, the transmission of vibration and consequently the noise radiated may be greatly reduced. The reduction depends on the ratio of the driving (forcing) frequency of the source to the natural frequency of the resilient system. The natural frequency may be determined from static deflection under actual load as given in [Fig. 1](#). The higher the ratio between the two frequencies, the lesser is the transmissibility, which is defined as the ratio of the force transmitted through the resilient isolator to the exciting force applied to it. Transmissibility and the equivalent noise reduction for various frequency ratios are given in [Fig. 2](#). For satisfactory operation, a ratio of 3 : 1 or more between the driving and natural frequencies is recommended.

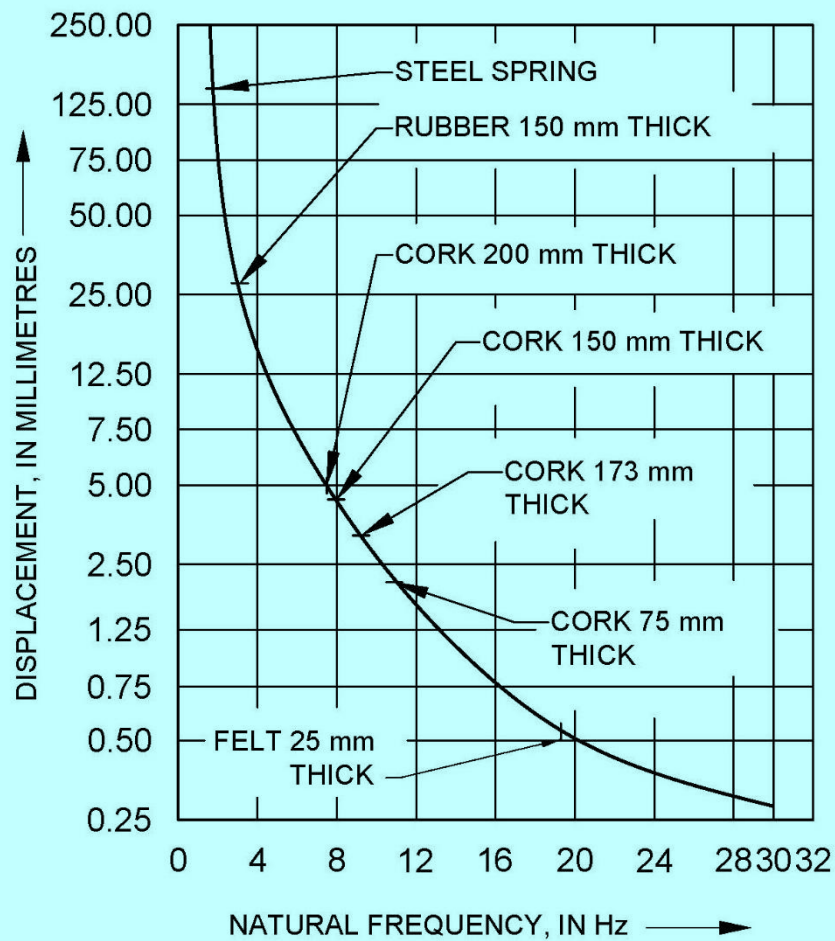


FIG. 1 RELATION BETWEEN STATIC DEFLECTION AND NATURAL FREQUENCY

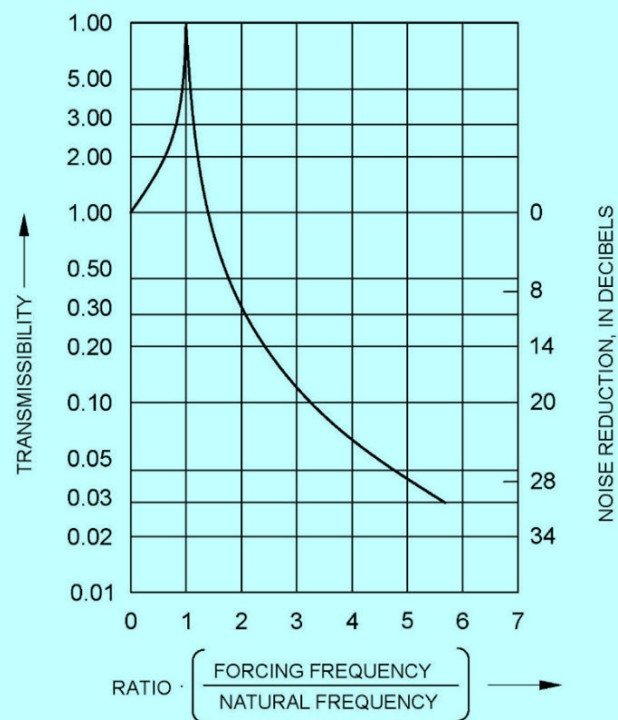


FIG. 2 TRANSMISSIBILITY AND EQUIVALENT NOISE REDUCTION FOR DIFFERENT RATIOS OF FORCING AND NATURAL FREQUENCIES

Materials for isolators and their position are given below:

- a) *Material for isolators* — Vibration isolators are usually made of resilient materials like steel in the form of springs and rubber:
- 1) Because of the large range of deflections obtainable in coil springs, they may isolate vibrations over a large spectrum of low frequencies. Metal springs transmit high frequency (from about two hundred to several thousand c/s) very readily. Transmission of these frequencies can be reduced by eliminating direct contact between the spring and the supporting structure. Rubber or felt pads may be inserted between the ends of the spring and the surfaces to which it is fastened;
 - 2) Rubber in the form of pads may be used to isolate very effectively engines, motors, etc. It may be used in compression or in shear. Some rubber mountings use rubber-in-shear as the primary elastic elements and rubber-in-compression as a secondary element which furnishes snubbing action if the mounting is subjected to an overload; and
 - 3) Felt or cork or both may be used as resilient mats or pads under machine bases. The load per unit area shall be chosen to produce enough deflection for the isolation required; and shall be such that at this deflection, it is not loaded beyond its elastic limit.
- b) *Position of isolators* — The normal position of the isolators is between the machine and its foundation. However, if the forcing frequency of the machine is low (less than 10 Hz) and vibration isolators with the requisite deflection for this location are not available, the machine may be bolted directly to an independent heavy inertia concrete base and the available vibration isolators used below the concrete base.
- 1) Large press and drop hammers which create serious impact vibration in heavy machine shops may be mounted rigidly on very massive blocks of concrete having weights several times greater than the weights of the supported machines. The inertia blocks may, in turn, be isolated from the building structure by large wooden blocks and with thick pads of cork;
 - 2) In critical installations (see Note), attempt should be made to locate the resilient mounts in a plane which contains the centre of gravity of the mounted assembly. It is also preferable to locate the mounts laterally as far away as possible from the centre of the machine;

NOTE — Critical installations are those installations where transmission of vibration from these installations seriously hamper the normal working.

- 3) Rigid mechanical ties between vibrating machine and building structure, short-circuit or reduce the effectiveness of isolators. Loose and flexible connections should be inserted in all pipes and conduits leading from the vibrating machine. Where flexible connections are impracticable, bends should be inserted into the pipes or the pipes themselves should be supported on vibration mounts for a considerable distance from the source; and
- 4) *Flexibility of foundation* — The effect of flexibility of the foundation on the isolator transmissibility shall be considered in the selection of practical vibration isolating mountings. The simplified vibration isolation theory assumes a completely rigid foundation. However, in practice, this can never be achieved. The foundation is never actually completely rigid. Generally, the relatively low stiffness of the isolation system permits the assumption of the foundation to be rigid. However, if the stiffness of the isolator is allowed to become comparable to the foundation stiffness (or greater), the deflection of the isolator will become smaller and the foundation will also deflect with increased transmissibility and decreased isolator efficiency. In a dynamic sense, supporting foundation or floors should have natural frequency as high and be as stiff as possible compared to the system being isolated. Good design practice requires that the isolators should be designed assuming a rigid foundation with the stipulation that the selected machine isolation system frequency should be well below the foundation frequency. This point should specially be kept in mind when installing machines at upper levels in buildings because supported slabs generally have lower natural frequencies (low stiffness) than slabs on grade in basement or ground floor locations.

10.4.4 Noise Reduction by Enclosures and Barriers

10.4.4.1 Enclosures

Airborne noise generated by a machine may be reduced by placing the machine in an enclosure or behind a barrier. The enclosure may be in the form of close-fitting acoustic box around the machine such that the operator performs

his normal work outside the box and thus is not subjected to the high noise levels of the machine. The enclosure may be made of sheet metal lined inside with an acoustical material.

Where size of the machine, working area and the operation do not permit close-fitting enclosures, the machine may be housed in a room of its own. The inside of the enclosure should be lined with sound-absorbing materials to reduce the noise level of the contained sound. The bounding walls of the enclosures should also have adequate transmission loss to provide desired noise reduction.

10.4.4.2 Barriers

A partial reduction of noise in certain directions may be obtained by barriers or partial enclosures or partial height walls. Two-sided or three-sided barrier, with or without a top and invariably covered on the machine side with acoustic absorption material should face a wall covered with sound-absorbing material. If the top of the enclosure is open, the reduction may be increased by placing sound-absorbing material on the ceiling overhead.

10.4.5 Acoustical Absorption Devices

10.4.5.1 Acoustical treatment of ceilings and side walls

In order to reduce the general reverberant noise level in machine shops, acoustical material may be provided on as much of the available area on the ceiling and side walls. With this treatment 3 dB to 6 dB reduction of middle and high frequency noise may be achieved. While the noise level at the source, affecting the operator, may not be reduced materially, the treatment would bring down the built-up noise level away from the source in reverberant field.

10.4.5.2 Functional sound absorbers

For efficient noise reduction functional sound absorbers may be clustered as near the machines as possible. These units may be suspended and distributed in any pattern to obtain lower noise levels within the machine shop. Compared on the basis of equal total exposed surface areas, functional sound absorbers have higher noise reduction coefficients (NRC) than conventional acoustical materials placed directly on ceilings and walls.

10.5 Noise Control in Specific Industrial Occupancies

10.5.1 Manufacturing and Heavy Machinery Industries

10.5.1.1 Industrial units with high-noise equipment such as stamping presses, crushers, and rolling mills should install vibration isolation platforms or resilient mounts to control structure-borne noise.

10.5.1.2 Machines producing significant noise should be enclosed in acoustic enclosures to prevent noise propagation within the facility and reduce exposure to workers.

10.5.1.3 Partition walls between noisy production areas and quieter zones (such as offices or break rooms) should incorporate sound-insulating materials or be designed with adequate thickness to ensure minimal noise transfer.

10.5.2 Chemical and Pharmaceutical Plants

10.5.2.1 Noise-generating equipment such as pumps, compressors, and centrifuges should be fitted with mufflers or acoustic casings to minimize sound emissions.

10.5.2.2 Mechanical rooms containing heating, ventilation, and air conditioning systems and other service units should be acoustically treated and isolated from production spaces to prevent noise transmission.

10.5.2.3 Noise-sensitive spaces such as laboratories and quality control areas should either be located away from noisy zones or separated by floating floors or soundproof partitions to ensure operational efficiency.

10.5.3 Food Processing Units

10.5.3.1 Equipment such as blenders, grinders, and packaging machines should use vibration-isolating mounts to control structure-borne noise.

10.5.3.2 Temporary sound barriers may be installed around seasonal operations like bulk-loading zones during periods of high activity.

10.5.3.3 Control rooms and staff areas should be acoustically insulated to ensure noise relief, with soundproof partitions and doors installed where necessary.

10.5.4 Power Plants and Utilities

10.5.4.1 Power plants should install noise barriers or walls around high-noise zones such as turbine halls and generator rooms to minimize sound emissions beyond facility boundaries.

10.5.4.2 Cooling towers and exhaust stacks should be equipped with noise control devices like silencers to mitigate low-frequency noise emissions.

10.5.4.3 Control rooms and operational centres should be designed to maintain appropriate noise levels conducive to work, with double-wall partitions or air gaps considered for turbine or engine rooms.

11 LABORATORIES AND TEST HOUSES

11.1 Sources of Noise

11.1.1 Outdoor Noise

In a test house or laboratory, where research workers and scientists are engaged in performing sophisticated experiments, the external noise is mostly contributed by noise emitting buildings (workshops, machine rooms), airports, railway stations and general traffic noises. The outdoor sources of noise in a college laboratory include noises produced in a playground as well.

11.1.2 Indoor Noise

The following sources mainly contribute to indoor noises in research institutions/college laboratories:

- a) Workshops, machine rooms, cafeteria, etc;
- b) Air conditioning and exhaust fans;
- c) Noise produced within the test house or laboratory while performing experiments; and
- d) Typing or other machine noises, telephone service, lift, sanitary services, etc.

11.2 Recommendations

11.2.1 Site Planning

While planning for a laboratory or test house, care should be taken in the design that no noise emitting installations should exist in its neighbourhood. However, where outdoor noises exist, such as from local factory, heavy traffic airports, railway lines, sport grounds or busy markets, buildings should be kept as far away as possible from the source of noise.

11.2.1.1 The window and door openings towards the noise sources should be minimum. Minimum amount of glazing should be placed on walls directly facing the noise sources.

11.2.2 Internal Planning

11.2.2.1 Noisy places should be kept separate from the quiet ones. The location of laboratories or test houses should be so chosen that it is cut off from the noisy zones. Where there are offices attached to a laboratory,

provision should be made to treat the offices and to use acoustical partitions, to achieve a sound isolation D_w of at least 35 dB.

11.2.2.2 In a laboratory, mostly hard reflecting surfaces and bare furnishings are found, which produce very reverberant conditions. The noise condition still deteriorates when noise producing instruments are switched on or a heavy object is dropped on the floor. Under these conditions, sound absorbing treatment of the space is very essential. Sound absorbing ceilings are recommended to deaden such noises. Rubber buffers may also be fitted to the legs of furniture.

11.2.2.3 In large span laboratories or test houses where scientists and researchers are engaged in work and/or simultaneously busy in calculations or desk work requiring high degree of mental concentration, use of sound absorbing screens is recommended.

11.2.2.4 Noise reduction between the test house or laboratory and corridors or general circulation space should be well kept in mind and due care should be taken of the type of doors and the manner of their fittings, etc. Transmission of noise through service ducts, pipes, lifts and staircases should also be guarded.

Telephones should preferably be placed in a separate small enclosure or acoustically efficient telephone booth.

11.2.2.5 To isolate a laboratory or a test house from structure-borne noises originating from upper floor, sandwich type floor construction is recommended.

11.2.2.6 Wherever the provision of double glazed windows is necessary to reduce the heat losses, care should be taken to provide sealed double windows rather than double glazing in a single window.

NOTE — Double glazed windows for sound insulation should have a minimum gap of 100 mm between the two glasses.

12 MISCELLANEOUS BUILDINGS

12.1 Law Courts and Council Chambers

It is important that law courts and council chambers be protected from the intrusion of outdoor noise and from indoor noise arising both from ancillary offices and circulation spaces. The general recommendations on site planning given in 3 apply to law courts and municipal buildings, but in the larger buildings at least, further protection against outdoor noise can be obtained by planning offices and other rooms around the court rooms or chambers, and separating the offices from the central rooms by means of corridors. This arrangement is usually convenient to the function of the buildings.

12.1.1 The wall between the corridors and the central rooms should have a sound reduction index, R_w of not less than 50 dB (for example 230 mm brick) to insulate against airborne noise in the corridors. Entrances from halls or corridors into court rooms or council chambers should be through baffle lobbies with two sets of quiet action doors. Sound absorbing treatment on ceilings and upper parts of walls or entrance lobbies is recommended.

12.1.2 The whole of the floor of the court room or chamber including steps and seating areas set aside for the public should have a resilient floor finish to reduce the noise of footsteps and shuffling of feet. Any tip-up seats should be quiet in action.

12.1.3 Sound absorbing treatment applied for acoustic purposes serves also to reduce the build-up of noise within the room. It is desirable to provide sound absorbing ceiling in the court room.

12.2 Libraries, Museums and Art Galleries

Quiet conditions for reading and study are essential in these types of buildings and, since their occupancy is not noise producing, intruding noise is more noticeable and distracting. Every opportunity therefore should be taken to plan for noise defence, both in respect of siting of the building and internal planning. When possible, stack rooms, store rooms and administrative offices should be planned to screen reading rooms, print rooms and lecture rooms from noise sources. In public libraries, the reference library and lecture rooms should receive first consideration; the lending library, newspaper and periodical rooms have a higher background noise and are secondary in importance.

12.2.1 In large libraries, museums and art galleries echoes from lofty, large domed or concave ceilings are often a nuisance. Small noises such as footsteps, coughs, chair scraping and closing of books are reinforced by reverberation, and concave surfaces even when treated with a sound absorbent may focus these noises. Treated flat ceilings, if not too high, obviate these troubles. Books on shelves in libraries constitute a valuable wall absorbent.

12.2.2 Floor finishes are important. The impact noise of footsteps on marble, terrazzo or wood block flooring, and especially on hardwood strip and batten flooring, can be disturbing both within the room in which the noise is generated and the rooms below. On solid floors, resilient floor finishes, such as rubber, cork and linoleum on an underlay, are highly desirable. In the children's sections of libraries and museums they are essential. In existing buildings, rubber linoleum or vinyl asbestos tiles laid over the floor in the traffic areas are often a solution to the problem.

12.2.3 Reference libraries in universities, research establishments, office buildings and science buildings having machines and testing benches, should be planned in a quiet part of the building. Walls enclosing the library should normally have a sound reduction index, R_w of not less than 50 dB (for example 230 mm brick) and baffle lobbies should be planned between the library and halls and corridors. Walls facing on to corridors or other noisy areas should not have fanlights unless they are double glazed and non-operable.

12.3 Auditoria and Theatres

The sources of noise that have to be considered in concert halls, opera house, theatres and similar auditorium buildings are as follows:

- Outdoor noise entering through walls, roofs, doors, windows or ventilation openings;
- Noise from any other hall in the same building, especially if let out separately for revenue;
- Noise from foyers, service rooms and other ancillary rooms, particularly rehearsal rooms;
- Noise from air conditioning plant, etc, and the cross-transmission of other internal noises via ventilating duct system; and
- Impact generated noise due to rainfall on light weight metal roofs.

12.3.1 Because of greatly increased outdoor noise, all auditorium buildings now need more care in siting than formerly. For listening to speech or music, a very low background noise level is desirable; in concert halls especially, the quietest possible conditions should be provided because the pauses and moments of silence which are an essential element of music cannot otherwise be given full value. Therefore, sites at cross-roads or close to steel railway bridges, religious places or near churches where bell ringing is practiced, should be avoided unless very high standards of structural sound insulation are contemplated. Sites adjoining underground railways may also prove unsatisfactory at basement levels owing to low-pitched noise or rumble transmitted through the ground; special isolation measure need to be adopted for isolating large buildings from ground vibration of this sort.

12.3.2 Whenever possible, for concert halls and theatres on city sites a noise survey of the site should be made; a suitable sound reduction value for the structure of the building can then be chosen so as to keep down to certain maximum noise levels within the auditorium. The maximum octave-band sound pressure levels (SPL) recommended are given in [Table 8](#).

Table 8 Maximum Sound Pressure Levels (dB) Due to External and Mechanical Equipment Noise in Auditoria (Clause 12.3.2)									
SI No.	Type of Auditorium	Centre Frequency Hz							
		63	125	250	500	1 000	2 000	4 000	8 000
(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
i)	Concert halls [dBA-25]	51	39	31	24	20	17	14	13
ii)	Drama theatres and Cinemas [dBA-30]	55	44	35	29	25	22	20	18

12.3.3 The minimum standard of sound reduction index, R_w likely to be required for the envelope of an auditorium in a city to protect it against external noise is of the order of 65 dB for a concert hall or 55 dB to 60 dB for a theatre. This reduction should be provided on all sides, but it would be reasonable to make the R_w for the roof 5 dB to 10 dB less provided the building is not unduly exposed to noise from aircraft in flight. Surrounding the auditorium with ancillary rooms and foyers is an obvious and invaluable planning method of obtaining the required insulation against outdoor noise.

12.3.4 Ventilation intakes and returns are vulnerable features in the defence against external noise. They should be positioned so as to avoid exposure to noise, and in addition sufficient length of both inlet and outlet ducts should be provided with carefully designed silencers. The ventilation system should also be designed to avoid transmitting or adding to internal noise.

12.3.5 The most serious internal noise problem arises when there are two halls meant for separate use in the same building, especially if one of them is a concert hall. The latter is a very loud potential source of noise and requires a high standard of protection against extraneous noise. In these circumstances it is doubtful whether a 'single' wall can be adequate for insulating the two halls unless it is designed with a wide unbridged cavity. Separation by planning is preferable.

12.3.6 Other sources of internal noise are rehearsal rooms, scenery bays and workshops, stages of other halls where rehearsals or erection of stage sets might be in progress and foyers and bars where loud conversation might occur. The insulation of the internal walls should be adequate to protect the auditorium from these noise sources and the insulation should not be by-passed by openings, doorways, etc. The general noise due to banging of doors also needs to be taken care of and soft sealing materials should be provided for all doors to ensure quiet closing.

12.3.7 Mechanical equipment such as lifts shall be effectively isolated from the building structure to help avoid noise transmission.

12.3.8 Adequate damping should be provided under light weight metal roofs with an additional light weight under deck noise sound barrier to reduce rainfall generated noise.

12.3.9 For detailed acoustical design of auditoria and conference halls reference may be made to good practice [D-4(3)].

12.4 Cinemas

The main objective of the design should be to control noise from adjacent screens, the projection area, the foyer, and outside the cinema. The first of these, controlling noise from adjacent screens, is likely to be the most difficult with modern digital sound systems. As most cinemas are air-conditioned, there will be some noise from services. To ensure reasonable listening conditions, this should be limited to 30 dBA. This will provide some masking of the noise from adjacent screens, but a high performance partition will still be essential. Masonry or lightweight construction may be used, and a typical performance specification for a lightweight wall separating two screens is given in [Table 9](#). Cinema design, however, normally requires specialist acoustic advice.

Table 9 Typical Sound Insulation Specification for Wall Separating Two Cinema Screens

(Clause [12.4](#))

Sl No.	Octave band	Sound Reduction Index, R
(1)	Hz	dB
(1)	(2)	(3)
i)	63	38
ii)	125	44
iii)	250	50
iv)	500	61
v)	1 000	57
vi)	2 000	58
vii)	4 000	57
viii)	8 000	55

13 NOISE FROM BUILDING SERVICES

13.1 Mechanical, electrical, air conditioning, heating and mechanical ventilation, and other services are provided in almost all large buildings including residential, commercial and industrial buildings. Noise control measures should be incorporated during the design and installation of such services to adhere to the recommended outdoor and indoor noise criteria for the kind of occupancy. For detailed design of noise control for services, specialist advice should be sought.

Some basic design techniques for noise control in air conditioning, heating and mechanical ventilation system are given in [Annex J](#).

13.2 Control of noise from mechanical equipment can also be done by specifying noise control requirements while purchasing the equipment (see [Annex K](#)).

14 NEIGHBOURHOOD ACOUSTIC SHIELDING DURING INFRASTRUCTURE DEVELOPMENT

14.1 General

Infrastructure development projects often generate substantial noise that can adversely affect the well-being of residents. Prolonged exposure to noise pollution may lead to health issues such as sleep disturbances, hearing impairments and stress. Hence, effective acoustic shielding of neighbourhood is required to minimize such impacts. The following outlines the provisions for noise control during infrastructure development.

14.2 Pre-Development Noise Assessment

14.2.1 A baseline noise assessment should be conducted prior to the commencement of construction activities to determine ambient noise levels and identify noise-sensitive receptors (NSRs), including residential areas, schools, and healthcare facilities.

14.2.2 Noise modelling should be performed to predict noise levels for each phase of construction. The model shall take into account equipment type, frequency and duration of activities, and the topography of the site.

14.2.3 Predicted noise levels shall be compared with the limits prescribed. If expected levels exceed the permissible limits, additional mitigation measures should be proposed.

14.3 Noise Mitigation Planning

14.3.1 A noise mitigation plan (NMP) should be prepared prior to the commencement of any construction work. The NMP shall include:

- a) Identification of potential noise sources and their expected noise levels;
- b) Location of noise-sensitive receptors and baseline noise levels;
- c) Proposed noise control measures and their implementation timelines;
- d) Monitoring strategies for continuous noise assessment; and
- e) Community engagement strategies and complaint management procedures.

14.3.2 Construction activities that generate high levels of noise should be scheduled during permissible hours. Noise-intensive operations should be avoided during night time hours unless specific approval is obtained.

14.4 Noise Barriers

14.4.1 Temporary noise barriers should be installed at the perimeter of construction sites to shield NSRs from construction noise. Barriers should be designed to achieve a minimum noise reduction of 10 dBA. The height of the barriers should be at least 2.5 m or as determined by site-specific acoustic analysis.

14.4.2 Permanent noise barriers, such as earth berms, concrete walls, or vegetative buffers, may be considered for long-term infrastructure projects with expected noise impacts exceeding permissible limits.

14.4.3 Recommended guidelines for the design and performance evaluation of noise barriers are given in [Annex F](#).

14.5 Low-Noise Equipment and Practices

14.5.1 Low-noise equipment, such as electric or hydraulic machinery, should be used wherever feasible. Equipment should be fitted with silencers, mufflers or acoustic enclosures to reduce emissions.

14.5.2 Construction techniques that minimize noise generation, such as hydraulic splitting instead of pneumatic hammering for concrete demolition, should be employed wherever possible.

14.6 Maintenance of Equipment

14.6.1 A maintenance schedule for all construction equipment should be implemented to ensure optimal operating conditions and minimize noise emissions.

14.6.2 All equipment should be inspected regularly for defects, such as worn-out parts, loose fittings or damaged mufflers, that may increase noise levels. Defective equipment should be repaired or replaced immediately.

14.7 Noise Monitoring and Compliance

14.7.1 Continuous noise monitoring should be conducted at designated locations near NSRs throughout the construction period. Monitoring equipment should be as per accepted standard [D-4(4)].

14.7.2 Noise levels should be logged at regular intervals. The logged data shall be reviewed daily to ensure compliance with the permissible noise levels.

14.7.3 If noise levels exceed permissible limits, corrective actions, such as modifying work practices, deploying additional noise barriers, should be implemented within 24 h.

14.7.4 A noise monitoring report should be maintained throughout the project, including logged data, incidents of exceedances, corrective actions taken, and community complaints received.

14.8 Community Engagement and Communication

14.8.1 A community noise management plan should be established as part of the noise management plan, outlining strategies for engaging with the local community, informing them about construction activities, expected noise impacts, and mitigation measures.

14.8.2 A public contact point should be established for residents to report noise concerns. All complaints should be logged, investigated, and resolved promptly.

14.9 Vegetative Buffers

14.9.1 Where feasible, vegetative buffers consisting of dense rows of trees or shrubs should be planted around the construction site to serve as supplementary noise barriers.

14.9.2 Vegetative buffers should have a minimum width of 5 m and be composed of mixed species with dense foliage for optimal sound absorption.

14.10 Post-Construction Noise Assessment

14.10.1 Upon completion of construction activities, a post-construction noise assessment should be conducted to verify that ambient noise levels have returned to pre-development conditions or meet permissible limits.

14.10.2 Temporary noise barriers should be dismantled and disposed of responsibly. If permanent barriers or vegetative buffers were installed, they should be integrated into the final landscape design.

15 TRAFFIC NOISE BARRIERS

Noise barriers are one of the most effective ways to mitigate road traffic noise. Noise barriers can reduce the A-weighted noise levels depending on their design and height. If the barrier surface density exceeds 20 kg/m^2 , a reduction of 5 dB can be achieved by having a barrier tall enough to break the line of sight from the road to the receiver and an additional 1.5 dB reduction can be achieved for each additional metre of height. The efficiency of noise barrier largely depends upon its geometry. Besides the height of the barrier and its top element form, the cross section of the barrier contributes to its performance. Recommended guidelines for the design and performance evaluation of noise barriers are given in [Annex F](#).

ANNEX A

(Clause [2.4](#))

NOISE CALCULATIONS

A-1 GENERAL

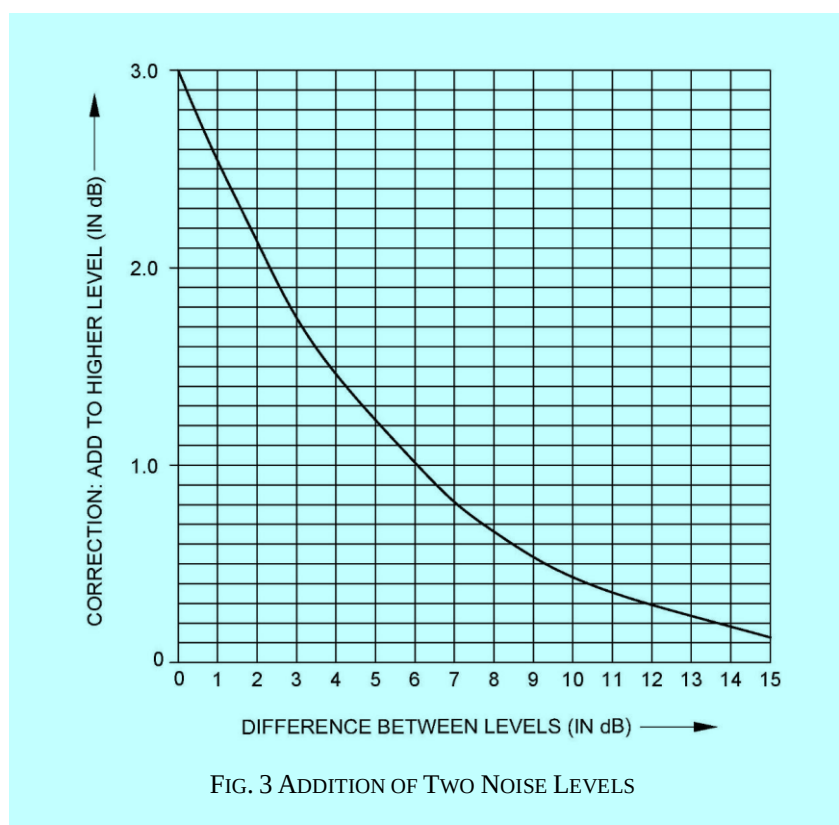
Some of the simpler types of noise calculation are described in this annex.

A-2 ADDITION OF TWO NOISE LEVELS

To determine the combined sound pressure level (L_c) resulting from the sound pressure levels of two or more noise sources (L_1 , L_2 , etc), it is necessary to calculate and add the mean square values of their individual sound pressures and then convert this back to a sound pressure level. This can be done using the following formula:

$$L_c = 10 \log_{10}(10^{L_1/10} + 10^{L_2/10})$$

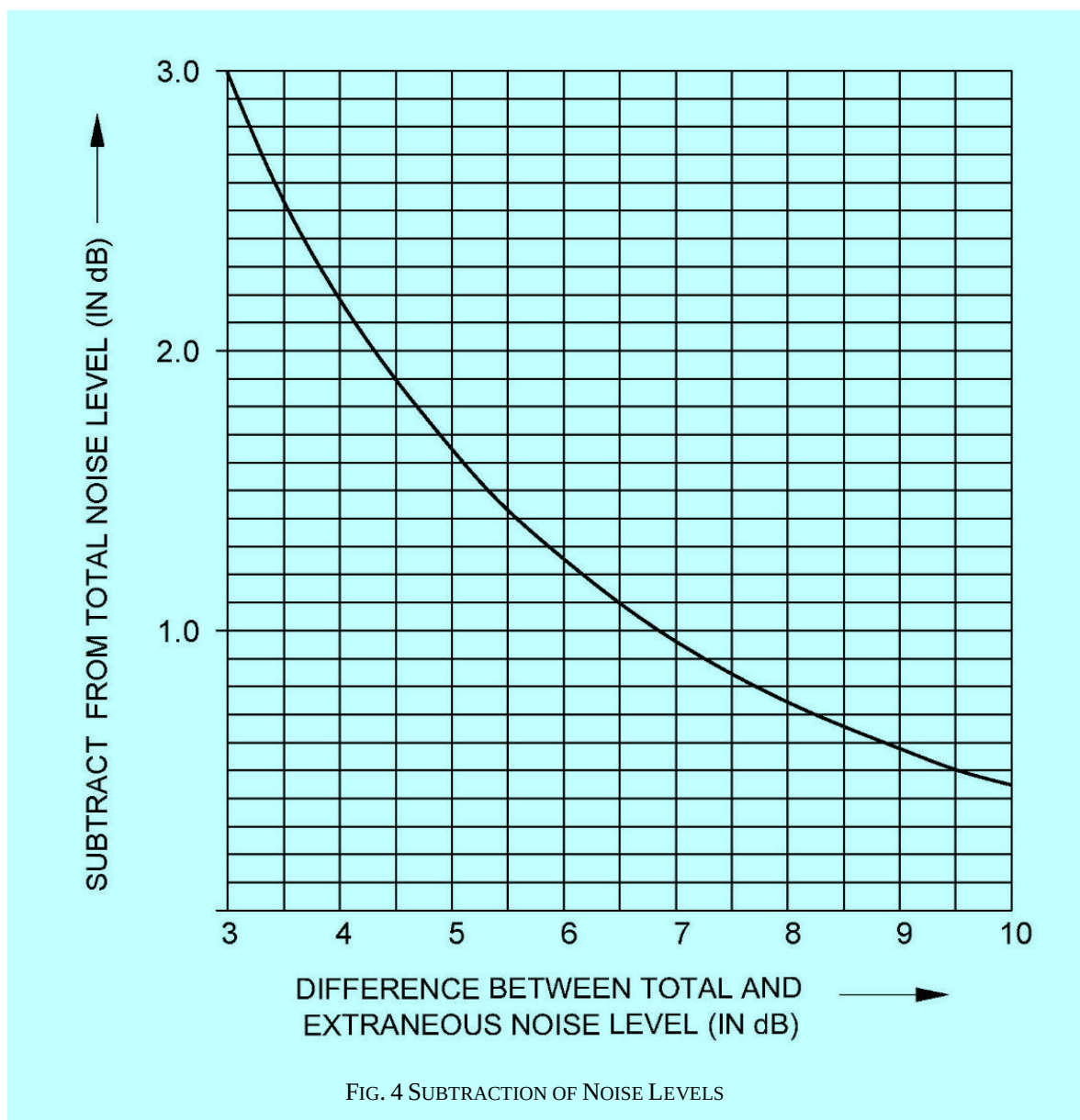
As the individual sound pressure levels are logarithms of the mean square sound pressures, they cannot simply be added arithmetically. [Fig. 3](#) shows a graphical method for adding the sound pressure levels from two independent sources to obtain the combined sound pressure level at a particular place. This graph may also be used for multiple sources by combining sources two at a time to produce virtual sources that can then be combined. The most accurate approach is to start with the lowest levels and work towards the highest.



The graph should be used with caution where the noise sources are not independent. For example, the sound pressure level from two large transformers fed with currents in phase will be very sensitive to the receiving position. This is because the effect of the constructive and destructive interference of the sounds from the two sources is very dependent on position.

A-3 SUBTRACTION OF TWO NOISE LEVELS

When measuring noise from a source, the true noise level of the source alone will be less than that shown by the meter if the level of extraneous noise is less than about 10 dB below the total noise level. An estimate of the true source level can be obtained from [Fig. 4](#).



A-4 NON-UNIFORM COMPOSITE PARTITIONS

[Figure 5](#) provides for calculating the overall sound insulation of a composite partition consisting of two parts having different sound-insulating properties, for example, a window in a wall. It may also be used to give an indication of the effect of gaps or holes in a partition by assigning a sound insulation value of 0 dB to the aperture.

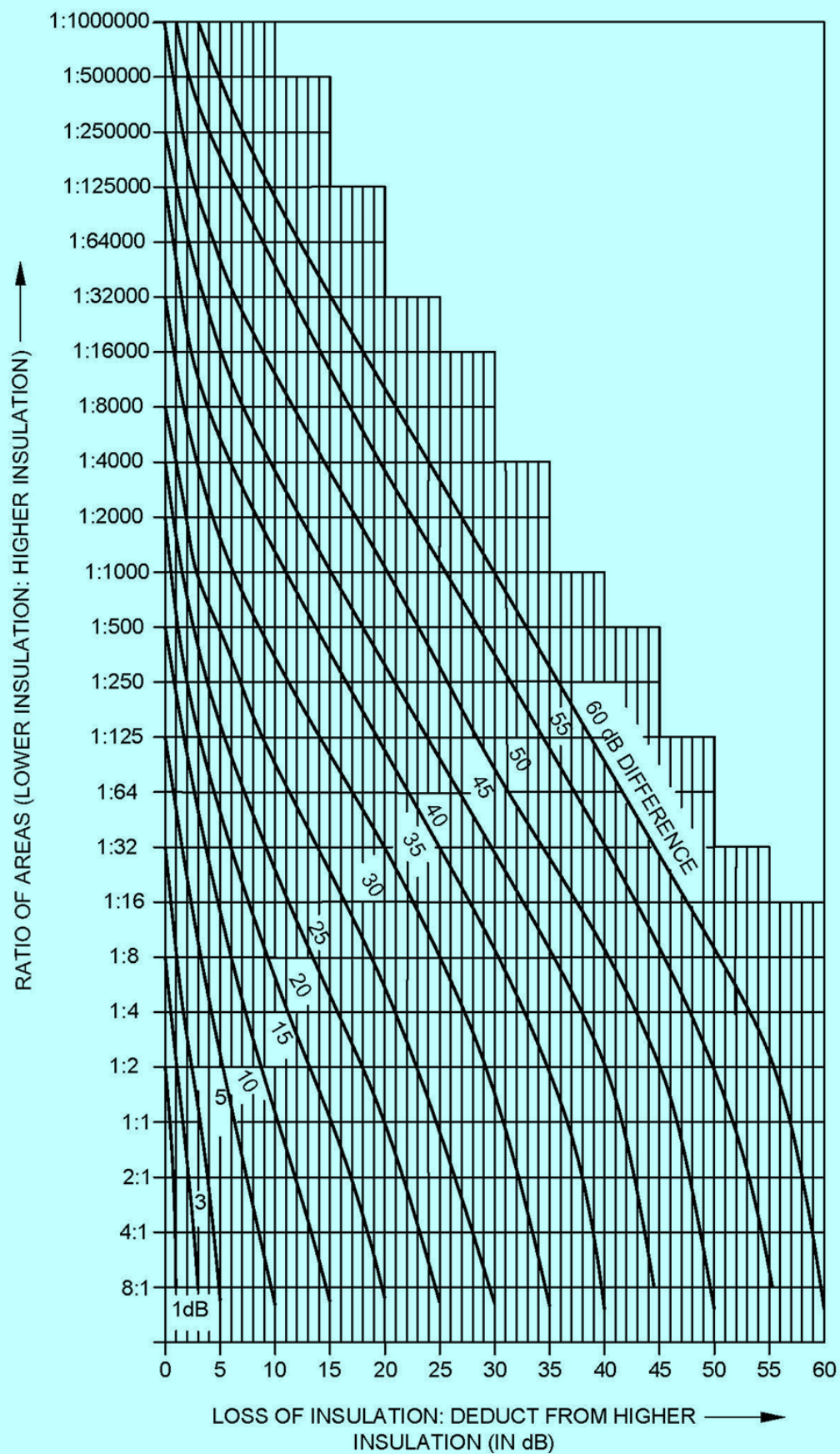


FIG. 5 SOUND INSULATION OF NON-UNIFORM PARTITIONS

A-5 A-WEIGHTING CALCULATIONS

The equivalent A-weighted level is often required when data on a noise source is available as a set of octave band or one-third octave band levels. The conversion can be done manually, using the standard A-weighting values (see Table 10) and the graph for combining levels (see Fig. 3). For all but the simplest situations it is more convenient to use a computer spreadsheet to do the conversion.

Table 10 Standard A-Weighting Values (dB)				
(Clause A-5)				
Sl No.	Third Octave Band Centre Frequency	A-Weighting	Third Octave Band Centre Frequency	A-Weighting
	Hz	dB	Hz	dB
(1)	(2)	(3)	(4)	(5)
i)	10	- 70.4	500	-3.2
ii)	12.5	- 63.4	630	-1.9
iii)	16	- 56.7	800	-0.8
iv)	20	- 50.5	1 000	0
v)	25	- 44.7	1 250	0.6
vi)	31.5	- 39.4	1 600	1.0
vii)	40	- 34.6	2 000	1.2
viii)	50	- 30.2	2 500	1.3
ix)	63	- 26.2	3 150	1.2
x)	80	- 22.5	4 000	1.0
xi)	100	- 19.1	5 000	0.6
xii)	125	- 16.1	6 300	-0.1
xiii)	160	- 13.4	8 000	-1.1
xiv)	200	- 10.9	10 000	-2.5
xv)	250	- 8.6	12 500	-4.3
xvi)	315	- 6.6	16 000	-6.6
xvii)	400	- 4.8	20 000	-9.3

A-6 REVERBERATION TIME CALCULATION

An estimate of the reverberation time (T) of a room can be obtained from the Sabine formula:

$$T = \frac{0.16 V}{\sum A_i}$$

where

V = volume of the room in cubic metres (m^3); and

A_i = equivalent sound absorbing area in the room, in square metres (m^2).

The A_i are the absorbing areas of each surface, or other permanent fixture in the room. Each A_i is determined by multiplying the area of that surface in square metre (m^2) by its absorption coefficient α_{si} . The surface of each significant fixture or feature of the room should be considered as well as the walls, ceiling and floor.

The total absorption is obtained by summing the individual A_i values. As the values of α_{si} are frequency dependent, this calculation should be repeated for each octave band of interest.

An allowance should also be made for people and furnishings in the room.

ANNEX B

(Clause 2.24)

NOISE RATING

B-1 Noise rating (NR) is a graphical method for assigning a single number rating to a noise spectrum. It can be used to specify the maximum acceptable level in each octave band of a frequency spectrum, or to assess the acceptability of a noise spectrum for a particular application. The method was originally proposed for use in assessing environmental noise, but was later also found suitable for describing noise from mechanical ventilation systems in buildings. To make a rating, the noise spectrum is superposed on a family of NR contours; the NR of the spectrum corresponds to the value of the first NR contour that is entirely above the spectrum. The data for drawing NR contours (from NR 0 to NR 75) is given in Table 11 for the frequency range 31.5 Hz to 8 kHz.

Table 11 Noise Rating Values										
(Clause B-1)										
Sl No.	Noise Rating	Octave Band Centre Frequency, Hz For Sound Pressure Levels dB _{re} 20 µPa								
		31.5	63	125	250	500	1 000	2 000	4 000	8 000
(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)
i)	NR75	106	95	87	82	78	75	73	71	69
ii)	NR70	103	91	83	77	73	70	68	66	64
iii)	NR65	100	87	79	72	68	65	62	61	59
iv)	NR60	96	83	74	68	63	60	57	55	54
v)	NR55	93	79	70	63	58	55	52	50	49
vi)	NR50	89	75	66	59	53	50	47	45	43
vii)	NR45	86	71	61	54	48	45	42	40	38
viii)	NR40	83	67	57	49	44	40	37	35	33
ix)	NR35	79	63	52	45	39	35	32	30	28
x)	NR30	76	59	48	40	34	30	27	25	23
xi)	NR25	72	55	44	35	29	25	22	20	18
xii)	NR20	69	51	39	31	24	20	17	14	13
xiii)	NR15	66	47	35	26	19	15	12	9	7
xiv)	NR10	62	43	31	21	15	10	7	4	2
xv)	NR5	59	39	26	17	10	5	2	– 1	– 3
xvi)	NR0	55	35	22	12	5	0	– 4	– 6	– 8

B-2 For computational methods, the curves are defined by the equation:

$$L = a + bN$$

where

L = octave band sound pressure level corresponding to NR level N ; and

a and b = constants for each frequency band, as given in Table 12.

NOTE — NR values cannot be converted directly to dBA values but the following approximate relationship applies:

$$NR = dBA - 6$$

B-3 Although the NR system is currently the preferred method for rating noise from mechanical ventilation system, other methods which are more sensitive to noise at low frequencies are available, but they are not yet widely accepted. Low frequency noise may be disturbing or fatiguing to occupants, but may not have much effect on the dBA or NR value.

Table 12 Values of a and b (Clause B-2)			
Sl No.	Octave Band Centre Frequency Hz	a	b
(1)	(2)	(3)	(4)
i)	31.5	55.4	0.681
ii)	63	35.4	0.790
iii)	125	22.0	0.870
iv)	250	12.0	0.930
v)	500	4.2	0.980
vi)	1 000	0.0	1.000
vii)	2 000	- 3.5	1.015
viii)	4 000	- 6.1	1.025
ix)	8 000	- 8.0	1.030

ANNEX C

(Clauses [2.26](#), [2.41](#), [2.44](#), [2.46](#), [2.54](#), [2.55](#), [2.56](#) and [2.57](#))

SPECIFICATION OF SOUND INSULATION

C-1 GENERAL

Sound insulating elements work mainly by reflecting sound energy back into the source room, not by absorbing it. The methods of measurement and the terms used are described in [C-2](#) to [C-4](#).

C-2 INSULATION AGAINST AIRBORNE SOUND

As per the standard tests, the insulation between a pair of rooms is measured either in third octave bands having centre frequencies which cover at least the range 100 Hz to 3 150 Hz, or in octave bands which cover at least the range 125 Hz to 4 000 Hz. The noise is produced by a loudspeaker in one of the rooms (called the source room) and at each frequency the average noise levels are measured in the source room (L_S) and in the adjacent receiving room (L_R). The difference between these two levels (D) is a measure of the sound insulation between the rooms regardless of the transmission path(s) the sound energy followed to travel between the rooms. The equation is as follows:

$$D = L_S - L_R$$

The actual level in the receiving room depends on the following:

- Sound insulation of the separating wall or floor;
- Area of the separating wall or floor;
- Volume of the receiving room;
- Amount of flanking transmission (that is the importance of transmission paths other than the separating wall or floor); and

- e) Amount of absorbing material (for example furniture) in the receiving room.

For field measurements, apart from the amount of absorption, these factors are a property of the building and should be taken into account by the measurement procedure. As the amount of absorbing material (for example soft furniture) in the room at the time of measurement is arbitrary, it should be allowed for separately. This is achieved by measuring the reverberation time (T) of the room in seconds (s), which is a measure of how long it takes a sound to die away after the source has been switched off. As the sound energy is dissipated as heat in the absorbing material (T), it is related to the total amount of absorption in the room. The receiving room level may then be corrected to the level it would be if the room had a standard reverberation time (T_0) which is typical of furnished rooms, and is taken to be 0.5 s. The corrected level difference is known as the standardized level difference, which has the symbol D_{nT} and is calculated using the following equation:

$$D_{nT} = L_S - L_R + 10 \log_{10}(T/T_0)$$

For laboratory measurements, the insulation of the separating wall or floor being tested is required in a way which is independent of the actual measuring laboratory. For this reason, laboratories are designed to have minimal flanking transmission and a different correction is applied to account for the other factors. This correction is $10 \log_{10}(S/A)$.

where

S = common area of the separating wall or floor, in m^2 ; and

A = equivalent absorption area in the receiving room, in m^2 .

The laboratory corrected level difference at each frequency is known as the sound reduction index, which has the symbol R and is calculated using the following equation:

$$R = L_S - L_R + 10 \log_{10}(S/A)$$

If the test wall or floor is mounted in a realistic way in the laboratory and flanking transmission will be low in the field, the sound reduction index may be used to predict its performance in the field. The relation between D_{nT} and R is given by the following equation:

$$D_{nT} = R - 10 \log_{10}(3S/V)$$

where

S = area of the separating wall or floor in the field, in m^2 ; and

V = volume of the receiving room in the field, in m^3 .

This equation shows that if the source and receiving rooms have different volumes, D_{nT} will depend on which is used as the source room; using the larger room as the source room will give lower value.

C-3 INSULATION AGAINST IMPACT SOUND

The procedure to measure the impact insulation of floors is rather different. Instead of a loudspeaker, a machine containing five small hammers is placed on the floor. While the hammers strike the floor at a rate of 10 blows a second, the resulting noise level (L_i) is measured in the receiving room below at each of the same frequency bands used for airborne insulation. In the field, the receiving room levels are again 'corrected' to a standard reverberation time (T_0) of 0.5 s to give the standardized impact sound pressure level, L_{nT} , which is calculated as follows:

$$L_{nT} = L_i - 10 \log_{10}(T/T_0)$$

In the laboratory, the noise level depends mainly on the characteristics of the floor being tested and the amount of absorption (A m^2) in the laboratory. It is therefore appropriate to correct the noise level to a standard area of absorption. The area used is $10 m^2$. The resulting normalized impact sound pressure level is given the symbol L_n and calculated as follows:

$$L_n = L_i + 10 \log_{10}(A/10)$$

C-4 RATING SOUND INSULATION

Measurements of insulation against both airborne and impact sounds yield values in a number of frequency bands. To make this information more manageable, rating methods such as those in accordance with [D-4(1)] are used to reduce the frequency band values to single figure ratings. These single figure ratings should be good predictors of subjective assessments of insulation. However, this is not always the case and it is prudent to examine the full measurement data in critical situations. The impact insulation measured on a floor with a carpet is likely to be overestimated by this method.

The more common indices used to describe sound insulation are summarized in [Table 13](#).

Table 13 Common Indices Used to Describe Airborne and Impact Sound Insulation (Clause C-4)						
SI No.	Airborne (A) Impact (I)	Lab (L) Field (F)	Measured Values		Single Number Quantity	
			Name	Symbol	Name	Symbol
(1)	(2)	(3)	(4)	(5)	(6)	(7)
i)	A	F	Standardized level difference	D_{nT}	Weighted standardized level difference	$D_{nT,w}$
ii)	A	L	Sound reduction index	R	Weighted sound reduction index	R_w
iii)	I	F	Standardized impact sound pressure level	L'_{nT}	Weighted standardized impact sound pressure level	$L'_{nT,w}$
iv)	I	L	Normalized impact sound pressure level	L'_n	Weighted normalized impact sound pressure level	$L'_{n,w}$

ANNEX D

(Clause [3.3](#))

OUTDOOR NOISE REGULATIONS IN INDIA

D-1 Government notifications are issued from time to time on the allowable ambient noise levels in general and specifically in different zones of various metropolitan cities of India.

D-2 Noise regulations and notifications are also issued from time to time specifying the maximum permissible sound levels from equipment commonly used in and around the residential areas and around sensitive buildings, specifically with regard to noise levels from electricity generating sets, construction equipment and HVAC utility equipment installed outdoors.

D-3 These regulations should be referred to by the designer for the design of measures for control of external noise.

ANNEX E

(Clause [3.5](#))

INTEGRATION OF NOISE BARRIERS AND NOISE ABATEMENT MEASURES IN ZONAL AND URBAN PLANNING

E-1 APPLICABILITY

E-1.1 These provisions establish requirements for incorporating noise barriers and noise abatement measures in the preparation of zonal plans to control noise pollution in urban and semi-urban areas.

E-1.2 The provisions apply to all zonal plans developed by municipal corporations, urban development authorities, and town planning bodies.

E-2 IDENTIFICATION AND SEGREGATION OF ZONES

E-2.1 Noise-Sensitive Zones (NSZs)

Zonal plans should identify noise-sensitive zones that include residential areas, educational institutions, healthcare facilities, and public parks, among others.

E-2.2 Noise-Generating Zones (NGZs)

Zonal plans should also identify noise-generating zones that include industrial areas, major transportation corridors, commercial zones, and recreational spaces prone to high noise levels.

E-2.3 Buffer Zones

Buffer zones should be established between NSZs and NGZs to minimize noise impacts. Buffer zones may include green belts, mixed-use developments, or other suitable land uses. The width and nature of buffer zones should be defined based on local conditions and practical feasibility.

E-3 NOISE BARRIERS

E-3.1 Requirement for Noise Barriers

Noise barriers should be required along major transportation corridors, such as highways and railway lines, adjacent to NSZs. The need for noise barriers should be determined based on the predicted noise levels and potential impact on NSZs.

E-3.2 Design criteria for noise barriers may be as per [Annex F](#).

E-3.3 Placement of Noise Barriers

Noise barriers should be placed as close as possible to the noise source or the boundary of NSZs to maximize effectiveness.

E-4 NOISE ABATEMENT MEASURES IN ZONAL PLANS

E-4.1 Noise Mapping and Assessment

Zonal plans should include noise mapping to identify areas with high noise levels and predict future noise impacts based on planned developments.

E-4.2 Green Belts and Vegetative Buffers

Zonal plans should integrate green belts and vegetative buffers as supplementary noise reduction measures wherever feasible. These green belts should consist of dense rows of trees or shrubs and should be positioned strategically to serve as natural sound barriers.

E-4.3 Building Design Considerations

For buildings located in or near NGZs, zonal plans should encourage the use of noise-reducing building materials and construction techniques, such as double-glazed windows, soundproofing walls, and appropriate building orientation. For detailed information on noise reduction for town planning schemes, reference may be made to good practice [D-4(2)].

E-5 REGULATION OF NOISE EMISSION SOURCES

E-5.1 Construction Noise Control

Zonal plans should specify noise control measures for construction sites, including the use of low-noise equipment, temporary barriers, and restricted working hours in NSZs.

E-5.2 Industrial and Commercial Noise Control

Zonal plans should regulate emissions to ensure that industries and commercial establishments comply with noise limits, monitored periodically for compliance.

E-6 MONITORING AND COMPLIANCE

E-6.1 Noise Monitoring Provisions

Zonal plans should include provisions for noise monitoring to assess compliance with noise standards. Monitoring stations should be established in key locations within each zone to regularly measure noise levels.

E-6.2 Review and Enforcement

Zonal plans should be reviewed periodically to ensure the effectiveness of noise abatement measures. Non-compliance with noise regulations should result in penalties, including fines or mandatory implementation of additional noise control measures as per local regulations.

E-7 COMMUNITY ENGAGEMENT AND PUBLIC AWARENESS

E-7.1 Public Consultation

The preparation of zonal plans should involve public consultation to incorporate community concerns related to noise pollution and to inform stakeholders about proposed noise abatement measures.

E-7.2 Awareness Programs

Zonal plans should include provisions for public awareness programs to educate residents and businesses on noise pollution control practices and compliance requirements.

ANNEX F

(Clauses [3.8.7](#), [14.4.3](#), [15](#) and [E-3.2](#))

DESIGN REQUIREMENTS AND PERFORMANCE EVALUATION OF NOISE BARRIERS

F-1 DESIGN AND GEOMETRY REQUIREMENTS

F-1.1 The barrier should be built as close as possible to either the source or receiver. Noise barriers should be tall enough and long enough to avoid degradations in barrier performance.

F-1.2 Barriers do not necessarily have to be of constant height as varying the height of the barrier may also help to alleviate the monotonous appearance of long length of barriers.

F-1.3 The ‘line of sight’ between the source and receiver should be cut off completely by the barrier. Barrier design also plays a major role where effective noise barriers should be high and long enough to break the line-of-sight between the sound source and the receiver.

F-1.4 Capped barriers of same height as conventional barriers may offer greater noise reductions than conventional barriers. A specified noise reduction may be achieved by a capped barrier of lower height than a conventional barrier. If high quality absorptive materials are available, the relative screening performance of cylinder-profile and mushroom-profile caps can be improved by up to 8 dBA relative to a plane screen of the same overall height. T-profile caps exhibit similar improvement when the uppermost surfaces are treated with the same absorptive materials. Double cylinder-profile caps are comparatively more efficient than single cylinder-profile caps. When absorptive, these provide equivalent average screening performance to T-profile configurations with the same overall height. A 3 m high T-shaped barrier provides the same performance as a 10 m high plain barrier.

F-2 MATERIALS AND CONSTRUCTION

F-2.1 Barriers should be constructed using solid, non-porous materials having minimum density of 20 kg/m².

F-2.2 The materials used for fabrication of noise barriers include

- a) Steel (painted, galvanized, stainless);
- b) Aluminium;
- c) Plastics or polymer sheet concrete;
- d) Brick or glass fibre reinforced concrete (GFRC);
- e) Proprietary-made acoustic panels; and
- f) Landscaped earth berms.

F-2.3 Sound absorbing material may be applied on the source side of the barrier to reduce the build-up of sound pressure level. An absorptive treatment may be appropriate to reduce the impact of reflected noise.

F-3 DESIGN MORPHOLOGY AND ADDITIONAL CONSIDERATIONS

F-3.1 Strategic planting may be considered to reduce the monotonous appearance of long length of barriers. The design morphology for noise barriers is shown in [Fig. 6](#).

F-3.2 The height of a noise barrier is directly proportional to its noise reduction capacity, wherein higher barriers result in greater levels of noise reduction.

F-3.3 The insertion loss for barriers of heights between 3 m to 6 m typically ranges between 5 dB to 12 dB.

F-3.4 Barriers with a thickness exceeding 3 m shall be classified as thick barriers.

F-3.5 Noise barriers may become part of the surrounding landscape over time and can have potential impacts on ecosystems, road users, and residents living alongside the road. The loss of sunlight and visual impact should also be considered while designing the noise barriers.

F-3.6 Special attention should be given to the selection of materials used in the construction of noise barriers, particularly in areas experiencing extreme weather conditions.

F-3.7 The design of noise barriers should ensure minimal maintenance, except for routine cleaning or repairs due to damage, over the intended lifespan.

F-3.8 The Insertion loss can be estimated by using the Kurze and Anderson model given by the following equation:

- a) $IL = 5dB + 20\log\left(\frac{\sqrt{2\pi N}}{\tanh\sqrt{2\pi N}}\right) dB \text{ upto } N = 12.5$
- b) $IL = 20 \text{ dB for } N > 12.5$, N is the Fresnel number

Maekawa’s curve is one of the most established methods for predicting the insertion loss behind the barriers.

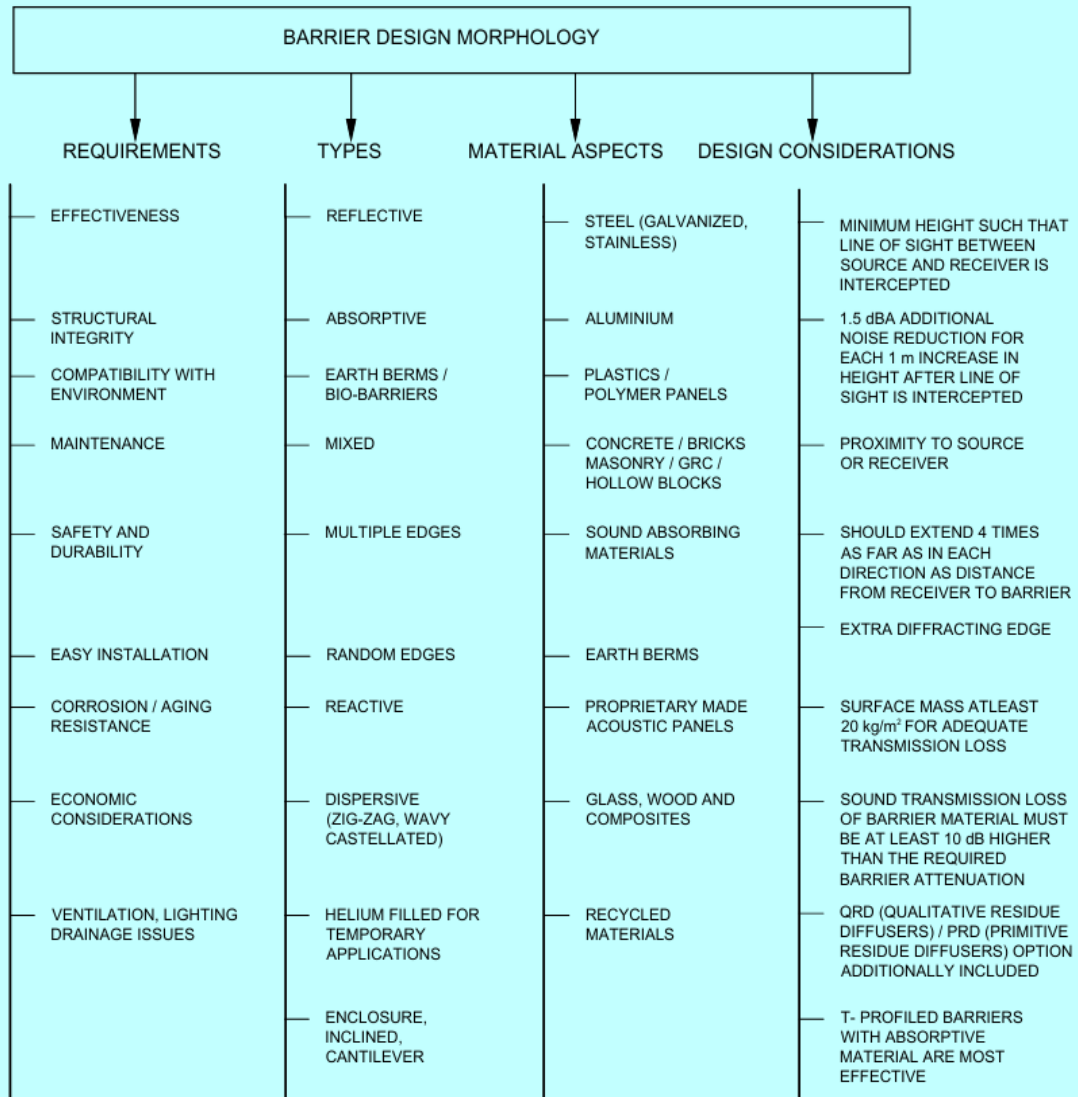


FIG. 6 DESIGN MORPHOLOGY FOR NOISE BARRIERS

F-4 PERFORMANCE EVALUATION OF NOISE BARRIERS

F-4.1 To ensure the effectiveness of a noise barrier, thorough testing should be conducted to verify its performance. The intrinsic acoustic properties of the noise barriers are tested as follows.

F-4.1.1 Diffuse Sound Field Method for Testing Sound Absorption

This method is based on the change of reverberation time inside a reverberation chamber after installing a noise barrier testing sample at the room's surface. This testing method is characterized by a single-number rating of the sound absorption evaluation $DL_{\alpha, \text{NRD}}$, which is described by the following equation:

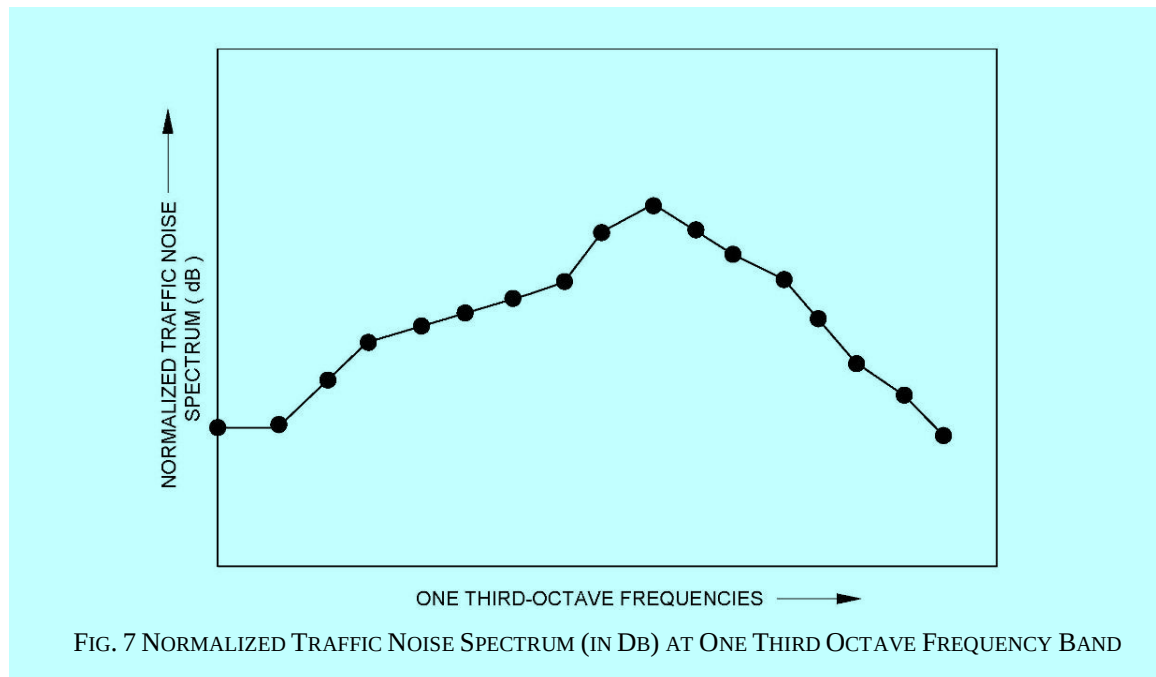
$$DL_{\alpha, \text{NRD}} = -10 \log_{10} \left[1 - \frac{\sum_{i=1}^{18} \alpha_{\text{NRD},i} 10^{0.1 L_i}}{\sum_{i=1}^{18} 10^{0.1 L_i}} \right]$$

where, α_{NRD} is the absorptance of the test sample and L_i is the relative A-weighted sound pressure levels (dB) of the normalized traffic noise spectrum. The classification of $DL_{\alpha, \text{NRD}}$ shall be done on the basis of [Table 14](#).

Table 14 Categories of the Absorptive Performance
(Clause F-4.1.1)

Sl No.	Category	$DL_{\alpha, \text{NRD}}$
(1)	(2)	(3)
i)	A0	Not determined
ii)	A1	< 4
iii)	A2	4 to 7
iv)	A3	8 to 11
v)	A4	12 to 15
vi)	A5	> 15

The value of $DL_{\alpha, \text{NRD}}$ ranges from 0 to 20. $DL_{\alpha, \text{NRD}}$ value greater than 8 is required for most highway projects. Close ‘corridors’, high barriers (> 3 m), and reverberant locations require noise barriers with $DL_{\alpha, \text{NRD}}$ greater than 12. Figure 7 represents the normalized traffic spectrum in one-third octave band. The spectra is normalized to an overall value of 0 dB as its shape is more important rather than the absolute value.



F-4.1.2 Diffuse Sound Field Method for Testing Airborne Sound Insulation

In this method, the noise barrier testing sample is inserted in an opening between two adjacent reverberation rooms, the ‘source room’ and the ‘receiving room’. This testing method is characterized by a single number rating, DL_R for summarizing the airborne sound insulation in third-octave bands and can be given by:

$$DL_R = -10 \log_{10} \left[\frac{\sum_{i=1}^{18} 10^{0.1L_i} 10^{-0.1R_i}}{\sum_{i=1}^{18} 10^{0.1L_i}} \right]$$

where, R is the airborne sound insulation of the test specimen. Table 15 represents the classification of DL_R .

Table 15 Categories of the Airborne Sound Insulation Performance (Clause F-4.1.2)		
SI No.	Category	DL_R
(1)	(2)	(3)
i)	B0	Not determined
ii)	B1	$= < 15$
iii)	B2	15 to 24
iv)	B3	$= > 24$

B3 is required for the majority of highway developments. B2 may be acceptable for low barriers (less than 2 m), however, the rating level of at least 20 dB is required.

F-4.1.3 Direct Sound Field Method for Testing Sound Reflection

This method is based on a comparison between the sound energy reflected from the noise barrier and the sound energy emitted towards it. The third-octave band values are then averaged and weighted according to the standardized traffic noise spectrum L_i , to get a single number rating for the NRD, DL_{RI} as:

$$DL_{RI} = -10 \log_{10} \left[\frac{\sum_{i=1}^{18} RI_i 10^{0.1L_i}}{\sum_{i=1}^{18} 10^{0.1L_i}} \right]$$

where, RI_i is the reflectance in i-th third-octave bands

F-4.1.4 Direct Sound Field Method for Testing Airborne Sound Insulation

This testing method is based on a comparison between the sound energy arriving at the microphone positions with and without the noise barrier. This testing method is characterized by a single number rating, DL_{SI} . The single-number rating by a weighted average with the standardized traffic noise spectrum L_i is described by:

$$DL_{SI} = -10 \log_{10} \left[\frac{\sum_{i=1}^{18} 10^{0.1L_i} 10^{-0.1SI_i}}{\sum_{i=1}^{18} 10^{0.1L_i}} \right]$$

F-4.2 These single-number ratings can be used for the performance evaluation of noise barriers or noise reducing devices. The *in-situ* method is comparatively more reliable and yields more realistic values than the laboratory method and additionally facilitates fault detection and ascertains the acoustic durability of the NRD.

ANNEX G

(Clauses [3.9](#) and [4.5](#))

SPECIAL PROBLEMS REQUIRING EXPERT ADVICE

G-1 GENERAL

Certain design problems require reliable advice of a kind which is not easy to find in published material. The advice of an expert should be sought for these kinds of problems, some examples of which are given in [G-2](#) to [G-9](#).

G-2 ACOUSTIC TEST ROOMS

The design of rooms in which acoustic measurements are carried out, such as reverberation chambers, free-field anechoic rooms and audiometric test rooms, usually requires the advice of an expert.

G-3 PERFORMING SPACES

The design of theatres, opera houses, concert halls and similar performing spaces usually requires expertise in room acoustics and noise control. The intrusion of quite low levels of noise may seriously interfere with the enjoyment of the performance and distract the performers. The requirements for low noise levels often mean that more room has to be allocated for low velocity ventilation ductwork and the impact on the design of the ventilation system is often substantial.

G-4 BROADCASTING AND RECORDING STUDIOS

Broadcasting and recording studios have requirements similar to those of performing spaces. For some infrequent intrusive noises, the requirements are sometimes relaxed on the grounds that a retake of a recording can be done, but this can result in higher operating costs.

G-5 AIRCRAFT NOISE

As there are many variables affecting the level of aircraft noise heard on the ground, expert advice is almost always required. Contours of daytime $L_{Aeq,T}$ levels are available from most major airports. Where measurements of façade insulation are necessary a standard test method may be referred.

G-6 GROUND-BORNE NOISE

Projects involving ground-borne noise from underground trains usually require expert advice.

G-7 LOW-FREQUENCY NOISE

Projects involving low-frequency noise usually require expert advice as accurate measurement is difficult and there is a shortage of reliable data below 100 Hz.

G-8 ACTIVE NOISE CONTROL

Active noise control is the reduction of noise by cancellation with a similar noise (anti-noise) generated by electro-acoustic means. The technique is still under development, but commercial systems are available which successfully reduce low frequency noise from mechanical ventilation systems.

G-9 NOISE SURVEYS

Noise surveys are carried out for a variety of reasons, for example:

- a) Before construction, to establish the existing noise climate at the site of a proposed development where reliable prediction is impracticable, as an aid to the design of the building envelope, either to protect against external noise or contain internally produced noise;
- b) During construction, to monitor noise from building activity, either to assess the likely nuisance to the local community or the risk of hearing damage to the work force;
- c) At the end of a building contract to check the insulation of the building envelope, or the noise levels produced by the services;
- d) As part of a planning requirement; and
- e) To provide objective evidence to support or defend a legal action.

The expense of carrying out a comprehensive noise survey of any kind is likely to be high, so the cost-effectiveness of a full or partial survey should be weighed against alternatives such as prediction. A survey will generally be more accurate and can take account of factors such as prevailing wind conditions.

ANNEX H

(Clause [4.4](#))

AIRBORNE AND IMPACT SOUND INSULATION

H-1 GENERAL

Airborne sound refers to sources which produce sound by directly setting the air around them into vibration. Impact sound refers to sources which produce sound by impulsive mechanical excitation of part of a building (for example by footsteps, electric light switches, slamming doors). Many sources of impact sound also produce significant levels of airborne sound. The term structure-borne sound has no very precise meaning as the structure can be excited by both airborne and impact sources; it is often used to refer to sound that travels for long distances via the structure, especially in connection with vibrating machinery linked directly to the structure.

H-2 DIRECT AND INDIRECT TRANSMISSION

[Figure 8](#) shows diagrammatically a pair of rooms in a house where the construction consists of solid walls, etc, bonded together. Sound travelling from room 1 to room 2 may travel via the direct path a-a and by the many indirect, or flanking, paths shown. The term flanking transmission is usually used to mean transmission paths involving the structure, while the term indirect transmission includes flanking paths and airborne paths through gaps and ducts, etc. The indirect paths may limit the sound insulation attainable no matter how much the direct sound is reduced by the separating wall or floor. The indirect transmission can be reduced by measures such as the following:

- a) Increasing the mass of the flanking walls;
- b) Increasing the mass of the partition and bonding it to the flanking walls;
- c) Introducing discontinuities in the indirect paths;
- d) Erecting independent wall linings adjacent to the flanking walls to prevent energy entering the flanking construction; and
- e) Sealing any air gaps and paths through ducts.

[Figure 9](#) shows a number of indirect paths that have been found in offices.

It is important to remember that standard test laboratories are designed to minimise transmission by all paths other than the direct path. This makes it difficult to relate the results of laboratory measurements to those likely to be obtained in the field.

H-3 AIRBORNE SOUND INSULATION

H-3.1 General

The sound insulation of structural elements such as walls and floors always varies with frequency, the insulation rising in general as the frequency rises.

H-3.2 Terminology

Results from field measurements are usually expressed in terms of the weighted standardized level difference, while laboratory measurements are usually expressed in terms of the sound reduction index. In the absence of significant flanking transmission, the numerical difference between the weighted standardized level difference and the sound reduction index of a wall or floor is usually small for furnished rooms in dwellings, and so either quantity may be used in considering principles; for this purpose, it is, therefore, convenient to use the general term insulation.

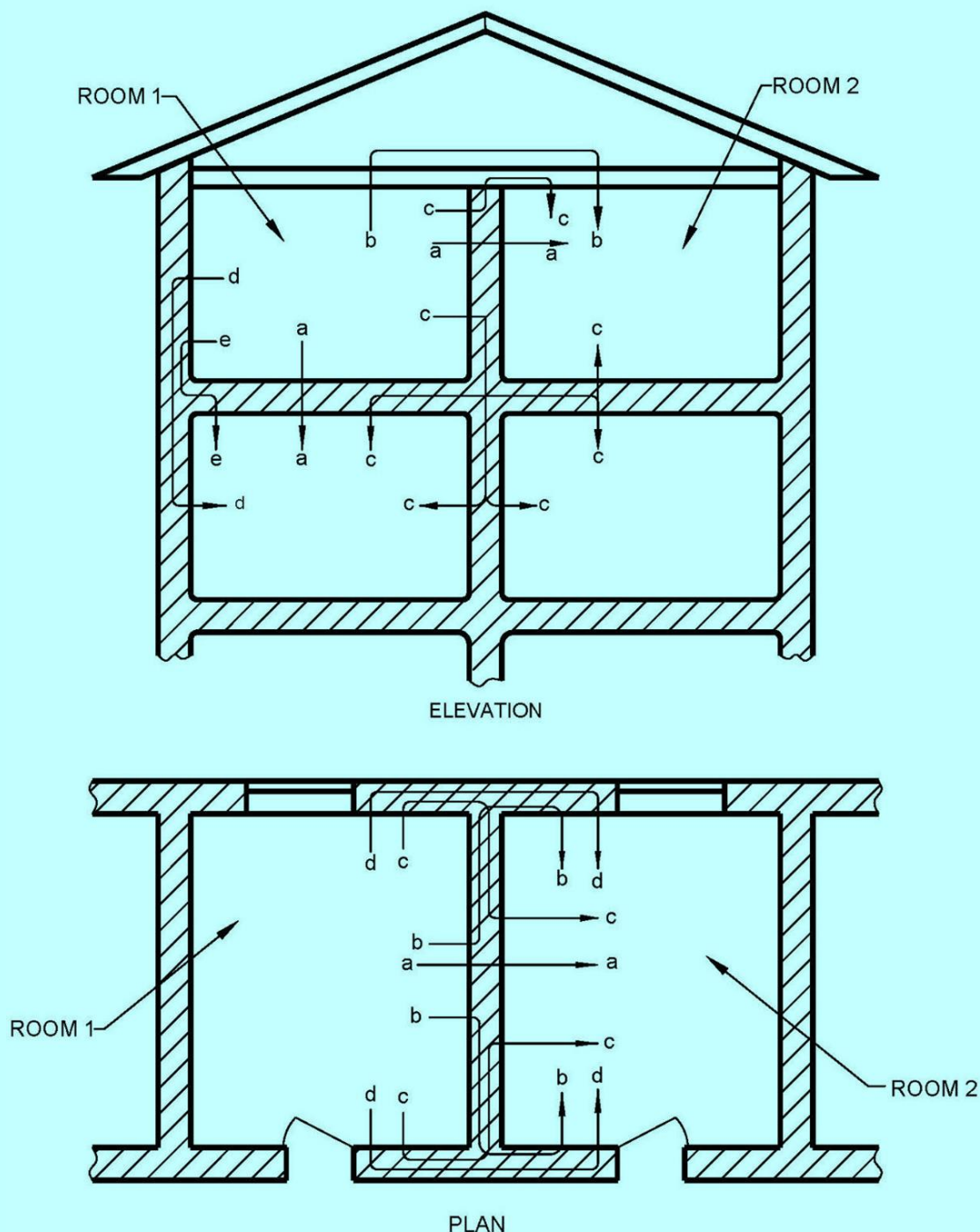


FIG. 8 TRANSMISSION PATHS (VIA THE STRUCTURE) OF NOISE ORIGINATING IN ROOM 1 (DIAGRAMMATIC)

H-3.3 Mass Law

An approximate empirical relationship has been established between sound insulation and mass for single leaf constructions as shown in Fig. 10. This so called ‘Mass Law’ gives a useful first approximation to the behaviour of a single sheet or plate. In practice, the sound insulation predicted by the mass law may not be attained because of factors such as the coincidence effect, which is outlined in H-3.4. Results for specific materials vary around the value given by the Mass Law relationship, and so measured data should be used when available.

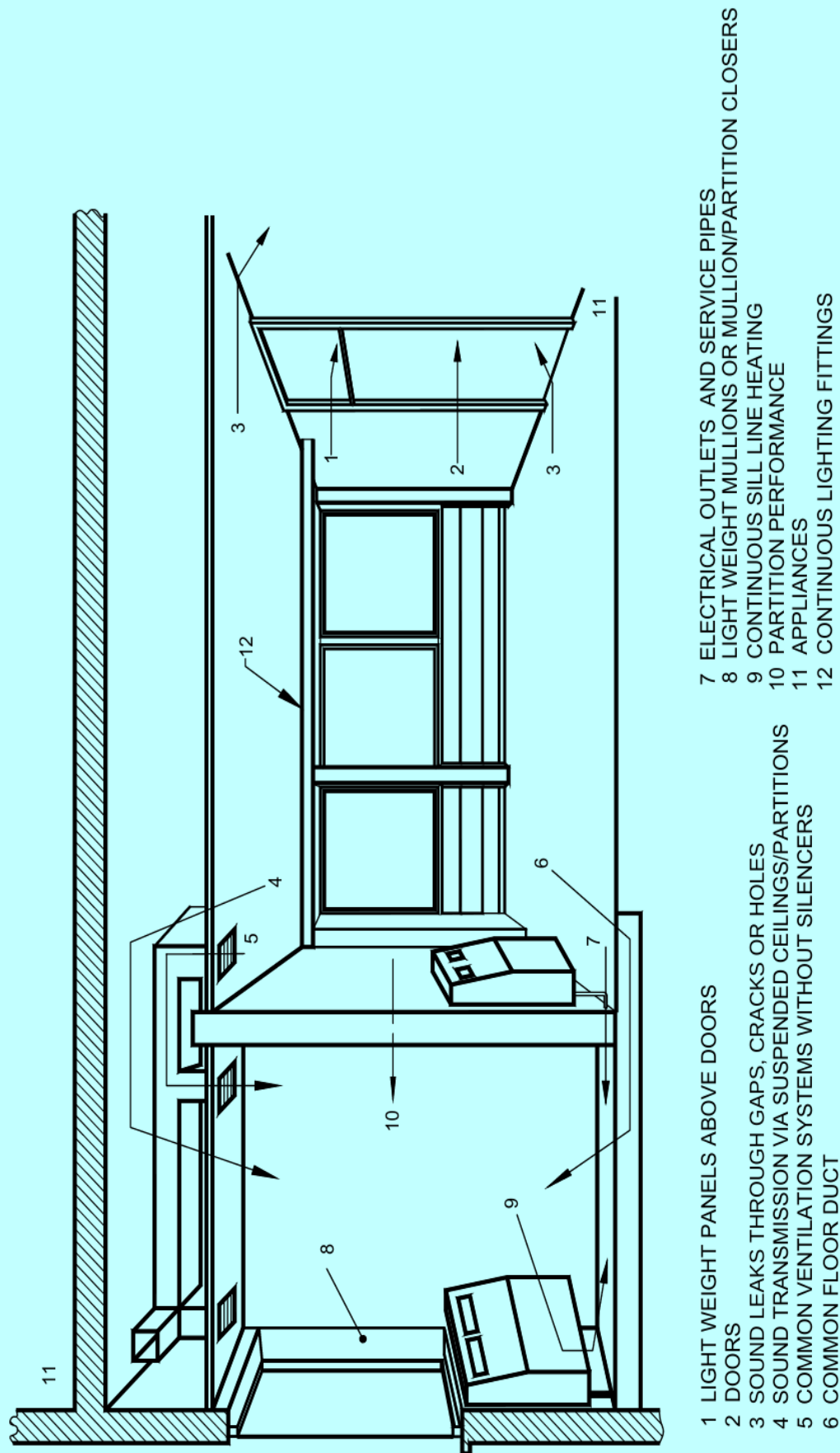
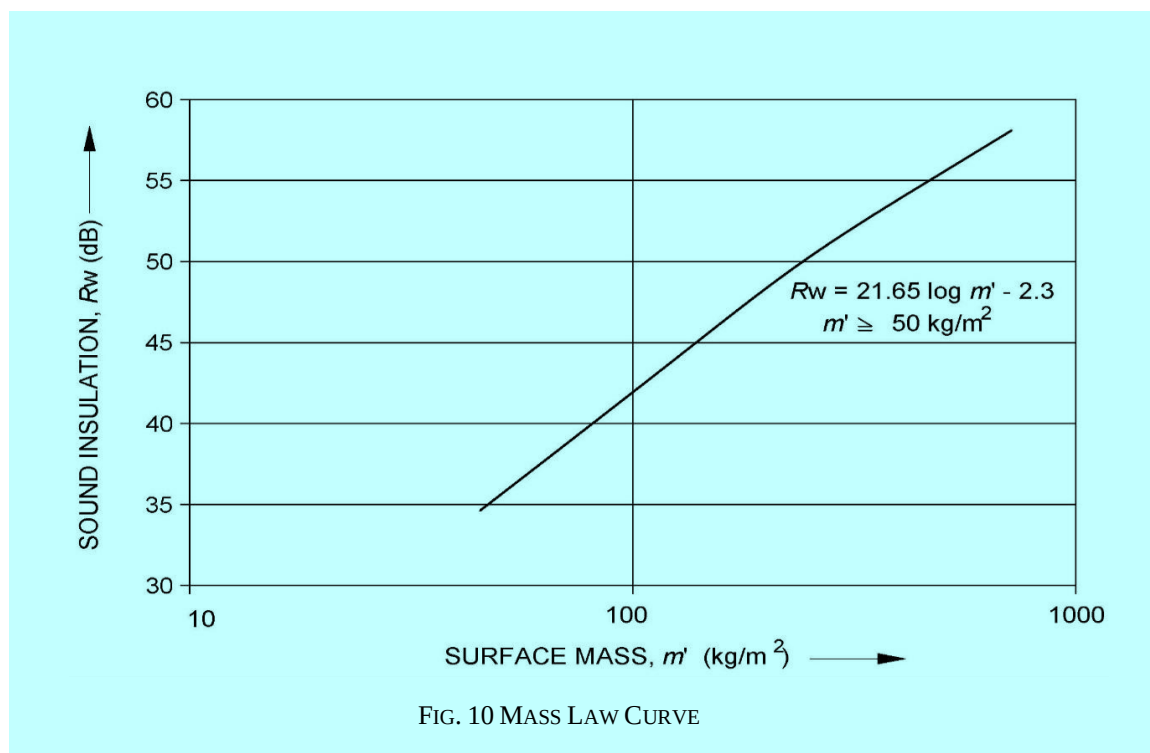


FIG. 9 INDIRECT SOUND LEAKAGE PATHS



H-3.4 Coincidence Effect

The coincidence effect occurs when the wavelength of the wave impressed on the panel by the incident sound wave is close to the wavelength of free bending waves in the panel. The effect of coincidence is to lower the sound insulation of a construction by as much as 10 dB below the level expected from its mass per unit area over a limited frequency range. The coincidence effect can be pronounced with thin light weight partitions, resulting in loss of insulation at middle and high frequencies. Reducing the stiffness without a corresponding reduction of mass can raise the critical frequency above 3 150 Hz, and so improve the insulation over the important 100 Hz to 3 150 Hz range. An increase of stiffness will have the reverse effect.

It is possible to design lightweight stud partitions so that they perform to their maximum effect in the speech frequency region between 250 Hz and 2 000 Hz, that is between the mass-spring-mass and coincidence regions respectively.

The worst coincidence dips occur in materials such as plate glass and rigid metal sheets. Heavily damped materials such as lead sheets are least affected.

H-3.5 Mass-Spring-Mass Frequency

A double leaf wall can perform better than a single leaf wall of similar mass because the sound has to pass through two barriers. If the two leaves are not connected to each other, the insulation values of the two leaves may be added together. However, in practice the leaves are often connected by ties or studs, and the full insulation cannot be achieved. Even where the two leaves are isolated from each other, the full benefit can only be obtained above a certain frequency that depends on the cavity width. This is because the air in the cavity behaves like a spring connecting the leaves together, and causes a resonance at the mass-spring-mass frequency. Below this frequency, the two leaves behave more like an equivalent single leaf.

Making the cavity width wide can reduce the mass-spring-mass frequency, as in the case of sound insulating secondary glazing. The mass-spring-mass frequency (F_0) may be estimated from the following equation:

$$F_0 = 59.6 \sqrt{\frac{1}{d} \left(\frac{1}{m_1} + \frac{1}{m_2} \right)}$$

where

- m_1 and m_2 = surface masses of the two leaves, in kilograms per square metre (kg/m^2); and
- d = cavity width, in metres (m).

H-3.6 Impact Sound Control

A structure that receives an impact or has a vibrating source in contact with it behaves more like an extension of the source rather than an intervening element between source and listener. For this reason, a relatively small amount of impact energy may produce a loud sound and, if the structure is continuous, the sound may travel a long distance. Control is usually obtained by inserting a resilient surface at the point of contact with the source (for example laying a carpet on a floor) or by introducing a structural discontinuity.

Floating floors, which are an example of the latter approach, are a common method of controlling impact sound from footsteps. However, it should be noted that an effective floating floor may result in increased sound from impacts on the source side of the floor.

H-4 AIRBORNE INSULATION VALUES OF WALLS AND AIRBORNE AND IMPACT INSULATION VALUES OF FLOORS

[Table 16](#) and [Table 17](#) give examples of common types of wall and floor construction with sound insulation in the ranges shown. The insulation indices are for field measurements accessed in accordance with good practice [D-4(5)]. The insulation values given are necessarily approximate since examples of nominally identical constructions may show variations of several decibels. All the figures represent values expected in the field, that is, in actual buildings. Many are based directly on field measurements, though other (in the absence of representative field measurements) have been assessed from laboratory data, with an allowance for typical flanking conditions in normal buildings. Variation in the amount of indirect transmission may affect significantly the insulation between two rooms separated by a given barrier. For example, the sound insulation of some types of floor may be reduced by indirect transmission along the walls supporting them, particularly if these walls are of lightweight masonry and carried past the floor.

Table 16 Airborne Sound Insulation of Walls and Partitions (Clause H-4)		
Sl No.	Sound Insulation $D_{nT,w}$ dB	Type of Wall or Partition
(1)	(2)	(3)
i)	26 to 33	a) 1 mm steel sheet panels fixed to steel frame members to form demountable partition units 50 mm overall thickness. Mineral wool cavity insulation b) Plywood or wood fibre board 12 mm thick nailed both sides of 50 mm × 50 mm timber framing members spaced at 400 mm centres c) Paper faced strawboard or wood wool 50 mm thick panels plastered both sides d) Chipboard hollow panels 50 mm thick tongued and grooved edges, hardboard faced. Joints covered with wood trim
ii)	33 to 37	a) Lightweight masonry blockwork. Plaster or drylining on at least one side. Overall mass per unit area not less than 50 kg/m ² b) Laminated plasterboard at least 50 mm thick fixed to timber perimeter framing, any suitable finish. Approximate mass per unit area 35 kg/m ² c) Timber stud partitions any size timbers greater than 50 mm × 50 mm, 400 mm centres, cross noggins, 9.5 mm plasterboard lining on both sides, any suitable finish d) Metal stud partition, 50 mm studs 600 mm centres, clad both sides with 12.5 mm plasterboard, joints filled and perimeters sealed. Approximate mass per unit area 18 kg/m ² e) 50 mm lightweight masonry blockwork, plastered both sides to 12 mm thickness or drylined with 9.5 mm plasterboard

Table 16 (Continued)

Sl No.	Sound Insulation $D_{nT,w}$ dB	Type of Wall or Partition
(1)	(2)	(3)
iii)	37 to 43	a) Lightweight masonry blockwork, plaster or dry lining on at least one side. Overall mass per unit area not less than 75 kg/m² b) Metal stud partition, 50 mm studs 600 mm centres, clad both sides with 12.5 mm plasterboard, joints filled and perimeters sealed. 25 mm mineral fibre quilt hung between studs. Approximate mass per unit area 18 kg/m²
iv)	43 to 50	a) Masonry wall, joints well filled. Either plaster or dry lining on both sides. Overall mass per unit area not less than 150 kg/m² b) 100 mm metal stud partition, 'C' section studs not greater than 600 mm spacing, not less than nominal 50 mm web depth. Clad on both sides with two layers of plasterboard of not less than 25 mm combined thickness. Mineral fibre quilt hung between studs. Approximate mass per unit area 35 kg/m² c) 75 mm x 50 mm timbers framing using staged studs at 300 mm spacing with 25 mm stagger forward and back. Frame clad with two layers of 12.5 mm of plasterboard on both sides. Mineral fibre quilt hung between studs. Approximate mass per unit area 36 kg/m²; d) 50 mm x 25 mm timber stud partition to form a 25 mm cavity, clad on both sides with minimum 38 mm wood wool slabs having their outer faces screeded or plastered; and e) Solid autoclaved aerated concrete block 215 mm thick plaster or dry lined finish on both sides, blockwork joints well filled. Overall mass per unit area not less than 160 kg/m²
v)	50 to 54	a) Two separate frames of timber studs not less than 89 mm x 38 mm, or boxed metal studwork with 50 mm minimum web depth studs at 600 mm maximum centres. A 25 mm mineral wool quilt suspended between frames. Frames spaced to give a minimum 200 mm overall cavity. Clad on outside of each frame with a minimum of 30 mm plasterboard layers (for example 19 mm plus 12.5 thickness). Approximate mass per unit area 54 kg/m² ¹⁾ b) Either in situ or pre-cast concrete wall panel not less than 175 mm thick and not less than 415 kg/m². All joints well filled ¹⁾ c) Brick wall nominal 230 mm thickness, weight (including plaster) not less than 380 kg/m². Plaster or dry-lined finish both sides. Brick work joints well filled ¹⁾ d) 'No fines' concrete 225 mm thickness, weight (including plaster) not less than 415 kg/m². Plaster or dry-lined finish both sides ¹⁾ e) Cavity lightweight aggregate block (maximum density of block 1 600 kg/m³) with 75 mm cavity and wall ties of the butterfly wire type. Dry lined finish on both sides. Joints in blockwork well filled. Overall mass per unit area not less than 300 kg/m² ¹⁾ f) Dense aggregate concrete block cavity wall with 50 mm cavity and wall ties of the butterfly wire type. Dry lined finish on both sides.

Table 16 (Concluded)

Sl No.	Sound Insulation $D_{nT,w}$ dB	Type of Wall or Partition
(1)	(2)	(3)
		Joints in block work well filled. Overall mass per unit area not less than 415 kg/m ² ¹⁾
		g) Autoclaved aerated concrete block cavity wall consisting of two leaves, 100 mm blocks not less than 75 mm apart, with wall ties of the butterfly type. Plaster or dry line finish on both sides. Joints in blockwork well filled. Overall mass per unit area not less than 150 kg/m ² ¹⁾
vi)	54 to 60	a) Two separate frames of timber studs not less than 100 mm × 50 mm spaced at 600 mm maximum centres. A 50 mm mineral wool quilt in each frame between studs. Frames spaced to give a minimum 300 mm overall cavity. Each frame clad on outside with three layers of 12.5 mm plasterboard nailed to framing. Approximate mass per unit area 51 kg/m² ¹⁾ b) Two separate frames of boxed 'C' section galvanized nominal 150 mm steel studs 100 mm apart with a 400 mm overall cavity. 50 mm mineral wool quilt fixed to the back of one frame, each frame clad on outside with three layers of 12.5 mm plasterboard by self drilling or tapping screws. Approximate mass per unit area 47 kg/m² ¹⁾ c) Solid masonry with an overall mass per unit area of not less than 700 kg/m² fully sealed both sides ¹⁾ d) Dense aggregate concrete block solid wall 215 mm thick plaster finish to both surfaces. Overall mass per unit area not less than 415 kg/m² ¹⁾ e) Cavity lightweight aggregate block (maximum density of block 1 600 kg/m³) with 75 mm cavity and wall ties of the butterfly wire type. Plaster finish on both sides. Joints in blockwork well filled. Overall mass per unit area not less than 300 kg/m² ¹⁾ Dense aggregate concrete block cavity wall with 50 mm cavity and wall ties of the butterfly wire type. Plaster finish on both sides Joints in blockwork well filled. Overall mass per unit area not less than 415 kg/m² ¹⁾
NOTES 1 Construction details and workmanship are important if the levels of sound insulation indicated are to be achieved. 2 Where plasterboard is specified it is assumed that the surface mass will be at least 6.5 kg/m² for 9.5 mm thick board, at least 8.5 kg/m² for 12.5 mm thick board, and at least 14.5 kg/m² for 19 mm thick board. If less dense plasterboard is used, the thickness should be increased.		
¹⁾ When considering these constructions for separating walls, expert advice should be sought.		

Table 17 Airborne and Impact Sound Insulation of Floor Constructions

(Clause [H-4](#))

Sl No.	Sound Insulation dB	Type of Floor Construction
(1)	(2)	(3)
i)	$D_{nT,w} = 49$ to 54 $L'_{nT,w} = 56$ to 65	a) A concrete floor having mass per unit area not less than 365 kg/m², including any screed or ceiling finish directly bonded to the floor slab; together with a floating floor or resilient floor covering equivalent to rubber or sponge rubber underlay or thick cork tile (for example carpet and underlay or sponge rubber backed vinyl flooring) b) A solid floor consisting of: 1) a solid slab; or 2) concrete beams and infilling blocks; or 3) hollow concrete planks; together with a floating floor. A ceiling finish is required for a beam and block floor. In each case the slab should have a mass per unit area of at least 300 kg/m², including any screed or ceiling finish directly bonded to it Where a floating floor is laid over a floor of beams and hollow infill blocks or hollow beams along the top of the structural floor, it should be sealed and levelled before the resilient layer is put down. It is also essential to have due regard for conduits and pipework which should be laid and covered so as to prevent any short circuit of the floor's isolating properties If precast units are used as a structural floor, it is essential that the joints are filled to ensure that the sound insulation performance is maintained The resilient material is laid to cover completely the structural floor and turned up against the surrounding wall along all edges. The resilient layer is usually of mineral fibre, or a special grade of expanded polystyrene. When the screed is laid, it is important that none of the mix finds its way through the resilient layer to the structural floor, as this will short circuit the isolation between the two decks and significantly reduce the sound insulation A floor consisting of boarding nailed to battens laid to float upon an isolating layer of mineral fibre capable of retaining its resilience under imposed loading. With battens running along the joists, a dense fibre layer can be used in strips. The ceiling below to be of metal lath and plaster not less than 29 mm thick, with pugging on the ceiling such that the combined mass per unit area of the floor, ceiling and pugging is not less than 120 kg/m². This construction will only give values for $D_{nT,w}$ of 50 dB to 53 dB, and a value for $L'_{nT,w}$ of 75 dB c) A floor consisting of 18 mm tongued and grooved chipboard on 19 mm plasterboard laid on battens running parallel to the joists and supported on 25 mm thick mineral wool of about 90 kg/m³ to 140 kg/m³ density; 100 mm of fibre

Table 17 (Concluded)

Sl No.	Sound Insulation dB	Type of Floor Construction
(1)	(2)	(3)
		absorbent (as used for insulation in roof spaces) laid between the joists on top of the plasterboard ceiling ¹⁾
		d) A floor consisting of 18 mm tongued and grooved chipboard on 19 mm plasterboard floating on a 25 mm thick mineral wool layer of about 60 kg/m ³ to 80 kg/m ³ density; this on a 12.5 mm plywood platform; 100 mm of fibre absorbent laid between the joists on top of the plasterboard ceiling ¹⁾ ; and
ii)	$D_{nT,w} = 32 \text{ to } 36$ $L'_{nT,w} = 80 \text{ to } 85$	Timber joist floor consisting of 22 mm tongued and grooved floor boarding or equivalent fixed directly to floor joists. Ceiling of 12.5 mm plasterboard and skim with no floor covering
NOTES		
1 Construction details and workmanship are important if the levels of sound insulation indicated are to be achieved.		
2 Where plasterboard is specified it is assumed that the surface mass will be at least 8.5 kg/m ³ for 12.5 mm thick board, and at least 14.5 kg/m ² for 19 mm thick board. If less dense plasterboard is used, the thickness should be increased.		
¹⁾ In these types of floor construction, the ceiling may be 19 mm plus 12.5 mm plasterboard. It is imperative that the resilient layer is not punctured by nails.		

In many cases, simple solid partitions give insulation values according to their mass (see [H-3.3](#)). Moreover, with partitions of this type there is usually little variation between field and laboratory test results unless the laboratory insulation exceeds 45 dB. Exceptions may occur in buildings that have not been specially designed to minimise common cavities and strongly coupled elements in lightweight panelling. The examples given are not exhaustive. Flanking structures are not listed since these can vary widely and are often dependent upon other factors such as thermal insulation, which are outside the scope of this NBCS.

ANNEX J

(Clause [13.1](#))

BASIC DESIGN TECHNIQUES FOR NOISE CONTROL IN AIR CONDITIONING, HEATING AND MECHANICAL VENTILATION SYSTEM

J-1 When selecting fans and other related mechanical equipment and when designing air distribution systems to minimise the sound transmitted from different components to the occupied spaces that they serve, the following recommendations should be considered:

- Design the air distribution system to minimise flow resistance and turbulence. High flow resistance increases the required fan pressure, which results in higher noise being generated by the fan. Turbulence increases the flow noise generated by duct fittings and dampers in the air distribution system, especially at low frequencies;
- Select a fan to operate as near as possible to its rated peak efficiency when handling the required quantity of air and static pressure. Also, select a fan that generates the lowest possible noise but still meets the required design conditions for which it is selected. Using an oversized or undersized fan that does not operate at or near rated peak efficiency may result in substantially higher noise levels;
- Design duct connections at both the fan inlet and outlet for uniform and straight air flow. Failure to do this may result in severe turbulence at the fan inlet and outlet and in flow separation at the fan blades. Both of these may significantly increase the noise generated by the fan;

- d) Select duct silencers that do not significantly increase the required fan total static pressure;
- e) Place fan-powered mixing boxes associated with variable volume air distribution systems away from noise-sensitive areas;
- f) Minimise flow-generated noise by elbows or duct branch take-offs, whenever possible, by locating them at least four to five duct diameters from each other. For high velocity systems, it may be necessary to increase this distance to up to ten duct diameters in critical noise areas;
- g) Keep airflow velocity in the duct as low as possible (7.5 m/s or less) near critical noise areas by expanding the duct cross-section area. However, do not exceed an included expansion angle of greater than 15°. Flow separation, resulting from expansion angles greater than 15°, may produce rumble noise. Expanding the duct cross-section area will reduce potential flow noise associated with turbulence in these areas;
- h) Use turning vanes in large 90° rectangular elbows and branch take-offs. This provides a smoother transmission in which the air can change flow direction, thus reducing turbulence;
- j) Place grilles, diffusers and registers into occupied spaces as far as possible from elbows and branch take-offs;
- k) Minimise the use of volume dampers near grilles, diffusers and registers in acoustically critical situations;
- m) Vibration isolates all vibrating reciprocating and rotating equipment if mechanical equipment is located on upper floors or is roof-mounted. Also, it is usually necessary to vibration isolate the mechanical equipment that is located in the basement of a building as well as piping supported from the ceiling slab of a basement, directly below tenant space. It may be necessary to use flexible piping connectors and flexible electrical conduit between rotating or reciprocating equipment and pipes and ducts that are connected to the equipment;
- n) Vibration isolates ducts and pipes, using spring and/or neoprene hangers for at least the first 15 m from the vibration-isolated equipment;
- p) Use barriers near outdoor equipment when noise associated with the equipment will disturb adjacent properties if barriers are not used. In normal practice, barriers typically produce no more than 15 dB of sound attenuation in the mid-frequency range; and
- q) Table 18 lists several common sound sources associated with mechanical equipment noise. Anticipated sound transmission paths and recommended noise reduction methods are also listed in the [Table 19](#). Airborne and/or structure-borne sound can follow any or all of the transmission paths associated with a specified sound source.

Table 18 Sound Sources, Transmission Paths and Recommended Noise Reduction Methods [Clause J-1(q)]		
Sl No.	Sound Source	Path No. (see Table 19)
(1)	(2)	(3)
i)	Circulating fans, grilles, registers, diffusers, unitary equipment in room	1
ii)	Induction coil and fan-powered VAV mixing units	1, 2
iii)	Unitary equipment located outside of room served; remotely located air-handling equipment, such as fans, blowers, dampers, duct fitting, and air washers	2, 3

Table 18 (Concluded)		
Sl No.	Sound Source	Path No. (see Table 19)
(1)	(2)	(3)
iv)	Compressors, pumps, and other reciprocating and rotating equipment (excluding air-handling equipment)	4, 5, 6
v)	Cooling towers; air-cooled condensers	4, 5, 6, 7
vi)	Exhaust fans; window air conditioners	7, 8
vii)	Sound transmission between rooms	9, 10

Table 19 Sound Transmission Paths and Recommended Noise Reduction Methods			
<i>[Clause J-1(q) and Table 18]</i>			
Sl No.	Path No.	Transmission Paths	Noise Reduction Methods
(1)	(2)	(3)	(4)
i)	1	Direct sound radiated from sound sources to ear Reflected sound from walls, ceiling and floor	Direct sound can be controlled only by selecting quiet equipment Reflected sound is controlled by adding sound absorption to the room and to equipment location
ii)	2	Air and structure-borne sound radiated from casings and through walls of ducts and plenums is transmitted through walls and ceiling into rooms	Design duct and fittings for low turbulence; locate high velocity ducts in noncritical areas; isolate ducts and sound plenums from structure with neoprene or spring hangers
iii)	3	Airborne sound radiated through supply and return air ducts to diffusers in room and then to listener by Path 1	Select fans for minimum sound power; use ducts lined with sound-absorbing material; use duct silencers or sound plenums in supply and return air ducts
iv)	4	Noise transmitted through equipment room walls and floors to adjacent rooms	Locate equipment rooms away from critical areas; use masonry blocks or concrete for equipment room walls and floor
v)	5	Vibration transmitted via building structure to adjacent walls and ceilings, from which it radiates as noise into room by Path 1	Mount all machines on properly designed vibration isolators; design mechanical equipment room for dynamic loads; balance rotating and reciprocating equipment

Table 19 (Concluded)

Sl No.	Path No.	Transmission Paths	Noise Reduction Methods
(1)	(2)	(3)	(4)
vi)	6	Vibration transmission along pipes and duct walls	Isolate pipe and ducts from structure with neoprene or spring hangers; install flexible connectors between pipes, ducts, and vibrating machines
vii)	7	Noise radiated to outside enters room windows	Locate equipment away from critical areas; use barriers and covers to interrupt noise paths; select quiet equipment
vii)	8	Inside noise follows Path 1	Select quiet equipment
ix)	9	Noise transmitted to an air diffusers in a room, into a duct, and out through an air diffuser in another room	Design and install duct attenuation to match transmission loss of wall between rooms
x)	10	Sound transmission through, over, and around room partition	Extend partition to ceiling slab and tightly seal all around; seal all pipe, conduit, duct and other partition penetrations

J-2 Room Criteria (RC)

Room Criteria (RC) is the rating methodology specifically intended for rating the heating ventilation and air-conditioning system-related noise in buildings and estimate the subjective response of an occupant. The subjective character or quality of the spectrum is determined by a calculation of the Quality Assessment Index (QAI). The QAI is a measure of the degree to which the shape of the spectrum under evaluation deviates from the shape of the RC reference curve.

NOTE — The ASHRAE-developed HVAC noise rating method, RC Mark II rating is a two-dimensional expression, which takes the form, RC xx(yy). The first term, 'xx', is the value of the RC reference curve corresponding to the arithmetic average of the levels in the 500 Hz, 1 000 Hz, and 2 000 Hz octave bands; it is a quantitative descriptor, in the sense that it identifies the level of the spectrum in the principal speech frequency region. The second term, 'yy', is a qualitative descriptor that identifies the character of the sound as perceived by the listener: (N) for neutral, (LF: 16 Hz to 63 Hz) for low-frequency dominant (rumble), (MF: 125 Hz to 500 Hz) for mid-frequency dominant (roar), and (HF: 1 000 Hz to 4 000 Hz) for high-frequency dominant (hiss). There are also two sub-categories of the low-frequency descriptor: (LF_A), denoting a substantial, clearly noticeable degree of sound-induced vibration of room surfaces in lightweight interior building construction, and (LF_B), denoting a moderate, but perceptible degree of sound-induced vibration.

ANNEX K

(Clause [13.2](#))

SUGGESTED EQUIPMENT NOISE DATA SHEET

K-1 It is recommended that an equipment noise data sheet be furnished to intending bidders of mechanical equipment such as air conditioning, heating and mechanical ventilation machinery or diesel generating units specifying noise requirements at the time of request for quotation. Following is a sample noise data sheet suggested for the purpose:

SAMPLE OF EQUIPMENT NOISE DATA SHEET FOR NOISE SPECIFICATION TO BE SENT TO SUPPLIERS				
Equipment Description _____ Type _____ Item No. _____				
Octave-Band Centre Frequency Hz	Desired Sound Pressure Level, L_p	Supplier to Complete		
		Actual	Special Design	Special Noise Control Measures Recommended
(1)	(2)	(3)	(4)	(5)
63				
125				
250				
500				
1 000				
2 000				
4 000				
8 000				

NOTES

- 1 The measurements of SPL shall be at a distance of 1.0 m from the equipment and 1.5 m above grade or floor. The measurement method shall be described and the point of maximum levels furnished.
- 2 Complete column 3 for actual levels of standard equipment.
- 3 Complete column 4 for special design for low noise (if such alternative is available).
- 4 Complete column 5 for noise control measures such as enclosure.
- 5 Indicate if the equipment meets the specified noise levels without modification (Yes/No).
- 6 If no, additional costs required :
For Column 4 _____
For Column 5 _____

It will be observed from the col (3), col (4) and col (5) that the buyer would get quotation for supply of standard equipment at a price P-1, whose noise characteristics would be as per col (3). Col (4) would indicate acoustical performance for a special design at a price P-2. Col (5) would indicate the acoustical performance if the owners were to provide special noise control measures for the installation (whose broad details and approximate estimated cost is also furnished by the vendor).

LIST OF STANDARDS

The following list records those standards which are acceptable as ‘good practice’ and ‘accepted standards’ in the fulfilment of the requirements of this NBCS. The latest version of a standard shall be adopted at the time of enforcement of this NBCS. The standards listed may be used for conformance with the requirements of the referred clauses in this NBCS. In the following list, the number appearing in the first column within parentheses indicates the number of the reference in this Section.

	<i>IS No.</i>	<i>Title</i>
(1)	IS 11050 (Part 1) : 2023	Rating of sound insulation in buildings and of building elements Airborne sound insulation (<i>first revision</i>)
(2)	IS 4954 : 1968	Recommendations for noise abatement in town planning
(3)	IS 2526 : 1963	Code of practice for acoustical design of auditoriums and conference halls
(4)	IS 15575 (Part 1) : 2016/ IEC 61672-1 : 2013	Electroacoustics — Sound level meters Specifications (<i>first revision</i>)
	IS 15575 (Part 2) : 2023/ IEC 61672-2 : 2013	Electroacoustics — Sound level meters Pattern evaluation tests (<i>second revision</i>)
(5)	IS 11050 (Part 1) : 2023 (Part 2) : 2023	Rating of sound insulation in buildings and of building elements Airborne sound insulation (<i>first revision</i>) Impact sound insulation (<i>first revision</i>)

NATIONAL BUILDING CONSTRUCTION STANDARDS

PART D BUILDING SERVICES

SECTION 5 INSTALLATION OF LIFTS, ESCALATORS, MOVING WALKS AND POWER-DRIVEN PARKING SYSTEMS

5A LIFTS

BUREAU OF INDIAN STANDARDS

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National Building Code Sectional Committee, CED 46

FOREWORD

This NBCS (Part D/Subsection 5A) deals with the requirements for planning, design, installation, operation, maintenance and inspection of lifts (passenger lifts, goods lifts, hospital lifts, service lifts and dumb waiter) so as to ensure safe movement of people with satisfactory performance in buildings.

This Section was first published in 1970 and was subsequently revised in 1983, 2005 and 2016 as part of National Building Code of India. This Section covers the requirements for installation and maintenance of lifts and escalators in buildings. This Section should be read with Part F 'Fire and Life Safety' of this NBCS from fire safety requirements point of view. The major changes in the first revision of 1983 were addition of outline dimensions of different types of lifts and requirements of escalators in buildings. Emphasis was laid on coordination between the engineer/architect and the lift manufacturer to arrive at the number and position of lifts for attaining optimum efficiency in serving the building with safety.

The significant changes incorporated in the 2005 revision included addition of new clauses/recommendations on building management system; addition of new clauses on fireman's lift, infrared light curtain safety, and braille button for the visually impaired people and the updation of provisions as per the revised Indian Standards on lifts on which this section was based.

In the 2016 revision, the erstwhile Section 5 on lifts and escalators was divided into two subsections namely:

5A Installation of lifts

5B Installation of escalators and moving walks

The significant changes incorporated in the 2016 revision included the introduction of high-speed lifts and lifts for tall buildings, the use of lifts for fire-fighting and emergency evacuation, machine room less (MRL) lifts, the introduction to destination control system, seismic resistance aspects in lifts, performance measurements for lifts and introduction of new clause on operation and maintenance of lifts.

In this revision, publication has been renamed as National Building Construction Standards (NBCS). As a result of experience gained since the implementation of the 2016 version of this NBCS and feedback received as well as revisions of Indian Standards on which this Section was based, a need was felt to revise this Section.

The significant changes incorporated in this revision (Subsection 5A) include the following:

- a) Definitions of various terms relating to energy dissipation and accumulation types, compensation means, hydraulic and electric systems, and safety mechanisms have been included to enhance clarity in performance measurement, seismic considerations, and maintenance aspects of lifts;
- b) Additions have been introduced in the planning and design guidelines to address factors such as absenteeism in office buildings, provision of cafeterias and specialty facilities, population considerations for residential and hotel buildings in [4](#);
- c) Additions have been made in the earthing provisions to elaborate on system earthing methods (TN, TT, IT), their sub-classifications, applications, and the importance of protective equipotential bonding and earth fault protection, in [6.2.3](#);
- d) Provisions on flood sensors have been introduced for installation in lift pits to enable early detection of water ingress, alarm activation and safe shutdown of lifts, in [8.7](#);
- e) Provisions of destination control systems (DCS) for call assignments, improving passenger flow and operational efficiency have been included, in [9.11](#);
- f) Introduction of double-deck lift systems, with two cabins arranged one above the other, optimizing shaft use and improving efficiency in high-traffic areas and for two independent lift cabins in a single shaft solution to address up-peak and inter-floor traffic demands, in [11.7](#);

- h) Introduction of passenger evacuation strategies (in [11.8.1](#) and [11.8.2](#)) for both independent and joined cabin systems, enhancing safety during emergencies; and
- j) Expanded requirements for machine room levels based on lift speed and shaft dimensions.

The information contained in this Section is largely based on the following Indian Standards:

<i>IS No.</i>	<i>Title</i>
IS 14665	Electric traction lifts:
(Part 1) : 2000	Guidelines for outline dimensions of passenger, goods, service and hospital lifts
(Part 2/Sections 1) : 2000	Code of practice for installation, operation and maintenance, Section 1 Passenger and goods lifts
(Part 3/Sections 1) : 2000	Safety rules, Section 1 Passenger and goods lifts
(Part 5 : 1999)	Specification & Inspection manual
IS 14671 : 1999	Code of practice for installation and maintenance of hydraulic lifts
IS 15330 : 2020	Requirements of lifts for persons with disabilities (<i>first revision</i>)
IS 15785 : 2007	Installation and maintenance of lift without conventional machine rooms & Code of practice

Assistance has also been derived from the following publications in preparation of this Subsection:

<i>Other Standards/Other Publication</i>	<i>Title</i>
ISO 8100-30 : 2019	Lifts for the transport of persons and goods & Part 30: Class I, II, III and VI lifts installation
ISO 4190-5 : 2006	Lift (elevator) installation & Part 5: Control devices, signals and additional fittings
ISO 3008-2 : 2017	Fire-resistance tests & Part 2: Lift landing door assemblies
ISO 3864-1 : 2011	Graphical symbols & Safety colours and safety signs & Part 1: Design principles for safety signs and safety markings
ISO 8100-33:2022	Lifts for the transport of persons and goods Part 33: T-type guide rails for lift cars and counterweights
ISO 18738-1 : 2012	Measurement of ride quality & Part 1: Lifts (elevators)
ISO/TS 18870 : 2014	Lifts (elevators) & Requirements for lifts used to assist in building evacuation
EN 81-20 : 2020	Safety rules for the construction and installation of lifts & Lifts for the transport of persons and goods & Part 20: Passenger and goods passenger lifts
EN 81-77 : 2022	Safety rules for the construction and installations of lifts & Particular applications for passenger and goods passenger lifts & Part 77: Lifts subject to seismic conditions
EN 81-50 : 2020	Safety rules for the construction and installation of lifts & Examinations and tests & Part 50: Design rules, calculations, examinations and tests of lift components
CIBSE Guide D: 2015	Transportation Systems in Buildings 2015, The Chartered Institution of Building Services Engineers, London, U.K.

<i>Other Standards/Other Publication</i>	<i>Title</i>
CTBUH 2020	Emergency Evacuation: Elevator Systems Guidelines, Council on Tall Buildings and Urban Habitat, Illinois, USA

All standards, whether given herein above or cross-referred to in the main text of this Subsection, are subject to revision. The parties to agreement based on this Subsection are encouraged to investigate the possibility of applying the most recent editions of the standards.

As the regulation of land development and buildings is a State subject and is dealt by the States and local bodies such as urban local bodies within their jurisdiction through building regulatory documents like building regulations, building byelaws, and fire regulations, this document is voluntary in nature and is non-binding. The implementation depends on adoption by concerned parties or stipulations in a contract or by appropriate adoption by the concerned authorities.

For the purpose of deciding whether a particular requirement of this Subsection is complied with, the final value, observed or calculated, expressing the result of a test or analysis, shall be rounded off in accordance with IS 2: 2022 *Rules for rounding off numerical values (second revision)*. The number of significant places retained in the rounded off value should be the same as that of the specified value in this Subsection.

Important Explanatory Note for Users of this NBCS

In any Part/Section of this NBCS, where reference is made to **‘good practice’** in relation to **design, constructional procedures or other related information**, and where reference is made to **‘accepted standard’** in relation to **material specification, testing or other related information**, the Indian Standards listed at the end of the Part/Section shall be used as a guide to the interpretation.

At the time of publication, the editions indicated in the standards were valid. All standards are subject to revision and parties to agreements based on any Part/Section are encouraged to investigate the possibility of applying the most recent editions of the standards.

In the list of standards given at the end of a Part/Section, the number appearing within parentheses in the first column indicates the number of the reference of the standard in the Part/Section. For example:

a) Good practice [D-5A(1)] refers to the Indian Standard given at serial number (1) of the List of Standards given at the end of this Part D/Subsection 5A, that is, IS 17900 (Part 1) : 2022 ‘Lifts for the transport of persons and goods Part 1: Safety Rules’

NATIONAL BUILDING CONSTRUCTION STANDARDS

PART D BUILDING SERVICES

SECTION 5 INSTALLATION OF LIFTS, ESCALATORS, MOVING WALKS AND POWER-DRIVEN PARKING SYSTEMS

5A LIFTS

1 SCOPE

1.1 This NBCS (Part D/Subsection 5A) covers the requirements for planning, design, installation, operation, maintenance and inspection of lifts (passenger lifts, goods lifts, hospital lifts, automobile lifts, service lifts and dumbwaiters) so as to ensure safe movement of people and goods with satisfactory performance.

1.2 This Subsection gives information that should be exchanged among the architect/engineer, the consulting engineer and the lift manufacturer from the stage of planning to installation including maintenance.

NOTE ð The provisions given in this Subsection are primarily for electric traction lifts; however, most of these provisions are also applicable to hydraulic lifts {see good practice [D-5A(1)] and [D-5A(2)]}.

2 TERMINOLOGY

2.0 For the purpose of this subsection, the following definitions shall apply.

2.1 General Terms Relating to Lifts

2.1.1 Accessibility ð Accessibility in the context of this standard is the quality of a building or structure or a lift which enables all people irrespective of their age, gender, or abilities to access it and use its features equally and independently.

2.1.2 Apron ð Smooth vertical part extending downwards from the sill of the landing or car entrance.

2.1.3 Automatic Rescue Device (ARD) ð Device that operates automatically in case of failure or loss of power supply to move the lift car to a nearest landing.

2.1.4 Automobile Lift ð A lift designed primarily for the transport of vehicles which may however, carry a lift attendant and/or occupants of the vehicle.

2.1.5 Available Car Area ð Area of the car, which is available for passengers or goods during operation of the lift.

2.1.6 Bottom Car Run-by ð The distance between the car buffer striker plate and the striking surface of the car buffer when the car is in level with the bottom terminal landing.

2.1.7 Balancing Weight ð Mass which saves energy by balancing all or part of the mass of the car.

2.1.8 Bottom Counterweight Run-by ð The distance between the counter weight buffer striker plate and the striking surface of the counterweight buffer when the car is in level with the top terminal landing.

2.1.9 Buffer ð A device designed to stop a descending car or counter weight beyond its normal limit of travel by storing or by absorbing and dissipating the kinetic energy of the car or counterweight.

The buffer shall be one of the following types:

- a) Energy dissipation type; and
- b) Energy accumulation type:
 - 1) Linear type
 - 2) Non-Linear type

2.1.9.1 Energy dissipation type ð These are usually hydraulic buffer that dissipate the energy of the impact in the form of heat during the travel of the buffer. The hydraulic buffer falls in this category.

These types of buffers may be used whatever may be the rated speed of the lift. For example, oil buffer.

2.1.9.1.1 Oil buffer stroke ð The oil displacing movement of the buffer plunger or piston, excluding the travel of the buffer plunger accelerating device.

2.1.9.2 Energy accumulation type ð These can take the form of simple mechanical springs or polymer buffers which store the absorbed energy of the impact in the form of strain energy.

Energy accumulation type buffers, with linear and non-linear characteristics, shall only be used if the rated speed of the lift does not exceed 1 m/s.

2.1.9.2.1 Spring buffer load rating ð The load required to compress the spring by an amount equal to its stroke.

2.1.9.2.2 Spring buffer stroke ð The distance, the contact end of the spring can move under a compressive load until the spring is compressed solid.

2.1.10 Call Indicator ð A visual and audible device in the car to indicate to the attendant the lift landings from which calls have been made.

2.1.11 Car Bodywork ð The enclosing bodywork of the lift car which comprises the sides and roof and is built upon the car platform.

2.1.12 Car Door Electric Contact ð An electric device, the function of which is to prevent operation of the driving machine by the normal operating device of the lift unless the car door is in the closed position.

2.1.13 Car Frame ð The supporting frame or sling to which the platform of the lift car, its safety gear, guide shoes and suspension ropes are attached.

2.1.14 Car Platform ð The part of the lift car which forms the floor and directly supports the load.

2.1.15 Control ð The system governing starting, stopping, direction of motion, acceleration, speed and retardation of moving member.

2.1.15.1 Variable voltage motor control (Generator generator field control) ð A system of control which is accomplished by the use of an individual generator for each lift wherein the voltage applied to the driving machine motor is adjusted by varying the strength and direction of the generator field.

2.1.15.2 Electronic control ð A system of control which is accomplished by the use of electronic devices for driving the lift motor at variable speed.

2.1.15.3 Alternating current variable voltage (ACVV) control ð A system of speed control which is accomplished by varying the driving and braking torque by way of voltage variation of the power supply to the driving machine induction motor.

2.1.15.4 Alternating current variable voltage variable frequency (ACVVVF) control ð A system of speed control that is accomplished by varying the voltage and frequency of the power supply to the driving machine induction motor.

2.1.15.5 Solid-state d.c. variable voltage control ð A solid-state system of speed control which is accomplished by varying the voltage and direction of the power supply to the armature of driving machine d.c. motor.

2.1.16 Counterweight ð A weight or series of weights to counterbalance the weight of the lift car and part of the rated load.

2.1.17 Compensation Means ð Includes all ropes/coated synthetic belts/elastomeric coated ropes/chain to counterbalance suspension means.

2.1.18 Deflector Sheave ð An idler pulley used to change the direction of a rope lead.

2.1.19 Door (Lift Landing Door and Lift Car Door) — The door located at each floor entrance of the lift shaft, providing access to the lift car. It remains closed when the lift is not present at that level for safety, which opens and closes in synchronization with the car door, ensuring passengers' safe entry and exit.

2.1.19.1 Door, centre opening sliding ð A door which slides horizontally and consists of two or more panels which open from the centre and are usually so interconnected that they move simultaneously.

2.1.19.2 Door, mid-bar collapsible ð A collapsible door with vertical bars mounted between the normal vertical members.

2.1.19.3 Door, multi-panel ð A door arrangement whereby more than one panel is used such that the panels are connected together and can slide over one another by which means the clear opening can be maximized for a given shaft width. Multipanels are used in centre opening and two speed sliding doors.

2.1.19.4 Door, single slide ð A single panel door which slides horizontally.

2.1.19.5 Door, two speed sliding ð A door which slides horizontally and consists of two panels, one of which moves at twice the speed of the other.

2.1.19.6 Door, vertical bi-parting ð A door which slides vertically and consists of two panels or sets of panels that move away from each other to open and are so interconnected that they move simultaneously.

2.1.19.7 Door, vertical lifting ð A single panel door, which slides in the same plane vertically up to open.

2.1.19.8 Door, swing ð A swinging type single panel door which is opened manually and closed by means of a door closer when released.

2.1.20 Door Closer ð A device which automatically closes a manually opened door.

2.1.21 Door Operator ð A power-operated device for opening and closing doors.

2.1.22 Dumb Waiter ð A lift with a car which moves in guides in a vertical direction; has a net floor area not exceeding 1 m², total inside height of 1.2 m, whether or not provided with fixed or removable shelves; has a capacity not exceeding 250 kg and is exclusively used for carrying materials and shall not carry any person.

2.1.23 Direct Acting Lift ð Hydraulic lift where the ram or cylinder is directly attached to the car or its sling.

2.1.24 Down Direction Valve ð Electrically controlled valve in a hydraulic circuit for controlling the descent of the car.

2.1.25 Drive Control System ð System controlling and monitoring the running of the lift machine.

2.1.26 Electrical and Mechanical Interlock ð A device provided to prevent simultaneous operation of both up and down relays or power contactors.

2.1.27 Electro-Mechanical Lock ð A device which combines in one-unit, electrical contact and a mechanical lock jointly used for the landing and/or car doors.

2.1.28 Electrical Anti-Creep System ð Combination of precautions for hydraulic lifts against the danger of creeping.

2.1.29 Electric Safety Chain ð The total of the electric safety devices connected in such a way as to stop the lift when one of them is activated.

2.1.30 Evacuation Lift ð A lift designed to be used for the evacuation of persons, in automatic mode, or under the direction of building management, or by trained evacuation person(s).

2.1.31 Evacuation Exit Lobby (EEL) ð Destination floor(s) for evacuating persons in the building using lift(s), determined by the fire evacuation plan and which has a safe area(s).

2.1.32 Existing Lift ð Lift which is in service at the disposal of its owner.

2.1.33 Floor Levelling Switch ð A switch for bringing the car to level at slow speed in case of double speed or variable speed machines.

2.1.34 Floor Stopping Switch ð A switch or combination of switches arranged to bring the car to rest automatically at or near any pre-selected landing.

2.1.35 Full Load Pressure ð Static pressure exerted on the piping, jack, valve block, etc, with the car and rated load being at rest at the highest landing level.

2.1.36 Geared Machine ð A machine in which the power is transmitted to the sheave through worm and worm wheel or spur reduction gearing.

2.1.37 Gearless Machine ð A lift machine in which the motive power is transmitted to the driving sheave from the motor without intermediate reduction gearing and has the brake drum mounted directly on the motor shaft.

2.1.38 Goods Lift ð Lift mainly intended for the transport of goods and on which only the operator and the persons necessary for loading/unloading of the goods are permitted to ride.

2.1.39 Guide Rails ð Rigid components which provide guiding for the car, the counterweight or the balancing weight.

2.1.40 Guide Rails Fixing ð The complete assembly comprising the guide rails bracket and its fastenings.

2.1.41 Guide Shoe ð An attachment to the car frame or counterweight for the purpose of guiding the lift car or counter weight frame.

2.1.42 Headroom/Overhead ð The vertical distance from the level of the top lift landing to the bottom of the machine room slab/ceiling.

2.1.43 Hoisting Beam ð A beam, mounted immediately below the machine room ceiling/machinery space ceiling, to which lifting tackle can be fixed for raising or lowering parts of the lift machine.

2.1.44 Hospital Lift ð A lift normally installed in a hospital, dispensary or clinic and designed to accommodate one bed or stretcher along its depth, with sufficient space around to carry a minimum of three attendants in addition to the lift operator.

2.1.45 Hydraulic Lift ð Lift in which the lifting power is derived from an electrically driven pump transmitting hydraulic fluid to a jack, acting directly or indirectly on the car (multiple motors, pumps and/or jacks may be used).

2.1.46 Indirect Acting Lift ð Hydraulic lift where the ram or cylinder is connected to the car or the car sling by suspension means [ropes or coated steel belts (CSBs)].

2.1.47 Jack ð Combination of a cylinder and a ram forming a hydraulic actuating unit.

2.1.48 Home Lift ð A lift specifically designed for a private home taking into consideration the following:

- a) compact design in view of the limitations of space in a private residence and light duty;
- b) usage of the lift restricted primarily to the residents of the private home;
- c) special facilities to meet the needs of elderly and persons with disabilities, including wheelchair users;
- d) quiet, smooth, jerk-free movement of the lift; and
- e) controls to have ease of operation.

2.1.49 Landing Call Push ð A push button fitted at a lift landing, either for calling the lift car or for actuating the call indicator.

2.1.50 Landing Door ð The hinged or sliding portion of a lift well enclosure, controlling access to a lift car at a lift landing.

2.1.51 Levelling ð Operation which achieves the accuracy of stopping at landings.

2.1.52 Levelling Accuracy ð Vertical distance between car sill and landing sill during loading or unloading of the car.

2.1.53 Levelling Devices

2.1.53.1 Levelling device, lift car ð Any mechanism which either automatically or under the control of the operator, moves the car within the levelling zone towards the landing only and automatically stops it at the landing.

2.1.53.2 Levelling device, two-way automatic maintaining ð A device which corrects the car level on both under run and over-run and maintains the level during loading and unloading.

2.1.53.3 Levelling device, two-way automatic non-maintaining ð A device which corrects the car level on both under run and over run but will not maintain the level during loading and unloading.

2.1.54 Levelling Zone ð The limited distance above or below a lift landing within which the levelling device may cause movement of the car towards the landing.

2.1.55 Lift ð Lift means a hoisting mechanism equipped with a lift car which moves in a substantially vertical direction, is worked by power and is designed to carry passengers or goods/vehicle or both.

2.1.56 Lift Car ð The load carrying unit with its floor or platform, enclosing bodywork and car door.

2.1.57 Lift Landing ð That portion of a building or structure used for discharge of passengers or goods or both into or from a lift car.

2.1.58 Lift Machine ð Unit which drives and stops the lift, including any motor, gear, brake, sheave and drum (traction or positive drive lift) or comprising the pump, pump motor and control valves (hydraulic drive lift).

2.1.59 Lift Pit ð The part of the well situated below the lowest landing served by the car.

2.1.60 Lift Well ð Space in which the car, the counterweight or the balancing weight travel. This space is usually bounded by the bottom of the pit, the walls and the ceiling of the well.

2.1.61 Lift Well Enclosure ð Any structure which separates the lift well from its surroundings.

2.1.62 Laminated Glass ð Assembly of two or more glass layers, each of which is bonded together with one or more plastic or liquid interlayers.

2.1.63 Load Bearing Member ð Steel wire rope or steel wire strand inside of an elastomeric coated belt.

2.1.64 Machine Room ð Fully enclosed machinery space with ceiling, walls, floor and access door(s) in which machinery as a whole or in parts is placed.

2.1.65 Machine Room Less Lift ð This special lift does not need a separate machine room.

2.1.66 Machinery ð Equipment such as: control cabinet(s) and drive system, lift machine, main switch(es) and means for emergency operations.

2.1.67 Machinery Space ð Volume(s) inside or outside of the well where the machinery as a whole or in parts is placed, including the working areas associated with the machinery.

NOTE ð A machinery cabinet with its associated working area(s) is considered as a machinery space.

2.1.68 Maintenance ð All the necessary operations to ensure the safe and intended functioning of the installation and its components after the completion of the installation and throughout its life cycle.

NOTE ð Maintenance shall include:

- a) Lubrication, cleaning, etc;
- b) Checks;
- c) Rescue operations;
- d) Operations of setting and adjustment; and
- e) Repair or changing of components which can occur due to wear or tear and do not affect the characteristics of the installation.

2.1.69 Minimum Breaking Load (MBL) ð Specified value in kN below which the measured breaking load is not allowed to fall in a breaking load test.

2.1.70 Mast ð Structure that supports and guides the cage.

2.1.71 Mast Section ð Indivisible piece of mast, between two adjacent mast joints.

2.1.72 Mast Tie ð Connection system between the mast and any building structure, providing lateral support for the mast.

2.1.73 Non-Return Valve ð Valve which allows flow in one direction only.

2.1.74 Operation ð The method of actuating the control of lift machine.

2.1.74.1 Automatic operation ð A method of operation in which by a momentary activation of a call button the lift car is set in motion and caused to stop automatically at any required lift landing.

2.1.74.2 Non-Selective collective automatic operation ð Automatic operation by means of one button in the car for each landing level served and one button at each landing, wherein all stops registered by the momentary actuation of landing or car buttons are made irrespective of the number of buttons actuated or of the sequence in which the buttons are actuated. With this type of operation, the car stops at all landings for which buttons have been actuated making the stops in the order in which the landings are reached after the buttons have been actuated but irrespective of its direction of travel.

2.1.74.3 Selective collective automatic operation ð Automatic operation by means of one button in the car for each landing level served and by up and down buttons at the landings, wherein all stops registered by the momentary actuation of the car button made as defined under non-selective collective automatic operation, but wherein the stops registered by the momentary actuation of the landing buttons are made in the order in which the landings are reached in each direction of travel after the buttons have been actuated. With this type of operation, all 'up' landing calls are answered when the car is travelling in the up direction and all 'down' landing calls are answered when the car is travelling in the down direction, except in the case of the uppermost or lowermost calls which are answered as soon as they are reached irrespective of the direction of travel of the car.

2.1.74.4 Single automatic operation ð Automatic operation by means of one button in the car for each landing level served and one button at each landing so arranged that if any car or landing button has been actuated, the actuation of any other car or landing operation button will have no effect on the movement of the car until the response to the first button has been completed.

2.1.74.5 Group automatic operation ð Automatic operation of two or more non-attendant lifts equipped with power-operated car and landing doors. The operation of the cars is coordinated by a supervisory operation system including automatic dispatching means whereby selected cars at designated dispatching points automatically close their doors and proceed on their trips in a regulated manner.

Typically, it includes one button in each car for each floor served and up and down buttons at each landing (single buttons at terminal landings). The stops set up by the momentary actuation of the car buttons are made automatically in succession as a car reaches the corresponding landings irrespective of its direction of travel or the sequence in which the buttons are actuated. The stops set up by the momentary actuation of the landing buttons may be accomplished by any lift in the group, and are made automatically by the first available car that approaches the landing in the corresponding direction.

2.1.74.6 Signal operation ð Same as collective operation, except that the closing of the door is initiated by the attendant.

2.1.74.7 Continuous pressure operation ð Operation by means of buttons or switches in the car and at the landings any of which may be used to control the movement of the car as long as the button or switch is manually pressed in the actuating position.

2.1.75 Operating Device ð A car switch, push button or other device employed to actuate the control.

2.1.76 Overhead Beams ð The members, usually of steel, which immediately support the lift equipment at the top of the lift well.

2.1.77 Over Speed Governor ð Device which, when the lift attains a predetermined speed, causes the lift to stop, and if necessary, causes the safety gear to be applied.

2.1.78 One-Way Restrictor ð Valve which allows free flow in one direction and restricted flow in the other direction.

2.1.79 Passenger Lift ð A lift designed for the transport of passengers.

2.1.80 Passenger ð Any person transported by a lift in the car.

2.1.81 Position and/or Direction Indicator ð A device which indicates on the lift landing or in the lift car or both, the position of the car in the lift well or the direction in which the lift car is travelling or both.

2.1.82 Pawl Device ð Mechanical device for stopping involuntary descent of the car, and maintaining it stationary on fixed supports.

2.1.83 Positive Drive Lift ð Lift which is directly driven (not reliant on friction) by drum and ropes.

NOTE ð In this NBCS, the term 'positive drive lift' includes drum drive.

2.1.84 Preliminary Operation ð Energizing of the machine and the brake/hydraulic valve as preparation to a normal run when the car is in the door zone and doors are not closed and locked.

2.1.85 Pressure Relief Valve ð Valve that limits the pressure to a pre-determined value by exhausting fluid.

2.1.86 Programmable Electronic System in Safety Related Applications for Lifts (PESSRAL) ð System for control, protection or monitoring based on one or more programmable electronic devices, including all elements of the system such as power supplies, sensors and other input devices, data highways and other communication paths and actuators and other output devices, used in safety-related applications

2.1.87 Pulley Room ð Room not containing the machine, in which pulleys are located and in which the overspeed governor should also be housed.

2.1.88 Pulley Space ð Space(s) inside or outside of the well where pulleys are placed.

2.1.89 Rated Load (Lift) ð In a normal operation the maximum load for which the lift car is designed and installed to carry safely at its rated speed.

2.1.90 Rated Speed (Lift) ð In a normal operation the mean of the maximum speed attained by the lift car in the upward and downward direction with rated load in the lift car at the time of normal operation.

2.1.91 Refuge Space ð An area of refuge is a place on the lift car roof or in the pit within the lift well designed to hold the technician/service person in safety when the car is at its highest or lowest position of the lift well, respectively.

2.1.91.1 Bottom refuge space ð When the lift car is at its lowest position, at least one clear area on the pit where refuge may be taken.

2.1.91.2 Top refuge space ð When the lift car is at its highest position, at least one clear area on the car roof where refuge may be taken.

2.1.92 Retiring Cam/Coupler ð A device which prevents the landing doors from being unlocked by the lift car unless it stops at a landing.

2.1.93 Roping Multiple ð A system of roping where, in order to obtain a multiplying factor from the machine to the car, multiple falls of rope are run around sheave on the car or counterweight or both. It includes roping arrangement of 2 to 1, 3 to 1, etc.

2.1.94 Re-levelling ð Operation, after the lift has stopped, to permit the stopping position to be corrected during loading or unloading.

2.1.95 Rescue Operations ð Specific actions required to safely release persons entrapped in the car and well by competent persons.

2.1.96 Residual Breaking Load (RBL) ð Measured value of suspension member after normal use at the end of expected life span which is detected when discard criteria have been reached.

2.1.97 Restrictor ð Valve in which the inlet and outlet are connected through a restricted passage way.

2.1.98 Rupture Valve ð Valve designed to close automatically when the pressure drops across the valve, caused by the increased flow in a pre-determined flow direction, exceeds a pre-set amount.

2.1.99 Rack and Pinion Lift ð Lift which uses a toothed rack and pinion as the load suspension system.

2.1.100 Safety Gear ð A mechanical device attached to the lift car or counterweight or both, designed to stop and to hold the car or counterweight to the guides in the event of free fall, or, if governor operated, of over-speed in the descending and/or ascending direction. Any anticipated impact force shall be added in the general drawing or layout drawing.

2.1.100.1 Instantaneous safety gear ð Safety gear in which the full gripping action on the guide rails is almost immediate.

2.1.100.2 Progressive safety gear ð Safety gear in which retardation is effected by a braking action on the guide rails and for which special provisions are made so as to limit the forces on the car, counterweight or balancing weight to a permissible value.

2.1.101 Service Lift ð A passenger cum goods lift meant to carry goods along with people.

NOTE ð Typically in an office building this may be required to carry food or stationeries, in a residential building to carry luggage or accommodate a stretcher and in a hotel to be used for food trolleys or baggage. There is a need in such lifts, to take care of the dimensions of the car and the door clear opening in line with the type of goods that may have to be carried based on mutual discussion between the supplier and the customer. Also, such lifts shall have buffer railings in the car at suitable height to prevent damage to the car panels when the goods are transported. Typically, such lifts, if provided with an automatic door, may use some means to detect trolleys and stretcher movement in advance to protect the doors against damage. The car floor load calculations and car area of such a lift is as in the case of a passenger lift except that these are not meant to carry heavy concentrated loads.

2.1.102 Sheave ð A rope wheel, the rim of which is grooved to receive the suspension ropes but to which the ropes are not rigidly attached and by means of which power is transmitted from the lift machine to the suspension ropes.

2.1.103 Slack Rope Switch ð Switch provided to open the control circuit in case of slackening of rope(s).

2.1.104 Suspension Ropes ð The ropes by which the car and counter weight are suspended.

2.1.105 Safety Component ð Component provided to fulfil a safety function when in use.

2.1.106 Safety Integrity Level (SIL) ð Discreet level (one out of a possible three) for specifying the safety-integrity requirements of the safety functions allocated to the programmable electronic safety-related system, where safety-integrity level 3 has the highest level of safety integrity and safety-integrity level 1 has the lowest.

2.1.107 Safety Circuit ð Circuit containing contacts and/or electronic components which is regarded to fulfill demands of an electric safety device.

2.1.108 Safety Rope ð Auxiliary rope attached to the car, the counterweight or balancing weight for the purpose of tripping a safety gear in case of suspension failure.

2.1.109 Shut-off Valve ð Manually operated two-way valve which may permit or prevent flow in either direction.

2.1.110 Single Acting Jack ð Jack in which displacement in one direction is by fluid action and in the other by influence of gravity.

2.1.111 Special Tool ð Tool unique to the equipment required in order to keep the equipment in a safe operating condition or for rescue operations.

2.1.112 Stopping Accuracy ð Vertical distance between car sill and landing sill at the moment when a car is stopped by the control system at its destination floor and the doors reach their fully open position.

2.1.113 Suspension Means ð Includes all ropes/elastomeric coated belts suspending car/counterweight/balancing weight and is engaged to car/counterweight/balancing weight side termination.

2.1.114 Suspension Member ð Is one rope/elastomeric coated belt in the system to suspend car/counterweight and is engaged to car/counterweight side terminations.

2.1.115 Terminal Slow-Down Switch ð A switch when actuated shall compulsorily cut off the high speed and switch on the circuitry to run the lift in levelling speed before reaching on terminal landings.

2.1.116 Terminal Stopping Switch Normal ð Switch for cutting off all the energizing current in case of car travelling beyond the top or bottom terminal landing or a switch that cuts off the energizing current so as to bring the car to a stop at the top or bottom terminal landing level in the respective direction of travel.

2.1.117 Terminal Stopping Device Final ð A device which automatically causes the power to be removed from an electric lift driving machine motor and brake, independent of the functioning of the normal terminal stopping device, the operating device or any emergency terminal stopping device, after the car has passed a terminal landing.

2.1.118 Top Counterweight Clearance ð The shortest vertical distance between any part of the counterweight structure and the nearest part of the overhead structure or any other obstruction when the car floor is level with the bottom terminal landing.

2.1.119 Travel ð The vertical distance between the bottommost and topmost lift landings served by the lift.

2.1.120 Traction Lift ð Lift whose suspension means are driven by friction on the driving sheave of the machine.

2.1.121 Travelling Cable ð Flexible electric cable containing multiple cores between the car and a fixed point.

2.1.122 Unintended Car Movement ð Non-commanded movement of the car with doors open within the door zone away from the landing, excluding movements resulting from loading/unloading operation.

2.1.123 Unlocking Zone ð Zone, extending above and below the landing level, in which the car floor has to be, to enable the corresponding landing door to be unlocked.

2.1.124 User ð Person making use of the services of a lift installation which includes passengers, persons waiting at the landings and authorized persons.

2.2 Terms Relating to Performance Requirements for Lifts

2.2.1 A95 ð Values of acceleration or vibration within defined boundaries or limits, in which 95 percent of found values are equal to or less than.

NOTE ð This value is used statistically to estimate typical levels.

2.2.2 Acceleration δ Rate of change of z-axis velocity, attributed to lift motion control.

NOTE δ It is expressed in metre per second squared (m/s^2).

2.2.3 Axis of Measurement δ Orthogonal reference axes for the measurements, where for lifts of conventional configuration:

- a) *X-axis* $\bar{\imath}$ Axis perpendicular to the plane of the car front door (that is, back to front);
- b) *Y-axis* $\bar{\imath}$ Axis perpendicular to X and Z (that is, side to side); and
- c) *Z-axis* $\bar{\imath}$ Axis perpendicular to the car floor (that is, vertical).

2.2.4 Equivalent Sound Pressure Level (L_{Aeq}) δ Average sound pressure level, using frequency weighting A and time weighing δ ast δ determined within defined boundaries.

2.2.5 Jerk δ Rate of change of z-axis acceleration, attributed to lift motion control. It is expressed in meter per second cube (m/s^3).

NOTE δ The passenger perception of vertical ride quality during jerk is represented by the assessment of vertical vibration during non-constant acceleration.

2.2.6 Lift Ride Quality δ Sound levels in the car and vibration of the car floor, relevant to passenger perception, associated with lift motion.

2.2.7 Peak to Peak Vibration Levels δ Sum of the magnitudes of two peaks of opposite sign separated by a single zero crossing.

2.2.8 Sound δ A-weighted sound pressure level measured in decibels (dB).

2.2.9 Sound Pressure Level ($L_{p,A}$) δ Sound pressure level using frequency weighting A as defined in Part D δ Building Services, Section 4 Acoustics, Sound Insulation and Noise Control δ of this NBCS:

$$L_{p,A} = 10 \log (p_A^2/p_0^2) \text{ dBA}$$

NOTES

- 1 The reference sound pressure level (p_0) is 20 δ Pa (2×10^{-5} Pa).
- 2 The measured sound pressure, p_A , is in Pascals (Pa), using frequency weighting A.

2.2.10 V95 δ Value of velocity, within defined boundaries or limits, in which 95 percent of found values are equal to or less than the defined limits.

NOTE δ This value is used statistically to estimate typical levels.

2.2.11 Velocity δ Rate of change of z-axis displacement, attributed to lift motion control.

NOTE δ Velocity is reported as speed and direction of travel. It is given in meters per second (m/s).

2.2.12 Vibration δ Variation with time of the magnitude of acceleration, when the magnitude is alternately greater and smaller than the average acceleration of the lift when no lift motion acceleration is present.

NOTES

- 1 It is expressed in metre per second squared (m/s^2).
- 2 The deprecated unit Gal (Galileo) is sometimes used: 1 Gal = 0.01 m/s^2

2.3 Terms Relating to Planning and Design of Lifts

2.3.1 Door Closing Time (t_c) δ Time period measured from the instant that car doors start to close until the doors are closed.

NOTE δ The door closing time is expressed in second (s).

2.3.2 Door Opening Time (t_o) ð Time period measured from the instant that car doors start to open until they are open to a specified width.

NOTES

- 1 Measurement may be completed at the instant that the doors are either 800 mm open or until the doors are fully open.
- 2 The opening width used shall be reported.
- 3 The door opening time is expressed in second (s).

2.3.3 Performance time/Door-to-Door Time (t_{pref}) ð Period of time between the instant the car doors start to close and the instant that the car doors are open to a specified width at the next adjacent floor.

NOTES

- 1 The performance time is expressed in seconds (s).
- 2 Measurements may be completed at the instant that the doors are 800 mm open or until the doors are fully open.

2.3.4 Handling Capacity (C_h) ð Maximum sustainable number of passengers per specified time period that a single lift or a group can transport for a specific traffic mix under specified loading constraints.

NOTES

- 1 Handling capacity is usually expressed as a number of passengers per five minutes (C_h) or as a percentage of population per 5 min (percent C_h).
- 2 Typical loading constraints are limiting the number of passengers in the cars and are determined by comfort and/or safety considerations that may reflect cultural and/or national norms.
- 3 The average waiting time increases rapidly and passengers are constantly left behind the departing lift, when passenger demand exceeds handling capacity.

2.3.5 Interval (INT) ð Time period between successive car arrivals at the main terminal floor with cars loaded to any value.

2.3.6 Nominal Travel Time/Theoretical Time of Travel (t_{nt}) ð Time period in seconds for a lift to travel from the lowest floor to the highest floor without any stops at rated speed.

NOTE ð The nominal travel time is expressed in second (s).

2.3.7 Passenger Arrival Rate ð Percentage of a building's population arriving within a 5 min period.

2.3.8 Passenger Transfer Time (t_p) ð Average time for a single passenger to enter or leave the lift car.

NOTES

- 1 The passenger usually moves out faster than they move in. Transfer time is an average of both of them.
- 2 The passenger transfer time is expressed in second (s).

2.3.9 Passenger Waiting Time (WT) ð Period of time from when a passenger either registers a call on a landing, or joins a queue, until the responding lift that will accommodate the passenger begins to open its doors at the boarding floor.

NOTES

- 1 The passenger waiting time continues if a passenger does not enter the responding lift, that is, it is full (a refusal).
- 2 The passenger waiting time is zero, if the responding lift doors are open or opening when the passenger arrives.
- 3 There may be difference between waiting times measured at a site and waiting times produced by a simulation due to the difficulty of accurately measuring waiting times at an actual site.

2.3.10 Round Trip Time (t_{rt}) ð Average period of time for a single lift car trip during up-peak traffic conditions, measured from the time the car doors open at the main terminal until the car doors reopen at the main terminal after serving the registered car calls.

NOTE ð The round-trip time is expressed in second (s).

2.3.11 Single Floor Flight Time (t_{fl}) ð Period of time measured from the instant that the car doors are locked until the lift is level at the next adjacent floor.

2.3.12 Single Floor Transit Time (t_v) ð Period of time required to transit two adjacent floors at rated speed.

2.3.13 Sky Lobby ð A sky lobby is the main floor for local groups in the upper part of a very tall building.

2.3.14 Door Closing Delay Time (t_{cd}) ð Delay after passenger clearance before start of door closing.

NOTE ð The door closing delay time is expressed in second (s).

2.3.15 Express Zone — Building zone situated between an entrance floor and served floors where lift travel nonstop.

2.3.16 Door Pre-Opening Time/Advance Door Opening Time (t_{pre}) ð Time period measured from the instant that car doors start to open until the lift is level at the landing.

NOTES

1 Door pre-opening may compensate for lift levelling, which is the final (slow) approach of the lift to a landing.

2 The pre-opening time is expressed in second (s).

2.3.17 Incoming Traffic ð Component of traffic where passengers travel from entrance floors to the populated floors.

2.3.18 Inter-Floor Traffic ð Component of traffic where passengers travel between the populated floors.

2.3.19 Outgoing Traffic ð Component of traffic where passengers travel from the populated floors to entrance floors.

2.3.20 Population (U) ð Maximum population a target building is going to be designed.

2.3.21 Start Delay Time (t_{sd}) ð Period of time from the instant the car doors are close until the lift starts to move.

NOTE ð The start delay time is expressed in second (s).

2.3.22 Flight Time (t_f) ð Period of time from the instant the lift starts to move until the lift is level at the next stop floor.

NOTE ð Flight time is usually calculated by assuming ideal lift kinematics based on the rated speed, rated acceleration and jerk.

2.3.23 Entrance Floor ð Floor with building entrance or utility floor, that is, restaurant, which attract people from populated floors.

2.3.24 Entrance Bias ð Proportion of traffic attributed to a specific entrance floor.

2.3.25 Transit Time ð Period of time from when a responding lift begins to open its doors at the boarding floor until the doors begin to open again at the destination floor.

NOTE ð The transit time commences when a passenger arrives, if the responding lift doors are open or opening.

2.4 Terms Relating to Seismic Operation of Lifts

2.4.1 Design Acceleration (a_d) ð The horizontal acceleration to be used for calculation of forces (moments acting on lift systems and arising from seismic events).

NOTE ð The value of (a_d) shall be obtained by the lift manufacturer from the building engineer/structural engineer.

2.4.2 Normal Operation ð Operation mode in which the lift performs when not in seismic mode or in seismic stand-by mode.

2.4.3 Retaining Device ð Mechanical device securely fixed to a structural member of the lift car, counterweight frame designed to retain the lift car and counterweight within its guide rails during seismic activity.

2.4.4 Seismic Mode ð Special mode in which the lift operates after detection of seismic trigger level.

2.4.5 Seismic Stand-By Mode ð Special mode in which the lift operates after detection of primary earthquake wave without the activation of the seismic detection system.

2.4.6 Seismic Trigger Level ð Seismic acceleration which is used to activate a seismic detection system.

2.4.7 Snag Point ð The point of interference between flexible elements (for example, ropes, chains, travelling cable, etc) and fixed elements (for example, guide rail brackets, guide rail clip bolts, fishplates, vanes and similar devices).

2.5 Terms Related to Maintenance of Lifts

2.5.1 Competent Maintenance Person ð Designated person, suitably trained, qualified by knowledge and practical experience, provided with necessary instructions and supported within their maintenance organization to enable the required maintenance operations to be safely carried out.

2.5.2 Installation ð Completely installed passenger lift or good passenger lift or accessible goods only lift or service lift.

2.5.3 Installer ð Natural or legal person who takes responsibility for the design, manufacture, installation and placing on the market of lifts.

2.5.4 Maintenance Organization ð Company or part of company where competent maintenance person(s) carry out maintenance operation on behalf of the owner of the installation.

2.5.5 Manufacturer ð Natural and legal person who takes responsibility for the design, manufacture and placing on the market either of safety components for lifts or complete lift.

2.5.6 Owner of the Installation ð Natural or legal person who has the power of disposal of the installation and takes responsibility for its operation and use.

2.5.7 Inspector ð Inspector means an inspector of lifts or any other officer appointed and approved by State/Union Government to maintain *Lift Act/Rules* or any *other Act/Rules* connected with safety of lifts and framed by State/Union Government

2.5.8 Authorized Person ð Person with the permission of the natural or legal person who has the responsibility for the operation and use of the lift, to access restricted areas (machinery spaces, pulley rooms and lift well) for maintenance, inspection or rescue operations.

2.5.9 License ð A permission granted by competent authority to engage in a business or occupation or in an activity otherwise unlawful.

3 GENERAL

3.1 Conformity with Lifts Act and Rules

3.1.1 The installation shall generally be carried out in conformity with *Lifts Act* and *Rules*, wherever they are in force.

3.1.2 It is the responsibility of the owner of the premises where the lift will be installed, to obtain necessary permission from the Authority before and after the installation of lifts and for subsequent operation of lifts.

3.1.3 A license for public use is a safety provision, issued by state authorities under *Lifts Act* and *Rules*, wherever they are in force and shall be obtained as per the laid down statutory requirement.

3.2 Conformity with *The Electricity Act* and CEA Regulations Thereunder

All electrical work in connection with installation of lifts shall be carried out in accordance with the provisions of the *Electricity Act*, 2003 as amended up-to-date along with the CEA regulations framed thereunder and shall also comply with the other provisions of Part D ð Building Services, Section 2 Electrical and Allied Installations ð of this NBCS.

3.3 Conformity with Indian Standards

All materials, fittings, appliances, etc, used in electrical installation shall conform to Indian Standard specifications wherever these exist. In case of materials for which Indian Standard specifications do not exist, the materials shall be approved by the competent authority. For detailed specification for lifts, reference shall be made to accepted standards [D-5A(2)].

3.4 Conformity to Accessibility Requirements

All lifts installed for public use shall meet the accessibility requirements in accordance with good practice [D-5A(3)].

3.5 Conformity with Fire Regulations

The installation shall be carried out in conformity with the state fire acts/local fire regulations, wherever they are in force (*see also* Part F ‘Fire and Life Safety’ of this NBCS for guidance purpose).

3.6 Considerations for Selection of Lifts

The considerations for selection of lifts shall be based on the following criteria:

- a) *The Intended use of lift* – The number of lifts and their capacities (that is, load and speed) required for a given building shall be suitably decided to meet the intended requirement. The passenger lifts shall meet the requirements of handling capacity and waiting time for passenger, depending on the lift’s expected usage and building type.
- b) *System performance* – System performance criteria shall be based on building type (residential, commercial, hotel, hospital, etc).
- c) *Accessibility requirements* – See [3.4](#).
- d) *Environmental conditions* – Lifts directly exposed to atmospheric conditions, that is, weather (for example, those meant for external applications), or any other adverse condition shall be appropriately designed and protected for that particular condition.
- e) *Type of main drive for lift* – Whether electric traction, geared or gearless or hydraulic depending on speed, stops/travel height and capacity requirement.
- f) *Civil engineering requirements* – Machinery location that is, machinery to be located in machine room or machinery to be kept inside lift well thereby eliminating conventional machine room.
- g) *Seismic considerations* – Whether the lift is to be protected against seismic forces or whether the lift is to be rescued to a landing on detection of earthquake or both.

3.7 Maintenance

The considerations relating to maintenance shall be as follows:

- a) The lift installation shall receive regular cleaning, lubrication, adjustment and adequate servicing by authorized competent persons at such intervals as per type of equipment and frequency of service demand. It is desirable and normal for the lift supplier to be entrusted with the servicing during the guarantee period of new lift.
- b) In order that the lift installation is maintained at all times in a safe condition, a proper maintenance schedule shall be drawn up in consultation with the lift manufacturer and rigidly followed. The provision of a log book to record all items relating to general servicing and inspection is recommended for all lifts.
- c) Any accident arising out of operation or maintenance of the lifts shall be duly reported to the authority in accordance with the rules laid down.
- d) Lifts are required by statutory regulations to be examined at regular intervals as specified by lift acts, by a competent person.
- e) The company entrusted with maintenance contract shall have valid licence to maintain the lifts. The persons assigned for maintenance work shall be appropriately qualified and experienced as required by *Lift Acts and Rules*.

3.8 Energy Efficiency and Sustainability

Design options like space restrictions, reliability and safety, riding comfort have been the major market and technological drivers. The following should be encouraged for reducing power consumption and promoting sustainability in buildings:

- a) Use of energy efficient a.c. variable voltage variable frequency (VVVF) motor drive or equivalent. Lifts with 1-speed and 2-speed motor control are not recommended for passenger lifts because of high power consumption, poor passenger comfort and tripping hazard;
- b) When the lift has answered the last call and stopped at a landing and no further landing call is registered, the car and landing doors shall close. If there is no further landing call after pre-determined period but not less than 90 s, the light and fan inside the car shall both be automatically switched off. Car lights and fan shall switch on automatically before the lift doors start to open or the lift is set in motion;
- c) Under normal operating status, at least one lift car of a lift bank shall operate under a standby or sleep mode during off-peak period when the traffic demand on the vertical transportation system is low. During low demand periods, even completely shutting down one or more lifts within a group can be a good energy saving option, without compromising the quality of service;
- d) Where a number of lifts are installed together, their controls are interconnected to optimize their operation. By efficiently delivering passengers with the least number of trips, starts and stops, the energy consumed is significantly reduced;
- e) Use of energy saving LED lamps for car lighting in place of conventional lamps;
- f) Use of gearless type machines to reduce transmission losses;
- g) Improvement in total power factor of the motor drive of a lift at the isolator connecting lift to the building's electrical supply circuit;
- h) Use of regenerative drives to recycle energy rather than wasting it as heat. The regenerated energy may be used for charging batteries, staircase lighting, lobby lighting, etc;
- j) Use of high efficiency motors such as permanent magnet synchronous motors, or induction motors having minimum efficiency class equivalent to IE2 as per accepted standard [D-5A(4)]; and
- k) Adoption of materials and practices that are environmentally friendly and sustainable.

4 PLANNING AND DESIGN GUIDELINES

This clause provides guidelines for traffic analysis and minimum vertical transportation requirements; and is not a substitute for detailed understanding of the complex theory and the required practical experience.

4.1 The planning guidelines, design considerations, and precautions to be exercised during design of passenger and goods lifts operated by electric traction are provided hereunder. Manufacturer may be consulted for other types of lifts such as hydraulic, home, automobile lifts, etc.

Design of lifts necessarily calls for coordination among various parties concerned, namely the client, the architect/engineer, the consulting engineers and the lift manufacturer. These guidelines give the information from the stage of planning till the design that should be exchanged between parties. It is essential that all the parties involved in the planning should have a clear understanding of the basis and the theory of planning. All parties involved need to recognize that it is highly impractical to correct badly planned lifts in a building as no changes can be carried out to the building core, if the number of hoist-ways is inadequate or of wrong size or wrongly positioned.

These provisions specify requirements for the architects/engineers, builders/developers and tenderers while finalizing specification of lifts. Two basic considerations, namely, the quantity of service required and the quality of service desired, determine the number and type of lifts to be provided in a particular building. The quantity of service factor, that is, how many people might use the lift system over a defined period of time is represented by the handling capacity. The quality of service factor, that is, how well the lift system deals with its passengers is represented by passenger waiting time and lobby queuing. These factors are interrelated and depend, among other things, on the type of building and its use and on the type of occupier. Both these factors require proper study into the character of the building, extent and duration of peak periods, frequency of service required, type and method of control, type of landing doors, etc.

[Figure 1](#) shows the flow chart to arrive at a reasonable vertical transportation solution for a building.

The adequacy of the lift group in a building is a major contributing factor to the success or failure of a building. Users of lifts would be intolerant of long queues and long waiting times. The planning and selection of the vertical transportation equipment is a specialized activity. Although the basic calculations are relatively simple, the theory on which they are based is complex. The difficulty in planning a lift installation is not only in calculating its probable performance but also in estimating the passenger demand and usage patterns. The architect/engineer doing the planning work should establish the lift system at a very early stage in consultation with the lift manufacturer/consulting engineer and not after the rest of the building has been designed.

This Subsection defines the calculation approach for conventional two button dispatching and does not define calculation methods for destination control systems or hybrid systems. It also does not define an approach to simulation methods as consensus on how to apply simulation has not been established.

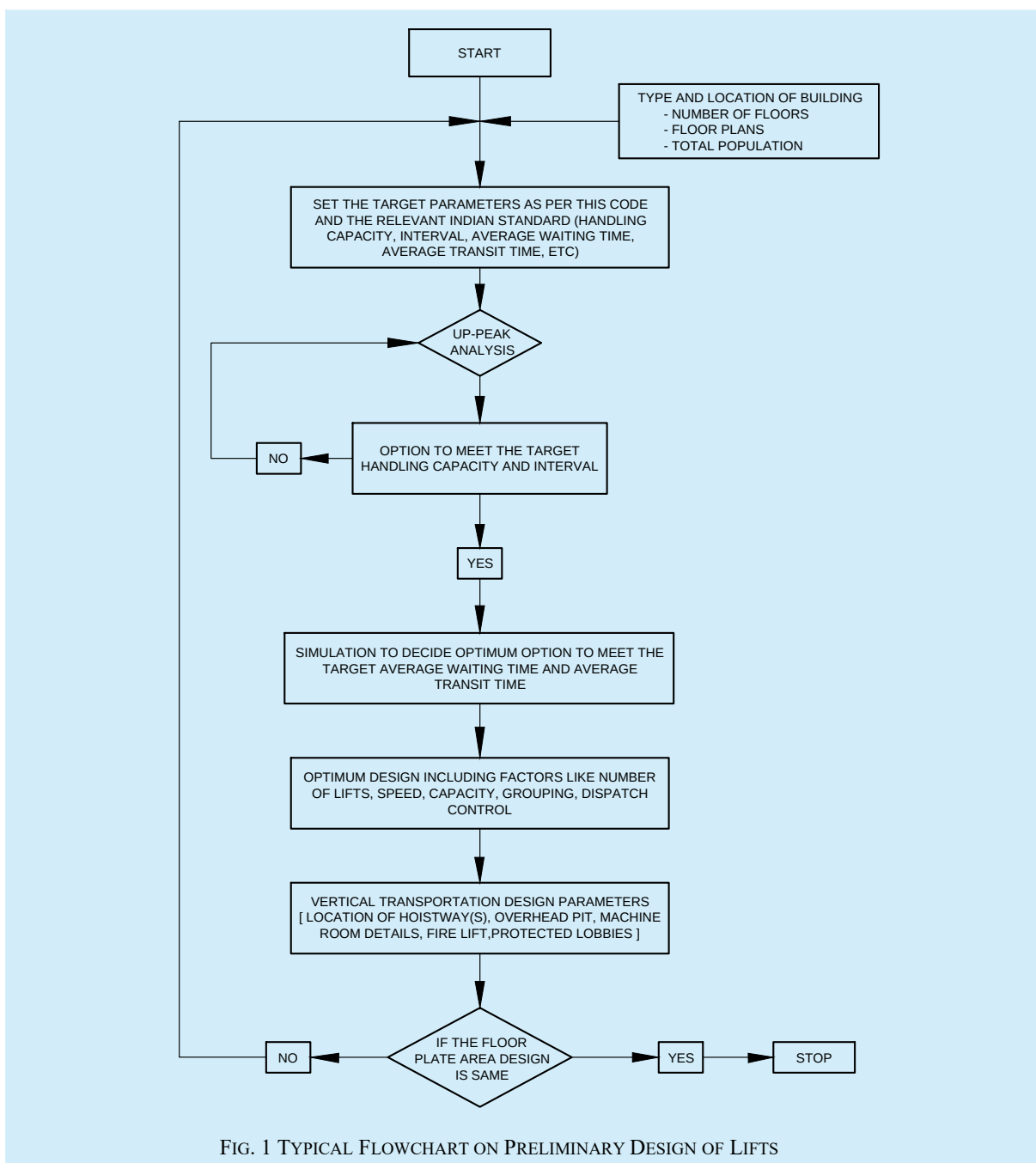


FIG. 1 TYPICAL FLOWCHART ON PRELIMINARY DESIGN OF LIFTS

4.2 Preliminary Lift Planning

4.2.1 Two models can be used for lift traffic analysis and design:

- a) The first model uses a calculation method based on mathematical formulae. This classical model uses the pure up-peak condition as this provides a well-defined, simple traffic pattern amenable to mathematical analysis. These calculations can be carried out manually, using a spreadsheet or by using computer based programme; and
- b) The second model is based on discrete digital simulation of the movement of lifts in a building and the passenger dynamics. This simulation model allows very complex situations to be analysed and is generally capable of better evaluations. However, simulation approaches and algorithms vary with software ranging from crude to very sophisticated. With the varied approaches and lack of consensus on simulation approaches and how to apply simulation this standard does not define standards for simulation, however passenger generation for simulation model should be for 120 min and should cover a minimum of 3 intervals (each of 5 min) to co-relate with practical situation.

Generally, if a lift system (using a conventional landing call system) is sized correctly for the up-peak traffic pattern, all other traffic patterns should be adequately served. The notable exceptions to this, are:

- 1) hotels where check-in/check-out clashes with meal times;
- 2) hospitals;
- 3) buildings that open at specified times (example, stock exchanges, movie theatres, stadiums);
- 4) buildings with 24 Ĩ 7 operation and shift changes (example, BPO/ITES buildings);
- 5) buildings with amenities located at higher floors (example, roof top restaurant, cafeterias, health clubs);
- 6) residential buildings (sometimes two-way traffic is considered);
- 7) buildings with multiple entry levels;
- 8) parking towers;
- 9) schools where students change floors or buildings at the beginning and end of classes; and
- 10) mixed-use buildings.

Methods of calculating the traffic handling capabilities of lifts were first devised for office buildings. In due course detailed modifications were devised to suit other applications without altering the basic principles. The application to office buildings is the most frequently used, and is outlined in [4.2.2](#) to [4.2.9](#) as a general approach to lifting.

Subsequently, traffic conditions other than up peak are examined as also building types other than office. It is recommended that calculations based on the classical model should be carried out in order to obtain an understanding of the various factors. A simulation may then be performed in order to cover special situations or to obtain information not provided by the calculation method.

4.2.2 Population

4.2.2.1 The first point to be ascertained from the owner/developer or proposed occupier is the total building population. If a definite population figure is not available, then an estimation should be made using floor areas, more specifically, the net usable area (NUA). The NUA excludes circulation areas (stairs, corridors, foyers, etc), structural intrusions (columns, ducts, etc) and facilities (pantry, kitchens, toilets, conference rooms, training rooms, server rooms, storage areas, etc).

4.2.2.2 The number of people occupying the net usable area can vary according to the usage pattern or building type. Because of the criticality of establishing the population, it is recommended that the architect/engineer should prepare a typical test fit to establish an estimate of the population. The test fit estimates can be validated from population counts from buildings with similar usage patterns and types.

4.2.2.3 Absenteeism

While the absenteeism is subjected to input from the developer/tenants, studies have shown that in many office buildings it is unlikely that all the total population is present on any one day. Where this is known to be the case the building population to be used in a design may be reduced by 10 percent to 20 percent on account of persons working from home, holiday, sickness, travelling, vacant posts, etc. It would be impractical to consider absenteeism beyond 20 percent of total population. The designer shall note that the absenteeism shall be applied only for office buildings. Absenteeism factor shall not be considered for residential buildings, hotels or hospitals.

People arriving in morning adopts a bell curve arrival pattern. Designers shall not adopt any additional diversity towards people arriving before peak period/after peak period. The designers need to understand the clear circulation and movement pattern of the occupants (and goods) of a building and design to facilitate the circulation and movement. [Table 1](#) gives typical values for the density of population.

Table 1 Occupancy Area Per Person in Office Buildings (Clause 4.2.2.3)			
Sl No.	Building Type	Open Plan/ Hall Seating	Cellular Plan/Cabins
(1)	(2)	(3)	(4)
i)	Premium	10 m ²	12 m ²
ii)	Regular	8 m ²	10 m ²
iii)	Low end	5 m ²	8 m ²

For certain specialized building functions such as BPO, areas ranging from 6 m² to 10 m² per person may be considered.

4.2.3 Quantity of Service

It is measured in terms of the total number of passengers handled during the peak 5 min period of the day. This is calculated by determining the number of trips made by the lifts over the peak 5 min period and then multiplying it by the average number of passengers carried in each trip. The target handling capacity as a percentage of the building population is usually the estimated value of passenger arrival rate for the 5 min up-peak period. The minimum recommended quantity of service for office building is given in [Table 2](#).

This would vary depending on various factors like starting time (fixed or flexi-time), nearness to fast access routes like mass rapid transport systems, express ways, major roads, etc.

Table 2 Recommended Quantity of Service in Office Buildings (Clause 4.2.3)		
Sl No.	Type of Occupancy	Quantity of Service Percent
(1)	(2)	(3)
i)	Multi tenancy	10 to 15
ii)	Single tenancy	15 to 25

4.2.4 Quality of Service

It is very difficult to get real indications of passenger waiting time and lobby queuing from the classic calculation models. Therefore, interval is considered as the measure. During pure up-peak traffic, the interval or the average time between successive arrivals of the lift cars at the main lobby is generally considered as an indicator of passenger waiting time. [Table 3](#) gives the quality of service based on interval for office buildings.

If interval is used as an indicator of quality of service, it should be noted that passenger average waiting time depends on the car occupancy, the handling capacity *vis-a-vis* the passenger arrival rates, the control systems, etc. The average waiting time approximates 75 percent to 85 percent of the interval for average car occupancy of 80 percent of the actual lift capacity provided the handling capacity is equal to or better than the peak arrival rate. Beyond 80 percent car loading passenger average waiting time increases exponentially and cannot be approximated from interval. [Table 4](#) gives the recommended quality of service measured as average waiting time for office buildings.

Table 3 Recommended Quality of Service in Office Buildings (Based on Interval)

(Clause [4.2.4](#))

Sl No.	Quality of Service	Interval s
(1)	(2)	(3)
i)	Excellent	Less than 25
ii)	Very good	25 to less than 30
iii)	Good	30 to less than 35
iv)	Fair	35 to less than 40
v)	Poor	40 to less than 45
vi)	Unsatisfactory	45 and above

Table 4 Recommended Quality of Service in Office Buildings (Based on Average Waiting Time)

(Clause [4.2.4](#))

Sl No.	Class of Building	Average Waiting Time s
(1)	(2)	(3)
i)	Premium	< 25
ii)	Regular	25 to 35
NOTES 1 The class of building may be as defined by the developer. 2 Service lifts shall be provided in addition to required passenger lifts. 3 Lift designer shall take into consideration the shutdown of lifts for maintenance, especially for office building operating 24 Ĩ 7.		

4.2.5 Traffic Analysis Equations

The classical method to size a lift installation requires the determination of the time, in second, that it takes for a single lift to make a round trip around the building during the up-peak traffic condition. This is called the round trip time (*RTT*) and is given by:

$$RTT = 2Ht_v + (S + 1)(T - t_v) + 2Pt_p$$

where

N = average highest reversal floor = $N - \sum_{i=1}^{N-1} (i/N)^P$;

H = average no of stops = $N \left[1 - \left(1 - \frac{1}{N} \right)^P \right]$;

S = average passengers carried (= 0.8 × maximum actual car capacity);

P = single floor transit time = average inter floor distance/rated speed;

t_v = single floor transit time = average inter floor distance/rated speed;

T = door operating time ($t_c + t_o$) + single floor flight time t_f (1); and

t_p = passenger transfer time.

The derivation of the round trip time equation assumes the following:

- Traffic pattern corresponds to a pure up-peak;
- Passengers arrive according to a rectangular probability distribution;
- Lifts, on an average, fill to 80 percent of the actual car capacity by numbers irrespective of the weight of each passenger or the space that maybe occupied;
- All floors are equally populated;
- Rated speed is achieved in a single floor jump;
- Inter-floor heights are equal; and
- Door dwell time does not exceed the calculated passenger transfer time.

Interval, $INT = RTT/L$, where L is the number of lifts in a single group.

$$5 \text{ min handling capacity, } HC = \frac{300 \times P \times L \times 100 / \text{Population}}{RTT}$$

or

$$\frac{300 \times P \times 100 / \text{Population}}{INT}$$

The down peak round trip time is estimated as:

$$RTT_{\text{downpeak}} = Nt_v + (0.5S + 1)(T - t_v) + 2Pt_p$$

The mid-day round trip time is estimated as:

$$RTT_{\text{mid-day}} = 2Ht_v + 2S(T - t_v) + 4Pt_p$$

An estimation of interfloor performance can be obtained using:

$$AWT_{\text{interfloor}} = INT_{\text{uppeak}} + \left(0.22 + 1.78 \frac{A_{\text{interfloor}}}{HC_{\text{uppeak}}} \right)$$

where

$A_{\text{interfloor}}$ = number of passengers arriving during the interfloor period.

4.2.6 Configuration of Lifts: Number

Determination of number of lifts depends on the required quality and quantity of service. It should be noted here that grouping of lifts and location should be given adequate attention.

If a bank of two lifts or more is required to meet the anticipated traffic requirements, they have to be oriented to facilitate grouping. Individual lifts will not achieve the required quantity and quality of service.

4.2.7 Configuration of Lifts: Capacity

Determination of car capacity depends on the required quality and quantity of service. A larger car, while improving the handling capacity might adversely impact the quality of service.

4.2.8 Configuration of Lifts: Speed

In addition to the speeds derived from the calculations for handling capacity and interval, the nominal travel time is used for selecting suitable rated speeds. The higher the building, the faster lifts are needed. The recommended values for nominal travel time in case of an office, commercial or hotel building are given in [Table 5](#).

4.2.9 Configuration of Lifts: Layout

The shape and size of a passenger lift car bears a distinct relation to its efficiency in traffic handling. A study of the most suitable proportions for passenger lifts reveals that the width of the lift well entrance is the basic element in the determination of the best proportions. In other words, the width of the car is determined by the width of the entrance and the depth of the car is regulated by the loading per square metre permissible under this standard.

Centre opening doors are the most practicable and the most efficient entrance units for passenger lifts. Given the same door speed, the centre opening is much faster than the side opening type. For passenger lifts, wider cars are preferred as the ingress and egress efficiency is higher. For service/freight/stretchers deeper cars are preferred, so as to easily carry the stretcher/goods.

While sizing the hoist-ways, the possible construction inaccuracies have to be considered. Further, where high speeds lifts are being considered the hoist-ways need to be designed to address the impact of the wind tunnel effect.

Table 5 Recommended Nominal Travel Time for Office, Commercial or Hotel Buildings <i>(Clause 4.2.8 and Table 11)</i>		
Sl No.	Level	Nominal Travel Time s
(1)	(2)	(3)
i)	Excellent	15 to 25
ii)	Good	> 25 to 35
iii)	Satisfactory	> 35 to 45

4.3 Planning for Specific Building Features

4.3.1 Special Building Facilities

Facilities like cafeteria, food courts, restaurants, gymnasium, etc in the building impact the circulation patterns in a building and should be considered for lift traffic design.

It is advisable to have cafeteria on ground floor or lower floor with separate access means other than the main lifts. When the cafeteria/food court is located in mid-way or top of the building, the probable number of trips per person is on higher side, lunch peak needs to be analysed considering multiple trips.

For office buildings having specialty restaurants, clubs on higher levels, it is recommended to have dedicated lifts with separate lobby to avoid conflict between the different constituents.

4.3.2 Basement Service

Buildings are sometimes designed with car parks or other facilities at basement levels below the terminal floor. Such floors, if they are served by the main lifts have an adverse impact on the up-peak, down-peak and mid-day traffic patterns. During up-peak conditions, this may cause lifts to arrive at the main terminal already partly full causing confusion. In such a situation people have a tendency to push both up and down buttons resulting into deterioration of the lift service.

If basement floors are served by only a part of a lift group, passengers experience difficulty in selecting the correct lifts out of a group that will serve the basement. It is therefore better to provide basement service to the main

lobby by a separate group of 'shuttle' lifts so as to avoid compromising the traffic handling capability of the main lift group.

4.3.3 Multiple Entry Levels

Some buildings have main entry points at more than one level. The effect of more than one main terminal is disruptive and adversely impacts effective circulation and movement. If there are more than one entrance levels, means should be provided to bring all the routes to a single terminal floor. If this is not possible, then the lift system planning should take into account the extra times incurred in stopping and loading at multiple entry floors. Another difficulty is in deciding whether the building population will use each entrance equally. In the absence of any guidance, the solution is to assume an entrance bias with an additional 10 percent and size the lifts to meet the additional required handling capacity.

The loss of lift efficiency because of multiple entry levels could be as high as 15 percent per additional entry level. Both basement service and multiple entry level buildings with lifts serving all floors can also impact the security of the building.

4.3.4 Non-Smoking Buildings

When buildings are defined as non-smoking buildings, designers need to factor in the additional load put on the lifts on account of the additional trips that smokers make to go outside the building. It is appropriate to provide pressurized smoking zones at the floor level rather than require smokers to make additional trips.

4.3.5 Reserved Lifts

When lifts have to be reserved for VIP personal movement, designers should not consider these lifts as part of the lift group.

4.3.6 Zoning (Vertical)/Sky Lobbies in Very Tall Buildings

As the number of floors served increases, the values of H and S also increase, adversely impacting the round trip time and performance of the lift system. This has led to the concept of zoning in tall buildings. In stacked zoning, a tall building is effectively divided into horizontal layers or stacks. RTT , quality and quantity of service can be arrived at separately for each zone. The desired level of service can be attained by adjusting two parameters, in this case – the number of lifts and the number of floors in the zone. The round trip time for each zone is given by:

$$RTT = 2Ht_v + (S + 1)(T - t_v) + 2Pt_p + 2[t_{\text{express}} - t_f(1)]$$

where

t_{express} = flight time from the main terminal to the express zone terminal/sky lobby.

The round trip time for shuttle lifts serving only 2 floors – main terminal and sky lobby is given by:

$$RTT = 2T + 2Pt_p$$

4.4 Planning for Specific Building Types

4.4.1 Residential Buildings

The peak traffic condition occurs during the morning down-peak when many adults and children are leaving for work and school at the same time. Population assumptions would also need to consider the service staff, drivers of cars, etc. For high rise buildings the delivery of newspapers, milk and disposal of garbage also requires to be considered.

In the luxury housing segment, where the number of ratio of service staff to residents is fairly high, separate service lifts may be required for door deliveries and service staff movement. Typical average daily population for a residential building is given in [Table 6](#).

The passenger handling capacity shall be as given in [Table 7](#). The target handling capacity for residential building is lower as compared to office building with an understanding that the traffic for residential building would be comparatively less. The absenteeism factor shall not apply to residential buildings. The intervals shall be as per values in [Table 8](#). [Table 9](#) gives the recommended quality of service measured as average waiting time for residential buildings and [Table 10](#) gives the nominal travel times.

Sizing of lifts should consider the requirement to shift the stretchers, heavy material, etc and it is recommended that at least one lift in each building should be a deep car which can accommodate a regular ambulance stretcher.

Table 6 Typical Average Population for Residential Buildings

(Clause [4.4.1](#))

Sl No.	Class of Building	Number of Bedrooms Per Apartment				
		1	2	3	3 With Servant Quarters	4 With or Without Servant Quarters
(1)	(2)	(3)	(4)	(5)	(6)	(7)
i)	<i>High end apartments:</i>					
	a) Residents	2 to 3	3 to 4	5	5	5 to 6
	b) Resident service staff	0	0	0	1	1
	c) Floating service staff	1	2 to 3	3 to 4	3 to 4	4 to 5
ii)	<i>Mid end apartments:</i>					
	a) Residents	3	4	5	5	5 to 6
	b) Resident service staff	0	0	0	1	1
	c) Floating service staff	1	1	2 to 3	2 to 3	3 to 4
iii)	<i>Low end apartments:</i>					
	a) Residents	4 to 5	5 to 6	NA	NA	NA
	b) Floating service staff	0.5	0.5	NA	NA	NA

NOTE δ Unless separate lifts are provided for service staff, due consideration for service staff shall be given while calculating the required number of lifts. The class of building shall be as defined by the developer.

Table 7 Recommended Handling Capacity for Residential Buildings

(Clause [4.4.1](#))

Sl No.	Class of Building	Handling Capacity Percent
(1)	(2)	(3)
i)	High end building	> 8
ii)	Mid end building	6 to 8
iii)	Low end building	5 to 7

Table 8 Recommended Quality of Service for Residential Buildings (Based on Time Interval) <i>(Clause 4.4.1)</i>		
Sl No.	Class of Building	Interval s
(1)	(2)	(3)
i)	High end building	0 to 60
ii)	Mid end building	61 to 80
iii)	Low end building	81 to 100

Table 9 Recommended Quality of Service in Residential Buildings (Based on Average Waiting Time) <i>(Clause 4.4.1)</i>		
Sl No.	Class of Building	Average Waiting Time s
(1)	(2)	(3)
i)	High end building	0 to 30
ii)	Mid end building	31 to 45
iii)	Low end building	46 to 60

Table 10 Recommended Nominal Travel Time for Residential Buildings <i>(Clause 4.4.1 and Table 11)</i>		
Sl No.	Level	Nominal Travel Time s
(1)	(2)	(3)
i)	High end building	25 to 35
ii)	Mid end building	36 to 45
iii)	Low end building	46 to 60
NOTE — In case of super tall buildings considering limit of lift speed as 10 m/s, the travel time may be longer than specified above.		

The class of building shall be as defined by the developer.

4.4.2 Service Apartments and Hostel

Average population as 2 person per flat to 3 person per flat/room may be considered. The quantity and quality of service can be considered as per high end residential building, as the peak demand would be during morning hours as maximum of population would be leaving for office, college, etc. The system shall be designed to cater to a peak arrival rate of 8 percent to 10 percent.

4.4.3 Hospitals

Factors to be considered include number of staff and shift patterns, number of visitors and visiting hours, location of operation theatres, facilities, delivery of housekeeping supplies, waste disposal, evacuation procedures and segregation of sterile areas. When carrying out traffic analysis for hospitals, designers should consider establishing average car loading by volume rather than by weight. Arrival rates may approximate 10 percent to 15 percent and interval may be 30 s to 50 s.

Lifts shall be taken into consideration according to the evacuation strategy of building. Lifts shall be considered when vertical evacuation is planned for patients who need to be evacuated with stretcher.

The sizing of the lift car and doors should be such that a standard hospital stretcher and attendant can be easily accommodated. Where the hospital bed is likely to be moved the lift car should be able to accommodate the standard hospital bed including the auxiliary support equipment like oxygen cylinders, etc.

In addition to passenger lifts, service lifts is required for the movement of housekeeping staff, catering, etc. Dedicated service lift may be considered for movement of medical waste, garbage, etc.

In large hospitals, it is recommended that the lifts for the patients/hospital beds (or stretchers) should be separated from the lifts for staff/visitors.

NOTE — For hospital buildings, designers should also consider that in an emergency a number of patients would require to be evacuated on stretchers.

4.4.4 Hotels

4.4.4.1 The most demanding time is during the check-in and check-out period and two-way traffic occurs during this period with guests going to and from rooms and restaurants and in and out of the hotel. Calculations should be made assuming an equal number of up and down stops during this period. Average room occupancy may range from 1.5 persons to 2 persons and arrival rates from 10 percent to 15 percent.

The following shall also be considered in the calculations:

- a) Handling capacity > 12 percent;
- b) Interval < 40 s; and
- c) Average waiting time : 30 s to 40 s (peak check-in/checkout time).

4.4.4.2 Security considerations need to be taken into account whilst establishing the circulation and lift requirements. Escalators should be employed for heavy short range movements such as from the lobby to banquet/function level. Adequate number of service lifts need to be provided for service movement of housekeeping and room service staff as well as movement of material. Way finding signage shall be helpful to locate lift lobby for guests.

4.4.4.3 While hotel operator guideline should be the base for lift designing, modern hotels that has public amenities like, roof top restaurants, bars, event spaces, casino, etc. There should be dedicated lifts with separate lift lobby to access these levels, this helps in bifurcating the public traffic from hotel resident guests and thus ensure the security and performance of guest lifts.

4.4.4.4 Banquets and conventional centre

Escalators shall be an ideal means for the movement of mass people; however, lifts shall need to be considered to cater specially-abled people. The lifts shall be designed to cater to the minimum arrival rate of 12 percent. When escalators are provided, lift shall be designed for a minimum for 30 percent of total capacity for the factors that all the people may not use escalators. For banquets and conventional centres with higher capacities, lifts shall be designed for a fairly larger capacity as people would travel in groups and expected in special attire. It is recommended that the minimum capacity of lifts shall be 1 360 kg/20 pax with minimum door width of 1 000 mm. Freight lifts may be required depending on the functioning and requirements.

4.4.4.5 Consideration has to be given to valet circulation for vehicle retrieval.

4.4.5 Retail — Malls with Multiplexes

Pedestrian movement in retail establishments like malls is generally centred on escalators, and lifts do not play a major part. However, provision should be made for movement of shopping trolleys, wheel chairs, perambulators and persons with limited mobility from one level to another. A commonly applied solution is the installation of inclined moving walks. Where lifts are provided, assumptions of lift car sizing should take into account space occupied by trolleys and shopping bags. Where multiplexes are located on the upper floors, the last movie of the day gets over after the mall is closed at which time the escalators might not be available. In such cases the lift provision has to be adequate to handle the egress of the multiplex population with recommended average waiting

time of below 40 s catering to a demand population of minimum 8 percent for 1/3rd of total screens for exiting at different intervals. Staircase factor can be considered for a simulation model.

4.4.6 Airports/Railway Stations

While moving walks and escalators greatly improve the building circulation, the use of lifts has to be considered for persons with limited mobility and for movement of baggage trolleys from one level to another. The RTT equation of 4.2.5 may be used, but care may need to be taken in the assumptions of lift car occupancy levels taking into account space occupied by trolleys.

4.4.7 Multi-Level Car Parking

Multi-level car parks may be standalone public car parks or attached to office, retail or residential complexes. These car parks may be fully automated where drivers would leave the cars inside or on a trolley at the entry floor and the car is parked and retrieved automatically. The other alternative would be for the use of car lifts and/or ramps.

When the movement of the cars is dependent on car lifts, detailed study has to be carried out to establish the required number of car lifts ensuring that average car retrieval/parking time does not exceed 2 min. The sizing of the car lifts has to be adequate to fit the largest vehicle that is intended to be transported as well as adequate space to enable opening of the doors to enable evacuation of passengers in the eventuality of an entrapment. Designers will also need to take into account the probability of queues developing and provide for holding lanes. For luxury/high end residential buildings the car parking provision may exceed the total resident count of building. Where there are no separate shuttle lifts for parking floors, designer need to consider the movement of drivers in passenger lifts. When parking count exceeds the count of residents, the entrance bias should be calculated with assumption of certain percentage of total parking from each entry level.

4.4.8 Multi-Level Car Parking Passenger Traffic

When car lifts or ramps are the means for parking the cars, then means would need to be provided for the movement of passengers from and to the parking floors. If the main building lifts also serve the parking floors, it is to be noted that, performance will be adversely impacted due to multiple entry floors. Additional entry floors will also affect the security of the building. It is therefore recommended that separate lifts should be considered to move passengers. If separate parking lifts are provided, the basic RTT equation of 4.2.5 may be used. Average vehicle occupancy may be considered as 1.5 per car for office car parks, 4 for airports and retail and 2 elsewhere. For high end residential building or private building, usually drivers would be accessing parking level, hence for such luxury building 1 person (driver) per car can be considered.

4.4.9 Schools and Other Educational Institutions

In schools and other educational institutions, the traffic flow would consist of peak demand for short duration that would exist just before the start or after finishing of a class or lecture. It is unlikely that an economical solution can be implemented for such high peak requirements. Therefore, the design of the building has to be such that heavy stair usage is facilitated. Mostly for college buildings, demand time would be before and after each lecture/tutorial or seminar; lifts shall be designed to cater to the demand population of 12 percent to 15 percent with an interval of 30 s to 50 s. A higher staircase factor may be considered for traffic analysis.

4.4.10 Data Centre

Building population for data centre is usually low, however data centre has requirement of freight lifts. The design of freight lift shall be based on the input from architect/owner on the class of loading, dimension details and weight of equipment to be taken in freight lifts. As the freight lift hoist-way size is very big and usage is not extensive, the lift car size and door should consider all possible designs to reduce hoist-way sizes and increase saleable area.

4.4.11 Buildings with 24 Ĩ 7 Operations

In buildings operating 24 Ĩ 7 (round the clock), the peak traffic conditions would not be a typical up-peak but will occur during shift changes. The handling capacity should take into account the incoming, outgoing and shift change traffic as well as cafeteria traffic. Calculations shall be made assuming both up and down stops during this period. Particular attention should be paid while designating the drop off points, these should be assigned to

a level so as to not increase the burden on the system. Designers need to consider that the lifts in such buildings would have a significantly higher number of starts/stops per hour than a conventional building. Lift designer shall take into consideration the shut-down of lifts for maintenance, especially for office building operating 24 Ĩ 7.

4.4.12 Observatory

The height at which the observatory is located and the number of people expected at the observatory level shall decide the speed and the number of lifts. The circulation logic is crucial in locating the observatory lifts. Also, if the observatory is connected with lounge/cafe/souvenir shops, etc, separate service lifts may need to be provided.

4.5 Other Considerations

Designers need to be aware that the door opening and closing times, acceleration and deceleration times, etc would vary between suppliers and equipment types and can impact the overall lift performance. The dispatch algorithms and controller responses would also vary between the suppliers and the equipment types and can impact the actual operational results.

Lifts will breakdown as well as require to be shut-down for regular maintenance and repairs. Lift availability might be hampered during renovation of the building as a lift might be taken up for movement of material and debris. Designers should take into consideration the impact of such non-availability of lifts. The provision of well-located and easily accessible stairs can considerably lessen the demands on the lifts and therefore architect/engineer should consider this aspect in the layout.

4.5.1 Lift Speed

For passenger lifts in a building, the general recommendations as given in [Table 11](#) may be followed.

Table 11 Lift Speed		
(Clause 4.5.1)		
Sl No.	No. of Floors	Speed m/s
(1)	(2)	(3)
i)	Up to 6	0.6 to 1.0
ii)	7 to 15	1.0 to 1.5
iii)	16 to 20	1.5 to 1.75
iv)	21 to 30	1.75 to 2.5
v)	31 to 45	3.0 to 4.0
vi)	46 to 60	4.0 to 6.0
vii)	Above 60	6.0 and above
NOTES		
1 Above table is considering buildings with average floor height of 3.0 m.		
2 For office buildings average floor height is 4.2 m (see Table 5 and Table 10).		
3 Finalizing the lift speed for any building depends on detailed traffic analysis because number of floors, area per floor, area per person and class of building are crucial factors, which serve as inputs for such traffic analysis.		
4 Finalizing the speed for hotel depends on the number of keys per floor and similarly for an observatory depends on the number of expected footfall.		
5 For any building, the speed of the lifts will have to be validated against the total population and travel height of the building.		

4.5.2 Quiet Operation of Lifts

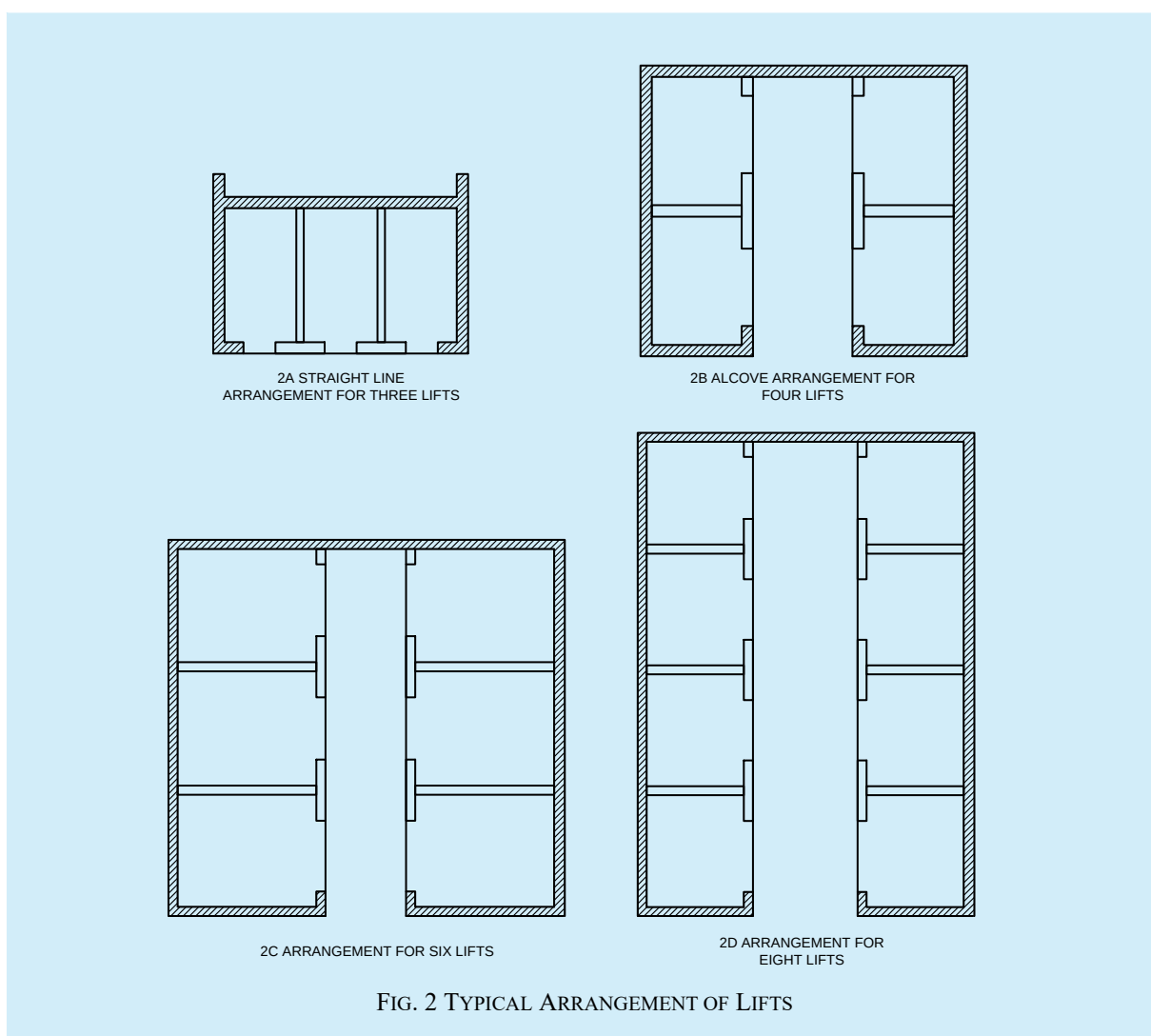
Every precaution should be taken with passenger lifts to ensure quiet operation of the lift doors and machinery. The insulating of the lift machine and any motor generator from the floor by rubber cushions or by a precast concrete slab with rubber cushions, prevents transmission of most of the noise. In this connection, reference may be made to good practice [D-5A(5)] and Part D 'Building Services, Section 4 Acoustics, Sound Insulation and Noise Control' of this NBCS for some useful recommendations.

5 LIFT ARRANGEMENTS AND PLANNING DIMENSIONS

The appropriate aspect of lift installation shall be discussed during the preliminary planning of the building with all concerned parties, namely, client, architect, consulting engineer and/or lift manufacturer. This enables the lift manufacturer to furnish the architect and/or consulting engineer with the proposed layout or vice-versa.

5.1 Exchange of Information

5.1.1 The guidelines laid down together with [Fig. 2](#) will enable the preliminary scheme for the installation to be established. [Figure 2](#) shows only some of the typical arrangements and variations are possible with respect to number of lifts and the layout of lifts especially when destination control system (DCS) is used instead of conventional group control system.



Although the recommended outline for the various classes of lifts enable the general planning details to be determined by the architect/engineer, these should be finally settled at the earliest possible stage by detailed investigation with the purchaser's representative reaching agreement with the lift manufacturer (where

necessary) before an order is finally placed. This will enable a check to be made and information to be exchanged on vital matters such as:

- a) number, capacity, speed and disposition of the lifts necessary to give adequate lift service in the proposed building;
- b) provision of adequate access to the machine room;
- c) loads which the lift will impose on the building structure, and the holes to be left in the machine room floor and cut-outs for wall boxes for push-buttons and signals;
- d) necessity for and type of insulation to minimize the transmission of vibration and noise to other parts of the building;
- e) special requirements of local authorities and other requirements set out in the planning permit;
- f) need for the builder to maintain accuracy of building as to dimensions and in-plumb;
- g) periods of time required for preparation and approval of relevant drawings for manufacturing and the installation of the lift equipment;
- h) requirements for fixing guide brackets to the building structure;
- j) time at which electric power will be required before completion to allow for testing;
- k) requirements for electrical supply feeders, etc;
- m) requirements for scaffolding in the lift well and protection of the lift well prior to and during installation of equipment; and
- n) delivery and storage of equipment.

5.1.2 Information to be provided by Architect or Engineer

As a result of preliminary discussions, the drawings of the building should give the following particulars and finished sizes:

- a) number, type and size of lifts and position of lift well;
- b) particulars of lift well enclosure;
- c) size, position, number and type of landing doors;
- d) number of floors served by the lift;
- e) height between floor levels;
- f) number of entrances;
- g) total headroom;
- h) provision of access to machine room;
- j) provision of ventilation and, if possible, natural lighting of machine room;
- k) height of machine room;
- m) depth of lift pit;
- n) position of lift machine, above or below lift well;
- p) size and position of any trimmer joists or stanchions adjacent to the lift well at each floor;
- q) size and position of supporting steel work at roof levels;
- r) size and position of any footings or grillage foundations, if these are adjacent to the lift pit; and
- s) in the case of passenger lifts whether the lift cage is required to carry household luggage, such as refrigerator, steel almirah, etc.

5.2 The lift lobby should be designed appropriately since this has bearing on the traffic handling especially when more number of lifts are involved. In a dual line arrangement (lifts opposite to each other) the lobby should be between 1.5 times and 2.5 times the depth of one car. Typically, the greater the number of lifts, the bigger the

multiplier to be used. As an example, a quadruplex may use 1.5 to 2, where as an octuplet will need 2 to 2.5. For in-line (single line) arrangements, the lobby can be typically half of the above recommendations.

5.3 The architect/engineer should advise the lift manufacturer, if the authority has any special requirements regarding lifts in buildings in the administrative area concerned.

5.4 The architect/engineer should inform the lift manufacturer of the dates when the erection of the lift may be commenced and is to be completed so that sufficient time is allowed for the manufacture and erection of the lift.

5.5 When submitting application for a building permit to the local Authority, the building plans shall include the details of lifts (number of lifts duly numbered, location, type, type of doors, passenger capacity and speed).

5.6 Positioning of Lifts

A thorough investigation should be made for assessing the most suitable position for lift(s) while planning the building. It should take into account future expansions, if any. Though each building has to be considered individually for purposes of location of lifts, factors influencing the locations of passenger and goods lifts are given in [5.6.2](#) to [5.6.4](#).

5.6.1 Arrangement of Lifts

The lifts should be easily accessible from all entrances to the building. For maximum efficiency, they should be grouped near the centre of the building. It is preferable not to have all the lifts out in straight line and, if possible, not more than four lifts should be arranged in this manner. If more than 4 lifts have to be grouped together in a straight line, then destination control systems would require to adopt. Further, the corridor should be wide enough to allow sufficient space for waiting passengers as well as for through passengers.

In some cases, when there are more than three lifts, the alcove arrangement is recommended. With this arrangement, the lift alcove leads off the main corridor so that there is no interference by traffic to other groups or to other parts of the ground floor. This arrangement permits the narrowest possible corridors and saves space on the upper floors. Walking distance to the individual lift is reduced and passenger standing in the centre of the group can readily see all the lift doors and landing indicators. The ideal arrangement of the lifts depends upon the particular layout of the respective building and should be determined in every individual case. Some typical recommended arrangements are given in [Fig. 2](#).

5.6.2 Passenger Lifts

5.6.2.1 Low and medium class flats

Where a lift is arranged to serve two, three or four flats per floor, the lift may be placed adjoining a staircase, with the lift entrances serving direct on to the landings. Where the lift is to serve a considerable number of flats having access to balconies or corridors, it may be conveniently placed in a well-ventilated tower adjoining the building.

5.6.2.2 Office buildings, hotels and high class flats

In general, the arrangement as recommended in [5.6.1](#) should be followed. However, in case that is not possible, it is desirable to have at least a battery of two lifts at two or more convenient points of a building. If this is not possible, it is advisable to have at least two lifts side by side at the main entrance and one lift each at different sections of the building for inter-communication. When two lifts are installed side by side, the machine room shall be suitably planned with sufficient space for housing the machine equipment. The positioning of lifts side by side gives the following advantages:

- a) All machines and switch gear may be housed in one machine room;
- b) Lifts can be inter-connected more conveniently from an installation point of view; and
- c) Greater convenience in service owing to the landing openings and each floor being adjacent.

5.6.2.3 Shops and departmental stores

Lifts in shops and stores should be situated so as to secure convenient and easy access at each floor.

5.6.2.4 For buildings with more than 12 floors, where passenger and service lifts are provided in one lobby, it is recommended to have group control for all the lifts.

5.6.3 Goods Lifts

The location of lifts in factories, warehouses and similar buildings should be planned to suit the progressive movement of goods throughout the buildings, having regard to the nature of position of the loading platforms, railway sidings, etc. The placing of a lift in a fume or dust laden atmosphere or where it may be exposed to extreme temperatures, should be avoided, wherever possible. Where it is impossible to avoid installing a lift in an adverse atmosphere, the electrical equipment should be of suitable design and construction to meet the conditions involved.

Normally goods lifts have lower speeds than passenger lifts for the same travel because traffic conditions are less demanding, and more time is required for loading and unloading. As loads for goods lifts increase in size and weight, so the operation of loading and unloading becomes more difficult. Therefore, it is usual to require greater accuracy of levelling as the capacity of the goods lift increases.

A large capacity goods lift operating at high speed is often a very uneconomical solution. The inherent high cost is enhanced due to the very small demand for such equipment, much of which is custom made. The high capital cost of the lift, building work and electrical supply equipment usually shows a much smaller return as an investment than more normal size of lifts.

5.6.4 Hospital Bed Lifts

Hospital bed lifts should be situated conveniently near the ward and operating at the entrances. There shall be sufficient space near the landing door for easy movement of stretcher/bed. It is convenient to place the passenger lifts in a hospital, near the staircases.

5.7 General Arrangement Drawings (GAD)

The general arrangement drawings should be prepared by the lift manufacturer. The lift manufacturer requires sufficient information for the preparation of working drawings and is usually obtained from architect's drawings supplemented by any information obtained from the site and by collaboration with the other contractors.

General arrangement drawings showing the layout of lifts duly numbered, details of builders work, for example, holes in walls for guide fixing, holes in machine room floor for ropes and conduits, recesses for landing sills, supports for lift machine and loads imposed on the building should be submitted by the lift manufacturer to the architect/engineer for written approval.

5.8 Additional Requirements for Passenger, Goods and Service Lifts

5.8.1 Refuge Spaces and Clearances in the Pit or Bottom Car Clearances

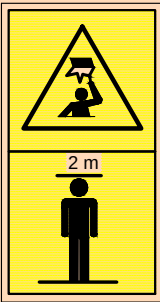
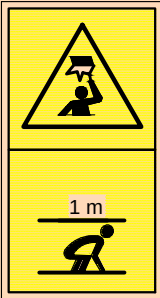
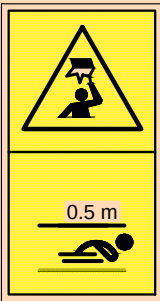
When the car is at its lowest position, at least one clear area where a refuge space can be accommodated as per [Table 12](#) shall be provided on the pit floor having a vertical clearance as specified.

When the car is at the lowest position the following conditions shall also be satisfied:

- a) The free vertical distance between the bottom of the pit and the lowest parts of the car shall be at least 0.50 m. This distance may be reduced for any part of the apron or parts of the vertically sliding car door(s) to a minimum of 0.10 m within a horizontal distance of 0.15 m to the adjacent wall(s).
- b) The free vertical distance between the highest parts fixed in the pit, for instance a tensioning device for compensation means being in its highest position, jack supports, pipes and other fittings and the lowest parts of the car, shall be at least 0.30 m.

Table 12 Dimensional Requirements of Designated Area in the Pit

(Clause [5.8.1](#))

Sl No.	Type	Posture	Pictogram	Horizontal Dimensions of the Refuge Space mm I mm	Height of the Refuge Space mm
(1)	(2)	(3)	(4)	(5)	(6)
i)	1	Upright posture		400 I 500	2 000
ii)	2	Crouching		500 I 700	1 000
iii)	3	Laying		700 I 1 000	500

The following conditions shall also apply:

- 1) When the car is at its lowest position at least one clear area where a refuge space can be accommodated shall be provided on the pit floor.
- 2) If more than one person is necessary in the pit to carry out inspection and maintenance work, an additional refuge space shall be provided for each additional person.
- 3) In the case of more than one refuge space, they shall be of the same type and not interfering with each other.
- 4) A sign in the pit readable from the entrance(s) shall clearly indicate the allowed number of persons and the type of posture considered for the refuge spaces(s) accommodation.

NOTES

- 1 Dimensional requirements of designated area in the pit may vary depending upon manufacturer.
- 2 Designated manufacturer shall coordinate and upgrade the requirements on order specific.
- 3 [Table 17](#) shall be referred for the recommended dimensions of pit depth.

5.8.2 Refuge Spaces on Car Roof and Clearances in Headroom or Car Top Clearances

When the car is at its highest position, at least one clear area where a refuge space can be accommodated as per [Table 13](#) shall be provided on the car roof having a vertical clearance as specified.

When the car is at its highest position, the clear distance between the lowest parts of the ceiling of the well (including beams and parts situated under the ceiling) and:

- the highest pieces of equipment fixed on the roof, except for those covered in (b) and (c) below, shall be at least 0.50 m in any vertical or inclined direction within the projection of the car;
- the highest part of the guide shoes or rollers, of the compensation means terminations and of the header or parts of vertically sliding doors, if any, shall be at least 0.10 m in any vertical direction within a horizontal distance of 0.40 m within the projection of the car; and
- the highest part of the balustrade shall be at least:
 - 0.30 m within a horizontal distance of 0.40 m within the projection of the car and 0.10 m on the outside of the balustrade; and
 - 0.50 m in any inclined distance beyond 0.40 m within the projection of the car.

Table 13 Dimensional Requirements of Designated Area on Car Roof and Clearance on Head Room
(Clause [5.8.2](#))

Sl No.	Type	Posture	Pictogram	Horizontal Dimensions of the Refuge Space	Height of the Refuge Space
(1)	(2)	(3)	(4)	mm Ĩ mm (5)	mm (6)
i)	1	Upright posture		400 Ĩ 500	2 000
ii)	2	Crouching		500 Ĩ 700	1 000

The following conditions shall also apply:

- At least one clear area where a refuge space can be accommodated shall be provided on the car roof;
- If more than one person is necessary in the car roof to carry out inspection and maintenance work, an additional designated area shall be provided for each additional person;
- In the case of more than one refuge space, they shall be of the same type and not interfering with each other; and

- 4) A sign on the car roof readable from the landings giving access to the car roof shall clearly indicate the allowed number of persons and the type of posture considered for the refuge space(s) accommodation.

NOTES

- 1 Dimensional requirements of designated area in the pit may vary depending upon manufacturer.
- 2 Designated manufacturer shall coordinate and upgrade the requirements on order specific.
- 3 [Table 17](#) shall be referred for the recommended dimensions of pit depth.

5.9 Additional Requirements for Dumb Waiters

5.9.1 Top Car Clearance

The top car clearance shall be sufficient to avoid any protruding part fixed on the top of the car coming in direct contact with the ceiling or diverting sheave.

The clearance shall be calculated taking into account the following and shall not be less than the sum of the following four items:

- a) bottom counterweight run-by;
- b) stroke of the counterweight buffer used;
- c) dimensions of the portion of the diverting sheave hanging underneath the ceiling in the lift well; and
- d) 150 mm for compensating for gravity stopping distance and future repairs to the rope connections at counterweight and at the car or at the suspension points.

5.9.2 Bottom Car Clearance

The bottom car clearance shall be maintained in such a way that the counterweight shall not come in contact with the ceiling or any part hanging underneath the ceiling, when the car completely rests on fully compressed buffers, provided the buffers are spring type mounted on solid concrete or steel bed.

In case of wooden buffers the bottom car clearance shall be maintained in such a way that the total downward travel of the car from the service level of the immediate floor near the pit, shall not be more than the top counterweight clearance, when the wooden buffers are completely crushed.

5.9.3 Top Counterweight Clearance

The top clearance for the counterweight can be calculated taking into account the following and shall not be less than the sum of the following three items:

- a) car run-by;
- b) compression of the buffer spring or height of the wooden block used as buffer; and
- c) 150 mm to compensate for gravity stopping distance for counterweight and any future repairs to rope connections at the counterweight at the car ends or at the suspension points.

5.9.4 Run-by for Cars and Counterweights

- a) *Minimum bottom run-by* ð The bottom run-by for cars and counterweights shall not be less than 150 mm; and
- b) *Maximum bottom run-by* ð In no case shall the maximum bottom run-by exceeds 300 mm.

5.10 Planning for Dimensions

5.10.1 General

The dimensions of lift well shall be chosen to accommodate the doors inside the well which is the normal practice. In special cases, the door may be accommodated in a recess in the front wall, for which prior consultation shall be made with the lift manufacturer.

5.10.2 Plan Dimensions

All plan dimensions of lift well given hereunder, are the minimum clear plumb sizes. The architect/engineer, in conjunction with the builder, shall ensure that adequate tolerances are included in the building design so that the specified minimum clear plumb dimensions are obtained in the finished work.

NOTE 8 The words 'clear plumb dimensions' should be noted particularly in case of high rise buildings.

Rough opening in concrete or brick walls to accommodate landing doors depend on design of architrave. It is advisable to provide sufficient allowances in rough opening width to allow for alignment errors of opening at various landings. When more than one lift is located in a common well, a minimum allowance of 150 mm for separator beams shall be made in the widths.

Where the governor operated counterweight safety is required under conditions stipulated in good practice [D-5A(1)], the tabular values under [5.10.3.1](#) should be revised in consultation with the lift manufacturer.

For outline dimensions of lifts having more than one car entrance, lift manufacturers shall be consulted.

5.10.3 Outline Dimensions

5.10.3.1 The outline dimensions of lift shaft, machine-room and machine room less lifts, pit depth, overhead height and raw door opening size for lifts to which this Subsection applies are specified in [Table 14](#) to [Table 32](#) as indicated below:

a) Machine room lifts:	
1) Recommended dimensions of passenger lifts and service lifts	Table 14
2) Recommended dimensions of stretcher/hospital bed lifts	Table 15
3) Recommended dimensions of goods lifts with machine room	Table 16
4) Recommended dimensions of machine room height, overhead, pit depth for passenger, goods, stretcher/hospital lifts with machine room	Table 17
5) Recommended dimensions of automobile lifts with machine room 8 Through type car	Table 18
6) Recommended dimensions of dumbwaiter lifts	Table 19
7) Entrance type for horizontal and vertical sliding doors	Table 20
8) Recommended dimension of goods lifts with machine room 8 Series A : Vertical sliding door (entrance Type 4 and 5)	Table 21
9) Recommended dimension of goods lifts with machine room 8 Series A : Vertical sliding door (entrance Type 6)	Table 22
10) Recommended dimension of goods lifts with machine room 8 Series B : Horizontal sliding doors (entrance Type 3)	Table 23
11) Recommended dimension of goods lifts with machine room 8 Series B : Vertical sliding doors (entrance, Type 4 and 5)	Table 24
12) Recommended dimension of goods lifts with machine room 8 Series B : Vertical sliding doors (entrance Type 6)	Table 25
13) Recommended dimension of automobile lifts with machine room 8 Trough type car 8 Series B : Vertical sliding Door (entrance Type 4)	Table 26
b) Machine room-less lifts:	
1) Recommended dimensions of passenger lifts MRL	Table 27
2) Recommended dimensions of stretcher/hospital lifts MRL	Table 28
3) Pit depth for passenger/stretcher/hospital lifts MRL	Table 29
4) Overhead height for passenger and stretcher/hospital lifts MRL	Table 30
5) Recommended dimensions of goods lifts MRL	Table 31
6) Recommended dimensions of automobile lifts MRL	Table 32

Typical car and landing dimensions for horizontal sliding doors and vertical sliding doors for Series A and Series B are as shown in Fig. 8 and Fig. 9 respectively.

5.10.3.2 Minimum floor to floor height

Minimum floor to floor height for landings on same side for horizontally sliding door is $f + 750$ mm and for vertically bi-parting doors is $1.5 f + 250$ mm, where f is clear entrance height in mm.

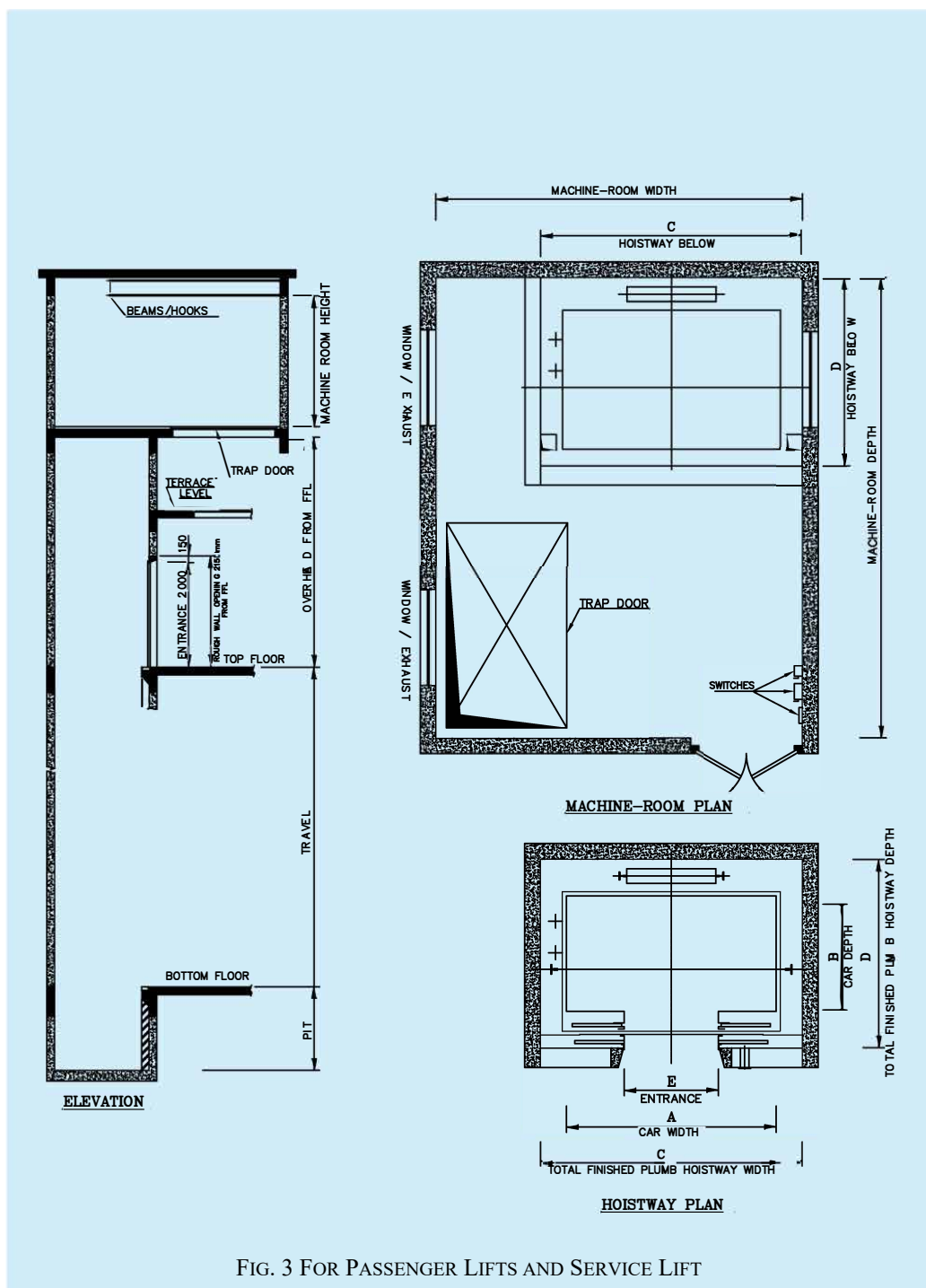


FIG. 3 FOR PASSENGER LIFTS AND SERVICE LIFT

Table 14 Recommended Dimensions of Passenger Lifts and Service Lifts

(Clause 5.10.3.1)

All dimensions in millimetres.

Sl No.	No. of Passengers	Rated Load	Car Size		COPD Door	Passenger and Service Lifts with Machine Room: Shaft Sizes													
						Rated speed < 1 m/s		1.0 m/s Ò Rated Speed Ò2.5 m/s		3.0 m/s Ò Rated Speed Ò3.5 m/s		4.0 m/s Ò Rated Speed Ò5.0 m/s		Rated Speed = 6.0 m/s		7.0 m/s Ò Rated Speed Ò8.0 m/s		9.0 m/s Ò Rated Speed Ò10.0 m/s	
						Shaft Size		Shaft Size		Shaft Size		Shaft Size		Shaft Size		Shaft Size		Shaft Size	
			kg	Width	Depth	Width	Widt h	Depth	Widt h	Depth	Widt h	Depth	Widt h	Depth	Widt h	Depth	Widt h	Depth	Widt h
(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)	(13)	(14)	(15)	(16)	(17)	(18)	(19)	(20)
i)	4	272	1 100	700	800	1 900	1 300	1 900	1 300	ï	ï	ï	ï	ï	ï	ï	ï	ï	ï
ii)	6	408	1 100	1 000	800	1 900	1 700	1 900	1 700	ï	ï	ï	ï	ï	ï	ï	ï	ï	ï
iii)	8	544	1 400	1 100	800	1 900	1 750	1 900	1 900	ï	ï	ï	ï	ï	ï	ï	ï	ï	ï
iv)	10	680	1 300	1 350	800	1 900	2 150	1 900	2 150	ï	ï	ï	ï	ï	ï	ï	ï	ï	ï
v)	13	884	2 000	1 100	900	2 600	1 900	2 600	1 900	2 800	2 200	ï	ï	ï	ï	ï	ï	ï	ï
vi)	14	952	1 600	1 400	900	2200	2100	2200	2100	2 400	2 400	ï	ï	ï	ï	ï	ï	ï	ï
vii)	15	1020	1450	1 650	1000	2 600	2 000	2 600	2 000	2 250	2 650	2 600	2 150	2 600	2 250	2 600	2 300	2 600	2450
viii)	16	1 088	1 600	1 600	1 000	2 450	1 900	2 600	2 100	2 400	2 600	2 600	2 350	2 600	2 450	2 600	2 500	2 600	2650
ix)	18	1 224	2 000	1 400	1 000	ï	ï	2 600	2 350	2 800	2 400	2 600	2 350	2 600	2 350	2 700	2 300	2 700	2450
x)	20	1 360	2 000	1 500	1 000	ï	ï	2 600	2 350	2 800	2 500	2 600	2 350	2 600	2 350	2 700	2 400	2 700	2550
xi)	22	1 496	2 100	1 600	1 100	ï	ï	2 700	2 300	2 900	2 600	2 700	2 350	2 700	2 450	2 800	2 500	2 800	2550

Table 14 (Concluded)

Sl No.	No. of Passengers	Rated Load	Car Size		COPD Door	Passenger and Service Lifts with Machine Room: Shaft Sizes													
						Rated speed < 1 m/s		1.0 m/s Ò Rated Speed Ò2.5 m/s		3.0 m/s Ò Rated Speed Ò3.5 m/s		4.0 m/s Ò Rated Speed Ò5.0 m/s		Rated Speed = 6.0 m/s		7.0 m/s Ò Rated Speed Ò8.0 m/s		9.0 m/s Ò Rated Speed Ò10.0 m/s	
						Shaft Size		Shaft Size		Shaft Size		Shaft Size		Shaft Size		Shaft Size		Shaft Size	
			kg	Width	Depth	Width	Width	Depth	Width	Depth	Width	Depth	Width	Depth	Width	Depth	Width	Depth	Width
(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)	(13)	(14)	(15)	(16)	(17)	(18)	(19)	(20)
xii)	24	1 632	1 800	2 000	1 100	ï	ï	2 600	2 600	2 600	3 000	3 000	2 350	3 000	2 450	3 050	2 550	3 050	2550
xiii)	26	1 768	2 350	1 600	1 200	ï	ï	2 600	2 700	3 150	2 600	2 600	2 350	2 600	2 450	2 650	2 550	2 650	2550
xiv)	29	1 972	2 350	1 700	1 200	ï	ï	ï	ï	3 150	2 700	3 000	2 500	3 000	2 550	3 050	2 600	3 050	2700
NOTES																			
1 All dimensions given above for lifts having centre opening power operated doors (COPD) with counterweight at rear, are recommended dimensions primarily for architects and building planners. Any variations mutually agreed between the manufacturer and the purchaser is permitted. However, variation in: a) Car inside dimensions shall be within the minimum and maximum area limits specified in accordance with Table 37; b) Entrance width on higher side is permitted; and c) Entrance width on lower side is permitted up to 100 mm subject to minimum of 700 mm.																			
2 The minimum size of the lift car and all other requirements relating to accessibility, in all public buildings shall be in accordance with good practice [D-5A(3)].																			

Table 15 Recommended Dimensions of Stretcher/Hospital Lifts

(Clause 5.10.3.1)

All dimensions in millimetres.

Sl. No	No. of Passengers	Rated Load	Rounded off Rated Load	Car Size		2P TSP D Door	Hospital Bed / Stretcher Lifts with Machine Room: Shaft Sizes													
							Rated Speed Ò1 m/s		0.6 m/s Ò Rated Speed Ò2.5 m/s		3.0 m/s Ò Rated Speed Ò3.5 m/s		4.0 m/s Ò Rated Speed Ò5.0 m/s		Rated Speed = 6.0 m/s		7.0 m/s Ò Rated Speed Ò 8.0 m/s		9.0 m/s Ò Rated Speed Ò10.0 m/s	
							Shaft Size		Shaft Size		Shaft Size		Shaft Size		Shaft Size		Shaft Size		Shaft Size	
				Width	Depth	Width	Depth	Width	Depth	Width	Depth	Width	Depth	Width	Depth	Width	Depth	Width	Depth	
(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)	(13)	(14)	(15)	(16)	(17)	(18)	(19)	(20)	(21)
i)	15	1 020		1 000	2 400	800	1 800	3 000	1 800	3 000	2 100	3 400								
ii)	20	1 360		1 300	2 400	1 200			2 300	3 000	2 650	3 400	2 350	3 000	2 350	3 000	2 500	3 000	2 500	3 000
iii)	26	1 768	1 800	1 400	2 550	1 200			2 650	3 050	2 650	3 550	2 500	3 050	2 500	3 050	2 600	3 150	2 650	3 150
iv)	29	1 972	1 500	1500	2 700	1 400			2 650	3 200	3 100	3 700	2 600	3 200	2 600	3 200	2 700	3 200	2 700	3 200
v)	36	2 448	2 500	1 800	2 700	1 400			2 800	3 150	2 900	3 200	2 900	3 200	2 900	3 200	3 000	3 200	3 000	3 200
NOTES																				
1 All dimensions given above for lifts having two panel telescopic doors (2P TSPD) with counterweight at side, are recommended dimensions primarily for architects and building planners. Any variations mutually agreed to between the manufacturer and the purchaser is permitted. However, variation in: a) Car inside dimensions shall be within the maximum area limits specified in accordance with Table 37; b) Entrance width on higher side is permitted; and c) Entrance width on lower side is permitted up to 100 mm subject to minimum of 700 mm. 2 Car depth of 2 100 mm may be considered in residential/commercial buildings, where use of only ambulance stretcher is envisaged. The hoist-way depth maybe reduced accordingly.																				

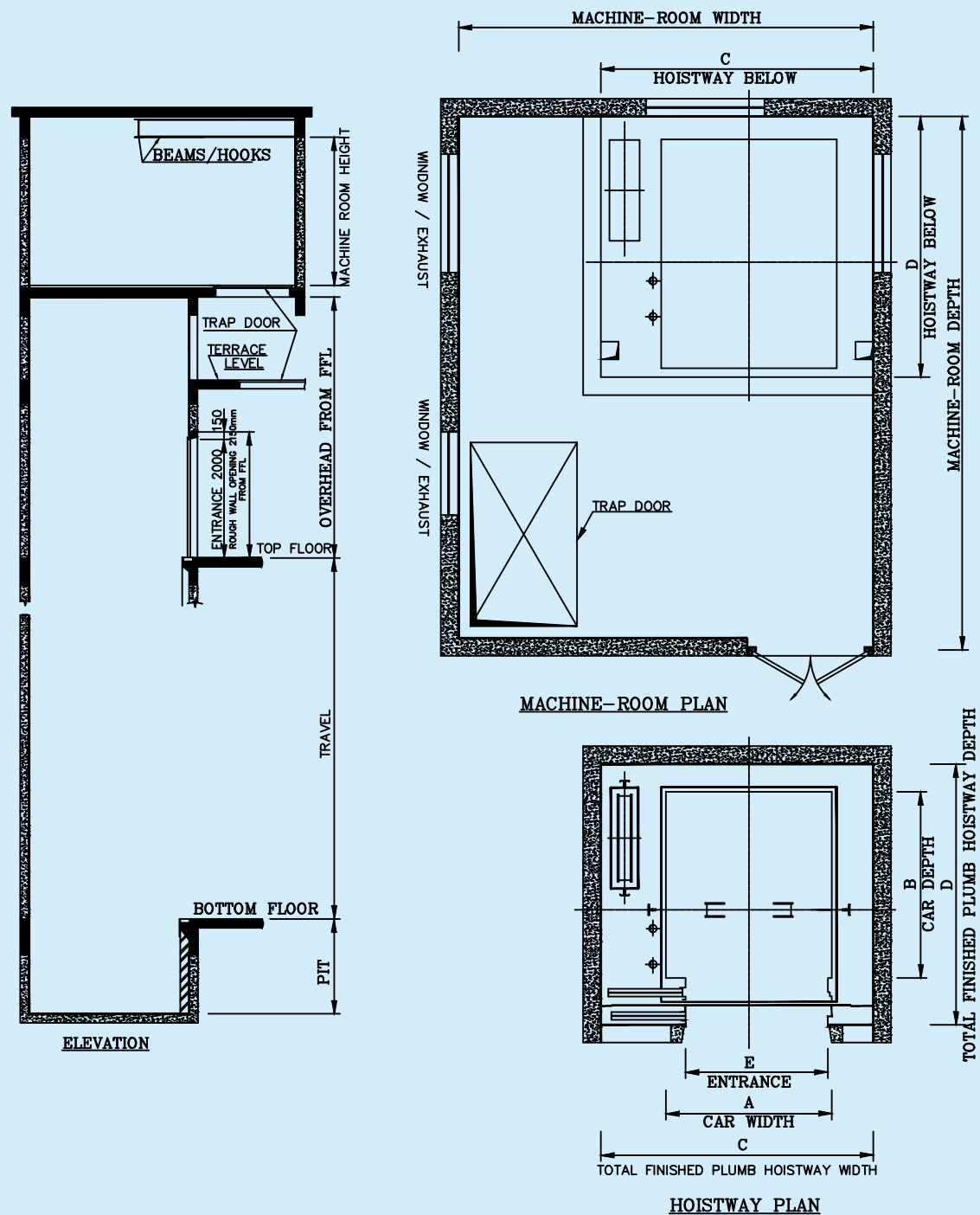


FIG. 4 (TABLE 15) FOR STRETCHER/HOSPITAL LIFTS

Table 16 Recommended Dimensions of Goods Lifts with Machine Room

(Clause [5.10.3.1](#))

All dimensions are in millimetres.

Sl No.	Rated Load	Car Size		COPD Door	0.6 m/s Ò Rated Speed Ò2.5 m/s		3.0 m/s Ò Rated Speed Ò3.5 m/s		4.0 m/s Ò Rated Speed Ò 5.0 m/s		Rated Speed = 6.0 m/s		7.0 m/s Ò Rated Speed Ò 8.0 m/s		9.0 m/s ÒRated Speed Ò10.0 m/s	
					Shaft Size		Shaft Size		Shaft Size		Shaft Size		Shaft Size		Shaft Size	
		kg	Width	Depth	Width	Depth	Width	Depth	Width	Depth	Width	Depth	Width	Depth	Width	Depth
(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)	(13)	(14)	(15)	(16)	(17)
i)	1 000	1 400	1 800	2P-tel-1 200	2 550	2 200	ï	ï	ï	ï	ï	ï	ï	ï	ï	ï
ii)	1 500	1 700	2 000	4P-1 500	2 750	2 550	ï	ï	ï	ï	ï	ï	ï	ï	ï	ï
iii)	2 000	1 700	2 500	4P-1 500	2 800	3 050	2 800	3 050	2 900	3 050	3 000	3 050	3 000	3 050	3 000	3 050
iv)	2 500	2 000	2 500	4P-1 800	3 100	3000	3 000	3 050	3 100	3 050	3 200	3 050	3 200	3 050	3 200	3 050
v)	3 000	2 000	3 000	4P-1 800	3 100	3 550	3 000	3 550	3 100	3 550	3 200	3 550	3 200	3 550	3 200	3 550
vi)	4 000	2 500	3 000	4P-2 100/ 6P-2 100	3 700	4 000	3 700	3 650	ï	ï	ï	ï	ï	ï	ï	ï
vii)	5 000	2 500	3 600	4P-2 100/ 6P-2 100	3 850	4 250	3 850	4 250	ï	ï	ï	ï	ï	ï	ï	ï
NOTES 1 All dimensions given above for lifts having centre opening power operated doors (COPD) with counterweight at side, are recommended dimensions primarily for architects and building planners. Any variations mutually agreed to between the manufacturer and the purchaser are permitted. However, the minimum rated load for the goods lift shall be based on a load of not less than 3.45 kN/m² of the net inside car area. 2 Car inside dimensions shall be within the maximum area limits specified in accordance with Table 32.																

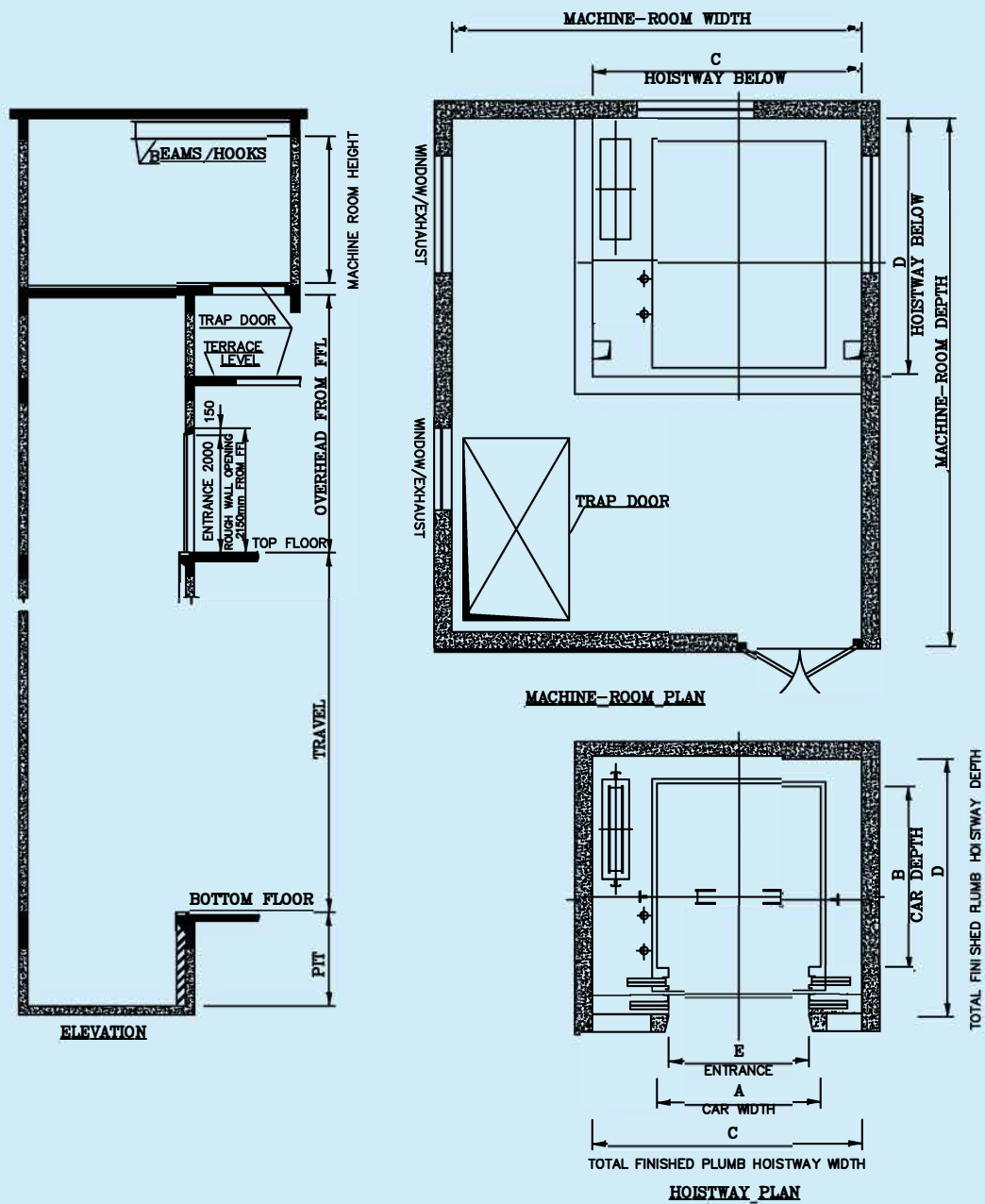


FIG. 5 (TABLE 16) FOR GOODS LIFTS

Table 17 Recommended Dimensions of Machine Room Height, Overhead, Pit Depth for Passenger, Goods, and Stretcher/Hospital Lifts with Machine Room																							
(Clauses 5.8.1, 5.8.2 and 5.10.3.1)																							
All dimetres. are in millimetres.																							
For Speed up to 3.5 m/s																							
Sl No.	No. of Passengers	Rated Load kg	Dimensions of Pit Depth, Overhead Height (HR) and Machine Room Height(MR)																				
			0.7 m/s			1.0 m/s			1.6 m/s			2 m/s			2.5 m/s			3 m/s			3.5 m/s		
			Pit Depth	HR	MR	Pit Depth	HR	MR	Pit Depth	HR	MR	Pit Depth	HR	MR	Pit Depth	HR	MR	Pit Depth	HR	MR	Pit Depth	HR	MR
(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)	(13)	(14)	(15)	(16)	(17)	(18)	(19)	(20)	(21)	(22)	(23)	(24)
i)	4	272	1 350	CH + 1 900	2 500	1 500	CH + 1 950	2 500	1 600	CH + 2 400	2 500	2 200	CH + 2 800	2 500	ī	ī	ī	ī	ī	ī	ī	ī	ī
ii)	6	408	1 350	CH + 1 900	2 500	1500	CH +1950	2500	1 600	CH + 2 400	2 500	2 200	CH + 2 800	2 500	ī	ī	ī	ī	ī	ī	ī	ī	ī
iii)	8	544	1 350	CH + 1 900	2 500	1500	CH +1950	2500	1 600	CH + 2 400	2 500	2 200	CH + 2 800	2 500	ī	ī	ī	ī	ī	ī	ī	ī	ī
iv)	10	680	1 350	CH + 1 900	2 500	1500	CH + 1950	2500	1 600	CH +2 400	2 500	2 200	CH + 2 800	2 500	2 250	CH + 2 800	2 500	ī	ī	ī	ī	ī	ī
v)	13	884	1 350	CH + 1 900	2 500	1500	CH +1950	2500	1 600	CH +2 400	2 500	2 200	CH + 2 800	2 500	2 500	CH + 3 000	2 500	3 200	CH + 3 250	3 500	4 500	CH + 3 550	3 500
vi)	14	952	1 350	CH + 1 900	2 500	1500	CH +1950	2500	1 800	CH + 2 400	2 500	2 200	CH + 2 800	2 500	2 500	CH + 3 000	2 500	3 200	CH + 3 250	3 500	4 500	CH + 3 550	3 500
vii)	16	1 088/ 1 000	1500	CH+ 1 900	2 500	1500	CH +1950	2500	1 800	CH + 2 400	2 500	2 200	CH + 2 800	2 500	2 500	CH + 3 000	2 500	3 200	CH + 3 250	3 500	4 500	CH + 3 550	3 500
viii)	18	1 224	ī	ī	ī	1 500	CH + 1950	2500	1 800	CH + 2 400	2 500	2 200	CH + 2 800	2 500	2 500	CH + 3 000	2 500	3 200	CH + 3 250	3 500	4 500	CH + 3 550	3 500
ix)	20	1 360	ī	ī	ī	1 500	CH + 1 950	2500	1 800	CH + 2 400	2 500	2 200	CH +2 800	2 500	2 500	CH + 3 000	3 000	3 200	CH+ 3 250	3 500	4 500	CH+ 3 550	3 500
x)	22	1 496/ 1 500	ī	ī	ī	1 500	CH + 1 950	2500	1 800	CH + 2 400	3 000	2 200	CH + 2 800	3 000	2 500	CH + 3 000	3 000	3 200	CH+ 3 250	3 500	4 500	CH+ 3 550	3 500
xi)	24	1 632	ī	ī	ī	1500	CH + 1950	2500	1 800	CH + 2 400	3 000	2 200	CH + 2 800	3 000	2 500	CH + 3 000	3 000	3 200	CH+ 3 250	3 500	4 500	CH+ 3 550	3 500
xii)	26	1 768	ī	ī	ī	1 500	CH + 1 950	3000	1 800	CH + 2 400	3 000	2 200	CH + 2 800	3 000	2 500	CH + 3 000	3 000	3 200	CH+ 3 450	3 500	4 500	CH+ 3 550	3 500
xiii)	29	1 972/ 2 000	ī	ī	ī	1 600	CH + 1 950	3000	1 800	CH + 2 400	3 000	2 200	CH + 2 800	3 000	2 500	CH + 3 000	3 000	3 200	CH+ 3 450	3 500	4 600	CH+ 3 550	3 500
xiv)	ī	2 500	ī	ī	ī	1 800	CH + 1 950	4000	1 800	CH + 2 400	4 000	2 200	CH + 2 800	4 000	2 500	CH + 3 000	4 000	3 200	CH+ 3 450	4 000	4 600	CH+ 3 550	4 000
xv)	ī	3 000	ī	ī	ī	1 800	CH + 1 950	4000	1 800	CH + 2 400	4 000	2 200	CH + 2 800	4 000	2 500	CH + 3 000	4 000	3 200	CH+ 3 450	4 300	4 600	CH+ 3 600	4 300
xvi)	ī	4 000	ī	ī	ī	1 800	CH + 2200	4300	1 800	CH + 2 400	4 300	2 200	CH + 2 800	4 300	2 500	CH + 3 000	4 300	3 200	CH+ 3 450	4 300	4 600	CH+ 3 600	4 300
xvii)	ī	5 000	ī	ī	ī	1 800	CH + 2200	4500	1 800	CH + 2 400	4 500	2 200	CH + 2 800	4 500	2 500	CH + 3 000	5 000	3 200	CH+ 3 450	5 000	4 600	CH+ 3 600	5 000

Table 17 (Concluded)

Sl No.	No. of Passengers	Rated Load	Passenger, Goods, Stretcher/Hospital Lifts with Machine Room: Dimensions of Pit Depth, Overhead Height (HR) and Machine Room Height(MR)																				
			4 m/s			5 m/s			6 m/s			7 m/s			8 m/s			9 m/s			10 m/s		
			Pit Depth	HR	MR	Pit Depth	HR	MR	Pit Depth	HR	MR	Pit Depth	HR	MR	Pit Depth	HR	MR	Pit Depth	HR	MR	Pit Depth	HR	MR
(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)	(13)	(14)	(15)	(16)	(17)	(18)	(19)	(20)	(21)	(22)	(23)	(24)
i)	14	952	4 600	CH + 3 450	4 000	4 700	CH + 3 450	4 000	5 500	CH + 3 800	4 300	5 900	CH + 5 000	4 300	6 500	CH + 5 800	4 300	7 100	CH + 6 500	5 000	7 500	CH + 6 900	5 000
ii)	16	1 088/1 000	4 600	CH + 3 450	4 000	4 700	CH + 3 450	4 000	5 500	CH + 3 800	4 300	5 900	CH + 5 000	4 300	6 500	CH + 5 800	4 300	7 100	CH + 6 500	5 000	7 500	CH + 6 900	5 000
iii)	18	1 224	4 600	CH + 3 450	4 000	4 700	CH + 3 450	4 000	5 500	CH + 3 800	4 300	5 900	CH + 5 000	4 300	6 500	CH + 5 800	4 300	7 100	CH + 6 500	5 000	7 500	CH + 6 900	5 000
iv)	20	1 360	4 600	CH + 3 450	4 000	4 700	CH + 3 450	4 000	5 500	CH + 3 800	4 300	5 900	CH + 5 000	4 300	6 500	CH + 5 800	4 300	7 100	CH + 6 500	5 000	7 500	CH + 6 900	5 000
v)	22	1 496/1 500	4 600	CH + 3 450	4 000	4 700	CH + 3 450	4 000	5 500	CH + 3 800	4 300	5 900	CH + 5 000	4 300	6 500	CH + 5 800	4 300	7 100	CH + 6 500	5 000	7 500	CH + 6 900	5 000
vi)	26	1 768	4 600	CH + 3 450	4 000	4 700	CH + 3 450	4 300	5 500	CH + 3 800	4 300	5 900	CH + 5 000	4 300	6 500	CH + 5 800	4 300	7 100	CH + 6 500	5 000	7 500	CH + 6 900	5 000
vii)	29	1 972/2 000	4 600	CH + 3 450	4 000	4 700	CH + 3 450	4 300	5 500	CH + 3 800	4 300	5 900	CH + 5 000	5 000	6 500	CH + 5 800	5 000	7 100	CH + 6 500	5 000	7 500	CH + 6 900	5 000
viii)	ï	2 500	4 600	CH + 3 450	4 300	4 700	CH + 3 450	5 000	5 500	CH + 3 800	5000	5 900	CH + 5 500	5 000	6 500	CH + 6 300	5 000	7 100	CH + 7 000	5 000	7 500	CH + 7 500	5 000
ix)	ï	3 000	4 600	CH + 3 600	4 300	4 700	CH + 3 600	5 000	5 500	CH + 4 000	5 000	5 900	CH + 5 500	5 000	6 500	CH + 6 300	5 000	7 100	CH + 7 000	5 000	7 500	CH + 7 500	5 000

CH ï Car clear inside height measured from car finished floor level to false ceiling.

NOTES

1 Recommended dimensions for pit depth, overhead and machine-room for different lift speeds are given in the table above. These dimensions may differ in practice as per individual manufacturer’s design depending upon load, speed and drive. However, the pit depth and overhead shall be such as to conform to the requirements of bottom clearance and top clearance in accordance with the accepted standards [D-5A(1)].

2 In case of goods lifts minimum pit depth required is 1 600 mm.

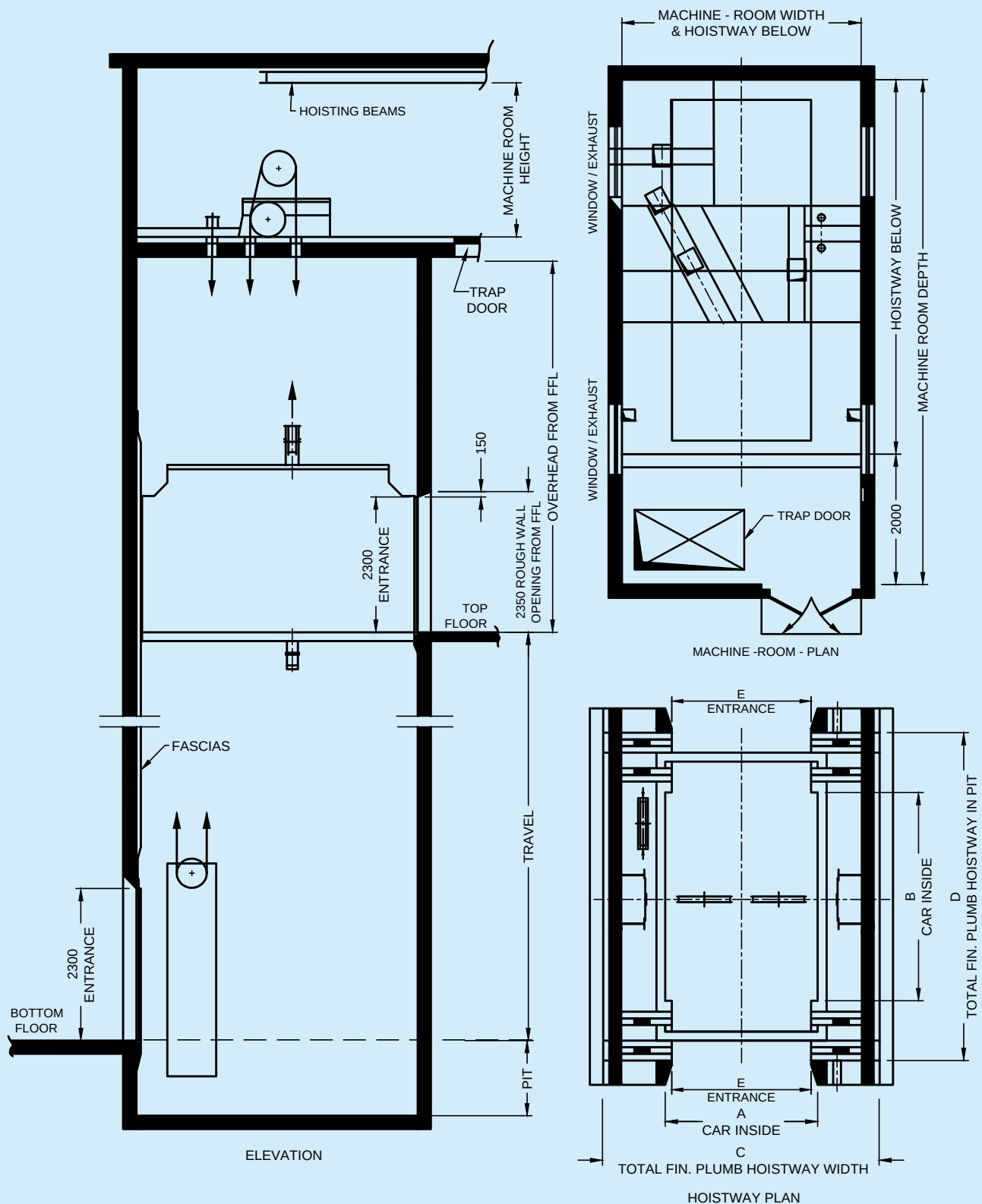
3 In case of lift speeds 3.5 m/s and 4 m/s with rear counterweight the pit depth requirement could be less by about 500 mm to 1 000 mm.

Table 18 Recommended Dimensions of Automobile Lifts with Machine Room-Through Type Car
(Rated Speed = 0.6 m/s or 1.0 m/s)

(Clause [5.10.3.1](#))

All dimensions in millimetres.

Sl No.	Rated Load	Car Size		COPD Door	Shaft Size		Car Height	Overhead Height	Pit Depth	Machine Room Height
	Kg	Width	Depth	Width	Width	Depth	CH	HR	PH	MR
(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)
i)	2 500	2 500	5 000	4P - 2 400	4 000	5 780	2 200	5 200	1 800	4 000
ii)	3 000	2 700	5 400	4P - 2 400	4 000	5 880				
iii)	4 000	3 000	5 800	6P - 3 000	4 300	6 700			2 000	4 500
iv)	5 000	3 000	6 000	6P - 3 000	4 300	6 900				
NOTES										
1 All dimensions given above for lifts having centre opening power operated doors (COPD) with counterweight at side, are recommended dimensions primarily for architects and building planners. Any variations mutually agreed to between the manufacturer and the purchaser are permitted. However, the minimum rated load for the automobile lift shall be based on a load of not less than 1.45 kN/m ² of the net inside car area.										
2 Dimensions of machine room height, pit depth and overhead may differ in practice as per individual manufacturer's design depending upon load, speed and drive. However, the pit depth and overhead shall be such as to conform to the requirements of bottom clearance and top clearance in accordance with the accepted standards [D-5A(6)].										
3 The lift car width should be selected such that in case of emergency while the automobile is in the car-lift the driver should be able to open the automobile door and come out of the automobile.										



All dimensions in mm.

FIG. 6 (TABLE 18) FOR RECOMMENDED DIMENSIONS OF AUTOMOBILE LIFTS

Table 19 Recommended Dimensions of Dumbwaiter Lifts

(Clause [5.10.3.1](#))

All dimensions in millimetres.

Sl No.	Load kg	Car Inside			Lift Well		Entrance E	Machine Room Extension
		A	B	H	C	D		
(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
i)	100	700	700	800	1 200	900	700	D + 1 000
ii)	150	800	800	900	1 300	1 000	800	D + 1 000
iii)	200	900	900	1 000	1 400	1 100	900	D + 1 000
iv)	250	1 000	1 000	1 200	1 500	1 200	1 000	D + 1 000
NOTES								
1 Entrance width δE is based on assumption of provision of vertical bi-parting doors (no car door is normally provided).								
2 D + 1000, where δD is shaft depth.								
3 Head room $\geq 3\ 000$ (with M/C Room) Max speed 0.34 m/s.								
4 Head room MRL $\geq 4\ 050$ to 4 450 (Max speed 0.34 m/s).								

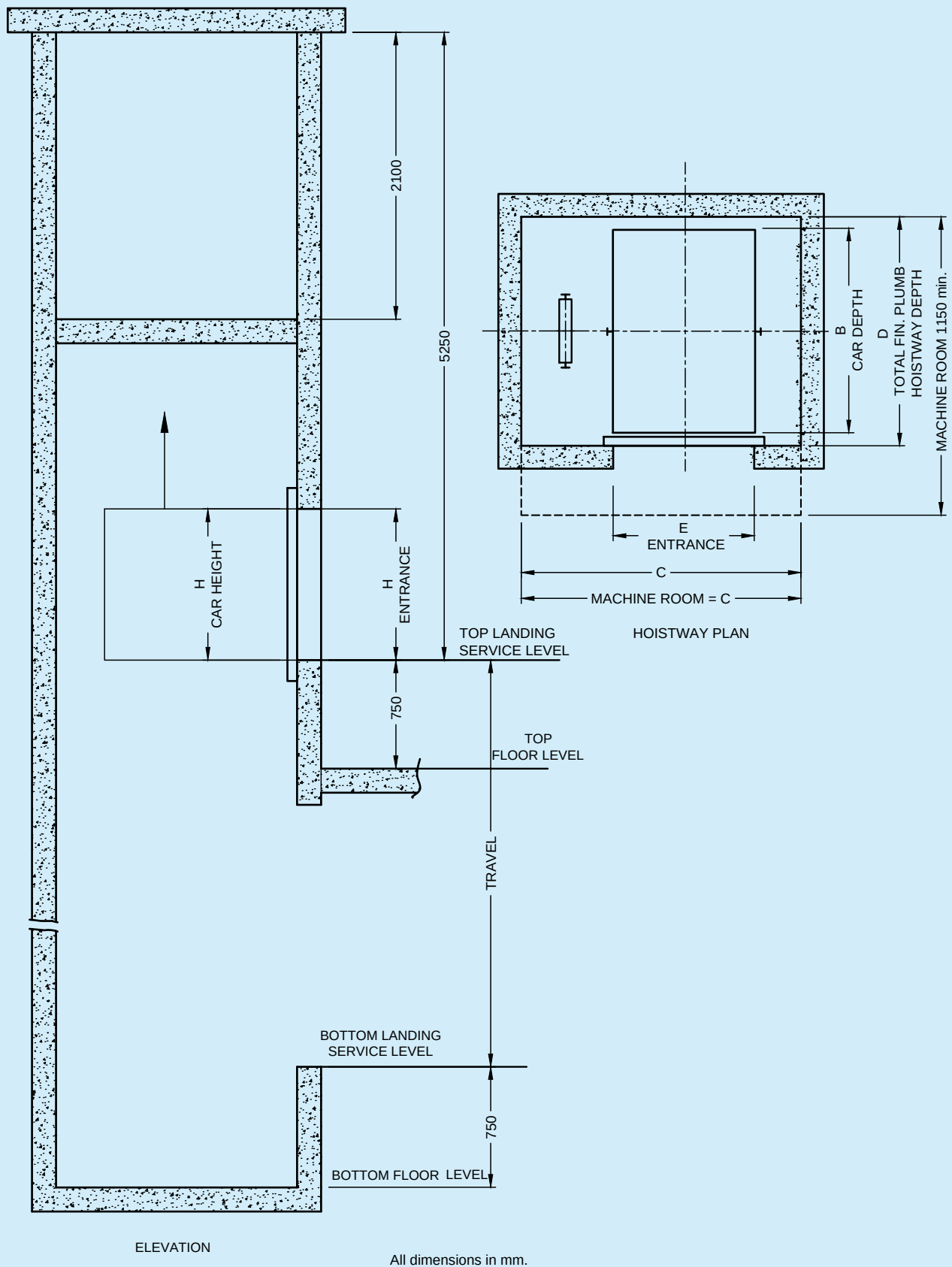


FIG. 7 (TABLE 19) FOR DUMBWAITER LIFTS

Table 20 Entrance Type: Horizontal and Vertical Sliding Doors (Clause 5.10.3.1 , Table 21 , Table 22 , Table 24 , Table 25 and Table 26)					
SI No.	Type	Horizontal Sliding Doors		Vertical Sliding Doors	
		Landing Door	Car Door	Landing Door	Car Door
(1)	(2)	(3)	(4)	(5)	(6)
i)	Type 1	Telescopic door (side opening)	Telescopic door (side opening)	ï	ï
ii)	Type 2	Centre opening door	Centre opening door	ï	ï
iii)	Type 3	Centre telescopic door	Centre telescopic door	ï	ï
iv)	Type 4	ï	ï	Bi-parting (central opening)	Telescopic upwards
v)	Type 5	ï	ï	Bi-parting pass type (central opening, overlapping)	Telescopic upwards
vi)	Type 6	ï	ï	Telescopic upwards	Telescopic upwards

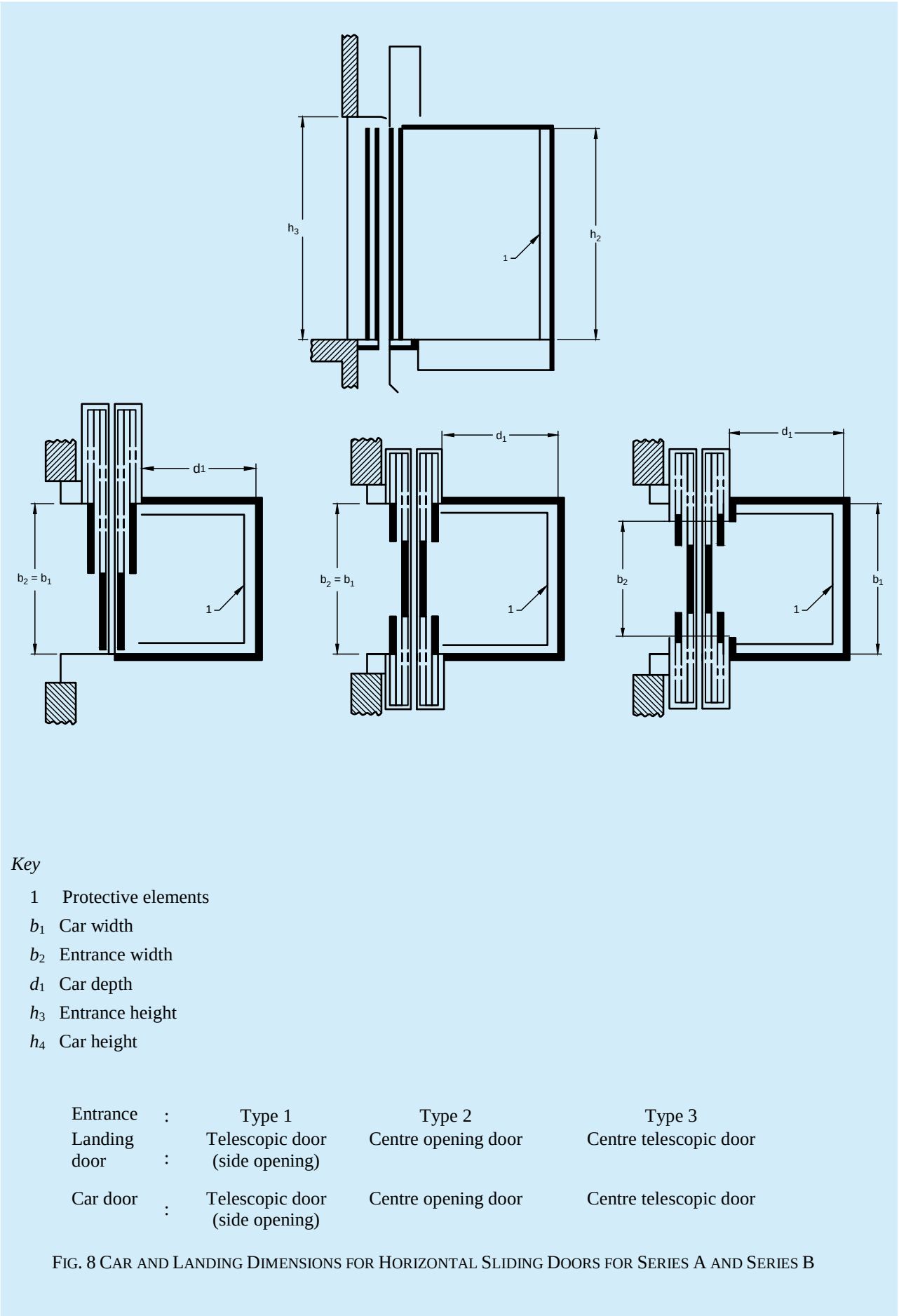
Two types of loading (load per unit area) are addressed:

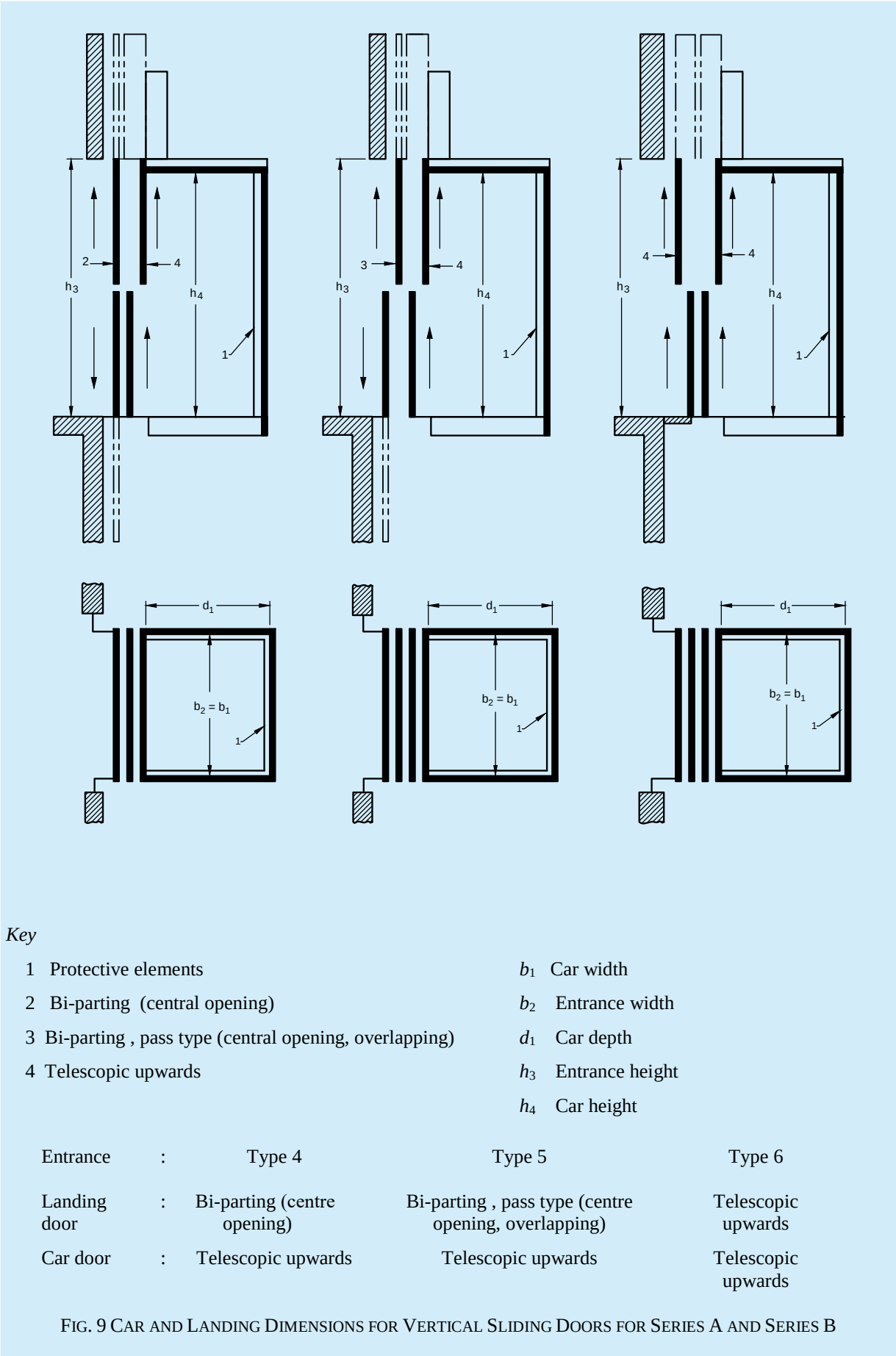
Series A ð Passenger and goods (freight) lifts with loading conforming to [Table 6](#) and [Table 7](#) of accepted standard [D-5A(1)].

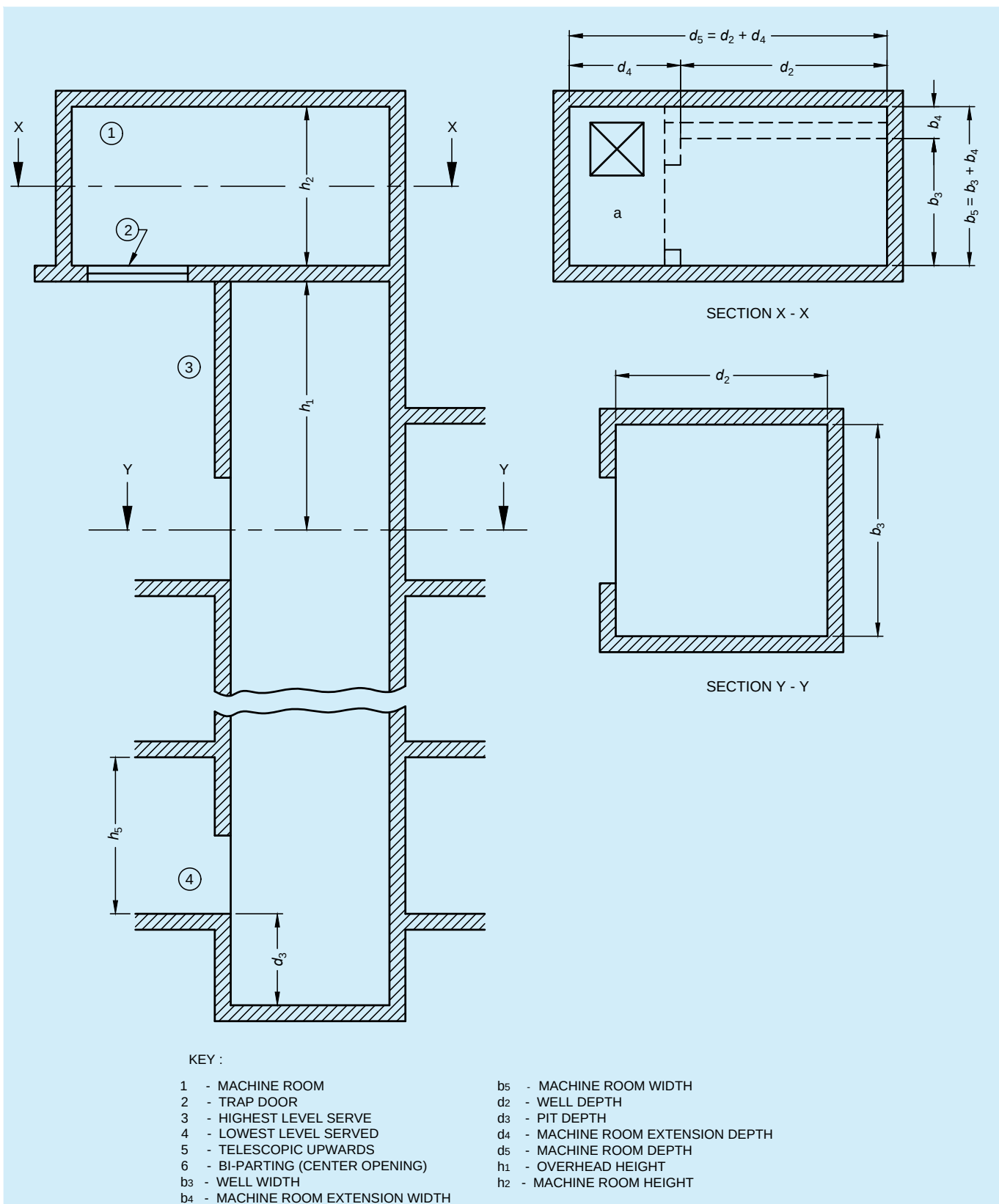
Series B ð Goods (freight) and automobiles lifts which have lesser loading as compared to that for Series A.

The lift well and machine room dimensions of electric lift for Series A and Series B are as shown in [Fig. 10](#), and lift well and machine room dimensions of hydraulic lift for Series A and Series B are as shown in [Fig. 11](#).

Automobile lifts with machine room for through type car Series B with vertical sliding door is shown in [Fig. 12](#).

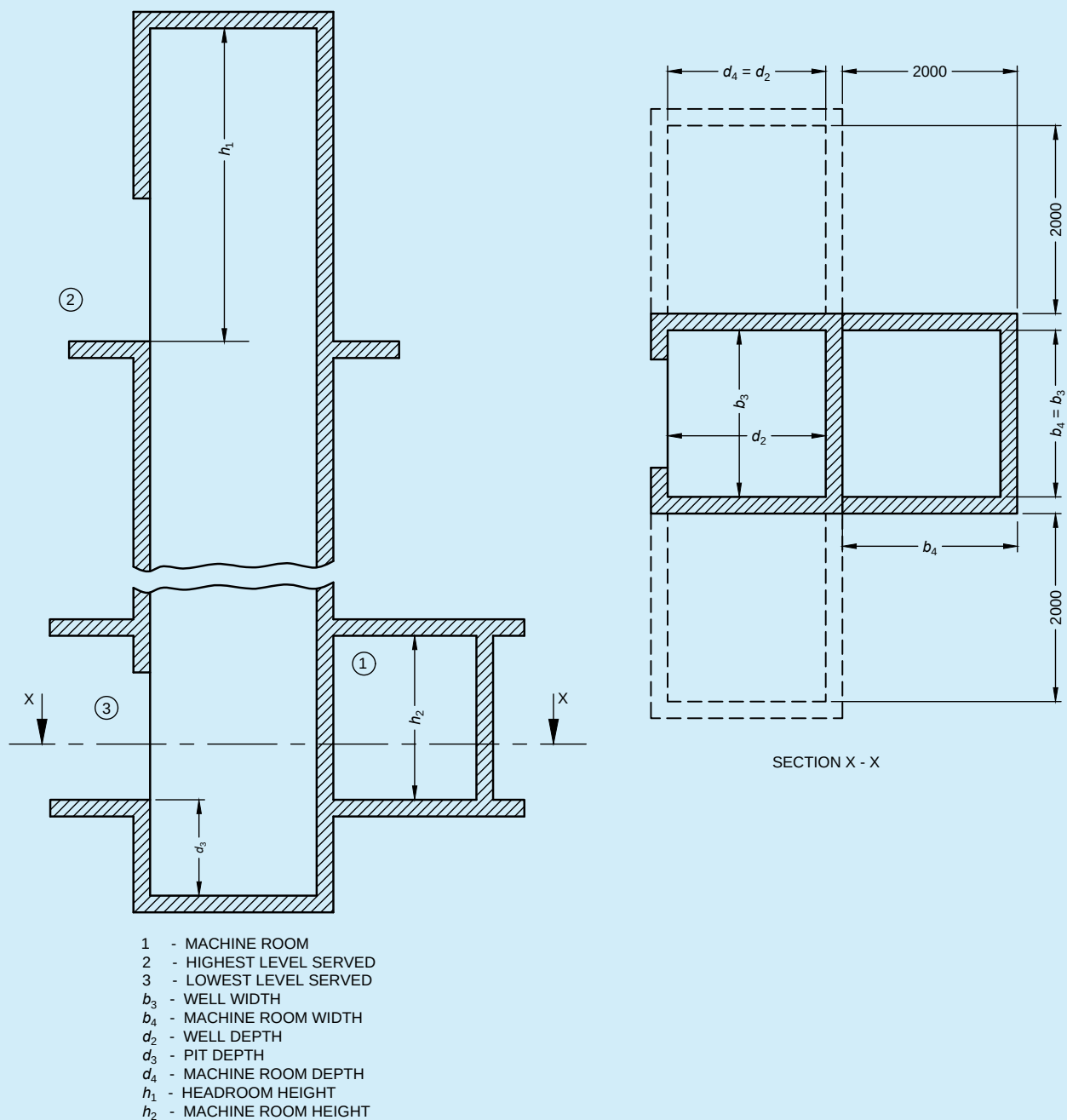






IT IS NECESSARY TO HAVE AN ACCESS DOOR TO THE MACHINE ROOM ALTHOUGH THIS IS NOT INDICATED ON THE SKETCH

FIG. 10 LIFT WELL AND MACHINE ROOM DIMENSIONS: ELECTRIC LIFTS FOR SERIES A AND SERIES B



IT IS NECESSARY TO HAVE AN ACCESS DOOR TO THE MACHINE ROOM ALTHOUGH THIS IS NOT INDICATED ON THE SKETCH

FIG. 11 LIFT WELL AND MACHINE ROOM DIMENSIONS: HYDRAULIC LIFTS FOR SERIES A AND SERIES B

Table 22 Recommended Dimensions of Goods Lifts with Machine Room ð Series A cars ð Vertical Sliding Door (Entrance Type 6)

(Clause [5.10.3.1](#))

All dimensions are in mm.

Series A Car with Vertical Sliding Door

Sl No.	Rated Load	Rated Speed	Car Size		Door	Shaft Size		Car Height	Entrance Height	Overhead Height	Pit Depth	Machine Room Size		Machine Room Height	Min Floor to Floor Height
	kg	m/s	Width (b ₁)	Depth (d ₁)	Width (b ₂)	Width (b ₃)	Depth (d ₂)	CH (h ₄)	EH (h ₃)	HR (h ₁)	PH (d ₃)	Width (b ₅)	Depth (d ₅)	MR (h ₂)	MM (h ₅)
(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(8)	(8)	(9)	(10)	(11)	(12)	(13)	(14)
i)	1 000	0.5 to 1.0	1 400	1 800	1 400	2 400	2 300	2 200	2 000	4 800	1 600	(b ₃ + 1 000)	(d ₂ + 2 500)	2 500	(h ₃ + f)
ii)	1 500		1 700	2 000	1 700	2 700	2 500	2 200	2 000	4 800	1 600			2 500	
iii)	1 600		1 700	2 100	1 700	2 700	2 600	2 200	2 000	4 800	1 600			2 500	
iv)	2 000		1 700	2 500	1 700	2 850	3 000	2 200	2 000	4 800	1 600			2 500	
v)	2 500		2 000	2 500	2 000	3 100	3 000	2 200	2 000	4 800	1 600			2 500	
vi)	3 000	0.5	2 000	3 000	2 000	3 150	3 600	2 200	2 000	4 800	1 800			3 000	
vii)	3 500		2 500	2 550	2 500	3 750	3 100	2 200	2 000	4 800	1 800			3 000	
viii)	4 000		2 500	3 000	2 500	3 750	3 600	2 200	2 000	4 800	1 800			3 000	
ix)	5 000		2 500	3 600	2 500	3 750	4 100	2 200	2 000	4 800	1 800			3 000	

NOTES

1 For width and depth of Machine size dimensions and minimum floor to floor height requirement refer below:

- Machine room Width is (b₃+b₄), where b₃ is width of shaft size and b₄ as 1000 mm to be considered;
- Machine room Depth is (d₂+d₄), where d₂ is depth of shaft size and d₄ as 2500mm to be considered; and
- Minimum floor to floor height (h₅) for landings for vertical sliding door is (h₃ + f), where h₃ is clear entrance height and dimension f depends upon the type of door. Refer Table 20 for vertical sliding door types.

2 In case counterweight safety is applicable check for dimensions with lift manufacturer.

3 Recommended dimensions for pit depth, overhead and machine-room for different lift speeds are given in the table above. These dimensions may differ in practice as per individual manufacturer's design depending upon load, speed and drive. However, the pit depth and overhead shall be such as to conform to the requirements of bottom clearance and top clearance in accordance with the accepted standards [D-5A(1)].

4 All dimensions given above for machine room lifts are recommended dimensions for architects and building planners. Any variations mutually agreed to between the manufacturer and the purchasers are permitted. However, the minimum rated load for the goods lift shall be based on Class of loading for the net inside car area. Refer [8.3.1.1](#) for Class of loading.

(Clause 5.10.3.1)

All dimensions are in millimetres.

Series B Car with Horizontal Sliding Door

Sl No.	Rated Load	Rated Speed	Car Size		Door	Shaft Size		Car Height	Entrance Height	Overhead Height	Pit Depth	Machine Room Size		Machine Room Height	Min Floor to Floor height
	kg	m/s	Width	Depth	Width	Width	Depth	CH	EH	HR	PH	Width	Depth	MR	MM
			(b ₁)	(d ₁)	(b ₂)	(b ₃)	(d ₂)	(h ₄)	(h ₃)	(h ₁)	(d ₃)	(b ₅)	(d ₅)	(h ₂)	(h ₅)
(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)	(13)	(14)	(15)	(16)
i)	2 000	0.5 to 1.0	1 800	3 100	1 250	2 700	3 750	2 500	2 100	4 800	1 800	b ₃ + 1 200	d ₂ + 2 000	3 000	(h ₃ + 750)
ii)	2 500		2 100	3 300	1 500	3 000	3 950	2 500	2 100	4 800	1 800	b ₃ + 1 200	d ₂ + 2 000	3 000	
iii)	3 500	0.25 to 0.5	2 400	3 600	1 750	3 300	4 250	2 500	2 100	5 000	2 000	b ₃ + 1 200	d ₂ + 2 000	3 000	
iv)	5 000		3 000	3 600	2 200	4 000	4 250	2 500	2 100	5 000	2 000	b ₃ + 1 200	d ₂ + 2 000	3 000	

NOTES

1 Minimum floor to floor height (h₅) for landings on same side for horizontal sliding door is (h₃ + 750), where d₃ is clear entrance height.

2 In case counterweight safety is applicable check for dimensions with lift manufacturer.

3 Recommended dimensions for pit depth, overhead and machine-room for different lift speeds are given in the table above. These dimensions may differ in practice as per individual manufacturer's design depending upon load, speed and drive. However, the pit depth and overhead shall be such as to conform to the requirements of bottom clearance and top clearance in accordance with the accepted standards [D-5A(1)].

4 All dimensions given above for machine room lifts are recommended dimensions for architects and building planners. Any variations mutually agreed to between the manufacturer and the purchasers are permitted. However, the minimum rated load for the goods lift shall be based on class of loading for the net inside car area. Refer 8.3.1.1 for Class of loading.

Table 24 Recommended Dimensions of Goods Lifts with Machine Room Series B Vertical Sliding Doors (Entrance, Type 4 and Type 5)

(Clause [5.10.3.1](#))

All dimensions are in millimetres.

Series B Car with Vertical Sliding Door															
Sl No.	Rated Load	Rated Speed	Car Size		Door	Shaft Size		Car Height	Entrance Height	Overhead Height	Pit Depth	Machine Room Size		Machine Room Height	Min Floor to Floor Height
	kg	m/s	Width	Depth	Width	Width	Depth	CH	EH	HR	PH	Width	Depth	MR	MM
			(b ₁)	(d ₁)	(b ₂)	(b ₃)	(d ₂)	(h ₄)	(h ₃)	(h ₁)	(d ₃)	(b ₅)	(d ₅)	(h ₂)	(h ₅)
(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)	(13)	(14)	(15)	(16)
i)	2 000	0.5 to 1.0	1 800	3 100	1 800	2 700	3 550	2 500	2 500	5 300	1 800	b ₃ + 1 200	d ₂ + 2 000	3 000	(h ₃ + f)
ii)	2 500		2 100	3 300	2 100	3 000	3 850	2 500	2 500	5 300	1 800	b ₃ + 1 200	d ₂ + 2 000	3 000	
iii)	3 500	0.25 to 0.5	2 400	3 600	2 400	3 300	4 050	2 500	2 500	5 300	2 000	b ₃ + 1 200	d ₂ + 2 000	3 000	
iv)	5 000		3 000	3 600	3 000	4 000	4 050	2 500	2 500	5 300	2 000	b ₃ + 1 200	d ₂ + 2 000	3 000	
NOTES 1 For minimum floor to floor height requirement refer below: Minimum floor to floor height (h₅) for vertical sliding door is (h₃ + f), where f is clear entrance height and dimension f depends upon the type of door. Refer Table 20 for vertical sliding door types. 2 In case counterweight safety is applicable check for dimensions with lift manufacturer. 3 Recommended dimensions for pit depth, overhead and machine-room for different lift speeds are given in the table above. These dimensions may differ in practice as per individual manufacturer’s design depending upon load, speed and drive. However, the pit depth and overhead shall be such as to conform to the requirements of bottom clearance and top clearance in accordance with the accepted standards [D-5A(1)]. 4 All dimensions given above for machine room lifts are recommended dimensions for architects and building planners. Any variations mutually agreed to between the manufacturer and the purchasers are permitted. However, the minimum rated load for the goods lift shall be based on Class of loading for the net inside car area. Refer 8.3.1.1 for Class of loading.															

Table 25 Recommended Dimensions of Goods Lifts with Machine Room Series B Vertical Sliding Doors (Entrance, Type 6)

(Clause [5.10.3.1](#))

All dimensions are in millimetres.

Series B Car with Vertical Sliding Door																	
Sl No.	Rated Load	Rated Speed	Car Size		Car Area	Loadin g	Door	Shaft Size		Car Height	Entrance Height	Overhead Height	Pit Depth	Machine Room Size	Machine Room Height	Min Floor to Floor height	
	kg	m/s	Width	Depth	m ²	kN/m ²	Width	Width	Depth	CH	EH	HR	PH	Width	Depth	MR	MM
			(b ₁)	(d ₁)			(b ₂)	(b ₃)	(d ₂)	(h ₄)	(h ₃)	(h ₁)	(d ₃)	(b ₅)	(d ₅)	(h ₂)	(h ₅)
(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)	(13)	(14)	(15)	(16)	(17)	(18)
i)	2 000	0.5 to 1.0	1 800	3 100	5.580	3.515	1 800	2 750	3 550	2 500	2 500	5 300	1 800	b ₃ + 1 200	d ₂ + 2 000	3 000	(h ₃ + f)
ii)	2 500		2 100	3 300	6.930	3.538	2 100	3 050	3 850	2 500	2 500	5 300	1 800	b ₃ + 1 200	d ₂ + 2 000	3 000	
iii)	3 500	0.25 to 0.5	2 400	3 600	8.640	3.973	2 400	3 350	4 050	2 500	2 500	5 300	2 000	b ₃ + 1 200	d ₂ + 2 000	3 000	
iv)	5 000		3 000	3 600	10.800	4.540	3 000	4 050	4 100	2 500	2 500	5 300	2 000	b ₃ + 1 200	d ₂ + 2 000	3 000	
NOTES																	
1 Minimum floor to floor height (h ₅) for vertical sliding door is (h ₃ + f), where h ₃ is clear entrance height and for dimension f depends upon the type of door. Refer Table 20 for vertical sliding door types.																	
2 In case counterweight safety is applicable check for dimensions with lift manufacturer.																	
3 Recommended dimensions for pit depth, overhead and machine-room for different lift speeds are given in the table above. These dimensions may differ in practice as per individual manufacturer's design depending upon load, speed and drive. However, the pit depth and overhead shall be such as to conform to the requirements of bottom clearance and top clearance in accordance with the accepted standards [D-5A(1)].																	
4 All dimensions given above for machine room lifts are recommended dimensions for architects and building planners. Any variations mutually agreed to between the manufacturer and the purchasers are permitted. However, the minimum rated load for the goods lift shall be based on case of loading for the net inside car area. Refer 8.3.1.1 for class of loading.																	

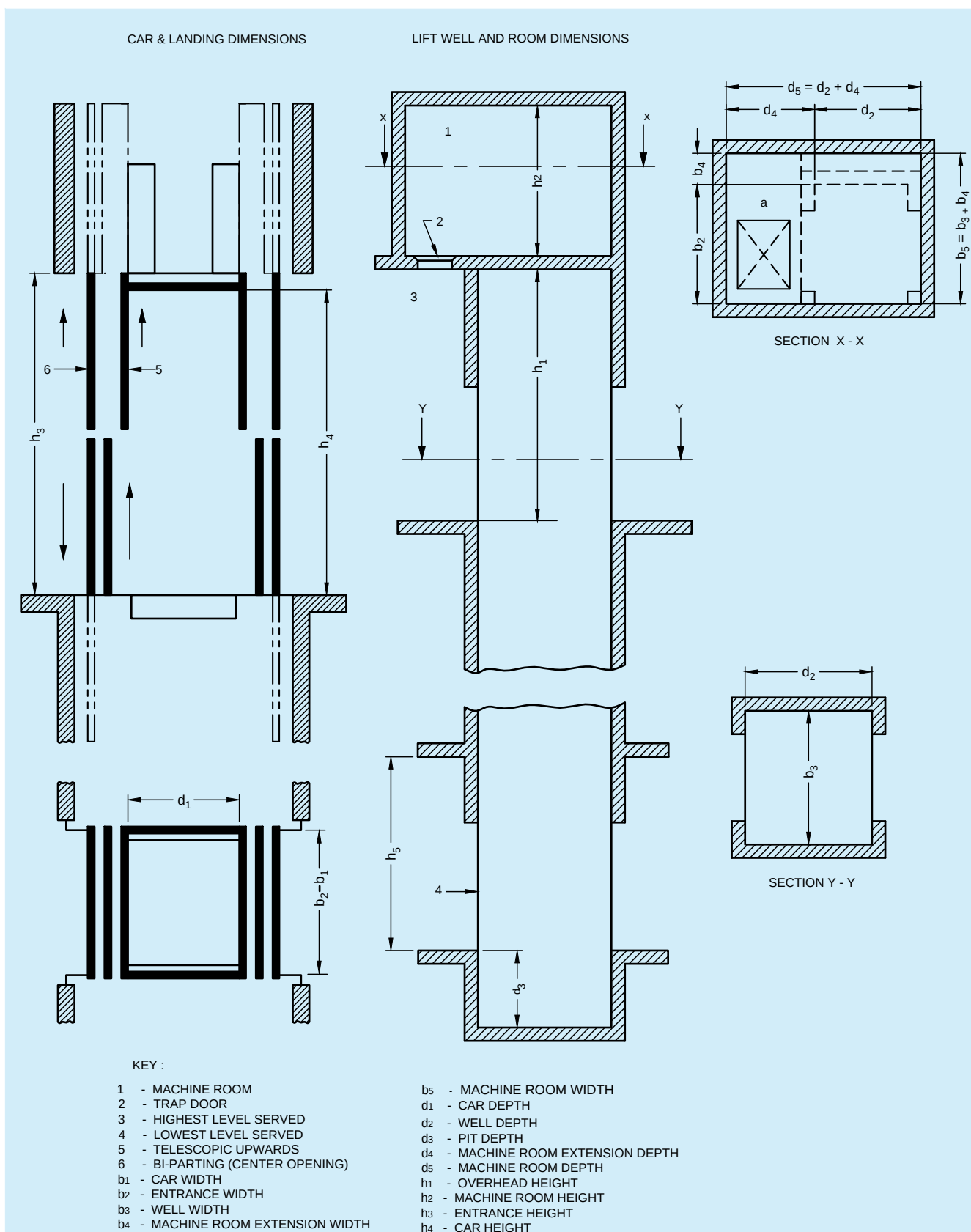


FIG. 12 (TABLE 26) AUTOMOBILE LIFTS WITH MACHINE ROOM -THROUGH TYPE CAR SERIES B VERTICAL SLIDING DOOR

Table 26 Recommended Dimensions of Automobile Lifts with Machine Room ð Through Type Car Series B Vertical Sliding Door (Entrance, Type 4)

(Clause [5.10.3.1](#))

All dimensions are in millimetres.

Sl No.	Rated Load	Rated Speed	Car Size		Door Size	Shaft Size		Car Height	Entrance Height	Overhead Height	Pit Depth	Machine Room Size		Machine Room Height	Below Lintel Height	Min Floor to Floor Height
	kg	m/s	Width (b ₁)	Depth (d ₁)	Width (b ₂)	Width (b ₃)	Depth (d ₂)	CH (h ₄)	EH (h ₃)	HR (h ₁)	PH (d ₃)	Width (b ₅)	Depth (d ₅)	MR (h ₂)		MM (h ₅)
(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)	(13)	(14)	(15)	(16)	(17)
i)	2 500	0.5 and 1.0	2 500	5 300	2 500	3 600	5 910	2200	2 000	5 000	1 600	(b ₃ + 1 000)	(d ₂ + 2 500)	2 500	2 075	(h ₃ + f)
ii)	3 000		2 700	5 400	2 700	3 900	6 010	2200	2 000	5 000	1 800			3 000	2 075	
iii)	4 000		3 000	5 800	3 000	4 200	6 410	2200	2 000	5 000	1 800			3 000	2 075	
iv)	5 000		3 000	6 000	3 000	4 200	6 610	2200	2 000	5 000	1 800			3 000	2 075	

NOTE ð For width and depth of Machine size dimensions requirement refer below,

1 For width and depth of machine size dimensions requirement refer below:

- Machine room Width is (b₃+b₄), where b₃ is width of shaft size and b₄ as 1 000 mm to be considered;
- Machine room Depth is (d₂+d₄), where d₂ is depth of shaft size and d₄ as 2 500 mm to be considered; and
- Minimum floor to floor height (h₅) for vertical sliding door is (h₃ + f), where ð is clear entrance height and Dimension ð depends upon types of door. Refer [Table 20](#) for vertical sliding door types.

2 Car height normal range is 2 200 mm to 2 700 mm.

3 Overhead height for all values of car height CH shown above.

4 In case counterweight safety is applicable check for dimensions with lift manufacturer.

5 All dimensions given above for machine room lifts used for auto mobiles are recommended dimensions for architects and building planners. Any variations mutually agreed to between the manufacturer and the purchasers are permitted. However, the minimum rated load for the automobile lift shall be based on a load of not less than 1.45 kN/m² of the net inside car area.

6 Dimensions of pit depth and overhead may differ in practice as per individual manufacturer's design depending upon load, speed and drive. However, the pit depth and overhead shall be such as to conform to the requirements of bottom clearance and top clearance in accordance with the accepted standards [D-5A(1)].

7 The lift car width should be selected such that in case of emergency while the automobile is in the lift car the driver should be able to open the automobile door and come out of the automobile.

Table 27 Recommended Dimensions of Passenger Lifts MRL COPD

(Clause [5.10.3.1](#))

All dimensions in millimetres.

Sl No.	No. of Passengers	Rated Load	Car Size		2 P COPD Door	1.0 m/s Ò Rated Speed Ò 2.0 m/s		Rated Speed = 2.5 m/s		Rated Speed = 3.0 m/s	
						Shaft Size		Shaft Size		Shaft Size	
		kg	Width	Depth	Width	Width	Depth	Width	Depth	Width	Depth
(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)
i)	4	272	800	950	700	1 600	1 300	ï	ï	ï	ï
ii)	6	408	1 000	1 100	800	2 000	1 550	ï	ï	ï	ï
iii)	8	544	1 300	1 100	800	2 150	1 550	ï	ï	ï	ï
iv)	10	680	1 300	1 350	800	2 250	1 800	2 300	2 150	ï	ï
v)	13	884	1 500	1 450	900	2 250	1 900	2 550	2 150	2 550	2 250
vi)	15	1 020	1 500	1 600	1 000	2 550	2 050	2 650	2 150	2 650	2 250
vii)	16	1 088	1 500	1 650	1 000	2 550	2 050	2 650	2 200	2 650	2 300
viii)	20	1 360	1 700	1 800	1 000	2 550	2 350	2 850	2 350	2 850	2 350
ix)	22	1 496	1 800	1 900	1 000	2 700	2 400	2 800	2 450	2 750	2 450
x)	24	1 632	1 500	2 400	1 000	2 500	2 900	2 750	2 850	2 750	2 850
xi)	26	1 768	1 900	2 000	1 000	2 850	2 700	2 850	2 700	2 850	2 700
xii)	29	1 972	1 900	2 200	1 200	2 900	2 750	3 100	2 200	ï	ï

NOTES

1 All dimensions given above for lifts having centre opening power operated doors (COPD) with counterweight at side, are recommended dimensions primarily for architects and building planners. Any variations mutually agreed to between the manufacturer and the purchaser are permitted. However, variation in:

- a) car inside dimensions shall be within the maximum area limits specified in accordance with [Table 37](#);
- b) entrance width on higher side is permitted; and
- c) entrance width on lower side is permitted up to 100 mm subject to minimum of 700 mm.

2 The minimum size of the lift car and all other requirements relating to accessibility, in all public buildings shall be in accordance with good practice [D-5A(3)] .

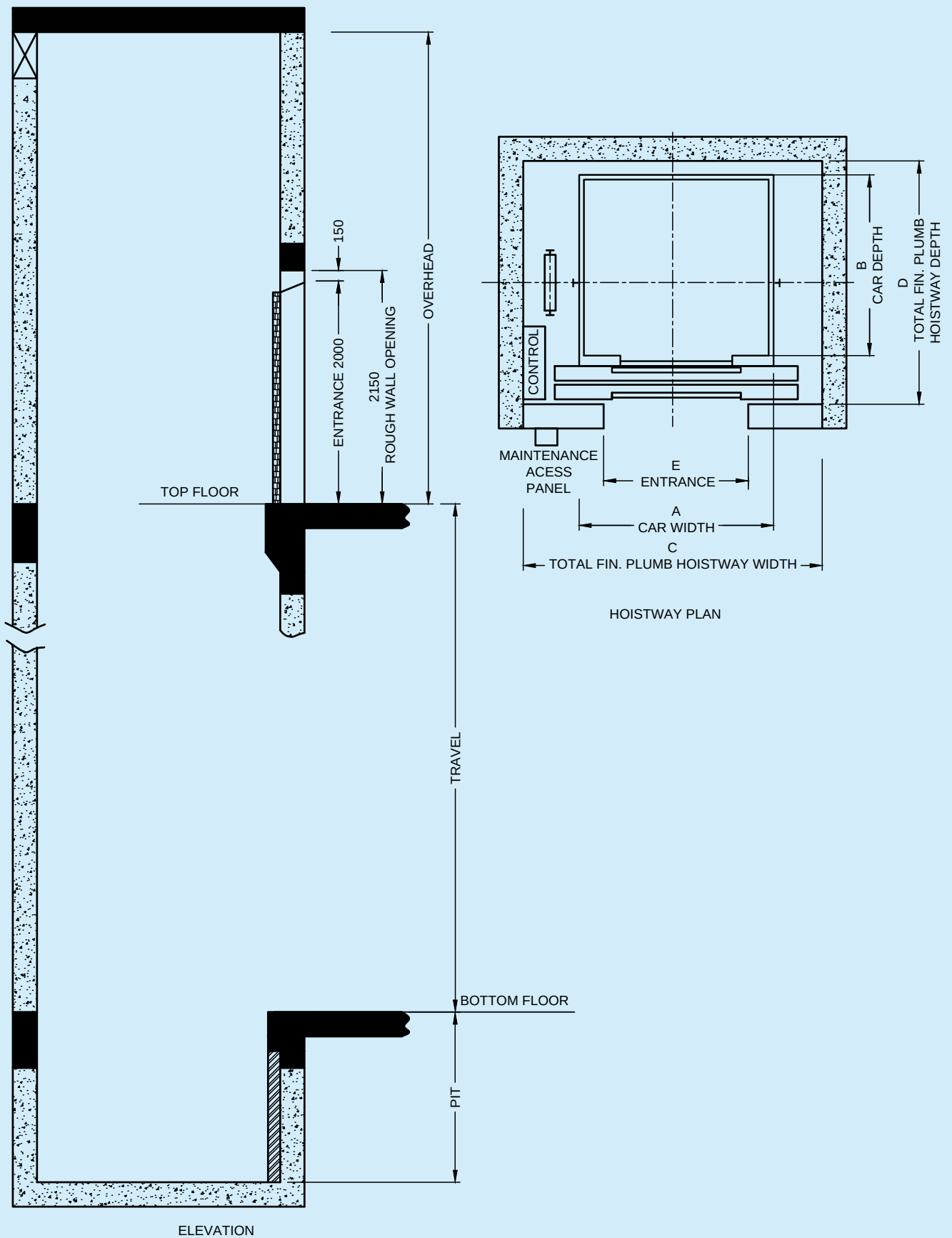


FIG. 13 (TABLE 27) FOR MRL LIFTS COPD

Table 28 Recommended Dimensions of Stretcher/Hospital lifts MRL (2P Telescopic Door/4P COPD, Speed 1.0 m/s to 2.0 m/s)

(Clause 5.10.3.1)

All dimensions in millimetres.

Sl No.	Rated Load		Car Size		Door Size	Shaft Size	
	Persons	kg	Width	Depth		Width	Depth
(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
i)	15	1 020	1 000	2 400	1 000	2 150	2 900
ii)	20	1 360	1 300	2 400	1 000	2 350	2 950
iii)	24	1 632	1 500	2 400	1 000	2 400	2 950
iv)	26	1 768	1 600	2 400	1 000	2 550	3 000
v)	29	2 000	1 500	2 700	1 000	2 450	3 250

NOTES

1 All dimensions given above for lifts having 2 panel telescopic or 4 panel centre opening power operated doors (4P COPD) with counterweight at side, are recommended dimensions primarily for architects and building planners. Any variations mutually agreed to between the manufacturer and the purchaser are permitted. However, variation in:

- a) car inside dimensions shall be within the maximum area limits specified in accordance with Table 37;
- b) entrance width on higher side is permitted; and
- c) entrance width on lower side is permitted up to 100 mm subject to minimum of 700 mm.

2 Car depth of 2 100 mm may be considered in residential/commercial buildings where use of only ambulance stretcher is envisaged. The hoist-way depth may be reduced accordingly.

Table 29 Pit Depth for Passenger/Stretcher/Hospital Lifts MRL

(Clause 5.10.3.1)

All dimensions in millimetres.

Sl No.	No. of Passengers	Pit Depth for Speed					
		1 m/s	1.6 m/s	1.8 m/s	2 m/s	2.5 m/s	3 m/s
(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
i)	4	1 600	1 800	2 000	2 200	ï	ï
ii)	6	1 600	1 800	2 000	2 200	ï	ï
iii)	8	1 600	1 800	2 000	2 200	ï	ï
iv)	10	1 600	1 800	2 000	2 200	ï	ï
v)	13	1 700	1 800	2 000	2 200	2 500	3 100
vi)	15	1 700	1 800	2 000	2 200	2 500	3 100
vii)	16	1 700	1 800	2 000	2 200	2 500	3 100
viii)	20	1 800	1 800	2 000	2 200	2 500	3 100
ix)	22	1 800	1 800	2 000	2 200	2 500	3 100
x)	24	1 800	1 800	2 000	2 200	2 500	3 100
xi)	26	1 800	1 800	2 000	2 200	2 500	3 100
xii)	29	1 800	1 800	2 000	2 200	2 500	ï

NOTE ð Recommended dimensions for pit depth for different lift speeds are given in the table. These dimensions may differ in practice as per individual manufacturer's design depending upon load, speed and drive. However, the pit depth shall be such as to conform to the requirements of bottom clearance and top clearance in accordance with the accented standards IS-5A(6).

Table 30 Overhead Height for Passenger and Stretcher/Hospital Lifts MRL

(Clause [5.10.3.1](#))

All dimensions in millimetres.

Sl No.	No. of Passengers	Overhead Height for Speed				
		1 m/s	1.5 m/s $\hat{O}$ Rated Speed $\hat{O}$ 1.75 m/s	2 m/s	2.5 m/s	3 m/s
(1)	(2)	(3)	(4)		(5)	(6)
i)	4	CH + 1 900	CH + 2 400	ï	ï	ï
ii)	6	CH + 1 900	CH + 2 400	ï	ï	ï
iii)	8	CH + 1 900	CH + 2 400	CH + 2 600	ï	ï
iv)	10	CH + 1 900	CH + 2 400	CH + 2 600	CH + 3 200	ï
v)	13	CH + 1 900	CH + 2 400	CH + 2 600	CH + 3 500	CH + 3 700
vi)	15	CH + 1 900	CH + 2 400	CH + 2 600	CH + 3 600	CH + 3 800
vii)	16	CH + 1 900	CH + 2 400	CH + 2 600	CH + 3 700	CH + 3 900
viii)	20	CH + 2 300	CH + 2 600	CH + 2 700	CH + 3 800	CH + 4 000
ix)	22	CH + 2 300	CH + 2 600	CH + 2 700	CH + 3 800	CH + 4 000
x)	24	CH + 2 500	CH + 2 600	CH + 3 200	CH + 4 100	CH + 4 300
xi)	26	CH + 2 600	CH + 2 600	CH + 3 250	CH + 4 250	CH + 4 450
xii)	29	CH + 2 600	CH + 2 600	ï	ï	ï

NOTE ð Recommended dimensions for overhead height for different lift speeds are given in the table, where CH is the clear car height from car finished floor to car roof. These dimensions may differ in practice as per individual manufacturer's design depending upon load, speed and drive. However, the overhead shall be such as to conform to the requirements of bottom clearance and top clearance in accordance with the accepted standards [D-5A(6)].

(Clause 5.10.3.1)

All dimensions in millimetres.

Sl No.	Rated Load	Car Size		Door Size	Telescopic Door		4P COPD		2P COPD		Overhead Height	Pit Depth
					Shaft Size		Shaft Size		Shaft Size			
		kg	Width	Depth	Width	Width	Depth	Width	Depth	Width	Depth	HR
(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)	(13)
i)	500	1 100	1 200	900	ï	ï	ï	ï	2150	1650	CH + 1900	1 600
ii)	1 000	1 400	1 800	1 100	ï	ï	ï	ï	2 600	2 200	CH + 1900	1 600
iii)	1 500	1 700	2 000	1 500	ï	ï	2 650	2 550	ï	ï	CH + 2400	1 800
iv)	1 600	1 400	2 400	1 300	2 700	3 000	3 200	3 050	ï	ï	CH + 2700	1 800
v)	2 000	1 700	2 500	1 500	2 550	3 300	2 600	3 300	ï	ï	CH + 2700	1 800
vi)	2 500	2 000	2 500	1 800	3 150	3 050	3 200	3 050	ï	ï	CH + 2700	1 800
vii)	3 000	2 000	2 700	1 800	ï	ï	3 350	3 300	ï	ï	CH + 3000	1 800
viii)	4 000	2 100	3 400	1 800	ï	ï	3 400	4 000	ï	ï	CH + 3100	2 100
ix)	5 000	2 500	3 500	2 200	ï	ï	3 800	4 100	ï	ï	CH + 3200	2 200
NOTES												
1 CH is the clear car height from car finished floor level to car roof (goods lifts will normally not have any false ceiling).												
2 Normal range for CH is 2 100 mm to 2400 mm.												
3 Door height is less than or equal to the car height CH.												
4 In case counterweight safety is applicable check for dimensions with lift manufacturer.												
5 All dimensions given above for lifts having telescopic or centre opening power operated doors (COPD) with counterweight at side, are recommended dimensions primarily for architects and building planners. Any variations mutually agreed to between the manufacturer and the purchaser are permitted. However, the minimum rated load for the goods lift shall be based on a load of not less than 3.45 kN/m ² of the net inside car area.												
6 Recommended dimensions for overhead height and pit depth for different lift speeds are given in the table. These dimensions may differ in practice as per individual manufacturer's design depending upon load, speed and drive. However, the overhead height and pit depth shall be such as to conform to the requirements of bottom clearance and top clearance in accordance with the accepted standards [D-5A(1)].												
7 Car inside dimension shall be within maximum area limit specified in accordance with Table 38.												

All dimensions in millimetres.

Sl No.	Rated Load	Car Size		6P COPD Door Size: Opposite Entrances	Shaft Size		Overhead Height	Pit Depth
	Kg	Width	Depth	Width	Width	Depth	HR	PH
(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
i)	2 500	2 500	5 000	4P-2 500	3 800	5 480	5 000	2 500
ii)	3 000	2 500	5 500	2 200	3 800	6 450	5 000	2 500
iii)	4 000	3 000	6 000	2 700	4 300	6 950	5 000	2 500
NOTES								
1 Car height normal range is 2 100 mm to 2700 mm.								
2 Overhead height shown is required for all values of car height CH shown above.								
3 In case counterweight safety is applicable check for dimensions with lift manufacturer.								
4 All dimensions given above for machine room less lifts with 6 panel centre opening power doors (6P COPD) used for automobiles are recommended dimensions for architects and building planners. Any variations mutually agreed to between the manufacturer and the purchaser are permitted. However, the minimum rated load for the automobile lift shall be based on a load of not less than 1.45 kN/m ² of the net inside car area.								
5 Dimensions of pit depth and overhead height may differ in practice as per individual manufacturer's design depending upon load, speed and drive. However, the pit depth and overhead height shall be such as to conform to the requirements of bottom clearance and top clearance in accordance with the accepted standards [D-5A(1)].								
6 The lift car width should be selected such that, in case of emergency while the automobile is in the lift car the driver should be able to open the automobile door and come out of the automobile.								

6 CIVIL AND ELECTRICAL REQUIREMENTS FOR LIFTS

6.1 Civil Requirements

Lift well enclosures, lift pits, machine rooms and machine supports besides conforming to the essential requirements given below should form part of the building construction and comply with the lift manufacturer's drawings.

6.1.1 Requirements of Lift Well Enclosure

The requirements of lift well enclosures are as given hereunder:

- a) *Totally enclosed well* ð Lift wells shall be totally enclosed by imperforate walls, floor and ceiling, constructed for protection against all weather and of dust free surface material or should be painted to minimize dust circulation on to moving apparatus and from being pumped by the car movement into machine room or on to landings and shall be rendered minimum 120 min fire-resistant.

The lift well may have openings for landing doors, inspection and emergency doors, vent openings for escape of gases and smoke in case of fire, ventilation openings, openings in the machine room floor for connecting lift components such as ropes, cables, etc.

- b) *Partially enclosed well* ð In case of partially enclosed lift wells used for applications, for example, observation lifts in connection with galleries or atriums, tower buildings, etc, the well does not need to be totally enclosed, provided the height of the enclosure at places normally accessible to persons is sufficient to prevent such persons:
- 1) being endangered by moving parts of the lift; and
 - 2) interfering with the safe operation of the lift by reaching lift equipment within the well either directly or with handheld objects.

The height of the enclosure shall be minimum 3.5 m at the landing door side. The height of enclosure at other sides shall be 2.5 m minimum when the horizontal distance between the moving parts of the lift and enclosure is 0.5 m minimum. The height of enclosure at other sides may be reduced to 1.1 m progressively if the horizontal distance between the moving parts of the lift and enclosure increases to 2 m. The enclosure shall be located within 150 mm of the edge of the floor. The enclosure shall be imperforate and made of non-combustible material.

- c) *General requirements related to the lift well* ð The well shall be exclusively used for lifts. It shall not contain ducts, cables, or devices, etc, other than those for the lift. Permanently installed adequate lighting shall be provided comprising sufficient number of lamps fixed throughout the lift well such that the illumination level 1 m above the car roof within its vertical projection at any position of the car in the well and 1 m above pit floor everywhere shall be at least 50 lux and at least 20 lux at remaining locations, even when all the doors are closed (so that a person can stand, work and/or move between the working areas).

NOTE ð Additional lamps may be fixed on the car top as a part of the well lighting system to achieve this.

Well lighting switches (or equivalent) shall be located both in the pit and in machine room or in controller (in case of machine room less application) so that the complete well lighting can be operated from either location that is, it shall be an arrangement of 2-way switch in machine room/controller (in case of MRL) and pit for controlling well lighting. Well lighting switch shall be in the machine room close to the main switch and pit lighting switch in lift well close to and easily accessible from lowest landing door. Each point of well lighting shall consist of bulkhead with bulb, 3-pin socket and switches to control supply.

Should a lift entrance open out into an area exposed to the weather, the entrance should be protected by a suitable canopy and the ground level sloped up to the lift entrance to prevent rain or drainage water entering the lift well through the clearances around the landing doors. Any push buttons exposed should be of weatherproof type.

The natural or artificial lighting of the landings in the vicinity of landing doors shall be at least 50 lux at floor level, such that a user can see ahead when they are opening the landing door to enter the lift, even

if the car light has failed. The approach of the landing door on each floor shall be kept lit during the whole time the lift is available for use at night, and during the day time, if so required, due to insufficient natural light.

When the distance between consecutive landing door sills exceeds 11 000 mm, intermediate emergency landing(s) with lift landing door(s) shall be provided such that the distance between any two consecutive landings is not more than 11 000 mm. Rescue to these landings is permissible in case of automatic rescue device operation.

No counterweight shall be allowed to travel in any lift well, or part of any lift well other than that to which it belongs.

In the case of a lift well which is common to more than one lift and where the lift car or the counterweight of one lift is working in juxtaposition to the lift car or counterweight of another lift, such lift cars or counterweights shall be guarded carefully and adequately in order to protect persons working in the lift well or on the lift cars from accidental contact with such cars or counterweights in any part of their travel.

- d) *Construction of the lift well* ð The side walls of the lift well may be made of reinforced cement concrete at least 150 mm thick; or brick or similar fire resisting materials provided tie beams are made available for lift guide rail fixings at such heights as specified by the lift manufacturer, to provide satisfactory anchoring arrangement for fixing guide rails and other equipment to the lift well.

The inner surface of the lift well and its enclosure facing any lift-car entrance shall, so far as practicable, be kept smooth and flush devoid of projections or recesses. Where any projection or top of the recesses cannot be rendered flush, they shall be levelled on the underside to an angle not less than 60° from the horizontal, by means of metal plates, cement rendering or other fire resisting material.

Where a lift car levelling device is operative with the lift car door open, such interior surfaces shall always form a smooth and flush surface below each landing level for a depth of at least the depth of the car-levelling zone plus the distance through which the lift car may travel of its own momentum when the power is cut-off.

The horizontal distance between the sill of the car and sill of the landing doors shall not exceed 35 mm. The distance between the lift well enclosure on the sides facing any lift-car entrance and the sill edge of the car shall not be more than 35 mm in the landing zone below the landing gate. If such distance is more than 35 mm in the lift well enclosure, the same shall be finished with suitable and smooth plaster work or fascia plates so as to make the surface thereof devoid of all projections and recesses. In case the enclosure wall on the sides facing the lift-car entrance is more than 130 mm from the sill edge of the lift-car platform, the lift-car door of such lift shall be provided with means to prevent it from being opened except when the lift-car is at the landing served by such car entrance.

Sufficient space shall be provided between the guides for the car and the side walls of the lift well enclosure to allow safe and easy access to the parts of the safety gears for their maintenance and repairs.

All landing openings shall be protected by gates or doors which shall extend to the full height and full width of the landing openings. These openings shall not be less than 2 000 mm in clear height and 700 mm clear in width (see [5.10.3.1](#)) when the gates or doors are fully opened. There shall not be any common wall between lift well and water tank.

- e) *Strength of the walls of the well* ð The structure of the well shall be able to support at least the loads which may be applied by the machine, by the guide rails at the moment of safety gear operation, by eccentric load in the car, by the action of buffers, by the application of anti-rebound devices, by loading and unloading the car, etc.

The walls of the well shall withstand a force of 1 000 N, being evenly distributed over an area of 0.09 m² in round or square section and applied at right angles to the wall at any point on either face,

- 1) without permanent deformation greater than 1 mm; and
- 2) without elastic deformation greater than 15 mm.

Glass panels, plane or formed shall be made of laminated glass. Such panels and their fixings shall withstand horizontal static force of 1 000 N applied on an area of 0.09 m² at any point, from either side, without permanent deformation.

- f) *Partition walls between lifts in a common lift well* δ In case of common lift well for multiple lifts, the partition wall(s) of the common lift well may be either be of RCC or brick or steel or glass which may be provided with openings for escape of air displaced due to movement of lift in the adjacent well. For high speed lifts the partition between the lifts maybe of suitable wire grill/expanded metal of following specifications:
- 1) It shall be made of at least 2.2 mm thick steel wire or expanded metal;
 - 2) A ball of 25 mm diameter shall not pass through the wire grill; and
 - 3) It shall be so fixed as not to deflect more than 15 mm when subjected to a force of 450 N applied horizontally at any point from either side.

6.1.2 Requirements of the Pit

Pit shall be soundly constructed and maintained in a dry (waterproofed) and clean condition. Where necessary, provision shall be made for permanent drainage. The floor of the pit shall be able to support beneath each guide rail, force due to mass of guide rails plus the reaction due to safety gear operation. The floor of the pit shall be able to support beneath the car buffer supports, force in Newton, equal to 4 times the static load being imposed by the mass of fully loaded car:

$$4.g (P + Q)$$

where

g = acceleration due to gravity;

P = mass of empty car and components supported by the car that is, part of travelling cable, compensating rope/chain, if any, etc, in kg; and

Q = rated load, in kg.

The floor of the pit shall be able to support beneath the counterweight buffer support, a force in Newton, equal to 4 times the static load being imposed by the mass of the counterweight:

$$4.g (P + q.Q)$$

where

g = acceleration due to gravity;

P = mass of empty car and components supported by the car, that is, part of travelling cable, compensating rope/chain, if any, etc, in kg;

q = live load balancing factor (normally, $q = 0.5$); and

Q = rated load, in kg;

The floor of the pit shall be able to support upward thrust of the tied down rope compensation device, if provided. If accessible spaces do exist below the well, the base of the pit shall be designed for an imposed load of at least 5 000 N/m² and the counterweight shall be equipped with safety gear. In addition, it is recommended to provide double slab for the lift pit.

NOTES

1 Lift wells should preferably not be situated above a space accessible to persons.

2 Enclosing projected area of the lift well under the pit in accessible spaces will not nullify the above provisions.

There shall not be any common wall/slab between lift pit and any water reservoir.

The pit shall be accessible by an access door. A permanent ladder inside the well may be provided close to the

last landing *in-lieu* of access door, if the pit depth does not exceed 2.5 m.

NOTE δ In case of structural limitations for providing pit access door at pit floor level, alternative arrangement to access pit may be provided which shall allow access to pit with complete safety.

A physical means of protection shall be provided to prevent accidental contact with the counterweight.

The pit shall be equipped with a stop switch which is easily accessible from the lowest landing and the pit floor. If more than one switch is required, they shall be wired in series.

6.1.3 Requirements Associated with Machine Room and Pulley Room

It must be noted that generally lifts have machine rooms immediately over the lift well and this should be arranged, whenever possible, without restricting the overhead distance required for normal safety precautions. In case where machine room provision on top is a limitation, either machine room less lift or basement drive or side drive lift can be considered. The spaces and tolerances shall be in accordance with the following:

a) *Requirements associated with machine room and pulley room (where lift machinery is placed in the machine room)*

- 1) The lift machine, controller and all other apparatus and equipment of a lift installation, except such apparatus and equipment having functional location in the lift well or other positions, shall be placed in the machine room which shall be adequately lit and rendered fire-resistant and weather-proof.

If parts of different lifts are present in one machine room and/or pulley room each lift shall be identified with a number, letter or colour consistently used for all parts (machine, controller, over speed governor, switches, etc). If the machine room contains several lifts, it is necessary to have one switch per car to control the supply to the circuit for lighting and socket outlets of the car. This switch shall be located close to the corresponding main power switch.

- 2) Machine room should be considered as plant space, and conditions provided to permit reliable operation of electrical switchgear and rotating machinery and be conducive to good maintenance. The machine room shall be either air conditioned or adequately ventilated so as to maintain the ambient temperature therein between + 5 °C and + 40 °C. The machine room shall be such that the motors and equipment as well as electric cables, etc are protected as far as possible from dust and humidity. Machine room should also be protected from all weather and if ventilation louvers are provided, they should be designed and located to prevent entry of rainwater in the machine room. Preferably, metal or wooden louvers should be provided.

The well, machinery spaces and pulley rooms shall not be used to provide ventilation of rooms other than those belonging to the lift.

- 3) Rescue instructions with required tools and tackles, if any, shall be made available in the machine room.
- 4) Machine room floor shall be provided with a trap door, if necessary. The size of the trap door shall be as per manufacturer's recommendation. Since the purpose of the trap door is to lift or lower the machinery, controller, etc between machine room and the top landing, it may be necessary to provide trap door of same size in the terrace slab which coincides with trap door in machine room, depending on the design of the building.

The trap door when closed shall be able to support 2 000 N on an area of 0.20 m × 0.20 m at any position. The trap door shall not open downwards. Hinges, if any, shall be of a type which cannot be unhooked. Trap doors shall be locked. When the trap door is in the open position, precautions shall be taken to prevent the fall of persons (for example, by a guardrail) and prevent the trapdoor from closing such as to cause a crushing hazard (for example, by counterbalance).

- 5) Lighting shall be provided to give at least 200 lux around the controller and machine. At least one plug socket point with button shall be provided in the machine room for every lift. The switch for the machine room light shall be fixed near the entrance of the machine room. The machine room walls, ceiling and floor should be finished in dust-resisting materials, tiles, etc or painted as a

minimum to stop dust circulation, which otherwise could damage rotating machinery and cause failure of switchgear.

- 6) The machine room shall be so designed as to allow free and easy access to all parts of the equipment and the width of the clear space around the machine from any two sides shall in no case be less than 600 mm. Provision shall be made to allow the removal and replacement of various units. The entrance of the machine room shall be minimum 1 000 mm wide and 2 000 mm high to allow removal and replacement of machinery therein. The floor of the machine room shall be designed and constructed to carry safely at any point the heaviest part of the equipment and withstand the loads and forces to which they are intended to be subjected to. If the floor does not extend to the enclosing wall, the open sides shall be adequately guarded by suitable means for example, rails, guards, etc.
- 7) The height of the machine room shall be sufficient to allow any portion of the equipment to be accessible and removable for repairs and replacement. Distance between the machine lifting point and the machine room hook shall be at least 1 000 mm and there shall be a clear vertical distance of at least 300 mm above the rotating parts of the machine. Machine room height shall not be less than 2 100 mm clear from the floor of the access area and working area in any case. This clear height for movement is taken to the underside of the lowest striking point and measured from the floor of the access area.
- 8) The machine room shall be locked and shall be accessible to only those who are concerned with the operation and maintenance of the machinery or equipment. Machine room doors shall not open towards the inside of machine room. They shall be provided with a key-operated lock, capable of being reclosed and relocked without a key and be openable from inside the machine room without a key, even when locked. Approach way to machine room from top landing level shall be direct, easy and safe to walk by way of stairs. If stairs are not possible, sufficiently wide and deep permanent structural steps may be used. These structural steps shall be provided with adequate hand holds and shall not be too steep. Minimum width of the steps shall be 1 000 mm.
- 9) A danger notice board shall be displayed permanently on the outside of the machine room door, trap door and near the machinery. Refer [8.1\(e\)](#) for the warning signs.
- 10) There shall not be any common wall/slab between machine room and any water reservoir.
- 11) The machine room shall not be used as a storeroom or for any purpose other than housing the machinery/components connected with the lift installation. The machine room shall not act as a passage to any other room or utility.

The lift well, machine and pulley rooms may, however contain:

- i) Equipment for air conditioning or heating of these spaces, excluding steam heating and high-pressure water heating. However, any control and adjustment devices of the heating apparatus shall be located outside the lift well, machine room and pulley room; and
- ii) Fire detectors and/or suppression system, with a high operating temperature (for example, above 80 °C), appropriate for the electrical equipment and suitably protected against accidental impact.

For requirements of sprinkler, when sprinkler systems are used, refer **9.24.4** of good practice [D-5A(7)].

NOTE δ Such smoke, fire detection and sprinkler systems are the responsibility of the building management.

- 12) Machine room floor shall be designed to carry an imposed (live) load of not less than 10 kN/m² over the whole area and any load which may be imposed thereon by the equipment used in the machine room or by any reaction from any such equipment both during periods of normal operation, maintenance and repair.
- 13) One or more beams or hooks with the indication of the safe working load, as appropriate, shall be provided in the machinery spaces. Position of the beams or hooks shall be according to the layout

provided by lift manufacturer. Hoisting beams or hooks shall be tested for safe working load (SWL) before indicating the lifting capacity on it.

- 14) The total load on overhead beams used for supporting machinery shall be assumed as equal to all equipment resting on the beams plus twice the maximum load suspended from the beams.
- 15) The factor of safety for all overhead beams and supports based on ultimate strength of the material and load in accordance with (14) shall not be less than the following:
 - i) For steel : 5
 - ii) For reinforced concrete : 7

The deflection of the overhead beams under the maximum static load calculated in accordance with above shall not exceed 1/1 500 of the span.

- 16) The place in which the overhead pulleys, over speed governors and similar machinery are fixed shall be easily accessible for maintenance and repair purposes. It shall be lit adequately and safe working procedure shall be established for maintenance and repair works.
- 17) It is recommended that emergency exit is provided in case of large machine rooms having four or more lifts.
- 18) Where the machine room occupies a prominent position on roof of a building, provision should be made for lightning protection in accordance Part D of Building Services, Section 2 Electrical and Allied Installations of this NBCS and applicable standards.

b) *Requirements associated with machine spaces (where machinery is placed in the well)*

- 1) Access to the interior of the machinery and pulley spaces shall be capable of being properly lit by a permanent electric light fixture(s) with intensity of at least 200 lux at floor level, easy to use, and completely safe in all circumstances without necessitating entry into private premises.
- 2) Machinery spaces inside the well shall be designed and constructed to withstand the loads and forces they are subjected to.
- 3) In the case of partially enclosed wells located at the exterior of the building, the machinery shall be suitably protected against environmental influences.
- 4) Provision should be given to access the machine for maintenance purpose either from car top or otherwise.
- 5) The clear height for movement shall not be less than 2 000 mm.
- 6) The machinery spaces shall be suitably ventilated. The electric equipment of the machinery shall be protected as far as it is reasonably practicable from dust, harmful fumes and humidity.
- 7) The working area and machinery spaces shall be provided with permanently installed electric lighting with at least 200 lux at floor level.
- 8) One or more beams or hooks with the indication of the safe working load, as appropriate, shall be provided in the machinery spaces. Position of the beams or hooks shall be according to the layout provided by lift manufacturer. Hoisting beams or hooks shall be tested for safe working load (SWL) before indicating the lifting capacity on it.
- 9) The total load on overhead beams used for supporting machinery shall be assumed as equal to all equipment resting on the beams plus twice the maximum load suspended from the beams.
- 10) The factor of safety for all overhead beams and supports based on ultimate strength of the material and load in accordance with (9) shall be not less than the following:
 - i) For steel : 5

- ii) For reinforced concrete : 7

The deflection of the overhead beams under the maximum static load calculated in accordance with above shall not exceed 1/1 500 of the span.

c) *Requirements associated with machine spaces (where machinery is placed outside of the well)*

- 1) Machinery spaces outside of the well shall be constructed to withstand the loads and forces to which they are subjected;
- 2) The total load on beams used for supporting machinery shall be assumed as equal to all equipment resting on the beams plus twice the maximum load suspended from the beams;
- 3) The factor of safety for all beams and supports based on ultimate strength of the material and load in accordance with (2) above shall be not less than the following:
 - i) For steel : 5
 - ii) For reinforced concrete : 7

The deflection of the overhead beams under the maximum static load calculated in accordance with above shall not exceed 1/1 500 of the span.

- 4) Access to the interior of the machinery and pulley spaces shall be:
 - i) capable of being properly lit by a permanent electric light fixture(s) with intensity of at least 200 lux at floor level; and
 - ii) easy to use and completely safe in all circumstances without necessitating entry into private premises.

The clear height for movement shall not be less than 2 000 mm.
- 5) The machinery shall be located inside a cabinet consisting of imperforate walls, floor, roof and lockable door(s).
- 6) The machinery cabinet shall be suitably ventilated. It shall be such that the machinery is protected as far as it is reasonably practicable from dust, harmful fumes and humidity.
- 7) The inside of the machinery cabinet shall be provided with permanently installed electric lighting with at least 200 lux at floor level. The switch placed inside shall be close to the door(s), at an appropriate height, to control lighting of the cabinet.

d) *Dimensional tolerances*

- 1) *Lift well dimensions* ð Plan dimensions of lift wells given by the lift manufacturer, represent the minimum clear plumb sizes. The purchaser's representative, in conjunction with the builder, should ensure that adequate tolerances are included in the building design so that the specified minimum plumb dimensions are obtained in the finished work. Dimensions in excess of these minimum plumb dimensions for lift well and openings (but not less) can be accommodated by the lift manufacturer up to certain maximum values beyond which changes in design may be necessary involving additional expense or work by the builder. The purchaser's representative should take these factors into account when specifying the lift well structural dimensions on the basis of the constructional tolerance appropriate to the building technique.
- 2) *Landing door openings* ð It is very important that finished landing openings should be accurate to design size and plumb one above the other for the full travel of the lift. In constructing the structural openings in concrete walls to lift wells it is not possible to achieve a degree of accuracy vertically which will allow doors and frames to be inserted in the opening without some form of masking or packing to overcome inaccuracies. Provisions should therefore be made in design by increasing the nominal height from design finished floor level and width of openings to each jamb and head. In addition, the alignment of the outer face of the front wall of the lift well is of importance when architrave of fixed dimensions are called for and in this case the alignment of the outer face from

floor to floor should not vary to a greater extent than can be accommodate by the subsequent front wall finish, the architrave being set accurately plumb.

- 3) *Structural limits for lift wells at any level* δ If the net plumb well (dimensions A and B in [Fig. 14](#)) and the nominal structural entrance openings (dimensions C and D in [Fig. 14](#)) are defined by plumb lines, the actual wall should not encroach on these dimensions.

Dimension K (inside face of wall of [Fig. 14](#)) should fall within the following limits:

- | | | |
|---------------------------------|---|--|
| i) For wells up to 30 m | : | 0 mm to 25 mm |
| ii) For wells up to 60 m | : | 0 mm to 35 mm |
| iii) For wells up to 100 m | : | 0 mm to 50 mm |
| iv) For wells taller than 100 m | : | 0 mm to 50 mm for any 100 m portion of the well. |

When architraves are to be supplied by the lift manufacturer dimension L (side of structural opening of [Fig. 14](#)) should fall within the limits of 0 mm and 25 mm and dimension M (outer face of the front wall of [Fig. 14](#)) should not vary to a greater extent than can be accommodated by the subsequent front wall finish, the architrave being set accurately plumb. When the entrance linings are supplied by the builder, corresponding provision should be made for the finished openings to be accurately plumb one above the other for the full travel of the lift end to design size.

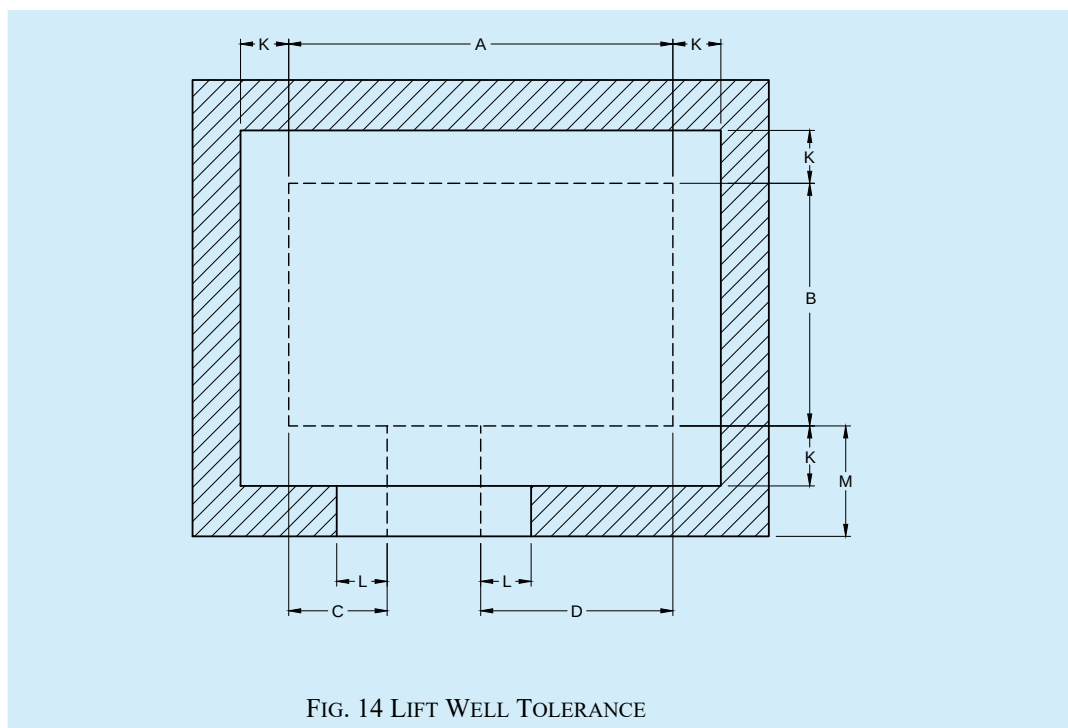


FIG. 14 LIFT WELL TOLERANCE

6.2 Electrical Installation Requirements

The lift manufacturer should specify, on a schedule, particulars of full load current, starting current, maximum permissible voltage drop, size of switches and other details to suit requirements. The main power cable and lighting cable shall be separate and routed through respective lift shaft for every lift. Every electric supply line other than the earthed or earthed neutral conductor of any system or the earthed external conductor of a concentric cable, shall be protected by a suitable switchgear by its owner. It is important that the switchgear at the intake and in the machine room which are provided by the electrical contractor of owner are of correct size. No form of δNO VOLTôtrip relay should be included anywhere in the power supply of the lift.

6.2.1 Main Switchgear

For each lift, a main switchgear capable of breaking the supply to the lift on all the live conductors shall be provided. This switchgear shall be capable of interrupting the highest current involved in normal conditions of use of the lift.

This switchgear shall not cut the circuits feeding the following:

- a) Car lighting and ventilation;
- b) Socket outlet on the car roof;
- c) Lighting of machinery spaces;
- d) Socket outlet in the machinery and pulley spaces and in the pit;
- e) Lighting of the lift well including pit;
- f) Alarm device; and
- g) Automatic rescue device.

The switchgear shall be located:

- 1) in the machine room, where it exists;
- 2) where no machine room exists, in the distribution box (DB) on top floor near the controller or meter room; or
- 3) at the emergency and test panel, when control panel is mounted in the well.

The following points shall be taken into consideration while fixing location of distribution box:

- i) It shall be installed in common areas where there is no hindrance to reaching and operating, especially during emergency conditions; and
- ii) Suggested height above floor level shall be such that it is beyond the reach of children. But the replacement/reset of protective devices shall be in an accessible position without the aid of any tackle, for example, at an average height between 1.8 m and 2.1 m.

If the emergency panel is separate from test panel, the switchgear shall be at emergency panel. If the main switch is not easily accessible from the controller cabinet, then the cabinet shall be provided with an isolating switch.

The main switchgear shall have stable open and closed positions and shall be capable of being locked-off in the open position with the use of a padlock or equivalent, to ensure no inadvertent operation takes place.

For a single lift, this switchgear should be fixed adjacent to the machine room entrance in the machine room or in distribution box (DB) near controller (in case of MRL). In a machine room common to more than one lift, each main switch should be conveniently situated with respect to the lift it controls. Main switch for each lift shall be conspicuously marked.

The supply to the car light should be from a separate circuit and controlled by a switchgear in the machine room or in distribution box (DB) near controller (in case of MRL). For multiple lifts with a common machine room a separate supply and switch gear should be provided for each car. The car lighting and well lighting supply should be independent of the power supply mains. Switch gear for car light and well light of each lift shall be separate and conspicuously marked.

Any capacitors to correct the power factor shall be connected before the main switch of the power circuit.

6.2.2 Supply Cables Wiring and Apparatus

The main power supply cables providing the electricity supply to a lift installation shall be flame retardant, low smoke, low halogen with cross-linked polyethylene (FRLSH XLPE) insulated armoured cables. The cables shall be terminated in the main switchgear panels/distribution box using cable glands such that the armour of the cables is positively connected to the metal body of switchgear panels/distribution box. For each lift, one $415\text{ V }^{+6\text{ percent}}_{-10\text{ percent}}$, 3-phase, 4-wire, 50 Hz and one single phase $230\text{ V }^{+6\text{ percent}}_{-10\text{ percent}}$, 3-wire, 50 Hz supply shall be independently provided. Other than this, power circuit at $230\text{ V }^{+6\text{ percent}}_{-10\text{ percent}}$ for power and lighting sockets in the lift machine room and the lift shaft shall be provided. This is to facilitate maintenance and other related works with lighting and socket outlets while the lift is switched off.

The cables for lift shall be run in respective lift wells. Cables which are not part of lift installation shall not run in the lift well.

Fixed electrical wiring for circuit voltage up to 250 V located in lift wells shall be one of the following:

- a) PVC wires encased in rigid conduit (or trunking) made either of metal or plastic;
- b) PVC insulated PVC sheathed cables; and
- c) PVC wires in flexible metallic or plastic conduits not exceeding 1 500 mm in length for limit switches, interlocks, push buttons and similar devices.

Control cables on lift cars shall run in rigid/flexible metallic/plastic conduit or trunking. Short runs of flexible conduits may be used, where they are securely fastened in place and not exposed to oil and grease. Short runs of flexible cord may be used as flexible connections between fixed wirings on the car and the switch on car door, provided that the cord is securely fastened in position and so located as not to be subjected to mechanical damage.

Wiring in machine room shall be run in rigid metallic trunking. The trunking in machine room floor shall be flush with the finished floor to avoid trip hazard. The following additional requirements may also be taken care of:

- 1) Dedicated power supply feeder shall be given for all lifts, which shall be separate from other building services. This feeder shall be distinctly identified and shall have controlled access such that it cannot be switched off by unauthorized person
- 2) The lift sub-circuit from the meter room should be separate from other building services. Each lift should be capable of being isolated from the mains supply. This means of isolation shall be with lock-out and tag-out facility. The electric supply cables shall run in a route safe from fire, subsequently within the respective lift shafts to the machine room.
- 3) For banks of interconnected lifts, a separate sub-circuit is required for the common supervisory system, in order that any car may be shut down without isolating the supervisory control of the remainder lifts.
- 4) When the alarm system is connected to a transformer or trickle charger, the supply should be taken from the machine room lighting.
- 5) All electrical supply lines and apparatus in connection with the lift installation shall be so constructed and shall be so installed, protected, worked and maintained that there may be no danger to persons therefrom.
- 6) All metal casings or metallic coverings containing or protecting any electric supply lines of apparatus shall be efficiently earthed.
- 7) No bare conductor shall be used in any lift car as it may cause danger to persons.
- 8) All cables and other wiring in connection with the lift installation shall be of suitable grade for the voltage at which these are intended to be worked and if metallic covering is used it shall be efficiently earthed.
- 9) Suitable caution notice shall be affixed near every equipment or apparatus in which voltage used exceeds 250 V.
- 10) A single trailing cable for lighting control and signal circuit shall be permitted, if all the conductors of this trailing cable are insulated for maximum voltage running through any one conductor of this cable.
- 11) *Emergency signal or intercom* ð The lift car shall be provided either with an emergency signal that is operative from the lift car and audible outside the lift well or with intercom.
- 12) When an alarm bell is to be provided, each car is fitted with an alarm push button which is wired to a terminal box in the lift well at the ground floor by the lift manufacturer. This alarm bell, to be supplied by the lift manufacturer (with indicator for more than one lift) should be fixed in an agreed position and wired to the lift well. The supply may be from a battery (or transformer) fixed in the machine room or, when available, from the building fire alarm supply.

- 13) An intercom system or similar device, powered by the emergency supply, shall be installed for communication between inside the car, someone outside the lift (security/reception/BMS room) and the place from which the emergency operation is carried out, if the lift travel exceeds 15 m or if direct acoustic communication in- between these locations is not possible. When an intercom is to be provided in the lift car the lift manufacturer should fit it in the car and provide wiring from the car to a terminal box adjacent to the lift well. Alternatively, a hands-free unit may be provided. The type of intercom should be stated in the enquiry.

6.2.3 Earthing

Earthing provides safety of human beings, livestock and apparatus against earth faults. Any system is characterized by the type of distribution system, which include types of systems of live conductors and types of system earthing.

The subject of earthing covers the problems associated with connection of exposed conductive part of installations to the main earthing terminal of that installation and to achieve automatic disconnection of supply before the fault voltage creates a shock hazard. The terms earth and earthing have been used interchangeably, to denote a low impedance return path of the fault current. The earth is mainly used for fixing the voltage of system neutrals and is not used as a part of return circuit.

6.2.3.1 System earthing

Earthing of system is designed, primarily, to preserve the security of the system by ensuring that the potential on each conductor is restricted to value consistent with the level of applied insulation. It is equally important that earthing should ensure efficient and fast operation of protective gears, in the case of earth faults. Commonly used system earthing are TN, TT and IT.

6.2.3.2 TN system

A system having one or more points of the source of energy directly earthed, the exposed conductive-parts (and extraneous conductive parts) of the installation are connected by means of protective conductors to the earthed point(s) of the source, that is, there is a metallic path for earth fault currents to flow from the installation to the earthed point(s) of the source. TN systems are further sub-divided into TN-S, TN-C-S and TN-C systems.

- a) *TN-S system* ð A system with separate neutral and protective conductor throughout the system (see Fig. 15).

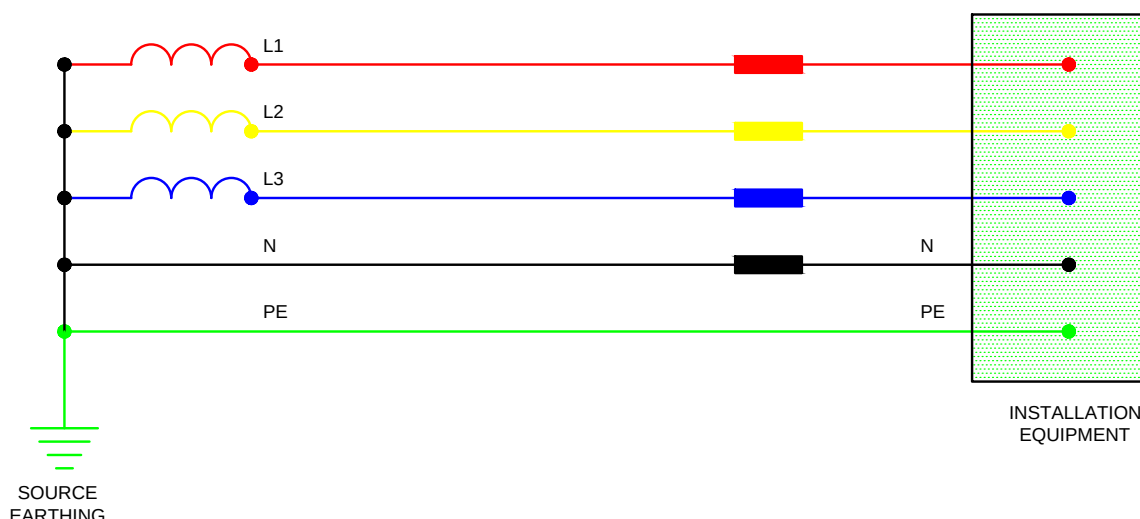


FIG. 15 TN-S SYSTEM WITH SEPARATE NEUTRAL AND EARTH CONDUCTOR FROM THE SOURCE

- b) *TN-C-S system* ð A system in which neutral and protective conductors are combined in a single conductor in part of the system. The PEN conductor referred to as a combined neutral and earth (CNE or PEN) conductor, is earthed at the source and at multiple points. Multiple earthing of the PEN conductor ensures that if the conductor becomes open circuit due to any reason, exposed-conductor parts

remain connected to earth. Under such conditions, the supply voltage between the installation line and neutral conductor is substantially reduced (see Fig. 16).

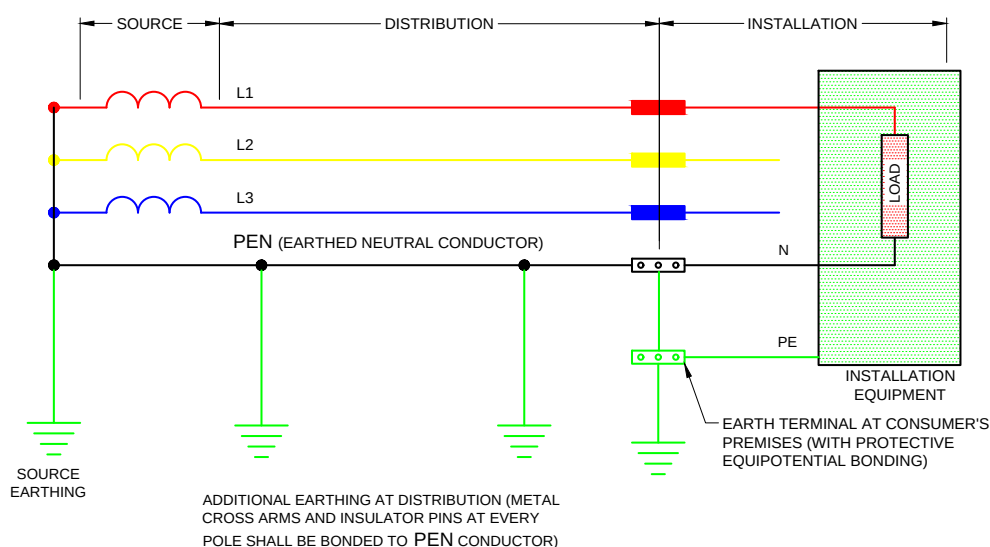


FIG. 16 TN-C-S SYSTEM WITH PROTECTIVE MULTIPLE EARTHING (PME) USED IN PUBLIC DISTRIBUTION

- c) *TN-C system* ð A system in which neutral and protective conductors are combined in a single conductor throughout the system. TN-C is not permitted inside consumer premises (see Fig. 17).

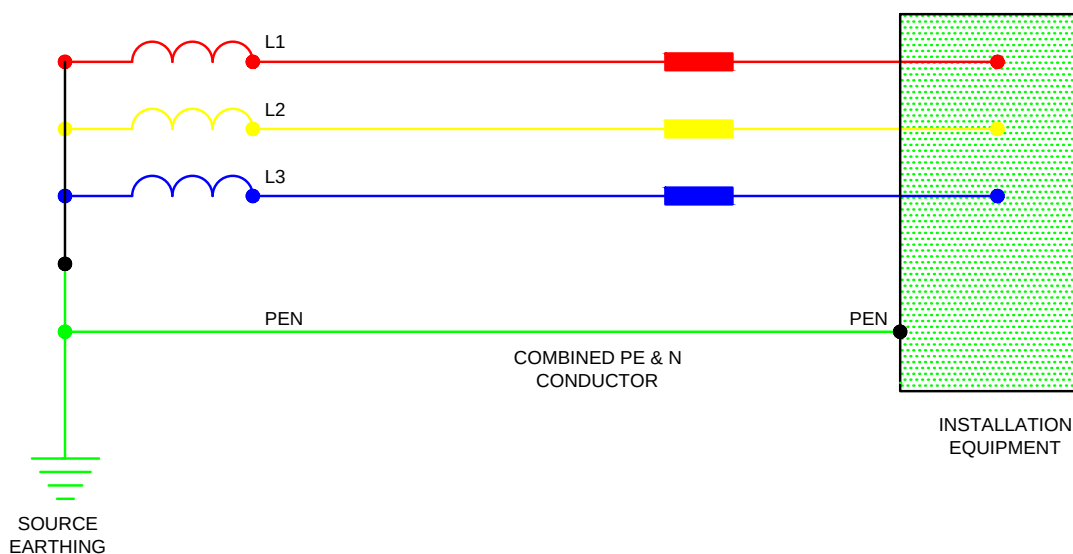


FIG. 17 TN-C SYSTEM (NEUTRAL AND PROTECTIVE FUNCTION COMBINED IN A SINGLE CONDUCTOR THROUGHOUT SYSTEM)

6.2.3.3 TT system

In this system one or more points of the source of energy are directly earthed and the exposed and extraneous conductive parts of the installation are connected to a local earth electrode, electrically independent of the source earth. Every consumer installation shall be protected with an RCD for earth fault protection at the origin of installation (see Fig. 18).

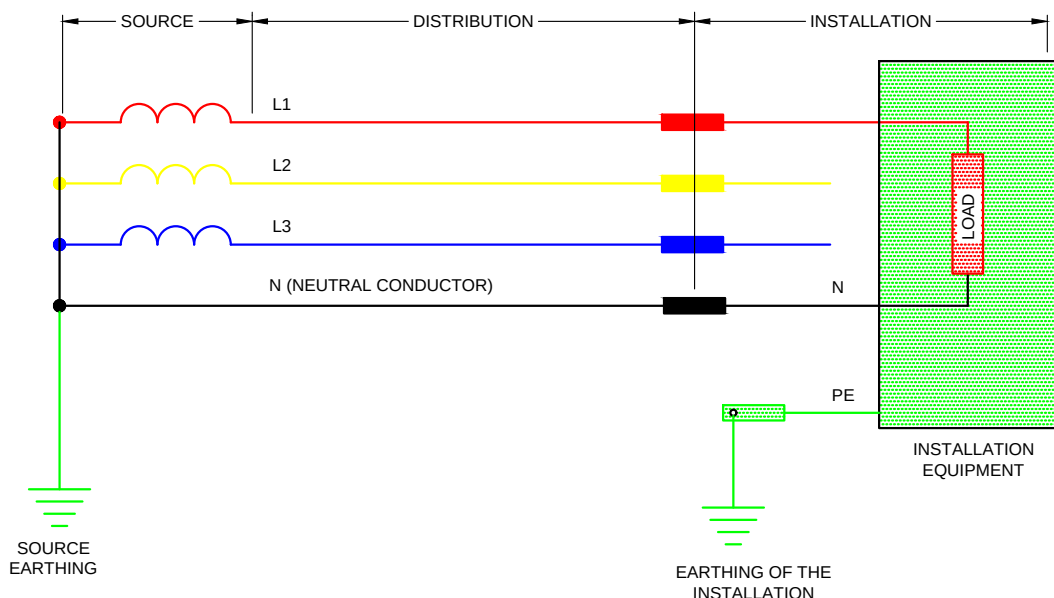


FIG. 18 TT SYSTEM

6.2.3.4 IT system

The live parts have no direct connection/isolated from earth or one point is connected to earth through an impedance. The exposed-conductive-parts of the electrical installation are earthed independently or collectively. The IT systems are generally used for special applications such as operation theatre in hospitals. It is not intended for use in lifts (see Fig. 19).

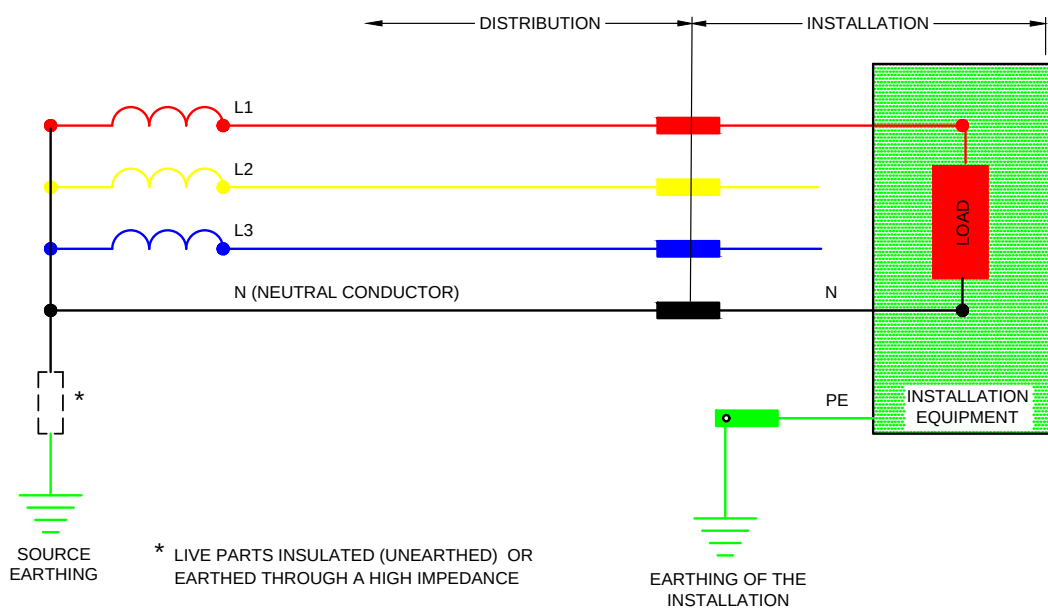


FIG. 19 IT SYSTEM

6.2.3.5 Basic purpose of earth fault protection

The occurrence of an earth fault in an installation creates two possible hazards:

- Voltages appear between exposed conductive parts and extraneous conductive parts and if these parts are simultaneously accessible, these voltages constitute a shock hazard. This condition is known as indirect contact.
- The fault current that flows in the phase and protective conductors of the circuit feeding the faulty equipment may be of such a magnitude as to cause an excessive temperature rise in those conductors, thereby creating a fire hazard.

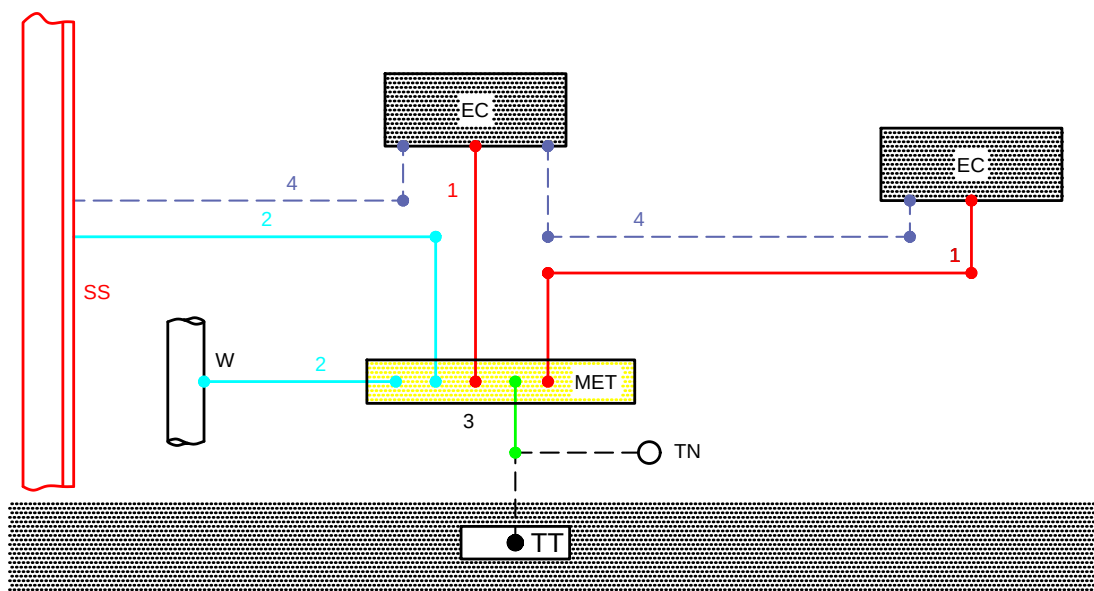
The protective measure known as protective equipotential bonding and automatic disconnection of the supply is intended to provide a high degree of protection against both hazards. The choice of protective device for disconnection is influenced by the type of system of which the installation is a part, because either:

- 1) the earth fault loop impedance shall be low enough to allow adequate earth fault current flow to cause an overcurrent protective device (for example, a fuse or circuit breaker) in the faulty circuit to operate in a sufficiently short time; or
- 2) where it is not possible to achieve a low enough earth fault loop impedance, disconnection may be initiated by fitting a residual current device (RCD).

6.2.3.6 Protective equipotential bonding

Protective equipotential bonding is through connecting extraneous conductive parts in a building to the main earthing terminal of the installation by protective bonding conductors. Where such conductive parts originate outside the building, they shall be bonded near their entry point within the building.

Typical illustration of earthing and protective conductors is shown in [Fig. 20](#).



Key

EC	Exposed conductive part (part board body)
MET	Main earthing terminal
SS	Structural steel (extraneous conductive part)
W	Metallic water pipe (extraneous conductive part)
TT	Earth electrode in a TT or IT system
TN	Other means of earthing in a TN system, for example, connection to earthed point of the point of the power system
1	Circuit protective conductor
2	Main protective bonding conductor
3	Earthing conductor
4	Supplementary protective bonding conductors (additional bonding, if required)
1,2,3,4	Protective conductors

FIG. 20 ILLUSTRATION OF EARTHING AND PROTECTIVE CONDUCTORS

In each building the earthing conductor, the main earthing terminal and the following conductive parts shall be connected to the protective equipotential bonding (see Fig. 21):

- metallic pipes supplying services into the building, for example, gas, water;
- structural extraneous conductive parts, if accessible in normal use, metallic central heating and air-conditioning systems; and
- metallic reinforcements of constructional reinforced concrete, if reasonably practicable.

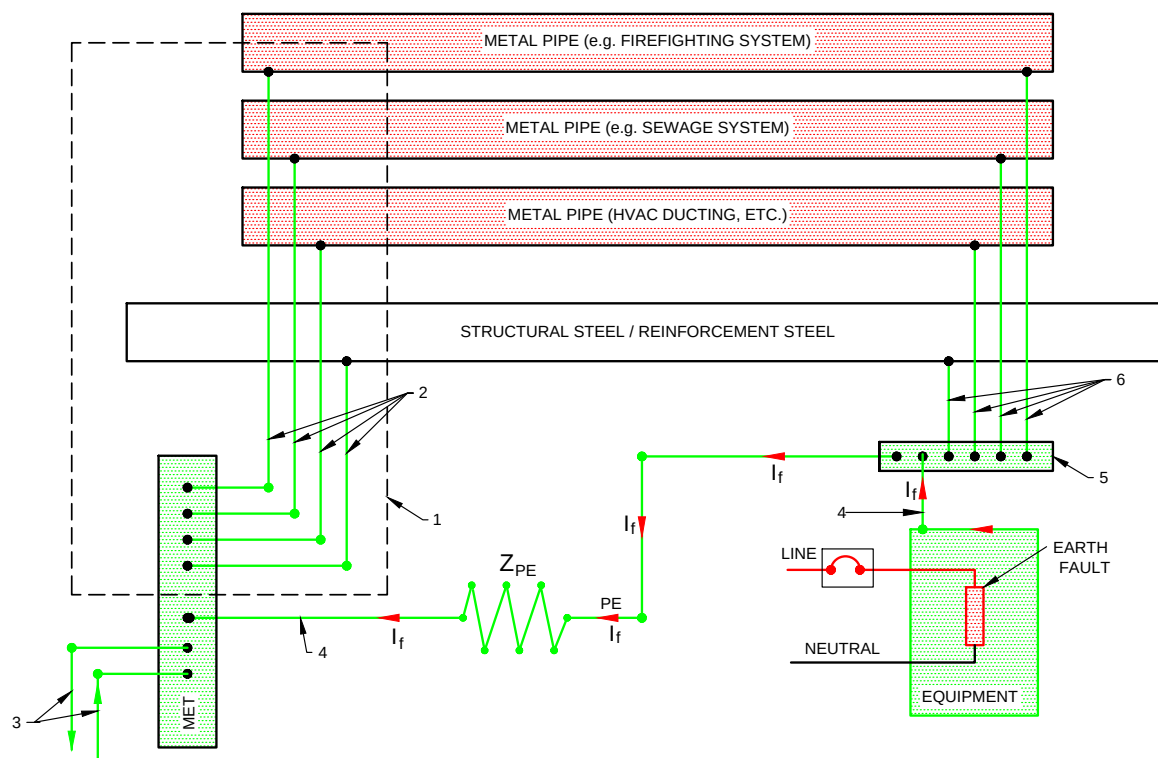


FIG. 21 ILLUSTRATION OF MAIN AND ADDITIONAL PROTECTIVE EQUIPOTENTIAL BONDING

The connection of various conductors in an earthing arrangement shall be proper. The connection shall be by exothermic welding, pressure connectors, clamps or other suitable mechanical connectors. Mechanical connectors shall be installed in accordance with the manufacturer's instructions. Where a clamp is used, it shall not damage the electrode or the earthing conductor.

Two separate copper protective earthing conductors shall be provided from main earthing terminal (MET) to lift controller. Refer to Table 33 for cross sectional area of protective conductor.

Table 33 Cross Sectional Area of Protective Conductor		
(Clause 6.2.3.6)		
Sl No.	Cross-sectional Area of Line Conductor, S	Minimum Cross-Sectional Area of the Corresponding Protective Conductor
	mm ² Cu	mm ² Cu
(1)	(2)	(3)
i)	$S \leq 16$	S
ii)	$16 < S \leq 35$	16
iii)	$S > 35$	0.5 S

Earthing system above 230 V should be designed as a protective multiple earthing (PME) system with separate protective conductor (TN-S). Protective multiple earthing system shall be implemented to reduce the earth fault loop impedance and to ensure reliable disconnection of supply during an earth fault by the over current protective device (OCPD), for example MCB/MCCB. In PME system, neutral of the supply transformer and the non-current carrying metal parts in the system are interconnected by the common earth grid, in addition to the separate protective conductor.

Hence, it is recommended to have TN-S system with PME (as shown in Fig. 22), where the neutral of the source and exposed conductive parts in the installation are solidly connected to the MET and an earthing grid. In this interconnected system, earth fault current is returned to the neutral, through the interconnected system consisting of the PE conductor and earth grid. Protective earth (PE) conductor shall run in parallel and proximity to line conductors.

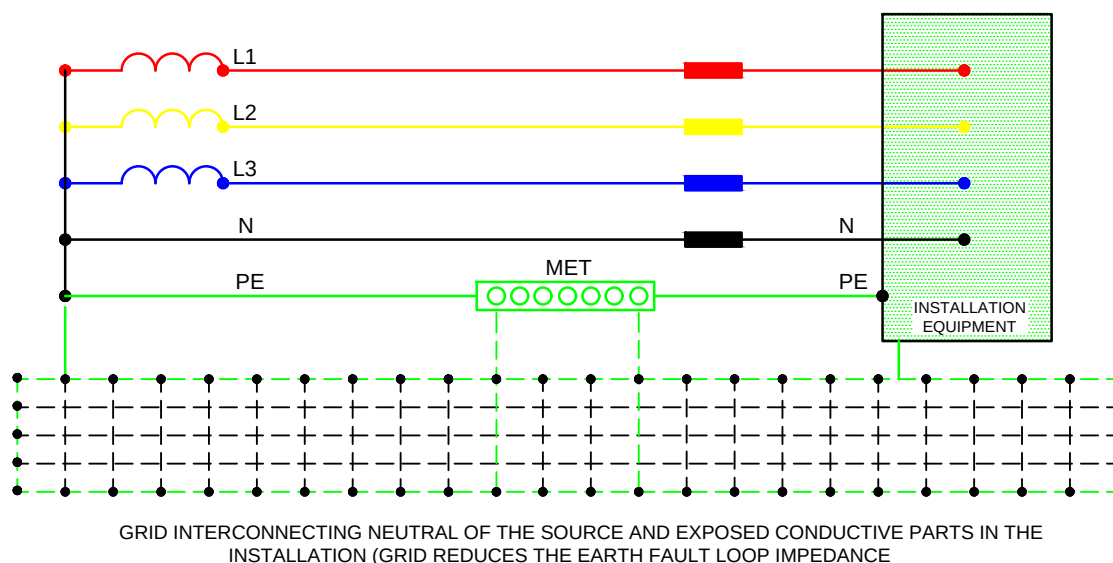


FIG. 22 SYSTEM WITH SEPARATE EARTH CONDUCTOR (TN-S) WITH PROTECTIVE MULTIPLE EARTHING (PME),
RECOMMENDED FOR INDUSTRIAL AND COMMERCIAL USE

The continuity resistance of the earth return path through the earth grid shall be less than one ohm.

6.2.3.7 Automatic disconnection of power supply in case of a fault

Automatic disconnection of supply is a protective measure in which:

- basic protection is provided by basic insulation of live parts or by barriers or enclosures; and
- fault protection is provided by protective equipotential bonding and automatic disconnection in case of a fault in accordance with 4.2.11.3 to 4.2.11.6 of the good practice given in [D-5A(8)].

A protective device shall automatically interrupt the supply to the line conductor of a circuit or equipment in the event of a fault of negligible impedance between the line conductor and an exposed-conductive-part or a protective conductor in the circuit or equipment within the disconnection time specified in 4.3.11.2.2, 4.3.11.2.3 and 4.3.11.2.4 of the good practice given in [D-5A(8)].

If automatic disconnection cannot be achieved in the time required, supplementary protective equipotential bonding shall be provided as specified in 4.2.15.2 of the good practice given in [D-5A(8)].

In general, every circuit is provided with a means of overcurrent protection. If the earth fault loop impedance is low enough to cause these devices to operate within the specified times (that is, sufficient current can flow to earth under fault conditions), such devices may be relied upon to give the requisite automatic disconnection of supply. If the earth fault loop impedance does not permit the overcurrent protective devices to give automatic disconnection of the supply under earth fault conditions, the first option is to reduce that impedance. It may be permissible for this to be achieved by the use of protective multiple earthing or by additional earth electrodes.

An alternate approach is to be adopted for the complete safety of the operating personnel and equipment from the hazards that may result from earth faults. This is to use residual current devices with appropriate settings to clear the faults within the permissible time, based on the probable contact potential. This method is equally applicable where earth loop impedances cannot be improved.

6.2.3.8 Residual current device (RCD)

RCD's can be used for fault protection and additional protection. RCD is used for automatic disconnection of supply during earth fault, in case of failure of this protection by OCPD/MCB. The good practice given in [D-5A(8)] recommends that in locations where fault protection is achieved by 500 mA RCD, earth fault loop impedance test is not required.

RCD of $I_{qn} \geq 30$ mA shall be used as an additional protection against electric shock in locations where the chance of shock is high.

In addition to giving protection against earth fault or shock protection, RCDs may also give fire risk protection, the degree of protection being related to the sensitivity of the device and magnitude of leakage current. The good practice given in [D-5A(8)] recommends use of 300 mA RCD for protection against fire in building due to leakage current.

Residual current device (RCD), when provided, right type and sensitivity shall be selected.

The RCD shall,

- a) have maximum rating of I_{qn} 500 mA as tripping current suitable for frequency converters with a 3-phase supply against earth faults; and
- b) be sensitive for sinusoidal currents up to 1 000 Hz, and insensitive for currents more than 1 000 Hz.

NOTE δ National Electrical Code of India 2023 recommends Type B RCD for applications which may create smooth DC residual current, for example, three-phase variable frequency drive.

6.2.3.9 Reinforced protective earthing conductors for currents exceeding 10 mA in the protective earthing conductor

For lifts with ACVF of higher KVA ratings if the current in the protective earthing conductor exceeds 10 mA the following applies:

- a) where the ACVF has only one protective earthing terminal, the protective earthing conductor shall have a cross-sectional area of at least 10 mm² Cu or 16 mm² Al, through its total run; or

NOTE δ Refer 5.4.3.4 of the good practice given in [D-5A(8)].

- b) where the ACVF has a separate terminal for a second protective earthing conductor, a second protective earthing conductor of at least the same cross-sectional area as required for fault protection shall be run from a point where the protective earthing conductor has a cross-sectional area not less than 10 mm² Cu or 16 mm² Al.

6.2.3.10 Additional protection

Additional protection by means of a residual current device (RCD) with a rated residual operating current not exceeding 30 mA shall be provided for:

- a) socket outlets;
- b) control circuits for landing controls and indicators and the safety chain with voltage higher than 50 V AC; and
- c) circuits on the lift car with voltage higher than 50 V AC.

This requirement is to protect persons from direct shock risk. It is different to automatic disconnection of supply offered by residual current device (RCD) > 30 mA as they are intended to protect persons from indirect shock risk to persons in contact with earthed part. Due to the inherent leakages from electronic devices such as power supplies and VFD's, an RCD of 30 mA as a fault protective measure could create unwanted tripping due to

excessive PE conductor current. Refer 5.3.2.2.1.3 of the good practice given in [D-5A(8)]. Therefore, 3-phase power supply for the lift should never be protected with a 30 mA residual current device (RCD). The 30 mA RCD trips easily when the lift starts to drive and prevents the lift to run. If the construction-time power supply is for some reason protected with a 30 mA residual current device (RCD), suitably sized isolation transformer needs to be provided by the electrical contractor.

6.2.3.11 Earth fault loop impedance

The impedance of the earth fault current loop (phase to earth loop) starts and ends at the point of earth fault. [Table 34](#) gives the recommended values. The earth fault loop comprises the following, starting at the point of fault:

- the circuit protective conductor;
- the consumer's earthing terminal and earthing conductor, and for terre-neutre (TN) systems, the metallic return path;
- for terra-terra (TT) and isolated-terra (IT) systems, the earth return path;
- the path through the earth neutral point of the transformer;
- the transformer winding; and
- the line conductor from the transformer to the point of fault.

Table 34 Recommended Maximum Earth Fault Loop Impedance Values (Z_s) for MCBs (Clause 6.2.3.11)				
SI No.	MCB Rating Amp	Z _s		
		Type B MCB	Type C MCB	Type D MCB
(1)	(2)	(3)	(4)	(5)
i)	6	5.11	2.56	1.28
ii)	10	3.07	1.53	0.77
iii)	16	1.92	0.96	0.48
iv)	20	1.53	0.77	0.38
v)	25	1.23	0.61	0.31
vi)	32	0.96	0.48	0.24
vii)	40	0.77	0.38	0.19
viii)	50	0.61	0.31	0.15
ix)	63	0.49	0.24	0.12
x)	80	0.38	0.19	0.10
xi)	100	0.31	0.15	0.08
xii)	125	0.25	0.12	0.06

6.2.3.12 Verification of effectiveness of automatic disconnection

The effectiveness of the measures for protection against indirect contact by automatic disconnection of supply shall be verified according to 6.2.3.6 and 6.2.3.7 of the good practice given in [D-5A(8)].

Ensuring effectiveness of automatic disconnection of earth fault is essential in every lift. For this, earth fault loop impedance must be measured and compared with the characteristics of OCPD/MCB or test the disconnection

time of RCD which is installed as an earth fault protective measure. This test shall be carried out before the lift is released for public usage.

NOTES

- 1 In TN systems, fault protection (protection against indirect contact) shall be offered by:
 - a) Overcurrent protective device (MCB/MCCB); and
 - b) Residual current protective devices (RCDs).
- 2 In TT systems, RCDs shall be used for fault protection.
- 3 Where an RCD is used for fault protection:
 - a) The circuit shall also be protected by an overcurrent protective device that is, standalone RCD shall not be used;
 - b) The disconnection time shall be according to [Table 35](#); and
 - c) Fault protection is provided in this case also if the fault impedance is not negligible.

Table 35 Minimum Disconnection Time (Clause 6.2.3.12)					
SI No.	Voltage V	Disconnection Time s			
		TN System		TT System	
		a.c.	d.c.	a.c.	d.c.
(1)	(2)	(3)	(4)	(5)	(6)
i)	$50 < U_0 \leq 120$	0.8	See Note	0.3	See Note
ii)	$120 < U_0 \leq 230$	0.4	5	0.2	0.4
iii)	$230 < U_0 \leq 400$	0.2	0.4	0.07	0.2
iv)	$U_0 > 400$	0.1	0.1	0.04	0.1
NOTE 3 Disconnection may be required for reasons other than protection against electric shock.					

The terminal for the earthing of the frame of the motor, the winding machine, the frame of the control panel and similar electric appliances which normally carry the main current shall be at least equivalent to a 10 mm diameter bolt, stud or screw.

The terminal for the earthing of the metallic cases and covers of door interlocks, door contacts, call and control buttons, stop buttons, car switches, limit switches, junction boxes and similar electrical fittings which normally carry only the control current shall be at least equivalent to 5 mm brass screw (such terminal being one specially provided for this purpose) and the earth conductor size shall be at least equivalent to 0.5 mm² copper.

The earth conductor shall be secured to earthing terminal in accordance with the recommendations made in good practice [D-5A(9)] and also in conformity with the latest provisions of *Electricity Act*, 2003 and the CEA Regulations framed thereunder from time to time.

The exposed metal parts of electrical apparatus installed on a lift car shall be sufficiently bonded and earthed.

Where screwed conduit screws into electric fittings carrying control current making the case and cover electrically continuous with the conduit, the earthing of the conduit may be considered to earth the fitting. Where flexible conduit is used for leading into a fitting, the fitting and such length of flexible conduit shall be effectively earthed.

7 FIRE PROTECTION REQUIREMENTS FOR LIFTS AND EVACUATION LIFTS

7.1 Fire Protection Requirements of Lifts in High Rise Buildings

7.1.1 Buildings of Height 15 m and Above

Following requirements over and above those specified in [6](#) and [8](#) are applicable to all lifts provided in buildings having height more than 15 m:

- a) All materials of construction in load bearing elements, stairways and corridors and facades shall be non-combustible.
- b) The interior finishing materials shall be of very low flame spread.
- c) Walls of the lift well enclosure for a lift or group of lifts shall have a fire rating of 2 h. The lift well shall have a vent at the top, of area not less than 0.2 m² per lift.
- d) *Landing doors* ð Lift landing doors shall be imperforate. Collapsible doors shall not be permitted. Lift landing doors shall have a minimum fire resistance rating of 1 h for integrity in accordance with accepted standard [D-5A(10)].
- e) *Lift car door* ð Lift car doors shall be imperforate. Collapsible car doors shall not be permitted.
- f) Telephone or other communication facilities shall be provided in the lift car and the lift main lobby. Communication system for lifts shall also be connected to the fire control room of the building, if provided. For lifts used by persons with disabilities, the facilities shall be provided in accordance with good practice [D-5A(3)].
- g) Photo luminescent safety signs shall be posted and maintained on every floor at or near the lift indicating that in case of fire, occupants shall use the stairs unless instructed otherwise. The sign shall have the plan of the respective floor showing location of the stairways. The plan shall indicate the direction to and the level of refuge floors. All signs posted and maintained on every floor of buildings open to and used by the public shall comply with the requirements of accessible signage given in good practice [D- 5A(3)].
- h) All lifts (fireman's lifts/non fireman's lifts) shall be provided with Phase I operation as per [7.1.1\(i\)](#) (grounding operation).
- j) The grounding operation may be initiated by individual switches for lifts or a common switch for a group of lifts or by a signal from smoke detector/fire alarm system/BMS room of the building where available.
- k) *Fireman's lift* ð The fireman's lift is provided in a building for the purpose of aiding firemen in evacuating trapped persons in the building and to take the equipment for fighting fire to upper levels with minimum delay. Some lifts out of all the lifts shall be identified as fireman's lifts.

The number of required fireman's lifts and their locations in a building will vary depending on the size, design, complexity of the building. Some considerations are as follows:

- 1) There shall be at least one fireman's lift per building;
- 2) If there are multiple wings in the building, there shall be at least one fireman's lift per wing;
- 3) If there are multiple banks of lifts in the building, there shall be at least one fireman's lift per bank of lifts;
- 4) If the building height is up to 600 m and it is zoned height-wise and it does not have fireman's lift serving every floor of the building, then there shall be at least one fireman's lift per zone which shall serve the main level/fire access level and shall serve all the landings in the respective zones; and
- 5) If the building height is more than 600 m and it does not have any single lift serving all the floors, that is, it has all lifts serving only respective zones, the fireman's lift shall be provided in each zone separately, serving all landings in respective zones, with a transfer landing for transferring from one zone to another.

Considering all above the fireman's lift(s) shall be identified on the building plan by the concerned firefighting authority. This shall be respected.

To be effective in firemen operation the fireman's lift(s) shall be identified on the building plan and duly displayed in fire command centre. The fireman's lift shall fulfil the following requirements:

- i) The fireman's lift may be used by the occupants in normal times;
- ii) The fireman's lift shall be provided with a fireman's switch. The switch shall be a two position (ON/OFF) switch fixed at the evacuation floor (normally main entrance floor) to activate fireman's mode. The switch shall be housed in a box with glass cover along with suitable label and fixed adjacent to the lift at the entrance level. When the switch is turned on, all landing operating panels shall become inoperative, and the lift shall operate only from car operating panel or priority control device. When the switch is off, the lift will return to normal working;
- iii) The fireman's lift shall be provided with an audio and visual signal in the car;
- iv) The fireman's lift shall have a floor area of minimum 1.43 m². It shall have load capacity of not less than 544 kg (8 persons lift);
- v) The fireman's lift shall be provided with power operated (automatic) doors of minimum 0.8 m width;
- vi) The speed of the fireman's lift shall be 1.0 m/s or more such that it can reach the top floor from main floor/fireman access level within 1 min. In case the building is zoned, the fireman's lift shall operate from the lowest served landing to the topmost served landing in 1 min;
- vii) Reliable back up source of power supply shall be provided for all fireman's lifts through automatically operated change-over switch. In case of failure of normal electric supply, it shall automatically changeover over to alternate supply. The route of wiring shall be safe from fire;
- viii) The normal power feeders and backup power feeders shall be provided with 2 h fire resistance rated enclosures until they reach the machine room/controller;
- ix) Suitable arrangements such as providing slope in the floor of lift lobby shall be made at all the landings to prevent water used during firefighting from entering the lift shafts;
- x) For requirements of sprinkler refer 9.24.4 of accepted standard [D-5A(7)];
- xi) The words 'FIREMAN LIFT' shall be conspicuously displayed in fluorescent paint/colour on the lift landings at each floor level;
- xii) *Operational requirements of fireman's lift* ð The lift shall be provided with the following operational controls, Phase I and Phase II:

A) *Phase I: Return to evacuation floor* ð Shall start when the fireman's switch at the evacuation floor is turned to the 'ON' position or the signal from smoke detector/fire alarm system (if provided by the building management system) is on. All lifts controlled by this switch/signal shall cancel all existing car calls and separate from landing calls and no landing or car calls shall be registered. The audio and visual signal shall be turned on. All heat and smoke-sensitive door reopening devices shall be rendered inoperative.

If the lift is travelling towards the evacuation floor, it shall continue driving to that floor.

If the lift is travelling away from the evacuation floor, it shall reverse its direction at the nearest possible floor without opening its doors and return non-stop to the evacuation floor.

If the lift is standing at a floor other than the evacuation floor, it shall close the doors and start travelling non-stop to the evacuation floor.

When at the evacuation floor the lift shall park with doors open.

The continuous audio signal is turned off after this return drive. The visual signal shall remain activated until the lift is restored to normal operation.

NOTE 3 If the building is designed for alternative evacuation floor, in case of fire at main floor the lifts shall park at the alternative evacuation floor with doors open.

- B) *Phase II: Operation of the lift* 3 The Phase II is started after Phase I, if the fireman's switch is 'ON'

If the lifts are grounded by the signal from smoke detector/fire alarm system, for Phase II to begin it shall be necessary to turn the fireman's switch 'ON'

The lift does not respond to landing calls but registers car calls. All heat and smoke sensitive door reopening devices are rendered inoperative.

When the car call button is pressed the doors start closing. If the button is released before the doors are fully closed, they re-open. The car call is registered only when the doors are fully closed.

After registering a car call the lift starts driving to the call. If more than one car call is registered, only the nearest call is answered, and the remaining calls will be cancelled at the first stop.

At the floor the doors are opened by pushing the door open button. If the button is released before the doors are fully open, they re-close.

The lift returns to normal service when it stands at the evacuation floor with doors open and the switch is turned 'OFF' thereafter.

The operation of fireman's lift shall be by means of a full set of push buttons in the car. Other operating systems shall be rendered inoperative.

7.1.2 Buildings of Height 60 m and Above

The following requirements over and above those specified in [7.1.1](#) are applicable to the lifts and lift enclosures provided in buildings having height more than 60 m:

- a) *Fireman's lifts* — Following additional requirements apply to all fireman's lifts in the building:
- 1) The fireman's lift shall have loading capacity of not less than 1 000 kg and floor area not less than 2.35 m²;
 - 2) Electrical equipment within the fireman's lift well and on the car, located within 1.0 m of any wall containing a landing door, shall be protected from dripping and splashing water or provided with enclosures classified to at least IPX3 according to good practice [D-5A(11)];
 - 3) The electrical switchgear placed less than 1 m above lift pit floor shall be protected to IP67 according to good practice [D-5A(11)]. The socket outlet and lowest lamp shall also be located at least 0.5 m above the highest permissible water level in the pit;
 - 4) Suitable means shall be provided in the lift pit to ensure that water will not rise above the level of the fully compressed car buffer;
 - 5) Means shall be provided to prevent the water level in the pit from reaching equipment which could create a malfunction of the fireman's lift; and
 - 6) Alternative source of power supply shall be provided for all fireman's lifts through automatically operated changeover switch. In case of failure of normal electric supply, it shall automatically change over to alternative supply. The route of wiring shall be safe from fire.

- b) There shall be fire command centre (FCC) and/or building management system (BMS) room in the building. CCTV cameras shall be fixed in the lift lobbies and the display screen(s) shall be placed in the FCC or BMS room.

7.2 Evacuation Lifts

Buildings, particularly super high rise (height more than 200 m) buildings, may be designed for a systematic evacuation of persons in case of emergencies that are relevant to a specific building. There are many reasons why a building may need to be evacuated, for example, a fire, explosion, chemical or biological attack, flooding, storm damage, earthquake, etc. Not all of these are relevant to every building hence those which are not relevant can be disregarded. Designers of buildings shall determine if a particular hazard is sufficiently great as to require addressing. This clause deals primarily with fire situation although similar logic can be used in case of other situations as well.

It defines intended use of the lift(s) by persons in order to assist the evacuation, either partial or complete, of a building in case of a fire emergency in the 'golden period' that is, after detection of the fire and before arrival of the fire brigade at the affected building; so long as use of the lift(s) is safe, thereby preventing possible human casualties, especially in case of complex high-rise buildings.

The lifts may be planned only supplementary to, not as a replacement of, exits. In fact, using lifts instead of stairs may well increase evacuation times in many building designs, especially low and medium rise buildings. However, lifts may be allowed to play a positive role in assisting and improving efficiency of the building evacuation strategy especially for persons with disabilities. The building designers should decide, if they want to use lifts for evacuation purpose. The evacuation lift may be used by the occupants in normal times. It may be noted that in general a fireman's lift may be suitable as an evacuation lift. Three evacuation operation methods are possible for lifts to be used for evacuation of persons. One or a combination thereof may be selected based on the building type and fire evacuation strategy planned for that building. These are:

- a) Automatic evacuation operation;
- b) Remote assisted evacuation operation; and
- c) Manually assisted evacuation operation.

To ensure that the lift(s) is available in the event of an evacuation, following principles shall be considered:

- 1) The evacuation strategy and process has been defined consistent with the operation of the evacuation lift(s) as described here;
- 2) Building requirements mentioned in this document are followed; and
- 3) Until the evacuation of persons is complete, the evacuation strategy and process ensure that the evacuation lift(s) is dedicated to their evacuation.

The number and size of lifts required, or the size of door opening shall be determined on a case-by-case basis by the building designer for calculating the number of lifts required, and the fire evacuation plan for the building, in collaboration with lift specialists and fire officials.

In case of an emergency, the usual way for the occupants of a building to escape to an exit floor by taking to the building stairs shall continue to be available.

7.2.1 Pre-requisites to Using Lifts for Evacuation in Case of Fire

7.2.1.1 Building interface and requirements for implementing an evacuation

Evacuation operation is designed with the assumption that the following building provisions are in place:

- a) The building shall be constructed and shall not sustained structural damage, for example, from fire, explosion, flood, lightning strike, earthquake, storm, etc;
- b) The automatic sprinkler system required is installed except that sprinklers in lift machine rooms and in hoist ways are prohibited;

- c) The building shall be provided with a means to protect the lift(s) from the effects of fire and smoke at every landing, a safe area directly outside the lift door, and a fire-resistant structure;
- d) The lift lobby at each floor shall be enclosed and separated from the remainder of the floor by a smoke barrier. The lift lobby door shall be 2 h fire rated and shall close automatically upon initiation of building fire alarm. The lift lobby shall be large enough to accommodate at least 25 percent of the floor population at 0.28 m² per person;
- e) The lift lobbies and staircase shall be pressurized, if required;
- f) The lift lobbies shall be provided with CCTV cameras with the monitor located in FCC/BMS room;
- g) Fire/smoke/temperature detectors shall be provided in the building in a thoughtful manner in all areas of the building including lift shaft, machine room, lift lobbies, etc;
- h) Activation of any of those detectors which signifies unsafe operating environment for the evacuation lift(s) shall initiate Phase I operation of the fireman's/non fireman's lift; and suspend the evacuation operation;
- j) The fire alarm system may have a provision to automatically send public announcement messages to the residents of the floor with active alarm and to the residents of two upper and two lower contiguous floors;
- k) Power supplies shall be secure and reliable; the provision of a secondary supply is essential. The cable providing power to the lift(s) shall be fire protected to the same fire protection level as given to the lift well structure;
- m) The secondary power supply shall be sufficient to run the evacuation lift at the rated load and rated speed for a minimum period equal to the planned evacuation time consistent with the evacuation strategy;
- n) The car lighting shall be also supplied by secondary power supply;
- p) The normal power feeders and backup power feeders shall be provided with 2 h fire resistance rated enclosures until they reach the machine room/controller;
- q) The source of the secondary supply, the level of independence between the primary and secondary supplies;
- r) The source of the secondary power supply and automatic switch gear shall be located in a fire protected area;
- s) The building shall be designed to minimize the risk of flooding into the lift or lift well. To this end sprinkler discharge, burst pipes, fire hose, etc should not be located to discharge towards the lift and any water close to the lift should be directed away from it by sloping floors, etc;
- t) Water sensor shall be provided in the pit to shut down the lift in case the operation becomes unsafe due to accumulation of water in the pit;
- u) Interior exit stairwell doors if locked, shall automatically unlock permitting exit and re-entry upon setting of alarm;
- v) The lift shall be maintained, and the evacuation operation shall be tested at suitable regular intervals;
- w) The building shall have a fire safety and evacuation plan including procedures for evacuation using stairs and lifts, the role of wardens, a routine training and drills for occupants;
- y) Training shall include the message that lifts shall not be used for evacuation in buildings not having lifts provided with special occupant evacuation operation;
- z) The machine room and lift lobbies shall be maintained with minimal fire load;

- aa) Responsibility for safe evacuation shall rest with and shall be in the control of a trained person(s) located at the building premises; and
- bb) Those with a disability shall be evacuated with the help of specially trained persons or with their personal assistant.

7.2.1.2 Detecting fire, smoke and temperature

Methods of fire detection are well established. If the lifts are to operate a safe evacuation service, good monitoring of the fire in relation to lift equipment is essential.

To ensure the safety of users and lift equipment; the lift machine room, lift well (including lift pit and overhead areas) and landing areas shall be provided with means to detect and monitor for the presence of smoke and heat. Temperature in any safe area provided for persons to wait (lobby, refuge, etc) and the lift well, machine room, etc, should be continuously monitored to determine if it is safe and remains safe for persons and equipment. When an unsafe temperature or presence of smoke is detected, the information should be sent to the BMS or FCC. The BMS or FCC should determine what is to be done next and send appropriate signals to the lift, for example remove lift from service at the respective floor x with doors parked open or closed as required. The equipment used for this sensing and monitoring of heat and smoke should be specified by those responsible for the design of the building fire and smoke detection system and not the lift manufacturer.

7.2.1.3 Location of an event in relation to lifts

In detecting an event like fire it will be important to know its location in relation to any lift intended for evacuation use. Experts in fire detection systems and lift experts should be consulted to determine the best solution for the design in question. The more the level of sophistication of detection system employed, the more precise will be the level of information available to make decisions and more meaningful these decisions will be. The building designer shall determine the degree of sophistication required taking into account the importance of the building, type of occupancy, etc.

7.2.1.4 Emergency power provisions

Secondary (emergency) power shall be provided that has sufficient capacity to run all the evacuation lifts at full speed for the required evacuation time. The route of the wiring shall be safe from fire.

7.2.2 Operational Decisions for Evacuation

7.2.2.1 Types of evacuation

The evacuation strategy developed should lead to an obvious evacuation type. Lifts will not be able to evacuate all floors at once, so to some degree a systematic evacuation will be governed by the capacity of lifts and the number of persons to be evacuated. Lift engineers can calculate the number of persons that can be moved for a given set of circumstances but the first thing to determine is the number of persons to be evacuated within the required evacuation time. Fire engineers and other experts shall determine these figures. It shall also be determined what proportion of wheel chair and others with impaired mobility may be there and if there is likely to be a floor or building area with a particular concentration of them.

In many instances some sort of phased evacuation will be desirable while at other times a full evacuation may be essential. Phased evacuation involves the evacuation of certain areas first, usually those floors areas at greatest risk followed by other key areas. Also, there could be horizontal evacuation, partly or fully, for example, in case of hospitals.

7.2.2.2 Evacuation process

There are 3 types of lift operation methods to hasten the building evacuation process.

- a) *Automatic evacuation operation* ð The evacuation operation is completely automatic;
- b) *Remote assisted evacuation operation* ð The evacuation operation is controlled remotely by the remote evacuation assistant and

- c) *Manually assisted evacuation operation* – The evacuation operation is controlled by an assistant in the car – manual evacuation assistant

The type of evacuation operation(s) to be deployed in a building shall depend on the specific building fire evacuation strategy. The evacuation service shall be operational between the floors as decided by the building designer. The logic of operation shall be mutually agreed between the building designer, the lift designer and the fire expert.

7.2.2.2.1 Evacuation service (Phase 1)

7.2.2.2.1.1 Initiation of evacuation service

Evacuation service shall be initiated by a two-position switch having START-STOP functions provided at the emergency exit lobby (EEL) for the manually assisted evacuation operation and/or a switch at the remote-control panel located in FCC/BMS room for the remote assisted evacuation and/or another switch at the remote-control panel for the automatic evacuation operation.

This switch shall enable either starting (START) or stopping (STOP) the evacuation process. When this switch is turned to START position, the lift system shall be ready to evacuate all floors served by that group of lifts.

7.2.2.2.1.2 On activation of the evacuation service, the lift shall behave as described below:

- a) All registered car and landing calls shall be cancelled and no new calls shall be registered;
- b) In the car the door re-open button, the door close button, and the door protective device shall remain operative; and
- c) The lift, occupied or unoccupied, shall proceed to the EEL as follows; except in case of automatic evacuation operation wherein the lift if unoccupied, shall proceed to the affected floor directly as per the pre-determined priority logic for serving the affected floors; and park there with doors closed till a landing call from the affected floor is initiated:
 - 1) A car stationary at a landing in no call condition, shall close the doors if open and travel nonstop to the active EEL;
 - 2) Visual indication in car operating panel shall display “Evacuation Service”
 - 3) A car travelling away from the active EEL shall make a normal stop at the nearest landing without opening the doors, make an audible message such as “Evacuation Service” and return to the active EEL;
 - 4) A car travelling towards the active EEL shall continue its travel nonstop to EEL;
 - 5) If the lift has already started stopping at a level, it shall continue to make a normal stop and without opening the doors, with message displayed “evacuation service” proceed to the active EEL;
 - 6) On reaching at the active EEL, the evacuation lift car shall open its door and shall provide an audio-visual indication “Exit Lift”. The audible signal shall have a sound level of between 35 dBA and 55 dBA adjustable to suit the site conditions with a typical recommended level of 50 dBA;
 - 7) If the door is not closed at the active EEL after 20 s, the protective device shall be made inactive, the doors shall attempt to close, and an audible signal shall sound in the car until the doors are closed;
 - 8) To start the Evacuation Service:
 - i) in case of automatic evacuation operation, any landing call at the affected floor(s) shall initiate the travel of the lift to that floor as per priority logic decided;
 - ii) in case of remote assisted evacuation operation, the person-in-charge at the remote-control panel shall initiate the travel of the lift to the affected floor; and

- iii) in case of manually-assisted evacuation operation, pressing the car call button at car operating panel (COP) shall initiate the travel of the lift to the affected floor.

7.2.2.2.2 Evacuation service (Phase 2)

After the evacuation service (Phase 1) is completed and any type of evacuation operation signal is active, the lift shall operate in Phase 2 as follows:

7.2.2.2.2.1 Automatic evacuation operation

In automatic evacuation operation the lift shall operate as per the detail below:

- a) When automatic evacuation signal is active and when no higher priority signal is active, the lift shall operate in automatic evacuation operation and shall serve registered landing calls;
- b) The priority of answering the landing calls shall be based on the distance from the active EEL, with the furthest landing call getting highest priority;
- c) On arrival at a landing other than the active EEL, the car shall open the doors and give a voice announcement as “Evacuation Service. Enter the Car”;
- d) Once the doors are closed, the car shall proceed towards the active EEL. If the car is empty, it may respond the next landing call without visiting the active EEL.
- e) The car if not full, may serve other landing calls on the way to the active EEL. If the car makes stops on the way to the active EEL, it shall provide audio-visual indication “Do Not Exit”. The audible signal shall have a sound level of between 35 dBA and 55 dBA adjustable to suit the site conditions with a typical recommended level of 50 dBA.
- f) Actuation of the landing call booking device shall not prevent a loaded car from closing its doors and leaving the floor.
- g) On arriving at the active EEL, the evacuation lift car shall open its door and shall provide an audio-visual indication “Exit Lift”. The audible signal shall have a sound level of between 35 dBA and 55 dBA adjustable to suit the site conditions with a typical recommended level of 50 dBA.

7.2.2.2.2.1.1 Deactivation of automatic evacuation operation:

Automatic evacuation operation shall cease when:

- a) the automatic evacuation signal is deactivated; or
- b) the remote assisted evacuation signal is activated; or
- c) the manually assisted evacuation signal is activated; or
- d) the suspend service signal is activated; or
- e) The lift receives Phase 1 signal of the fireman's control.

In these cases, the car shall return to the active EEL.

7.2.2.2.2.2 Remote assisted evacuation service:

Remote assisted evacuation is the evacuation under the control of an authorized person “remote evacuation assistant” who controls the lift from remote evacuation control panel located in BMS/FCC room in the same building.

NOTE 8 Remote assisted evacuation operation shall be subjected to cybersecurity. Cybersecurity of industrial control systems are specified in the IEC 62443 series of standards and in the accepted standard [D-5A(12)].

In the remote assisted evacuation operation, control from outside the lift car that is, from a remote location, is permitted provided there are communication means between the remote-control location and the lift car. The use

of remote assisted evacuation and the communication means shall be selected based on building evacuation strategy.

Activation of the remote assisted evacuation operation shall override the automatic evacuation signal, if any.

7.2.2.2.2.1 Remote assisted evacuation operation

After the completion of Phase 1 of evacuation service, if the remote assisted evacuation signal is active and no higher priority signal is active, the lift shall operate as per details below:

- a) Information sharing between the lift and the remote evacuation control panel shall be opened by the lift control;
- b) After communication sync is established, the remote evacuation control panel is activated to operate in remote control mode;
- c) The alarm button and door open button in the car operating panel shall remain operative; and other controls of the lift shall be operable solely from the remote evacuation control panel;
- d) The remote evacuation control panel shall have displays indicating the registered landing calls, car position and direction of travel, and car door status;
- e) A call registered through remote evacuation control panel shall cause the lift doors to close and the car to travel to the required landing;
- f) If the door closing is prevented by a door protective device, a separate door close command from remote evacuation control panel shall bypass the protective device and the doors shall attempt to close;
- g) At any time, it shall be possible to register a new call from the remote evacuation control panel. The previous call shall be cancelled. The car shall travel in the shortest time to the newly registered landing;
- h) On arrival at a landing, the lift shall open the doors and shall provide an audio-visual indication "Evacuation Service. Enter the Car". The audible signal shall have a sound level of between 35 dBA and 55 dBA adjustable to suit the site conditions with a typical recommended level of 50 dBA; and
- j) On arriving at the active EEL, the evacuation lift car shall open its door and shall provide an audio-visual indication "Exit Lift". The audible signal shall have a sound level of between 35 dBA and 55 dBA adjustable to suit the site conditions with a typical recommended level of 50 dBA.

7.2.2.2.2.2 Deactivation of remote assisted evacuation

Remote assisted evacuation operation shall cease when:

- a) The remote assisted evacuation signal is deactivated; or
- b) The remote evacuation control panel becomes inactive; or
- c) The connection between the remote evacuation control panel and the lift is interrupted for more than 30 s; or
- d) The audible or video communication system is interrupted for more than 30 s; or
- e) The manually assisted evacuation signal is activated; or
- f) The suspend service signal is activated; or
- g) The lift receives Phase 1 signal of the fireman's control.

In these cases, the car shall return to the active EEL.

7.2.2.2.2.3 Manually assisted evacuation operation

Manually assisted evacuation operation shall override the automatic evacuation operation or remote assisted evacuation operation.

Manually assisted evacuation is the evacuation under the control of an authorized trained person(s), who controls the lift from car operating panel and assists occupants to evacuate the building.

After the completion of Phase 1 of evacuation service, if the manually assisted evacuation operation signal is active, the lift shall operate as per details below:

- a) Registered landing calls shall be indicated in the car operating panel by blinking car call acceptance light of the given landings;
- b) Where multiple lifts in a group are in evacuation operation, all registered landing calls shall be indicated in each lift. If one of the lifts serves the landing call, the landing call shall be cancelled from the other lift(s) as well;
- c) The active EEL location shall be indicated in the car;
- d) All heat and smoke sensitive door protection devices shall be made inactive;
- e) Constant pressure on any car call button or on the door close button shall cause the car door to close. Closing shall start only after 1 s to 2 s delay. If the button is released before the car door is fully closed, the doors shall automatically reopen;
- f) Car call shall be registered when the car door is fully closed, and the car shall start to travel to the destination landing;
- g) If car doors are closed by the door-close button and car call is not registered within 15 s, the car shall travel to the active EEL and open its doors;
- h) At any time, it shall be possible to register a new call from within the car. The previous car call shall be cancelled, and the car shall travel to the newly registered landing; and
- j) On arrival at any destination, the car shall automatically open its selected door and remain at the landing until the car door is closed by the attendant and a new car call is registered or until timeout expires.

7.2.2.2.3.1 Deactivation of manually assisted evacuation operation

Manually assisted evacuation operation shall stop when the car is at the EEL; and when the manually assisted evacuation operation signal is no longer active or when the suspend service signal is active or when the lift receives Phase 1 signal of the fireman's control.

7.2.2.3 The following decision for evacuation may be taken based on below:

- a) *If evacuation is required (building management)*

This is a decision to be made by other experts and those responsible for building management. In large buildings evacuation may take some time and it shall not be assumed that continued use of a particular lift or lifts is guaranteed. Serious thought shall be given to how information will be displayed to those managing the building especially where the BMS system is capable of providing a large amount of information. It should be possible to see at a glance if lifts are operating correctly and what services they are running. There shall always be provision made to allow the building management or authorities to override any automatic evacuation signal that is generated by a smart BMS. Irrespective of the systems employed the evacuation decision cannot be made by a lift system and therefore other experts in this field shall determine what risks will constitute the need for evacuation and the type of evacuation (partial or full).

- b) *If lifts are to be used for the evacuation*

If for some reason a significant number of lifts are not available for use then it may be prudent to remove all lifts from automatic evacuation service rather than have too few. If too few lifts are in service, delays will result that could be dangerous or cause panic. In such circumstances it is probably desirable to run a limited service for the evacuation of those who have difficulty negotiating stairs. Any such service

needs to be managed by lift attendants who will drive the lift, manage possible crowding and assist those who need it. The decision to operate some form of service when not all lifts are available shall be taken by those managing the building evacuation.

7.2.2.5 Building evacuation information

When evacuation is required, updated information shall be provided through the building systems informing persons concerned including those in waiting areas which lifts to use during the particular emergency. This information shall be in both audible and visual format. Any safe areas/waiting areas should also have an emergency communication system.

7.2.3 Specifications of Automatic Evacuation Lift

7.2.3.1 Determining the number and size of lifts

To calculate the number and size of lifts required for an adequate evacuation service, the handling capacity of a given lift or group of lifts shall be determined considering possibility that a lift or lifts may not be available for some reason such as planned maintenance, repairs, etc. This analysis can be done by the lift designer. Also, due consideration shall be given to the requirement of accommodating a wheel chair in the lift car.

7.2.3.2 Protection of lift equipment

Lift shafts and machine rooms or machinery spaces located outside the shaft should be fully enclosed. The temperature in the enclosures should be monitored and maintained to acceptable levels for the equipment, as determined by the lift supplier in consultation with the building management. Protection from water may be considered, if required as per overall evacuation strategy.

7.2.3.3 Removal or suspension of lifts from evacuation service

Where a lift receives a command from a BMS or manual signal to stop or suspend service to a floor or area of the building, any stop shall be a controlled stop at a safe area. A controlled stop means allowing the lift to slow down and stop at a floor in the normal manner. Where a lift or lifts are instructed to suspend service the lift(s) shall inform the FCC and any BMS once it is no longer available for service.

7.2.3.4 Remote lift car surveillance

At times of emergency, it is vital to be able to see that lifts do not contain trapped passengers who may be incapacitated. A means to display the entire floor area of the car shall be provided with information to the occupants that the lift car is under surveillance. At least one viewing terminal shall be located in the FCC and clearly marked "Lift Car Surveillance" with the lift designation identified.

7.2.3.5 Communication system requirements

As a minimum a 3-way communication system shall be available for passenger use to permit direct communication between the lift car and safe area and FCC. Operation of the communication device in the lift car shall be simply by means of a single button, operation of which shall connect the system to the FCC. Further operation of the device in the lift car shall be hands free.

7.2.3.6 Initiation of evacuation service

Evacuation service can be initiated by a device such as a button or lever, key, etc located near the lift shaft at the evacuation landing, or from a signal sent by a BMS or hazard detection system. Where a button or other manual activated device is used it shall be protected against unauthorized use. Its purpose shall be clearly marked with a symbol and/or the words "LIFT EVACUATION SERVICE".

The evacuation service shall be operational between the floors as decided by the building designer. The logic of operation shall be mutually agreed to between the building designer and the lift designer. However, any lift will be automatically removed or suspended from evacuation service if, where provided, it is turned to firemen operation.

7.2.3.7 Cancellation of evacuation service

On cancellation of a signal from the device that initiated the service and receipt of a signal from an evacuation service cancellation switch, all lifts which are not at the main landing shall complete any allocated task and return to the main landing.

On arrival at the main landing the doors shall open and then the lift evacuation for that particular lift shall be cancelled. Any lift already at the main landing shall on opening its doors cancel its evacuation service.

7.2.4 Information to be Provided to the Building Owner/Operator

Detailed instructions shall be provided to the building owner/operator by the building designer in the form of a manual. The manual shall explain the evacuation strategies to be used, how any detection system operates, how they shall be maintained and how the lift will operate on evacuation service. It should also provide advice on periodic checks the owner/operator can make to ensure the system is working correctly and explain the importance of a suitable testing and maintenance system being in place. Training being very important aspect of this new concept, the manual shall include such details of training as who should be covered by training, how the training should be carried out, what should be the frequency, etc.

8 MINIMUM TECHNICAL AND SAFETY REQUIREMENTS

8.1 General Requirements

The following general requirements shall be adhered to:

- a) Passenger and goods lifts shall comply with the safety requirements and/or protective measures as given below. In addition the passenger and goods lifts shall be designed according to good practices [D-5A(2)];
- b) All labels, notices, markings and operating instructions shall be permanently affixed, indelible, legible and readily understandable (if necessary aided by signs or symbols). *See also* good practice [D-5A(3)]. They shall be of durable material, placed in a visible position, and written in the accepted language(s);
- c) The minimum factor of safety for any part of the lift shall not be less than five. Higher factor of safety for various parts shall be applicable in accordance with good practice [D-5A(13)];
- d) All passenger, goods, stretcher, automobile, dumbwaiter lifts shall have conformity with *Lifts Acts and Rules* in force, *The Electricity Act*, 2003, CEA Regulations and all relevant and applicable Indian Standards, applicable fire regulations, fire prevention act and rules in force;
- e) *Notices* — The following notices shall be placed at respective places:
 - 1) A notice (*see* [Fig. 23](#)) bearing the following (typical) minimum inscription shall be fixed to the outside of doors or trap-doors (excluding landing doors and doors of emergency and test panels) giving access to machine and pulley rooms. Use of notice boards as per relevant standards is recommended, that is danger board shall be as per accepted standard [D-5A(14)]. Use of non-standard notice boards should be avoided; and

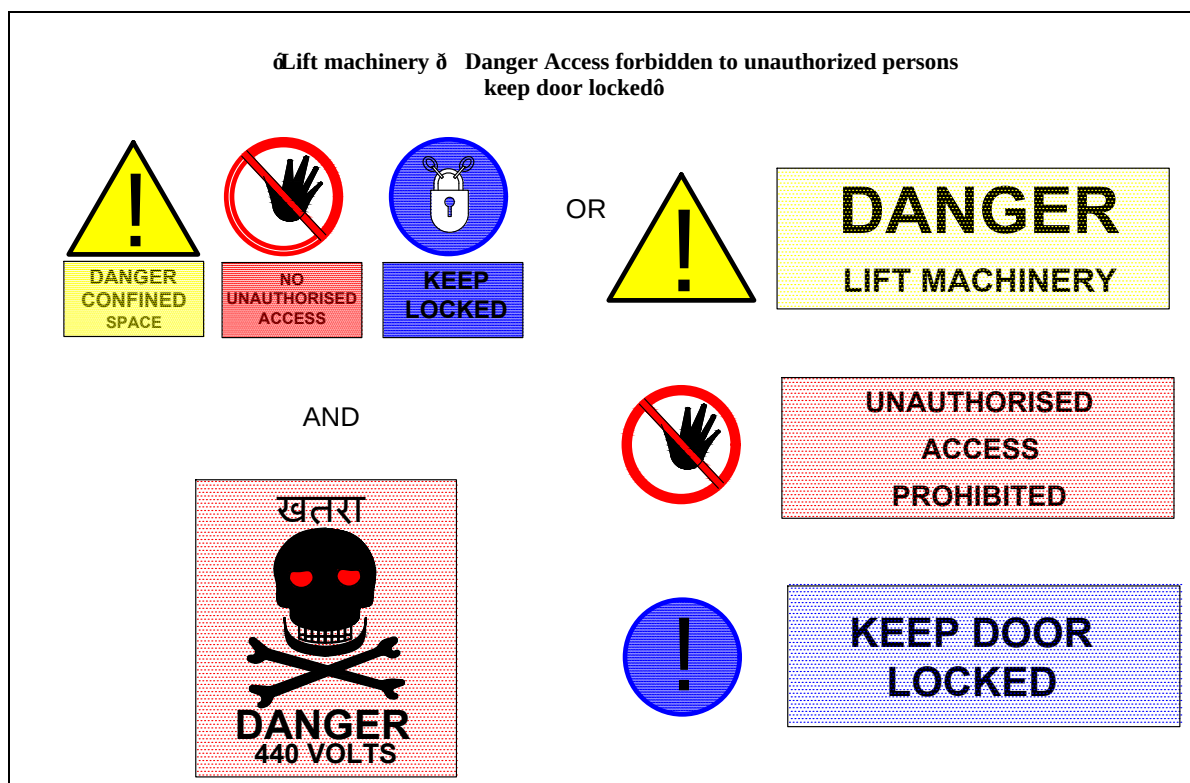


FIG. 23 TYPICAL INSCRIPTIONS ON DOORS

- 2) In the case of trap-doors, a permanently visible notice (see [Fig. 24](#)) shall indicate necessary warning to those using the trap-door.

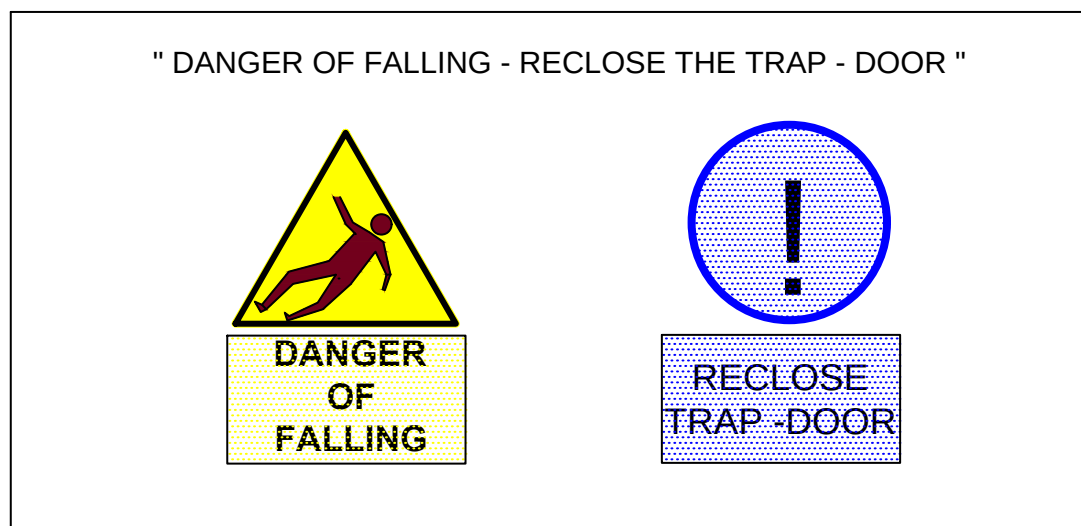


FIG. 24 TYPICAL NOTICE ON TRAP DOORS

- 3) Outside of the well, near the access doors and emergency doors, if any, there shall be a notice as shown in [Fig. 25](#).

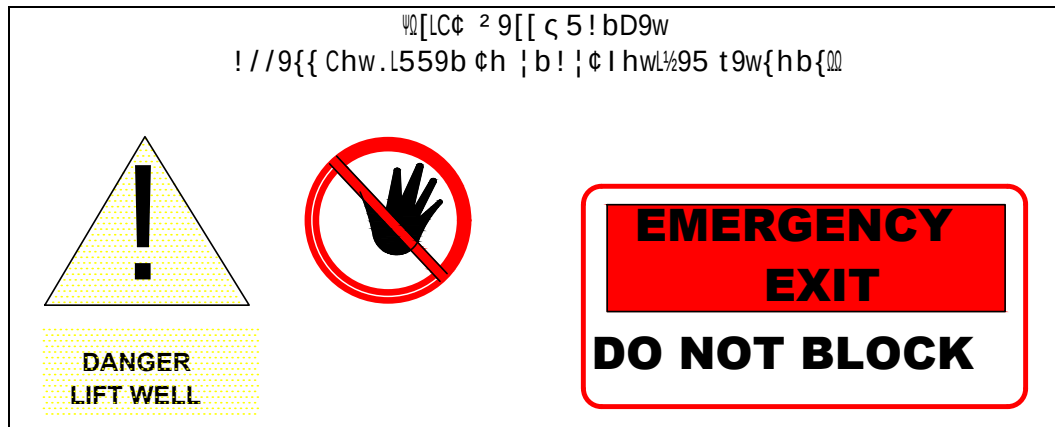


FIG. 25 TYPICAL NOTICE ON ACCESS DOORS AND EMERGENCY DOORS

- 4) A notice board with following instructions as applicable, and/or other applicable instructions shall be placed in conspicuous position in the lift car:
 - i) Maximum capacity of lift _____ kg/ _____ persons;
 - ii) Close the car and landing door/gate properly, on entering, or while leaving the lift car;
 - iii) Do not open the lift-car gate when the lift-car is moving. The gate should only be opened after the lift-car has stopped opposite a landing gate;
 - iv) In case of danger, press the alarm button, but do not try to open the car door. Wait inside, until the lift car is brought opposite a landing, and do not attempt to leave the lift car until the landing door is opened fully; and
 - v) Children under 12 years of age shall not use the lift, unless accompanied by an adult.
- 5) The [Table 36](#) is a typical list of Do's and Don'ts that should be displayed in the lift car, and in case of lifts open to public use, in an accessible format complying requirements of accessible information board/signage, given in good practice [D-5A(3)]:

Table 36 List of Do's and Don'ts	
[Clause 8.1(e)(5)]	
Do's	Don'ts
a) Press call button lightly	a) Do not overload the lift beyond maximum capacity of the lift
b) Know your destination. Push lift call for direction you want to go	b) Do not press call button again and again
c) Stand aside for exiting passenger	c) Do not force-open the doors, it is dangerous
d) Pay attention to floor indicator	d) Do not stand in between the doors
e) Push and hold the door open button, if door needs to be held open	e) Do not fiddle with the equipment
f) Wait for the next car, if the lift is full	f) Do not step in until lift has stopped at landing and doors are fully open
g) Take the stairs, if there is a fire	g) Do not crowd in the middle of car
h) In case of emergency call customer service cell for rescue operation	h) Do not lean against car panels
j) Children of 12 years or below and pets shall always be accompanied by an adult inside the lift	j) Do not panic in case of power failure
	k) Do not attempt to jump out, if the lift has stopped out of floor level
	m) Do not try force open the doors when the lift is in motion
NOTE 8 No other notices should be fixed in the lift car.	

8.2 Power and Control Systems

8.2.1 Features Associated with Power Systems

The features associated with power systems are listed below.

8.2.1.1 Industrial switchgear

Switchgear for controlling lift power systems is characterized by its high duty cycle and its high rupturing capacity. Switchgear shall be robust enough and shall be so designed as to withstand the high duty cycle and high rupturing capacity introduced during the operation of the lifts.

8.2.1.2 Levelling accuracy

The levelling tolerances in accordance with good practice [D-5A(6)] are those which can be reasonably expected between no load and full load in either direction. With the VVVF drives good levelling accuracy can be expected.

8.2.1.3 Corrective levelling

This should only be used, when it is impossible otherwise to achieve the required levelling tolerances or on long travel lifts to maintain the required levelling tolerances during loading and unloading.

8.2.1.4 Levelling with variable voltage

A variable voltage system is one using continuous regulation which minimizes speed differences due to load variation. Therefore, the actual levelling speed is of less importance than the general refinement of its regulation control. In fact, no levelling speed as such may be identifiable.

8.2.1.5 Overload tests

A lift is designed to operate and transport the contract load, at the required duty cycle, and should not by intention or habitually be used to carry overloads. During test as a safeguard to cover variable supply and temperature conditions, a lift is checked for the car to complete one round trip with contract load plus 10 percent at nominal supply voltage and nominal ambient temperature. There is also a static test with contract load plus 25 percent to check that the brake will sustain the car. It is unnecessary to specify an additional overload test or capacity and in fact it is detrimental to the normal running efficiency and safety of the lift to do so.

8.2.1.6 Occasional extra load

It is not good practice to request that a lift should be designed to carry an occasional extra load. It is tantamount to specifying an excessive overload test which is detrimental to the normal running efficiency and safety of the lift.

8.2.2 Description of Operation Systems

The following description relates to the associated operating systems.

8.2.2.1 Methods of control systems

The methods of control systems are as follows [see [8.2.2.2\(a\)](#)]:

- a) Attendant and dual control; and
- b) Automatic push button operation.

8.2.2.2 Types of control systems

The features of control systems shall be as described below:

- a) *Automatic push button operation* — Automatic control is a method of operation by which a momentary pressure on a push button sets the car in motion and causes it to stop automatically at any required lift landing. This is the simplest control system and it is sometimes referred to as push button control. A

car answers a landing or car call, whichever is actuated first by momentary pressure provided the lift is not in use. Momentary pressure of a car push button will send the car to the designated floor. The car always responds to a car push button in preference to a landing push button. With this type of control, a red landing signal light or direction arrow indicates that the car is in use that is the lift is travelling. This type of control is recommended for single passenger lifts serving up to 4 floors and goods lifts.

For special purposes, the following two systems may be considered:

- 1) Dispatch from landings as an additional feature for a goods lift with manually operated doors. The call is registered by pressing the car push button and when the doors are closed the car will travel to the designated floor.
 - 2) Automatic with attendant control as an additional feature on goods lifts with a key operated switch in the car to transfer the control from normal automatic to attendant operation. There is also a visual call indicator with buzzer in the car to indicate to the attendant the landing floors at which push buttons have been pressed when the car is under attendant control.
- b) *Collective control* — Collective control is a generic term for those methods of automatic operation by which calls made by pressing push buttons in the car and at lift landings are registered and answered by the car stopping in floor sequence at each lift landing for which calls have been registered irrespective of the order in which the calls have been made, and until all calls have had attention. Collective control of any form is usually not suitable for goods lifts except where loading is not expected to fill the car and additional loads can be taken at other stops.
 - c) *Single push button collective control* — Single push button collective control has a single push button at each landing. It is not recommended, as the direction in which it is desired to travel cannot be registered by the intending passenger.
 - d) *Down collective control* — Down collective is a control system where landing calls are registered from a single push button, irrespective of the car being in motion or the landing door being open and calls are stored until answered. Any number of car calls can be registered and the car will stop in sequence in the down direction at each of the designated floors. The car will travel in the up direction to the highest call registered stopping only in response to car calls. It will then travel downwards answering calls in floor sequence. If only one call has been registered the car travels to the floor of call.

This system is suitable where there is traffic between the ground and upper floors only and no inter floor traffic. Two or three car banks have interconnected control. With this type of control the following signals are included:

- 1) A landing signal light indicates that the call has been registered and will be answered.
 - 2) Illuminated car position indicator above car entrance.
- e) *Directional collective control for one car* — Directional collective control for one car is a control system having $\uparrow$ and $\downarrow$ push buttons at intermediate landings whereby the call is registered for the intended direction of travel. Calls from the car or landing push buttons are registered and stored until answered. The car will answer calls in floor sequence in one direction of travel. Calls for the opposite direction of travel are answered when the direction of travel is reversed.

This system is suitable for single lifts serving four or more floors with inter floor traffic, such as small office blocks, hotels and blocks of flats. With this type of control the following signals are included:

- 1) A landing signal light for each landing push button indicates that the call has been registered and will be answered.
 - 2) Illuminated car position indicator above the entrance in the car.
 - 3) Arrow shaped signal lights in the back of the car or on the landing to indicate to the entering person in which direction the car is going to depart.
- f) *Directional collective control for two or three cars* — Directional collective control for two or three cars is a system covering a control in which the two or three cars in a bank are interconnected. One push button unit with $\uparrow$ and $\downarrow$ push buttons or floor buttons (in case of car control from floor) are

required at each landing and the call system is common to all lifts. If for architectural balance, in the case of a three car bank, extra push button units are required, these should be specified. Each landing call is automatically allocated to the best placed car. The control is designed so that cars are effectively spaced and thus give even service. When a car reaches the highest floor to which there is a call its direction of travel is automatically reversed when it next starts. One or more cars will return to the parking floor. Automatically bypassing of landing calls when a car is fully loaded is an essential feature for three-car banks. It is also necessary for two-car banks in offices. Other cars will continue to provide service to all floors. When three-car banks serve 7 or 8 floors and over, some form of automatic supervisory control [see 8.2.2.2(g)] is generally necessary in the interest of efficiency. With this type of control the following signals are included:

- 1) A landing signal light for each landing push button to indicate that the call has been registered and will be answered.
 - 2) Illuminated car position indicator above the entrance in the car.
 - 3) Arrow shaped signal lights in conjunction with an audible single stroke gong or an indication on the landing call push button station above each landing entrance to indicate to the waiting person(s) which car is going to stop and in which direction it will continue its course.
- g) *Group supervisory control* — A bank or group of intensive traffic passenger lifts requires a supervisory system to coordinate the operation of individual lifts which are all on collective control and are interconnected. The very nature of intensive service calls for a sophisticated automatic supervisory control system so as to match the speed capacity of these lifts. The supervisory system regulates the dispatching of individual cars and provides service to all floors as different traffic conditions arise minimizing such unproductive factors as idle cars, uneven service and excessive waiting time. The system will respond automatically to traffic conditions such as ‘UP’ and ‘DOWN’ peaks, balanced or light traffic and provides for other specialized features. If desired, a master station can be provided in the lift lobby which gives by indicators, visual information regarding the pattern under which the system is operating. Where the system is based on a definite programme, control means are provided for altering the type of traffic programme. There are other facilities, such as the removal of any lift from service.
- h) *Destination control system* — In case of a bank of lifts having more than 6 lifts or as decided by user, destination control system (DCS) may be provided. The passengers enter their destination floors on the destination operating panels (DOPs) provided on the lift landings. The lift controller allocates a lift to every call registered which is displayed on the DOP. The passenger is expected to board the assigned lift and travel to his destination floor. He need not register his destination floor in the lift car operating panel (COP) again. In fact, COP in the lift car may not be provided with floor buttons in case of DCS control system. The principle behind the DCS is to group passenger with same destinations together thus reducing the number of stops of the lift reducing round trip time and boosting traffic.
- j) *Number of DCS terminals* — Number of terminals at every floor shall be established based on the 5 min peak traffic, however it shall not be less than 2 on every floor.

It is recommended that the DCS input terminals barring one for people having special needs is not located in the position of the conventional hall buttons. It is recommended that the DCS input terminals are positioned outside the lift lobby or at the entrance to the lift lobby or integrated with the turnstiles.

The advantages of DCS system over conventional collective control are:

- 1) enhanced passenger handling capacity;
- 2) less transit time, few intermediate stops; and
- 3) passengers do not have to struggle through crowd to book the calls.

In addition to the DOPs the lift landings shall be provided with lift identifier. Indicators shall be provided on the car jams indicating the car stops.

Hybrid destination control (HDCS) combines features of the DCS and conventional control system. The main floor is provided with DOPs, lift identifiers and the lift car is provided with the conventional COP as well. Thus, while availing benefit of traffic boosting HDCS provides convenience of call booking from inside the lift car as well.

8.2.2.3 Features of operation systems

The features associated with the operating systems are as follows:

- a) *Car preference* ð Sometimes it is necessary to give a special personal service or a house service. When this service is required and for whatever purpose, it should be specified as 'car preference' is activated by a key operated switch in the car. The operation is then from the car only and the doors remain open until a car call is registered for a floor destination. All landing calls are bypassed and car position indicators on the landing for this lift are not illuminated. The removal of the key when the special operation is completed, restores the control to normal service.
- b) *Landing call automatic bypass* ð For collective operation, automatic bypassing of landing calls can be provided. This device will bypass landing calls when a car is fully loaded but the calls are not cancelled.
- c) *Basement service* ð For lifts with collective control when service is required below the main parking floor, which is usually the ground floor, to a basement and/or a sub-basement, the lift manufacturer should be informed of the type of service required, as special technical considerations are then usually necessary.
- d) *Hospital service* ð Lifts for carrying beds and stretchers require a car preference switch so that an attendant can have complete control of the car when required. This requirement should be specified as 'car preference' and it will function as described in [8.2.2.1](#). Otherwise such lifts can have the same control systems as for normal passenger lifts, the choice depending on the number of floors served, the service required and the number of lifts.
- e) *Manually operated doors (without closers)* ð A 'door open' alarm should be provided to draw attention to a car or landing door which has been left open.
- f) *Automatically power closed doors* ð For passenger operation when the car arrives at a landing the doors will automatically open and then close after lapse of a time interval. This time interval can be overruled by the pressure of a push button in the car to give instant door closing. An 'open door' push button is provided in the car to reverse closing motion of the doors or hold them open.
- g) *Controlled power closed doors* ð When there are conditions that particularly affect the safety of passengers or damage to vehicles or trucks, the closing of the doors should only be made by the continuous pressure of push buttons in the car or on landings. A 'door open' alarm should be provided to draw attention to a car or landing door which has been left open. This means of operation is required for some forms of goods lifts.
- h) *Safe operation of doors* ð The safety of passengers passing through lift entrances is fully covered by the provision of good practice [D-5A(13)]. No modification of these provisions should be specified.
- j) *Directorial service* ð There are many forms of giving special service for individuals, but they should always be avoided. They range from key operated switches at preferred landings to the complete segregation of one out of a bank of lifts. It is obvious that any preferential treatment of this nature can seriously jeopardize the efficiency of the service as a whole. When a bank of say three lifts is installed to meet the anticipated traffic requirements and then, when the building is occupied, one lift is detached permanently for directors' service, the traffic handling can be reduced by a half rather than a third. When preferential service is imperative, then the car preference feature should be available [see [8.2.2.3\(a\)](#)].
- k) *Indication of car arrival* ð All lift cars are illuminated when available (in service). It is recommended that this illumination be used to signal the arrival of a car at a landing in preference to special signals such as 'LIFT HERE' signs since signal lamps can fail when the lift is still operating satisfactorily.

The following is the practice adopted for vision panels in doors:

- 1) For lifts with manually operated car and landing doors, vision panels are provided in all doors.
- 2) For lifts with power operated car doors and manually operated landing doors, vision panels are provided in the landing doors only.

- m) *Service switches* – When switches are provided to take cars out of service, that is because the remaining cars in the group can cater for the required passenger traffic, it is essential that such switches should not stop the fireman's control from being operative in the event of the lift being designated as a fireman's lift. Service switches should not be confused with maintenance switches which are only used when it is dangerous to attempt to operate the lift because maintenance work is actually in progress. A control station fitted on top of the car is regarded as a maintenance switch.
- n) *Fire switch* — When required by the fire authority a fire switch has to be provided, the function of which is to enable the fire authority to take over the complete control of one or more lifts in an installation.
- p) *Push buttons and signals* – It is most important that the purpose of every push button and signal should be clearly understood by all passengers.
- q) In public places where visually impaired are expected to use the lifts it is recommended to provide Braille/tactile buttons.

8.3 Capacity and Loading

The minimum rated load of a passenger lift car corresponding to the net inside car area shall bear the relationship given in Fig. 26; the net inside car area being evaluated as shown in Fig. 27. The net inside car areas for various passenger capacities with corresponding minimum rated loads have been given in Table 37 and for goods lifts in Table 38.

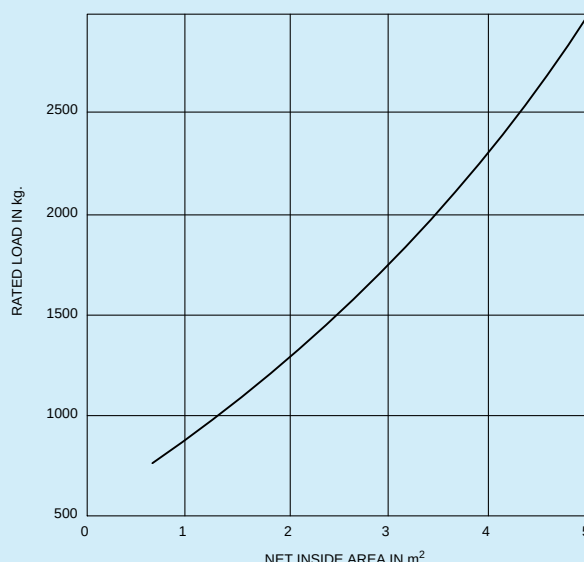


FIG. 26 MINIMUM RATED LOADS FOR PASSENGER LIFT CAR

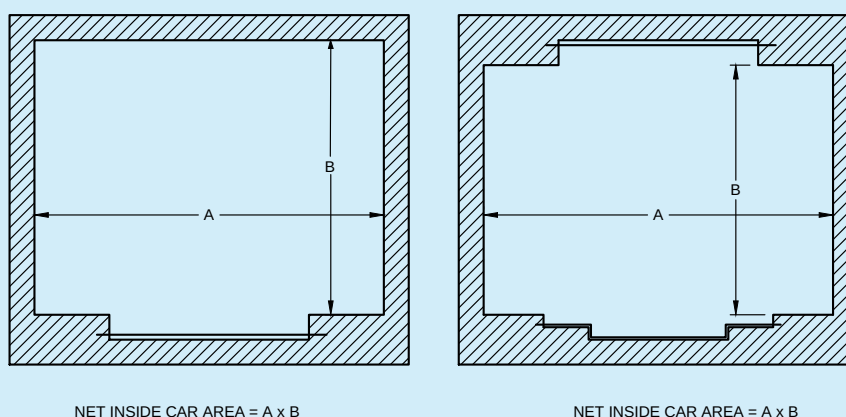


FIG. 27 NET INSIDE CAR AREA FOR PASSENGER LIFTS

NOTES

1 The graph shown in [Fig. 26](#) is based on the following formula:

$$W = 35.05 A^2 + 325.66 A$$

where

W = rated load, in kg; and

A = net inside areas, in m^2 .

2 For rated loads exceeding those covered by the graph, value may be evaluated from the formula given above.

3 To avoid the possibility of serious overloading of bed lifts, they shall be treated as passenger lifts.

4 The minimum rated load for goods lifts shall be based on a load of not less than 3.45 kN/m^2 of the net inside car area.

5 The minimum rated load for automobile lifts shall be based on a load of not less than 1.45 kN/m^2 of the net inside car area.

6 When the load in a goods lift consists of pallets or similar single piece loads loaded by means of a power truck, it is necessary during loading and unloading to take into account the total load on the car platform the capacity of the brake and the resistance to slipping of the ropes on the sheave of a traction machine. Similarly, guide rails, guide fixing, car frame and platform shall be designed to withstand the horizontal thrust imposed by power trucks, motor vehicles and the like.

7 For motor vehicle and other special purpose lifts for transportation of light weight and heavy volume loads, the lift car inside dimensions mutually agreed to between the manufacturer and the purchaser will be permitted with the provision of a device to prevent the lift from starting from a landing in case of overloading.

8 A load plate giving the rated load of the lift shall be filled in each lift car in a conspicuous position. For passenger lifts, the rated load shall be given in number of person and kg. For goods lifts, the rated load shall be given in kg or other convenient units and in persons. For the purpose of this clause, a person shall be regarded as weighing 68 kg.

Table 37 Minimum and Maximum Net Car Areas for Various Rated Loads δ Passenger Lifts

([Table 14](#), [Table 15](#), [Table 27](#), [Table 28](#) and [Clause 8.3](#))

Sl No.	No. of Passengers	Rated Load W kg	Minimum Available Net Inside Area m^2	Maximum Available Net Inside Area A m^2
(1)	(2)	(3)	(4)	(5)
i)	4	272	0.68	0.77
ii)	5	340	0.85	0.95
iii)	6	408	1.00	1.12
iv)	7	476	1.16	1.28
v)	8	544	1.31 (1)	1.45
vi)	9	612	1.46	1.60
vii)	10	680	1.61	1.76
viii)	11	748	1.77	1.91
ix)	12	816	1.92	2.05
x)	13	884	2.06	2.20
xi)	14	952	2.23	2.34
xii)	15	1 020	2.35	2.47
xiii)	16	1 088	2.48	2.61
xiv)	17	1 156	2.62	2.74
xv)	18	1 224	2.75	2.87
xvi)	19	1 292	2.88	3.00

Table 37 (Concluded)

Sl No.	No. of Passengers	Rated Load W kg	Minimum Available Net Inside Area m ²	Maximum Available Net Inside Area A m ²
(1)	(2)	(3)	(4)	(5)
xvii)	20	1 360	3.01	3.13
xviii)	21	1 428	3.14	3.25
xix)	22	1 496	3.26	3.37
xx)	23	1 564	3.38	3.49
xxi)	24	1 632	3.50	3.61
xxii)	25	1 700	3.62	3.73
xxiii)	26	1 768	3.74	3.84
xxiv)	27	1 836	3.85	3.95
xxv)	28	1 904	3.96	4.07
xxvi)	29	1 972	4.08	4.18
xxvii)	30	2 040	4.17	4.29
xxviii)	31	2 108	4.25	4.39
xxix)	32	2 176	4.34	4.50
xxx)	33	2 244	4.44	4.61
xxxi)	34	2 312	4.53	4.71
xxxii)	35	2 380	4.62	4.81
xxxiii)	36	2 448	4.72	4.92

NOTES

1 The maximum available car area is calculated as per formula:

$$W = 35.05A^2 + 325.66A,$$

where

W = the rated load, in kg; and

A = the maximum area, in m².

2 Beyond 36 passengers add 0.109 m² in maximum available area for each extra passenger.

3 Beyond 36 passengers add 0.104 m² in minimum available area for each extra passenger.

4 Hospital lifts shall be treated as passenger lifts and shall be designed according to above table.

5 Minimum net car inside area for fireman lift shall be 1.43 m².

For planning the quantity of lifts (passenger/service/hospital bed/ stretcher lifts) and deciding their main specification in any project, reference shall be made to [4](#). For enhanced personal space in the lift car without impacting the rated load and the car area as specified; while planning for the lifts, passenger occupancy may be considered less than the 80 percent specified in [4](#), thereby providing larger area per person in the lift car. The suggested number of passengers may be displayed in the lobby and in the car alongside the statutory rating plate required as per [5.4.2.3.2](#) of good practice [D-5A(1)], by the user.

**Table 38 Maximum Net Car Areas for Various
Rated Loads — Goods Lifts**
([Table 31](#), [Clauses 8.3](#) and [8.3.1](#))

Sl No.	Rated Load	Maximum Available Net Inside Area A
	kg	m ²
(1)	(2)	(3)
i)	250	0.71
ii)	500	1.34
iii)	750	1.91
iv)	1 000	2.43
v)	1 250	2.92
vi)	1 500	3.38
vii)	1 750	3.81
viii)	2 000	4.22
ix)	2 250	4.62
x)	2 500	5.00

NOTES

1 The maximum available area in above table is calculated as per formula:

$$W = 35.05A^2 + 325.66A$$

where

W = the rated load, in kg
A = the maximum available area, in m².

2 Beyond 2 500 kg, add 0.16 m² for each extra 100 kg.

3 Refer to Notes under [Fig. 27](#).

8.3.1 For goods lifts, the available area of the car may be greater than the value determined from [Table 38](#), but shall not exceed the value in m² determined by dividing the rated load in kg by (3.45 kN).

For motor vehicle lifts, the available area of the car may be greater than the value determined from [Table 38](#), but shall not exceed the value in m² determined by dividing the rated load in kg by (1.45 kN).

8.3.1.1 Class of loading

a) General loading

This applies where the load is distributed, the weight of any single piece of freight or of any single loaded hand truck is not more than 25 percent of the rated load of the lift, and the load is handled on and off the car platform manually or by means of hand trucks.

For this class of loading, the rated load of the lift shall be not less than the load (including any truck) to be carried, and shall in no case shall be based on less than 3.45 kN/m² of inside net platform area.

b) Motor vehicle loading

This applies where the lift is used solely to carry passenger automobiles up to the rated capacity of the lift.

For this class of loading, the rated load of the lift shall be not less than the load to be carried, and in no case shall be based on less than 1.45 kN/m² of inside net platform area.

c) *Concentrated load with/without powered/manual industrial truck*

This applies where the weight of the concentrated load including a powered industrial or hand truck if used, is more than 25 percent of the rated load; during loading and unloading the maximum weight on the car platform may exceed the rated load up to a maximum of 150 percent of rated load; and where the load to be carried does not exceed the rated load.

For this class of loading, the car platform and car frame shall be designed for 150 percent of rated load. The driving-machine motor, safety gear, brake and traction relation shall be designed to sustain and level the full 150 percent of rated load. The guide rails, guide fixings, car frame and platform shall be designed to withstand the horizontal thrust imposed by power trucks, motor vehicles and the like.

The rated load of the lift shall be not less than the load (including any truck) to be carried and shall in no case be based on less than 3.45 kN/m² of inside net platform area.

d) *Other loading with heavy concentrations*

This applies where the weight of the concentrated load including a powered industrial or hand truck, if used, is more than 25 percent of the rated load; where the static load during loading and unloading does not exceed the rated load; and where the load to be carried does not exceed the rated load.

For this class of loading; the guide rails, guide fixings, car frame and platform shall be designed suitable for the forces due to the heavy concentrated load and its location in the car. The rated load of the lift shall be not less than the load (including any truck) to be carried, and shall in no case be based on less than 3.45 kN/m² of inside net platform area.

8.4 Other Technical and Safety Requirements

8.4.1 All lifts shall be provided with one or more car door(s) and landing doors, passenger lifts shall be provided with power operated doors which are imperforate.

8.4.2 Power operated car doors on automatically operated lifts shall be so designed that their closing and opening is not likely to injure a person. The power operated car door shall be provided with a sensitive device which shall automatically initiate reopening of the door in the event of a passenger being struck or is about to be struck by the door, while crossing the entrance during closing movement. The effect of the device may be neutralized:

- a) During the last 58 mm of travel of door panel in case of side opening doors;
- b) When panels are within 58 mm of each other in case of centre opening doors; and
- c) During emergency fire operation.

The force needed to prevent the door from closing shall not exceed 150 N and this measurement shall not be made in the first third of the travel of the door. In order to achieve this, it is desirable that all power operated doors have a full length (covering at least 1 600 mm of car door height from bottom) infrared or equivalent light curtain safety to retract the door in the event of coming across any obstacle during closing of the door.

8.4.3 Single speed and two speed drives which are poor in levelling accuracy and energy consumption shall not be used for new lifts in view of availability of latest technology energy efficient variable voltage variable frequency (VVVF) drive systems with improved levelling accuracy.

When a lift moves up or down in negative load conditions, that is empty car/less than 50 percent load in UP direction or more than 50 percent load/full load in DOWN direction and during lift system deceleration, the lift motor generates energy which is lost in the form of heat in set of resistors in traditional non-generative drives. The additional heating could also result in additional cooling costs to the building owner depending on the size, duty and number of lifts in a building.

The regenerative drive converts the energy generated by the motor and feeds it back to the building internal electrical grid thereby eliminating the need of using braking resistors. This energy can be utilized to feed other loads or users connected to the same network. The amount of energy savings due to regeneration depends on various system parameters and configurations such as car load, speed, length of run, traffic pattern and system efficiency. The efficiency of the regenerative drive in both motoring and generating modes should not be less than 0.8. Regenerative drives lower overall building operating costs, delivering significant annual savings to

building owners and tenants year-after-year during the life of the lift. These drives help to reduce peak power demand and energy consumption. As a result, both the fixed costs based on peak power demand (kVA) and variable costs based on energy consumption (kWh) decrease. Depending on different parameters (traffic, load, speed, etc) energy saving can be up to 40 percent. It reduces the cost of the air conditioning system and eliminates excess heat in the building that would have been otherwise wasted.

If a stand-alone generator or uninterrupted power supply (UPS) is used to feed power supply to the lift system, the energy fed back to the stand-alone generator or UPS during negative load conditions or deceleration cannot be consumed. It is therefore imperative to connect sufficient load to the generator (other than lift load), which can consume the regenerated energy. If UPS is used, it shall be capable of handling the recuperated energy. If such measures are not implemented, it will lead to frequent breakdowns of the lift or failure of regenerative drive. Building owners should consult lift manufacturers to understand the amount of regenerated energy and implement suitable measures.

8.4.4 For passenger lifts with car call button control in car, an additional car operating panel with call buttons shall be provided for ease of access to call buttons when the car is wider than 1 600 mm and/or the rated capacity of the lift is 16 passengers and more. This additional car operating panel shall be suitably located in the lift car. The faceplate of an operating panel is recommended to be in contrast to its surrounding background.

8.4.5 Battery operated automatic rescue device (ARD) shall be provided on all lifts in accordance with [D-5A(1)] to take lifts to the nearest possible landing in case of power failure, which includes power failure of one or more phases. In case of lifts equipped with backup generator with automatic mains failure (AMF) panel, the provision of ARD is optional. For all lifts with ARD, an audio and visual indicator shall be provided inside the lift car to alert the persons trapped inside that they are being rescued. Capacity of batteries shall be such that minimum three rescue operations can be performed with full charged battery, without recharging. Emergency light and at least one fan inside the lift car shall be operational till the end of rescue operation. Levelling accuracy of ± 40 mm or better shall be achieved in the ARD Operation. The ARD should operate only when all safety circuits are operational. The safety circuits shall be checked and verified continuously during the rescue operation.

8.4.6 Minimum one stretcher lift shall be provided in each building having height more than 24 m so as to take care of medical as well as other emergencies. This lift shall have minimum car entrance width of 800 mm and car depth of 2 100 mm.

8.4.7 Every passenger lift shall be provided with an overload device in accordance with [D-5A(1)].

8.5 Painting at Works and on Site

Lift equipment will normally receive a protective coat of paint at works before dispatch to site. Further painting of lift equipment may be necessary and is normally in the form of a finishing coat and can take place on site. Alternatively, the further painting of the equipment may be carried out at works as a finishing coat with normal touching up after site erection as may be necessary. Any additional painting, due to site conditions during erection and/or final operating conditions in the premises, is subject to negotiation between the lift manufacturer and the purchaser. Decorative finishes are a subject for separate negotiation.

8.6 Emergency Manual Rescue

The emergency manual rescue operation with brake release should only be carried out in an emergency and by authorized persons who have received necessary instructions, because it is dangerous for any other persons to attempt to do so.

Before attempting to move the car, it is imperative that any person in the car be warned of the intention to move the car and that they shall not attempt to leave the car until they are advised that it is safe to do so. Any failure to carry out this precaution may render the person concerned guilty of negligence should an accident occur.

Before attempting to hand wind the lift machine, it is vital that the supply is switched off at the main switch. It is usually necessary to have two persons in the machine room: one to operate the brake release and the other to carry out the hand winding. The exceptions are small lift machines where the hand winding can be easily controlled by one man and larger machines which need two men to operate the hand winding alone with an additional man to control the brake release. In case of gearless machines especially, the brake shall be released and applied at very

short intervals (of the order of 0.5 s to 1 s) so as to avoid the lift car attaining dangerously high speed due to the lift car and create unbalance in the absence of the gear box with large reduction ratio.

If the car is stuck in the lift well and cannot be moved when an attempt is made to move it in a downward direction, then no attempt at hand winding should be made because the car safety gear may have set. Any further procedure should be carried out under the instruction of a qualified lift mechanic. Provided the car is free to be moved in the downward direction, then it should be hand wound to the nearest floor. There is a preference to move the car in down direction. However, this may not always be practical owing to the distance involved and the time taken to complete the movement. In addition, the amount of out of balance load on the counterweight side, due to the size of car and the small number of persons inside it, may make it necessary to wind the car upwards.

In the case of high speed lifts the direction of hand winding will usually be governed by the effort required to move the car because of the absence of a large gear reduction ratio. It is essential that all detail operations be carried out according to the manufacturer's instructions for the lift concerned and these should be clearly stated and permanently displayed in the form of a notice in the machine room.

8.7 Flood Sensors

Water flood sensors shall be installed in the lift pit to monitor ingress of water levels. A warning hooter/alarm shall be provided to alert of any potential ingress of water in lift pit area. Two water level sensors to be installed in the pit to provide two input signals to the control system; one input is for warning and the other input is for shutdown of lift. When the first switch is activated, an alarm/hooter is to be generated, so that maintenance personnel can understand the situation of water ingress in the pit and eliminate the danger in advance by removing the water safely. In case action is not taken with first switch alarm and when the second switch is activated due to the rise in water level position which is going to affect the operation of the lift (for example, the potential submerged tension device switch), the lift shall stop at the nearest landing level and an alarm needs to be triggered again (see [Fig. 28](#)).

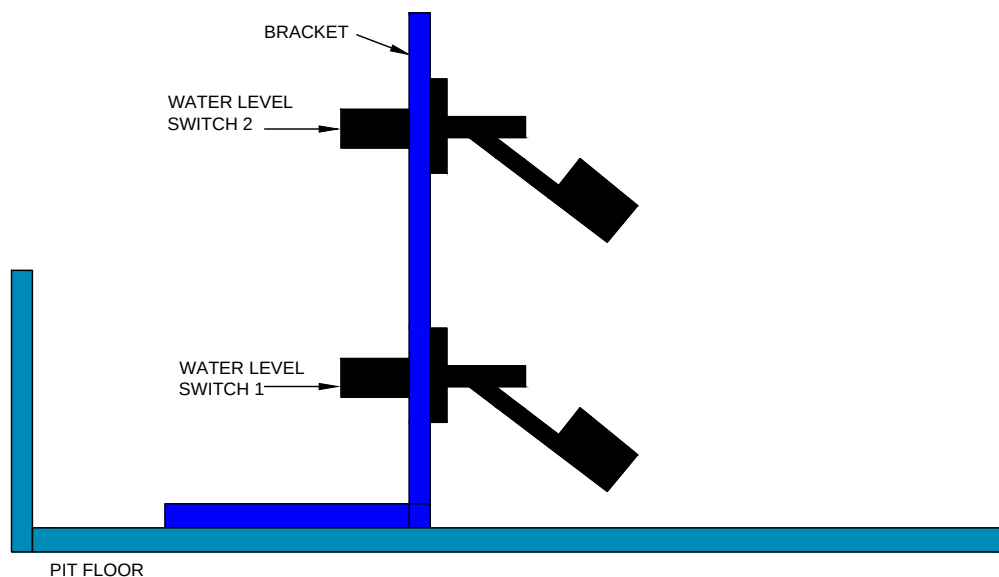


FIG. 28 FLOOD SENSORS (TYPICAL REPRESENTATION OF ARRANGEMENT)

The water level Switch 1 is recommended to be installed 50 mm to 100 mm above the pit floor to detect the low-level water accumulation. The water level Switch 2 is recommended to be installed at 200 mm to 300 mm above the pit so as to ensure that the water level detection switch lies below the lift's any electrical switch such as buffer and safety switches which monitor the safe operation of the lift.

The pit water ingress signal from water level detection switch may be used to connect to a water pump to remove water from pit.

9 SPECIAL TECHNICAL REQUIREMENTS

The objective of this clause is to define some special type of lifts that are used currently in buildings such as lifts without conventional machine rooms (MRL lifts), lifts used in private apartments (home lifts), hydraulic type of lifts and lifts required for differently abled persons, etc.

Additional requirement for lifts to be installed in hazardous (as classified) locations shall be considered. The electrical equipment or systems shall comply to Part D Building Services, Section 2 Electrical and Allied Installations. Also, equipment shall need valid statutory approvals as applicable.

9.1 Lifts without Conventional Machine Rooms (MRL Lifts)

9.1.1 MRL lifts are a special type of lifts where hoisting machine is placed inside the lift shaft and control system may be inside and/or adjacent or at close proximity to top landing door, eliminating the need for conventional machine room.

9.1.2 The MRL lifts shall be in conformance with good practice [D-5A(15)] apart from other relevant provisions of this Section.

9.1.3 MRL lifts may be provided with emergency door on the top portion of the hoist-way which is equipped with electromechanical lock wired in series with lift safety chain and of size 700 mm $\times$ 700 mm minimum, so as to access the machinery space in case of emergency or safety device being activated on or above the top landing.

9.2 Lifts Used in Private Apartments/Houses/Villas (Home Lifts)

9.2.1 The home lift is designed especially for private home having up to 5 stops (maximum rise 12 m), where the usage of the lift is restricted primarily to the residents of the private home. Unlike conventional lifts which allow virtually unlimited access to members of the general public, in case of home lifts non-residents shall have limited access.

9.2.2 The rated capacity of the home lift shall be minimum 208 kg (3 persons) and maximum 272 kg (4 persons) and lift car speed shall not exceed 0.2 m/s.

9.2.3 The home lifts shall conform to good practice [D-5A(1)] and [D-5A(16)] apart from other relevant provisions of this Section.

9.2.4 Electric MRL type home lifts, shall in addition conform to the technical requirements specified in good practice [D-5A(1)].

9.2.5 Hydraulic type home lifts shall in addition conform to the technical requirements specified in good practice [D-5A(1)].

9.3 Hydraulic Lifts

9.3.1 The hydraulic lift is a special type of lift where the lift car is directly or indirectly driven by action of one or more hydraulic jacks.

9.3.2 The hydraulic lifts shall conform to good practice [D-5A(1)] apart from other relevant provisions of this Section.

9.3.3 These lifts should preferably be hole-less type of lifts.

9.4 Lifts Accessible for Persons with Disabilities

9.4.1 All lifts open to public shall adhere to the accessibility provisions in terms of size of the car, door width, control panel, flooring, finishes, handrails, communication systems and information, given in good practice [D-5A(3)].

9.4.2 Lifts for differently abled persons shall conform to good practice [D-5A(17)] apart from other provisions of this Section.

9.5 Seismic Resistance in Lifts

When seismic protection of the passengers and equipment is a consideration, the safety requirements and protective measures given in good practice [D-5A(18)] may be adopted, the specified requirements being applicable when the units are subject to seismic conditions where the specified design acceleration (a_d) $\geq 1 \text{ m/s}^2$.

NOTE δ The design acceleration is a function of ground acceleration, soil behaviour, importance factor, etc. Thus, the value of (a_d) applicable to the building and to be considered for lift design/installation shall be provided by the building architect or structural designer to the lift manufacturer.

9.6 Automobile Lifts

Where required, lifts suitable for moving passenger vehicles such as light motor vehicles (LMV), small utility vehicles (SUV), etc, may be provided. The minimum requirements to be considered while providing such lifts are as given in [9.6.1](#) to [9.6.8](#).

9.6.1 The minimum and maximum car area requirements for passenger lifts as defined in **6.1** of good practice [D-5A(1)] may not be applicable for automobile lifts. For this class of loading, the rated load shall be based on not less than 1.45 kN/m^2 of inside net platform area.

9.6.2 Strengthening stiffeners may be added to car panels to have sufficient strength to avoid permanent or temporary deflection of car panels beyond limits in case the vehicle touches the lift car panels while moving in or out. Under no circumstance, it should result into unsafe condition.

9.6.3 The hall button for calling the lift may be provided in the approach way at such a location that the button is accessible to the vehicle driver in the driving position without having to get off the vehicle. Optional light ray detection system or card reader system may be provided for automatic detection of vehicles and calling the lift to the floor.

9.6.4 Minimum entrance width of such lifts may be 2 400 mm and entrance height of 2 300 mm. The minimum car inside dimensions may be 2 500 mm wide and 3 300 mm deep and typically with entrance doors on both sides of the lift. This will enable the car to be driven inside and can be taken out in the same direction, without any need to reverse the same. Minimum load carrying capacity shall not be less than 2 500 kg.

9.6.5 Barricades may be provided outside the lift entrance door(s), so as to limit the size of the vehicle and preventing oversized vehicles from entering the lifts and possibly damaging the equipment.

9.6.6 There shall be sufficient place on both sides of the vehicle, once the vehicle is inside the lift. The lift car width should be selected such that in case of emergency while the automobile is in the lift car the driver should be able to open the automobile door and come out of the automobile.

9.6.7 The car operating panel, where the car call can be given, may be provided at such a location that is accessible by the vehicle driver in the driving position without having to get off the vehicle. The time duration for which the door remains open upon arrival at landing floor or upon stopping at landing floor shall be sufficient enough so as to enable smooth movement of the vehicles in and out of the lift car. Lift car entrance door(s), shall be equipped with light curtain device in accordance with **19.1.4** of good practice [D-5A(6)].

9.6.8 Overload warning device shall be installed in accordance with **6.2** of good practice [D-5A(1)].

9.7 Air Conditioning System for Lift Cabin (Car)

Where the lift car is equipped with air conditioning system, following additional requirements are recommended:

- a) The whole equipment, its container, condenser, etc, should be typically made of corrosion resistant material, with average designed life span of 10 years.
- b) The air conditioning system should work on single phase electric power supply of maximum 230 V a.c. and should be well within the current carrying capacity of trailing cable cores provided for the air conditioning system. The power supply for the air conditioning system shall be provided from RCCB of suitable rating. This RCCB shall be located in the lift machine room with proper identification and lock out tag out facility.

- c) The air conditioning system should be provided with suitable air filters which should ensure clean air inside lift car. The filters should be easily accessible and serviceable from top of the lift car in a safe and easy manner.
- d) The lift air conditioning should typically include functions such as:
 - 1) Cooling, with adjustable temperature setting provided on the unit itself or through remote control unit, which may be maintained in the lift car.
 - 2) Heating, with adjustable temperature setting provided on the unit itself or through remote control unit, which may be maintained in the lift car.
 - 3) Ventilation, with adjustable air flow having less than 52 dBA noise level in the car.
 - 4) De-humidification, with effective condensation water management system.
- e) The lift air conditioning should be equipped with effective condensation monitoring and condensed water recycling system so as to avoid water spillover or leakage on top of the lift car and other lift components, as detailed below:
 - 1) Condensation water monitoring and dealing with it is a very important feature any lift air conditioner shall have. Condensation water leakage or spillover on the lift car top may cause damage to the lift and may lead to electrocution or even fire.
 - 2) When the condensation water level in the reservoir reaches high level and remains at that or above that level for three minutes, the compressor should stop working and the air conditioner should go into ventilation mode. In the ventilation mode the air conditioner fan still works, water pump still runs, the water is pumped to the condenser for condensation water to be vaporized. When the water goes below 80 percent, the compressor may restart and the air conditioner may go into cooling mode.
 - 3) When the condensation water level reaches 100 percent, and remains at that level for 1 min, to avoid the water spill over and damage to lift components, the air conditioner should power off by itself. All components of air conditioner should stop working. This air conditioning unit shut down should not be automatically resettable and should require manual intervention. Also, it should be possible for this condensation water to be drained out by opening the drain plug provided. This facility of draining off excess condensation water shall be safely accessible either from lift car or lift pit.
- f) It shall be possible to manually switch *ON* and *OFF* the lift air conditioning unit either by key switch provided in the lift car or by remote control.

9.8 Lift Interface with Building Management System (BMS)

Where required, a means to interface, monitor and control be provided by lift manufacturer, which should become the part of building management system. The following information and controls, but not limited to those, may be provided either through potential free contacts or building automation (BA) interface device suitable for providing necessary lift information over other accepted communication protocols. With potential free contacts only lift monitoring is possible. It shall not be possible to control the lift.

a) *Monitoring signals:*

- 1) Floor position of the lift for each lift.
- 2) Direction of travel for each lift.
- 3) Load condition of each lift specially indicating overload status.
- 4) Lift door status of each lift, whether open or closed.
- 5) Lift modes such as attendant, automatic, independent operation, emergency fire service, seismic mode, emergency medical service and inspection mode for each lift.

b) *Control signals:*

- 1) Bring the lift down to lobby floor in case of emergency or VIP movement.
- 2) Park and shutdown lifts based on building and traffic requirements, so as to save power.
- 3) Force the lift into emergency fire service, seismic mode or emergency medical service, etc through command from BA system based on respective sensors such as smoke detectors, seismic sensor, etc input to BMS.

- c) *Reports* — Software tools may be developed to report out performance of each lift such as average response time, number starts per hour, average lift uptime, round trip time, etc with the flexibility to provide additional reports based on building requirements.

The above-mentioned requirement for potential free contacts, other standard protocols, etc, may be optional, if the full-fledged lift management system software is provided by lift manufacturer for monitoring and control of lifts in the building. This software may be custom built providing above mentioned requirements. The wiring between lift machine room to building management systems shall be planned and carried out by the builder along with other wiring in the building.

9.9 Closed Circuit Camera Inside Lifts for Monitoring (CCTV)

Where required, closed circuit camera is provided either by lift manufacturer or by third party supplier, inside the lift car for security purposes. Suitable warning mentioning 'you are under CCTV surveillance' shall be posted inside the lift car. These cameras may be located at ceiling level and at any location which cannot be easily accessed, noticed or tampered with.

The cable used for connecting in car camera to monitoring system, provided by supplier, shall be of trailing grade. Ordinary co-axial or twisted pair wires shall not be lashed or tied along with trailing cables. Trailing cables are specialized cables used in lift applications. Wireless CCTV may also be considered alternatively.

If the camera signal is disconnected, either because of fault or intentionally, the lift monitoring system should get an alert indicating the same. Responsibility of monitoring, recording, organizing and maintaining CCTV shall be duly decided in advance.

9.10 Emergency Medical Service (EMS) for Hospitals

Optional emergency medical car operation may be provided especially for hospitals and operation theatre lifts and the lift behaviour in this mode is as described below. Activation of the emergency medical hall switch activates the in-car buzzer and illuminates a lamp in the car and hall. This forces the car to cancel car calls, reassigns hall calls, and return non-stop to the defined medical landing. Upon arrival, the door opens and the buzzer is deactivated.

The door remains open until the switch is turned off or the doors have been open for at least 60 s. If the medical switch is toggled, this operation remains in effect for additional 60 s. The second phase of EMS operation can only be activated when the doors are open at the medical landing. Once the car is on emergency medical operation fire service operation cannot be initiated, even though the fire service lamp and buzzer are activated.

If the car is performing fire service operations, emergency medical operations cannot be initiated, even though the medical lamp and buzzer are activated. The car automatically enters the emergency medical car operation after the doors are open at the medical return landing. If the emergency medical car switch is not activated within 60 s, the medical operation is cancelled.

To move the car, the emergency medical car switch needs to be activated. Prior to activation, the door cannot be closed and car calls cannot be registered. An active car switch allows car calls to be registered, which automatically closes the door and moves the car to the target landing. At the target landing the door opens and remains open until another car call is registered. The door close button has no effect on this operation.

If the car medical switch is deactivated away from the medical return landing with the doors open, the car remains at the landing not allowing car calls to be registered or the door to be closed. The car shall be returned to the medical landing and the emergency medical car switch deactivated to remove the car from emergency medical car operation.

9.11 Destination Control System (DCS)

The destination control system (DCS) uses passenger destination information to improve traffic flow and travel time in busy mid- and high-rise buildings. Instead of using standard hall call buttons, passenger enters specific floor number he wants to travel to, before entering the lift. The system groups passengers traveling to nearby floors to the same car.

9.11.1 Conventional destination control system requires keypads or touch screens at all floors to book the calls. This conventional DCS may be additionally provided the car operating panel (COP) in the lift car to improve passenger comfort. Alternatively, the COP in the lift car may be hidden. Traditional hall lanterns are no longer needed.

9.11.2 Hybrid destination control system requires keypads or touch screens only at the main floor allowing up-peak boosting of traffic. Traditional hall button stations and hall lanterns need to be provided at all other floors. A full COP is required in the lift. With both conventional and hybrid DCS provided with touch screens and/or keypads, hot button(s) may be provided to facilitate call booking to common floors as fast as possible.

9.11.3 Identification of Lifts

To avoid confusion between identification of lifts and floor nomenclature, particularly when braille is considered, it is recommended that the lift identification should be alphabetic and floor nomenclature to be numeric.

9.11.4 DCS Car Loading for Simulation

While 80 percent is considered as the average loading for traffic analysis with conventional controls, the car loading for simulation with DCS shall not exceed 80 percent

9.11.5 It is advisable that DCS input keypads at main floor to be placed at the entrance of lift lobby and not at conventional location (between two lifts) to avoid multiple calls by same person. When DCS keypads are integrated with turnstiles, additional DCS keypad to be provided in the lobby to re-register the call in case a person fails to enter lift.

9.11.6 Conventional control system allows maximum of 8 lifts in group operation, having not more than 4 lifts in row, however DCS provides an advantage of increasing number of lifts in group operation, which can be extended to 10 lifts in a group to 16 lifts in a group. The designer should note that arrangement for lifts shall be limited to 5 lifts in row.

9.12 Special Environments

Standard equipment is suitable for use inside normal residential, commercial and industrial buildings but when unusual environments are likely to be encountered, the advice of the lift manufacturer should be sought at the earliest possible stage to enable the most economic satisfactory solution to be found. Special mechanical protection and or electrical enclosures may be necessary as well as compliance with statutory or other regulations and with the purchaser's particular requirements, which should be fully considered at the time of enquiry.

Examples of situations which necessitate special consideration are:

- a) Exposure to weather, for example, car parks;
- b) Low temperatures, for example, cold stores;
- c) High temperatures, for example, boiler plant;
- d) Hosing-down, for example, for hygiene or decontamination;
- e) Corrosive atmosphere, for example, chemical works;
- f) Dusty atmospheres, for example, gas plants;
- g) Explosive and inflammable atmosphere, for example, gas plants, and petroleum and polyester industries; and
- h) Nuclear/heavy water plants.

10 PERFORMANCE REQUIREMENTS FOR LIFTS AND EVACUATION LIFTS

10.1 It is necessary to have uniformity in the definition, measurement, processing and expression of vibration and noise signals that comprise lift ride quality. The aim of such uniformity is to benefit users by reducing variability in the results of lift ride quality measurements caused by differences in the methods of acquiring and quantifying the signals. Experience indicates that evaluation of vibration in terms of peak-to-peak levels is of particular relevance to passenger comfort. It is considered necessary to provide a dual form of expression, quantifying both the maximum peak-to-peak and A95 peak-to-peak vibration levels.

10.2 Measurement of Lift Ride Quality

Ride quality of all lifts shall be measured in accordance with accepted standard [D-5A(19)]. The selection and calibration of measuring and test equipment shall also be in accordance with this standard.

Measurements shall be made during normal building conditions, after the lift has been running under normal daily operation for at least one hour. Vibration measurements shall be made at the geometric centre of the car.

11 SPECIAL TECHNICAL REQUIREMENTS FOR SUPER TALL BUILDINGS

The provisions given hereunder describes the different physical phenomena that occur in super high-rise buildings (> 200 m high) and influence vertical transportation solutions.

Although there are no major differences regarding suspension and guidance systems between high-rise and low-rise lifts, there are other aspects to be considered such as system technology and passenger comfort levels. Control system artificial intelligence levels, safety requirements, load-bearing component ratings, rope weight, energy consumption, and thus cost increase exponentially with the height of the building.

11.1 Air Pressure Effect

In high rise buildings, air pressure is different on ground level and on top of the building. Practically, pressure change on the ear becomes uncomfortable for some persons when lift is landing faster than 10 m/s. The thresholds are lower for small children, elderly persons and passengers having flu. Therefore, practically, as a standard solution lifts are offered up to 10 m/s maximum speed.

11.2 Piston Effect

If high speed lifts, especially those having rated speed ≥ 4 m/s, are housed in single shaft, the displaced air due to movement of the lift car tries to escape through the gaps between the lift car walls and the shaft walls in turbulent flow. This phenomenon known as piston effect causes the ride quality in the lift to deteriorate. Noise level in the lift car may increase along with vibrations which will be felt by the passengers. The air being displaced past the car generates whining and whistling noises inside the car. Buffeting when passing doors may be audible and passenger may feel uncomfortable. Both lift car doors and landing doors may whistle at high speeds.

Therefore, high speed lifts should be preferably housed in common shaft with ventilation holes in the common wall(s), if any. These holes allow the air displaced due to the movement of the lift car to escape in the adjoining hoist-way. Thus, the velocity of the escaping air, which depends on the ratio of the lift car area to lift shaft area, reduces to mitigate the piston effect ensuring good ride comfort for the passengers. The size of the vent and the pitch at which the vents need to be provided depend on lift car size in relation to lift shaft size, as well as the speed of the lift and the ride comfort expected.

It is recommended to eliminate common wall between the shafts for lifts with speed ≥ 4 m/s. Shaft should be provided with divider beams at every bracket level for the support of the guide rails. Metallic wire mesh shall be provided for separation of shafts.

If a lift with speed ≥ 4 m/s is housed in an enclosed shaft which is standalone, it is recommended to increase the shaft size by 25 percent to 30 percent.

NOTE δ The percentage increase in shaft is given as a range to have flexibility for builders and lift manufacturers to accommodate the lift and reduce the piston effect.

In addition, high speed lifts especially those having rated speed more than 6 m/s may have cars with aerodynamically designed exteriors. This special construction of the car however increases the requirements of the overhead and pit for which lift manufacturer should be consulted. To reduce hammering sounds caused by the passage of the lift past the landing doors, shaft in front of the lift car door may be clad with facia plates equal to full shaft width between the landing door sill of the upper floor and door header of the lower floor door to create an even, flush surface. Lift cars also need to be specially constructed to absorb noise and thus reduce transmission of noise in the car.

11.3 Stack Effect

Stack effect is defined as the vertical, natural air movement throughout a high-rise building caused by the difference in temperatures between the inside air and the outside air. Positive stack effect is characterized by a strong draft from the ground floor to the roof. Positive stack effect is more significant in cold climates because of the greater difference in temperature between the inside and the outside of the building. The colder the weather and the taller the building, the greater will be the stack effect. Negative stack effect can also occur in the reverse direction in hot climates but is not as significant because the difference in temperature is not as great.

The magnitude of stack effect is a function of:

- a) building height;
- b) air tightness of exterior walls;
- c) air leakage between floors; and
- d) difference in temperature between inside air and outside air.

The stack effect has following effects:

- 1) *Differential pressure on the lift landing doors* — The stack effect generates differential pressure on the landing doors. In case of buildings with airtight separation of each storey the differential pressure on a particular landing door from inside and outside of the lift shaft is small, however, the pressure on the door of the first landing and that on the last landing may vary a lot.

In case of buildings with no air partitions the converse is true. The differential pressure on a particular landing is significant and is the maximum at the topmost and bottom most landings which are farthest from the neutral pressure level. At the neutral pressure level, which occurs somewhere in the mid section of the building height, the differential pressure on the landing door from inside and outside of the lift shaft is nil. The differential pressure on the landing door may cause difficulty in the operation of the door; and

- 2) *Smoke movement* — The stack effect influences the smoke movement. In case of positive stack effect, the air moves out from core of the building towards the exterior of the building above the neutral pressure level and it moves in from the exterior of the building to the core of the building below the neutral pressure level. In case of negative stack effect, reverse phenomenon occurs. This air movement naturally affects the smoke movement in case of fire and must be considered.

Since stairwells, lift shafts and lifts tend to contribute to the stack effect it is essential that all lift lobbies are well sealed, and lifts parked with door closed to minimize stack effect. Especially in case of fire the stack effect needs to be controlled to prevent the spread of smoke in the building. On the other hand, the stack effect may be beneficially used to quickly drive smoke out of the building in case of fire.

11.4 Building Sway

The wind induced sway of the building may cause the lift suspension means and/or compensation ropes and lift car travelling cables to sway. Buildings have natural frequencies of oscillations. Similarly, lift ropes also have natural frequencies of oscillations which depend on the position of the car in the hoist-way amongst other parameters. When natural frequencies of the ropes and the building match, a large amount of energy is transferred to the ropes, and they start swaying violently. This results into reduced ride comfort for the passengers in the lift and increases possibility of ropes getting stuck to snag points in the shaft and damaging shaft equipment. In general, the rope and travelling cable resonance phenomenon should be considered in the lift design for over 200 m tall buildings and slender buildings over 150 m tall, for example, TV towers. It is a fact that the resonance

cannot be completely avoided in high rise buildings, hence the need for the solutions to cope with the phenomenon. If resonance is not considered in the lift design, it could increase the lift down (out-of-service) time and decrease lift ride comfort.

The main purpose of the sway management is to improve lift safety and service time by reducing rope sway by optimizing tension of compensation ropes, providing shaft protection equipment to prevent ropes from entangling and damaging the lift equipment. A sway detector may be used to control the lift operations. The output signals from the sway detector are used to reduce the speed of the lifts or park them at a safe floor.

In order that the sway analysis is done, the building designer should be in touch with the lift designer since early design stage of the building and share information on the building natural frequencies, maximum amplitude of building displacement with probability of occurrence, etc. The lift designer would then be able to calculate the rope/cable sway and decide on the preventive measures to be taken in the lift design.

11.5 Ride Comfort

A robust car guide system with meticulously aligned guide rails is essential for reducing vibrations to an acceptable comfort level. Guide rails shall be rated for heavy-duty usage. Irregularity in the guidance system causes an unpleasant and uncomfortable transverse vibration within the car at high speeds. High-quality roller guide system is required for smooth ride. In addition, the lift guide system shall be able to withstand the effects of the building settling and the response of the building's structure to thermal and climatic changes.

11.6 Jump Lifts

In super high-rise buildings, there may be a need to start lift erection before the machine room is finished to compress total time required to complete the building. The self-climbing lift technique may be used to build the lift as the building construction progresses. Jump lift uses a temporary machine room that is moved upward as construction progresses. Typically, when the building construction reaches the seventh floor, the jump lift may start operating and start serving lower floors. Exterior hoists can be removed sooner, enabling the facade to be enclosed. This means that lower floors are ready for finishing much earlier and the entire building can be completed sooner. Since all transportation takes place in a dry, windproof shaft, construction can continue safely and unhindered, even in bad weather. The change-over to final lift is very quickly done.

11.7 How to Handle High-Density Population in Buildings

Due to continued exponential growth in urbanization, thickly populated buildings are becoming common in all metro and Tier-I cities. It has become quite essential to improve the efficiency of vertical transportation systems within the available shafts in the buildings, significantly beyond the conventional single deck system (see [Fig. 29](#)).

The following two different types of systems are possible solutions:

- a) 2 cabins running independently in one single shaft; or
- b) 2 cabins mounted one above the other.

These systems are compliant with the latest good practices as per [D-5A(1)] and [D-5A(20)].

11.7.1 The benefits of these systems are as follows:

- a) optimal use of shaft with 2 lift cabins arranged one above the other;
- b) connection of two main landing levels with the highest efficiency;
- c) handling of high traffic volume and short travel times between two main landing levels; and
- d) simultaneous filling of both lift cars.

In combination with other lift systems, double deck is generally used as a shuttle lift. It is particularly suitable for passenger transport from main entrance levels to defined terminal landings, for example outdoor viewing platforms or transfer landings, so-called sky lobbies.

Two independent cars in a single shaft solution, is used where both up-peak and inter-floor traffic is heavy.

The advantages come into their own when each of the landings in the access area or terminal landings are interconnected via escalators to split up the flows of passengers. In this case, both lift cars can be loaded and unloaded simultaneously.

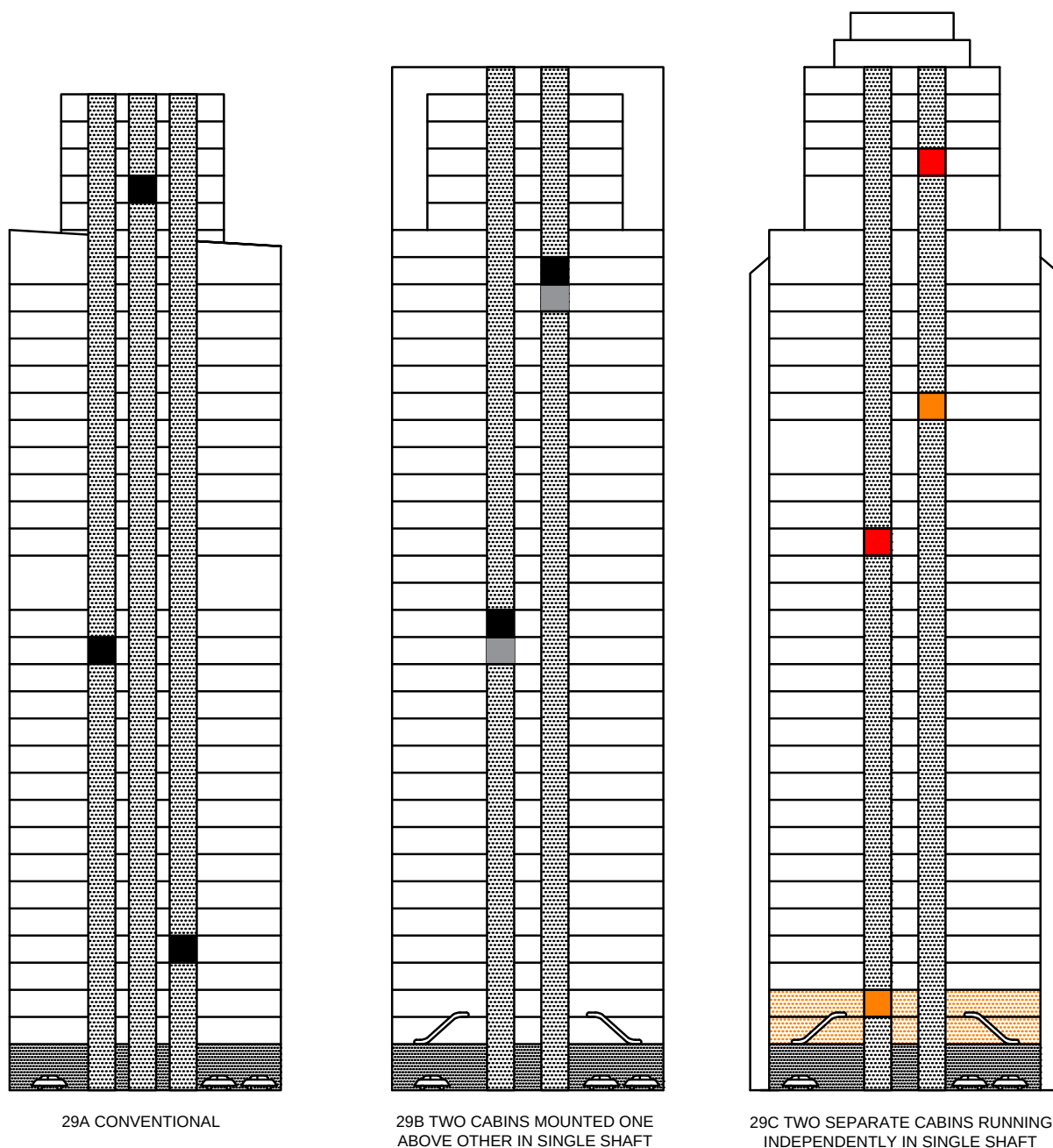


FIG. 29 DIFFERENT TYPES OF VERTICAL TRANSPORTATION SYSTEMS

11.7.1.1 The benefits for new installation projects include:

- significantly more handling capacity with fewer lift shafts compared to conventional lifts;
- 2 independent cars and 2 cars one above the other in one shaft reduce the building core by approximately 50 percent;
- reduction in construction cost in both labour and material for the developer and contractor; and
- increase in the rentable floor space and the rental income for the owner.

11.7.1.2 The benefits for modernization projects include:

- a) increased handling capacity with two lift cabins in one shaft;
- b) reduction in waiting time by around 30 percent to 40 percent over the conventional single deck system; and
- c) replace single-deck lifts that can no longer handle the building capacity and passenger comfort demands.

Whether used in new buildings or as part of a modernization project, these 2 car systems in one shaft can transport up to 40 percent to 50 percent more passengers.

11.7.2 An overview of the system with 2 separate cabins moving independently inside the shaft is given below:

- a) The lift group must be planned as a multiple car system. The cars must operate independently and use the same shaft. The lift cars are arranged one above the other and run independently;
- b) Both cars use the same guide rails. The cars can move towards each other and are also able to move away from each other;
- c) Each of the 2 lifts has its own drive system and counterweight. Each counterweight runs on its own guide rails. Both lifts have their own safety equipment where required. Both cars use the same landing doors;
- d) The suspension of the cars is dependent on the lift speed and travel height. It can be 1 : 1 or 2 : 1;
- e) Measures for rope compensation can be designed with compensation chains or belts to reach the required traction. For speeds above 2.5 m/s tensioned compensation rope systems should be used;
- f) The call assignments will be undertaken by destination control system. The destination call will be set onto an input station (terminal) in the lobby in front of the lifts or in its proximity. Therefore, pushbuttons for call assignment in the car are not required for passenger use. Hidden COP is provided for fireman or special operation. The cars can serve floors with adequate distance directly above each other;
- g) The minimum required safety distance depends on contract speed of the lifts. An independent working safety controller guarantees the prevention of a collision in any operating condition;
- h) Depending on the type of suspension and shaft dimensions, the drives are distributed in one, two or three machine room levels; and
- j) This system is designed for safety integrity level (SIL) 3 as defined in PESSRAL.

NOTE 8 Programmable electronic systems in safety related applications for lifts (PESSRAL) is included in the accepted standard [D-5A(21)].

11.7.3 An overview of the system with 2 cabins joined and mounted one upon the other is given below:

- a) The lift group must be planned as a single car system with 2 entrances (upper and lower). The cars are joined to operate simultaneously and use the same shaft.
- b) Both cars use the same guide rails, counter-weight rails, landing doors, machine, drive, controller system, rope system and safety equipment.
- c) The suspension of the cars is dependent on the lift speed and travel height. It can be 1 : 1 or 2 : 1. The suspension of both cars is same.
- d) Measures for rope compensation can be designed with compensation chains or belts to reach the required traction. For speeds above 2.5 m/s tensioned compensation rope system should be used.
- e) The call assignments will be undertaken by destination control system (DCS). The destination call will be set onto an input station (terminal) in the lobby in front of the lifts or in its proximity. Therefore, pushbuttons for call assignment in the car are not required for passenger use. Hidden COP is provided for fireman or special operation. Depending on the speed, the drives are distributed in one or two machine room levels.

11.8 Fireman's Lift

Minimum one number of single deck lift (15 persons, 1 000 kg capacity) serving all floors of the building is to be provided meeting all the requirements as per concerned fire authority such as 1 h fire rated landing doors, minimum speed criteria of travel time less than 60 s for travel from the main lobby to the topmost landing (see Fig. 30).

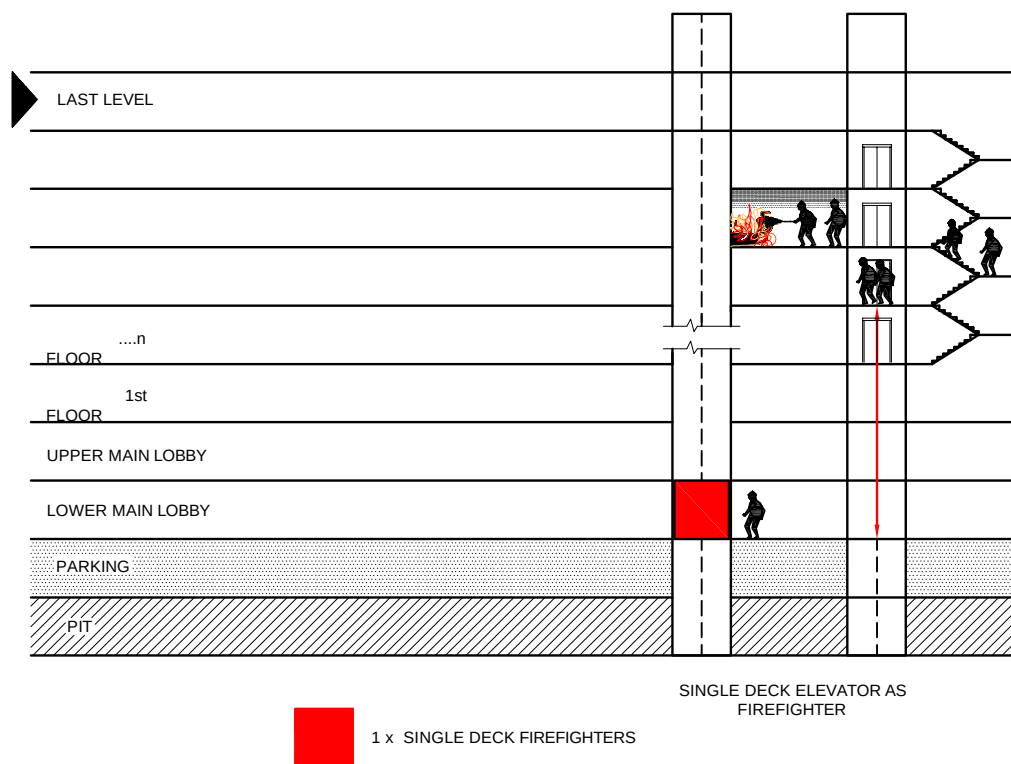
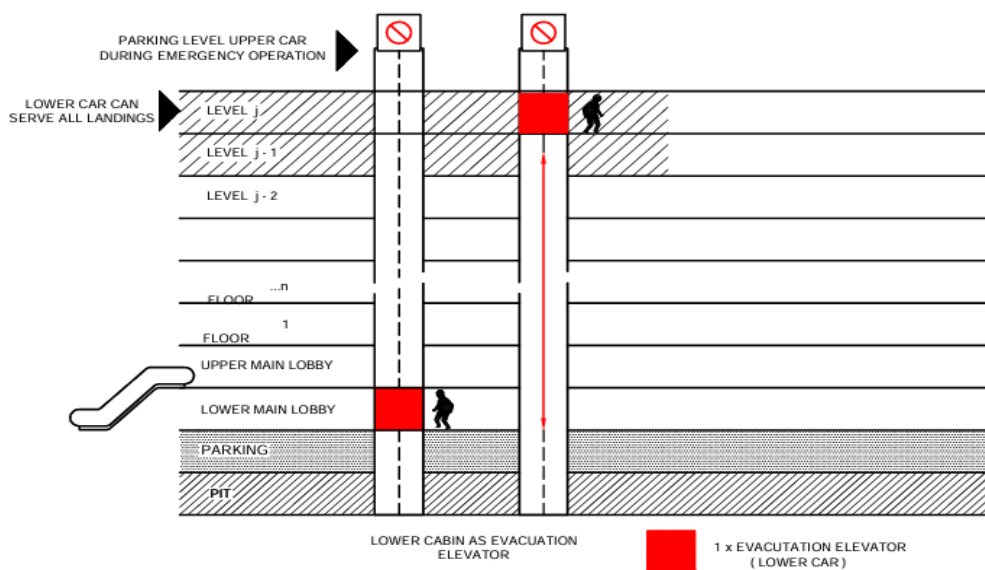


FIG. 30 FIREMAN'S STRATEGY

11.8.1 Passenger Evacuation Strategy for 2 Cabins Running Independently Inside a Single Shaft

a) Case (I) (see Fig. 31)



NOTE (I) OEM to be consulted for suitable depth of the pit.

FIG. 31 GIVE WAY IN LIFT OVERHEAD ROOM FOR UPPER CAR AND OPERATION ONLY WITH LOWER CAR

- f) Floors served (no. of regular openings, emergency openings, blocked openings);
- g) Method of control (simplex or group control);
- h) Position of machine room (if machine room-less, machine below, hydraulic);
- j) Machine room height;
- k) Clear overhead/clear top clearance;
- m) Pit depth;
- n) Sizes of hoist-way (width & depth), in mm;
- p) Position of counterweight;
- q) Car size (width & depth), in mm;
- r) Construction, design and finish of car bodywork;
- s) Car entrances:
 - 1) Number of entrances;
 - 2) Size (width & height); in mm
 - 3) Type of door (centre opening, telescopic, 4 panel, vertical bi-parting); and
 - 4) Whether through opening.
- t) Car light;
- u) Call indicator;
- v) Position indicator in car;
- w) Lift landing entrances:
 - 1) Number of entrances;
 - 2) Size (width & height); in mm
 - 3) Type of door or gates or shutters (for goods lifts) (centre opening, telescopic, 4 panel, vertical bi-parting); and
 - 4) whether through opening.
- y) electric supply:
 - 1) Power in volts a.c.;
 - 2) Phase; and
 - 3) Cycles (wire system).
- z) whether neutral wire available for control circuit;
- aa) temporary guarding of hoist-way and car cabin;
- bb) lighting volts a.c./d.c.;
- cc) are premises subject to *Lifts Act and Rules*;
- dd) disable friendly requirements [such as voice announcement of floor levels, accessible control panel (good contrast with the background wall, braille, raised letter and font), railing on three sides, two way accessible communication system];
- ee) earthquake protection requirements; and
- ff) additional items.

12.2 Additional Items

The enquiry should state any additional items required beyond those specified in good practice, such as interface requirements, dismantling of existing lifts, fireman's control, seismic design consideration, sway/deflection requirements, gravity distortion, etc.

12.3 Finishes

Finishes should be specified at the enquiry stage or provisional sums should be included for them. Finishes to be considered may include interior car finish, false ceiling design, false ceiling lighting, flooring material and weight, blower, ventilation details, car wall cladding details (weight to be considered), multimedia display details (like screen type, size, location, etc), door cladding details, transom details (flushed with the wall, material type, etc). *See also* good practice [D-5A(3)].

12.4 Inclusions and Exclusions

A number of peripheral items are associated with lifts installation, of which some should always be provided by the builder and some are best included by the manufacturer. The requirements vary to some extent with the type of installation.

It is important that the limits of responsibility are clearly understood, and the enquiry documents should be specific in this respect.

The lifts manufacturer should include such items mentioned in [12.1](#).

The lifts manufacturer should exclude the supply and fixing of the items or as per the contract conditions as follows:

- a) Builder's work such as forming lift well, pit and machine room, and building wall inserts;
- b) Machine room floor including any reinforcement necessary for load bearing;
- c) Hoisting hook/lifting beam fixing, where necessary;
- d) Any necessary tanking, lining or reinforcement of the pit;
- e) Diving beams for multiple wells and inter well pit screen (or as per the contract);
- f) Trap door;
- g) Intermediate beams for bracket fixing;
- h) Door cladding (or as per the contract);
- j) Cutting away and making good;
- k) Working lights/permanent electricity supplies; and
- m) Site painting of steel work other than lift material (if any).

For more detailed discussion of the requirements for site preparation and work by other trades, reference should be made to good practice [D-5A(6)] and to other relevant clauses of this Subsection.

Facilities for the use of the main contractor's crane should be provided to assist in installing heavy equipment in addition to other unloading facilities on site in the course of erection.

The main contractor should be instructed to include these facilities in his own quantities.

Where the lift manufacturer agrees to use mobile platforms in place of lift well scaffolding, the general contractor should provide 400/440 V 3-phase and 200/220 V single-phase supply in the lift shaft to operate such equipment, the supply to terminate at the position in the lift well required by the lift manufacturer.

These mobile platforms are limited in use for erection personnel and the transportation of light equipment only, but use of crane will also be necessary to assist in the installation of the heavy machinery and also in the initial installation of the mobile platform equipment.

12.5 Site Programme

The enquiry should indicate as accurately as possible the contract programme as it affects the manufacturer, in particular the target date for completion, construction schedule, site preparedness, the availability of crane for hoisting, coordination details (with other contractors like BMS interfacing, CCTV interfacing, etc).

13 ACCEPTANCE OF TENDER AND SUBSEQUENT PROCEDURE

13.1 General

The procedure indicated below particularly relates to the most usual case, where the lift manufacturer is the sub-contractor.

13.2 Order

The building owner places an order with the selected contractor for supply and installation of lifts. If alternative schemes have been offered, the order should clearly indicate which has been accepted.

13.3 Programme

The contractor shall submit a detailed programme after acceptance of the order showing his intended method, stages of the works, together with the period of time that has been estimated for each and every stage of progress.

The programme shall cover each lift separately, including dates, such as:

- a) date of order of equipment and materials;
- b) date of commencement and completion of every stage of the works in line with the building construction programme;
- c) date of expected completion of builder's work requirements;
- d) date of delivery of equipment and materials to site; and
- e) date of requirement of temporary and permanent electricity supply and date of completion, commissioning and testing.

The period between order and delivery of material falls into two stages: first the finalizing of details and secondly the actual production of the equipment which depends on the first stage. Within the first stage, other dates may need to be considered, such as:

- 1) all relevant building information available;
- 2) submission of lift manufacturer's drawings;
- 3) approval of drawings; and
- 4) final selection of finishes.

13.4 Drawings to be Submitted after Placing of Order

The lift manufacturer shall prepare layout drawings based on the civil/structure drawings, order requirements and submit to building owner within mutually agreed time frame. The same shall be briefed to building owner.

The drawings shall indicate clearly the position and sizes of all holes and cuttings, the loads on beams and structures, and all other requirements in relation to 'Lift Installation' namely:

- a) lift machine room and associated equipment, including sub-floor where appropriate;
- b) all structural openings, such as landing entrances (including structural dimensions), ventilation openings, etc;
- c) all bases, plinths, channels, holes, grouting-in of fixings, etc;
- d) lifting beams or other facilities for supporting lifting tackle in the machine room and lift well, permanent means of access to the lift pit;
- e) hoisting facilities and access required for delivery of equipment to the machine room, etc;
- f) details of structural steelwork for lift machinery in the lift machine room;
- g) details of shaft dividing steelwork for supporting guard brackets, etc and inter-well screens/wall for multiple wells; and

- h) method of fixing guide rails.

13.5 Approval of Drawings

The building owner should go through the drawings submitted by lift manufacturer and ensure that it is prepared in line with civil/structural drawing and order requirements provided.

If any addition/alteration is noted in the drawing, the same shall be updated in the drawing by the lift manufacturer and resubmitted to the building owner for approval. The drawing shall be approved by building owner, once it has been clearly understood.

13.6 Selection of Finishes

Where the contract provides for the purchaser's choice of decorative finishes, colours, etc, the decisions should be communicated by the purchaser's representative as early as possible, preferably not later than the time of approval of drawings.

13.7 Electricity Power Supply for Lifts

Operation of the machine requires electrical power supply from a comparatively early stage of installation for the most efficient working, and shall be provided accordingly. Whilst temporary supply may be sufficient for erection purposes, the quality of power should be ensured by provider. Final testing and setting up can only be carried out with the permanent power supply. For this reason, the timely provision of the permanent supplies is important.

14 CO-ORDINATION OF SITE WORK

14.1 Preparatory Work on Site

In accordance with the general arrangement drawing for the lift, the representative of the lift company would brief the building owner or his nominated representative about the preparatory work that is required to be carried out before commencement of the lift installation.

The building owner will ensure that the preparatory work is done in accordance with the general arrangement of lift and also the state lift rules are followed for the construction of the lift shaft as well as machine room. Once the preparatory work is completed, the building owner or his nominated representative, will ask the lift company to cross check and see that everything is in order. The lift installation will start after receiving all necessary statutory approvals applicable as per the state laws.

The representative of lift manufacturer shall check the lift shaft and bring the shortfalls, if any, to the notice of the building owner till the preparatory work is completed in compliance with the requirement of lift supplier. The following works shall be completed by the building owner before commencement of lift erection.

14.1.1 Machine Spaces/Head Room

- a) Adequate ventilation for machine room with rain protected windows or louvers with exhaust fans or air-conditioning system so that the ambient temperature of the machine spaces is maintained between + 5 °C and + 40 °C (taking into account the heat dissipated by the equipment, ambient temperature, humidity, direct sunlight, air quality and air tightness of buildings due to energy saving requirements);
- b) Suitable access door preferably with louvers to machine spaces with RCC steps or mild steel staircase with hand railing (not applicable for MRL);
- c) Machine spaces access door shall be of ms, in two panels, hinged type construction, opening outward and clear entrance not less than 1 000 mm wide & 2 000 mm high with proper locking arrangement. Caution notices of 440V and restricted entry shall be marked on the entrance (not applicable for MRL). A trap door with steel enclosure shall be provided for each machine spaces as per GAD (not applicable for MRL).
- d) Hoisting beams/load hooks of required lifting capacity in machine room for machine room lift, one directly above the lift shaft and other directly on top of the trap door as specified in the lift general

arrangement drawing. The same shall be provided in the slab in the overhead area for machine room less lifts, as specified by the lift manufacturer in their general arrangement drawing. The safe working load (SWL) shall be specified after certification and tagged on the hook/beam in the machine room/ head room;

- e) Machine room slab shall be cast after the slab marking for rope pocket holes is done by manufacturers team. Reinforced concrete machine room slab of 230 mm thickness designed to sustain more than 1 000 kg/sqm uniformly distributed load all over its surface in addition to specific load shown in GAD (not applicable for MRL);
- f) Machine spaces walls, floor and ceiling to be treated and painted to minimize accumulation of dust (not applicable for MRL);
- g) RCC pedestal for fixing the machine bed frame (not applicable for MRL); and
- h) Fire extinguisher shall be provided by building contractor to prevent fire accidents.

14.1.2 Hoistway

- a) Hoistway dimensions mentioned in the GAD shall have minimum clear plumb dimensions in mm and hence actual wall shall not encroach on these dimensions;
- b) Hoistway shall be without any projections and recess from headroom to the pit floor;
- c) If any projections or recesses are there, then those shall be flushed and levelled on underside to an angle not less than 60° from the horizontal by means of metal plates or cement;
- d) Anything not related to lift equipment like service pipes, ventilation ducts, plumbing and mechanical systems, electrical conduits shall not be allowed to run in lift well and machine room;
- e) Front wall at the ground floor and top floor shall not be constructed until all lift equipment is set in position;
- f) The thickness of lift walls shall be minimum 230 mm in brick or 150 mm in RCC. The walls shall be constructed in consultation with structural engineers or consultant to verify suitability based on lift load/reactions;
- g) In case of brick walls, it would be recommended to have concrete solid block at locations where guide rail brackets are to be installed and in the overhead area in case of MRL Lifts. Similarly, in case of structural shaft suitable ISMC/ISMB shall be provided by the builder to fix guide rail brackets at all bracket fixing levels. The lift manufacturer shall provide a drawing indicating the locations of brackets in the lift shaft;
- h) Similarly, for fixing the landing entrances, the builder shall provide RCC lintels/RCC beams for fixing landing door header and sill. In case of structural shaft suitable metallic channels shall be provided in the entrance area at lintel and floor level;
- j) The shaft in masonry shall be fully plastered and given two coats of white wash or paint;
- k) All the structural shafts shall be completely covered and finished in accordance to the local codes before the start of installation;
- m) Suitable cut-outs near the entrance area as indicated in the lift general arrangement drawing shall be provided for fixing human interface system like landing operating panel, floor indicator, hall gong and lantern, emergency alarm, fireman switch and other devices like "THIS CAR NEXT" group indicators, etc;
- n) Scaffolding (bamboo/steel) shall be erected as required by the lift manufacturer with necessary safety measures;

- p) Adequate barricading shall be provided on all the lift entrance openings to avoid mishap;
- q) Smoke window with louvers shall be provided below overhead slab in the hoistway as per GAD; and
- r) Adequate measures shall be taken to prevent water from seeping into the hoistway at all times.

14.1.3 Pit

- a) Lift well pit floor and side wall of the lift shaft may be plastered and water proofed. Pit shall be free from any water leakage/seepage and debris. Building owner shall ensure the construction of the pit depth in accordance with the speed of the lift as specified by the manufacturer in the general arrangement drawing;
- b) Pit floor shall be strong enough to bear the load reactions as per GAD; and
- c) Pedestals shall be provided for fixing car and counterweight buffers to bear the reactions as mentioned in GAD.

14.1.4 Lighting and Power for Installation

- a) Lift shaft shall be provided with permanently installed electric lighting giving at least 50 lux, 1.0 m above the pit floor and above the pit platform (when provided) everywhere a person may stand, work and/or move between the working areas and at least 20 lux in all other areas, excluding shadows created by car or components, even when doors are closed;
- b) The lighting to be protected against mechanical damage;
- c) The lighting outside each landing to be provided with 50 lux illumination;
- d) Adequate power supply (single phase/3 phase) as required by the lift manufacturer, shall be provided for running winch machine and lift machine for scaffold-less installation; and
- e) Double earthing shall be provided in the machine room/control panel as required by the manufacturer.

14.2 Delivery of Material

The delivery of the material shall be made based on mutual agreement of the lift manufacturer and the building contractor. It is strongly recommended that the material should be delivered at site only when a suitable dry, lockable and weather-proof storage area is provided by the building contractor; the building shall be ready for installation as per GAD.

The scope of unloading and stacking materials shall be as per mutually agreed terms between the lift manufacturer and the building contractor. While unloading and shifting of materials due care shall be taken by both the parties to ensure safe unloading and shifting of materials.

Suitable equipment like hydra, fork lift, crane and trolleys shall be used for unloading and shifting of lift materials. Building contractor shall ensure safe passage for vehicles carrying materials, up to the storage area.

14.3 Storage

In order to enable efficient material distribution, it is important that the material shall be unloaded from the delivery truck within a distance of 50 m of the storage/unit location and have proper access from unloading area to storage area. The building contractor shall provide the storage area near the installation site, that is within 20 m distance from the installation site. The minimum storage space per lift shall be 35 m² and shall be increased adequately in consultation with the lift company for bigger capacity high speed lifts and tall buildings.

The storage room should be dry, well lit, weather protected and should have locking arrangement and provision of a 16 A power point is recommended. The storage area provided by the building owner should have PCC flooring. The storage room shall be kept ready before arrival of material at site and shall remain in possession of installation engineer till the erection of lift is completed and handed over to the owner.

The material shall be stacked in sequential manner, in accordance with the planned activities. The storage area should be provided at a suitable location near the vicinity of the lift shaft preferably above the ground floor level and should remain in possession of the lift supplier till the installation and commissioning work is completed. It is strongly recommended that the location of the store is not changed to avoid material damages.

14.4 Site Meetings

For the successful progress of the work, full cooperation among all agencies is essential. In large sites, regular meetings of such agencies are beneficial for coordination of work efficiently. Programme for the constructional work in that part of the building containing the lift should be made in consultation among all parties concerned.

14.5 Service of Other Trades

The lift erector will require the services of architect, civil, and electrical contractors, and other agencies as the work proceeds, and it is essential that the lift erector should give due notice to the building contractor of the demands to be made on other trades, so that they may plan accordingly. Mutually agreed, periodic meeting to review work progress with agencies concerned would ensure better coordination and efficiency.

14.6 Scaffolding, Fencing, Wherever Provided

Scaffolds and stationary work platforms shall be erected in accordance with adequate safety under the supervision of a competent person. When used, wooden or synthetic planks shall be scaffold-grade or approved by a qualified person for scaffold and stationary work platform use, and the same shall be capable of withstanding 4 times the working load.

The assembly and disassembly of scaffolds and stationary work platforms shall be done using a safety harness, rope grabber and lifeline as required. All temporary lighting shall be so installed so as not to make any contact with the metal frames of the scaffold. Also, the metallic scaffolding shall be double earthed. Prior to start of any lift installation work, all lift landing shall be barricaded adequately, preferably to full height, with display of warning notices informing presence of persons at work. When scaffold less method of installation is used, certified man-riding rope hoist machine shall be used with double fall protection system.

14.7 System Building Sites

If the building programme allows insufficient time for lift erection in conventional fashion after the well is completely built special procedures are needed. This applies particularly to industrial and multi-storeyed buildings. The building contractor should provide a suitable portable cover to the completed portion of the lift well in order to protect the lift erectors working below against the weather and falling objects.

When the top of the well has been reached it is normal to cap it immediately with a precast load bearing floor slab on to which is lowered the pre-assembled machine room equipment. It then remains for the building contractor to complete and weatherproof the machine room as swiftly as possible. On all such projects the close cooperation between the building contractor and the lift manufacturer is essential.

14.8 Connecting to Power Supply

The lift manufacturer should give prior intimation to the building contractor of the date the power supply to the lift is required, so that suitable arrangements for connection should be made.

15 INSPECTION AND ACCEPTANCE PROCEDURE FOR LIFTS

15.1 Reference

This suggested procedure for inspection of lifts is based on good practice [D-5A(2)], [D-5A(15)] and [D-5A(21)] for the traction lift.

15.2 Purpose

To ensure safe operation of new lifts on their completion and before being placed in service, they need to be subjected to acceptance inspection and tests in the field to determine that installation shall conform to the requirements of applicable standards. An inspection and test shall also be carried out on an existing installation after a major alteration.

15.3 General Requirements

These requirements shall be as follows:

- a) *Personal safety:*
 - 1) Use of relevant personal protective equipment is mandatory by the inspector and his support staff at the time of inspection; and
 - 2) Work should be stopped and correction(s) arrangement, if unsafe conditions are noticed at the installation.
- b) *Safety precautions during inspection:*
 - 1) Inspectors should have a proper working knowledge of the lifts;
 - 2) They should always ensure that the car is in his control while working on car roof and pit;
 - 3) They should at all times be alert of moving objects, and when on top of lift car for moving counter weight, hoist-way projection such as beams, adjacent moving cars, cams and other equipment attached thereto or mounted in the hoist-way;
 - 4) The overhead clearance should always be noted as number of fatal accidents occurs due to cars running into limited overhead spaces while inspections are made from top of the lift cars;
 - 5) Inspector should never enter in lift pits containing water;
 - 6) When working in the lift pit the inspector should always note the position of car; and
 - 7) Clear distance shall be kept from descending counterweight in the hoist-way of the lift being inspected and those in adjoining hoist-way.

15.4 Description of Installation

The following provides a template of typical description of installation:

Site name with details Length of travel (mm) Number of floors served Total: * Front: * Rear: * Car door: Glass/ SS Landing door: Glass / SS Rated load Rated speed m/s * Lift Type : Passenger/Goods Machine room/Machine room less/hydraulic Location of machine room (for MR/ hydraulic lifts) Give a tick mark in the appropriate box Above well Below well At side Roping Ratio: _____	Vendor Job No: Purchaser's identification number Power supply Permanent Temporary Voltage * Phase * Frequency * Wire (3 or 4) * MCCB rating * RCD rating * Are the above entries acceptable Yes No
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15.5 Visual and Functional Checks

A typical checklist of visual and functional checks involved in the installation of lifts is given in [Annex A](#).

16 MAINTENANCE OF LIFTS

16.1 Maintenance of the lift essentially covers all necessary operations to ensure safe and intended functioning of the installation and its components after completion of the installation and throughout its life cycle. To this effect, regular maintenance of the installation shall be carried out to ensure the safety of the installation, the ability to be maintained without causing injury or damage to health.

16.2 Information to the Owner of the Installation

16.2.1 The need for the owner to keep the installation in a safe operating condition. To fulfil this, owner shall use a maintenance organization with trained manpower, adequate infrastructure, lift spares and complying with the state requirements.

16.2.2 The need for the owner to take care of any state/local regulations and other requirements, where relevant, and their implications on maintenance.

16.2.3 The need for planned maintenance to be carried out by a maintenance organization, at the latest when the installation is put into service or if the installation is to remain unused for a long period of time, before being put into service first time thereafter.

16.2.4 The importance for the owner of the installation to have the same maintenance organization in the case of several installations having common well/spaces and/or machine room/spaces.

16.2.5 The need for the owner of a passenger/goods-passenger lift to keep the two-way means of communication efficient and linked to a 24 h rescue service for the whole of the time that the installation can be used.

16.2.6 The need for the owner to remove the passenger/goods-passenger lift from service when the two-way means of communication is out of order.

16.2.7 The need for the owner to put the installation out of service in case of dangerous situations.

16.2.8 The need for the owner of the installation to inform the maintenance organization, immediately about any perceived abnormal operation of the installation or abnormal change in its direct environment,

- a) immediately after putting the installation out of service in the case of a dangerous situation;
- b) after any rescue intervention by their authorized and instructed person(s);
- c) before any modification related to the installation and/or its environment;
- d) before any authorized third-party inspection or works other than maintenance works are carried out on the installation;
- e) before taking the installation out of service for a prolonged period; and
- f) before putting the installation back into service after a prolonged period of non-operating time.

NOTE ð The owner of the installation should obtain from the company carrying out the relevant modification, the maintenance instructions for the maintenance organization.

16.2.9 The owner needs to inform the access ways to areas reserved to maintenance persons, the need for the owner of the installation to inform the maintenance organization, about:

- a) the access ways to be used and fire evacuating procedures from the building;
- b) the place where the keys of the reserved areas can be found;
- c) if necessary, the persons who shall accompany the maintenance persons to the installation; and
- d) if necessary, personal protective equipment to be used in the access ways and, possibly, where such equipment can be found.

The information shall be made available also on site to the maintenance organization.

16.2.10 The need for the owner of the installation to ensure that the name and the telephone number of the maintenance organization are always available to the user of the installation, permanently affixed and clearly visible.

16.2.11 The need for the owner of the installation to ensure that the keys of machine and pulley room doors (trap doors) and of inspection and emergency doors (trap doors) are permanently available in the building and are used only by persons authorized to gain access.

16.2.12 The need for the owner of the installation to provide, in all circumstances, safe access to the building and to the installation for the maintenance organization involved in the rescue of persons.

16.3 Information to the Owner of the Installation Regarding Self-Monitoring

In addition to those examinations and tests which the owner of the installation entrusts to the maintenance organization, there is a need for the owner to carry out periodically, in their own interests, the following:

- a) A full ascent and descent travel to assess any changes in the quality of the ride or damage to the equipment. Typical items to be checked to ensure that they are in place, undamaged and functioning correctly are:
 - 1) landing doors and bottom door tracks;
 - 2) stopping accuracy;
 - 3) indicators that are not located in a reserved area;
 - 4) landing push controls;
 - 5) car push controls;
 - 6) door open controls;
 - 7) two-way means of communication in the car which provides permanent contact with a rescue service;
 - 9) normal car lighting;
 - 10) door reversal device; and
 - 11) safety signs/pictograms.
- b) For goods only and service lifts the checks to be carried out are the same, when relevant.

16.4 Information to the Owner regarding Lift Rescue Operations

The information to the owner shall be inclusive of that given in [16.4.1](#) to [16.4.5](#).

16.4.1 The need for person(s) authorized by the owner of the installation to rescue trapped passengers to be trained by the maintenance organization.

16.4.2 The need to ensure that the training is appropriate to the specific installation and updated.

16.4.3 The need to ensure his authorized person(s) rescue people only through landing doors.

16.4.4 The need to ensure that the maintenance organization is called when the owner's authorized person(s) is not able to move the car using the manual and/or electric emergency devices.

16.4.5 The need to inform their authorized person(s) about any condition for which only the maintenance organization shall carry out a rescue operation.

The information required for maintenance of lifts, to be known to the owner of the installation is included, as above detailed maintenance guide for lifts is included in accepted standard [D-5A(22)].

ANNEX A

(Clause 15.5)

TYPICAL CHECKLIST OF VISUAL AND FUNCTIONAL CHECKS INVOLVED IN THE INSTALLATION OF LIFTS

A-1 MACHINE ROOM

Sl No.	Requirement	Result	Remarks
(1)	(2)	(3)	(4)
i)	There shall be easy access from the top landing to the machine room	Accessible/Not accessible	
ii)	Steps along with handrail for the access to machine room to be provided	Provided/Not provided	
iii)	Machine room access door of M.S, in two panels, hinged type construction, opening outward and clear entrance not less than 1000 mm wide & 2000 mm high	Provided/Not provided	
iv)	Caution notice of 440 V and restricted entry marked on the entrance	Displayed/Not displayed	
v)	Locking arrangement provided and machine room shall be kept locked	Provided/Not provided	
vi)	The machine room used for the purpose of lift machinery only	Yes/No	
vii)	Machine room walls, floor and ceiling treated and painted to minimize accumulation of dust	Yes/No	
viii)	Machine room shall be maintained clean, without any seepage of water from roof, walls or any other opening	Yes/No	
ix)	The height of the machine room shall not be less than 2 100 mm at working areas	Yes/No	
x)	Machinery spaces and pulley rooms provided with permanently installed electric lighting with an intensity of at least 200 lux at floor level everywhere a person needs to work and 50 lux at floor level to move between working areas	Provided/Not provided	
xi)	Adequate ventilation for machine room with rain protected windows or louvers with exhaust fans or air-conditioning system so that the ambient temperature of the machine room is maintained between + 5 °C and + 40 °C	Yes/No	
xii)	Are the rope openings below the machine closed adequately as required	Yes/No	
xiii)	The hoisting beams or hooks are certified as per the safe working load required and marked	Provided/Not provided	

Sl No.	Requirement	Result	Remarks
(1)	(2)	(3)	(4)
xiv)	Trap door with metal cover to be provided to lift the machine to/or from the machine room and to be located vertically below the hook provided in the machine room ceiling	Provided/Not provided	
xv)	Rescue chart pasted in the machine room with proper identification	Yes/No	

A-2 POWER SUPPLY

Sl No.	Requirement		Result	Remarks
(1)	(2)		(3)	(4)
i)	Incoming power supply to be 415 V Ñ 10 percent or as specified in the contract agreement		R-Y : _____ Y-B : _____ R-B : _____ Ok/Not Ok	
ii)	Neutral ð Earth voltage	Idle condition within (0 to 3) V	Value: _____ Ok/Not Ok	
iii)	Neutral ð Earth voltage	Running condition within (0 to 3) V	Value: _____ Ok/Not Ok	
iv)	Separate supply with MCB	Three phase supply	Provided/Not provided	
		Single phase supply (for car lighting and hoistway lighting)	Provided/Not provided	
v)	Check whether the power cables are connected with lugs for termination		Connected/Not connected	
vi)	The correct capacity main switch (with lock out tag out provision) with protective device, provided as per the requirement, separately for lift power, light for machine room and hoistway. Three pin sockets with switch provided		Provided/Not provided	
vii)	Double grounding conductor of correct size is provided on the main switch. Earthing of lift motor, control panel, overspeed governor switch, etc provided		Provided/Not provided	
viii)	Hoistway lighting provided at least 50 lux, 1 m above the pit floor and above the pit platform (when provided), everywhere a person can stand, work and/or move between the working areas and at least 20 lux in all other areas, excluding shadows created by car or components, even when doors are closed		Provided/Not provided	

Sl No.	Requirement	Result	Remarks
(1)	(2)	(3)	(4)
ix)	Residual current protective device (RCD) with rated residual operating current not exceeding 30 mA (not applicable to main drive) provided	Provided/Not provided	

A-3 MACHINE AND BRAKE ASSEMBLY (FOR TRACTION LIFTS)

Sl No.	Requirement	Result	Remarks
(1)	(2)	(3)	(4)
<i>Make:</i> _____ <i>Sl No.:</i> _____ <i>Rating:</i> _____			
i)	The motor is smooth in operation without overheating	Satisfactory/Unsatisfactory	
ii)	The gear box (if provided) is smooth and without any oil leakage except little oozing	Satisfactory/Unsatisfactory	
iii)	The condition of machine grooves is such that the ropes/CSB do not slip when the traction sheave stops normally	Satisfactory/Unsatisfactory	
iv)	Is the machine fixation as per the installation manual	Yes/No	
v)	Are the electrical connections of the machine routed and well tightened	Yes/No	
vi)	Are the machine isolation pads installed correctly (if applicable)	Yes/No	
vii)	Ropes/elastomeric coated steel belts fixed by suspension means in the machine room are fixed as required and locked with nuts and split pins.	Yes/No	
viii)	Clear vertical distance of at least 300 mm is available above rotating equipment	Yes/No	
ix)	The lift provided with braking system which operates automatically in the event of loss of: a) the main power supply b) the supply to the control circuits	Provided/Not provided	
x)	The machine, governor, etc, identified with the correct contract number	Yes/No	
xi)	The manual brake release device is available and easily accessible	Yes/No	
xii)	Guards provided for moving parts such as machine sheave, etc to protect against pinching hazard	Provided/Not provided	

A-4 CONTROLLER

SI No.	Requirement	Result	Remarks
(1)	(2)	(3)	(4)
<i>Type:</i> _____ <i>SI No.:</i> _____ <i>Wiring drawing No:</i> _____			
i)	The controller fixed as per the GAD (general arrangement drawing)	Yes/No	
ii)	The controller identified with the correct contract number	Yes/No	
iii)	The controller should have an easy access to open and close with lock	Yes/No	
iv)	The required warning stickers pasted	Provided/Not provided	
v)	All extra cables neatly arranged	Yes/No	
vi)	Are the battery terminals covered	Yes/No	
vii)	The field and controller wiring properly routed	Yes/No	
viii)	The control panel wiring is neatly grouped and the insulation of wires is not more than one Mega Ohm with 500 volt megger	Yes/No	
ix)	Logbook maintained in up-to-date condition	Maintained/Not maintained	

A-5 OVER SPEED GOVERNOR

SI No.	Requirement	Result	Remarks
<i>Make:</i> _____ <i>Model:</i> _____ <i>SI No:</i> _____ <i>Tripping speed:</i> _____			
i)	Is the correct over speed governor provided and installed as per layout	Yes/No	
ii)	The sticker/painted indication represents the down direction of rotation which will facilitate the safety gear actuation	Yes/No	
iii)	Check whether the OSG is calibrated and sealed	Yes/No	
iv)	The governor is aligned, and the rope passes through the holes without any obstruction	Yes/No	
v)	The governor rope runs free is not in contact with any object in the hoist way	Yes/No	
vi)	The governor tripping mechanism is free and clean	Yes/No	
vii)	The electrical switch is provided and operates in both the directions	Yes/No	
viii)	All fixation bolts are provided and adequately tightened	Yes/No	

A-6 GENERAL INSPECTION ON CAR ROOF

Sl No.	Requirement	Result	Remarks
<i>Instructions before entering to the car top:</i> a) Allow the lift to move in down direction from the top terminal landing with no load b) Stop the lift from the controller by changing to maintenance mode, such that easy access to enter the car top from the top terminal landing c) Switch on the shaft lighting d) Use the door open key to open the top terminal landing door e) Switch on the car top stop switch from the landing to enable the 'STOP' mode f) Switch on the light on car top g) Switch to maintenance mode from the car top before entering the car top h) Put Stop switch to Run position and press UP/DN and common button to move the car UP or down in maintenance speed			
i)	Car top guard rail on car roof to be provided of 0.70 m where the distance between car and hoistway wall is > 0.30 m and ≤ 0.50 m; is 1.10 m where the distance exceeds 0.50 m and not required if the distance is < 0.30 m	Provided/Not provided	
ii)	The car roof has been provided with toe board of 100 mm and non-slip working area	Provided/Not provided	
iii)	Car roof has sufficient strength to resist a minimum force of 2 000 N at any position on an area of 0.30 m $\times$ 0.30 m without permanent deformation	Yes/No	
iv)	Lighting with proper protection is provided on the car roof of at least of 50 lux intensity, 1.0 m above the car roof	Provided/Not provided	
v)	Emergency light on car top with minimum intensity of 20 lux for 1 h operation and alarm initiation device provided	Provided/Not provided	
vi)	Wires and trunking are properly routed and does not hinder with any maintenance related equipment	Yes/No	
vii)	The car fan/blower is properly fixed with isolation	Yes/No	
viii)	Three pin plug sockets with switch is available and is in working condition	Yes/No	
ix)	The emergency stop switch is positioned not more than 0.75 m from the landing	Yes/No	
x)	Check whether the top car clearance meets the requirement Value required: _____	Value measured: _____ Ok/Not Ok	
xi)	Overhead dimension in W $\times$ H $\times$ D Value required: _____	Value measured: _____ Ok/Not ok	
xii)	If overhead height is less, then the indication to be provided with zebra painting in the overhead	Provided/Not provided	

Sl No.	Requirement	Result	Remarks
xiii)	Floor levelling switches are fixed as per requirement	Yes/No	
xiv)	Rope/CSB hitches on car roof are fixed as required and locked with nuts and split pins	Yes/No	
xv)	Car sheave; if available; is adequately guarded from accidental contact	Yes/No	
xvi)	The junction box is fixed securely and all wires are adequately protected	Yes/No	
xvii)	The car shoe liners/rollers are fixed as per requirement and adjusted properly	Yes/No	
xviii)	Oil tank are provided and fixed with proper oil levels	Yes/No	
xix)	Engagement of the car door coupler with the landing door at every landing	Ok/Not ok	
xx)	Engagement of shaft limit switches	Ok/Not ok	
xxi)	The governor rope and linkage is correctly connected to the car	Yes/No	
xxii)	The guide rail clips are fixed as per requirements and in the correct directions	Yes/No	
xxiii)	The safety gear assembly and connecting rod should be free from external disturbances	Yes/No	
xxiv)	The safety switches have been verified for correct operation during travel	Yes/No	
xxv)	The safety gear linkages is checked for operation and actuation of switches	Yes/No	
xxvi)	Clearance between the car door coupler and the landing sill to be as per the manufacturer design at every landing	Ok/Not ok	
xxvii)	Clearance between the landing rollers with the car sill to be as per the manufacturer design at every landing	Ok/Not ok	

A-7 PIT INSPECTION

Sl No.	Requirement	Result	Remarks
<i>Instructions before entering to the pit:</i> - Allow a person on the car top and ask him to move the lift to the topmost landing - Open the bottom terminal landing door with door open key - Switch on the pit stop switch from outside the pit - Enter into the pit using ladder or from access door			
i)	It should be clean and dry	Yes/No	

Sl No.	Requirement	Result	Remarks
ii)	Pit access door to be provided, if pit depth exceeds 2 500 mm. Either an access door or a ladder inside the well to be provided, where pit depth does not exceed 2 500 mm	Pit depth = mm Pit access door/ Ladder provided/ Not provided	
iii)	Oil trays are available at the bottom of each guide rail	Provided/Not Provided	
iv)	Pit stop switches to be provided, accessible from the lowest landing. When pit depth ≥ 1.6 m, two pit switches, one accessible from lowest landing and other from pit floor to be provided	Provided/Not Provided	
v)	Pit stop switches to be verified with the electrical circuit such that lift should not move further after the activation of pit stop switches	Ok/Not OK	
vi)	Inspection control station is permanently installed in the pit. If more than one inspection control station is switched to "inspection" it shall not be possible to move the car from any of them, unless the same push buttons on the inspection control are operated simultaneously	Provided/Not Provided Ok/Not Ok	
vii)	Counterweight screen should be provided and fixed, except where rope compensation sheave is provided	Provided/Not Provided	
viii)	Buffers provided for bottom limit of travel of car and counterweight are correctly fixed and not in bent condition. PU buffers not to be in degenerated condition	Ok/Not Ok	
ix)	The governor tension pulley is fixed at the correct height from the pit floor as per the manufacturer recommendation	Ok/Not Ok	
x)	The electrical switch for the tension pulley is verified, if provided	Ok/Not Ok	
xi)	Check whether the bottom car clearance meets the requirement Value required: _____	Value measured: _____ Ok/Not Ok	
xii)	Pit lighting to be provided of at least 50 lux, 1 m above the pit floor and above the pit platform (when provided), everywhere a person can stand or work	Ok/Not ok	
xiii)	Refuge space to be provided in the pit and signage readable from entrance, indicating allowed number of persons and type of posture to be provided	Provided/Not provided	
xiv)	Is the travelling cable below car secured and correctly fixed and the excess cable is not secured below car	Ok/Not ok	

Sl No.	Requirement	Result	Remarks
xv)	The travelling cable is adjusted and clear of the pit floor with the car at the lowest position	Ok/Not ok	
xvi)	The bottom guide shoes or rollers are fixed as per requirement and adjusted for smooth travel	Ok/Not ok	
xvii)	The car safety gear is properly adjusted as per the installation manual	Ok/Not ok	
xviii)	The compensation chain fixing arrangement is fixed as per requirement	Ok/Not ok	

A-8 EARTHING

Sl No.	Requirement	Result	Remarks
(1)	(2)	(3)	(4)
i)	Separate earth bar to be provided in the machine room	Provided/Not provided	
ii)	Earthing to be done for all controllers and the machine	Yes/No	
iii)	Earthing to be done for the shaft equipment and car top components	Yes/No	
iv)	Earthing from controller to car top components to be linked through the travelling cable	Yes/No	
v)	Routing of the earth from the controller to the main earth bar	Ok/Not ok	
vi)	Connection of the lift earth bar to main earth bar	Ok/Not ok	

A-9 INSPECTION FROM THE LIFT CAR

Sl No.	Requirement	Result	Remarks
(1)	(2)	(3)	(4)
Instructions: a) No one should be present in the pit as well as on the car top b) Make sure the lift is normal mode c) Enter the lift car from any of the landing			
i)	Car inside width Value required: _____	Value at site: _____ Ok/Not Ok	
ii)	Car inside depth Value required: _____	Value at site: _____ Ok/Not Ok	

Sl No.	Requirement	Result	Remarks
(1)	(2)	(3)	(4)
iii)	Height below false ceiling Value required: _____	Value at site: _____ Ok/Not Ok	
iv)	Door opening width Value required: _____	Value at site: _____ Ok/Not Ok	
v)	Door opening height Value required: _____	Value at site: _____ Ok/Not Ok	
vi)	Car capacity to be displayed inside the lift car	Yes/No	
vii)	Safety warning plate to be displayed (<i>see Note</i>)	Yes/No	
viii)	Functioning of car display	Functioning/ Not functioning	
ix)	The stopping accuracy of the car shall be ± 10 mm. If during loading and unloading levelling accuracy of ± 20 mm is exceeded, shall be corrected to ± 10 mm	Ok/Not Ok	
x)	DO button and DC button functional check: For power operated door: 1) Press the DO button at level - Door should open 2) Press the DO button during running - Door should not open 3) While closing press the DO button at level - Door should open 4) DC button at level when the door is in open condition - Door should close	Functioning/ Not functioning	
xi)	<i>For manual operated door:</i> 1) Lift should not move if any of the door car/landing in open condition 2) If door is opened during run, lift should stop and should not move further	Functioning/ Not functioning	
xii)	Car alarm functions even though there is no power supply to the lift	Functioning/ Not functioning	
xiii)	Functioning of intercom (<i>see Note</i>)	Functioning/ Not functioning	
xiv)	Functioning of car call buttons and landing call buttons (<i>see Note</i>)	Ok/Not ok	

SI No.	Requirement	Result	Remarks
(1)	(2)	(3)	(4)
xv)	Provision of fan/blower and its operations	Provided/ Not provided	
xvi)	Provision and working of permanently installed cabin lights that are continuously illuminated except when the car is parked with doors closed. Illumination shall be 100 lux	Provided/ Not provided	
xvii)	Emergency light provided shall come on automatically upon failure of the normal car lighting supply. Shall be capable of illumination intensity of at least 20 Lux for 1 h	Provided/ Not provided	
xviii)	Provided with ventilation apertures on the upper and lower parts of the car with imperforate car doors	Provided/ Not provided	
xix)	Doors are verified for smooth opening and closing at each level	Ok/Not Ok	
xx)	Riding comfort ð No appreciable jerk shall be noticed at the time of travelling, stopping/starting in the car	Ok/Not OK	
xxi)	Car door protective device shall automatically initiate re-opening of the power operated car door(s) in the event of person crossing the entrance during the closing movement	Ok/Not OK	
xxii)	Lift should not move for any safety failure	Ok/Not Ok	
xxiii)	Provision of grab bars inside the lift car (<i>see Note</i>)	Provided / Not provided	
xxiv)	Provision of buffer rail (applicable only for service lifts) at the rear panel of the lift car	Provided / Not provided	

NOTE ð Lifts open to public use shall also be inspected for requirements as per good practice [D-5A(3)], including those relating to lift closing time, finishes of interior surfaces, etc.

A-10 INSPECTION FROM THE FLOOR LANDINGS

SI No.	Requirement	Result	Remarks
(1)	(2)	(3)	(4)
i)	Provision and functioning of landing door de-locking at every landing	Functioning/Not functioning	
ii)	Condition of landing door at every floor	Ok/Not ok	
iii)	Functioning of landing display at every floor	Functioning/Not functioning	
iv)	Functioning of landing call button	Functioning/Not functioning	
v)	Functioning of duplex/group operation at every landing (if applicable)	Functioning/Not functioning	

A-11 RESCUE OPERATIONS

Sl No.	Requirement	Result	Remarks
(1)	(2)	(3)	(4)
i)	Is the automatic rescue device (ARD) installed and operating as per the instructions	Yes/No	
ii)	An audio-visual indicator is provided inside the lift car to alert the passenger(s) trapped inside the car that the lift is on emergency power or there is a mains power failure and they are being rescued	Yes/No	
iii)	Is the stopping accuracy of ± 40 mm achieved during the automatic rescue operation	Yes/No	
iv)	ARD operate in case of failure of any one phase, or any two phases or all the three phases. It also operates during phase reversals (in non-VF type lifts where phase failure relays are installed)	Yes/No	
v)	Emergency light is at least operational till the end of the ARD operation)	Yes/No	

A-12 FIRE OPERATION

Sl No.	Requirement	Result	Remarks
(1)	(2)	(3)	(4)
i)	Phase I a) When the fireman switch is activated, all the landing calls to be inoperative and the car shall report to the evacuation floor and the lift doors to be in opened condition b) If the lift is moving away from the evacuation floor, then it shall reverse its direction at the nearest floor landing without opening its door, and return back to the evacuation floor and remains there itself with the doors open	Functioning/ Not functioning	
ii)	Phase II (if applicable) a) It will be started once after the completion of Phase I along with the fireman switch in ON position b) In Phase II, lift should not respond to landing call and when the car call button is pressed, the door should start closing. If the button is released before the full close of doors, then the doors should open automatically c) After the full close of door, the lift should move on to the floor of registered car call d) The doors should only open if the open button is pressed after reaching the level. If the open button is released before the full open, then the doors should reclose automatically	Functioning/ Not functioning	

A-13 INSPECTION FOR HYDRAULIC LIFTS

SI No.	Requirement	Result	Remarks
(1)	(2)	(3)	(4)
<i>Instructions before inspecting hydraulic lift:</i> a) <i>Do not open or tighten the parts of a pressurized system</i> b) <i>Make sure that the energy source has been locked and tagged before starting the work</i> c) <i>Ensure that there is no pressure</i> d) <i>The car must be at the lowest position</i> e) <i>The piston must be at the bottom</i> f) <i>The shut-off valve must be closed</i> g) <i>Press the emergency lowering valve to release all system pressure</i>			
a)	Check that the oil level in the tank covers the motor-pump system (minimum 20 mm over the motor) when the cylinder is in upper end position and the oil level is 70/80 mm under the tank edge, when the cylinder is in lower extra travel position	Found Ok/Not ok	
b)	Check visually the connection pipe sealing, the joints of the flexible hoses and rigid pipes	Found Ok/Not ok	
c)	Check the rapture valve intervention by loading the car with rated load uniformly distributed and over-speeding the descending car	Found Ok/Not ok	
d)	Is the oil temperature, same as room temperature after the rapture valve intervention	Ok/Not ok	
e)	Check protection against accidental operation of emergency lowering valve	Ok/Not ok	
f)	When the main shut off valve is closed, activating the hand pump, the pressure on the manometer has to increase up to the adjusting valve	Ok/Not ok	
g)	Check the provision of motor thermistor	Provided/Not provided	
h)	Any abnormal noise during normal drive, both in UP and DN direction travel	Ok/Not ok	
j)	Check the working of electrical anti-creep system	Ok/Not ok	
k)	Emergency lowering with battery backup provided and working	Provided/Not provided	
m)	At least one electrically retractable pawl is provided, designed in its extended position to stop the downward moving car against fixed supports	Provided/Not provided	
n)	Check visually, the integrity of the hydraulic system after applying 200 percent full load pressure applied between the non-return valve and the jack for 5 min (observe for any leakage/drop in pressure)	Ok/Not ok	

A-14 SAFETY DEVICES

Sl No.	Requirement	Result	Remarks
(1)	(2)	(3)	(4)
Instructions: a) <i>The lift is in idle condition and the following safety devices get activated, then it should not move further</i> b) <i>If the lift is in running condition and the following safety devices get activated, then the lift shall stop and should not move further</i>			
i)	OSG ð Over speed governor	Functioning/ Not functioning	
ii)	Car door contact	Functioning/ Not functioning	
iii)	Landing door contact	Functioning/ Not functioning	
iv)	Car top stop switches	Functioning/ Not functioning	
v)	Pit stop switches	Functioning/ Not functioning	
vi)	Functioning of screen sensors and door safety edge: 1) Door should reopen, if we cut the screen 2) Door should reopen, if it hit any object	Functioning/ Not functioning	
vii)	Check whether the car and counterweight buffers installed as per the requirement	Yes/No	
viii)	Functioning of limit switches so as lift should not travel beyond the limit	Functioning/ Not functioning	
ix)	Brake on its own shall be capable of stopping the machine when the car is travelling downward at rated speed and with the rated load plus 25 percent	Ok/Not Ok	
x)	Ascending car overspeed Protection provided to detect overspeed of the ascending car and cause the car to stop or at least reduce its speed	Functioning/Not functioning	
xi)	Provision to prevent or stop unintended car movement away from landing with landing door not in the locked position and the car door not in the closed position. Stopping distance shall not exceed 1.20 m from the landing where the unintended car movement has been detected	Functioning/ Not functioning	
xii)	Safety gear test: 1) Lift car safety gear to be tested in down direction with 100 percent rated load in the lift car (during maintenance it shall be tested only in inspection speed), with test tripping arrangement 2) If provided, counterweight safety gear to be tested in up direction of lift car with no load with test tripping arrangement	Stopping distance: _____ Ok/Not ok	

Sl No.	Requirement	Result	Remarks
(1)	(2)	(3)	(4)
	3) Lift stopping distance depends upon the tripping speed of the governor and shall be within the range as per the Indian Standard		

A-15 STOPPING ACCURACY AND RUNNING CLEARANCE TEST

Instructions:

a) The stopping accuracy shall be within ± 10 mm of the finished floor level

b) The running clearance between the lift car threshold and landing door sill should be 35 mm

No. of floors: _____

Floors (Landing)	Stopping Accuracy		Running Clearance		
	UP	DN	LH	Centre	RH
1 st					
2 nd					
3 rd					
4 th					
5 th					
6 th					
é					
é					
Result Ok/Not ok					

A-16 LOAD TEST

Sl No.	Load in Percent	Direction of Travel	Voltage	Current
			V	A
(1)	(2)	(3)	(4)	(5)
<i>Instructions:</i> a) With balanced load, the current value in both the up and down directions to be same b) Check the current only after the lift reaches its rated speed				
i)	No load	UP		
		DN		
ii)	Balance load	UP		
		DN		
iii)	Full load	UP		
		DN		

LIST OF STANDARDS

The following list records those standards which are acceptable as 'good practice' and 'accepted standards' in the fulfilment of the requirements of this NBCS. The latest version of a standard shall be adopted at the time of enforcement of this NBCS. The standards listed may be used as a guide in conformance with the requirements of the referred clauses in this NBCS.

	<i>IS No.</i>	<i>Title</i>
(1)	IS 17900 (Part 1) : 2022	Lifts for the transport of persons and goods: Part 1 Safety rules
(2)	IS 17900 (Part 2) : 2022	Lifts for the transport of persons and goods: Part 2 Design rules calculations examinations and tests of lift components
(3)	IS 4963 : 2025	Accessibility in built environment for older adults and persons with disabilities ð Requirements (<i>second revision</i>)
(4)	IS 12615 : 2018	Line operated three phase a.c. motors (IE Code) 'efficiency classes and performance specification (<i>third revision</i>)
(5)	IS 1950 : 1962	Code of practice for sound insulation of non-industrial buildings
(6)	IS 14665 (Part 2/Sec 1 and Sec 2) : 2000	Electric traction lifts: Part 2 Code of practice for installation, operation and maintenance, Section 1 Passenger and goods lifts, Section 2 Service lifts
(7)	IS 15105 : 2021	Design and installation of fixed automatic sprinkler fire extinguishing systems ð Code of practice (<i>first revision</i>)
(8)	IS 732 : 2019	Code of practice for electrical wiring installations (<i>fourth revision</i>)
(9)	IS 3043 : 2018	Code of practice for earthing (<i>second revision</i>)
(10)	IS 17518 (Part 2) : 2021/ ISO 3008-2 : 2017	Fire-resistance tests: Part 2 Lift landing door assemblies
(11)	IS/IEC 60529 : 2001	Degrees of protection provided by enclosures (IP Code)
(12)	IS 17805 (Part 20) : 2024 ISO 8102-20 : 2022	Electrical requirements for lifts escalators and moving walks: Part 20: Cyber security
(13)	IS 14665 (Part 3/Sec 1 and Sec 2) : 2000	Electric traction lifts: Part 3 Safety rules, Section 1 Passenger and goods lifts, Section 2 Service lifts
(14)	IS 2551 : 1982	Danger notice plates (<i>first revision</i>)
(15)	IS 15785 : 2007	Installation and maintenance of lifts without conventional machine rooms ð Code of practice
(16)	IS 15259 : 2002	Installation and maintenance of home lifts ð Code of practice
(17)	IS 15330 : 2020	Requirements of lifts for persons with disabilities (<i>first revision</i>)
(18)	IS 17900-7-7	Lifts for the transport of persons and goods: Part 7 Lifts for special applications, Section 7 Lifts under seismic conditions (<i>under prepration</i>)
(19)	IS 17615 (Part 1) : 2021/ ISO 18738-1 : 2012	Measurement of ride quality : Part 1 Lifts elevators

	<i>IS No.</i>	<i>Title</i>
(20)	IS 17106 (Part 1) : 2019/ ISO 22201-1 : 2017	Lifts (elevators), escalators and moving walks — Programmable electronic systems in safety-related applications: Part 1 Lifts (elevators) (PESSRAL)
(21)	IS 17900 (Part 5) : 2026	Lifts for the transport of persons and goods: Part 5 Guide for inspection of lifts
(22)	IS 17900 (Part 6) : 2022	Lifts for the transport of persons and goods: Part 6 Guide for maintenance of lifts

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NATIONAL BUILDING CONSTRUCTION STANDARDS

PART D BUILDING SERVICES

SECTION 5 INSTALLATION OF LIFTS, ESCALATORS, MOVING WALKS AND POWER-DRIVEN PARKING SYSTEMS

5B ESCALATORS AND MOVING WALKS

BUREAU OF INDIAN STANDARDS

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National Building Code Sectional Committee, CED 46

FOREWORD

This NBCS (Part D/Subsection 5B) covers the requirements for planning and design, installation, operation, maintenance, and inspection of escalators and moving walks to ensure safe movement of people with satisfactory performance.

This Section was first published in 1970 and was subsequently revised in 1983, 2005 and 2016 as a part of National Building Code of India. This Section covers the requirements for installation of lifts and escalators in buildings. This Section should be read with Part F 'Fire and Life Safety' of this NBCS from fire safety requirements point of view. The major changes in the first revision of 1983 were addition of outline dimensions of different types of lifts and detailed requirements of escalators in buildings. Emphasis was laid on coordination between the engineer/architect and the lift manufacturer to arrive at the number and position of lifts for attaining optimum efficiency in serving the building with safety.

The significant changes with respect to lifts incorporated in the 2005 revision, included the addition of new clauses/recommendations on the building management system; addition of new clauses on fireman's lift, infrared light curtain, safety and Braille button for blind people and updation of provisions as per the revised standards on lifts on which this Section was based.

In the 2016 revision, the erstwhile Section 5 on lifts and escalators was divided into two Subsections namely:

- a) 5A Installation of lifts; and
- b) 5B Installation of escalators and moving walks.

This Subsection 5B addresses exclusively the provisions related to the installation of escalators and moving walks, which have been increasingly provided in buildings. It covers comprehensive provisions on all aspects, including requirements for planning and design, installation, operation, maintenance, and inspection of escalators to ensure the safe movement of people with satisfactory performance. This Subsection also covers the provisions on moving walks.

In this revision, this publication has been renamed as National Building Construction Standards (NBCS).

As a result of technical developments in the subject of escalators and moving walks, the experience gained since implementation of 2016 version of this NBCS and feedback received as well as the revision of related Indian Standards on the subject, a need was felt to revise this Subsection. The significant modifications made in this revision of the subsection include amongst others, the following key changes:

- a) The terminology has been updated addressing maintenance aspects and electrical safety system.
- b) The requirements of power supply including protection devices have been elaborated, in [6.5.3](#).
- c) Fire protection requirements for the installations have been elaborated, in [7](#).
- d) Technical requirements for outdoor escalators/moving walks and semi- outdoor escalators/moving walks have been included in [9.2](#) and [9.3](#).
- e) Inspection check list has been revised, under [14.6.2](#).
- f) [Annex A](#) on typical escalator site checklist/inspection list has been considerably revised.
- g) [Annex B](#) on typical check list for functional tests on safety switches and devices has been revised.

The information contained in this Subsection is largely based on the following Indian Standards/Special Publications:

IS 732 : 2019	Code of practice for electrical wiring installations (<i>fourth revision</i>)
IS 3043 : 2018	Code of practice for earthing (<i>second revision</i>)

IS 4591	Escalators and moving walks:
(Part 1/Sec 1) : 2020	Safety requirements (<i>first revision</i>)
(Part 1/Sec 2) : 2020	Guide for planning and selection (<i>first revision</i>)
(Part 1/Sec 3) : 2020	Inspection and test (<i>first revision</i>)
(Part 2) : 2022	Guide for maintenance
CEA : 2023	Central Electricity Authority Regulation: 2023
SP 30 : 2023	National Electrical Code of India 2023

All standards, whether given herein above or cross-referred to in the main text of this Subsection, are subject to revision. The parties to agreement based on this Subsection are encouraged to investigate the possibility of applying the most recent editions of the standards.

Assistance has also been derived from the following publications in the revision of this Subsection:

IBC 2024 International Building Code, International Code Council, Washington, USA

CIBSE Guide D : Transportation Systems in Buildings, 2025, The Chartered Institution of Building Services Engineers, London, U.K.

EN 115-1 : 2017 Safety of escalators and moving walks — Part 1: Construction and installation

As the regulation of land development and buildings is a State subject and is dealt by the States and local bodies such as urban local bodies within their jurisdiction through building regulatory documents like building regulations, building byelaws, and fire regulations, this document is voluntary in nature and is non-binding. The implementation depends on adoption by concerned parties or stipulations in a contract or by appropriate adoption by the concerned authorities.

For deciding whether a particular requirement of this Subsection is complied with, the final value, observed or calculated, expressing the result of a test or analysis, shall be rounded off in accordance with IS 2 : 2022 'Rules for rounding off numerical values (*second revision*)'. The number of significant places retained in the rounded off value should be the same as that of the specified value in this Subsection.

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Important Explanatory Note for Users of this NBCS

In any Part/Section of this NBCS, where reference is made to ‘**good practice**’ in relation to **design, constructional procedures or other related information**, and where reference is made to ‘**accepted standard**’ in relation to **material specification, testing, or other related information**, the Indian Standards listed at the end of the Part/Section shall be used as a guide to the interpretation.

At the time of publication, the editions indicated in the standards were valid. All standards are subject to revision and parties to agreements based on any Part/Section are encouraged to investigate the possibility of applying the most recent editions of the standards.

In the list of standards given at the end of a Part/Section, the number appearing within parentheses in the first column indicates the number of the reference of the standard in the Part/Section. For example:

a) Good practices [D-5B(1)] refers to the Indian Standard(s) referred to in serial number (1) of the list of standards given at the end of this Part/Subsection, that is, SP 30 : 2023 ‘National Electrical Code of India 2023’, IS 732 : 2019 ‘Code of practice for electrical wiring installation (*fourth revision*)’, IS 3043 : 2018 ‘Code of practice for earthing (*second revision*)’.

NATIONAL BUILDING CONSTRUCTION STANDARDS

PART D BUILDING SERVICES

SECTION 5 INSTALLATION OF LIFTS, ESCALATORS, MOVING WALKS AND POWER-DRIVEN PARKING SYSTEMS

5B ESCALATORS AND MOVING WALKS

1 SCOPE

1.1 This NBCS (Part D/Subsection 5B) covers the essential requirements for planning, installation, operation, maintenance, and inspection of escalators and moving walks to ensure safe movement of people with satisfactory performance.

1.2 This Subsection gives information that should be exchanged among the architect/engineer, the consulting engineer and escalator/moving walk manufacturer from the stage of planning to installation including maintenance.

2 TERMINOLOGY

For this Subsection, the following terms and definitions apply.

2.1 General Terms Relating to Escalators and Moving Walks

2.1.1 *Angle of Inclination* — Maximum angle to the horizontal in which the steps, the pallets, or the belt move.

2.1.2 *Operational or Service Brake* — Brake that usually acts on motor shaft and used to stop an escalators/moving walk under all normal conditions or under certain fault conditions.

2.1.3 *Balustrade* — Part of the escalator/moving walk which ensures the user's safety by providing stability, protecting from moving parts and supporting the handrail.

2.1.4 *Balustrade Decking* — Transverse member of the balustrade which meets the handrail guidance profile, and which forms the top cover of the balustrade.

2.1.5 *Brake Load* — Load on the step/pallet/belt for which the brake system is designed to stop the escalator/moving walk.

2.1.6 *Comb* — Pronged section at each landing that meshes with the grooves of the steps/pallets.

2.1.7 *Comb Lighting* — Small flush light panels located in the skirt panels on both sides of the escalators/moving walk unit at both upper and lower landing and immediately adjacent to the comb teeth to illuminate the comb and step tread and assist passengers boarding and alighting the escalators/moving walk.

2.1.8 *Comb Plate* — Platform at each landing to which the combs are attached.

2.1.9 *Comb Plate Switch* — Switch in safety circuit that opens when excessive force is detected in vertical and/or horizontal direction on the comb or comb plate of escalators/moving walk.

2.1.10 *Comb Teeth* — Series of teeth which ride the grooves of the escalators/moving walk step tread as the step passes underneath and are designed to allow them to break-off if a wedging action should occur at their point of contact with step tread.

2.1.11 *Auxiliary or Emergency or Safety Brake* — Mechanically operated brake which operates automatically to stop a fully loaded escalator if the drive chain breaks. It is typically situated on one side of the main drive shaft.

2.1.12 *Emergency Stop Switch* — Separate stop button usually located in adjacent walls, columns or within the balustrade providing the facility to stop the escalator/moving walk, in the event of emergency.

2.1.13 Inspection Door — Means of access to equipment areas and other spaces pertaining to an escalators/moving walk installation such as machinery spaces, etc, and with access usually restricted to authorized persons.

2.1.14 Electrical Safety Devices — Part of a safety circuit consisting of safety switches and/or fail-safe circuits.

2.1.15 Escalator — Power-driven, inclined, continuous moving stairway used for raising or lowering persons in which the user carrying surface (for example, steps) remains horizontal.

NOTE — Escalators are machines even when they are out of operation and shall not be considered or used as fixed staircases.

2.1.16 Exterior Panel — Part of the exterior side of the enclosure of an escalator or moving walk.

2.1.17 Handrail — Power-driven moving rail for persons to grip while using the escalator or moving walk.

2.1.18 Interior Panel — Panel located between the skirting or lower inner decking and the handrail guidance profile or balustrade decking.

2.1.19 Lower Inner Decking — Profile that connects the skirting with the interior panel when they do not meet at a common point.

2.1.20 Lower Outer Decking — Profile that connects the exterior panels with the interior panel.

2.1.21 Machinery — Escalator or moving walk machine(s) mechanisms and associated equipment.

2.1.22 Machinery Spaces — Space(s) inside or outside of the truss where the machinery as a whole or in parts is placed.

2.1.23 Maximum Capacity — Maximum flow of persons that can be achieved under operational conditions.

2.1.24 Moving Walk — Power-driven installation for the conveyance of persons in which the user carrying surface remains parallel to its direction of motion and is uninterrupted (for example, pallets, belt).

NOTE — Moving walks are machines even when they are out of operation and shall not be considered or used as a fixed access.

2.1.25 Newel — End of the balustrade.

2.1.26 Nominal Speed — Speed in the direction of the moving steps, pallets, or the belt, when operating the equipment under no load condition (that is, without persons), stated by the manufacturer as that for which the escalator or moving walk has been designed.

NOTE — Rated speed is the speed the escalator/moving walk moves under rated load conditions.

2.1.27 Pit — Recess in the floor to receive that portion of the lower landing and the lower end of the incline section which occurs below the floor line when there is no floor under the escalator such as in a basement.

2.1.28 Rated Load — Load which the equipment is designed to move.

2.1.29 Rise — Vertical distance between the upper and lower finished floor levels.

2.1.30 Safety Circuit — Part of the electric safety system consisting of electrical safety devices.

2.1.31 Skirting — Vertical part of the balustrade interfacing with the steps, pallets, or belt.

2.1.32 Skirt Deflector — Device to minimize the risk of trapping between the step and the skirting.

2.1.33 Electrical Safety System — Safety related part of the electrical control system as an arrangement of safety circuits and monitoring devices.

2.1.34 Maintenance — All the necessary operations to ensure the safe and intended functioning of the installation and its components after the completion of the installation, throughout its life cycle. See [15.1](#) for details.

2.1.35 Maintenance Organization — Company or part of company where competent maintenance person(s) carry out maintenance operation on behalf of the owner of the installation.

2.1.36 Competent Maintenance Person — Designated person, suitably trained, qualified by knowledge and practical experience, provided with necessary instructions, and supported within their maintenance organization to enable the required maintenance operations to be safely carried out.

2.1.37 Manufacturer — Entity who takes responsibility for the design, manufacture and placing on the market the machinery of escalators or moving walks.

2.1.38 Installer — Entity who takes responsibility for the design, manufacture, installation and placing on the market of escalator/moving walk.

2.1.39 Installation — Completely installed escalator or moving walk.

2.1.40 Owner of the Installation — Person who has the power of disposal of the installation and/or takes the responsibility for its operation and use.

2.2 Terms Related to Performance Requirements of Escalators and Moving Walks

2.2.1 Acceleration — Rate of change of velocity. It is expressed in m/s^2 .

2.2.2 Ride Quality — Sound levels and vibration of the steps/pallet, relevant to passenger perception, associated with escalator or moving walk operation.

2.2.3 Sound Pressure Level (L_p) — Ten times the logarithm to the base 10 of the ratio of the square of the sound pressure to the square of the reference sound pressure.

NOTE — The reference sound pressure level is $20 \mu\text{Pa}$ ($2 \times 10^{-5} \text{ Pa}$).

2.2.4 Velocity — Rate of change of displacement. Velocity is reported as speed and direction of travel. It is given in metre per second (m/s).

2.2.5 Vibration — Variation with time of the magnitude of acceleration. It is expressed in metre per second square (m/s^2).

3 GENERAL

3.1 Conformity with Act and Rules

3.1.1 The installation of escalators and moving walks has been governed in states by different Acts and Rules, as applicable, which are intended to ensure safe installation and operation of the same.

3.1.2 The installation shall be generally carried out in conformity with relevant Act and Rules wherever they are in force.

3.1.3 It is the responsibility of the owner of the premises where the escalators and moving walks will be installed, to obtain necessary permission from the Authority before and after the erection of escalators and moving walks and for their subsequent operation.

3.1.4 A license for public use is a safety provision issued by state authorities under applicable Act and Rules, wherever they are in force.

3.2 Conformity with Indian Electricity Act and Rules

All electrical work in connection with installation of escalators and moving walk shall be carried out in accordance with the provisions of *Electricity Act, 2003* Central Electrical Authority (Measures relating to Safety and Electric Supply) Regulations, 2023 and accepted standards [D-5B(1)] as amended up to date along with the rules and regulations framed thereunder and shall also comply with the other provisions of Part D 'Building Services, Section 2 Electrical and Allied Installations' of this NBCS.

3.3 Conformity with Indian Standards

All materials, fittings, appliances, etc, used in electrical installation shall conform to Indian Standard specifications, wherever these exist. In case of materials for which Indian Standard specifications do not exist, the materials shall be approved by the competent authority. The installation shall be according to best practices as in the accepted standards [D-5B(1)].

3.4 Conformity with Fire Regulations

The installation shall be carried out in conformity with the state fire acts/local fire regulations, wherever they are in force *See also* Part F 'Fire and Life Safety' of this NBCS for guidance purpose.

3.5 Safety Design Considerations and Selection of Escalator and Moving Walks

The design consideration for type and selection of escalator and moving walk shall be based on following criteria:

- a) *Location* — To ensure reliability, it is important to have design specifications to suit environmental conditions;
- b) *Physical Requirements* — Physical factors like the vertical and horizontal distance to be spanned shall be considered. The ability of the building infrastructure to support the heavy components is also a critical physical concern;
- c) *Type of Segment and Traffic Patterns* — The design consideration depends on segment type. Some typical segments are transit systems like metro, airport, shopping malls, hotels, etc. Traffic patterns shall also be anticipated in escalator and moving walk design. The carrying capacity of an escalator system shall match the expected peak traffic demand;
- d) *Safety Considerations* — Escalators and moving walks shall not be subjected to abuse, misuse, and negligence. There are a few potential hazards, both from escalators and moving walks themselves and those hazards that users may bring with them, which include loose clothing, type of footwear (especially soft footwear), carried goods and trolleys. The severity of these hazards may increase when used by vulnerable groups of people, such as the old, very young and those with impaired mobility. Proper design risk assessment and control are all critical to ensure reliability and, above all safety; and
- e) *Aesthetic Preferences* — To be suitably decided between the parties concerned.

3.6 Maintenance

The considerations relating to maintenance shall be as follows:

- a) The escalator or moving walk installation should receive regular cleaning, lubrication, adjustment, and adequate servicing by authorized competent persons at such intervals as the type of equipment and frequency of service demand. It is desirable and normal for the escalator or moving walk supplier to be entrusted with the servicing during the guarantee period of new equipment;
- b) In order that the escalator or moving walk installation is always maintained in a safe condition, a proper maintenance schedule shall be drawn up in consultation with the manufacturer and rigidly followed. Provision of a logbook to record all items relating to general servicing and inspection is recommended. Any accident arising out of operation or maintenance shall be duly reported to the Authority in accordance with the rules laid down;
- c) Escalators and moving walks are required by statutory regulations to be examined at regular intervals as specified by Acts and Rules, by a competent person. The company entrusted with maintenance contract shall have valid license to maintain the escalators and moving walks. The persons assigned for maintenance work shall be appropriately qualified and experienced as required by statutory Acts and Rules;
- d) Companies and organizations responsible for the safe use of escalator and moving walks should ensure that their employees are adequately instructed and informed about the safe use of the escalator or moving walk and the hazards arising from their unsafe use and are familiar with emergency procedures; and
- e) Proper barricading shall be done to ensure that the open pits of escalator shall be reached/accessed by only the maintenance team. The maintenance service provider shall be responsible for placing of the

barricades and signages before start of the maintenance work.

3.7 Energy Efficiency and Sustainability

The following technologies shall be encouraged for reducing power consumption by using energy efficient equipment and behavioural changes:

- a) As an option, escalators/moving walk should not run when there are no passengers to move. In this stop mode, only the control system and the passenger detection system are kept running. Another option is to reduce the speed to match the passenger demand, thus reducing energy consumption using variable voltage variable frequency (VVVF) drives, which provide very smooth, almost imperceptible speed transitions;
- b) Energy saving LED lamps for lighting in place of conventional lamps;
- c) Improvement in total power factor of the motor drive of an escalator or moving walk at the isolator connecting equipment to the building's electrical supply circuit; and
- d) Adoption of materials and practices that are environmentally friendly and sustainable shall be promoted.

4 DESIGN GUIDELINES FOR ESCALATORS AND MOVING WALKS

4.1 Escalators are desirable where the movement of people, in large numbers at a controlled rate in the minimum of space, is involved, for example, railway stations, shopping centres/malls, airports, etc. These encourage people to circulate freely and conveniently. Inclined moving walks are preferable where trolley movement is expected between levels, like department stores, airport, etc.

Typically, horizontal moving walks are used where medium to long distance travel is involved such as airports/metro stations/exhibition halls.

4.2 As the escalators and moving walks operate at a constant speed, serve only two levels, and have a known maximum capacity, the traffic study is rather easy, provided the population to be handled in a given time is known. It is easy to predict the rate at which the population can be handled.

Depending on the location of project and the expected tenancy, architect/owner shall provide the expected footfall in weekdays/weekends/special occasions (holidays, sale, etc) to the designer to consider the peak footfall while providing escalators.

[Figure 1](#) shows the brief idea about the steps to be followed for arriving at appropriate escalator and moving walk solution for an application.

Based on [Fig. 1](#), type, and location of building, expected population, and the serving floor details are required to work on the preliminary traffic analysis for escalators and moving walks.

For selection, installation and location of escalators and moving walks, the design of escalators and moving walks fall into four distinct duty categories, as shown in [Table 1](#).

When deciding the duty category, following points shall be considered:

- a) The duration and magnitude of peak demands that the equipment might be subjected to; and
- b) The number of passengers using the escalators or moving walks per day.

NOTE — Select an escalator or moving walk appropriate for intended duty. If in doubt specify a more robust escalator or moving walk, as applicable.

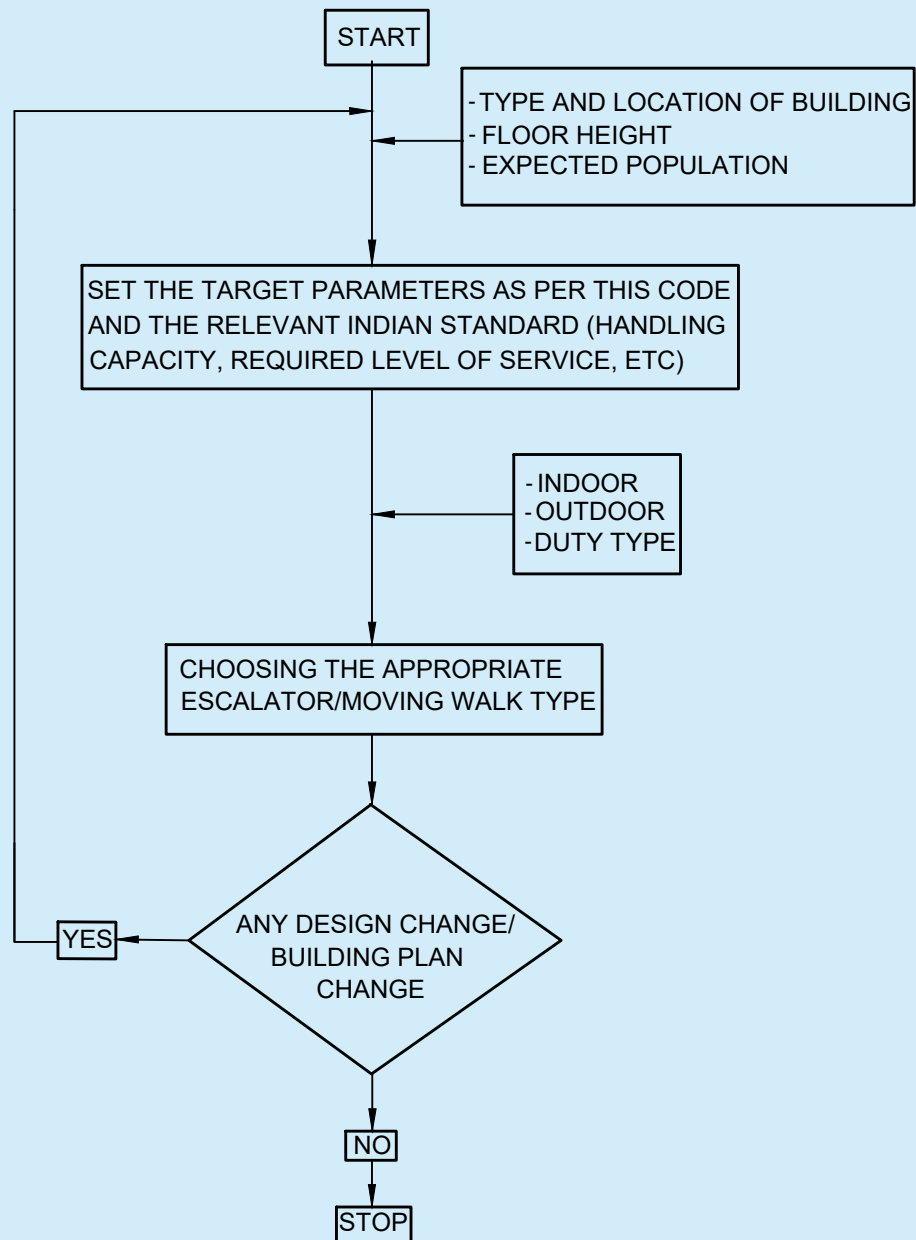


FIG. 1 TYPICAL FLOWCHART ON PRELIMINARY DESIGN

Table 1 Escalator Duty Categories

(Clause 4.2)

Sl No.	Duty Category	Typical Usage (Passengers/Day)	Typical Locations
(1)	(2)	(3)	(4)
i)	Light	Up to 3 000	Shops, museums, libraries, leisure facilities and multiplexes
ii)	Medium	Up to 10 000	Department stores, shopping centres, domestic airports and railway stations
iii)	Heavy	Up to 20 000	Major railway and metro stations, major international airports and critical locations such as underground railway systems
iv)	Intensive	Over 20 000	Same as heavy category, but at a higher usage

4.3 The most common rated speed of escalators is 0.5 m/s although speeds of 0.65 m/s and 0.75 m/s are available. Most escalators run at one speed only, although some escalators running in energy saving mode can switch to the higher speed on traffic detection.

Step widths of 600 mm, 800 mm and 1 000 mm for escalators are available, the latter allowing two columns of passengers to be carried.

Angle of inclination of an escalator from the horizontal shall not exceed 30°, but for rises not exceeding 6 m and rated speed not exceeding 0.5 m/s the angle of inclination is permitted to be increased up to 35°.

The rated speed of the escalator shall not exceed:

- 0.5 m/s for an escalator with an angle of inclination of more than 30° but within 35°; and
- 0.75 m/s for an angle of inclination up to 30°.

For normal peak periods, the recommended handling capacities for design purposes should be taken as 3 200 to 6 400 persons per hour depending upon the width of the escalator.

The number of persons that may be theoretically carried by the escalator in 1 h can be calculated as follows:

- The density factor (k) considers the step occupancy levels according to step width. For determination of theoretical capacity, it is assumed that one step with an average depth of 0.4 m can carry 1 person for a step width of 0.6 m, 1.5 persons for a step width of 0.8 m and 2 persons for a step width of 1.0 m;
- The theoretical capacity then is: $3\,600 \times (\text{rated speed in m/s} \times k)/0.4$

where

$k = 1, 1.5, \text{ or } 2$ for 0.6, 0.8 and 1.0 m step widths respectively.

- Some values calculated as per the above are given in [Table 2](#):

Table 2 Theoretical Escalator Handling Capacity
(Clause 4.3)

Sl No.	Step/Pallet Width m	Nominal Speed		
		m/s		
		0.5	0.65	0.75
		Persons/hour	Persons/hour	Persons/hour
(1)	(2)	(3)	(4)	(5)
i)	0.6	4 500	5 850	6 750
ii)	0.8	6 750	8 775	10 125
iii)	1.0	9 000	11 700	13 500

4.4 The handling capacity (C_e) of a standard escalator with a step depth of 400 mm is: $C_e = 150 V k$

where

V = speed along the incline in m/s; and

k = average density of people (persons per escalator step).

The density factor (k) considers the step occupancy levels. From experience, practical values of k that have been found satisfactory are:

- For 600 mm steps:
 $k = 0.5$ (one person per two steps of 600 mm);
- For 800 mm steps:
 $k = 0.75$ (one and half persons per two steps of 800 mm); and
- For 1000 mm steps:
 $k = 1.0$ (one person per step of 1 000 mm).

[Table 3](#) gives guidance to practical escalator handling capacities based on the densities assumed above in person per hour and horizontal speeds in meters per second.

Table 3 Practical Escalator Handling Capacity

(Clause [4.4](#))

SI No.	Step/Pallet Width m	Nominal Speed		
		m/s		
		0.5	0.65	0.75
		Persons/hour	Persons/hour	Persons/hour
(1)	(2)	(3)	(4)	(5)
i)	0.6	2 250 [3 600*]	2 925 [4 400*]	3 375 [4 900*]
ii)	0.8	3 375 [4 800*]	4 388 [5 900*]	5 063 [6 600*]
iii)	1.0	4 500 [6 000*]	5 850 [7 500*]	6 750 [8 200*]

*IS 4591 values are shown in parentheses, which assumes passenger densities averaging $k = 0.75, 1.00, 1.25$ respectively and suggests maximum values.

4.5 Moving walks can be horizontal or inclined. Horizontal moving walks are typically used for medium/long range travel in airports, exhibition centres and railway stations. Inclined moving walks are typically used for short range travel in shopping centres, stores and railway stations.

Inclinations range from 6° to 12° , although the safest maximum inclination is generally considered to be 10° . The typical running speeds for horizontal moving walks are 0.5 m/s, 0.65 m/s and 0.75 m/s. For moving walks with an inclination greater than 6° , the nominal speed shall be limited to 0.5 m/s. The speed is measured in the direction of movement of the pallets.

Nominal widths for moving walks up to an inclination of 6° are 800 mm, 1 000 mm and 1 400 mm and above 6° are 800 mm and 1 000 mm. A width of 1 400 mm allows two columns of passengers and the possibility for some passengers to walk along the moving walk.

The use of suitably designed shopping trolleys and baggage carts on moving walks is permitted. Shopping trolleys or baggage carts which are chosen for use on a moving walk shall be specified between the baggage cart manufacturer and the moving walk manufacturer. If non-specified shopping trolleys or baggage carts are available in the moving walk area, there is a serious risk of misuse. It is necessary to prevent access to the moving walk entrance. The width of the shopping trolley or baggage cart and its contents shall be at least 400 mm less than the nominal pallet/belt width. Passengers shall be able to leave the moving walk, even if shopping trolleys or baggage carts are on the moving walk.

[Table 4](#) provides the theoretical values of handling capacity in person per hour of a moving walk.

4.6 [Table 5](#) gives guidance to practical moving walk handling capacities in person per hour assuming a density of 2.0 persons/m².

Table 4 Theoretical Moving Walk Handling Capacity

(Clause 4.5)

Sl No.	Pallet Width m	Horizontal Moving Walk (0° Inclination)			Inclined Moving Walk For (Nominal Speed 0.5 m/s)		
		Nominal Speed (m/s)			Inclination		
		0.5	0.65	0.75	6°	10°	12°
(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
i)	0.8	3 000	3 700	4 400	3 000	3 700	3 000
ii)	1.0	3 800	4 600	5 400	3 700	3 700	—
iii)	1.4	5 100	6 400	7 700	5 100	—	—

Table 5 Practical Moving Walk (Horizontal and Inclined) Handling Capacity

(Clause 4.6)

Sl No.	Step/Pallet Width m	Nominal Speed		
		0.5	0.65	0.75
		Persons/hour	Persons/hour	Persons/hour
(1)	(2)	(3)	(4)	(5)
i)	0.8	2 880 [4 800*]	3 648 [5 900*]	4 320 [6 600*]
ii)	1.0	3 600 [6 000*]	4 560 [7 300*]	5 400 [8 200*]
iii)	1.4	5 000 [-----*]	6 350 [-----*]	7 560 [-----*]

*IS 4591 values are shown in parentheses.

NOTES

1 Use of shopping trolleys and baggage carts will reduce the capacity by approx. 80 percent.

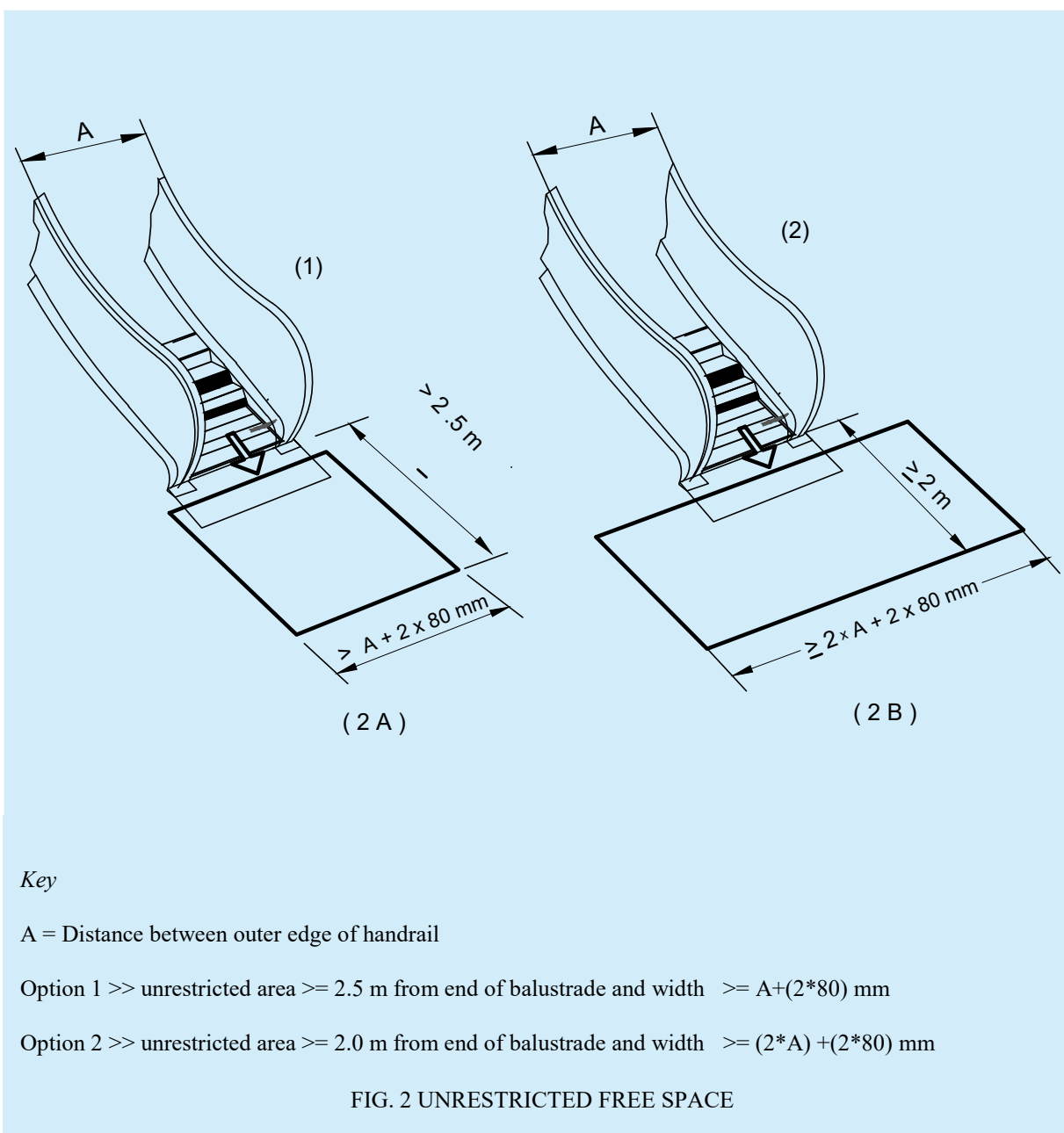
2 For moving walks with a pallet width in excess of 1.0 m the capacity is not increased as users need to hold the handrail, the additional width is to principally enable the use of shopping trolleys and baggage carts.

5 ARRANGEMENT AND PLANNING DIMENSIONS

5.1 Escalators are typically used for short range of movement between adjacent floors. They are usually situated in an obvious circulation path making it easy for pedestrians to board them. Escalators and moving walks should only be accessed from adjacent corridors/walkways, landing and lobby areas, where people do not obstruct other pedestrian circulation routes. Space should be available to accommodate queuing at the boarding point.

It is especially important that the boarding and alighting areas adjacent to an escalator or moving walk are not part of another circulation route, to provide a safe area for passengers to board and alight at landings. The area of this space is defined as a distance between the handrails plus 160 mm, multiplied by a depth of 2.0 m to 2.5 m, depending on the configuration of escalator or moving walk.

[Figure 2](#) illustrates the above clause with option 1 being a 2.5 m landing depth and option 2 being 2.0 m landing depth. Where successive units are installed, each successive escalator or moving walk should have its own individual unrestricted area. For succeeding escalators and moving walks the depth of an unrestricted area shall be determined in each individual case depending on, for example, type of use (persons only or persons with transport devices), number of intermediate exits, relative orientation, and theoretical capacity).



Where the exit of the escalator or moving walk is blocked by structural measures (for example shutters, fire protection gates, removable barriers), an additional STOP button for emergency situations shall be provided:

- within reach from inside the escalator/moving walk at 2.00 m to 3.00 m before the steps/pallets reach the comb intersection line; and
- Between 200 mm below and 400 mm above the handrail, measured from the top of the handrail to the middle of the actuator of STOP button.

NOTE — STOP button shall be with protection to prevent nuisance activation for example shroud guard. STOP switch with cover is not permitted.

[Figure 3](#) illustrates the position of additional STOP button.

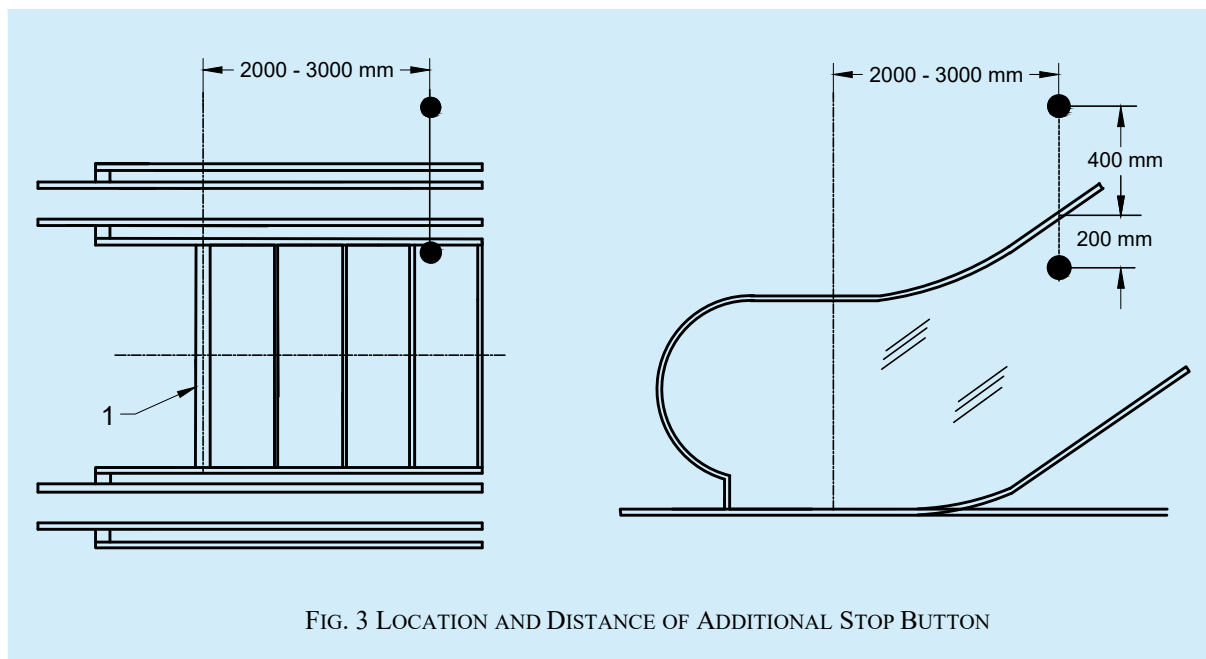


FIG. 3 LOCATION AND DISTANCE OF ADDITIONAL STOP BUTTON

The position of the shutters, fire protection gates, removable barriers, etc shall be protected with a contact in the safety circuit. As soon as the doors close, the escalator or moving walk should stop.

Escalators occupy more space than stairs to accommodate their inclination. There are few standard escalator arrangements as shown in Fig. 4. Type (a) and type (b) provide efficient circulation by providing the shortest transition path and time from one escalator and the next. Type (b) requires larger structural opening than type (a) and type (c) and present users with a higher risk of falling into the void. Type (c) is typical of a store as it allows the store to lengthen the circulation route past goods for sale. This configuration also takes up free space.

Some moving walks are adopted to receive and lock on shopping/baggage trolleys. In these cases, the unrestricted area at the landing should be increased to at least 5 m and number of flat/horizontal steps at the boarding and alighting points increased. The most significant effect is the increased footprint required for the equipment. Shopping/baggage trolleys are not permitted on escalator.

To encourage pedestrian confidence and to assist efficient and safe boarding/ alighting of escalators, the start and end of escalator should present a few horizontal/flat steps. At least two flat steps are provided for escalators speeds up to 0.5 m/s; at least three flat steps for speeds above 0.5 m/s and up to 0.65 m/s for escalator having rise more than 6 m; and at least four flat steps for speeds above 0.65 m/s. In locations where it is anticipated that escalators may be used by persons with impaired mobility or heavy passenger traffic, additional flat steps should be considered.

5.2 Ensuring Accessibility for All Users

5.2.1 The accessibility requirements shall be complied with.

5.2.2 The location of escalators and moving walks should be considered in relation to the position of adjacent fire protected lift shafts and lobbies, staircases, and their associated areas of rescue assistance, as at the time of an emergency these become non-functional. A surface of the escalator and moving walks should contrast visually with the approach and there should be audible signals or pre-recorded messages that indicate the start and finish of the escalator to help visually impaired people.

For important reasons of safety, special warning notices and tactile warning indicators shall be provided at the top and bottom of escalators. Escalators shall not be considered as the accessible route. Persons with locomotor disability cannot use them. Hence a lift should also be installed as an alternative accessible route.

For moving walks, the maximum inclination should be up to 6° for use by people with disabilities. People with disability may find it difficult to use moving walks. Hence, alternative like an accessible battery cart, may be provided to them in appropriate cases. Moving walks shall be free of projecting objects and obstacles up to a height of 2 100 mm. A minimum level of illumination of 100 lux shall be provided on moving walks.

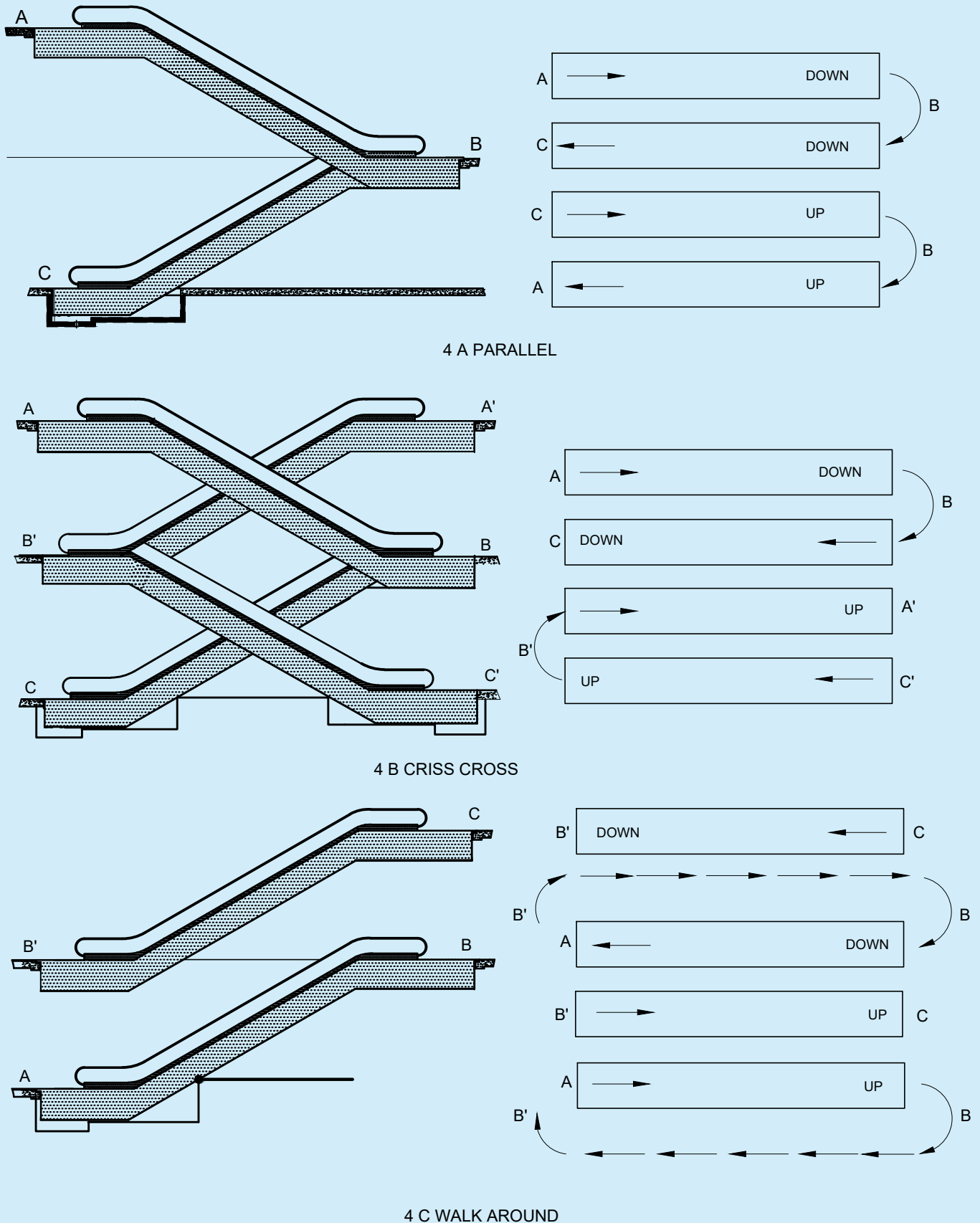


FIG. 4 ESCALATOR CONFIGURATION

6 SPACE REQUIREMENT

6.1 Free Space for Users

6.1.1 The clear height above the steps of the escalator or pallets or belt of the moving walk at all points shall not be less than 2.30 m (see X4 in [Fig. 5](#) and [Fig. 7](#)). The clear height shall extend to the end of the newel.

NOTE — The clear height of 2.30 m should also be applied to the unrestricted area.

6.1.2 To prevent collision, a minimum free area around the escalator or moving walk is defined as per [Fig. 5](#). The height X12, measured from the steps of the escalator or the pallets or the belt of the moving walk shall be at least 2.10 m (see X12 in [Fig. 5](#)). The distance between the outer edge of the handrail and walls or other obstacles (see Y10 in [Fig. 5](#)) shall under no circumstances be less than 80 mm horizontally and 25 mm vertically below the lower edge of the handrail (see Y12 in [Fig. 8](#)). The area is permitted to be smaller, if by appropriate measures, the risk of injury is minimized.

6.1.3 For escalators arranged adjacent to one another either parallel or criss-cross, the distance between the handrails shall be not less than 160 mm (see Y11 in [Fig. 5](#)).

6.1.4 Where building obstacles may cause injuries, appropriate preventive measures shall be taken.

At floor intersections and on criss-cross escalators or moving walks, a vertical deflector of not less than 0.30 m in height, not presenting any sharp cutting edges, shall be placed above the handrail level and extend at least 25 mm below the lower edge of the handrail, for example, as an imperforate triangle (see X5 in [Fig. 7](#) and [Fig. 9](#)).

It is not necessary to comply with these requirements when the distance Y9 between the outer edge of the handrail and any obstacle is equal to or greater than 400 mm (see [Fig. 5](#)).

6.1.5 In the case of successive escalators and moving walks with intermediate landing without exits, they shall be electrically interlocked to run in same direction and shall have the same transportation capacity. They shall also be electrical interlocked to stop the escalator carrying passengers into the intermediate landing without exits if the escalator carrying passengers away from the landing stops.

6.1.6 Where it is possible for people to come into contact with the outer edge of a handrail at a landing and may be drawn into a hazardous situation, such as toppling over a balustrade, appropriate preventive measures shall be taken (for example, see [Fig. 6](#)).

Some examples are:

- a) Prevention of entry into the space by the placement of permanent barriers.
- b) Increasing the height of the building structure of the fixed balustrade in the hazard area by at least 100 mm above the handrail level and positioned between 80 mm and 120 mm from the outer edge of the handrail.

The placement of permanent barriers shall be towards the newel end in the area where the handrail becomes flat after the curved portion of the newel. The permanent barriers shall be designed to prevent injury to travelling passengers.

6.1.7 The surrounding space of the escalator or moving walk shall be illuminated, especially in the vicinity of the combs.

NOTE — Information should be exchanged between the manufacturer and the customer.

6.1.8 It is permissible to arrange the lighting in the surrounding space and/or at the installation itself. The intensity of illumination at the landings including the combs shall be related to the intensity of illumination of the general lighting in the area. The intensity of illumination shall be not less than 50 lux at the comb intersection line measured at floor level.

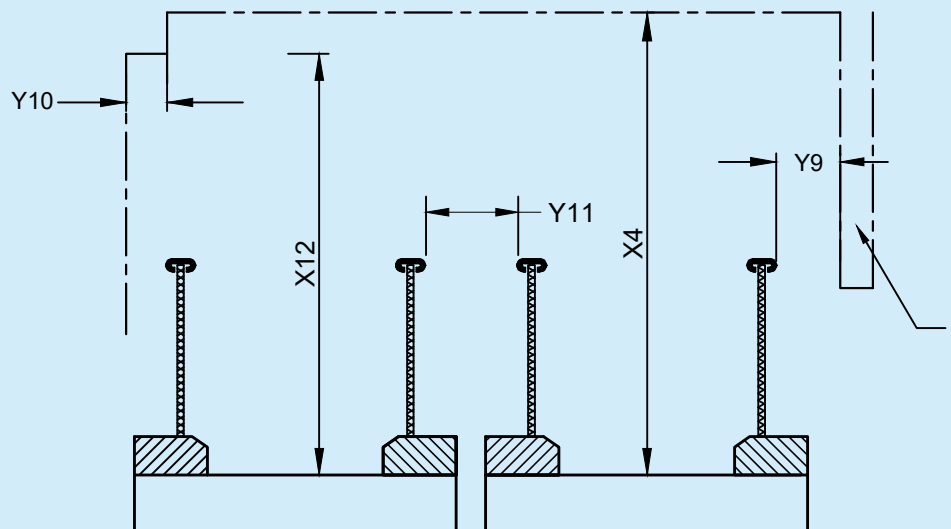
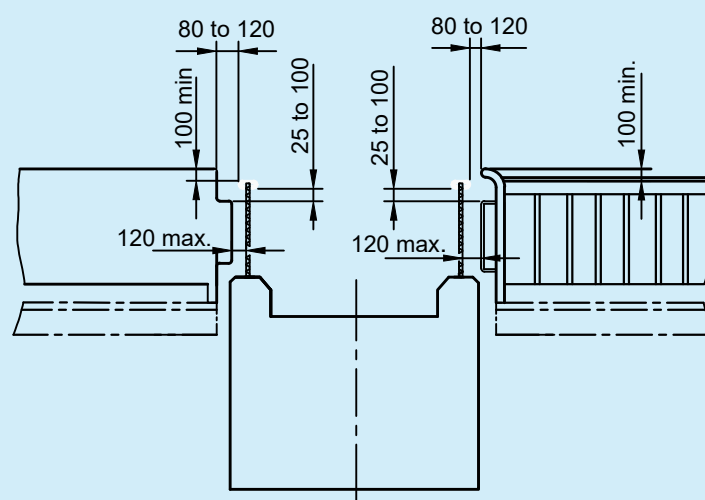


FIG. 5 CLEARANCES BETWEEN BUILDING STRUCTURE AND ESCALATOR / MOVING WALK



All dimensions in millimetres

FIG. 6 EXAMPLE OF BARRIERS AT LANDINGS

Key

1 Obstacle (for example, Column)

Sl No.	Principal Dimensions	Ref to Clause	Principal Dimensions	Ref to Clause
	Clause		Clause	
(1)	(2)	(3)	(4)	(5)
i)	Y9 ≥ 400 mm [Sl No. (xxii) of Annex A]	6.1.4	X4 ≥ 2 300 mm [Sl No. (xix) of Annex A]	6.1.1
ii)	Y10 ≥ 80 mm [Sl No. (xx) of Annex A]	6.1.2	X12 ≥ 2 100 mm [Sl No. (xxii) of Annex A]	6.1.2
iii)	Y11 ≥ 160 mm [Sl No. (xxi) of Annex A]	6.1.3	—	—

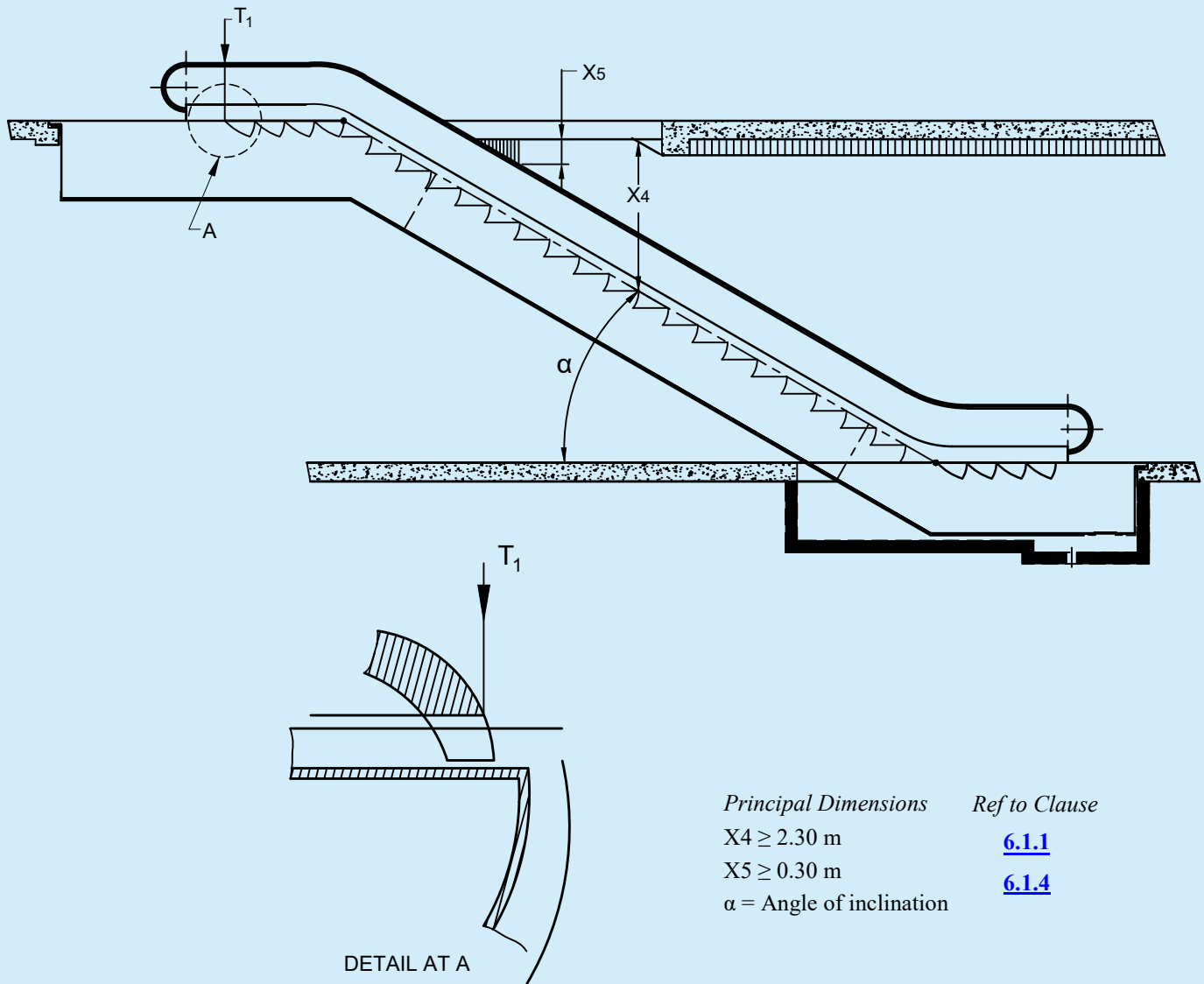
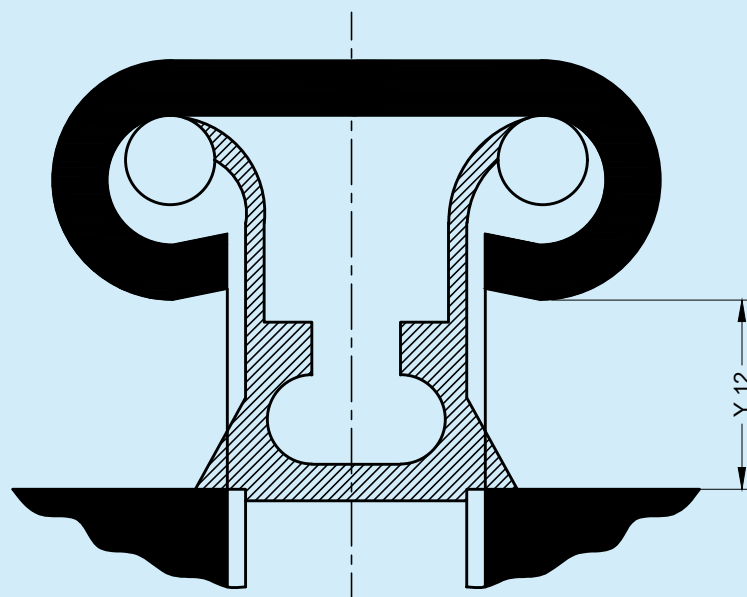


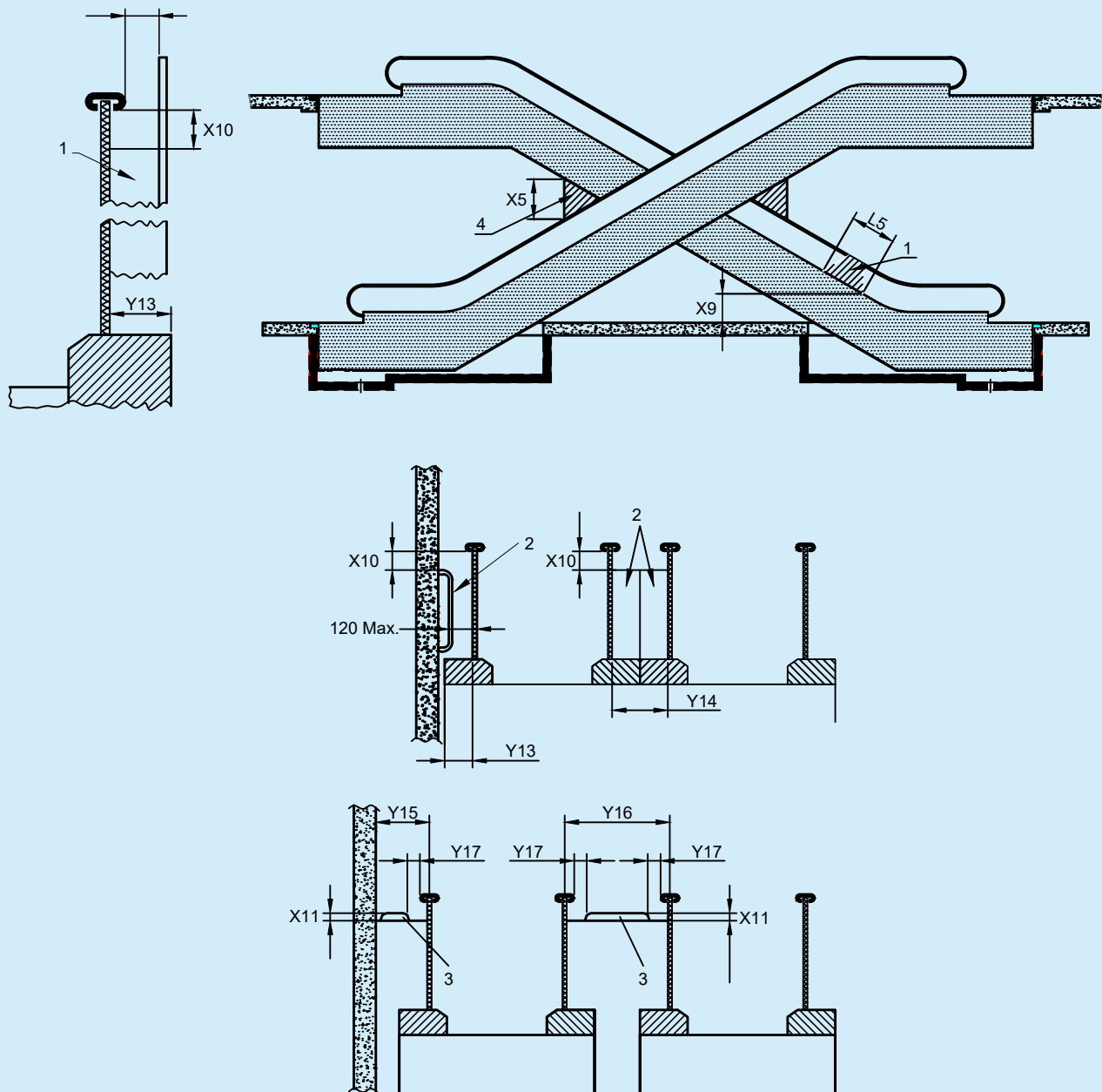
FIG.7 ESCALATOR (ELEVATION), PRINCIPAL DIMENSIONS



Principal Dimensions *Ref to Clause*

$Y12 \geq 25 \text{ mm}$ [6.1.2](#)

FIG.8 HANDRAIL (SECTIONAL VIEW)



Key

- 1) Anti-climbing device (see [6.3](#))
- 2) Access restriction device (see [6.3](#))
- 3) Anti-slide device (see [6.3](#))
- 4) Vertical deflector (see [6.1.4](#))

Principal Dimensions

Y13, Y14, Y15, Y16
Y17 $\geq$ 100 mm
X5 $\geq$ 0.30 m
X9 = (1 000 $\pm$ 50) mm
X10 = 25 mm to 150 mm
X11 $\geq$ 20 mm
L5 $\geq$ 1 000 mm

Ref to Clause

[6.3](#)
[6.3](#)
[6.1.4](#)
[6.3](#)
[6.3](#)
[6.3](#)
[6.3](#)

FIG. 9 ANTI-MISUSE DEVICES

6.2 Machinery Spaces Outside the Truss

The following shall be ensured:

- a) A safe access for persons to machinery spaces shall be provided;
- b) Machinery spaces shall be lockable and only accessible to authorized persons;
- c) Machinery spaces shall be provided with permanently installed electric lighting on the following basis;
 - 1) Minimum of 200 lux at floor level in working areas;
 - 2) Minimum of 50 lux at floor level in access routes leading to these working areas.
- d) Emergency lighting shall be installed to allow the safe evacuation of all personnel working in any machinery space.

NOTE — Emergency lighting is not intended for continuation of maintenance or other activities.

- e) The dimensions of machinery spaces shall be sufficient to permit easy and safe working on equipment, especially the electrical equipment. These shall be provided at least at a clear height of 2.00 m at working areas, and a clear horizontal area in front of the control panels and the cabinets. This area is defined as follows:
 - 1) Depth, measured from the external surface of the enclosures At least 0.70 m;
 - 2) Width, the greater of the following values 0.50 m or the full width of the cabinet or panel.

A clear horizontal area of at least 0.50 m × 0.60 m for maintenance and inspection of moving parts at points where this is necessary.

- f) The clear height for movement shall not be less than 1.80 m. The access ways to the clear spaces shall have a width of at least 0.50 m. This value may be reduced to 0.40 m where there are no moving parts. This full height for movement is taken to the underside of the structural roof beams and measured from both:
 - 1) the floor of the access area; and
 - 2) the floor of the working area.
- g) In machinery spaces the clear height shall under no circumstances be less than 2.0 m;
- h) 3 pin plug with socket for the maintenance propose; and
- j) Machinery spaces shall not accommodate equipment not related to the escalator or moving walk.

6.3 Balustrade

Balustrades shall be installed on each side of the escalator or moving walk. The dimensions of the balustrade shall be as follows:

In the inclined section the vertical height X_1 from step nose or pallet surface or belt surface to top of the handrail shall not be less than 0.90 m and not exceed 1.10 m (see [Fig. 10](#)).

The balustrades shall have no parts on which a person will normally stand.

Appropriate measures shall be taken to discourage people from climbing on the outsides of the balustrade if there is a danger of people falling from them.

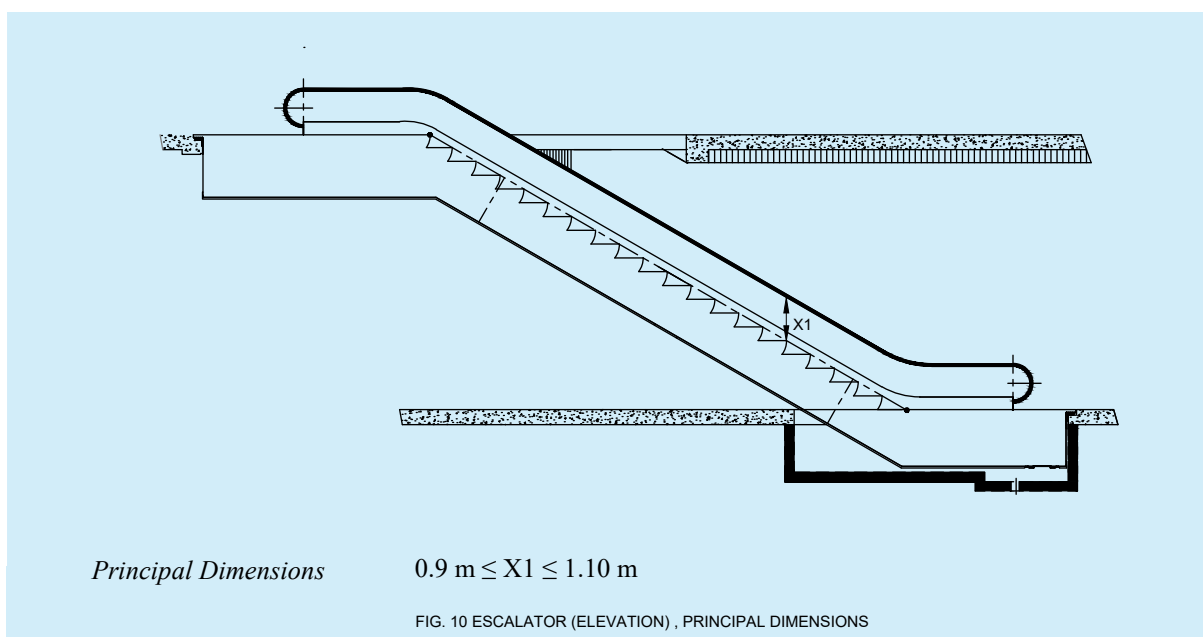
To ensure this, on escalators and moving walks, devices (*see* 1 in [Fig. 9](#)) shall be provided on the lower outer decking at a point $1\,000\text{ mm} \pm 50\text{ mm}$ above the floor level (*see* X9 in [Fig. 9](#)) where the bottom of the device intersects with the balustrade decking and shall extend to a length L5 of at least $1\,000\text{ mm}$ parallel with the balustrade decking where no stepping is possible. The device shall extend to at least a height in line with the top of the handrail not conflicting with the requirements of Y10 and Y12 (*see* [Fig. 5](#) and [Fig. 8](#)).

Where escalators or moving walks are located adjacent to walls, devices (*see* 2 in [Fig. 9](#)) to restrict access to the balustrade decking shall be provided at the top and bottom ends of these walls when the lower outer decking width Y13 exceeds 125 mm . On adjacent parallel arrangements, this protection shall be provided when the combined balustrade decking width Y14 exceeds 125 mm . The device shall extend to the height X10 (*see* [Fig. 9](#)).

All exposed fastener heads for the devices should be of the vandal resistant type.

Where handrail level balustrade decking are provided between escalators/inclined moving walks and adjacent walls, anti-slide devices (*see* 3 in [Fig. 9](#)) shall be provided on the balustrade decking when the distance Y15 between the structure of the building (wall) and the centreline of the handrail is greater than 300 mm . These devices shall consist of objects fastened to the balustrade decking, no closer than 100 mm to the handrail (*see* Y17) and spaced no greater than $1\,800\text{ mm}$ apart. The height X11 shall be not less than 20 mm . The devices shall have no sharp corners or edges (*see* [Fig. 9](#)).

The above mentioned also applies to adjacent escalators/inclined moving walks when the distance Y16 between the centrelines of the handrails is greater than 400 mm .



Exposure of one or both sides of escalators over a void, and the changes made to the balustrade/handrail design on escalators, can increase risks to users. The placement and type of escalator may introduce hazards, particularly when exposed over a void, leading to risks of falling at two areas:

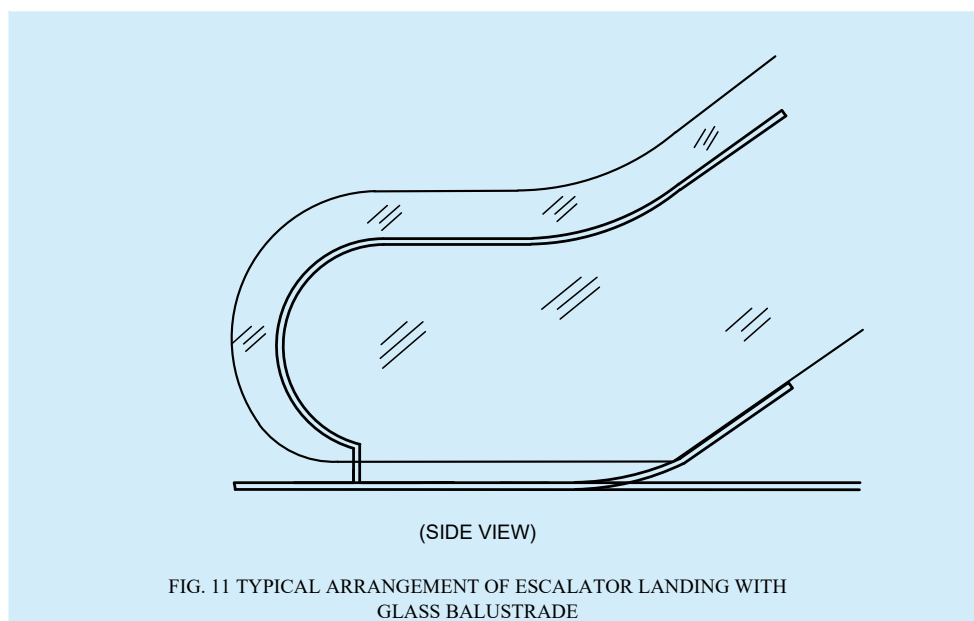
- a) *Escalator and Moving Walk Landings* — There is a risk for people to come into contact with the outer edge of a handrail and be drawn into the void; and

- b) *Escalator and Moving Walk Balustrades* — There is a risk for people to trip or accidentally overbalance and topple into the void.

A risk assessment should be conducted to identify hazards, potential misuse, affected individuals, and necessary preventive measures. Some examples of misuse are - sitting, standing, sliding on the handrail, holding onto the handrail from outside the balustrade, being lifted onto the handrail, walking up, and sliding down the decking.

Balustrades should balance fall risk and passenger support. Balustrade with 1 100 mm height offer firm grip and better fall protection than 900 mm. To mitigate falling risks, following risk mitigation measures shall be considered:

- 1) Avoiding voids to remove the hazard;
- 2) Using 1 100 mm high balustrades if voids are present; and
- 3) For escalators exposed over a void, falling risks increase, and additional protection beyond the handrail and balustrade may be needed. Where escalators are not bounded totally by adjacent walls or partitions and are therefore exposed over a void on one or both sides, persons are at risk of falling a substantial height. The building architects/developers might need to make provision of additional protection beyond the handrail and balustrade. Glass enclosures, shields, or additional guard rails shall be installed as shown in [Fig. 11](#) if high balustrades alone are insufficient.



NOTE — Based on the risk assessment carried out, fixing of external guarding, if required, shall be undertaken with professional consultation to ensure that the integrity of the escalator truss or balustrade is not compromised.

6.4 Installation

The following needs to be planned and carried out on site to ensure easy installation of escalators and moving walks.

6.4.1 Installation Timing

To prevent damage to the escalator during the construction phase, the timing of installation and assembly should be as close as possible to the opening date. It should, however, proceed completion of the floor to dispense with the need to take further measures to protect floor coverings. Ensure that the provisional floors along the transport route in the building may comfortably cope with the load weight.

6.4.2 Transport to the Assembly Site

A suitable unloading area shall be available in front of the building. The access routes to the building shall be level and negotiable by heavy duty rollers. Along the entire transport route to the assembly site, the free height shall not be less than the minimum specified in the manufacturer plan (also goes for pipes and other attachments fitted by others to the ceiling structure).

6.4.3 Access Openings, Transport Route and Load Capacity

Normally, escalators are moved in through openings at the side of the building at ground level. Optionally, suitable roof openings should be provided, to bring the escalators by crane. It is important that the route to the assembly location within the building is free of obstacles and level, and the ceiling can support a load. Otherwise, the appropriate weight distribution shall be catered to.

6.4.4 Special Delivery

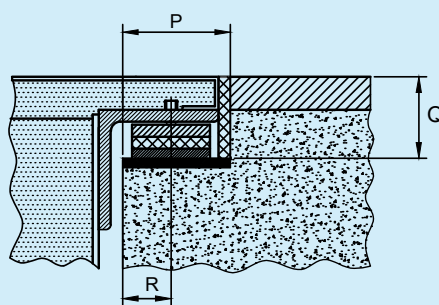
Some escalators are so long that they should be installed in sections. In this case, the escalator may have to be manufactured in split form and assembled on site.

6.4.5 Ceiling Openings

Manufacturer should notify the location and size of the required ceiling openings on the installation drawing. Builder shall supervise the precise implementation of these openings.

6.4.6 Top and Bottom Supports

While designing the support recess, the support loads shall be considered. They are shown on the installation drawing given by the manufacturer. At those locations where the supports of escalator (or moving walk) will be placed, the support shall be able to bear the weight of the escalator including traffic load. When preparing the supporting structures, the measurements indicated on manufacturer's installation plans shall be precisely adhered to (see [Fig. 12](#)).



NOTE - P, Q, R ARE BASED ON MANUFACTURER'S DRAWING

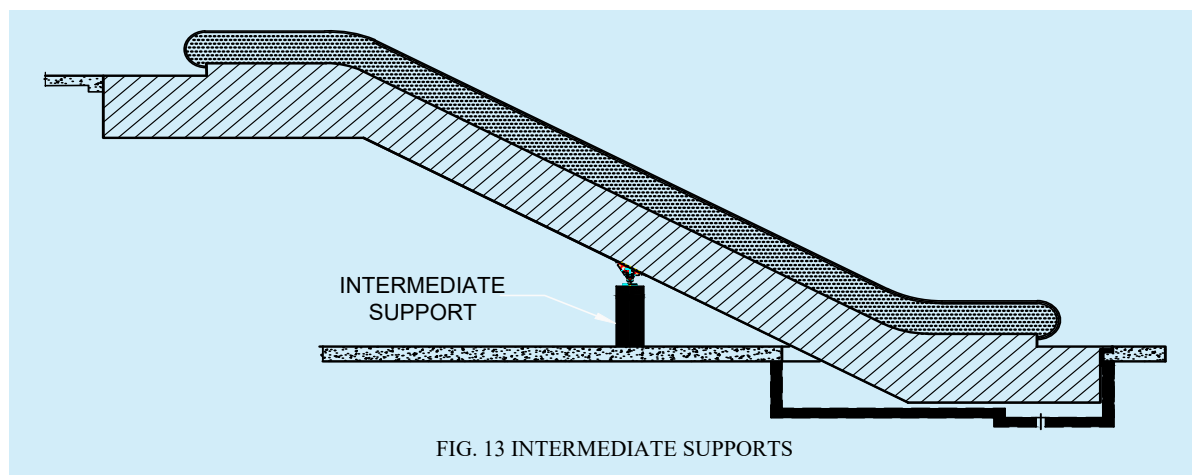
FIG. 12 TOP AND BOTTOM SUPPORTS

6.4.7 Intermediate Supports

Intermediate supports are needed on escalators with a considerable travel height (recommended above 5 000 mm) and on long moving walks. If the intermediate support is required, the manufacturer shall provide the location and load on intermediate support to the customer irrespective of the segment for which the escalator is provided, to include the additional space required in the construction plan (see [Fig. 13](#)).

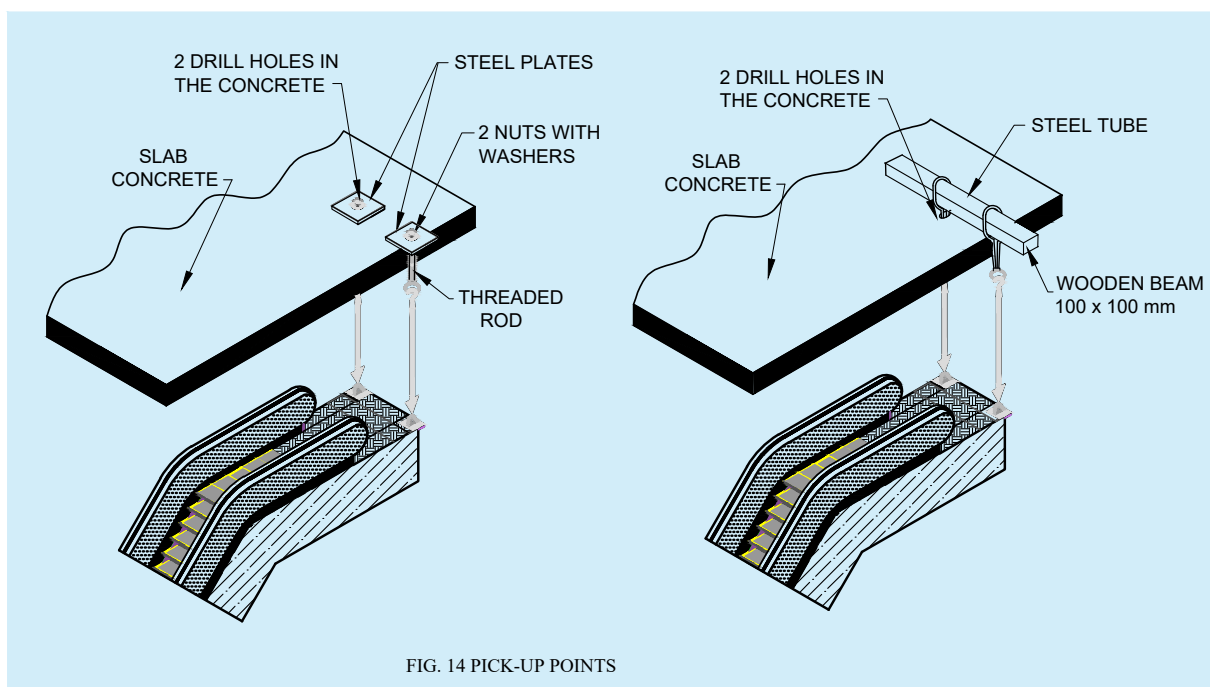
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6.4.8 Hoisting Arrangement

It shall be ensured that pick-up points for hoisting and supporting the escalator shall be fitted securely during assembly. These should be located exactly above the centre of the supporting points. For systems with several supporting structures, plan for additional pick-up points above the intermediate supports. All pick-up points shall be capable of taking a load strain (see [Fig. 14](#)).



6.4.8.1 Installation with a frame

If no assembly holes can be provided, the escalators and moving walks should be lifted to their final location by means of special sturdy girder construction. In this case, ceiling load strain shall bear the capacity as per the manufacturer's instruction.

6.4.8.2 Installation with lifting gear

Holes with a 100 mm diameter are to be made through the ceiling, according to manufacturer instructions. These are used by installation team for fixing the lifting gear. In this way escalator or moving walk is simply lifted upwards from floor to floor. The load strain of every pick-up point shall be maintained as per the manufacturer instruction.

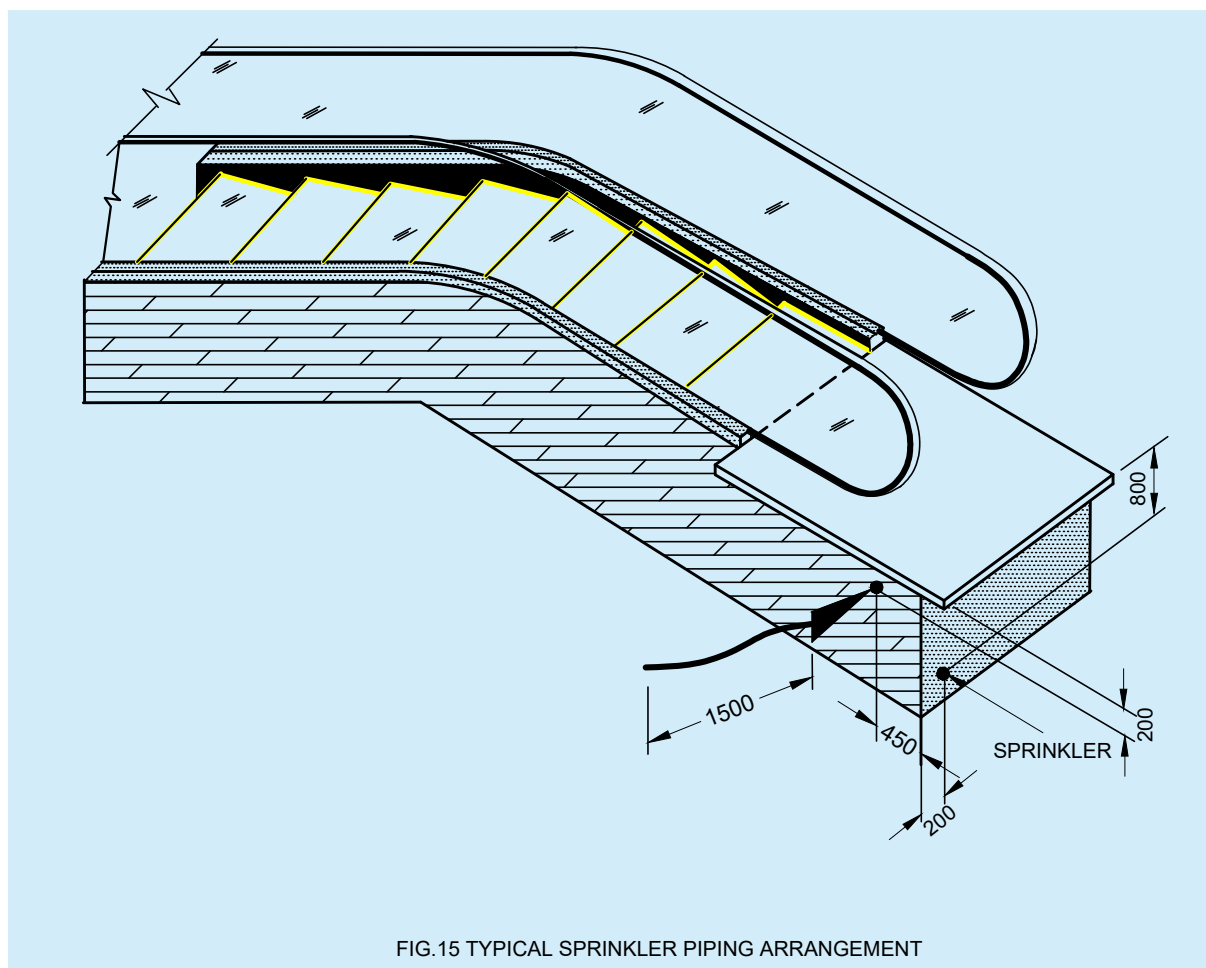
6.4.8.3 Installation with crane assembly

Where a suitable roof opening is available, or the site is located outdoor, lift the escalator or moving walk by crane and place it on the supports provided. The builder is responsible for ensuring for all local authority approvals.

6.4.8.4 Permanent hoisting hooks shall be provided by the builder in the top slab for hoisting the escalator as per the load requirement.

6.4.9 Sprinkler Piping

An optional safety feature is the installation of a sprinkler piping within the escalator or moving walk (see [Fig. 15](#) for typical sprinkler piping arrangement).



6.4.10 Oil Separator

A type-proofed oil separator is essential for escalators and moving walks which are exposed to the weather. At the construction site, a recess shall be provided for the oil separator in the escalator pit.

6.4.11 Railings

In the threshold areas of the escalators, railings should be installed by the builder. The distance to the escalator handrail from railing shall be at least 80 mm. For adjacent escalators, the distance between the edges of the handrails shall be at least 160 mm (see Y11 in [Fig. 5](#)).

6.5 Electrical Requirement

6.5.1 The electrical installations shall be in accordance with Part D 'Building Services, Section 2 Electrical and Allied Installations' of this NBCS and good practice [D-5B (2)].

6.5.2 Power Supply

6.5.2.1 The main power supply cables providing the electricity supply from main electrical panel of the building to escalator/moving walk equipment shall be PVC or XLPE insulated, flame retardant type, armoured cables. The cables shall be terminated in the main switchgear panels/distribution box using cable glands such that the armour of the cables shall be positively connected to the metal body of switchgear panels/distribution box. For each escalator/moving walk, one $415\text{ V}_{\substack{+6\text{ percent} \\ -10\text{ percent}}}$, 3-phase, 4-wire, 50 Hz supply and one $230\text{ V}_{\substack{+6\text{ percent} \\ -10\text{ percent}}}$, 1-phase, 3-wire, 50 Hz supply shall be provided outside the escalator/moving walk in the vicinity of the machine or in the return stations, or in the vicinity of the control devices. Main switchgear shall be capable of breaking the supply to the motor, to the brake releasing device and to the control circuit in the live conductors.

The electric lighting and the socket outlets shall be independent of the power supply to the machine being fed either by a separate cable or a branch cable which is connected before the main switch of the escalator or moving walk. It shall be possible to break the supply of all phases by means of a separate switch. This switch shall not cut the supply to the socket outlets or to the lighting circuits while the escalator/moving walk is switched off for inspection and maintenance. Size of the three-phase and single-phase cables shall be as specified by the manufacturer.

When separate supplies are provided for auxiliary equipment, such as, heating, balustrade lighting and comb lighting, it shall be possible to switch them off independently. The corresponding switches shall be located close to the main switch and be marked unambiguously.

6.5.2.2 The escalator/moving walk sub-circuit from the meter room should be separate from other building services. Each equipment should be capable of being isolated from the mains supply. This means of isolation shall be with lock out and tag out facility. The electric supply cables shall run in a route safe from fire.

6.5.2.3 All electrical supply lines and apparatus in connection with the escalator/moving walk installation shall be so constructed and shall be so installed, protected, worked, and maintained that there may be no danger to persons therefrom.

6.5.3 Protection

It is important that earthing should ensure efficient and fast operation of protective device, in the case of earth faults. Escalators or moving walks may be provided with or without variable voltage and variable frequency (V3F) drives. Due to its principle of operation, the V3F drive produces leakage current greater than 3.5 mA. Therefore, the cross-section of earthing conductor should be at least the cross-section of the active conductors and at least 10 mm². As an alternative, two ground conductors with the same cross-section as the active conductors (if < 10 mm²) may be used in parallel.

Suitable overcurrent protection device (MCB/MCCB) and residual current device (RCD) shall be provided outside the escalator/moving walk in the vicinity of the machine or in the return stations or in the vicinity of the control devices.

Protective measure of protective equipotential bonding and automatic disconnection of supply is intended to provide protection against shock and/or fire hazard. The choice of protective device, depends on the earthing network that is TN-S, TN-C-S, TT.

Residual current device (RCD) shall be required in case of fault protection against indirect contact or in case of TT network. When RCD is used, circuit shall also be protected by overcurrent protective device that is standalone RCD shall not be used.

6.5.3.1 MCB and RCD in 3-phase circuit:

- a) Irrespective of whether the escalator or moving walk is with or without V3F, MCB of characteristics 'D' shall be used.

- b) *RCD with V3F* — Special type of residual-current circuit breaker appropriate for this application should be used having following characteristics:
 - 1) Suitable for use with a frequency converter;
 - 2) Universal-current-sensitive (Characteristic *B*);
 - 3) Short-time delayed tripping (Type *S*);
 - 4) Tripping current at nominal frequency: 300 mA; and
 - 5) Tolerant against leakage currents of up to 2 000 mA at higher frequencies.
- c) *RCD without V3F* — RCD with below specifications shall be used:
 - 1) Type *A* or *AC* — details to be provided by manufacturer;
 - 2) Non-delayed; and
 - 3) Tripping current: 100 mA

Alternatively, residual current circuit breaker with overcurrent protection (RCBOP) which meets above specifications may be provided.

NOTE — RCD/RCBO is mandatory in case of TT network.

6.5.3.2 Over current protective device in the 1-phase circuit

For the lighting circuit, miniature circuit breaker (MCB) of characteristics ‘C’ shall be provided. In addition, RCD of the following specification shall be provided:

- a) Pulse-current-sensitive (Characteristic *A*);
- b) Non-delayed;
- c) Tripping current: 30 mA; and
- d) Rated current: 16 A.

Alternatively, RCBO which meets above specifications may be provided.

7 FIRE PROTECTION

Fire protection and building requirement differ with the application/use of building. For specific requirements for fire protection, Part F ‘Fire and Life Safety’ of this NBCS should be referred. It is recommended that escalators and moving walks are made of non-combustible material that do not create additional hazard in case of fire.

To limit the danger caused by overheating, ventilation for the rooms/spaces that contain the motors and gears shall be provided. These areas may contain fire alarm systems, extinguishing systems, and sprinkler heads if the equipment is adequately protected from incidental damage and do not create additional dangers for maintenance personnel to operate. In case sprinklers are used, their integration and fixation into the escalator and moving walk shall be done by taking into consideration the special needs of the machine.

Accumulation of materials (such as grease, oil, dust, paper) represent a fire hazard. Therefore, it should be possible to clean the underside enclosure. If such cleaning is not possible, other precautions (such as installation of sprinkler system) shall be provided to avoid the fire hazard. Outer and inner decking, truss, pallets/steps, track system shall be according to Class 3 of accepted standards [D-5B(3)]. In case of fire, upon receiving the signal, the escalator operation should stop immediately. It is recommended that as far as possible, escalators and moving walks are not used in case of emergency situations.

Escalator wellways may be equipped with a fire shutter(s), operated by an activation of temperature and smoke detectors, and shuts off the wellway of a floor and prevent the spread of fire up through the building.

8 MINIMUM TECHNICAL AND SAFETY REQUIREMENTS

8.1 General

The safety requirements of escalators and moving walks shall comply with good practice in accordance with this Subsection. Certain escalators and moving walks are subject to special operational and environmental conditions.

8.2 Supporting Structures (Truss) and Enclosure

The truss shall be designed to safely sustain steps and running gear in its guides. All mechanically moving parts of the escalator or moving walk shall be completely enclosed within imperforate panels or walls. Exceptions are the accessible steps, the accessible pallets, the accessible belt, and that part of the handrail available for the user. Apertures for ventilation are permitted. However, it shall not be possible to pass a straight rigid rod 10 mm in diameter through the enclosure and to touch any moving part through a ventilation aperture.

Angle of inclination of an escalator from the horizontal shall not exceed 30°, but for rises not exceeding 6 m and nominal speed not exceeding 0.5 m/s the angle of inclination is permitted to be increased to 35°.

Rise m	Speed m/s	Angle of Inclination	
		Standard	Option
≤ 6	≤ 0.5	30°	35°
≤ 6	> 0.5	30°	-
> 6	≤ 0.5	30°	-
> 6	> 0.5	30°	-

The angle of inclination of moving walks shall not exceed 12°.

8.3 Step, Pallet, or Belt

For escalators and moving walks, the nominal width shall not be less than 0.58 m and not exceed 1.10 m. For moving walks with an angle of inclination up to 6°, widths up to 1.65 m are permitted.

8.3.1 Step Treads and Pallets

The step depth in the direction of travel shall not be less than 0.38 m. The step height shall not exceed 0.24 m. The surface of the step treads shall have grooves in the direction of the movement, with which the teeth of the combs mesh. They shall be sensibly horizontal in the usable area of the escalator. The step risers shall be cleated, and the surface of the cleat shall be smooth. The ends of the step tread shall mesh with the cleating of the next step riser.

The width of the grooves shall be at least 5 mm and shall not exceed 7 mm. The depth of the grooves shall not be less than 10 mm. The web width shall be at least 2.5 mm but not exceed 5 mm.

8.3.2 Belts

The belts shall have grooves in the direction of travel with which the teeth of the comb mesh. The width of the grooves shall be at least 4.5 mm and shall not exceed 7 mm. The depth of grooves shall not be less than 5 mm. The web width shall be at least 4.5 mm but not exceed 8 mm. Splicing of tread way belt shall be such as to provide continuous unbroken tread way surface.

8.4 Driving Machine and Brake

8.4.1 A driving machine shall not drive more than one escalator or moving walk.

8.4.2 The speed of an unloaded escalator shall not deviate by more than ± 5 percent from nominal speed at nominal frequency and nominal voltage.

8.4.3 The driving machine shall be connected to main drive shaft, preferably by non-friction driving elements such as by gear wheels, shaft, coupling, multiplex chain, two or more single chains. Where friction elements are used such as trapezoidal belts (flat belts are not permitted) an auxiliary brake shall be used.

8.4.4 Braking System

Escalators and moving walks shall have an operational braking system by means of which they can be brought to rest with a uniform deceleration and maintained stationary; there shall be no intentional delay in the application of the braking system. If the control system starts a braking sequence immediately to bring the escalator/moving walk to a stop, this is not considered as an intentional delay. A device shall be provided to monitor the lifting of the braking system after starting the escalator/moving walk.

Operational braking shall be affected by an electro-mechanical brake or by other means. Where no electro-mechanical operational brake is used, an auxiliary brake shall be provided.

8.4.4.1 Auxiliary brake

Escalators and inclined moving walks shall be equipped with auxiliary brake(s) if:

- a) the connection between the operational brake and the driving sprockets of the steps/pallets or the drum of the belt is not accomplished by shafts, gear wheels, multiplex chains, or more than one single chain; or
- b) the operational brake is not an electro-mechanical brake; or
- c) the rise exceeds 6 m; or
- d) escalators or moving walks are part of a public transport/service system.

8.5 Steps and Pallets Drive

The steps of escalators shall be driven by at least two chains of which at least one shall be located on each side of the step. The pallets of moving walks may be permitted to be driven by only one chain if the parallel movement of the pallets in the usable area is ensured by other mechanical measures.

8.6 Belt Drive

The belt shall be driven by drums and be tensioned continuously and automatically. Springs working in tension are not permitted for the tensioning device. When weights are used for tensioning, they shall be safely retained should their suspension break.

8.7 Balustrade

Balustrades shall be installed to comply with the requirements of [6.3](#).

If glass is used for the interior panel, it shall be toughened glass. In case of splinter free one layer safety (tempered) glass, the minimum thickness shall be 6 mm. When multi-layer glass balustrades are used, they shall be laminated toughened glass, the thickness of at least one layer shall also be not less than 6 mm.

8.8 Skirting

The skirting shall be vertical, plain, and sufficiently rigid. Where the skirting of escalators or moving walks is placed beside the steps and pallets or the belt the horizontal clearance shall not exceed 4 mm at either side, or 7 mm for the sum of clearances measured at both sides at two directly opposite points.

Where the skirting of moving walks finishes above the pallets or the belt, the clearance shall not exceed 4 mm measured vertically from the tread surface. Motion of the pallets or the belt in lateral direction shall not cause a gap between the sides of the pallets or the belt and the vertical projection of the skirting.

8.8.1 Requirements for Skirt Deflectors

The skirt deflectors shall be designed with rounded edges. Fastening heads and joint connections shall not extend into the path of travel. They shall consist of a rigid and a flexible part (for example, brushes and rubber profiles) and shall be mandatorily provided in the escalators.

8.9 Handrails

Each balustrade shall be provided with a handrail moving on the top of each balustrade, in the same direction and at the same speed, within tolerance of $^{+2}_0$ percent, as the steps, pallets or belt. Each moving handrail shall extend at normal handrail height not less than 300 mm beyond the line of point of comb plate teeth at the upper and lower landings. Hand or finger guards shall be provided at the points where handrails enter the balustrade.

The width of handrail shall be between 70 mm and 100 mm. The handrail shall be a minimum of 80 mm horizontally and 25 mm vertically away from adjacent surfaces (see [Fig. 6](#)). The handrail shall be guided and tensioned in such a way that it does not leave its guides during normal use.

8.10 Landing

The landing area (that is, comb plate and floor plate) of escalator shall have a surface that provides a secure foot hold (combs are exempted from this) for a minimum distance of 0.85 m measured from the root of the comb teeth.

8.10.1 Configuration of Steps, Pallets, and Belts

8.10.1.1 At the upper and lower landing, the steps of the escalator shall be guided in such a way that the front edges of the steps leaving the comb and the rear edges of the steps entering the comb are moving horizontally for a length of at least 0.80 m measured from point T1 (see [Fig. 7](#) and detail A).

8.10.1.2 At the upper landings of moving walks with an inclination of more than 6° , the pallets or belt shall move for a length of at least 0.40 m at a maximum angle of 6° before entering or after leaving the comb.

8.11 Comb Plates

There shall be a comb plate at the entrance and at the exit of every escalator. The comb plate teeth shall be meshed with and set into the tread surface so that the points of the teeth are always below the upper surface of the treads. The comb plates shall be adjustable. Combs shall be easily replaceable.

8.12 Speed Governor

8.12.1 Where speed control devices are used for this purpose, they shall have switched off the escalator before the speed exceeds 1.2 times the nominal speed. This requirement may not be applied if the design prevents the excessive speed.

8.12.2 Escalators and inclined ($\alpha \geq 6^\circ$) moving walks shall be equipped in such a way that they stop automatically by the time the steps and pallets or the belt change from the pre-set direction of travel.

8.12.3 For operation and safety devices, electrical work, precautions and tests, reference may be made to accepted standards [D-5B(3)].

8.13 Inspection Control

Escalators or moving walks shall be equipped with inspection controls to permit operation during maintenance or repair or inspection by means of portable and manually operated control devices. At least one portable control device shall be provided for each escalator or moving walk.

8.14 Information for Use

8.14.1 General

All escalators and moving walks are required to be provided with documentation that shall include an instruction handbook relating to use, maintenance, inspection, periodic checks and rescue operations. Information for use shall cover, separately or in combination, transport, assembly and installation, commissioning, use (setting, teaching/programming, operation, cleaning, fault finding and maintenance) of the escalator and moving walk, and, if necessary, de-commissioning, dismantling and disposal.

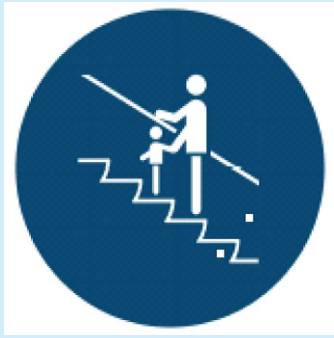
8.14.2 Signals and Warning Devices

8.14.2.1 Plates, inscriptions, and notices for use

- a) *General* — All signs, inscriptions and notices for use shall be of durable material, placed in a conspicuous position and written in clearly legible characters in English as well as in local language.
- b) *Safety signs near the entrances of escalators or moving walks* — The following mandatory action signs and prohibition signs for the user shall be fixed in the vicinity of the entrances:
 - 1) Small children shall be held firmly (see [Fig. 16](#));
 - 2) Pets shall be carried (see [Fig. 17](#));
 - 3) Use the handrail (see [Fig. 18](#));
 - 4) Push chairs not permitted (see [Fig. 19](#)); and
 - 5) Soft footwear prohibited (see [Fig. 20](#)).

When required by local conditions, prohibition signs, for example, ‘Transportation of bulky and heavy loads not permitted’ and mandatory action signs like ‘Use permitted only with footwear’, shall be necessitated.

The minimum diameter of the signs shall be 80 mm.



"SMALL CHILDREN SHALL BE HELD FIRMLY"
FIG.16 MANDATORY ACTION SIGN



"PETS SHALL BE CARRIED"
FIG.17 MANDATORY ACTION SIGN



"USE THE HANDRAIL"
FIG.18 MANDATORY ACTION SIGN



"PUSH CHAIRS NOT PERMITTED"
FIG.19 PROHIBITION SIGN



"SOFT FOOTWEAR PROHIBITED"
FIG. 20 PROHIBITION SIGN



"WHEELCHAIR USERS NOT PERMITTED"
FIG. 21

Stop devices shall be colored red either on the device itself or in its immediate vicinity and be marked with the inscription 'STOP'.

During maintenance, repair, inspection or similar work, the access to the escalator or moving walk shall be barred to unauthorized persons by devices which shall bear the notice 'NO ACCESS' or the 'NO ENTRY' indicator and be available in immediate area.

NOTE — The safety sign (see Fig. 21) shall be affixed prominently near the entrances of escalators or moving walks which are inclined above 6°.

- c) *Instructions for hand winding devices* — If a hand winding device is provided, operating instructions for use shall be available in the vicinity. The direction of travel of the escalator or moving walk shall be indicated clearly.
- d) *Notices on the access doors to machinery spaces outside the truss, driving and return stations* — On access doors to machinery spaces outside the truss, and driving and return stations, a notice shall be fixed with the inscription reading, 'MACHINERY SPACE — DANGER' and 'ACCESS PROHIBITED TO UNAUTHORIZED PERSONS'.

9 SPECIAL TECHNICAL REQUIREMENTS

If escalators or moving walks are intended to be operated under special conditions, such as directly exposed to the weather or explosive atmosphere, or in exceptional cases serve as emergency exits, appropriate design criteria, components, material and instructions for use should be used that satisfy the particular conditions.

9.1 Escalators or Moving Walks for Public Transport

For escalators and inclined moving walks for public transport which:

- a) are part of a public transport system including entrance and exit points; or
- b) are suitable for intensive use, regularly operating for approximately 140 h/week with a load reaching 100 percent of the brake load for a total duration of at least 0.5 h during any time interval of 3 h.

It is recommended to install auxiliary brakes also for rises less than 6 m. Auxiliary brakes shall be mandatory for public transport. The load conditions and additional safety features should be agreed to between the manufacturer and the owner reflecting the traffic levels which exist.

9.2 Semi Outdoor Escalator/Moving Walk

Semi-outdoor escalators and moving walks are installed outside the building while they have roofs above them to protect them from rain and snow. Wherever possible, the escalator/moving walks should be placed above the road level, with a ramp, to prevent water on the road directly entering the escalator/moving walk. These escalator/moving walk shall have:

- a) Aluminium die-cast step;
- b) Outdoor type handrail;

- c) Three-layered protective truss painting;
- d) Float switch at lower machine room;
- e) Cooling fan in control panel box;
- f) Control box of IP54 protection rating;
- g) Inverter with enclosure of IP54 protection rating;
- h) Safety switch IP65 protection rating;
- j) Comb heating (temperature below 4 °C) (if applicable);
- k) Step/pallet band heating (temperature below 0 °C) (If Applicable);
- m) Lubrication: Lifetime lubricated and sealed step chain roller bearings; and
- n) Water drainage system.

9.3 Outdoor Escalator/Moving Walk

Fully outdoor escalators and moving walks are installed outside the building without any roof or overhead covers. Wherever possible, the escalator/moving walk should be placed above the road level, with a ramp, to prevent water on the road directly entering the escalator or moving walk. These escalator/moving walk shall have:

- a) Aluminium die-cast step;
- b) Extrude aluminium floor plate panel;
- c) Outdoor type handrail;
- d) Hot-Dip galvanized truss (minimum thickness of 85 microns);
- e) Zinc plated track and handrail guide;
- f) Galvanized steel drive chain cover;
- g) Float switch at lower machine room;
- h) Control panel box cooling system:
 - 1) with inbuilt cooling fan; or
 - 2) with external cooling fan.
- j) Control box of IP54 protection rating;
- k) Inverter with enclosure of IP54 protection rating;
- m) Safety switches of IP65 protection rating;
- n) Stainless steel material grade 304 or higher for skirting, outer decking, handrail decking, balustrade (if not glass balustrade);
- p) Class E insulation of electromagnet;
- q) Class F insulation of motor;
- r) Comb heating (temperature below 4 °C) (if applicable);
- s) Step/Pallet band heating (temperature below 0 °C) (if applicable);
- t) Lubrication-Life time lubricated and sealed step chain roller bearing;
- u) Water drainage system;
- w) Provision of galvanized drip pans below the step chain throughout the length of chain with catch tray at return station;
- x) Automatic lubrication system with lubrication reservoir of adequate capacity;
- y) Provision of hose/pipe to drain water from oil/water separator;
- z) Provision for prevention of flooding of return station; and

- aa) Stainless steel fasteners, wherever provided.

9.4 Measures to Prevent Access of Shopping Trolleys and Baggage Carts

9.4.1 General

If there is a reasonably foreseeable risk that shopping trolleys and/or baggage carts may be taken onto escalators or moving walks, adequate measures shall be taken to eliminate risks and access shall be prevented, if the following conditions are given:

- a) *For escalators* — Where shopping trolleys or baggage carts are available in the area around; and where shopping trolleys or baggage carts are in an area not close to the escalator, but it is reasonably foreseeable that they are taken onto the escalator.
- b) *For moving walks* — Where shopping trolleys or baggage carts are not intended to be used on a moving walk.

9.4.2 Barriers

If barriers are used, the following requirements shall be fulfilled:

- a) Barrier shall be installed at the entrance only. An installation at the exit is not permitted in the unrestricted area;
- b) Design of the barrier shall not create another risk;
- c) Free entrance width between ends of the newels and barriers and between barriers itself, shall be at least 500 mm and less than the width of the type of shopping trolley or baggage cart to be used;
- d) Height of the barrier shall be between 900 mm and 1 100 mm;
- e) Barrier and its fixation should withstand the following load: At a height of 200 mm a horizontal force of 3 000 N applies. This force results from an impact of a chassis of a shopping trolley/baggage cart loaded with 160 kg moving with a speed of 1.00 m/s;
- f) Barrier shall be fixed preferably at the building structures. It is also permitted to fix it at the floor plate. In that case, when the defined forces apply, there shall be no permanent deformation and increased/additional gaps; and
- g) If the escalator/moving walk is dedicated to operating in both directions, the presence of the removable barrier shall be electrically monitored preventing wrong positioning of the barrier resulting in operation towards the barrier, except when under maintenance condition.

9.4.3 Fixed Devices

Where fixed devices such as guiding barriers and/or traffic columns are necessary in the unrestricted area, their design shall not create another risk. The following requirements shall be complied with:

- a) Guiding barriers or traffic columns shall have a minimum horizontal distance (radius) of 500 mm to any point of the handrail and shall be positioned outside of the centre line of the handrail;
- b) Minimum horizontal distance (radius) to any point of the handrail may be reduced to 300 mm, provided that the guiding barriers or traffic columns are positioned outside of the centre lines of the handrails and an additional barrier is installed between the guiding barriers or traffic columns and the vertical centre line of the balustrade newel;
- c) The additional barrier shall have a lateral distance between 80 mm and 120 mm of the handrail outer edge and shall close at least the area between the actual lowest point of the handrail entry into the newel and the balustrade decking profile and shall have filled inlets with gaps less than 25 mm;
- d) Minimum horizontal distance (radius) to any point of the handrail may be reduced to 180 mm, provided that the guiding barrier or traffic column is positioned outside of the outer edge of the handrail and an additional barrier is installed between the guiding barrier or traffic column and the vertical centre line of the balustrade newel;

- e) The additional barrier shall have a lateral distance between 80 mm and 120 mm of the handrail outer edge and shall close at least the area between the actual lowest point of the handrail entry into the newel and the balustrade decking profile and shall have filled inlets with gaps less than 25 mm; and
- f) Minimum horizontal distance (radius) to any point of the handrail may be reduced to 100 mm, provided that the guiding barrier or traffic column is of a round shape and is positioned outside of the outer edge of the handrail and an additional barrier is installed between the guiding barrier or traffic column and the vertical centre line of the balustrade newel.

In addition, the following specific requirements shall apply:

- 1) Devices and its fixation shall withstand a horizontal force of 1 kN/m applied to the top of the device;
- 2) Barrier shall be fixed preferably at the building structures. It is also permitted to fix it at the floor plate. In that case, when the defined forces apply, there shall be no permanent deformation and increased/additional gaps;
- 3) Height of traffic columns shall be at least the height of the handrail level;
- 4) Height of guiding barriers shall be at least the height of the balustrade decking;
- 5) If fixed devices are placed inside the unrestricted area, the size of the unrestricted area shall remain the same and in this case be extended in the length; and
- 6) Guiding barriers and traffic columns are not considered as structural measures that block the exit.

NOTE — For fixed devices in the immediate environment of the unrestricted area these requirements also apply.

9.5 Escalators and Moving Walks Subject to Seismic Conditions

9.5.1 Structural Requirements

The requirements given below are applicable:

- a) *General* — Escalators and moving walks, if specified to meet specified seismic conditions, shall comply with the relevant safety requirements and/or protective measures as listed below;
- b) *Pit/support* — Pit and support conditions are to be secured in that way that the escalator and moving walk under seismic condition is not subjected to constraint. One support should be executed as fixed support and the other supports are to be executed as movable support in horizontal direction. The supporting conditions need to be simply statically determinate;
- c) Escalator and moving walk shall be retained on the supports in vertical direction by adequate measures so that the unit shall not jump off the supports under seismic conditions;
- d) *Arrangement* — The length and the movability of the escalator and moving walk should be chosen in accordance with the building movement between two storeys of the building. The minimum allowable compensation length of the movable support of the escalator and moving walk has to be chosen same as the maximum allowed storey drift of the building; and
- e) *Mechanical safety devices for escalators and moving walks* — If the design of the escalator and moving walk does not allow the safe situation in the pit or support an additional safety device need to be applied so that the escalator and moving walk should not fall off the support.

9.5.2 Design Requirements

It is assumed that negotiations have been made for each contract between the customer and the supplier/installer about the peak ground acceleration to be considered. The building designer or owner shall provide the design acceleration which shall be documented in the information. Client and supplier/contractor should agree on one common acceleration value. The acceleration value needs to be communicated between all participants of the contract. The following to be considered in addition to the above:

- a) The design of the truss for the escalators and moving walks shall be sufficiently executed for the ambient seismic condition. The structural design shall provide a dissipation capacity to withstand a design-relevant earthquake of the region where it is located;

- b) *Drive and control unit* — Machinery [including control cabinet(s) and drive system, machine, main switch(es), and means for emergency operations], free-wheeling switch shall be designed and anchored to prevent overturning and displacement due to the forces imposed on them including forces generated by the design acceleration; and
- c) *Electrical installation and other equipment* — In case the building where the escalator and moving walk will be installed is equipped with a seismic detector/sensor, the electrical system of the unit shall provide an interface for the connection and shall stop the escalator and moving walk in case of seismic activity.

9.6 Escalator/Moving Walk Interface with Building Management System (BMS)

Where required, a means to interface, monitor and control be provided by the escalator/moving walk manufacturer, which should become the part of building management system. Following information and controls, but not limited to these, may be provided either through potential free contacts or building automation interface device suitable for providing necessary escalator/moving walk information over accepted communication protocols:

- a) *Monitoring signals:*
 - 1) Power status;
 - 2) Running up/down (Escalator);
 - 3) Running forward/backward (Moving walk);
 - 4) Under maintenance;
 - 5) Under fault; and
 - 6) Emergency 'STOP' pressed.
- b) *Remote operation* — Escalator or moving walk shall not be started or stopped from remote monitoring station using remote signal, unless the escalator/moving walk operation is visible/monitored from the remote station.

10 PERFORMANCE REQUIREMENTS FOR ESCALATORS AND MOVING WALKS

10.1 It is necessary to have uniformity in the definition, measurement, processing and expression of vibration and noise signals that comprise escalator and moving walk ride quality. The aim of such uniformity is to benefit escalator and moving walk users by reducing variability in the results of ride quality measurements caused by differences in the methods of acquiring and quantifying the signals. Experience indicates that passenger perception and sound pressure levels measured while travelling on an escalator or moving walk may be influenced by the presence of extraneous noise and vibration sources and by acoustic characteristics of the environment in which the unit is installed and operated. Additionally, the proximity of the escalator or moving walk to strong reflecting surfaces such as walls, ceilings or diagonally opposite units may also influence the sound pressure level measured. These influences may cause a sound pressure level measurement to significantly overestimate the sound level emitted exclusively by the unit. To address this issue, this Subsection defines the methodology for measuring vibration and sound pressure level that corresponds to passenger perception and additionally defines the methods that should be used, if further understanding of the result is required to quantify vibration and noise emitted by the unit as compared to the background or environmental contributions.

10.2 Measurement of Escalator and Moving Walk Ride Quality

Escalator and moving walk should be measured in both running directions except where the unit is designed to operate in one direction only. The measurements should be carried out under following conditions where, escalator and moving walk shall:

- a) be assembled completely, adjusted and operating in accordance to the requirement;
- b) have reached normal operating temperature;
- c) be measured in unloaded condition;
- d) be measured after starting sequence has been completed. If the unit operates at different speeds, all the speeds shall be measured;

- e) have finished star/delta changeover, if used, before starting the measurement;
- f) have vibration measurements made by aligning the axis of transducer to the axis of entire escalator or moving walk; and
- g) have the noise measured at measuring distance to the floor/surface of steps and pallets of $1.55 \text{ m} \pm 0.075 \text{ m}$.

NOTE — Accepted standard [D-5B(4)] shall be referred for Ride Quality measurements.

11 ESCALATOR AND MOVING WALK ENQUIRY OR INVITATION TO TENDER

11.1 A period of two to four weeks is normally sufficient for return of tenders. This should be extended if large numbers of escalators/moving walks or special requirements are involved. The enquiry documents should be kept to the essential minimum and should be strictly confined to material relevant to the escalators/moving walk works and to the project concerned.

When enquiring for and ordering the escalators/moving walks, the particulars given below should be furnished:

- a) Type(s) of escalator/moving walk;
- b) Number of escalators/moving walks;
- c) Location of escalators/moving walks (indoor or outdoor);
- d) Capacity/usage type (persons per hour/heavy duty, medium duty, light duty);
- e) Angle of inclination, in degree;
- f) Rise (travel), in mm;
- g) Nominal speed, in m/s;
- h) Step width, in mm;
- j) Number of flat steps (2 or 3 or 4);
- k) Balustrade details;
- m) Construction, design, finishes details;
- n) Escalator/moving walk lighting details;
- p) Electric supply — power volts a.c./d.c.; phase; cycles; wire system;
- q) Proposed date for commencement on site;
- r) Additional items, if required;
- s) Details of maintenance, schedule and circuit diagram where so specified;
- t) Details of voice announcements, visual contrast, tactile indicators and warning signage; and
- u) Number of horizontal steps/pallets.

11.2 Additional Items

The enquiry should state any additional items required beyond those specified in accepted standards [D-5B(3)], such as interface requirements, seismic design, etc.

11.3 Finishes

Finishes should be specified at the enquiry stage or provisional sums should be included for them. Finishes to be considered may include step lighting, handrail colour, handrail lighting, balustrade material, step colour, truss cladding details (weight to be considered).

11.4 Inclusions and Exclusions

Several peripheral items are associated with escalators/moving walk installation, of which some should always be provided by the builder, and some are best included by the manufacturer. The requirements vary to some extent with the type of installation.

It is important that the limits of responsibility are clearly understood, and the enquiry documents should be specific in this respect.

The escalators/moving walk manufacturer should include items (as applicable), such as:

- a) tracks;
- b) drive unit;
- c) controller;
- d) broken chain detection switch;
- e) stop switches at both ends;
- f) balustrade;
- g) handrail;
- h) steps/step drive assembly;
- j) saree/skirt guard;
- k) step demarcation lights;
- m) missing steps;
- n) comb plate;
- p) flat steps;
- q) comb plate switches/skirt switches;
- r) floor plate (anti-slip material);
- s) truss;
- t) guarantee of equipment; and
- u) scaffolding for installation.

The escalators/moving walk manufacturer should exclude the supply and fixing of the items (as applicable) or as per the contract conditions as follows:

- 1) Hoisting hook/beam fixing;
- 2) Truss cladding;
- 3) Temporary guarding;
- 4) Cutting and making goods;
- 5) Working lights/permanent electricity supplies; and
- 6) Site painting of steel work.

For more detailed discussion of the requirements for site preparation and work by other trades, reference should be made to relevant provisions of this NBCS and other good practices.

Facilities for the use of the main contractor's crane should be provided to assist in installing heavy equipment in addition to other unloading facilities on site in the course of erection. The main contractor should be instructed to include these facilities in his own bill of quantities.

11.5 Site Programme

The enquiry should indicate as accurately as possible the contract programme as it affects the manufacturer, in particular the target date for completion, construction schedule, site preparedness, the availability of crane for hoisting, co-ordination details (with other contractors like BMS interfacing, CCTV interfacing, etc).

12 ACCEPTANCE OF TENDER AND SUBSEQUENT PROCEDURE

12.1 Order

The owner places an order with the selected contractor for supply and installation of escalator/moving walk.

12.2 The general arrangement drawings supplied by owner with the tender documents,

- a) are provided for the supplier's guidance and information when tendering;
- b) should the arrangement and dimensions shown be inadequate to properly install the equipment as specified then necessary amendments shall be issued by client; and
- c) any amendments issued after the commencement of contract shall be subject to the agreement and approval.

12.3 Layout Drawing

The contractor shall furnish proposed layout giving particulars such as dimensions, steel work requirements and loading imposed on structure. Such drawings shall include following details:

- a) Position and sizes of all holes and cuttings;
- b) Loads on beams and structure;
- c) Escalator/moving walk pits and associated equipment;
- d) Requirement of end supports, intermediate support(s) with location and load;
- e) Hoisting facilities and access requirement for delivery of equipment;
- f) Builder's work; and
- g) Power and ventilation requirement.

12.4 Approval of Layout Drawing

The contractor shall commence actual production of equipment after approval of drawings submitted by contractor. Where the contract provides for the purchaser's choice of decorative finishes, colours, etc, the decisions should be communicated as early as possible, and preferably not later than the time of approval of drawings.

12.5 Programme of Works

The contractor shall submit a detailed programme showing intended method and stages together with period estimated for each stage. The programme shall include following:

- a) date of commencement and completion of every stage of the works in line with the building construction programme;
- b) date of expected completion of builder's work requirements;
- c) date of delivery of equipment and materials to site;
- d) date of requirement of temporary and permanent electricity supply; and
- e) date of completion, commissioning and testing.

The contractor shall ensure the work in accordance with the criticality of adherence of installation dimensions and supports as indicated in layout.

12.6 Electricity Supply to Escalators/Moving Walk

Operation of the machine under power is required from a comparatively early stage of installation for most efficient working, and power supply shall be furnished accordingly. Whilst temporary power supply may be sufficient for erection purposes, final testing and setting up should only be carried out with the permanent power supply connected. For this reason, the timely provision of the permanent power supply is important.

13 COORDINATION OF SITE WORK

13.1 Preparatory Work on Site

13.1.1 It is expected for the escalator/moving walk supplier to make periodic visits to the site to check progress on the construction and discuss relevant matters with the owner. The contractor should assure himself that all building work has been completed in accordance with his requirements. Immediately before the time for erection to commence, the contractor should check that site conditions are fit to permit erection to proceed.

13.1.2 Building works to be completed before the erection work, includes the following:

- a) Providing adequate access to the well-way to allow for setting the equipment in place;
- b) If the project is not ready for placement at the time of delivery, providing dry storage area located close to well-way;
- c) Providing protection of all floor openings adjacent to and in the general area of escalator/moving walk;
- d) Providing safety compliant stairwell access to floors;
- e) Providing clear unobstructed workspace adjacent to well-way;
- f) Providing safety compliant overhead and floor opening protection;
- g) Providing barricades to prevent the entrance of unauthorized personnel to construction area;
- h) Drying and cleaning of pit and floor openings with proper dimensions;
- j) Providing properly located supports of adequate strength for the truss at the landings including intermediate supports, if required;
- k) Providing suitable floor fill and patching of flooring including floor covering, adjacent to the landing plates of the escalator/moving walk; all floor finishing to be completed after installation of the landing plates; and
- m) Providing flooring with little slope in direction outwards of the escalator to prevent ingress of water, etc, into the escalator.

13.2 Delivery of Material

The contractor should advise the owner when equipment is ready for dispatch, so that the owner is able to make arrangements on site to receive and unload as near as possible to the well-way.

13.3 Storage

Adequate provision should be made by the owner for storage, protection and preservation against loss, deterioration of or damage to all material on the site. Attention is drawn to the adverse effect of damp conditions on electrical equipment.

13.4 Site Meetings

The contractor shall coordinate his proposed programme of work with that of the building contractor and all other contractors and sub-contractors to adhere to the latest overall construction programme and to achieve efficient and safe working on site. In large sites, regular meetings of all parties are beneficial for successful progress of work.

13.5 Service of Other Trades

The contractor may require the services of other trades as the work proceeds, and it is essential that the contractor should give due notice to the building contractor of the demands to be made on other trades, so that he may plan accordingly.

13.6 Connecting to Power Supply

The contractor should give prior notice to the owner of the date the power supply is required, so that suitable arrangements for connection can be made. During erection, necessary electrical power shall be provided for illumination, operation of required tools and hoist and for starting, testing and adjustment.

14 INSPECTION AND ACCEPTANCE PROCEDURES

14.1 General

Escalators and moving walks shall be inspected, before their first use; after major modifications; and at regular intervals. Such inspections and tests should be made by a competent person.

14.2 Constructional Inspection and Acceptance Inspection and Test

14.2.1 The constructional inspection and acceptance inspection and test shall be carried out at the jobsite on completion of the escalator or moving walk. During inspection and tests, precautions shall be taken so that escalators or moving walks are not set in motion without the knowledge and consent of the person(s) performing such work.

For the constructional inspection and acceptance inspection and test, the data specified in [14.4](#) should form part of a pre-completed test sheet. Furthermore, layout drawings, description of the equipment and wiring diagrams (current flow chart with legend or explanations, and a terminal connection chart) which permit a check of compliance with the safety requirements, shall be provided.

The constructional inspection comprises the examination of the completed installation for conformity with the required data and about proper workmanship as specified hereunder.

14.2.2 The acceptance inspection and test comprises,

- a) overall visual inspection;
- b) functional test;
- c) test of electric safety devices regarding their effective operation; and
- d) test of the brake(s) of the escalator or moving walk under no load, for compliance with the prescribed stopping distances.

Measurement of the insulation resistance of the different circuits between conductors and earth shall also be done. For this measurement, the electronic components shall be disconnected. This includes a test of the electric continuity of the connection between the earth terminal(s) in the driving station and the different parts of the escalator or moving walk, liable to be live accidentally.

14.3 Inspection and Test after Major Modifications

A major modification is a change of the location, a change of the speed, of the electric safety devices, of the braking system, of the drive, of the control, of the step band, of the truss and of the balustrades. Wherever applicable, the principles set forth for the constructional inspection and acceptance inspection and test should apply to the new environmental conditions, modified components and other components which are affected.

The replacement of parts with parts of same design is not considered to be a major modification.

14.4 Specific Data, Test Reports and Certificates

The following documentation should be held by the manufacturer:

- a) stress analysis of the truss or equivalent certificate by a stress analyst;
- b) proof by design calculation of sufficient breakage resistance of the parts directly driving the steps, pallets or the belt, for example step chains, racks;
- c) calculation of the stopping distances for loaded moving walks together with adjustment data;
- d) proof testing of steps or pallets;
- e) proof of the breaking strength of the step chain/pallet chain/belt;
- f) proof of sliding coefficients for skirting;
- g) proof of anti-slip properties of tread surfaces (steps, pallets, floor and comb plates without combs);
- h) proof of stopping distances and deceleration values;
- j) proof of electromagnetic compatibility; and
- k) test certificate for handrail breaking strength (for public service escalators and public service passenger moving walks).

14.5 Escalator Site Checks and Inspection List

Suitable site checklist/inspection list may be used. A typical list is given at [Annex A](#).

14.6 Escalator Commissioning and Acceptance Tests

14.6.1 Escalator Details

Typical details to be provided are given below:

- a) Name of station :
- b) Location :
- c) Escalator No. and job No. :
- d) Drawing No. :
- e) Rise :
- f) Type of balustrade : SS/Glass - Tick (✓)
- g) Manufacturer's Sl No. :

14.6.2 General Inspection

A typical general inspection list is given below:

<i>Sl No.</i>	<i>Item</i>	<i>Method/Requirement</i>	<i>Result</i>
(1)	(2)	(3)	(4)
i)	All unwanted material have been removed	Visual check	
ii)	Escalator (including steps, handrails, floor plate, cladding, etc) has been cleaned	Visual check	

<i>Sl No.</i>	<i>Item</i>	<i>Method/Requirement</i>	<i>Result</i>
(1)	(2)	(3)	(4)
iii)	Proper barricading of the area has been done prior to testing	Visual check	
iv)	Emergency signage to be fixed at the emergency stop; safety labels placed at top/bottom newel end	Visual check	
v)	Check provision of 'Danger Sign' plate on the escalator control panel	Visual check	
vi)	Wedge guard/deflector fixed for obstacles such as walls, pillars, floor intersection or criss-cross arrangement of escalators/moving walks	Visual check	
vii)	Clear height of 2.3 m above steps throughout the length of escalator	Measurement check	
viii)	Access restriction guard fixed between parallel escalators	Visual check	
ix)	Permanent barriers at entry and exit with gaps according to code requirement	Measurement check	
x)	Connection of hose/pipe to oil/water separator	Visual check	
xi)	Power cables and earthing provided at drive station	Visual check	

14.6.3 Electrical Test

A typical check list of electrical tests is given below:

<i>Sl No.</i>	<i>Item</i>	<i>Test Procedure</i>	<i>Recommended Result</i>	<i>Instrument / Sl No.</i>	<i>Measured Value</i>	<i>Result</i>
(1)	(2)	(3)	(4)	(5)	(5)	(7)
i)	Insulation resistance test	Shall be tested between lines connected together and earth at 500 V d.c.	> 0.5 MΩ (Mega Ohm)			

NOTES

- 1 Protective earthing of escalator shall be as per clauses 4.2.11.3.1.1 of good practice [D-5B(5)].
- 2 Verification of the effectiveness of the measures for fault protection by automatic disconnection of supply shall be according to 6.2.3.6 and 6.2.3.7 of good practice [D-5B(5)].
- 3 Where the current exceeds 10 mA under normal operating conditions, a reinforced protective conductor shall be used according to 5.4.3.7 of good practice [D-5B(5)].

14.6.4 Functional Tests on Safety Switches and Devices

A typical check list is given in [Annex B](#)

15 OPERATION AND MAINTENANCE

15.1 General

The installation shall be maintained in good working order in accordance with the installer's instructions. To this effect, regular maintenance of the installation shall be carried out to ensure the safety of installation. The safety of an installation shall consider the ability to be maintained without causing injury or damage to health.

Regular maintenance of the installation shall be carried out to ensure the reliability of the installation. The access and the associated environment shall be maintained in good working order in accordance with the installer's instructions.

The instructions for maintenance of an installation shall be provided by the manufacturer, when placed on the market, after risk assessment.

In order that the aim of the maintenance instructions should be achieved, they shall be formulated so that they should be clearly and easily understood by competent maintenance person. The competent maintenance person within the maintenance organization shall be continuously updated.

The installer/manufacturer shall provide maintenance instructions intended for the owner of the installation (see [15.3.2](#)) including the information intended for the maintenance organization (see [15.3.3](#)).

NOTE — The owner of the installation should be informed that the qualification of the maintenance organization is in conformity with the regulation applicable in the location in which the installation operates.

Maintenance aspects include:

- a) Lubrication, cleaning, etc;
- b) Checks;
- c) Passenger rescue operations;
- d) The operations of setting and adjustment; and
- e) Repair or changing of components which may occur due to wear and tear, and do not affect the characteristics of the installation.

However, cleaning of the external parts of the escalator or moving walk shall not be considered as maintenance.

The following shall not be considered as maintenance operations:

- 1) changing of major component, such as the machine, even if the characteristics of the new component are the same as the original;
- 2) replacement of installation;
- 3) modernization of the installation including changing of any characteristics of the installation (such as load, speed, etc); and
- 4) rescue operation carried out by fire brigade.

15.2 Elements to be considered for Maintenance Instructions

When preparing the content of the maintenance instructions (see [4.3](#) and [5](#)) the following elements shall be considered:

- a) Specifications and the intended use of the installation (type of installation, performance, etc);
- b) Environment in which the installation and its components are installed (weather conditions, vandalism, etc);
- c) Any restriction of use;
- d) Result of risk assessment (see [5](#)) for every working area and for every task undertaken;
- e) Specific maintenance instructions provided by the manufacturer of safety components; and

- f) In case of components other than safety components, where maintenance is necessary, the maintenance instructions provided by the manufacturer of these components.

15.3 Information to be Included in the Maintenance Instructions

15.3.1 General

The maintenance instructions shall contain information relating to the tasks of both the owner and the maintenance organization, respectively.

15.3.2 Information to the Owner of the Installation

The information relating to the tasks of the owner of the installation shall include the following:

- a) Need for the owner to keep the installation in a safe operating condition. To fulfil this, the owner shall use a maintenance organization complying with the requirements of this Subsection;

NOTE — It is recommended to inform the owner of the installation about the need to use a maintenance organization with adequate and proper insurance cover provided by an insurance company.

- b) Need for the owner to take care of applicable regulation(s) and other requirements, where relevant, and their implications on maintenance;
- c) Need for planned maintenance to be carried out by a maintenance organization, at the latest when the installation is put into service or if the installation is to remain unused for a long period of time before first being put into service;
- d) Importance for the owner of the installation to have the same maintenance organization in the case of several installations having common well/spaces and/or machine room;
- e) Need for the owner to put the installation out of service in case of dangerous situations;
- f) Need for the owner of the installation to inform the maintenance organization:

- 1) Immediately about any perceived abnormal operation of the installation or abnormal change in its direct environment;
- 2) Immediately after putting the installation out of service in the case of a dangerous situation;
- 3) After any rescue intervention by their authorized and instructed person(s);
- 4) Before any modification related to the installation and/or its environment or use;

NOTE — The owner of the installation should obtain from the company carrying out the relevant modification the maintenance instructions for the maintenance organization.

- 5) Before any authorized third-party inspection or works other than maintenance works are carried out on the installation;
- 6) Before taking the installation out of service for a prolonged period; and
- 7) Before putting the installation back into service after a prolonged period of non-operating time.
- g) Need for the owner of the installation to take into consideration the consequences of the risk assessment carried out by the maintenance organization;
- h) Need for the owner of the installation to make sure that the risk assessment for maintenance is carried out:
 - 1) If the maintenance organization is replaced;
 - 2) If the use of the building and/or the installation changes;
 - 3) After a major modification of the installation or of the building; and
 - 4) If it is the case, after an accident involving the installation.
- j) Need for the building owner to ensure, through a risk assessment, that:

- 1) their premises are safe and free from risk to health as far as is practicable. This includes access to the premises and installation equipment, and articles or substances used according to the regulation for the use of work equipment at the workplace;
- 2) the persons using the premises are informed about any remaining risks; and
- 3) any action to be done as a consequence of risk assessment carried out.

Regarding the access ways to areas reserved for maintenance persons, the need for the owner of the installation to inform the maintenance organization, about,

- i) the access ways to be used and fire evacuating procedures from the building;
- ii) the place where the keys of the reserved areas shall be found;
- iii) if necessary, the persons who shall accompany the maintenance persons to the installation; and
- iv) if necessary, personal protective equipment to be used in the access ways, and, possibly, where this equipment shall be found.

The information shall be made available also on site to the maintenance organization.

- k) Need for the owner of the installation to ensure that the name and the telephone number of the maintenance organization are always available to the user of the installation, permanently affixed and clearly visible;
- m) Need for the owner of the installation to ensure that the keys of machine room doors (trap doors) and of inspection and emergency doors (trap doors), if any, are permanently available in the building and are used only by persons authorized to gain access;
- n) Need for the owner of the installation to provide, in all circumstances, safe access to the building and to the installation for the maintenance organization involved in the rescue of persons;
- p) Need for the owner of the installation to keep the access to working areas and working rooms safe and free for the maintenance persons and to inform the maintenance organization about any hazard or change in the workplace and/or the access ways (lighting, obstructions, ground conditions, etc);
- q) In addition to those examinations and tests (for escalators/moving walk) which the owner of the installation entrusts to the maintenance organization, the need for the owner to carry out periodically, in their own interests, include the following:
 - 1) A full travel in both directions, when they exist, to assess any changes in the quality of the ride or damage to the equipment.
 - 2) Typical items to be checked to ensure that they are in place, undamaged and functioning correctly are:
 - i) all lighting and indicators;
 - ii) emergency stopping device;
 - iii) handrails;
 - iv) skirting/deflector devices;
 - v) combs;
 - vi) safety signs/pictograms;
 - vii) approximation of speed between handrail and steps/pallets;
 - viii) steps/pallets;
 - ix) balustrade and panels;
 - x) head guard and decking; and
 - xi) safe and unobstructed access to entry and exit areas.

15.3.3 Information for the Maintenance Organization

The information relating to the tasks of the maintenance organization shall include the following:

- a) Need to carry out the work of maintenance in conformity with the maintenance instructions and based on systematic maintenance checks. After these checks, the maintenance organization shall decide in conformity with the maintenance instructions what is required to be done. A list of typical examples of maintenance checks to maintain the installation is shown in [Annex C](#);

NOTE — Due to the fact that the components may be different in design and operation, it is therefore not possible to give specific guidelines in this Subsection.

- b) Need to update the original maintenance instructions, if the installation changes its intended use and/or the environmental conditions existing on the completion of the installation;

NOTE — The maintenance organization should be provided by the owner of the installation with the relevant maintenance instructions where modifications are carried out on the installation.

- c) Need for the maintenance organization to ensure that a risk assessment for any working area and for any maintenance operation has been carried out considering the installer's maintenance instructions and all information supplied by the owner of the installation;

- d) Need for the maintenance organization to inform the owner of the installation about any work to be carried out as a consequence of a risk assessment especially for the access and/or the environment related to the building/installation;

- e) Need to carry out a maintenance plan so that preventive maintenance is suitable for the installation and maintenance time is as short as reasonably practicable, without reducing the safety of persons, in order to minimize the non-operational time of the installation;

- f) Need to adapt the plan for maintenance to take account of any predictable failures, for example, those due to misuse, mishandling, deterioration, etc;

- g) Need to carry out maintenance operations by competent maintenance persons and provided with the necessary tools and equipment;

- h) Need to maintain the competency of maintenance persons;

- j) Need to carry out the maintenance periodically. In determining the frequency of maintenance interventions, the following non-exhaustive list should be considered:

- 1) Operating time and any non-operating periods of time;
- 2) Age and condition of the installation;
- 3) Location and type of building in which the installation is installed, as well as the needs of the users; and
- 4) Local environment where the installation is situated, as well as external environmental elements, for example, weather conditions (rain, heat, cold, etc) or vandalism.

- k) Need to provide a 24 h, all year-round call-out service for rescue of persons;

- m) Need to keep records of the result of each intervention due to a failure of the installation. These records shall include the type of failures to detect any repetition. They shall be available to the owner of the installation on request;

- n) Need to put out of service the installation, if the maintenance organization is aware of a dangerous situation, detected during the maintenance, which cannot be eliminated immediately, and to inform the owner of the installation of the need to keep it out of service until repaired;

- p) Need to be organized to provide the necessary spare parts for any repair;

- q) Possible need for attendance of a competent maintenance person(s), given reasonable notice, for any inspection carried out by an authorized third party or for building maintenance works to be carried out in the areas reserved for the maintenance organization;

- r) Need to inform in due time, the owner of the installation about necessary progressive upgrading of the installation; and

- s) Need to organize rescue operations, even with sub-contractor(s), and to make provision for circumstances such as fire, panic, etc.

15.4 Maintenance Contract

It is recommended that the maintenance is carried out by original equipment manufacturer (OEM). Any maintenance contract should include the following at a minimum, to ensure that escalators/moving walk is always maintained/operated in the safe working condition:

- a) All batteries used for emergency operations;
- b) All safety switches/devices;
- c) Handrail system components including handrails;
- d) Operational and auxiliary brake; and
- e) Combs.

15.5 Risk Assessment

15.5.1 General

Before an installation is placed on the market, it is necessary that the installer/manufacture carries out a risk assessment. Every risk shall be limited as much as reasonably possible by means of safety measures and suitable instructions. The instructions can never replace a safety measure which can be provided to reduce the risk.

It is necessary to determine the different intervention procedures of the maintenance operations and to determine the appropriate safety measures for each of these procedures.

The use of diagnostic systems may support fault finding, improve the maintainability of the installation and reduce the exposure of maintenance persons to hazards.

Safety in maintenance operations of the installation is ensured by adopting safety measures and providing instructions. Safety measures on the installation and in the building shall be provided by the installer and by the owner of the installation, respectively.

For any working area, it is necessary to identify the list of the specific hazards related to health and safety and to carry out a risk assessment for any maintenance operation, including access to the working area.

For this purpose, the following should be considered:

- a) Presence of one or more maintenance persons in a working area;
- b) Foreseeable actions of persons other than maintenance persons (for example, person switching on or off power circuits and dependent circuits or lighting circuits or trying to use the installation during maintenance operations, etc); and
- c) Possible states of the installation (normal or abnormal due to a foreseeable failure of its component parts, external disturbances, disturbance of its power supply, etc).

[Annex D](#) gives a list of examples of elements to be considered in any risk assessment for maintenance operations. However, several methods are available for the systematic assessment of risk.

15.5.2 Information for the Maintenance Organization

For safe maintenance and to provide relevant instructions, it is first necessary to identify the maintenance operations.

Maintenance operations are:

- a) those operations considered necessary for a correct and safe functioning of the installation and its components after the completion of the installation; and
- b) those operations considered necessary during the 'life' of some components, determining, as far as possible, the time or condition after which the functioning or the integrity of the component is no longer ensured even if correctly maintained.

In carrying out specific maintenance operations, if it is necessary to neutralize some safety functions (for example, an electric safety device), the hazard identification shall be considered for such a situation.

It is necessary to inform and warn the maintenance persons about: (a) residual risks, that is, those for which risk reduction by design and safeguarding techniques are not there or not completely effective; and (b) risks that arise from the necessary removal of certain guards to carry out specific maintenance operations. The maintenance instructions and warnings shall prescribe the procedures and operating modes intended to overcome these risks and, if it is necessary, to specify personal protective equipment, instruments, tools and provisions to be used.

15.6 Markings, Signs, Pictograms and Written Warnings

If the risk assessment of the maintenance organization indicates that additional specific warnings are required for the purpose of maintenance, these shall be affixed directly on the installation/component or, in the close vicinity, when this is not possible. Markings, signs, pictograms, and written warnings shall be readily understandable and unambiguous. Readily understandable signs and pictograms shall be used in preference to written warnings.

Signs or written warnings carrying only the words 'DANGER' shall not be used. Information affixed directly on the installation/component shall be permanent and legible.

Any markings, signs, pictograms, and written warnings affixed on the installation shall be renewed if they become illegible.

15.7 Format of the Maintenance Instruction Handbook

The maintenance instruction handbook for any installation shall contain at the front at least the following:

- a) Type of installation, with its serial number, to which the instructions apply;
- b) Title of the handbook;
- c) Date of issue;
- d) Name and address of the installer/manufacture; and
- e) Name of the publisher, when different from the installer/manufacture.

In the handbook:

- 1) all units used shall be SI units;
- 2) all pages shall be numbered; and
- 3) all references to other documents shall be in full.

Warnings shall state the hazard, the related risks, and the appropriate safety measure.

Type and size of print shall ensure the best possible legibility. Safety warnings and/or precautions shall be emphasised using colours or symbols and/or large print.

Documents giving instructions for maintenance shall be produced in durable form (that is, they shall be able to survive frequent handling) or triplicate copies shall be provided.

15.8 Maintenance Record and Annual Inspection

Maintenance organization is required to maintain a logbook at the equipment to record of maintenance and repair work carried out on the equipment. In addition, the layout diagram and latest wiring diagram is required to be maintained at the equipment by maintenance organization. Maintenance organization is required to carry out annual inspection of the installation and certify that escalator is in safe for operation. Owner of the installation is responsible for ensuring that any deficiencies found during the annual inspection are rectified before the equipment is put back in service.

ANNEX A

(Clause [14.5](#))

CHECK LIST FOR INSPECTION OF ESCALATOR/MOVING WALK AT SITE

SI No.	Reference No.	Item	OK	Not OK	NA
(1)	(2)	(3)	(4)	(5)	(6)
	1	Escalator/Moving Walk - External			
i)	1.1	General fire protection			
ii)	1.2	Geometry			
iii)	1.3	Handrails			
iv)	1.4	Entrance and exit ends			
v)	1.5	Lighting			
vi)	1.6	Caution signs			
vii)	1.7	Comb-plate			
viii)	1.8	Deck barricade			
ix)	1.9	Steps/Pallets and upthrust device			
x)	1.10	Operating devices			
xi)	1.11	Skirt obstruction devices			
xii)	1.12	Handrail-speed monitoring devices			
xiii)	1.13	Speed			
xiv)	1.14	Balustrades			
xv)	1.15	Ceiling intersection guards			
xvi)	1.16	Step/pallet/skirt clearance panels and performance index			
xvii)	1.17	Outdoor protection			
	2	Escalator/Moving Walk - Internal			
xviii)	2.1	Machinery space access, lighting, receptacle and condition			
xix)	2.2	Stop switch			
xx)	2.3	Controller and wiring			
xxi)	2.4	Drive machine and brake			
xxii)	2.5	Speed governor			
xxiii)	2.6	Broken drive chain and disconnected motor device			
xxiv)	2.7	Reversal stop switch			
xxv)	2.8	Broken step/pallet chain switch			
xxvi)	2.9	Step/pallet upthrust device			
xxvii)	2.10	Missing step/pallet device			
xxviii)	2.11	Step/pallet level device			
xxix)	2.12	Step/pallet, step/pallet chains and trusses			
xxx)	2.13	Handrails			
xxxi)	2.14	Name plate			
xxxii)	2.15	Response to smoke detector			
xxxiii)	2.16	Log card			
xxxiv)	2.17	Pit cleanliness			
xxxv)	2.18	Inspection box			

ANNEX B

(Clause [14.6.4](#))

TYPICAL CHECK LIST FOR FUNCTIONAL TESTS ON SAFETY SWITCHES AND DEVICES

SI No.	Check Point	Method	Value Required	Measured Value	Result
(1)	(2)	(3)	(4)	(5)	(6)
i)	Emergency stop switch	Push the button	OK/NOT OK	Top:	
			OK/NOT OK	Intermediate level:	
			OK/NOT OK	Bottom:	
ii)	Direction key switch (Start switch)		OK/NOT OK	Top:	
			OK/NOT OK	Bottom:	
iii)	Traffic bollard		OK/NOT OK		
iv)	Traffic light working	Check for correct movement indication. (Up/Down) visual check	OK/NOT OK	Top:	
			OK/NOT OK	Bottom:	
v)	Provision of comb light	Visual check	OK/NOT OK		
vi)	Sprinkler:				
	a) Connected to main system		Yes/No/NA		
	b) Number of sprinkler heads provided		Yes/No/NA		

ANNEX C

[(Clause [15.3.3\(a\)](#))]

TYPICAL EXAMPLES OF MAINTENANCE CHECKS TO MAINTAIN THE INSTALLATION

C-1 ESCALATORS AND MOVING WALKS

Sl No. (1)	Component (2)	Checks (3)
i)	Controller	Check cabinet is clean, dry and free from dust
ii)	Gear box	Check gear and associated parts Check lubrication
iii)	Drive motor	Check bearings for wear Check lubrication
iv)	Brake	Check braking system Check parts for wear
v)	Auxiliary brake	Check braking system Check parts for wear
vi)	Intermediate gear box	Check gear and associated parts Check lubrication
vii)	Main drive chain	Check for tension and wear Check lubrication
viii)	Step/Pallet chain	Check for tension and wear Check lubrication
ix)	Step/pallet	Check step/pallet and step/pallet wheels for integrity
x)	Conveyor belt	Check for condition and tension
xi)	Drive belt	Check for condition and tension
xii)	Clearances	Check step to step and step to skirting clearances
xiii)	Combs	Check condition Check meshing with steps, pallets or belt
xiv)	Comb plate	Check clearances and operation
xv)	Handrails	Check for free running and condition Check tension Check synchronization between step/pallet band and the handrail
xvi)	Track system	Check for condition and wear Check fixings
xvii)	Safety devices	Check operation
xviii)	Deflector devices	Check condition
xix)	Lighting	Check operation
xx)	Display	Check operation
xxi)	Signs/pictograms	Check condition
xxii)	Balustrade	Check condition of panels Check fixings of interior claddings

EXAMPLES OF ELEMENTS TO BE TAKEN INTO ACCOUNT THE RISK ASSESSMENT FOR MAINTENANCE OPERATIONS

D-1 ESCALATORS/MOVING WALKS

Elements	Machinery Spaces	On Step/Pallet Band	Inside Step/Pallet Band	Upper and Lower Landing	Control Cabinet	Machine Room (External Drives)
(1)	(2)	(3)	(4)	(5)	(6)	(7)
Access and entry	R	R	R	R	R	R
Inadequate lighting (including access)	R	R	R	R	R	R
Falls/slips	R	R	R	R	R	R
Falling on machine	NR	R	NR	R	NR	NR
Falling over balustrade	NR	R	NR	R	NR	NR
Contact with moving machinery	R	R	R	R	R	R
Indirect contact with machinery	R	R	R	R	R	R
Crushing and shearing (step to step or comb/step to skirting)	R	R	R	NR	NR	NR
Gaps in balustrade	NR	R	NR	NR	NR	NR
Intersection between floors and/or escalators	NR	R	NR	NR	NR	NR
Persons on the step band	NR	R	NR	R	NR	NR
Safety switches and emergency stopping device(s)	R	R	R	R	R	R
Inspection control	R	R	R	R	R	R
Intersection between fixed and moving parts	NR	R	NR	NR	NR	NR
Unintentional start/stop	R	R	R	R	R	R
Machine moving (other than on power)	R	R	R	R	R	R
More than one maintenance person working	R	R	R	R	R	R
Manual operation	R	NR	R	R	R	R
Falling objects	R	R	R	R	NR	R
Unexpected water/dirt	R	R	R	R	R	R
Contamination by oil and grease	R	R	R	R	NR	R
Dangerous substances	R	R	R	R	R	R
Fire	R	NR	R	NR	R	R
Missing step/pallet	R	R	R	R	NR	NR

‘R’ = Relevant and ‘NR’ = Not relevant

LIST OF STANDARDS

The following list records those standards which are acceptable as ‘good practice’ and ‘accepted standards’ in the fulfilment of the requirements of this NBCS. The latest version of a standard shall be adopted at the time of enforcement of this NBCS. The standards listed may be used for conformance with the requirements of the referred clauses in this NBCS.

In the following list, the number appearing in the first column within parentheses indicates the number of the reference in this Part/Section.

	<i>IS No.</i>	<i>Title</i>
(1)	SP 30 : 2023	National Electrical Code of India 2023
	IS 732 : 2019	Code of practice for electrical wiring installation (<i>fourth revision</i>)
	IS 3043 : 2018	Code of practice for earthing (<i>second revision</i>)
(2)	IS 12777 : 1989	Method for classification of flame spread of products
(3)	IS 4591	Escalators and moving walks:
	Part 1/Sec 1 : 2020	Safety requirements (<i>first revision</i>)
	Part 1/Sec 2 : 2020	Guide for planning and selection (<i>first revision</i>)
	Part 1 / Sec 3 : 2020	Inspection and test (<i>first revision</i>)
	Part 2 : 2022	Guide for maintenance
(4)	IS 17615 (Part 2) : 2021/ ISO 18738-2 : 2012	Measurement of ride quality: Part 2 Escalators and moving walks
(5)	IS 732 : 2019	Code of practice for electrical wiring installations (<i>fourth revision</i>)

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NATIONAL BUILDING CONSTRUCTION STANDARDS

PART D BUILDING SERVICES

SECTION 5 INSTALLATION OF LIFTS, ESCALATORS, MOVING WALKS AND POWER-DRIVEN PARKING SYSTEMS

5C POWER-DRIVEN PARKING SYSTEMS

BUREAU OF INDIAN STANDARDS

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National Building Code Sectional Committee, CED 46

FOREWORD

This NBCS (Part D/Subsection 5C) covers the requirements for planning and design, installation and operation, of car parking systems to ensure efficient operation of system with satisfactory performance.

This Section was first published in 1970 and was subsequently revised in 1983, 2005 and 2016 as a part of National Building Code of India. This Section covers the requirements for installation of lifts and escalators in buildings. This Section should be read with Part F 'Fire and Life Safety' of this NBCS from fire safety requirements point of view. The major changes in the first revision of 1983 were the addition of outline dimensions of different types of lifts and detailed requirements of escalators in buildings. Emphasis was laid on coordination between the engineer/architect and the lift manufacturer to arrive at the number and position of lifts for attaining optimum efficiency in serving the building with safety.

The significant changes with respect to lifts incorporated in the 2005 revision, included the addition of new clauses/recommendations on the building management system; the addition of new clauses on the fireman's lift, infrared light curtain, safety and braille button for blind people and updation of provisions as per the revised standards on lifts on which this Section was based.

In the 2016 revision, the erstwhile Section 5 on Lifts and Escalators was divided into two subsections namely:

5A Lifts

5B Escalators and moving walks

The growing demand for efficient and space-saving parking solutions for passenger cars in urban environments has led to the development of power-driven parking systems in the country that address the requirements of power-driven vehicle parking systems. These systems, which may be either automated or semi-automated, are crucial for optimizing land use in densely populated areas. Recognizing the importance of such systems, in this NBCS, a new subsection 5C 'Power-Driven Parking Systems' has been introduced to provide comprehensive guidelines for the planning, design, installation, operation, maintenance and inspection of parking systems assuming safe and efficient movement of vehicles.

In this revision, this publication has been renamed as National Building Construction Standards (NBCS).

This new Subsection on power-driven parking systems reflects the evolving technological advancements and urban planning needs and its integration within the lifts and escalators related building services chapter underscores the significance of vertical transportation solutions in modern building infrastructure.

The following important provisions are addressed in this new Subsection:

- a) *Automated parking systems* — Fully automated car parks allow drivers to leave their vehicles at designated entry points, from where the vehicles are transported and parked automatically. This system minimizes human intervention, reduces parking time and enhances space utilization;
- b) *Car lifts and ramps* — In semi-automated systems, the movement of cars depends on the use of car lifts with/without ramps. It is essential to conduct a detailed analysis to determine the optimal number of car lifts required, ensuring that the average car retrieval/parking time does not exceed 3 min; and
- c) *Design and safety* — Car lifts are to be designed to accommodate the largest vehicles intended for use, providing adequate space for door opening to facilitate passenger evacuation in case of emergencies. Additionally, designers need to account for potential queues and provide sufficient holding areas for vehicles.

This NBCS (Part D/Subsection 5C) deals with the requirements for planning, design, installation, operation, maintenance and inspection of parking systems so as to ensure safe movement of cars within and out of the building spaces and with satisfactory performance.

Assistance has also been derived from the following publications in the preparation of this Subsection:

- | | |
|------------------------|---|
| IBC 2024 | International Building Code, International Code Council, Washington, USA |
| DIN EN 14010 : 2009-12 | Safety of machinery — Equipment for power-driven parking of motor vehicles —
Safety and EMC requirements for design, manufacturing, erection and
commissioning stages |

As the regulation of land development and buildings is a State subject and is dealt by the States and local bodies such as urban local bodies within their jurisdiction through building regulatory documents like building regulations, building byelaws and fire regulations, this document is voluntary in nature and is non-binding. The implementation depends on adoption by concerned parties or stipulations in a contract or by appropriate adoption by the concerned authorities.

For the purpose of deciding whether a particular requirement of this Subsection is complied with, the final value, observed or calculated, expressing the result of a test or analysis, shall be rounded off in accordance with IS 2 : 2022 'Rules for rounding off numerical values (*second revision*)'. The number of significant places retained in the rounded off value should be the same as that of the specified value in this Subsection.

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Important Explanatory Note for Users of this NBCS

In any Part/Section of this NBCS, where reference is made to **‘good practice’** in relation to **design, constructional procedures or other related information**, and where reference is made to **‘accepted standard’** in relation to **material specification, testing, or other related information**, the Indian Standards listed at the end of the Part/Sec shall be used as a guide to the interpretation.

At the time of publication, the editions indicated in the standards were valid. All standards are subject to revision and parties to agreements based on any Part/Section are encouraged to investigate the possibility of applying the most recent editions of the standards.

In the list of standards given at the end of a Part/Section, the number appearing within parentheses in the first column indicates the number of the reference of the standard in the Part/Section. For example:

a) Good practice [D-5C(1)] refers to the Indian Standard given at serial number (1) of the List of Standards given at the end of this Part 5/ Subsection 5C, that is, SP 30 : 2023 ‘National Electrical Code of India 2023’.

NATIONAL BUILDING CONSTRUCTION STANDARDS

PART D BUILDING SERVICES

SECTION 5 INSTALLATION OF LIFTS, ESCALATORS, MOVING WALKS AND POWER-DRIVEN PARKING SYSTEMS

5C POWER-DRIVEN PARKING SYSTEMS

1 SCOPE

1.1 This NBCS (Part D/Subsection 5C) covers the requirements for planning, design, installation, operation, maintenance and inspection of permanently installed power-driven vehicle parking systems in buildings so as to ensure safe parking and movement of cars with satisfactory performance. These systems which may be automated or semi-automated are crucial for optimizing land use in densely populated areas.

1.2 This Subsection gives information that should be exchanged among the architect/engineer, the consulting engineer and the car parking solution providers/installers from the stage of planning of installation including up to the functioning and maintenance stage.

2 TERMINOLOGY

For the purpose of this Subsection, the following definitions shall apply.

2.1 General

2.1.1 *Grid* — One or more (sideways) working areas load handling devices standing vertically one above the other.

2.1.2 *Operating Panel* — Stationary or mobile device giving command for parking space request. Signal transmission for mobile controls can be based on different technologies (radio, infrared, RFID, bluetooth, etc).

2.1.3 *Parking Unit* — A parking unit refers to one or several load carriers connected together mechanically and moving as a single entity.

2.1.4 *Parking Equipment Attendant* — A person tasked with operating, cleaning and/or supervising power-driven parking equipment.

2.1.5 *Public Use* — Those parking systems designed and intended for access and use by the general public, as opposed to private or restricted use.

2.1.6 *Series* — One or more load-handling attachments placed horizontally and one behind the other as a multi-row arrangement.

2.1.7 *Transfer Area* — An area within a semi-automatic/fully automatic parking facility into which cars are parked in and/or from which they are removed. Normally only the driver should be allowed to enter in this area.

2.1.8 *Working Area* — Power-operated parking system for motor vehicles and cars, in which the transfer area is the same as the working area. A main vertical and horizontal sliding door (recommended for safety) is located in front of each transfer area which stores and makes the car available again in sequential operation. During the operation of the parking system, access of any personnel is prevented in this working area.

2.2 Terms Relating to Parking Systems

2.2.1 *Automatic Parking System* — A parking system that consists of parking equipment along with its linked ancillary equipment, such as main doors, working area doors and emergency doors all of which operate in an automatic/preprogrammed manner.

2.2.2 Electromechanical Parking Systems — A parking system that employs mechanical components such as gears, pulleys, ropes and chains to move vehicles horizontally or vertically into parking positions, often using automated platforms or carriages.

2.2.3 Hydraulic Parking Systems — A parking system that utilizes fluid-driven mechanisms, including hydraulic cylinders, to lift and lower vehicles into designated parking spaces, typically allowing for multiple vehicles to be stacked vertically in a compact space.

2.2.4 Locking Device — A mechanism that prevents the platform from leaving a defined position within the parking system.

2.2.5 Puzzle Parking System — A type of automated parking system in which vehicles are moved on platforms in vertical direction and horizontal direction (similar to a puzzle).

2.2.6 Safety Switch — A control switch with one or more break-contact elements connected via non-resilient members, ensuring full contact opening when the actuator is moved through the specified positive opening travel.

2.2.7 Safety Gear — A mechanical device designed to stop and hold the platform in case of over speeding with gravity within the parking system.

2.2.8 Stack Parking System — A type of parking system which moves the car in up and down position.

2.2.9 Wheel Track Width — The distance between the centre lines of the wheels on a single axle of a vehicle.

2.2.10 Wheel Base — The distance between the centres of the wheels on the front and rear axles of a vehicle.

2.3 Terms Relating to Semi-Automatic Parking Systems

2.3.1 Area of Parking — The area reserved for the car parking system where the system is supported at various load bearing points shall ensure firm support from flooring/ceiling.

2.3.2 Drive Way — The front open space in the parking system (minimum 6.0 m in length or the diagonal distance of car + 500 mm) where the car enters in 90° direction into the system.

2.3.3 Horizontal Movements — The lateral shifting of vehicles within a parking level, often using automated platforms or conveyors for efficient space utilization.

2.3.4 Restricted Access — Access restricted to authorized persons to use the parking facilities and are controlled through access code, magnetic card, key switch, etc.

2.3.5 Vertical Movements — The transporting of vehicles between different levels vertically, typically using car lifts or stacker systems to maximize vertical space.

3 GENERAL

3.1 Conformity with Act and Rules

3.1.1 The installation of parking systems, including stack parking, puzzle parking, tower parking and other automated parking systems, is governed by different acts and rules across states, intended to ensure the safe installation and operation of such systems.

3.1.2 The installation shall be carried out in conformity with relevant acts and rules wherever they are in force, ensuring compliance with all safety and operational guidelines.

3.1.3 A license for public use is a safety provision issued by state authorities under applicable acts and rules, wherever they are in force.

3.2 Conformity with the Electricity Act and Rules

3.2.1 All electrical work related to the installation of parking systems shall be carried out in accordance with the

provisions of the *Electricity Act, 2003* and *Central Electricity Authority (CEA) Regulations* as amended up to date, along with the Rules and Regulations framed thereunder. Additionally, the installation shall comply with the provisions of Part D ‘Building Services, Section 2 Electrical and Allied Installations’ of this NBCS.

3.3 Conformity with Indian Standards

3.3.1 All materials, fittings, appliances, etc used in the installation of parking systems shall conform to Indian Standard specifications wherever these exist. In cases where Indian Standard specifications do not exist, the materials shall be approved by the competent authority (see Part B ‘Building Materials’ of this NBCS). The installation shall adhere to best practices as outlined in relevant standards such as accepted standards [D-5C(1)], [D-5C(2)] and [D-5C(3)].

3.4 Conformity with Fire Regulations

3.4.1 The installation of parking systems shall be carried out in conformity with the applicable state fire acts/rules/local fire regulations, wherever they are in force. The installation shall be carried out in conformity with relevant acts and rules wherever they are in force, ensuring compliance with the same. See also Part F ‘Fire and Life Safety’ of this NBCS for guidance purpose.

3.5 Safety Design Considerations

The design consideration for type and selection of parking systems walk shall be based on following criteria:

- a) *Location* — To ensure reliability and longevity, the design specifications of parking systems should suit the environmental conditions of the installation site. Factors such as exposure to weather elements, seismic activity, susceptibility to stagnant rain water/flood prone zone and temperature variations should be considered when selecting and designing a parking system;
- b) *Physical Requirements* — The physical characteristics of the site, such as available space and the building's structural capacity, play a crucial role in selecting the appropriate parking system. The system should be designed to accommodate the vertical and horizontal dimensions of the space. The building infrastructure should also be evaluated to ensure it can support the weight and dynamic loads of the components of the parking system, apart from the various external loads, forces and effects (see Part C ‘Structural Design, Section 1 Loads and Load Effects’ of this NBCS);
- c) *Type of Segment and Traffic Patterns* — The design and selection of parking systems depend on the type of segment where they will be installed, such as residential buildings, commercial complexes, transit hubs, or public parking facilities. The expected traffic patterns and peak demand should be carefully analysed to ensure that the parking system's capacity matches the anticipated usage, preventing congestion and delays;
- d) *Safety Considerations* — Parking systems should be designed and installed with a focus on safety to prevent accidents and injuries. Safety features should include proper load limits, emergency stop functions, and clear signalling systems. Operators and users should adhere to the rules for safe operation, including compliance with all safety signals and instructions outlined in the operation and maintenance manual. Routine inspections should be conducted before operation to confirm the equipment's condition, and overloading should be strictly avoided to prevent mechanical failure and any local buckling damage; and
- e) *Aesthetic Preferences* — The aesthetic integration of the parking system with the surrounding environment should be determined in consultation with all relevant stakeholders. This may involve choosing materials, finishes and designs that complement the architectural style of the building or site.

4 PARKING SYSTEMS (OVERVIEW AND SPECIFICATIONS)

4.1 General

Parking systems are essential in urban environments where space is limited and the demand for efficient vehicle storage is high. These systems optimize the use of vertical and horizontal space, ensuring that vehicles are securely and conveniently parked. The choice of a parking system should be based on factors such as the type of

vehicles to be parked, the available space, the required capacity and specific site conditions, time taken in parking, maintainability and economy.

4.2 Stack Parking System

An optimized system in which vehicles are stacked vertically is known as stack parking system. Such systems are particularly useful in urban settings where land space is scarce. The stack parking system operates by lifting vehicles and placing them on platforms supported by vertical columns. They are ideal for residential and commercial buildings having limited space and are also suitable for above-ground installations where space constraints prevent extensive horizontal expansion. There are two types of stack parking systems, namely:

- a) *Dependent stack parking system* — A vertical parking arrangement where two/three cars are parked one above the other. The lower car should be removed to access the upper car. This system is suitable for both indoor and outdoor installations and offers a cost-effective solution by utilizing a single driving level to create two parking levels (see [Fig. 1A](#) and [Fig. 2A](#)). The dimensions and configurations for two-level and three-level dependent stack parking systems are provided in [Table 1](#).

Table 1 Dimension and Configuration of Dependent Stack Parking System						
[Clause 4.2(a)]						
SI No.	Dependent System	Minimum Height				
		mm				
(1)	(2)	(3)	(4)	(5)	(6)	(7)
i)	2 Level parking system:					
	a) Upper level	1 600	1 600	1 600	1 600	1 600
	b) Lower level	1 600	1 700	1 800	1 900	2 000
	c) Combined system	3 400	3 500	3 600	3 700	3 800
ii)	3 Level parking system:					
	a) Upper level	1 600	1 700	1 800	1 900	2 000
	b) Middle level	1 600	1 700	1 800	1 900	2 000
	c) Lower level	1 600	1 700	1 800	1 900	2 000
	d) Combined system	5 300	5 600	5 900	6 200	6 500

- b) *Independent stack parking system* — A parking system that allows two/three cars to be parked vertically above each other, with each parking level being independently accessible. This system maximizes space efficiency by providing multiple parking levels on a single driving level, making it an ideal solution for areas with limited space (see [Fig. 1B](#) and [Fig. 2B](#)). The dimensions and configurations for two-level and three-level independent stack parking systems are provided in [Table 2](#).

Table 2 Dimension and Configuration of Independent Stack Parking System				
[Clause 4.2(b)]				
SI No.	Independent System	Height		
		mm		
(1)	(2)	(3)	(4)	(5)
i)	2 Level parking system:			
	a) Required pit	1 850	2 000	2 250
	b) Upper level	1 650	1 800	2 050
	c) Lower level	1 650	1 800	2 050
	d) Combined system	3 550	3 850	4 350

Table 2 (Concluded)

Sl No.	Independent System	Height mm		
		(3)	(4)	(5)
ii)	3 Level parking system:			
	a) Required pit	3 450	3 750	3 850
	a) Upper level	1 550	1 700	1 750
	b) Middle level	1 550	1 700	1 750
	c) Lower level	1 550	1 700	1 750
	d) Combined system	4 950	5 400	5 550

4.2.1 Key Features and Components

The following are the key features and components of stack parking systems:

- Vertical support* — Supported on two sides by vertical columns with a pair of slides that hold the platform and allow movement along the columns.
- Balancing chain* — A chain fixed at both ends that supports the platform during travel, having sufficient strength to withstand impact loads in case of a sudden fall. This also ensures that platform remains in horizontal position during entire operation.
- Locking mechanism* — Lockers positioned on both sides of the vertical columns engage during downward movement. Locking mechanism restricts free fall of platform ensuring safety during entire travel length (see Note).
- Hydraulic cylinder* — The system uses hydraulic cylinders designed to bear the load of the vehicle on the platform during vertical transportation.
- Hydraulic power pack* — A single hydraulic power pack can serve multiple systems, with individual controls for each system.
- Non-return valve* — Controls the flow of hydraulic oil in one direction with a self-locking arrangement to prevent accidental movement.
- Foundation support* — The system is anchored by RCC foundations (minimum M25 grade) capable of sustaining the pull-out loads exerted through foundation bolts ensuring anchoring depth of (125 mm to 150 mm) and concrete depth of 200 mm.
- Ceiling or side supports* — The upper end of the system is supported by either ceiling or side supports.
- Four-column support* — The car's weight is transferred to four supporting pillars, typically driven by mechanical traction machines using ropes or chains.
- Braking system* — The movement is controlled by brakes, though the system lacks a secondary stopping mechanism if the brakes fail.
- Limit switches and sensors* — Limit switches control movement, but over-reliance can lead to accidents. Sensors below the platform detect vehicles, though potential blind spots exist.
- Operational safety* — An emergency safety switch is mandatory for every operating panel. The system requires a 'hold to run' operation to ensure no objects are within the parking area before activation.
- Security features* — Each operating device should have a key lock to prevent unauthorized use. The system's components should have an IP rating of 67 for outdoor installations.

NOTE — Until the Indian Standard on the subject is formulated, specialist literature like EN 14010 : 2003 'Safety of machinery — Equipment for power-driven parking of motor vehicles. Safety and EMC requirements for design, manufacturing, erection and commissioning stages' may be referred to.

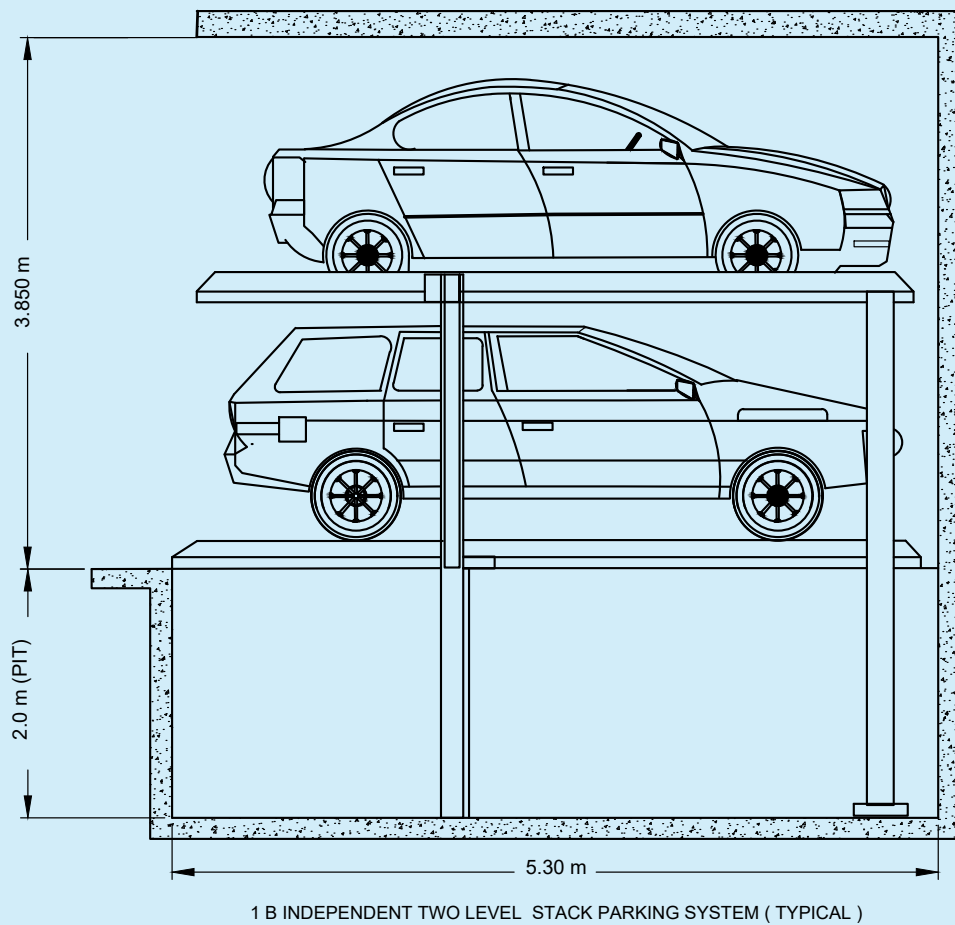
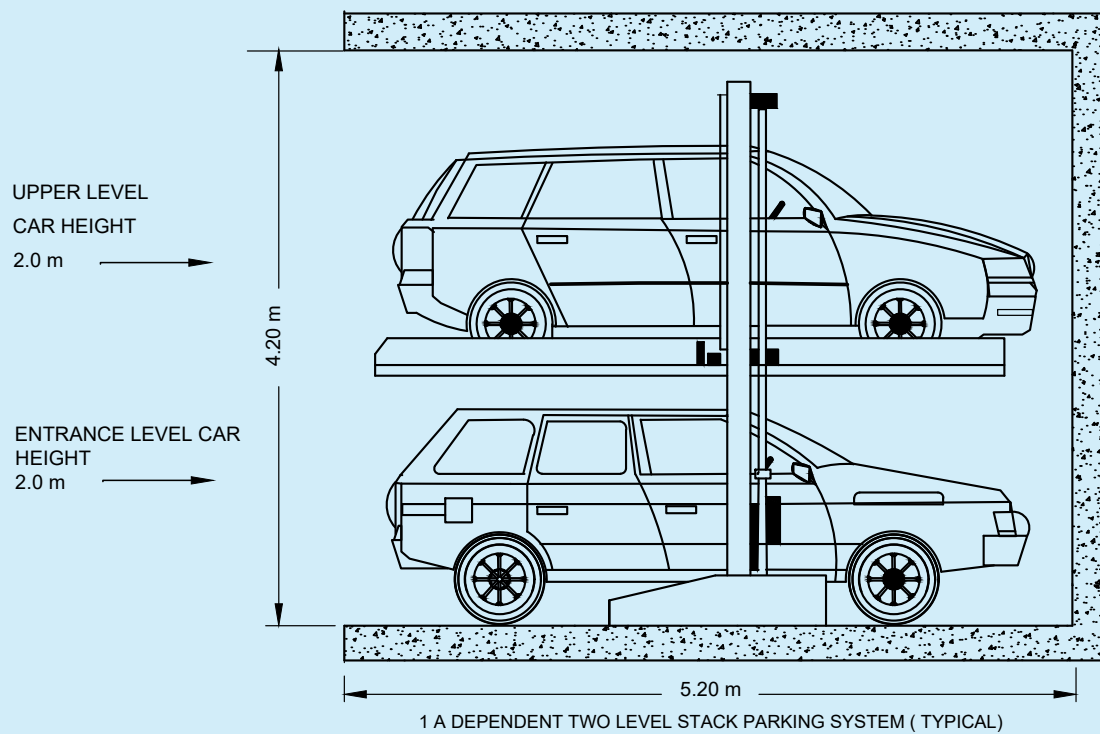
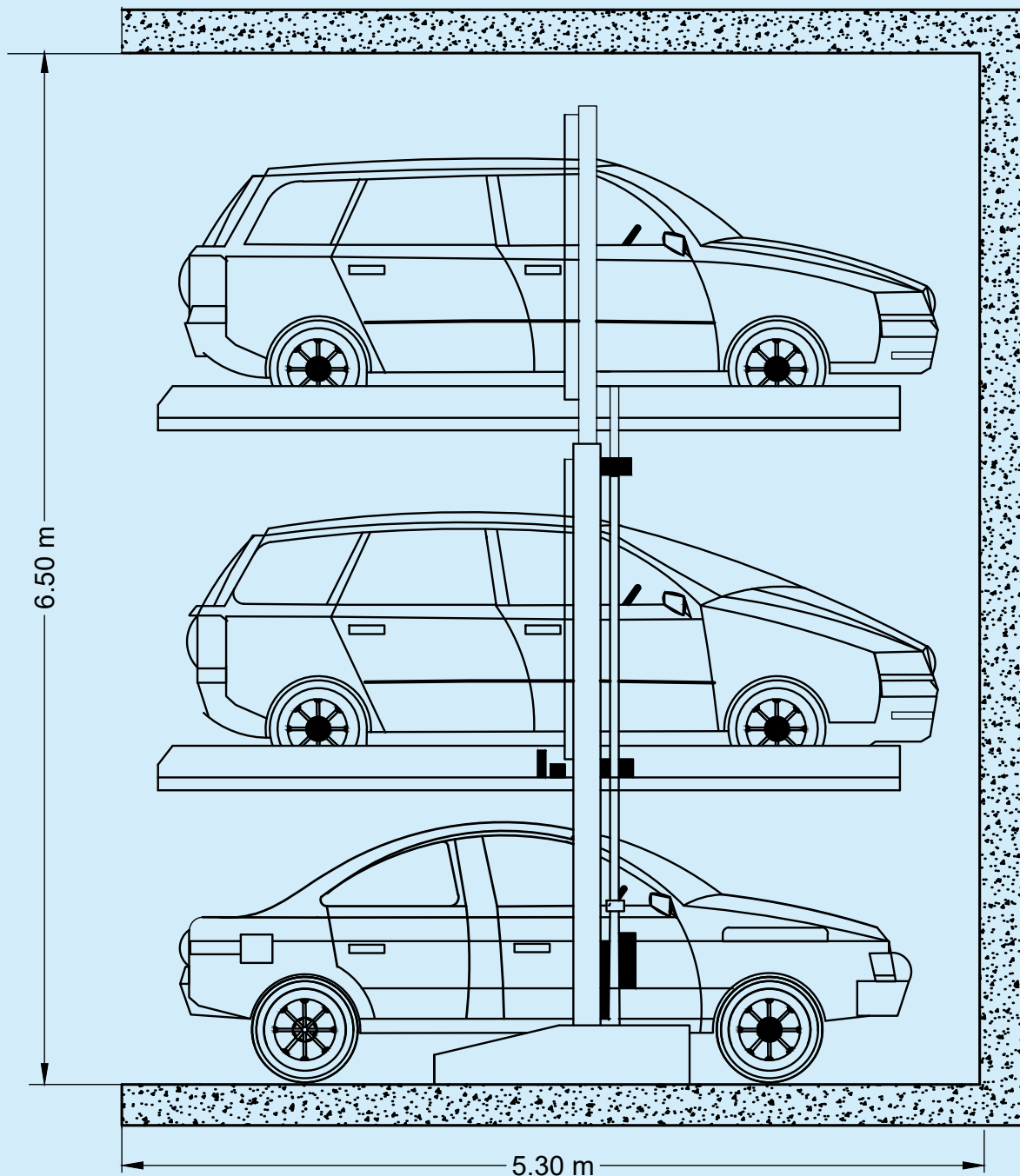
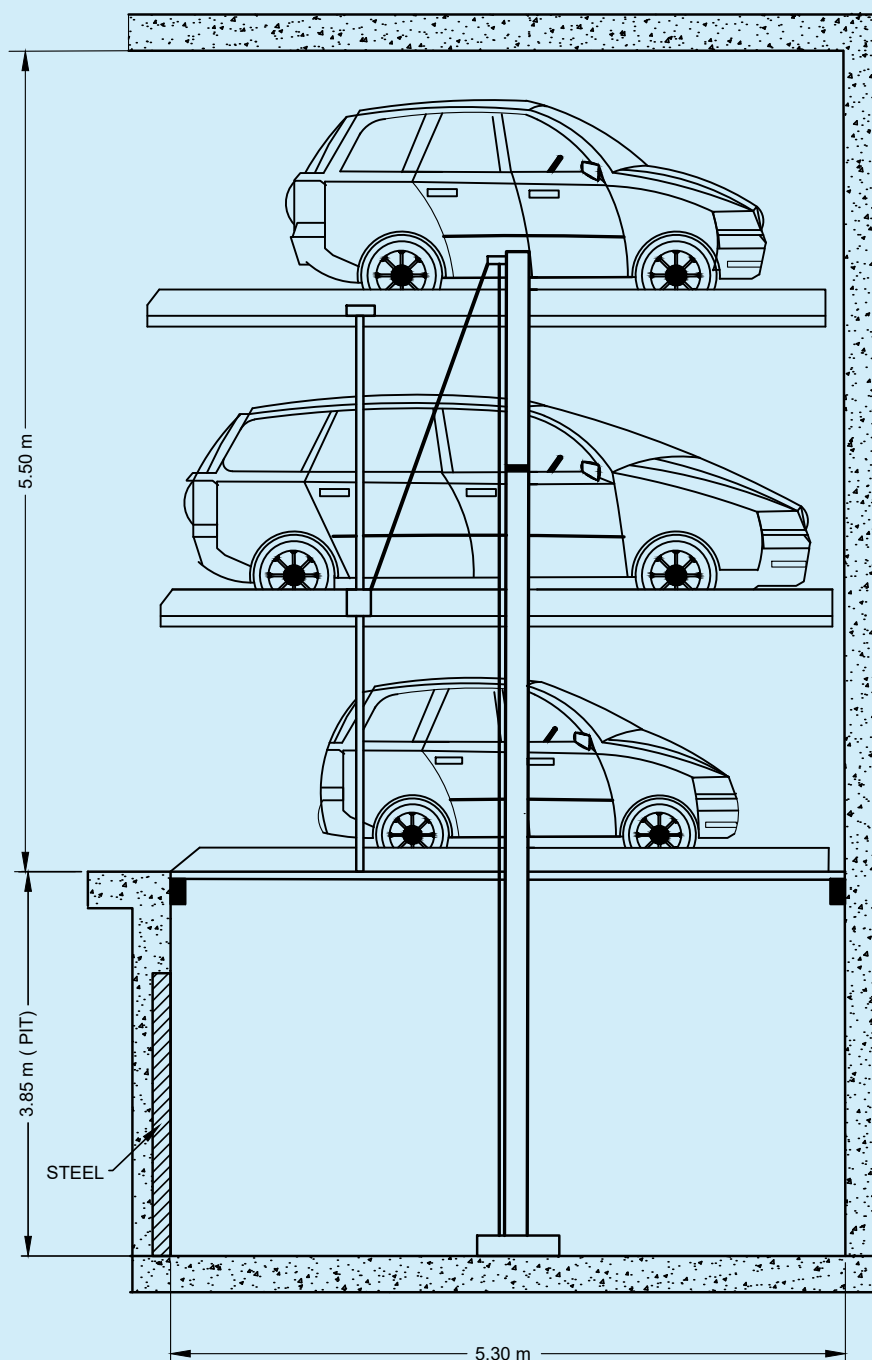


FIG. 1 STACK PARKING SYSTEM TWO LEVELS



2 A DEPENDENT THREE LEVEL STACK PARKING SYSTEM



2B INDEPENDENT THREE LEVEL PARKING SYSTEM

FIG. 2 STACK PARKING SYSTEM — THREE LEVELS

4.3 Puzzle Parking

Puzzle parking system is a type of semi-automatic parking system designed to maximize parking space efficiency by utilizing a combination of vertical and horizontal movements. The system mechanically arranges vehicles in a grid-like structure to optimize space utilization and minimize the need for manual parking. [Figure 3](#) to [Fig. 8](#) depicts various options of puzzle parking and the essential systems in them. The retrieval time for cars should be maintained within the time range specified in [Table 3](#) to ensure efficient operation and user convenience.

Table 3 Categorization of Retrieval Time (Clause 4.3)		
SI No.	Performance Level	Retrieval Time s
(1)	(2)	(3)
i)	Excellent	≤ 90
ii)	Very good	90 to 100
iii)	Good	100 to 120
iv)	Acceptable	120 to 140
v)	Poor	> 140
NOTE — These are under ideal conditions. However, it may vary depending upon the arrangement of parking systems.		

4.3.1 Key Features and Components

The features and components of the puzzle parking are described hereunder:

- a) *Vertical movement:*
 - 1) *Geared motor and machine* — The geared motor and machine for vertical movement shall be located at the side or rear of the system. Movement shall be facilitated using ropes or chains; and
 - 2) *Hydraulic cylinders* — Vertical movement using hydraulic cylinders may be located either vertically or horizontally, depending on the system design.
- b) *Horizontal movement* — Horizontal movement shall be controlled by a traction motor utilizing a chain sprocket mechanism.
- c) *Safety precautions:*
 - 1) *Entry to transfer rooms* — Access to transfer rooms should be controlled with sliding doors to ensure safety. Sensors may be used to halt the system in case of obstruction, but reliance solely on sensors may not fully prevent accidents; and
 - 2) *Flood sensors* — Flood sensors shall be installed in the pit to monitor water levels. A warning hooter shall be provided to alert any potential malfunction in the sump pump system.
- d) *Pit requirements:*
 - 1) *Waterproofing* — The pit shall be waterproofed to prevent water ingress. A sump shall be provided for efficient water removal;
 - 2) *Condition before installation* — The pit shall be handed over with a whitewash applied prior the installation of the puzzle parking system; and
 - 3) Entire area to be properly ventilated, illuminated and in hygienic conditions.
- e) *Maintenance and inspection* — Regular maintenance and inspection schedules shall be established to ensure all components are functioning correctly and system is in safe working condition.

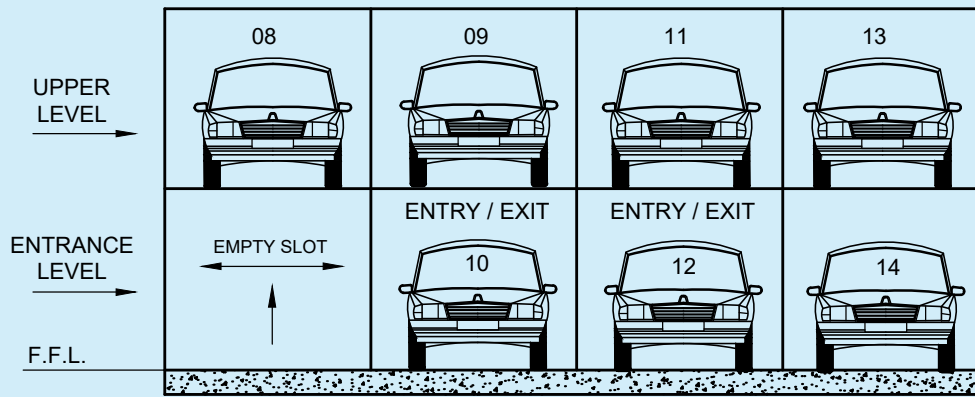
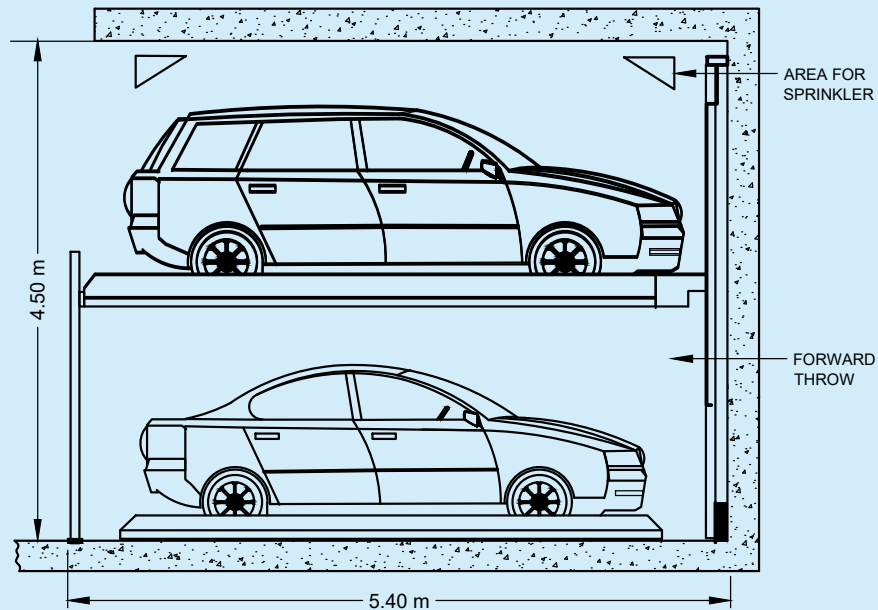
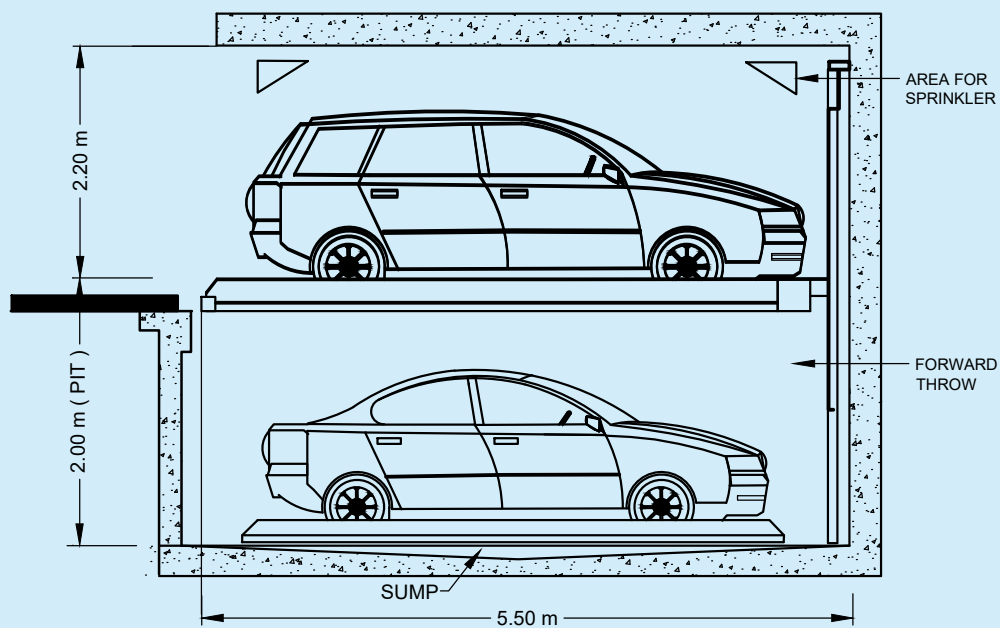


FIG. 3 SYSTEM SHOWING 2 LEVEL PARKING ABOVE GROUND



4A TWO LEVEL INDEPENDENT PUZZLE PARKING SYSTEM



4B TWO LEVEL INDEPENDENT PUZZLE PARKING SYSTEM WITH PIT

FIG.4 CROSS-SECTION SHOWING FREE SPACE FOR SERVICES LIKE FIRE SPRINKLERS

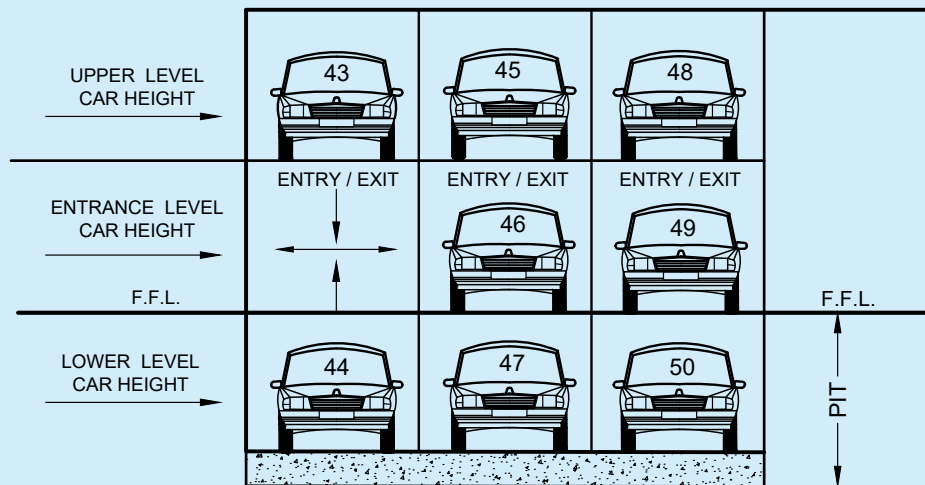


FIG. 5 SYSTEM SHOWING 3 LEVEL PUZZLE PARKING WITH PIT

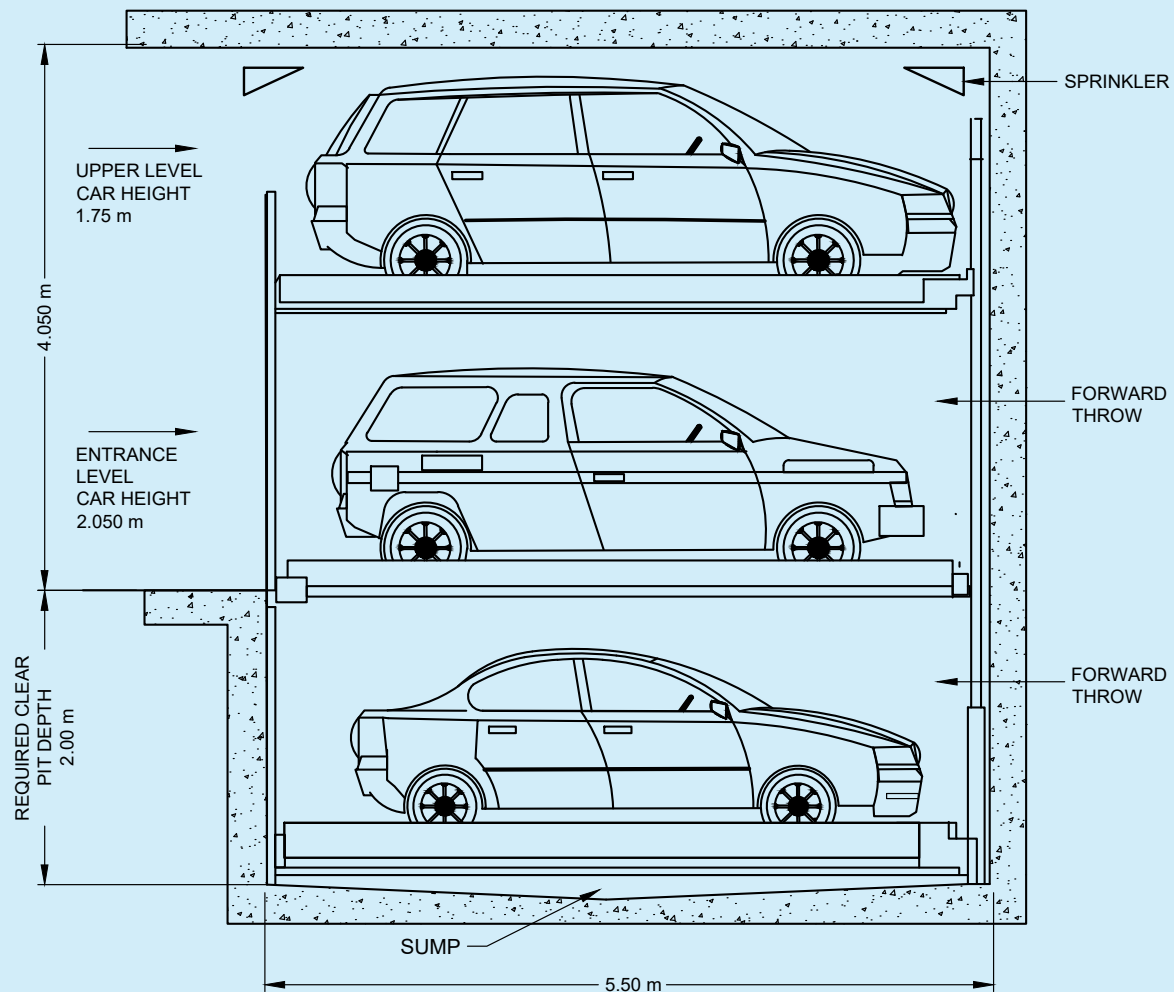


FIG. 6 CROSS-SECTION SHOWING FREE SPACE FOR SERVICES LIKE FIRE SPRINKLERS

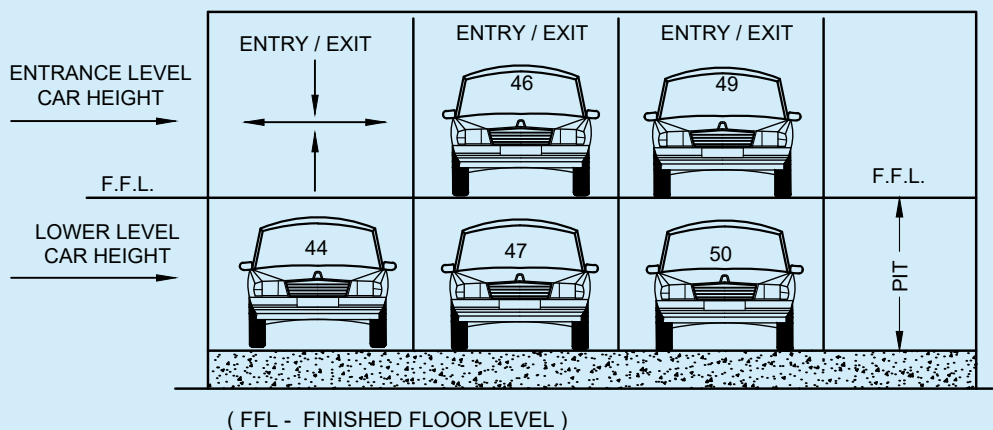


FIG. 7 ARRANGEMENT OF PUZZLE WITH PIT

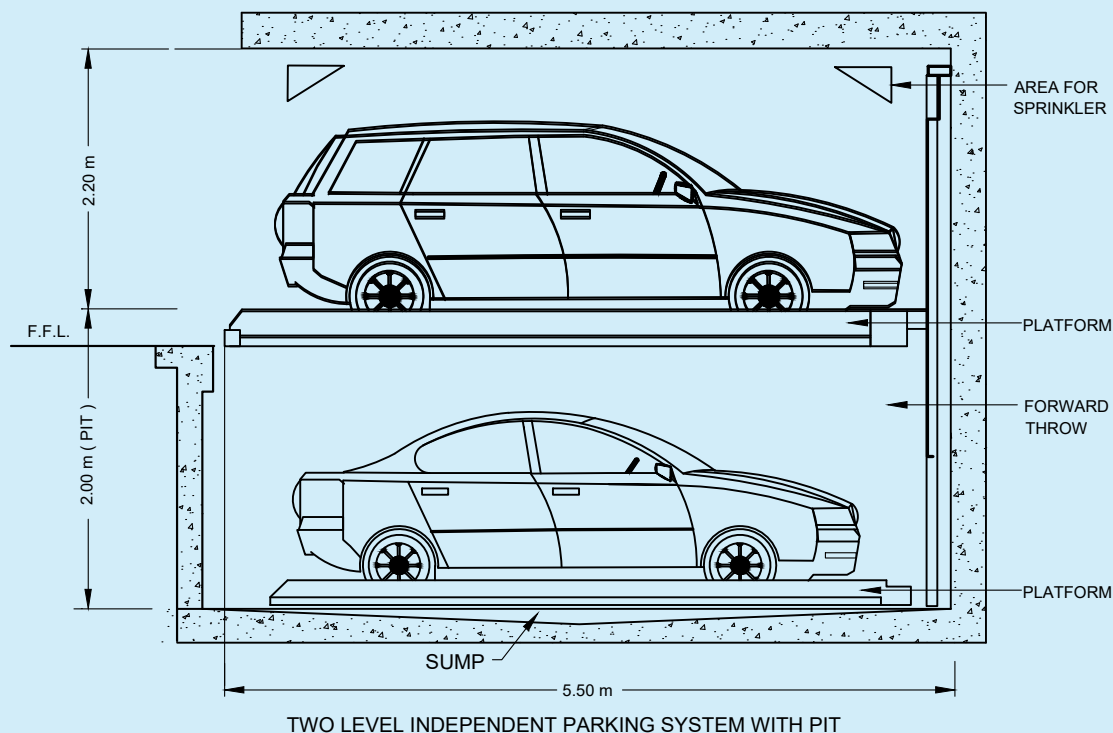


FIG. 8 CROSS-SECTION SHOWING FREE SPACE FOR SERVICES LIKE FIRE SPRINKLERS

4.4 Automatic Tower Parking System — Inside RCC Shaft and Free Standing Structure

Automatic tower parking system is a vertically oriented automated parking system designed to efficiently utilize space by stacking vehicles in a slender tower. The system combines mechanical and automation technologies to transport vehicles to the designated parking spots within the tower. Foundation shall be evaluated as per structural stability norms (*refer* Part C ‘Structural Design, Section 2 Soils and Foundations’ of this NBCS). Free standing tower, shall be enclosed within RCC framework for better stability.

4.4.1 Key Features and Components

The following shall be applicable:

- System configuration* — The system features a central lift with various configurations including 1/1, 1/2, 1/3, 2/2, 2/3 and 3/3 arrangements, as illustrated in the system's design, see [Fig. 9](#).
- The lift facilitates the vertical transportation of vehicles to different levels within the tower.
- Entrance room* — Its location shall be positioned for convenient access to vehicles entering the parking tower thereby allowing smooth entry and exit of vehicles, facilitating the parking process.
- Car handling and turning* — Turntables or turning devices to rotate vehicles to ensure proper alignment for exit direction.
- Car hold area* — There shall be adequate space in front of the entrance room to accommodate vehicles during the parking and retrieval process.
- Car handling capacity* — The system should be able to handle approximately 25 cars to 30 cars per hour, with a full cycle of filling or evacuating the tower completed in about 3 h.
- Parking capacity* — The tower should have a capacity to accommodate up to 90 cars for optimum retrieval time.

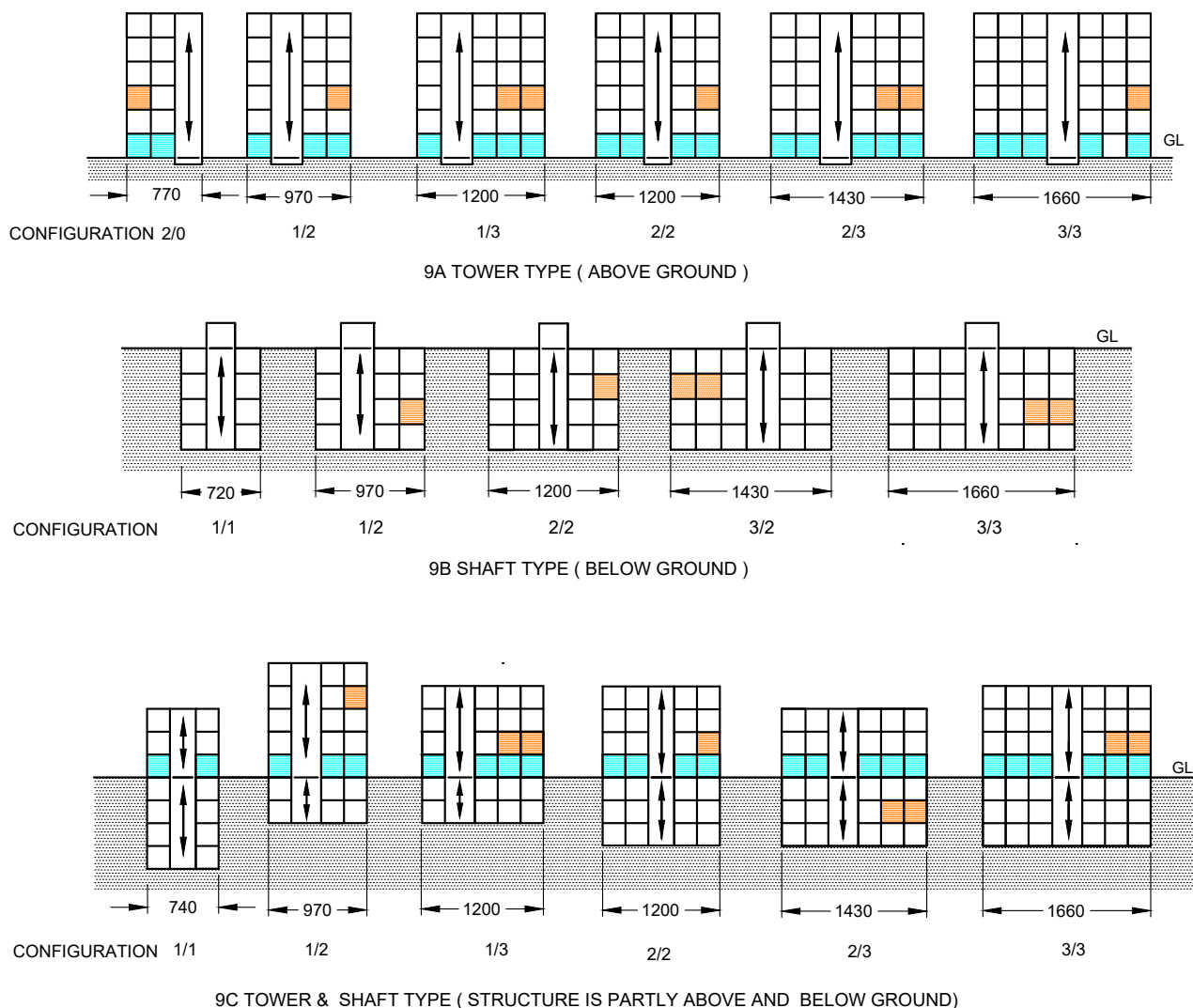


FIG. 9 VARIOUS ARRANGEMENTS DEPICTING THE SECTIONS/SLOTS FOR PARKING

- h) *Lift locking mechanism* — It should be ensured that the lift platform is securely locked before any vehicle transfer or retrieval operation is performed.
- j) *Platform guide rails* — Closed-type guide rails shall be employed to provide stability and ensure safe vehicle movement within the tower.
- k) *Platform locking mechanism* — Each platform guide rails should have built-in feature which includes an integrated locking mechanism to secure vehicles in place.
- m) *Tower structure stability* — Proper anchoring of the tower structure should be ensured for stability and safety during operation.
- n) *Illumination inside shaft* — Light points shall be installed every 6.0 m, controlled by 3-phase 16A switches to ensure adequate visibility inside the shaft.
- p) *Sound and vibration control* — The vertical structure should be connected to the RCC with sound isolation pads to mitigate noise and vibration transmission.
- q) *Shaft size* — The shaft dimensions shall be designed to accommodate large-size cars comfortably.
- r) *Finish of the structure* — The structure should be finished with hot-dip galvanization and is fire-resistant for up to 120 min to ensure durability and safety.
- s) *Entrance door* — Safety doors shall be able to withstand a force of 300 N applied perpendicular to the surface without permanent deformation or impairment of functionality. The force should be distributed evenly over an area of 500 mm².

4.5 Longitudinal Parking System

This is a parking system having lift at centre with front and back arrangement when there is more space available in length than in width. This system is thus considered and designed to maximize parking efficiency in narrow spaces, particularly in narrow broad strips between buildings, such as the area between the set-back lines and the building.

4.5.1 Key Features and Components

The following shall be applicable:

- a) *Vehicle accommodation:*
 - 1) *Car height and length* — The system shall be designed to accommodate various car heights and lengths, ensuring flexibility for different vehicle types; and
 - 2) *Capacity* — The slim parking system shall be capable of parking up to 30 cars, depending on the specific design and available space.
- b) *Design efficiency:*
 - 1) *Floor plan* — The system shall be designed with a very small floor plan area and a slim construction, maximizing the use of narrow spaces; and
 - 2) *Ramps and driveways* — No space-intensive ramps or driveways shall be required for this system, allowing for a more compact and efficient design.
- c) *Configurations and arrangements:*
 - 1) *Configurations* — The system shall offer configurations of 1/1 or 2/2, which are particularly suitable for the side margins of buildings;
 - 2) *Above or below ground* — The system should allow for both above-ground and below-ground arrangements, depending on the site's requirements (see [Fig. 10](#) and [Fig. 11](#));
 - 3) *Lengthwise transportation* — Vehicles shall be transported lengthwise with restricted levels to control the number of vehicles within the system; and

- 4) *Capacity* — The longitudinal parking system shall have a capacity of around 30 cars, accommodating the needs of various building types.

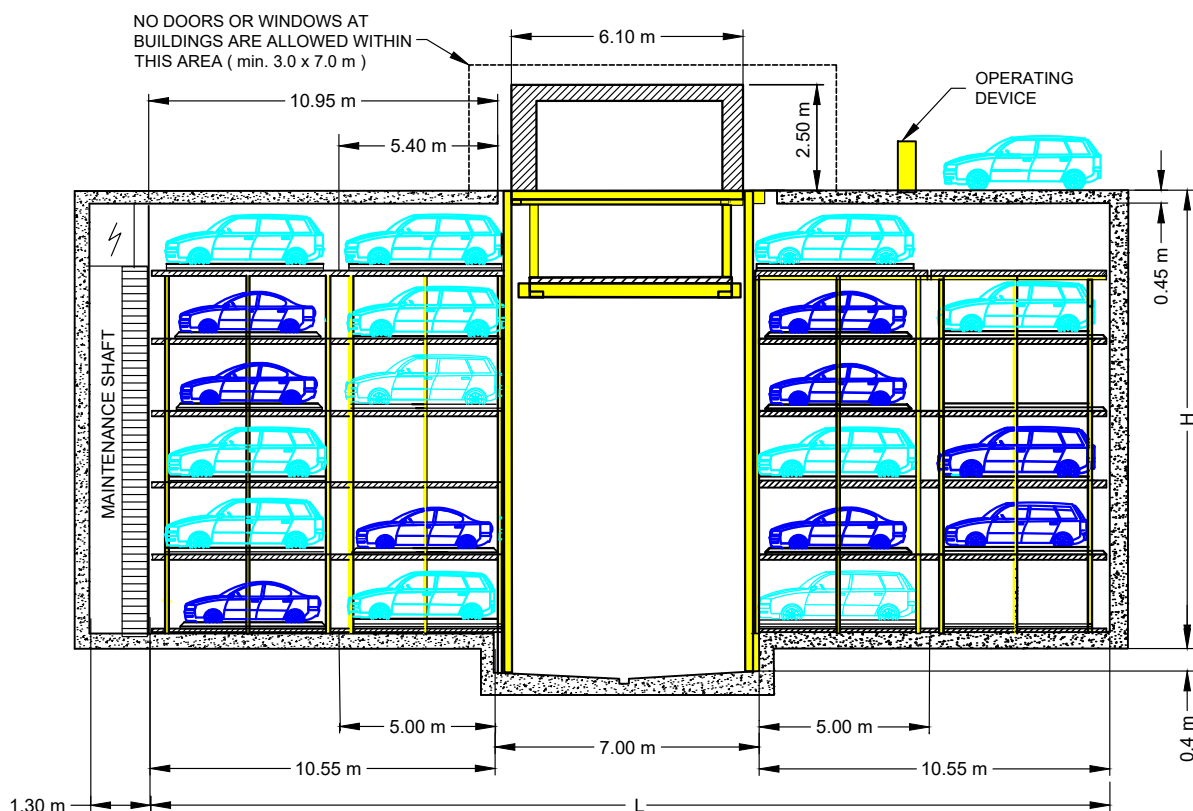


FIG. 10 ARRANGEMENT BELOW GROUND

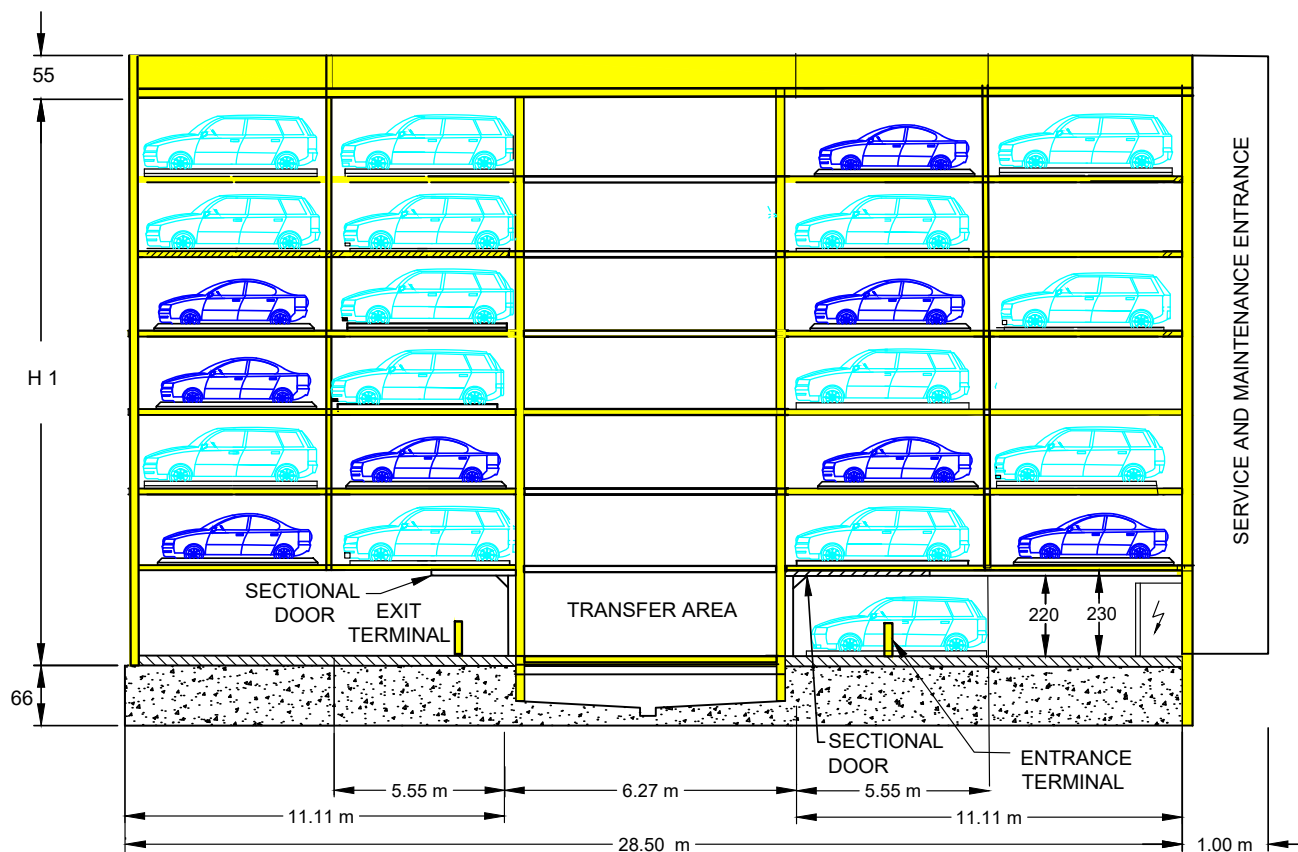


FIG. 11 ARRANGEMENT ABOVE GROUND

4.6 Chess Parking System

The chess parking system, also referred to as the level parker, shall be employed in scenarios where vertical space is optimized for vehicle storage. This system features a vertical lift serving multiple levels, with vehicles being horizontally circulated in a loop and transported to an entrance room or transfer area. See [Fig. 12](#) and [Fig. 13](#) for details of typical arrangement of the vehicles and the operation sequence.

4.6.1 Key Features and Components

- a) *Vertical lift* — The system shall be equipped with a vertical lift that serves various levels, enabling the vertical movement of vehicles within the parking structure;
- b) *Horizontal circulation* — Vehicles shall be circulated horizontally in a loop, allowing them to be transported efficiently to the designated entrance room or transfer area;
- c) *Synchronized movement* — All platforms within the loop shall move simultaneously in synchronization, ensuring the coordinated transportation of vehicles;
- d) *Performance considerations*:
 - 1) *Speed* — The chess parking system should be noted for its slower operational speed compared to other parking systems, as the synchronized movement of all platforms can reduce efficiency; and
 - 2) *Power requirements* — The system shall require more power due to the simultaneous movement of all platforms in the loop.
- e) *Design efficiency* — The following parameters to be considered for efficiency:
 - 1) *Space utilization* — The system shall be designed to maximize vertical space utilization, making it suitable for locations with single entry and exit with efficient horizontal space utilization; and
 - 2) *Loop configuration* — The loop configuration should be designed to ensure smooth and continuous movement of vehicles within the system.

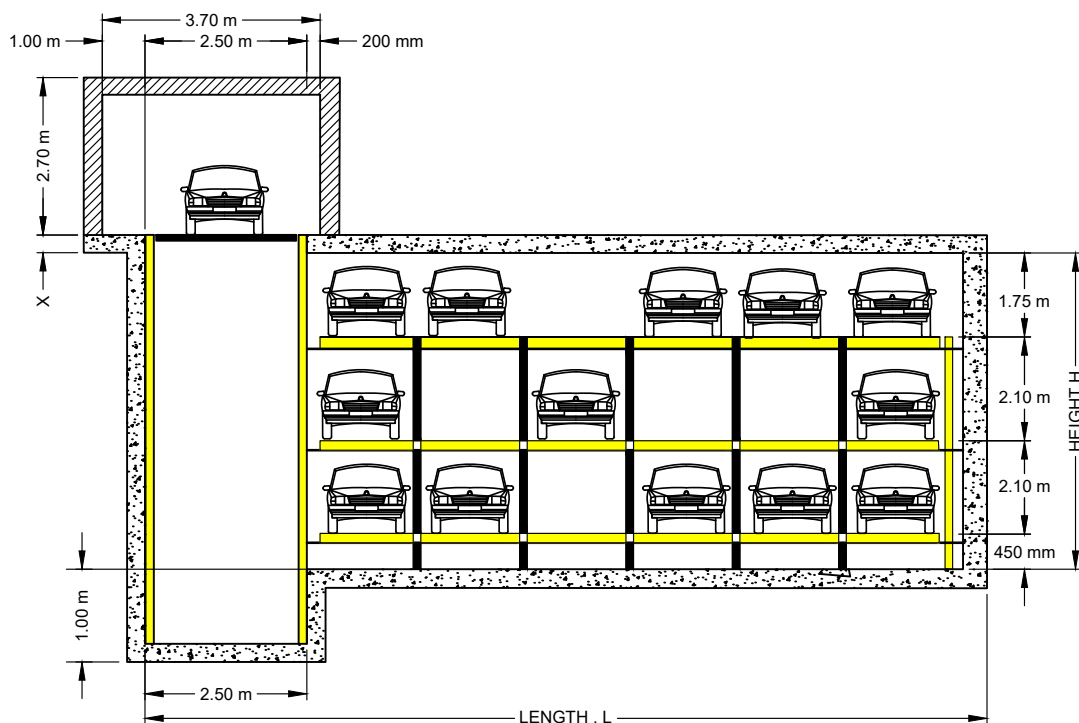


FIG. 12 CROSS SECTION SHOWING LIFT MOVEMENT AND PLATFORMS ARRANGEMENT

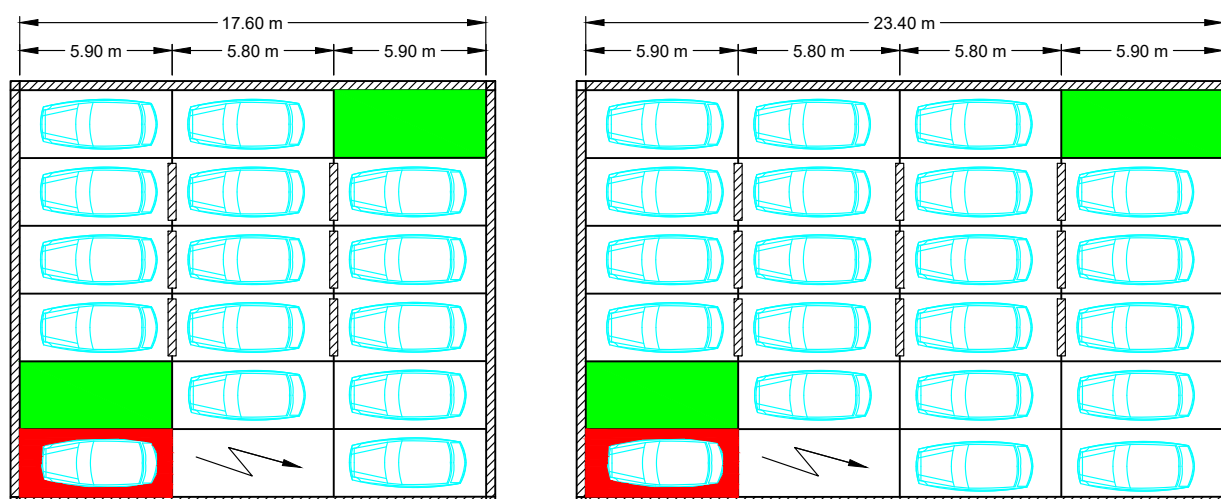


FIG. 13 ARRANGEMENT SHOWING MULTI-ROW ARRANGEMENT IN PLAN

4.7 Robotic Parking System

The robotic parking system is an advanced automated vehicle storage and retrieval system that shall utilize robotics technology to move vehicles efficiently within a parking facility. This system operates on the X, Y and Z axes, either with or without pallets, using steel or RCC structures. The system should ensure precise and swift car placement and retrieval, optimizing both space and time. [Figure 14](#), [Fig. 15](#) and [Fig. 16](#) depict the typical arrangement in robotic parking system.

4.7.1 With Pallets

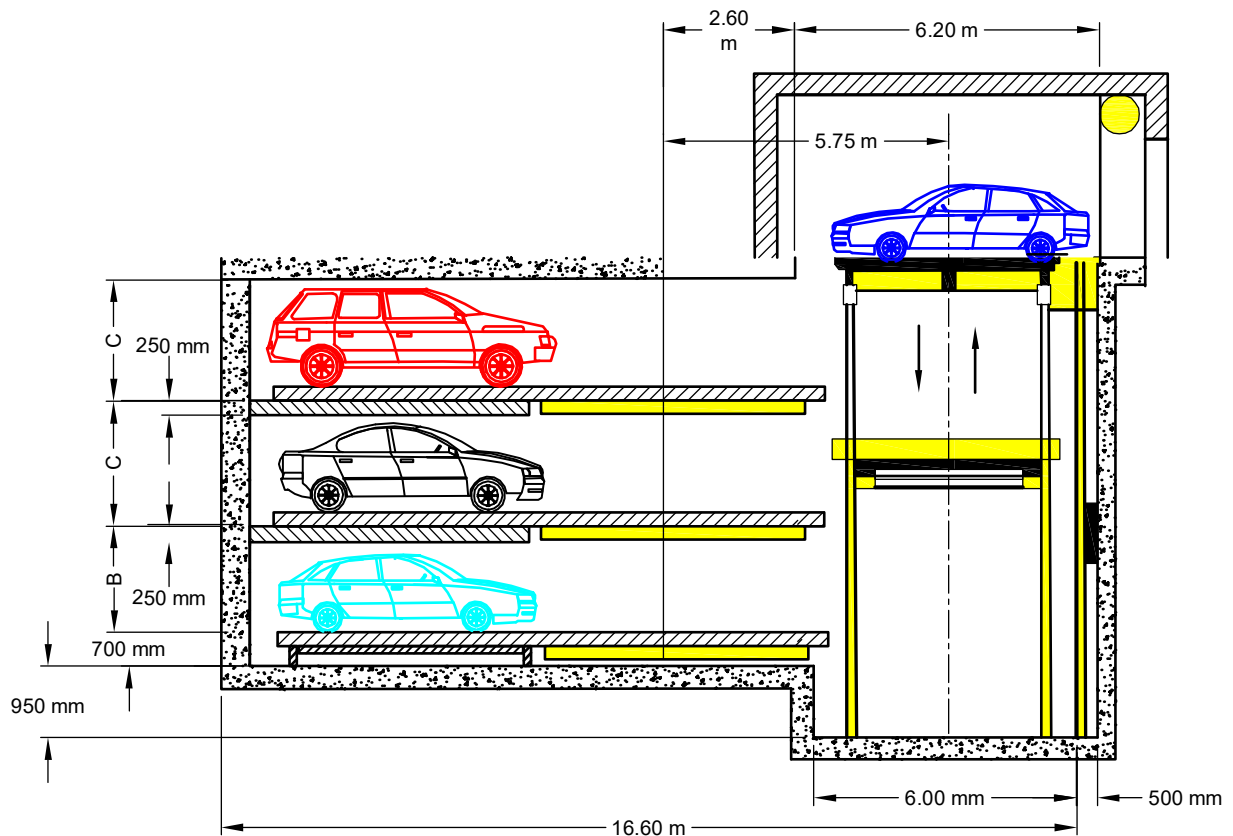
The following are the key features of this system:

- Parking arrangement* — Vehicles shall be parked on pallets, which shall remain engaged from the time of parking until retrieval. The system may be arranged above or below the ground, depending on the location of the entrance room;
- Structural requirements* — Steel structures used in this system shall be galvanized to ensure durability and protection against corrosion; and
- Installation flexibility* — This system shall be possible to install above ground, below ground, or with a combination of both, depending on site-specific requirements and design considerations.

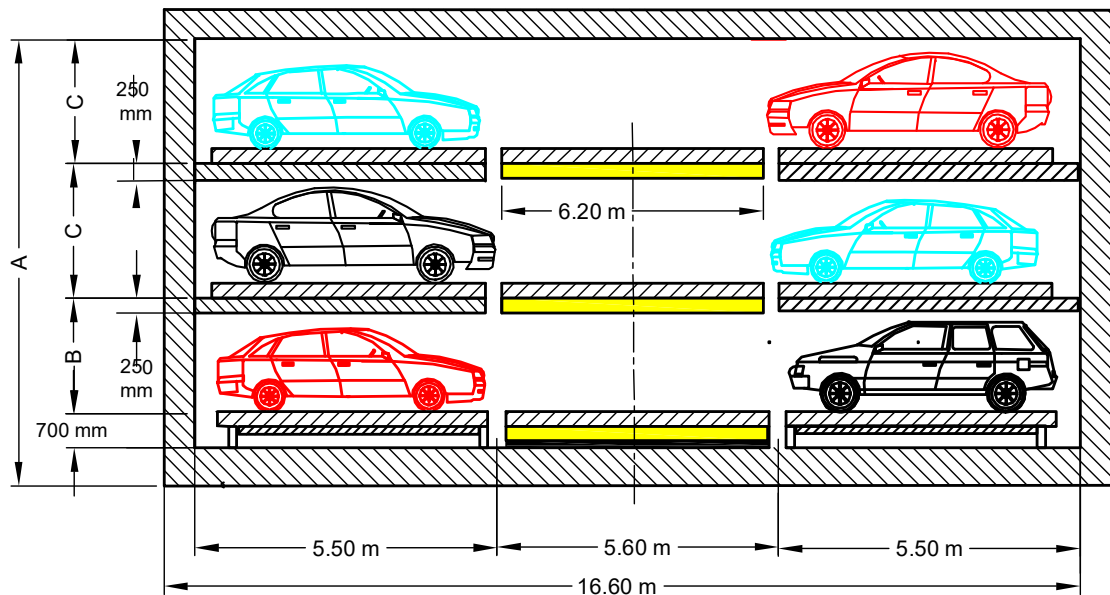
4.7.2 Without Pallets

The following are the key features of this system:

- System movement* — The system shall operate on the X, Y and Z axes, with vehicles parked directly on RCC slabs or steel floor without the use of pallets. The system shall be faster in service as the car is transferred by a lift to the robot, utilizing a shuttle and dolly mechanism for movement;
- Noise reduction* — Parking vehicles on RCC slabs should help reduce noise levels compared to pallet-based systems, providing a quieter operational environment;
- Fire safety* — The use of RCC slabs between levels shall aid in controlling the spread of fire from one level to another, enhancing the overall fire safety of the system; and
- Installation flexibility* — This system shall be possible to install above ground, below ground or with a combination of both, depending on site-specific requirements and design considerations.

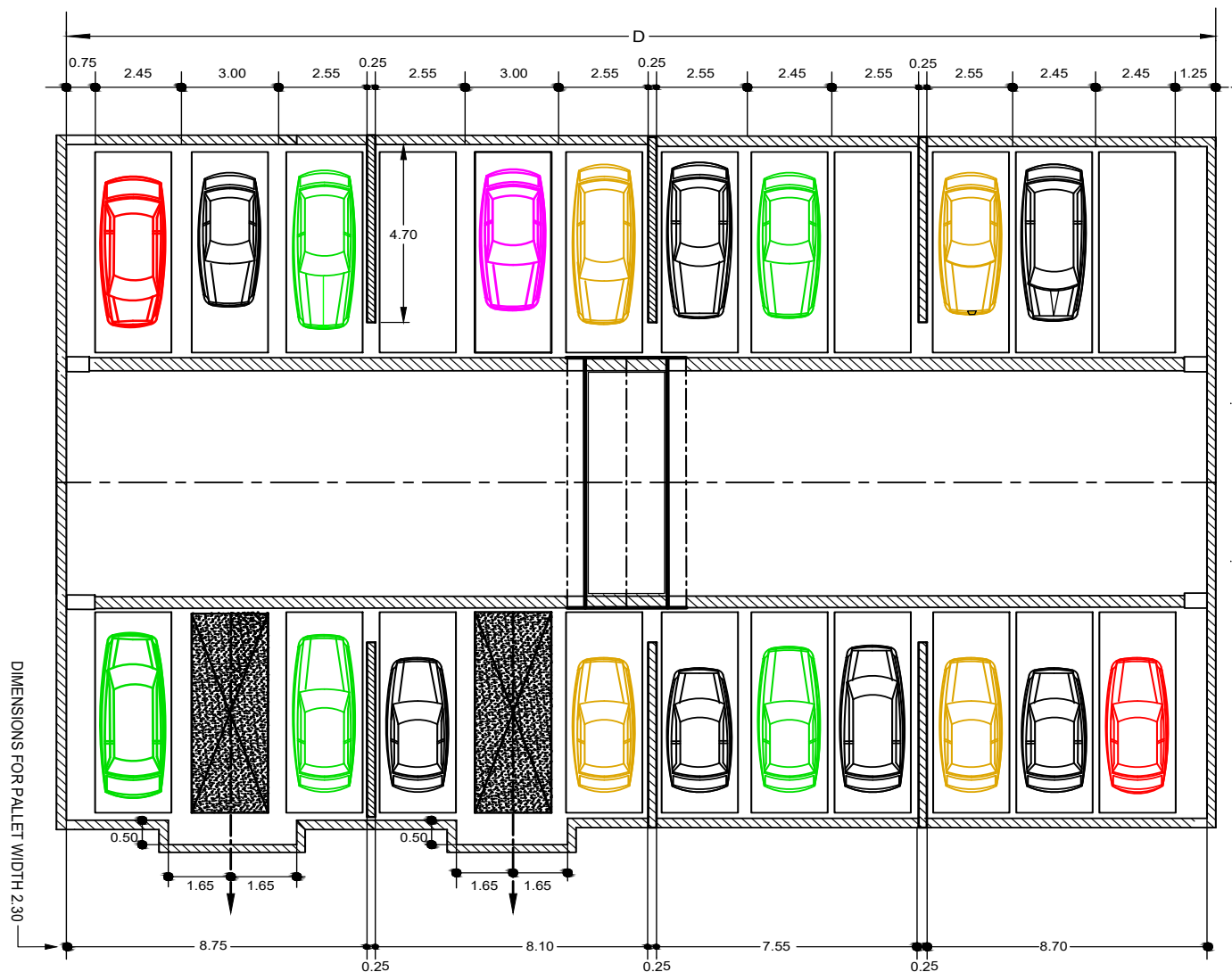


14 A WITH PALLETS



14 B WITHOUT PALLETS

FIG.14 ROBOTIC PARKING SYSTEM CROSS-SECTIONAL VIEW

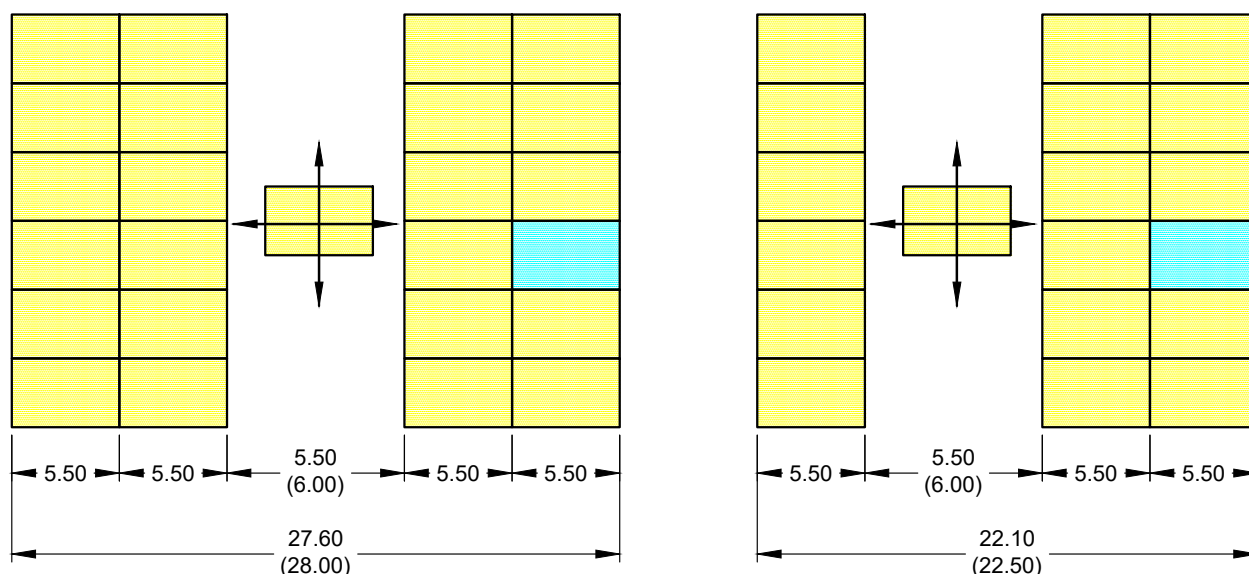


All dimensions in metres.

NOTE — For the control unit, space (at least length 2.40 m × width 1.60 m × height 2.20 m) must be available near the transfer area.

FIG. 15 ROBOTIC PARKING SYSTEM PLAN

MULTI-ROW ARRANGEMENT



All dimensions in metres.

NOTE — The multi-row arrangement allows an optimum utilization of the available space and/or land area and saves construction costs, particularly with the shaft variant.

(DIMENSIONS IN BRACKETS ARE FOR STORAGE AND RETRIEVAL
UNIT WITH TURNING DEVICE)

FIG. 16 POSSIBILITY OF MULTI-ROW ARRANGEMENT

4.8 Rotary Parking

The rotary parking system is a space-efficient vehicle storage solution designed for locations where ground space is limited, but vertical space can be utilized. This system operates by rotating vehicles (anticlockwise or clockwise) in a vertical loop, allowing multiple cars to be parked in a compact footprint.

4.8.1 Key Features

The following are important features of rotary parking:

- Anchoring and structure** — The rotary parking system shall be a free-standing type, securely anchored to the ground. This ensures stability and safe operation, even in areas with limited space or challenging terrain;
- Entry and exit** — The system shall provide entry and exit at the ground level for easy access by drivers. To ensure safety, the entry and exit points shall be protected by an entrance door or gate that automatically opens and closes as vehicles enter or exit the system;
- Rotational movement** — The parking platforms within the system shall rotate in both clockwise and anti-clockwise directions, allowing for flexible vehicle placement and retrieval. This bi-directional movement enhances the system's efficiency, enabling quicker access to parked vehicles; and
- Vehicle capacity and variations:**
 - Capacity options** — The rotary parking system may be available in various configurations, typically accommodating up to 20 cars, among other possible variations. These options allow for customization based on the specific needs and space constraints of the installation site; and
 - Vertical loop design** — Vehicles are rotated vertically in a loop, with each vehicle occupying a designated platform. The system shall be designed to minimize the time required for vehicle retrieval, ensuring that any vehicle within the system can be accessed quickly and efficiently.

4.9 Sliding Platforms

These parking platforms are used to slide the car sideways or in longitudinal directions. Rails are placed directly on the flooring. Cars are parked on the platform and moved sideways.

5 PLANNING AND DESIGN GUIDELINES FOR PARKING SYSTEMS

5.1 Design Criteria for Power-Operated Parking Equipment

When designing or specifying power-operated parking systems, several critical design criteria shall be taken into consideration to ensure the safety, durability and efficiency of the equipment. These criteria are essential to accommodate various types of vehicles, ensuring smooth operation and long service life of the parking system.

5.2 Nominal Load Capacity

The nominal load capacity of the parking system should be designed to handle vehicles with a minimum weight of 2 000 kg and not more than 3 500 kg. This range is typical for most passenger vehicles and ensures that the system can safely accommodate both lighter and heavier cars within this spectrum.

5.3 Wheelset Load Distribution

The load on the wheelsets should be distributed evenly, with 50 percent of the nominal load allocated to the front and rear wheelsets. This balanced distribution is crucial for maintaining stability and preventing uneven stress on the parking platform.

5.4 Individual Wheel Loads

Each wheel should bear approximately 25 percent of the nominal load, ensuring that the weight is evenly distributed across all four wheels. This distribution minimizes the risk of overloading any single wheel, which could lead to mechanical failure or uneven wear on the system.

5.5 Wheelbase Specification

The design should account for a wheelbase of 2.90 m, measured from the centre of the front wheel to the centre of the rear wheel. This specification ensures compatibility with a wide range of vehicle sizes and provides adequate spacing for secure parking.

5.6 Track Gauge Requirements

The track gauge, or the distance between the wheels in width, should range from a minimum of 1.5 m to a maximum of 2.10 m. This range accommodates different vehicle widths while ensuring that the parking system can handle various car models without compromising stability.

5.7 Tyre Footprint Consideration

The tyre footprint, or the area of the tyre that comes into contact with the parking platform, should measure approximately 0.18 m by 0.18 m. This specification helps distribute the vehicle's weight evenly across the platform, reducing the risk of damage to the system.

5.8 Forces in Direction of Travel

The design should account for forces in the direction of travel equivalent to 1/7 of the nominal load in kilonewton (kN). This consideration is critical to ensure that the system can handle the dynamic forces exerted when vehicles move on or off the platform.

5.9 Transverse Forces

Forces transverse to the direction of travel should be considered as 1/20 of the nominal load in kilonewton (kN). While this does not exclude other design criteria, it ensures that the system can withstand lateral forces, such as those generated by turning or shifting vehicles.

5.10 Load Considerations for Parking Equipment

When designing or specifying parking equipment, it is essential to consider the worst-case loads that may occur. These include loads from the stationary position of the vehicle on the platform as well as dynamic loads during transportation or when the vehicle is driving onto the platform. These considerations help ensure that the system is robust enough to handle peak stresses without failure.

5.11 Equipment Life Expectancy

The parking system should be designed for a minimum operational life of 20 years. This long-term durability requirement ensures that the system remains functional and reliable over an extended period.

5.12 Operation Cycle Requirements

The system should be capable of performing at least one parking-in and one parking-out operation per day, amounting to two cycles per platform or space. This cycle rate should be sustainable every day of the year, ensuring the system can meet the daily demands of a busy parking facility.

6 ARRANGEMENT AND PLANNING DIMENSIONS

6.1 Grid and Rows

When planning and designing parking systems, it is essential to define the maximum number of rows and grids to ensure optimal space utilization and system efficiency. The arrangement of grids and rows in a parking module impacts the capacity and operational effectiveness of the parking system (see [Fig. 17](#)).

6.1.1 Maximum Number of Rows

The parking system shall not exceed a maximum of four rows to maintain manageable distances for vehicle retrieval and to ensure safety and accessibility within the parking structure.

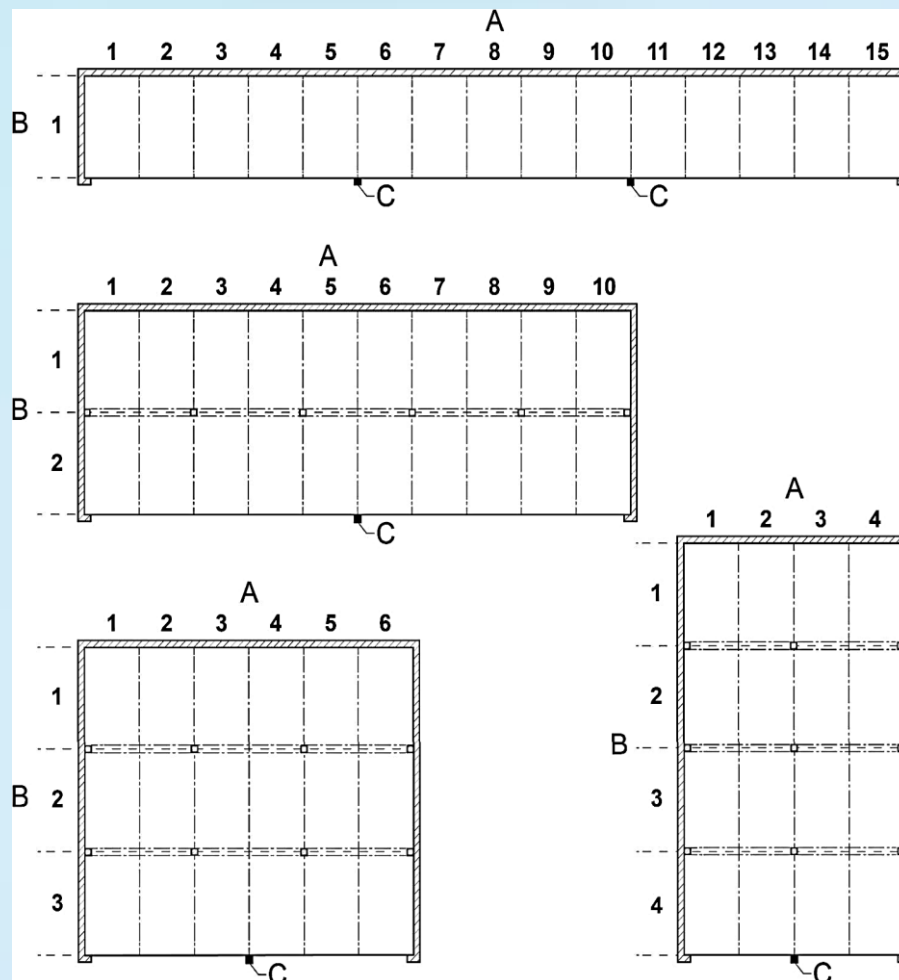
6.1.2 Maximum Number of Grids

The maximum number of grids shall be as given below:

- a) *One-row system* — A module with a maximum of 15 grids should be utilized. However, it is advisable to limit it to no more than 10 grids to maintain ease of access and operational efficiency;
- b) *Two-row system* — The module should contain no more than 10 grids to ensure that vehicles can be parked and retrieved with minimal obstruction;
- c) *Three-row system* — The system should be designed with a maximum of 6 grids to allow for sufficient manoeuvring space within the parking structure; and
- d) *Four-row system* — A limit of 4 grids per module should be adhered to for optimal operation and safety.

6.1.3 Control Panel Requirements

An additional control panel shall be installed after every maximum of 5 grids on either the left or right side of the parking system as shown in the [Fig. 17](#). This requirement ensures that users can conveniently operate the system and enhances the safety and usability of the parking setup.



Key

- Grid Horizontal parking spaces
- Rows Back-to-back parking arrangement
- C Indicates control panel

FIG. 17 GRID AND ROWS IN PARKING SYSTEM

6.2 Arrangement and Planning

The suggested guidance on the arrangement and planning of the parking requirements are given above; see also [Fig. 17](#).

6.2.1 Open Parking Spaces

Open-to-sky parking areas may be permitted in side margins or in the surrounding open spaces of a building.

6.2.2 Closed Parking Spaces

Parking areas within the building line, in basements or on lower ground or podium levels inside the building.

6.2.3 Automatic Parking Spaces

Unmanned areas where vehicles are transported from the entrance room by means of mechanical systems such as conveyors, lifts, etc and transported back to the exit room. The parking area may be clearly demarcated with a

reflective paint and driveways shall be connected to all the parking areas.

6.2.4 Car hold area

The space between the entrance from the connecting road to the actual parking area shall be considered as the storage space or car hold area.

6.2.5 Entrance and exit room

This room should be properly ventilated and illuminated.

6.2.6 Width of Parking Area

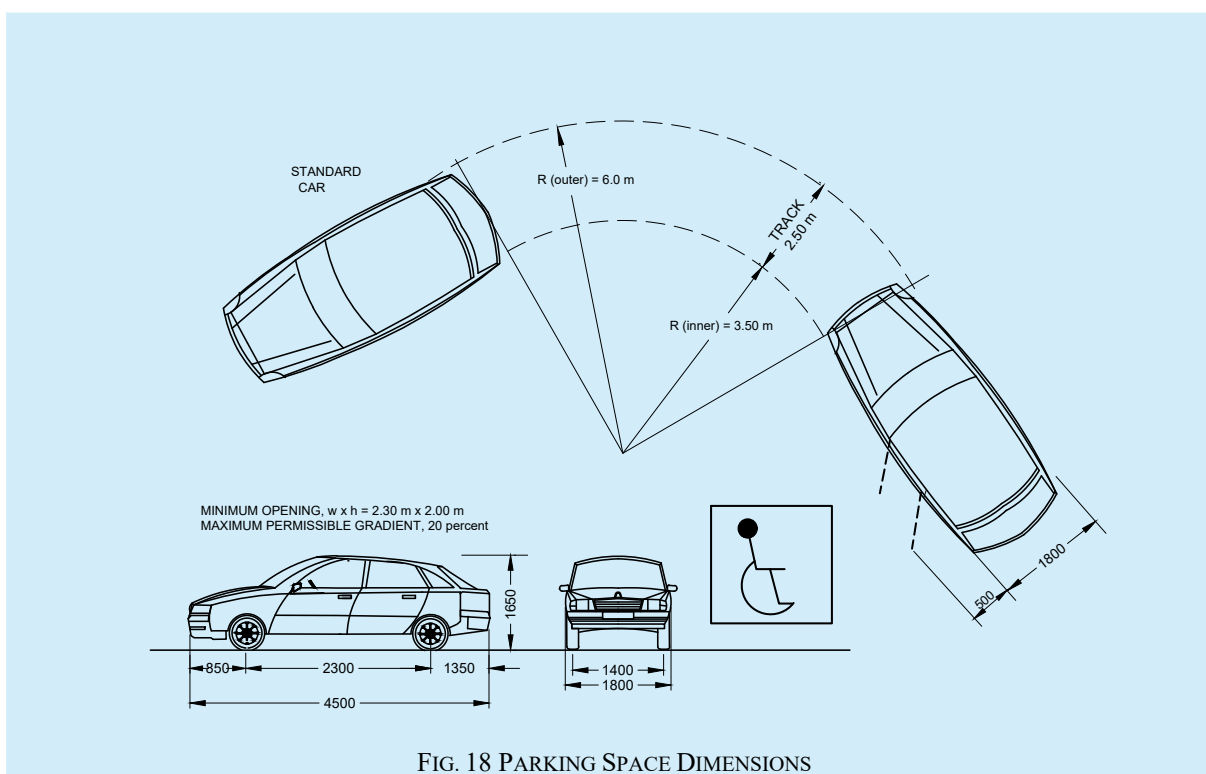
The clear width of a parking space shall be not less than 2.7 m for small cars, such as sedans and hatchbacks. The clear width of a parking space shall be not less than 2.9 m for large cars, such as sport utility vehicles (SUVs). Typical layouts of parking spaces are illustrated in [Fig. 20](#).

6.2.7 Ramps and Driveways

For curvatures and circular ramps or driveways, the inner radius shall be 5.0 m. Wider driveways with an inner radius of approximately 10.0 m should be considered for safety purposes (see [Fig. 18](#)).

6.2.8 Pedestrian Walkway

In large parking areas, a separate pedestrian walkway of 0.8 m next to driveways shall be considered, which should be raised above the main parking level or marked permanently and illuminated for visibility and safety.



6.2.9 Slope of the Ramps

The inclination of ramps shall not exceed 15 percent or a 1 : 8 ratio. The recommended slope is 1 : 10. The width shall not be less than 2.75 m, and at least 3.5 m in circular or curved areas for single-lane traffic. The joint at the start and end of the ramp shall have a slope of 3 percent with proper curvature to ensure smooth travel and prevent vehicles from hitting the bottom or nose of the car. For public parking, a slope not exceeding 10 percent (1 : 10 ratio) should be considered. An extension of at least 3.0 m in front of the ramp shall be provided (see [Fig. 19](#)).

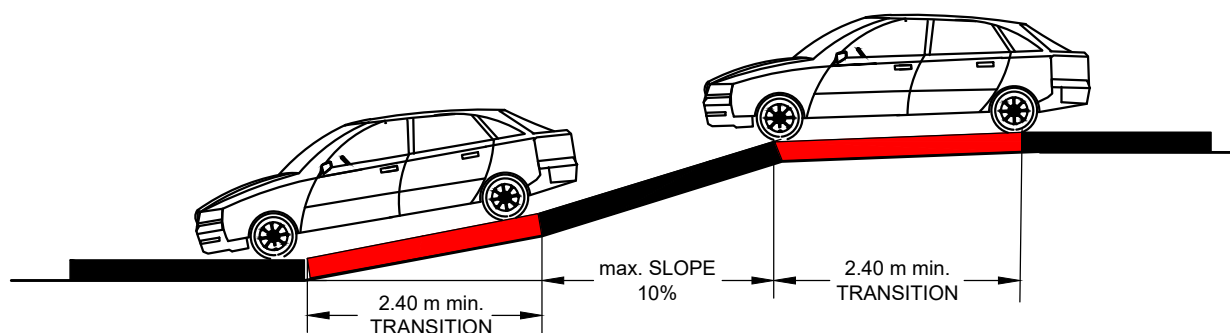


FIG. 19 TRANSITIONAL RAMP

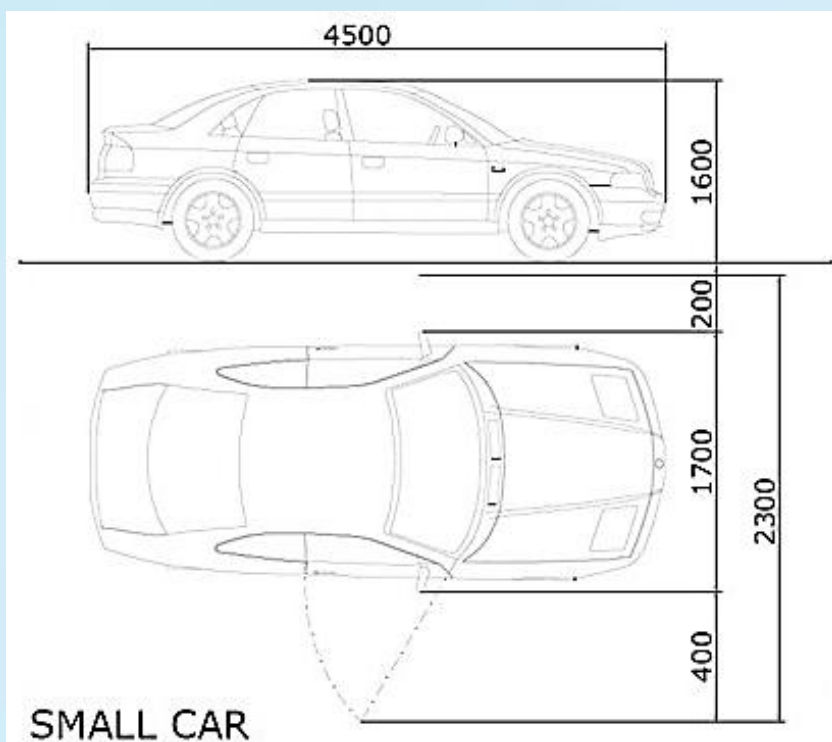
6.2.10 Size of Parking Space and Women Parking Space

The length of car parks shall be 5.0 m, and parking arranged alongside driveways shall be at least 6.0 m wide. The entire parking space may be designed as a column-less structure for safety.

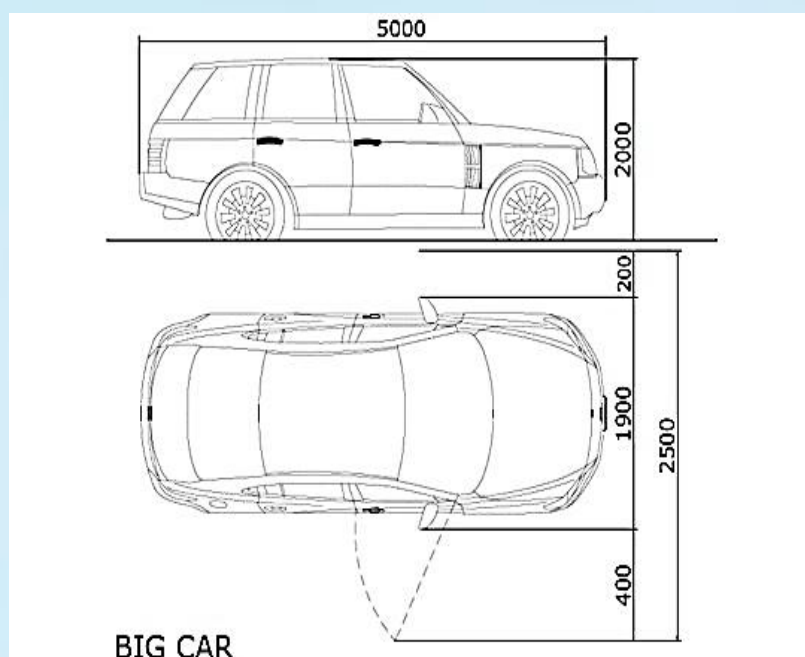
6.2.11 Parking Space

The minimum width for parking spaces shall be 2.3 m for small cars (see [Fig. 20A](#)) and 2.5 m for big cars (see [Fig. 20B](#)). For power-operated mechanized parking systems, at least 0.2 m of space on either side shall be added to the above basic widths. See also [Fig. 20](#):

- Small cars* — 2 300 mm × 5 000 mm (clear space + additional space of 400 mm for mechanized clearance);
- Big cars* — 2 500 mm × 5 500 mm (clear space + additional space of 400 mm for mechanized clearance); and
- Clearance for door opening* — Minimum of 400 mm on the driver's side for normal parking and 1 000 mm for persons with disabilities.



20A FOR SMALL CAR



20B FOR BIG CAR

FIG. 20 PARKING SPACE DIMENSION

6.2.12 Reserved Parking for Persons with Disabilities

Parking spaces for the persons with disabilities shall be as per and shall be considered as conventional parking having size 3 900 mm in width with 5 400 mm in length. One to two percent of the parking spaces shall be reserved as per local norms and the type of project, located close to the barrier-free entrance and as close as possible to the ramp and passenger lifts. The area shall be highlighted with signages.

6.2.13 Driveway Width for 90 Degree Entry

The driveway width shall be at least 6.5 m in front of mechanized systems.

6.2.14 Single Driveways/Passing Driveways

Single driveways where parking is not connected shall be at least 2.75 m wide and for two-way traffic, the width shall be at least 5.0 m.

6.2.15 Identification of Driveways

Parking systems and driving aisles shall be clearly marked. The signage indicating entry/exit direction of driveways shall be permanently marked and illuminated. See [Fig. 21](#) for guidance.



FIG. 21 MARKINGS IN DRIVEWAYS AND PARKING SPACES

6.2.16 Demarcation of Parking Space

Parking spaces shall be marked with numbers on the floor to facilitate easy identification of parking spots.

6.2.17 Reserved Parking for Women

Ten percent of the parking spaces in large parking area shall be reserved for women drivers and should be located near access ramps/roads, preferably in areas free of columns or other obstructions.

6.2.18 Clear Height for Pedestrian Passage

A clear passage height of 2.2 m in pedestrian areas shall be maintained and clearly marked with signages on the floor.

6.2.19 Escape Route

Every medium-sized and large-sized parking spaces shall have at least two independent escape routes on each floor. The second escape route may lead through ramp. In parking areas above ground, one escape route is sufficient. External staircases shall be arranged.

6.2.20 Stairway

At least two staircase or exit to the open air shall be provided from each floor. Permanent and easily recognizable signage/indications of exits shall be provided. General electric lighting shall be provided to ensure an illuminance of at least 20 lux in escape routes and driving aisles. Solar-powered illumination should be considered for environmental sustainability.

6.2.21 Emergency Lighting in Parking Area

Emergency lighting, with a backup power source independent of the mains supply, shall switch on automatically in the event of a power failure. The lighting shall be designed for at least 1 h of operation with an illuminance of at least 20 lux or shall be provided with fluorescent markings that offer adequate illumination for at least 1 h, leading to exits in an easily recognizable manner.

7 SAFETY REQUIREMENTS

7.1 Safety Doors

Parking systems employing pit type designs shall be protected by safety doors to ensure user safety and system integrity.

7.1.1 Strength of Safety Doors

The safety doors shall have mechanical strength in the closed position, capable of withstanding a force of 300 N applied at any point perpendicular to the door surface on either side, evenly distributed over an area of 500 mm². This force should not cause permanent deformation of the door or impair its normal function. The height of the safety doors shall be considered as 2.0 m unless restricted by a car height of less than 2.0 m. This requirement does not apply to emergency access doors for automatic systems. The size of the emergency access doors may be restricted to 0.65 m in width and 2.0 m in height. If the main door(s) is intended to be closed manually by the user, the control devices shall be located such that the door is visible from the operating position. Safety devices shall not be easily accessible to users. The safety device preventing the door from opening shall be designed as an interlock. The movement of the load-carrying means shall not be possible until the locking device(s) has/have engaged by at least 7 mm. The latch shall resist without permanent deformation or release of the latch when a force of 1 000 N is applied at the height of the latch in the direction of the door opening. The latch shall remain engaged and held permanently in position. Electromagnets, pneumatic and hydraulic devices shall not be used as retainers.

7.2 Operation of Parking Facilities

7.2.1 General Operation

The following lists the general operations in the semi-automatic parking facilities:

- a) The user parks the motor vehicle in the transfer area;
- b) The motor vehicle shall be automatically stored;
- c) The motor vehicle shall be automatically made available for retrieval; and
- d) The user retrieves the motor vehicle and drives out of the transfer area.

7.2.2 Design of Transfer Area

For permanent users, the design of the system shall allow the main door to be opened by signals from outside, such as remote control, magnetic card, induction loop, manually operated control devices, or mobile terminals. The particular door from which a 'PARKING IN' or 'PARKING OUT' call is registered shall open.

7.2.3 Checking the Vehicle Dimensions

The user shall ensure that the maximum vehicle dimensions and weights, as specified in the operating instructions, are not exceeded.

7.2.4 Closing the Main Door

The following illustrates the steps involved for:

- a) *For manually operated doors* — The user shall exit the motor vehicle and leave the transfer area through the main door. The user shall check that no person remains in the transfer area before manually closing the main door; and
- b) *For power-operated doors* — The user shall exit the motor vehicle and leave the transfer area through the main door. The user shall check that no person remains in the transfer area before closing the main door using the designated control device, such as a magnetic card or manually operated control device. Access during the closing process shall be monitored by a light barrier sensor.

7.3 Safe Practices

7.3.1 Monitoring by System Operator

The user should not remain in the car for an extended period and shall exit the car within 45 s. The operator shall ensure adherence to this practice for safety.

7.3.2 Automatic Malfunction Detection Devices

Semi-automatic parking facilities shall be equipped with one or more automatic malfunction detection devices designed and installed to minimize the risk of unsafe operation. When any fault or hazardous condition is detected, an appropriate audible and/or visible warning shall be provided to the user or operator. The equipment shall be immediately stopped using the emergency safety button located on the operation panel. Restarting the equipment shall only be possible by maintenance personnel and under the control of all safety devices.

7.3.3 Operating Modes

The changeover from automatic mode to any other operating mode shall only be carried out by authorized or instructed persons. The changeover to safety-related operating modes shall comply with all safety requirements.

7.3.4 Safety Monitoring and Testing

The following aspects shall be adhered to:

- a) *Type test* — Verification of the machine type to determine whether the parking system meets the requirements;
- b) *Individual testing* — Testing each parking system in order to ensure compliance with safety requirements before putting it in operation and handing over to end user;
- c) *Design test* — Checking whether the design of the parking system or its components meets the functional requirements;
- d) *Visual inspection* — Determining the presence of required safety features on the machine, system, or component, and ensuring that provided documents meet the functional requirements;
- e) *Measurements* — Ensuring that specified measurable parameters, such as geometric dimensions and safety distances, are met; and
- f) *Operational test* — Verifying that the parking system operates as intended and that all safety devices function correctly during a normal cycle or part of a cycle without load.

7.4 Safety Devices

Parking systems shall be equipped with the following safety devices to ensure the safe and efficient operation of the equipment:

- a) *Limit switches* — These shall be installed to detect and control the movement of the parking platforms in the upward direction, preventing them from exceeding their designated range of motion;
- b) *Non-return valve* — This shall be integrated into the hydraulic system to prevent the unintended lowering of the platform in the event of a hydraulic failure, ensuring that vehicles remain securely positioned;
- c) *Wheel stoppers* — Wheel stoppers shall be provided at the end of each parking bay or platform to prevent vehicles from rolling off the platform. These shall be of sufficient strength and size to securely hold the vehicle in place;
- d) *Locker* — These shall be fitted in the vertical guide rail element of the parking system to stop the downward movement in case of brake failure;
- e) *Electromagnetic locks* — These shall be used to securely lock the platforms or gates in place when the parking system is not in operation, preventing unauthorized movement or access;
- f) *Car height sensor* — A car height sensor may be installed to detect the height of the vehicle and ensure it is compatible with the parking system's designated height restrictions, preventing potential collisions with the platform or ceiling;

- g) *Car in-out sensor* — These sensors shall detect the presence of a vehicle entering or exiting the parking bay, ensuring that the system operates when it is safe to do so; and
- h) *Emergency stopper* — An emergency stop device should be provided at accessible locations within the parking area to immediately halt the operation of the system in the event of an emergency, preventing accidents or injuries.

7.5 Wiring Practices

Wiring practices for power-driven parking equipment, including the connections to the disconnection means from the power supply, shall adhere to the guidelines outlined in Part 1 of good practice [D-5C(1)] and [D-5C(2)]. The electrical equipment of the parking system shall comply with the good practice [D-5C(4)]. The earthing and bonding connections shall be as per the good practice [D-5C(3)] and lightning protection of the parking system shall comply with accepted standard [D-5C(5)].

7.5.1 Wiring methods both inside and outside of enclosures shall be clearly identified and marked to prevent incorrect connections that could result in hazardous conditions, such as unintended movement direction or disruption of safety devices.

7.5.2 Whenever possible, external wiring shall be positioned away from combustible materials and areas where it might be subject to mechanical damage. In situations where this cannot be avoided, the wiring shall be protected using appropriate means such as rigid conduits, flexible tubing, raceways or other suitable protective methods.

7.5.3 Electrical equipment shall be designed, marked, and arranged to minimize the risk of incorrect connections, which could lead to injury. Proper care shall be taken to ensure all connections are safe and correctly executed.

7.6 Essential Safety Guidelines for Operation of Parking Systems

Apart from the operational mechanisms, the following essential safety guidelines shall be observed for the successful and safe operation of various parking systems.

7.6.1 Maintenance Doors

Properly designed maintenance doors are critical to ensure safe access for inspection and repair without compromising system integrity. The following requirements shall apply:

- a) In fully automatic systems, a separate maintenance door shall be provided in addition to the main door. This door shall be openable only from outside by authorized persons with a key and from inside without a key;
- b) Maintenance doors shall not open inwards into the working area and shall be provided with appropriate electrical connection; and
- c) When the maintenance door is opened, the parking equipment shall automatically stop by means of a safety switch or equivalent device. Restarting of the equipment shall only be possible when the service door is closed and shall be carried out by authorized personnel.

7.6.2 Access and Inspection

Safe and reliable access routes are essential for regular inspection, servicing and emergency interventions. Accordingly, the following provisions shall apply:

- a) A separate staircase shall be externally installed, with access provided at every alternate level for inspection; and
- b) Emergency doors for accessing the system shall be provided with electrical/electronic controls to switch off the system.

7.6.3 Selector Switches

Selector switches shall be located outside the working area with free access for changeover from mains supply to alternate power source.

7.6.4 Safety Guards

Fixed enclosure guards or interlocking guards shall be provided to protect persons from hazards such as crushing, shearing, entanglement or entrapment.

7.6.5 Walkways, Control Stands and Platforms

Safe access routes shall be ensured during installation. Permanent access to control panels and parking areas shall be provided in accordance with applicable building safety norms.

7.6.6 Illumination of Working and Transfer Areas

Adequate lighting is vital to maintain visibility, safety and ease of operation in both working areas and transfer areas. The following requirements shall apply:

- a) Working areas where maintenance, inspection, cleaning, and lubrication are to be carried out shall be properly illuminated, with a minimum of 100 lux; and
- b) The transfer area shall be adequately illuminated between 40 lux and 60 lux. Lighting in the transfer area may automatically switch off after closing of the main door.

7.6.7 Safety Railings

Safety railings of minimum 1 m height shall be provided in stack and puzzle parking systems to prevent accidental access to open spaces around the system.

7.6.8 Operating Instructions

Clear operating instructions shall be displayed at suitable locations.

7.6.9 Restricted Entry

Unauthorized persons shall be prevented from entering the transfer and working areas during system operation.

8 VENTILATION AND FIRE SAFETY REQUIREMENTS FOR PARKING SYSTEMS

8.1 Mechanical Exhaust Air Systems

8.1.1 Mechanical exhaust air systems shall be required in closed medium-sized and large parking areas. The supply air openings shall be distributed to ensure sufficient ventilation throughout all parts of the parking areas. If the supply air openings are inadequate, a mechanical supply air system shall be provided.

8.1.2 The mechanical exhaust air systems shall be dimensioned to ensure that the half-hourly mean value of carbon monoxide (CO) concentration in the air, measured at a height of 1.5 m above the floor, does not exceed 100 ppm. This requirement shall be deemed satisfied if the exhaust air systems in parking areas with low ingress and egress traffic discharge at least 6 m³ of exhaust air per hour per m² of usable parking area. In other garages, the system shall discharge at least 12 m³ of exhaust air per hour per m² of usable parking area.

8.1.3 Mechanical exhaust air systems shall be equipped with at least two equally sized fans in each ventilation system, which together shall provide the required total volume flow during simultaneous operation. Each fan of a mechanical supply or exhaust air system shall be supplied from a separate circuit, to which no other electrical systems shall be connected. If the ventilation system is temporarily operated with only one fan, the fans shall be switched in such a way that in the event of failure of one fan, the other fan automatically switches on.

8.2 CO Monitoring and Warning Systems

8.2.1 Large enclosed parking areas with substantial incoming and outgoing traffic shall have CO measuring and warning systems. The CO warning systems shall be designed to prompt users to switch off their engines through acoustic signals and flashing signals if the CO content in the air exceeds 250 ppm. These CO warning systems shall be connected to a backup power source.

8.3 Safety Notices

8.3.1 In enclosed medium-sized and large parking areas, the following notices shall be prominently and permanently displayed at the entrance and on each floor:

"EXHAUST FUMES ENDANGERS HEALTH.
AVOID PROLONGED STAY."

8.4 Fire Safety Provisions

8.4.1 Fire extinguishing systems, smoke and heat extraction systems, and fire alarm systems should be provided in accordance with the requirements specified in Part F 'Fire and Life Safety' of this NBCS.

8.4.2 Automatic parking areas with more than twenty parking spaces shall be equipped with sprinkler systems. For automatic parking areas with less than twenty parking spaces, power-operated lifting platforms that allow vehicles to be stacked, and parking spaces separated from tramlines by closures, non-automatic fire extinguishing systems shall be provided. The type of system shall be determined on a case-by-case basis in consultation with the fire protection authority, ensuring that all parking spaces within the parking areas can be reached with an extinguishing agent in every operating state.

8.4.3 Enclosed medium-sized and large parking areas shall be equipped with fire alarm systems if they are connected with structural installations or rooms for which fire alarm systems are required.

8.4.4 Fire extinguishing systems may also be required for enclosed medium-sized parking areas if necessary for deploying fire lines for flammable substances or electrical lines connected to the fire tank.

8.4.5 A sprinkler system shall be installed in all parking areas, regardless of size (small, medium or large) and shall be designed to cover all parked vehicles effectively. This requirement also applies to automatic parking areas with more than twenty parking spaces. The sprinkler system should preferably not be positioned at the immediate centre (S2 as depicted in [Fig. 22](#)) of the top area with respect to the parking system. Instead, it should have some eccentricity from the centre position (S1 and S3 as depicted in [Fig. 22](#)) to enhance effectiveness in the event of a fire, ensuring better coverage and rapid response.

8.5 Ventilation Requirements

8.5.1 Natural ventilation for parking system installed at ground or above natural ventilation shall be considered.

8.5.2 Natural ventilation for parking system installed below ground for low traffic, natural ventilation may also be sufficient.

8.5.3 *Ventilation Openings and Shafts (for Fully Automatic Systems)*

Permanent ventilation openings or ventilation shafts, with a total free cross-section of at least 0.15 m² per parking space, shall be provided up to a height of 2 m to ensure proper air circulation. These provisions shall comply with the requirements outlined in the local building code to maintain safety standards and mitigate fire risks.

8.6 Restrictions on Parking Locations

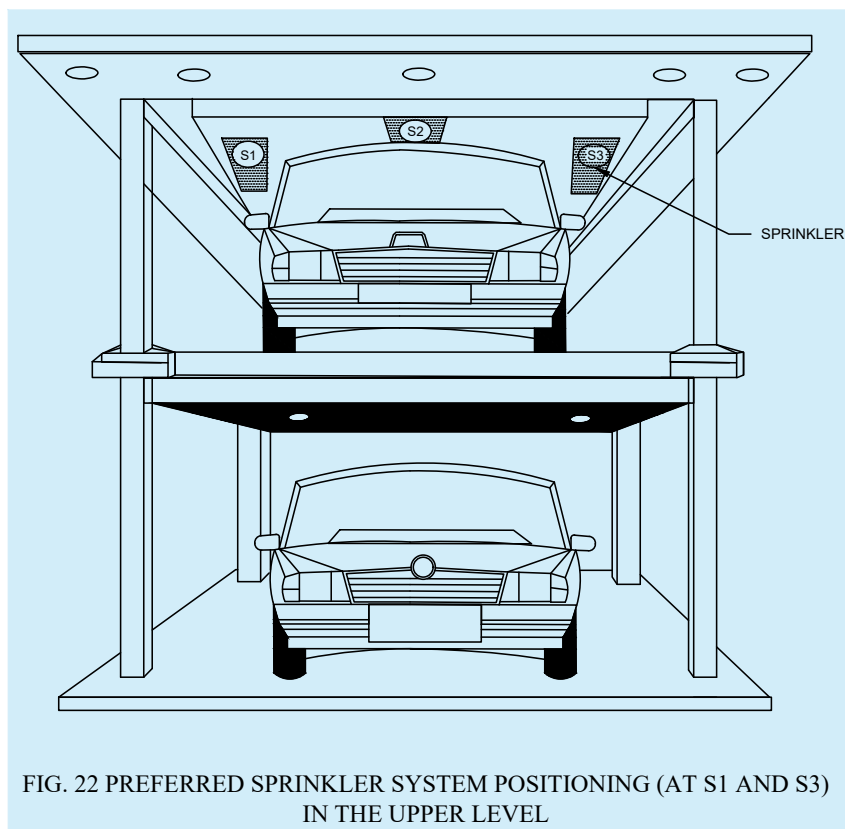
Motor vehicles shall not be parked in stairwells or generally accessible corridors, including ramps.

9 SPECIAL TECHNICAL REQUIREMENTS

This is considered for special applications of parking; for example, parking systems within the side margins of the existing buildings. These systems can be considered above or below the flooring within the side margins

9.1 Cantilever Systems

These systems are also installed within side margins, however due to its construction, the systems are not suitable for long life.



9.2 Suspended Systems

These systems are hung from top connected through building beams and are not safe.

10 PERFORMANCE REQUIREMENTS FOR PARKING SYSTEMS

10.1 Design Considerations

When designing or specifying parking equipment, consideration shall be given to the worst-case loads that may arise from both the stationary position of a motor vehicle on the platform or load-carrying equipment, as well as during the vehicle's movement on the load-carrying equipment, where such movement may cause higher stresses.

10.2 Service Life Definition

The parking system shall be designed to meet the following minimum service life criteria:

- | | | |
|----------------------------|---|--|
| a) Service life | : | 20 years |
| b) Cycle definition | : | 1 parking in and 1 parking out operation |
| c) Cycles per day | : | 2 cycles per platform or space |
| d) Operating days per year | : | 365 days |

LIST OF STANDARDS

The following list records those standards which are acceptable as ‘good practice’ and ‘accepted standards’ in the fulfilment of the requirements of this NBCS. The latest version of a standard shall be adopted at the time of enforcement of this NBCS. The standards listed may be used for conformance with the requirements of the referred clauses in this NBCS.

In the following list, the number appearing in the first column within parenthesis indicates the number of the reference in this Part/Section.

	<i>IS No.</i>	<i>Title</i>
(1)	SP 30 : 2023	National Electrical Code of India 2023
(2)	IS 732 : 2019	Code of practice for electrical wiring installations (<i>fourth revision</i>)
(3)	IS 3043 : 2018	Code of practice for earthing (<i>second revision</i>)
(4)	IS 16504 (Part 1) : 2019/ IEC 60204-1 : 2016	Safety of machinery — Electrical equipment of machines: Part 1 General requirements (<i>first revision</i>)
(5)	IS/IEC 62305-1 : 2010	Protection against lightning: Part 1 General principles
	IS/IEC 62305-4 : 2010	Protection against lightning: Part 4 Electrical and electronic systems within structures

NATIONAL BUILDING CONSTRUCTION STANDARDS

PART D BUILDING SERVICES

SECTION 6 INFORMATION AND COMMUNICATION ENABLED INSTALLATIONS

BUREAU OF INDIAN STANDARDS

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National Building Code Sectional Committee, CED 46

FOREWORD

This NBCS (Part D/Section 6) covers the essential requirements for information and communication enabled installations, technology systems and cabling installations in a building. It also covers the basic design and integration requirements for telecommunication spaces within building/buildings along with their cabling infrastructure, their pathway components and passive connectivity hardware. It also includes general requirements relating to installation of different wireless and wireline communication equipment, cable terminations, power connections and general guidelines required for planning and providing information and communication technology (ICT) services in the building at the planning and execution stages. The provisions given herein are basic requirements applicable to all residential and other buildings. These can be used at the time of upgradation of existing buildings for properly accommodating telecom systems/services. Buildings meant for data centres and those for housing telecom exchanges/facilities for offering public services may have various other considerations. For such buildings, provisions of this Section are indicative and may be a subject of actual requirements.

Telecommunication plays a vital role in modern society similar to electricity, water, gas, and transport systems. It is difficult to imagine a life without telecommunication technologies. In a building, a broad variety of telecommunication systems are expected to be installed. Buildings have provision for installation of various utilities. However, unlike traditional utilities, telecommunication systems are constantly evolving at a rapid pace.

Telecommunications connectivity can be delivered through cable, wire, optical fibre, fixed wireless and mobile wireless technologies. Each of these technologies when considered for use inside buildings, especially when the buildings are either commercial or multi-dwelling units or complexes, places its own requirement in terms of building space, power supply, internal extensions to various work areas/dwelling units. For example, for wire line services, broadband cable television, etc. The entry to the buildings/complexes will be through underground cables and the distribution of services further into the complexes will be from the bottom of the building to the upper stories. On the contrary, any wireless technologies, where antennas are to be installed at terrace along with a system which can distribute the signals through cables or in building solutions, the flow of cables will be from top storeys of the building to the bottom. Keeping in view that there are multiple telecom service providers (TSPs), there is growing need for provision of seamless wireless services to the users at public buildings such as airport, malls, hospitals, railway stations by sharing infrastructure without any discrimination.

The telecom facilities can be chosen by the user from the gamut of technologies and associated features as options are available based on the requirement, cost, service and maintenance convenience, future upgradation requirements, user preference, quality of service/experience etc. While providing telecom enabling infrastructure in the building, the provisions are to be made for making the infrastructure supportive for multiple technologies/products and the requirements of telecom service providers. Use of building management system (BMS), access control, surveillance, and creation of 'Smart' building will require significant telecom infrastructure to be created within the buildings which may be supported by M2M communication technologies. Individual/tenant users of the building may also have their own M2M/IoT systems with wireline or wireless (cellular, satcom, wifi, loRa, zigbee, and other wireless protocols) gateways. The telecom support infrastructure in the buildings/campuses needs to be responsive and accommodative of these requirements.

In order to facilitate installation/upgradation of telecom systems, proper planning and understanding of enabling provisions for telecom technologies and physical infrastructure are necessary. Modern telecommunication technologies such as distributed antenna system, in-building solutions (using technologies for example wifi 6/6E or higher versions, cellular 4G/5G and beyond) are also to be considered during the building planning stage itself and providing shared access to multiple telecom service providers as per demand by end users. The enabling infrastructure may include cable riser systems, conduits, cable trays, fibre management systems, passive and active infrastructure, distributed antenna system (DAS), etc. Appropriate space need to be earmarked for installation of equipment at the entry point of service and running the cables, etc, through shafts and horizontal conduits inside the walls, centre of the corridors and centre of the work space, etc. Thus, cabling pathways infrastructure should be designed to be of general nature but flexible enough to accommodate a variety of current and future telecom systems and emerging technologies.

Choice of service to be provided inside the complexes/buildings will depend on the users. As already mentioned above, the delivery of service can be spread across technologies and across various service providers. Therefore, the arrangement for telecommunication infrastructure needs to be made in such a manner that the requirements and the challenges of at least near future (say 10 years) can be met without disturbing or retrofitting the building infrastructure. This Section has been formulated to cover these aspects.

In the formulation of this Section, due weightage has been given to international coordination among the standards and practices prevailing in different countries, in addition to relating it to the practices in this field in this country. This has been done by deriving assistance from the publications of ISO, IEC, ITU, TEC, TRAI Regulations and DOT Orders/Guidelines, etc.

Considering all the above, this Section was introduced in 2016 as a part of National Building Code of India to address the essential requirements for information and communication enabled installations and the related.

In this revision, this publication has been renamed as National Building Construction Standards (NBCS).

In line with technological advancements and evolving connectivity requirements, significant modifications have been made to the section on ICT installations in buildings, in this version of NBCS. These updates aim to address current demands for faster data transfer rates, advanced wireless standards, and improved maintenance protocols to ensure robust and reliable ICT infrastructure. The significant changes incorporated in this revision include:

- a) The table on balanced twisted-pair cabling channel performance is updated to reflect maximum data transfer rates, aligning with present-day bandwidth demands and ensuring future readiness;
- b) A new provision on meshed bonded networks, which provide enhanced immunity to electromagnetic interference (EMI) and are recommended for environments with high-density ICT equipment, in [4.1.1](#);
- c) A new clause [A-6](#) is included on overview of in-building solutions (IBS), including distributed antenna systems (DAS), leaky feeder cables, and small cells, to enhance indoor wireless connectivity in areas with weak signals. It also includes key considerations such as coverage requirements, interference management, capacity planning, and aesthetic integration to ensure seamless indoor mobile connectivity;
- d) A new informative [Annex B](#) on street furniture that highlights the use of streetlights, utility poles, bus stops, traffic signals, etc to support ICT infrastructure; and
- e) A new [Annex C](#) on building related information and digital connectivity requirements has been introduced.

As the regulation of land development and buildings is a State subject and is dealt by the States and local bodies such as urban local bodies within their jurisdiction through building regulatory documents like building regulations, building byelaws, and fire regulations, this document is voluntary in nature and is non-binding. The implementation depends on adoption by concerned parties or stipulations in a contract or by appropriate adoption by the concerned authorities.

For the purpose of deciding whether a particular requirement of this Section is complied with, the final value, observed or calculated, expressing the result of a test or analysis, shall be rounded off in accordance with IS 2 : 2022 'Rules for rounding off numerical values (*second revision*)'. The number of significant places retained in the rounded off value should be the same as that of the specified value in this Section.

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Important Explanatory Note for Users of this NBCS

In any Part/Section of this NBCS, where reference is made to ‘**good practice**’ in relation to **design, constructional procedures or other related information**, and where reference is made to ‘**accepted standard**’ in relation to **material specification, testing, or other related information**, the Indian Standards listed at the end of the Part/Section shall be used as a guide to the interpretation.

At the time of publication, the editions indicated in the standards were valid. All standards are subject to revision and parties to agreements based on any Part/Section are encouraged to investigate the possibility of applying the most recent editions of the standards.

In the list of standards given at the end of a Part/Section, the number appearing within parentheses in the first column indicates the number of the reference of the standard in the Part/Section. For example:

a) Good practices [D-6(1)] refers to the Indian Standard(s) given at serial number (1) of the list of standards given at the end of this Part D/Section 6, that is, IS 9537 (Part 2) : 1981 ‘Specification for conduits for electrical installations: Part 2 Rigid steel conduits’

NATIONAL BUILDING CONSTRUCTION STANDARDS

PART D BUILDING SERVICES

SECTION 6 INFORMATION AND COMMUNICATION ENABLED INSTALLATIONS

1 SCOPE

1.1 This NBCS (Part D/Section 6) covers the essential requirements for information and communication enabled installations, technology systems, cabling installations, and other passive and active equipment in a building. It also covers the basic design and integration requirements for telecommunication spaces within building/buildings along with their cabling infrastructure, their pathway components and passive connectivity hardware considering future requirements including flexibility requirement for installation of active infrastructure by the service providers.

1.2 It also includes general requirements relating to installation of different communication equipment, cable terminations, power connections and general guidelines required for planning and providing information and communication technology (ICT) services in the building at the planning and execution stages. The provisions given herein are basic requirements applicable to all residential and other buildings. These can also be used at the time of upgradation of existing buildings for properly accommodating telecom systems/services. Buildings meant for data centres and those for housing telecom exchanges/facilities for offering public services may have various other considerations. For such buildings, provisions of this Section are indicative and may be a subject of actual requirements. Introduction of street furniture as a crucial enabler of ICT infrastructure, leveraging existing structures like streetlights, utility poles, and bus stops to support small cells and air fibre for enhanced connectivity has been made.

NOTE — In this Section, 'Telecom' has been used interchangeably with 'ICT', as most of the time there may be only telecommunication hardware present. Apart from the provision of telecommunication systems, Information Technology (IT) system in the building may be either common for the building or separate for the individual users. In case the entire building is used by the same user, the IT system may be common and thus IT space for Building Management System (BMS), CCTV, etc, can be shared with telecom facilities. In case where individual IT systems owned by several users are provided, separate spaces may be earmarked inside telecom spaces for individual IT infrastructure. Sometimes, individual users may also use the common IT facilities. However, the cabling, wiring, etc, for IT systems should use the same pathways, which are used for telecommunication hardware.

2 TERMINOLOGY

For the purpose of this Section, the following definitions shall apply.

2.1 Access Point — A hardware device or a computer's software that acts as a communication hub or as an interconnection port for users of wireless devices to connect them to a wired or wireless local area network (LAN).

2.2 Antenna — An electrical device designed to transmit or receive radio waves or more generally, electromagnetic waves for the purpose of radio frequency communication such as Wi-Fi, radio, television, satellite communication, radar communication and mobile communication. This is also called an aerial.

2.3 Attenuation — A general term that refers to any reduction in the strength of an electrical/electromagnetic signal.

2.4 Backbone — A high-capacity facility (for example, pathway, cable or conductors) which acts as a major pathway within a network from which further smaller capacity facilities get distributed.

2.5 Backbone Cabling Media Distribution and Building Pathway — A part of a building premises telecom cable distribution system that provides connection between telecommunications spaces. It typically provides building connections between floors in multi-storeyed buildings as well as campus connections in multi-building environments.

2.6 Building Management System — It is a low voltage automation or remote control system installed for managing different building sub-systems like, heating, ventilation, lighting, air conditioning, fire protection, access control, etc.

2.7 Bus Bar — It refers to thick strips of copper or aluminium, in electrical power distribution that conduct electricity within a switchboard, distribution board, substation, or other electrical apparatus.

2.8 Coaxial Cable — A type of wire that consists of a centre wire surrounded by insulation and then a grounded shield of braided wire. The shield minimizes electrical and radio frequency interference.

2.9 Consolidation Point (CP) — A location for interconnection between horizontal cables extending from building pathways and horizontal cables extending into furniture pathways.

2.10 Cross-Connect — A facility enabling the termination of cable elements and their interconnection or cross-connection.

2.11 Distributed Antenna System (DAS) — It is a system that uses network of physically or spatially separated antennas to extend radio coverage of any technology inside a building.

2.12 Direct to Home (DTH) — The direct reception of satellite programs using small dish antennae (personal dish) placed on a roof or window of a house.

2.13 Frequency — The measurement of the number of times that a repeated event occurs per unit of time.

2.14 Horizontal Cabling — It includes:

- a) the cabling between and including the telecommunications outlet/connector and the horizontal cross-connect; and
- b) the cabling between and including the building management system outlet or the first mechanical termination of the horizontal connection point and the horizontal cross-connect.

2.15 Horizontal Cabling Media Distribution and Building Pathway — It consists of the horizontal cabling in the building, the horizontal pathways supporting the horizontal cabling, and the telecommunications spaces that support the horizontal pathways. The use of the term horizontal in the name of the element does not require that the elements be placed or installed parallel to the ground or floor.

2.16 Horizontal Cross-Connect (HC) — A cross-connect of horizontal cabling to other cabling, for example, horizontal, backbone, and equipment.

2.17 In-Building Solution (IBS) — It is a telecom infrastructure solution designed to enhance indoor network coverage for mobile or fixed wireless or Wi-Fi network for seamless indoor connectivity. DAS can be a part of IBS.

2.18 Inside Plant (ISP) — Telecommunications infrastructure designed for installation, interior to the buildings.

2.19 Intermediate Cross-Connect (IC) — A cross-connect between first level and second level backbone cabling. First level backbone is a cable between a main cross-connect (MC) and intermediate cross-connect (IC) or horizontal cross-connect (HC). Second level backbone exists between an intermediate cross-connect (IC) and horizontal cross-connect (HC).

2.20 Internet of Things (IoT) — A network of all physical objects connected to internet primarily using M2M communication technologies.

2.21 Local Multipoint Distribution System (LMDS) — A broadband radio service designed to provide two-way transmission of voice, high-speed data and video (wireless cable TV) digitally through microwave.

2.22 Local Area Network (LAN) — A computer network covering a local area, like a home, office or small group of buildings.

2.23 Machine to Machine (M2M) Communication — It is a set of wired/wireless communication technologies which enable machine/device (for example sensor, meter, etc) to capture an event (for example temperature, motion, meter reading, etc) and relay through communication network to an application that translates the data from device to a meaningful information.

2.24 Main Cross-Connect (MC) — A cross-connect for first level backbone cables, entrance cables, and equipment cables.

2.25 Main Distribution Frame (MDF) — A structure where all the copper wires, fibre or coaxial cables for a network terminates for patching/cross-connection on separate or composite frames.

2.26 Metropolitan Area Network (MAN) — A network designed to carry data over an area larger than a campus, such as an entire city and its outlying area.

2.27 Multichannel Multipoint Distribution System (MMDS) — A method of delivering multiple voice/data/television signals digitally by microwave transmission to subscriber households.

2.28 Multi-user Telecommunications Outlet Assembly (MUTOA) — A grouping in one location of several telecommunications outlet/connectors.

2.29 Optical Fibre — It refers to the medium and the technology associated with the transmission of signals/information as light pulses along a glass fibre.

2.30 Optical Distribution Frame (ODF) — Optical fibre distribution frame where ends of optical fibre terminates for further patching/cross-connection.

2.31 Outside Plant (OSP) — Telecommunications infrastructure designed for installation, exterior to the buildings.

2.32 Propagation Delay — In a communications system, it refers to the time lag between the departure of a signal from the source and the arrival of the signal at the destination.

2.33 Public Mobile Network — A network that is established and operated by a telecom service provider for the specific purpose of providing mobile telecommunications services to the public using technologies such as GSM, CDMA and LTE.

2.34 Radio Frequency — A frequency range from 20 kHz and above, used for transmitting/receiving text, data, audio, or video signals.

2.35 Server — Any computer on a network that contains data or applications shared by users of the network on their client terminals.

2.36 Splicing — The process of permanent joining of bare fibre end to another fibre by means of splicing tools.

2.37 Smart City/Building/Home — An innovative city/building/home that uses information and communication technologies, and other means to improve quality of life, efficiency of operation and services and competitiveness, while ensuring that it meets the needs of present and future generations with respect to economic, social and environmental aspects.

2.38 Telecommunication Spaces — The rooms and areas where telecommunications cabling systems are terminated, cross-connected and interconnected to installed telecommunications equipment. Various examples of these based on the function and areas, are equipment room (ER), telecommunications room (TR), entrance facilities (EF) and telecommunication enclosures (TE).

NOTE — The definitions of telecommunication spaces and workspace (*see 2.46*) are used for calculation of infrastructure requirements, facilities, etc. This does not inhibit the utilization of space for any other purpose. Depending upon the requirements, number of telecommunication spaces in a given building may vary for the same area.

2.39 Telecom Service Provider (TSP) — A service provider who is duly authorized by Government of India or its designated entity to operate telecom services.

For example, access service providers, telecom infrastructure providers (IP1s), internet service providers (ISP), virtual network operators (VNOs), satellite communication service providers, public data offices (PDO) and public data office aggregators (PDOA), or any other wireline, wireless, direct to home (DTH), cable service providers, IPTV service providers and any other service provider authorized by the government from time to time.

2.40 Telecom Tower/Mast/Pole — A vertical structure for installation of antenna(e) that provide radio frequency air interface for telecommunications services. It can also be in the form of mast or multiple poles of varying height. Poles may be located at different points typically on the roof.

2.41 Telecommunication Media and Connecting Hardware — It consists of cables, equipment cords, patch cords, and connecting hardware components. All balanced twisted-pair, optical fibre, coaxial cabling and wireless systems are made up of such components. These cabling components and resulting cabling systems are used in outside plant (OSP) and premises cabling [also known as inside plant (ISP)] environments of the building telecommunication infrastructure.

2.42 Underground Cable Vault (UCV) — An underground cable vault (part of an underground duct system) used to facilitate placing, connecting, and maintaining telecommunications cables and associated equipment. It collectively refers to manholes, hand holes, and pull holes.

2.43 Very Small Aperture Terminal (VSAT) — It usually refers to satellite terminal used to transmit and receive signal from satellites with an antenna installed in user premises.

2.44 Wide Area Network (WAN) — It is a network that is capable of spanning a geographical area larger than a city.

2.45 Workstation — An electronic device that performs some information processing or display function and connects to the communications network. Typically, this may be a desktop computer with keyboard and display or laptop computer, but might also be a telephone, a printer, an access control terminal, or some data-gathering device.

2.46 Workspace — Any location and the space around where a workstation may be located. Typically, there can be several workspaces in a room.

3 GENERAL REQUIREMENTS FOR TELECOMMUNICATION SPACES AND CONNECTING HARDWARE

3.1 Telecommunication Spaces

3.1.1 General Considerations

The following shall be the considerations for telecommunication spaces:

- a) *Accessibility* — Telecom spaces that are intended to serve multiple users should be located in common spaces that should be accessible through a common corridor or outside door. The space for each telecom service providers may be separated to the extent feasible by partitions for security reason and controlled access to their equipment. The telecommunication spaces should be accessible, but the access should be controlled against the unauthorized access (for example, with a lock and key arrangement or by an electronic access system).
- b) *Acoustic noise levels* — Acoustic noise levels in telecom spaces should be kept to a minimum by not collocating noise-generating equipment (for example, photocopy equipment, high-speed printers, and mechanical equipment).
- c) *Administration* — All pertinent documentation of deployment of telecom equipment and cables should be maintained by owner or agent when the installation is completed. All telecom spaces shall have appropriate signs to identify the space and should be included within the security plan of the building. There must be sufficient space between the equipment of different service providers for maintenance and better workmanship.
- d) *Cable separation* — Telecom cables should be separated from possible sources of electromagnetic interference (EMI) and from possible radio frequency interference (RFI). For safety purposes, power cables should be separated from telecom cables.
- e) *Ceilings* — The general requirements for ceilings in telecom spaces should include the following:
 - 1) Generally, the minimum ceiling height shall be 2.4 m above finished floor. However, consideration may be given for having a 3 m ceiling height.
 - 2) If suspended ceilings are provided, necessary rodent protection may be provided.

- 3) The ceiling finish should minimize dust and be light coloured to enhance the room lighting.
- f) *Conduits, trays, slots, sleeves and ducts* — Slot/sleeve systems should be located in places where pulling and termination of cables is easy to achieve. Bend radius requirements and service loop guidelines shall be considered. Sleeves and slots should not be left open after cable installation. All sleeves and slots shall be fire-stopped. (See also Part F ‘Fire and Life Safety’ of this NBCS for guidance purpose).
- The size and number of conduits or sleeves used for backbone pathways depends on the usable floor space served by the backbone distribution system. However, sleeves in multiple of 100 mm are recommended to serve a TR, ER or EF. Multiple telecom spaces on the same floor shall be interconnected with a minimum of one 75 mm conduit or a pathway that provides equivalent capacity.
- g) *Dust and static electricity* — Dust and static electricity should be avoided by:
- 1) Placing active printers outside of telecom spaces.
 - 2) Treating floors, walls, and ceiling to minimize dust.
- h) *Electrical power* — When active equipment of TSPs or the building equipment (for example, PABX) are installed in TR/ER, etc, the telecom spaces shall be equipped to provide adequate electrical power with redundancy and sub-meters to each of the service providers at TRs and ERs. The recommendations are as under:
- 1) A power generator set should be considered if longer power interruptions are expected for which the battery backup is not adequate. Further, as telecom services are required on 24 × 7 basis, uninterrupted electrical power for telecom services shall be provided from local or central uninterrupted power supply (UPS) with adequate battery backup. Further, UPS arrangement may be extended to equipment in TE/corridors also.
 - 2) Separate socket outlets for equipment, tools, test instruments, etc.
 - 3) Separate electrical distribution panels that serve telecom equipment/spaces from those that serve other purposes are recommended. These should be clearly identified with proper sign writing.
- j) *Prevention from flooding* — Telecom spaces should be located above any threat of flooding. When locating telecom spaces where a threat of flooding is unavoidable, rack elevations may be designed so that active equipment and telecom components are placed above the threat level. Sometimes locating the ER/TR on a floor above ground floor may be safer. The locations that are below or adjacent to areas of potential water hazard (for example, restrooms, kitchens) should be avoided. Liquid carrying pipes (for example, water, waste, steam) should not be routed through, above, or in the walls encompassing the ICTs space. If due to unavoidable circumstances, liquid carrying pipes are passing through telecom spaces then these spaces may be preferably monitored for water seepage by deploying suitable water seepage detection system. Otherwise, regular watch may be kept to detect seepage.
- When air conditioning ducts are installed in telecom spaces, water dripping/leaking from AC vent should be controlled and, if required, monitored by water leaking system. Instructions for prevention from liquid cleaning agents, air purifying liquids, pest repellents and sprays usage should be indicated by proper sign writing.
- k) *Lighting* — Following are the important considerations for lighting of telecom spaces:
- 1) Locating light switches near the entrance(s) to the telecom space.
 - 2) Coordinating the lighting layout with the planned equipment layout (especially overhead cable trays) to ensure that lighting is not obstructed.
 - 3) Providing electrical power for the lighting and ICT equipment from different circuits.
 - 4) Placing at least one light or set of lights on normal power and one light or set of lights on emergency/UPS power.
 - 5) Using a light coloured finish on walls, floors, and cabinets to enhance room lighting.
 - 6) Providing task lighting illumination at the point of cable termination.
 - 7) Providing adequate lighting of spaces, which may be controlled by automated controllers such as building management system (BMS), if required, or ensured otherwise. Use of dimmers and

occupancy sensors is not recommended.

- m) *Location* — All telecom spaces shall be located in areas that are best suited (preferably centrally located to reduce the cable length) to serve the occupants of a floor or building. The following shall be observed when locating the spaces:
 - 1) Telecom spaces in multi-floor buildings should be aligned vertically.
 - 2) Telecom spaces should be located in areas that are dedicated to telecom use. Equipment that is not related to the support of telecom spaces (for example, piping, duct work, distribution of building power) shall not be located in or pass through a telecom space.
- n) *Environmental control* — When active devices which are heat producing equipment are present, original equipment manufacturer (OEM) recommended temperature and humidity range should be maintained by adequate arrangement for heat dissipation. If environmental parameters are exceeded, an alarm should be activated. A positive pressure is required to be maintained with a minimum of one air change per hour in the telecom space. More stringent requirements may apply based on the equipment needs in the telecom space. It is recommended to have sufficient arrangements of air conditioners for cooling of telecommunication equipment housed or to be housed in telecom room and equipment room.
- p) *Fire protection* — The telecom spaces shall be equipped with adequate fire detection, alarm and suppression systems. (See also Part F ‘Fire and Life Safety’ of this NBCS for guidance purpose). Fire retardant cables are recommended to be used inside the building to reduce incidences of fire.
- q) *Bonding and grounding* — All equipment and cable shields shall be properly bonded to the telecom bonding and grounding infrastructure of the space (see [4.1](#) for details).

3.1.2 Equipment Room (ER)

3.1.2.1 An equipment room is an environmentally controlled centralized space for telecom equipment that usually house equipment of higher complexity than telecommunication rooms and also usually houses a main or intermediate cross-connect. Any or all of the functions of a telecommunications room may be provided in an equipment room. ERs differ from TRs in the way, that, ERs are generally considered to serve a building, campus, tenant, or SP, whereas TRs serve a floor area of a building. In some cases, an ER may also contain the EF or it may serve as TR. ERs may be connected to backbone pathways that run both within and between buildings. Although an ER usually serves an entire building, many building designs may use more than one ER in order to provide separate facilities for different types of equipment and services or redundant facilities and disaster recovery strategies.

The initial assessment for design and specifications for an ER shall be based on detailed information about the site, including:

- a) User requirements;
- b) Telecom pathway locations;
- c) Service providers’ requirements;
- d) Environment/facility conditions; and
- e) Building requirements.

The functional earth terminals of ICT equipment shall be connected either directly to the Primary Bonding Busbar (PBB) or through an Secondary Bonding Busbar (SBB) as per [4.1](#). Where mesh bonding network is used, multiple connections (minimum four) from ICT to mesh-BN are recommended.

3.1.2.2 Space allocation and layout

The space allocation and layout may be determined by following methods:

- a) *Determining size based on area served* — When the telecom designer does not know what specific equipment will be used in an ER, the designer can use the amount of floor space that the room will serve, to determine the minimum size of the ER. If the usable floor space is also unknown, the usable floor space may be estimated by deducting 20 percent from the total floor area.

Generally, an area of 9.3 m² may be considered for calculating the work areas. If work areas are smaller

leading to possible increase in number of work areas in building, the size of the ER should be increased accordingly. The minimum recommended size for ER is 14 m². The general practice is to provide 0.07 m² of ER space for every 10 m² of usable floor space. For special use buildings, such as, hospitals and hotels, ER size requirements may vary.

- b) *Telecom service provider space requirements* — If equipment or cable terminations that are owned or maintained by a TSP are to be located in the ER, then location and amount of space required shall be determined as per the space requirement of TSP.
- c) *Telecom equipment locations* — As equipment for communication systems are located in the various equipment rooms throughout the building, these rooms shall be physically secure, aesthetically provisioned and conveniently located. The communication equipment may be located with due consideration of the modular provision of building design and criteria for expansion joints matching to electrical distribution and maximum fire safety distance requirements. The other considerations for telecom equipment location include the following aspects.
 - 1) Types of cables, their uses, bending radius, turning radius, conduit radius, etc;
 - 2) *Provision for future expansion* — In view of the evolving nature of the communication technologies, provision for future expansion should be ensured by way of stand-by arrangements. The distribution infrastructure should also be as flexible as possible. Small conduits will quickly fill up and may not accommodate some technologies, for example, low loss broadband or high capacity fibre optic cables. The most flexible distribution design involves cable trays for horizontal (floor level) distribution with large conduit sleeves for wall penetration, where necessary. Vertical trunk distribution should also be achieved with conduit sleeves; and
 - 3) *Maintainability* — As it is difficult to remove or shift just a few cables from a crowded or convoluted conduit system for operational or maintenance requirements, steps should be taken to ensure easy access and maintainability of cable system right in the initial architectural design stage.

3.1.3 Telecommunications Room (TR)

3.1.3.1 A telecommunications room is the area within a building that houses the telecommunications cabling system and associated equipment. This includes the mechanical terminators and/or cross-connects for floor-serving distribution facility for horizontal cabling and backbone cabling system. There should be at least one TR or TE per floor. TR or TE can also house active equipment, like LAN switches, routers, OLT, DAS, cable TV or DTH systems etc, or passive equipment, like optical splitters, FDMS. Most of these equipment are available in wall mount units. The switches/routers can also be shared between telecom equipment and other low voltage in-building systems. This reduces the demand on floor space requirements for TR/TE. Sometimes, a TSP with the consent of the building owner may require to provide additional equipment, such as for mobile signal booster, at a later stage, in TR/TE. Therefore, it is recommended to have additional space in advance, especially in tall buildings of five storeys or higher, preferably on top floor.

The functional earth terminals of ICT equipment shall be connected either directly to the PBB or through an SBB as per [4.1](#). Where mesh bonding network is used, multiple connections (minimum two) from ICT to mesh-BN are recommended.

3.1.3.2 Multiple rooms TEs are required if the cable length between the horizontal cabling (floor distribution) [HC (FD)] and the work area outlet location, including slack, exceeds 90 m (or more in case of optical fibre depending upon its type and application as indicated in [3.2.4.2](#)). If the usable floor space to be served exceeds 930 m², additional TRs should be considered. Following are the guidelines for TR size:

<i>Sl No.</i>	<i>Area Served</i>	<i>Dimension of TR</i>
(1)	(2)	(3)
i)	465 m ² or less	3 m × 2.4 m
ii)	More than 465 m ² and less than equal to 930 m ²	3 m × 3.4 m
NOTE — The size of 3 m × 2.4 m has been specified to allow a centre rack, cabinet, or enclosure configuration.		

In smaller buildings, less space is required to serve the telecom distribution needs of the occupants. The size guidelines for smaller buildings should be as below:

<i>Sl No.</i>	<i>Building Smaller Than</i>	<i>Served as</i>
(1)	(2)	(3)
i)	465 m ²	Shallow/Walk-in rooms (Walk-in rooms shall be at least 1.3 m × 1.3 m. Shallow rooms shall be at least 0.6 m deep by 2.6 m wide.)
ii)	93 m ²	Wall cabinets, self-contained cabinets, enclosed cabinets

NOTE — Installation of active equipment in shallow or walk-in rooms is not recommended because many types of equipment require environmental controls and a depth of at least 1 m.

3.1.4 Entrance Facilities (EF)

Building entrance facilities (EF) provide the point at which outdoor cabling of the telecom service provider interfaces with the intra-building/intra-campus backbone cabling. In cases, where the functions of the EF are combined with the functions of the ER in the same space, the room may house equipment that is owned and maintained by service providers. In these cases, requirements specified by the service providers shall be considered when designing the ER.

If a separate space is required for service providers, it should be in or adjacent to the EF and may require a mesh partition or locked cabinet. A space of at least 1.2 m × 1.83 m shall be considered for each service provider. An entrance facility necessarily be an enclosed space. However, having an enclosed area or access protected area is suggested at all occasions, considering safety requirements.

The duct from the cable vault (CV) of the telecom service provider generally consists of 100 mm conduits encased in reinforced concrete. The exact number of conduits will depend upon the size and use of the building. These conduits may be used for all communication systems serving the building including additional capacity and access cables and pathways capacity. To maintain the safety of the TSP cable, it may preferably enter the building above the ground and then feed the equipment room.

A separate 230 V, 16/6 A circuit with double outlets should be provided (for plugging in temporary test equipment or to power the tools) every 2.4 m along the EF and ER combined wall. All wall outlets are to be located approximately 300 mm above the finished floor levels with the breaker and clearly labelled. In addition, two dedicated 230 V, 16/6 A sockets should be provided for each (permanently installed) communication service rack. If the sockets of racks are fed from UPS power, appropriate levelling may be provided to distinguish between main power sockets and UPS power sockets.

3.1.5 Telecommunication Enclosures (TE)

A TE is simply a case or housing for telecom equipment, cable terminations, and cross-connect cabling for distribution of telecom services on a floor. The TE may also contain access points for wireless services. Although, TEs serve much in the same way as that of a TR, a minimum of one TE should be located on each floor, if no TR is considered on the floor. The TE door(s) may be hinged or removable. If the enclosure consists of metallic components, it shall be earthed to either PBB or SBB.

3.2 Telecom Media and Connecting Hardware

3.2.1 A cabling system consists of cables, equipment cords, patch cords, and connecting hardware components like cable patch panel, MDF and FDMS. All balanced twisted-pair, optical fibre, and coaxial cabling systems are made up of such components. There can be different topologies for cabling used in a building like star, tree, bus bar, ring, etc, or a combination of these to suit different technological requirements. Within a building, most of the wiring is between the work spaces and the equipment room. Wireless systems can also be part of the telecom media. However, connectivity to the network, which delivers telecom service, is generally provided using cables at the back end.

The selection of appropriate style of cabling is important and accordingly, some of the user requirements to be

considered before selecting the specifications and style of cabling and connectors for a telecom enabled building may include the following:

- a) Number of user work areas and telecom spaces used to serve the building occupants;
- b) Types of services and technology used for delivery of services;
- c) Number of telecom outlets/connectors desired at each user work area;
- d) Number and styles of user equipment (for example, telephony, LAN, building automation);
- e) Cabling system transmission performance expectations;
- f) Backbone distances and horizontal cable lengths involved in the building or campus;
- g) Future growth expectations (for example, 15 percent to 20 percent recommended minimum growth factor);
- h) Environmental conditions including electromagnetic interference; and
- j) Other types of low voltage systems and their network requirements.

3.2.2 The cables should have ferrule marking at the terminations as well as along the length of the cable at suitable interval. Similarly, copper pair/fibre pair should be identifiable at terminating end for easy identification and convenience during maintenance. For the purpose, a suitable chart should be available at locations, like inside cross-connect box. Further, all the documentation of pathways and cables should be properly maintained.

3.2.3 While designing the cabling system for large multi-tenant building, it should be kept in mind that the tenants may prefer subscribing to the services of multiple service providers. Thus, enough space should be available in ducts, pathways, etc, for laying additional conduits/trays, keeping in view the present and future demands. Capacity for pathways (that is, conduit, raceways, trays, baskets) may be planned so that initial installation achieves a fill of 40 percent and at final installation, fill of 60 percent, taking into account the expected future requirements.

3.2.4 Various cabling systems are described under [3.2.4.1](#) to [3.2.4.3](#). For the purpose of reference, details about connecting hardware components and classification of cables by fire safety properties are given in [Annex A](#).

3.2.4.1 Copper twisted-pair

The transmission performance of balanced twisted-pair cabling and telecom associated components are based on a number of factors within the cabling or component design. These performance levels use the terms category and class. Standards developed internationally utilize both class and category, depending on the specific cabling element, as described in [Table 1](#).

Table 1 Balanced Twisted-Pair Cabling Channel Performance (Clause 3.2.4.1)				
SI No.	ISO Category/Class	TIA Category	Frequency MHz	Maximum Data Transfer Rate
(1)	(2)	(3)	(4)	(5)
i)	Category 3/Class C	Category 3	16	10 Mbps
ii)	Category 5/Class D	Category 5e	100	1 Gbps
iii)	Category 6/Class E	Category 6	250	1 Gbps
iv)	Category 6A/Class EA	Category 6A	500	10 Gbps
v)	Category 7/Class F	NA	600	10 Gbps
vi)	Category 7A/Class FA	NA	1 000	10 Gbps
vii)	Category 8/Class G	Category 8	1 500 to 2 000	25/40Gbps
NOTE — For Category 8/Class G balanced twisted-pair cabling, the maximum permissible distance is restricted to 30 m.				

A large number of cable designs are used in the ICT industry, resulting in various names and acronyms for their identification. Specialist publication (*see* NOTE) may be referred which gives balanced twisted-pair cable designations using x/y designation, where x is the overall screen type and y is the individual pair screen type, which may be used. Details of various nomenclature based on this, using various categories of cables, are given in [Annex A](#).

NOTE — The specialist publication referred here is ISO/IEC 11801 : 2002 ‘Information technology — Generic cabling for customer premises’.

3.2.4.2 Optical fibre

Optical fibre cables are used in backbone and horizontal cabling applications, when the service is delivered on fibre to the user or inter/intra building communication is to be carried on fibre. For example, of former is gigabit passive optical network (GPON), which require fibre to the home (FTTH) and the example of latter is LAN implementation using optical network. Transmission of information through optical fibre cables is not degraded by crosstalk, ambient noise, lightning, and most electromagnetic interference (EMI). It has better loss characteristics compared to copper pair. Thus, it permits high data rate communication to be carried out to a much larger distances than copper pair.

Optical fibres are classified as either single-mode or multi-mode. Single-mode optical fibres have a relatively small diameter featuring a core of 8 μm to 11 μm and a cladding diameter of approximately 125 μm . Multi-mode optical fibre has a larger core diameter (for example 50 μm or 62.5 μm) with the cladding of approximately 125 μm . Multi-mode fibre has higher loss compared to single-mode fibre. However, it is generally recommended for in building/intra campus use as it cost less than the single fibre. The latter has low loss characteristics and can carry higher speed to longer distances due to which it is mainly used in long haul applications. However, depending upon the size of the building/campus or availability of various types of cables, combination of both types of cables or only single-mode cable may need to be used. Use of single-mode fibre has the advantage that future upgradation requirements are taken care of right in the beginning.

Optical fibre cables are available in various capacities in terms of number of fibres like 6 fibre, 12 fibre, 24 fibre, 48 fibre, 96 fibre, etc. Each fibre has a different colour which helps in easier identification during installation and operations. The colour scheme is repeated for every group of 12 fibres for higher fibre count cables.

Selection of optical fibre cable size for different segments of cabling depends upon various factors.

The colour code chart for fibre optic cable is as under:

<i>Sl No.</i>	<i>Fibre/Tube No.</i>	<i>Color</i>
(1)	(2)	(3)
i)	1	Blue
ii)	2	Orange
iii)	3	Green
iv)	4	Brown
v)	5	Grey
vi)	6	White
vii)	7	Red
viii)	8	Black
ix)	9	Yellow
x)	10	Purple
xi)	11	Rose
xii)	12	Aqua

NOTE — The specifications of the certain types of optical fibre cables are available in Department of Telecommunications, Telecom Engineering Centre (TEC), Govt. of India, the details of which can be seen from TEC website, www.tec.gov.in

When it is decided to use optical fibre cable, choice is to be made from among variety of standard cables. There are two broad categories of cables namely multi-mode optical fibre (OM) and single-mode optical fibre (OS). The former is suitable for much shorter distances compared to the latter as, data rate supported by cable is a function of cable length used. OM cables are preferred cables for campus applications as they are economical for short distances. OM cables are further sub-categorized into OM1, OM2, OM3 and OM4, which are differentiated based on data speed and distance combinations. Once the data speed and distance to be covered is decided, choice of cable category becomes simpler. When used inside the buildings, attenuation of signals is caused due to cable fibres facing sharp bends. Multi-mode cables show larger attenuation than single-mode cables on bends inside the building. Thus, bend insensitive and fire retardant optical fibre cable as per relevant TEC specification are recommended for use in the buildings.

Considering that backbone cable carries the combined traffic of all work areas and the growth of data use with time, provision for future expansion should be provided. It is recommended that planning should be done for 10 Gbps to 1 Tbps or above speeds depending upon anticipated requirements of the users. Therefore, when using multi-mode cable, choice may need to be made typically from among OM3 and OM4 cables.

A chart showing typical length and data rate support characteristics of various cables is given below:

<i>Sl No.</i>	<i>Subsystem Backbone</i>	<i>Lengths Up to</i>	<i>Data Rates Up to</i>
(1)	(2)	(3)	(4)
i)	Campus backbones (OM1 fibre)	2 000 m	155 Mb/s
ii)	Campus backbones (OM2 fibre)	550 m	1 Gb/s
iii)	Building backbones (OM2 fibre)	300 m	1 Gb/s
iv)	Building backbones (OM3 fibre)	300 m	10 Gb/s
		100 m	100 Gb/s
v)	Campus/building backbones (OM4 fibre)	550 m	10 Gb/s
		150 m	100 Gb/s
vi)	Campus/building backbones (OS fibre)	10 000 m	100 Gb/s

Enhanced versions of OM3/OM4 cables can also be examined for use. Once the cable category is selected, care should be taken that active equipment which feeds the fibre is equipped with the corresponding interface, to take care of, for example requirement of different optical power source and optical wavelengths for different types of cables. Further, from each TR drop fibre or cable of 2 to 4 fibres are also recommended.

3.2.4.3 Coaxial cable

The predominant coaxial cables are Series-6, Series-11, and radio grade (RG) 59. These coaxial cables have a characteristic impedance of 75 Ω . While the termination procedures may be similar, special attention shall be paid to the manufacturer's specific instructions for termination and connectors. The cables/passive components used for DAS/IBS system should cover frequency range from 400 MHz to 4 000 MHz. Coaxial RF telecom cables of various sizes (for example, 12.70 mm, 22.23 mm, 31.75 mm) or as required for efficient RF signal transmission be deployed to ensure comprehensive mobile signal coverage within the building.

Coaxial cable is used for computer networks, CATV and video systems. Historically, coaxial cable was designated as RG cable. Coaxial cables used in broadband applications are available in the market with following constructional variations:

- Centre conductor diameter;
- Centre conductor being solid or stranded;
- Dielectric composition;
- Outer braid's percent of coverage; and
- Impedance.

3.2.5 Wireless Systems

Wireless connectivity is also provided inside the building for enabling access to public mobile network or for enabling internet/intranet connectivity with flexibility of user terminal movement. Various wireless systems are described below:

- a) *Wireless LAN (WLAN) access point (AP)* — A wireless LAN (WLAN) access point (AP) is a network device located in areas of a building or campus and placed in relatively close proximity to where users interact with their wireless enabled network devices. APs allow wireless enabled devices (for example, computer, printer, mobile, tablets, laptop, IoT devices or any other device) to connect through Wi-Fi or related standards. Wireless networks here can even include the networks of low voltage automation system in the building as well as M2M/IoT smart buildings/smart homes/smart offices. There are various wireless standards for M2M/IoT device connectivity. AP network devices are typically mounted on walls or ceilings with structured cabling that provides a physical connection to a HC (FD), which provides further connectivity to private/public network through the facility installed in TR/TE.
- b) *Wi-Fi access points (WAP) with centralized controller* — Access point shall support at least IEEE 802.11 ax/be or any higher versions/ more modern standards. There may be numbers of WAPs to cover area/building and in such situation user may prefer centralized controller. Centralized controller can be hardware or software based and depending upon number of WAPs may be provided at each ER/TR/TE. These access points can also be used for connecting the low voltage automation systems (building management system and other in-building system). The access points are connected to the network as described in (a).

NOTE — The secure access to Wi-Fi network and authorization of users is critical and the same will need to be ensured by owner.

- c) *Distributed antenna systems (DAS)/In-building solutions (IBS)* — DAS/IBS are signal distribution systems for strengthening the public mobile wireless signal inside the buildings or installations. They require cable and passive antenna to be installed in corridors/rooms, normally along the centre line. The active equipment, wherever required can be installed in ER/TR/TE, on which cables to/from DAS/IBS are connected.
- d) *Wireless repeater* — It takes an existing signal from a wireless router or wireless access point and rebroadcast it with amplified signal. It is also called wireless range extender. This is used to mitigate low signal issues.

While planning and installing wireless devices, it is required to maintain enough spacing between them to avoid frequency overlap and interference by their own signal. The wireless coverage area radius is impacted by a number of factors including the following, and these factors like penetration losses, signal to noise ratio etc, are to be taken into account by using for which the site survey approach and planning. An appropriate planning tool should be considered for this purpose:

- 1) Building materials (concrete, sheetrock, wood, steel, etc);
- 2) Building configuration (closed, semi-closed or open space); and
- 3) Building furnishings (cabinets, partitions, furniture, etc).

Periodic monitoring is desirable for maintaining wireless coverage.

3.2.6 Connecting Hardware

The ends of cable terminating in work areas or on intermediate equipment like LAN switches, splitters, etc, need to be provided with connectors. There can be variety of equipment which may be connected to the cable ends. Hence, a careful planning should be done to identify types of connectors with flexibility to connect variety of equipment keeping in view also the fact that the cable ends are permanently wired on to connector panels in the work areas. Details of the typical connecting hardware are given in [Annex A](#).

3.3 Backbone Cabling Media Distribution and Building Pathways

The backbone distribution system provides interconnection between telecom spaces, that is, telecommunication rooms, equipment rooms and entrance facilities. It consists of the backbone cables, intermediate and main cross-

connects, mechanical terminations and patch cords or jumpers used for backbone-to-backbone cross-connection. This includes:

- Vertical cable connections between floors (risers);
- Cables between an equipment room and building cable entrance facilities; and
- Cables between buildings (inter building).

The elements of the telecommunications cabling system structure are:

- Horizontal cabling;
- Backbone cabling;
- Work area;
- Telecommunications rooms;
- Equipment rooms; and
- Entrance facilities.

[Figure 1](#) illustrates a representative model for the various functional elements that comprise a building campus/complex telecommunications cabling system. It depicts the relationship between the elements and how they are configured to create a total system.

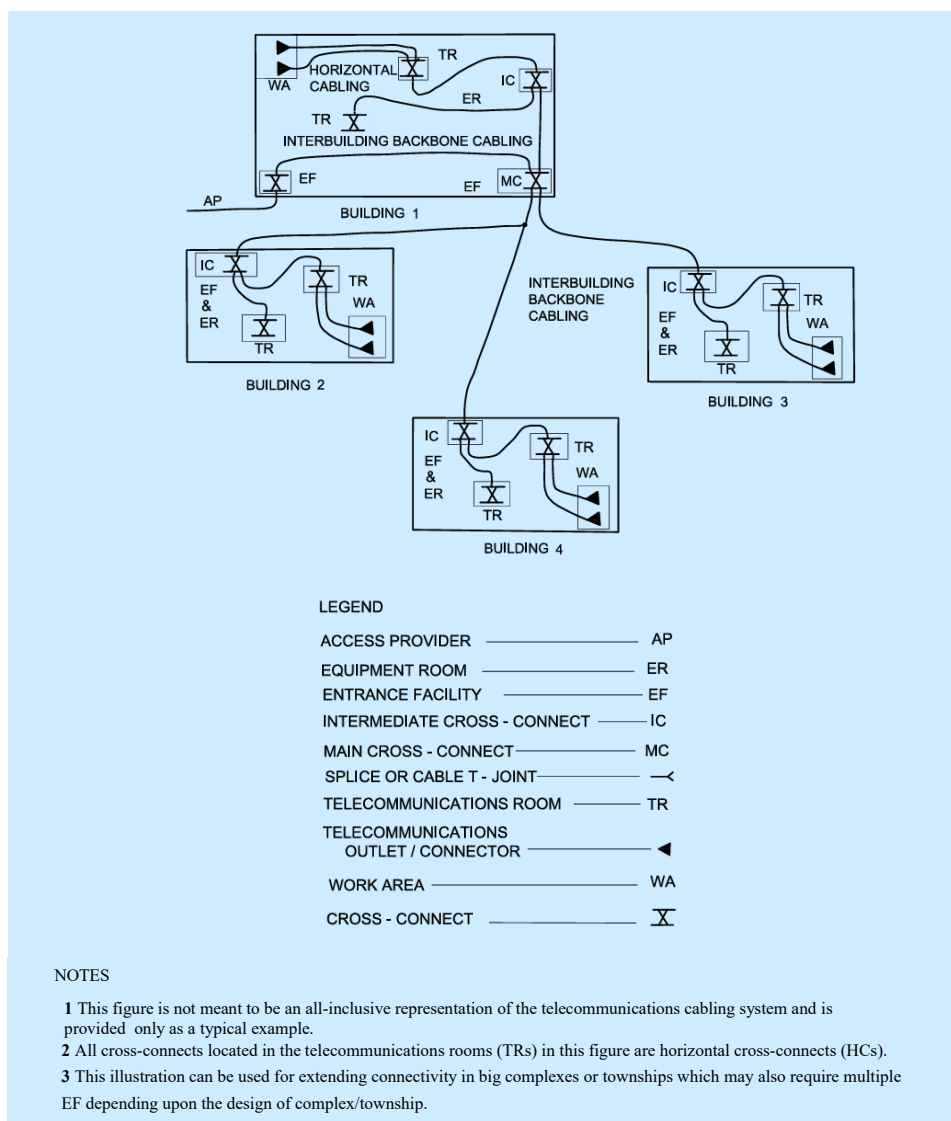


FIG. 1 TYPICAL TELECOMMUNICATIONS CABLING SYSTEM

3.3.1 Backbone Cabling and Topologies

Cabling between the building cross-connect (main or intermediate) and the horizontal cabling (floor distribution) [HC (FD)] is designated as backbone cabling. In a multi-building complex, inter-building cabling, though terminating in the EF, is designated as inter-building backbone cabling. The cable(s) can be of any type, for example, copper pair, optical fibre or coaxial.

3.3.1.1 The two primary options for connectivity are:

- a) Star topology, where the HC (FD) is connected directly to the main cross-connect (campus distributor) [MC (CD)] which is further connected to inter building backbone, where applicable; and
- b) Hierarchical star topology, where some or all of the HCs (FDs) are connected to an intermediate cross-connect (building distributor) [IC (BD)], which in turn, is connected to the MC (CD).

3.3.1.2 The best design is the star design between the building MC and the HCs. However, in some extremely large buildings (for example, high-rises), a hierarchical star may be an option for consideration. The trade-offs between different cable sizes and labour cost is to be considered to determine a suitable cost effective solution. The direct connections between HCs are generally avoided. Although this kind of connectivity might be of value in providing a redundant path, a user should design a link from HC to HC only in specific applications. The best design is the star design between the building MC and the HCs.

3.3.1.3 A typical star configuration in a building with MC (CD) or HC (FD) is given [Fig. 2](#).

3.3.1.4 An example of backbone hierarchical star topology with addition of IC (BD) is shown in [Fig. 3](#).

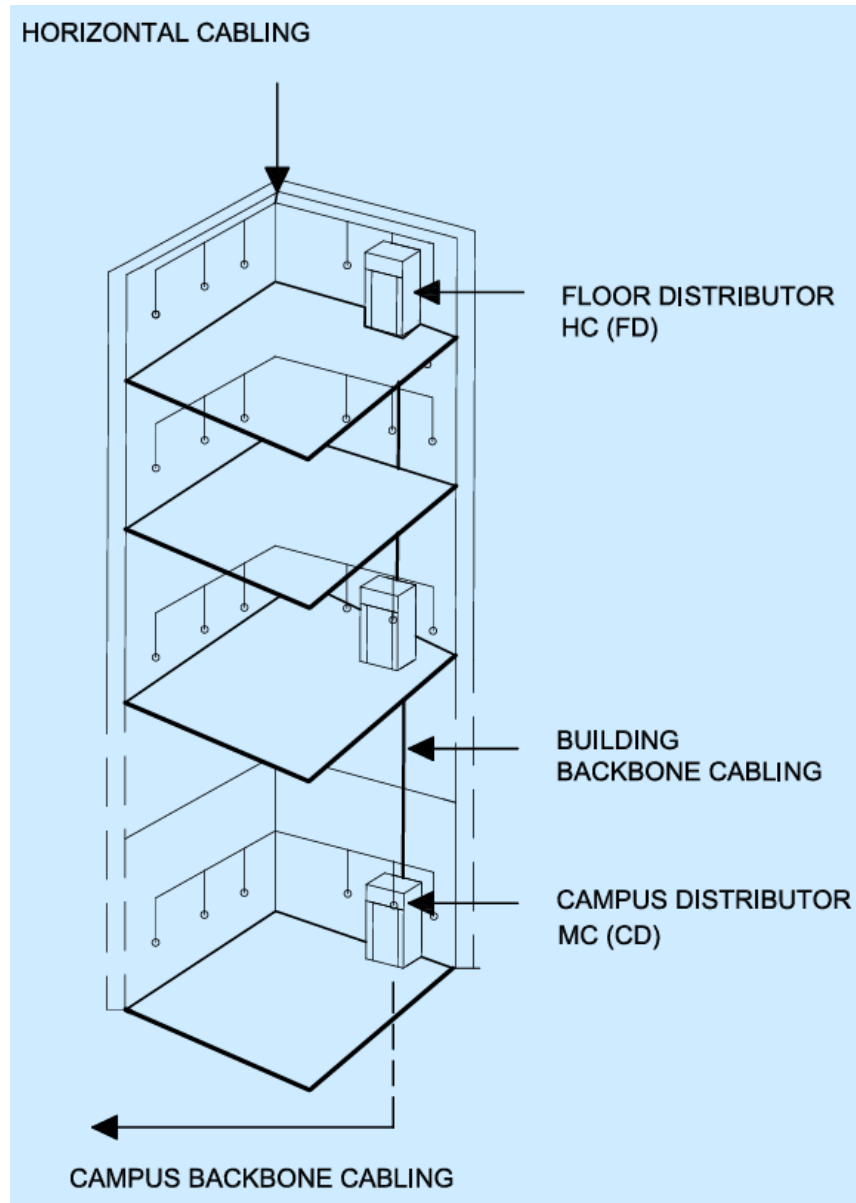


FIG. 2 TYPICAL CABLING STAR CONFIGURATION IN A BUILDING

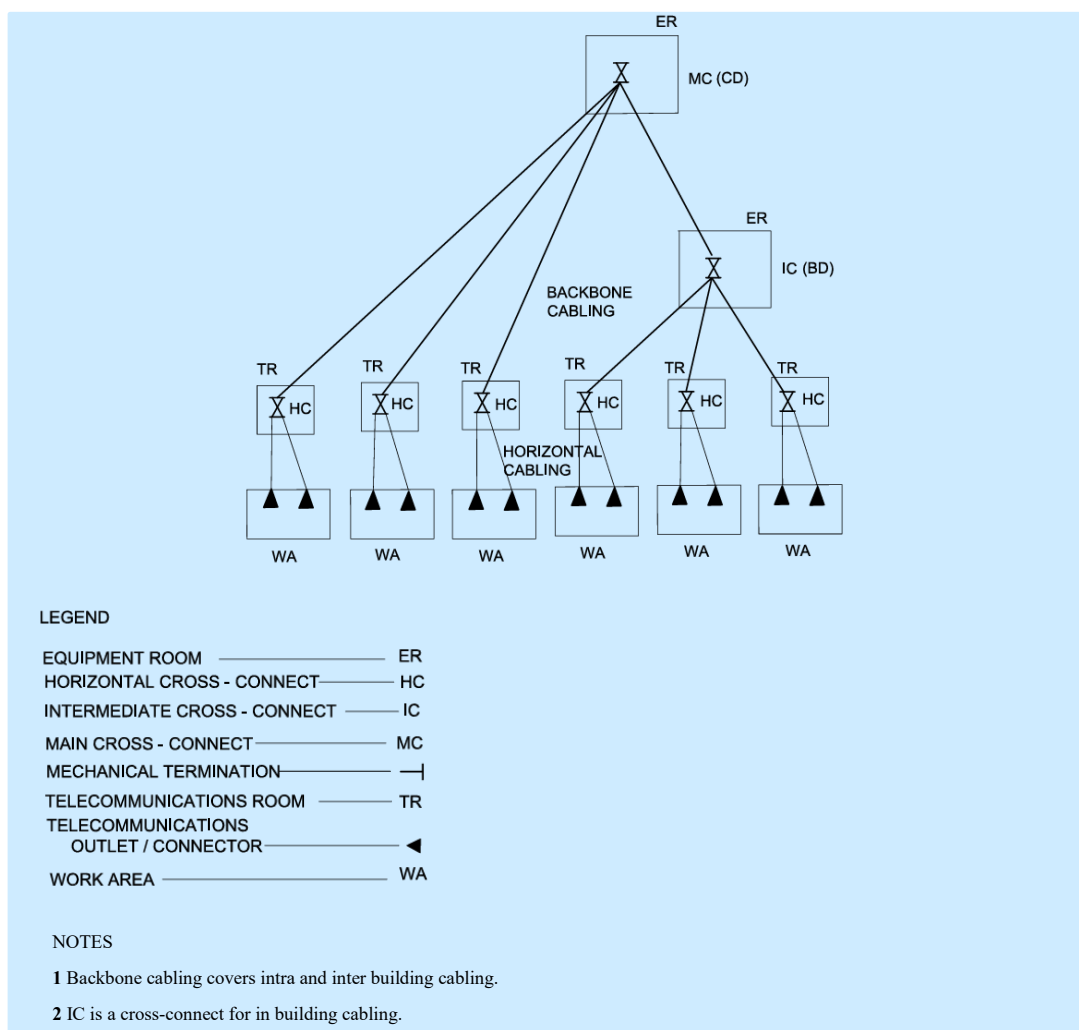


FIG. 3 EXAMPLE OF BACKBONE HIERARCHICAL STAR TOPOLOGY

3.3.2 Inter-Building Pathways

In a campus environment, inter-building pathways are required to connect separate buildings. For this, underground, buried, aerial and tunnel are the main pathway types used. As the complexes have varying conditions of land features and locations, size and use of building, choice if any or combination of the pathways need to be made based on local requirements. The cable running over the pathways generally terminates at the entrance facility of the building at both ends.

3.3.2.1 Underground inter-building backbone pathways

The pathway shall be fully covered and the cabling shall be installed within protective conduit or ducting for the entire external section of the cable route in all inter-building pathways. Provision should be made for existing as well as future demand. Underground pathways consist of conduit, ducts and troughs; possibly including manholes. The manhole can be planned normally, if:

- The cable requires joint;
- The distribution of copper/fibre pair is required midway; or
- Numbers of bends in the path are more than two.

Each manhole should be dimensioned to accommodate spare cable of 2 m to 4 m length in each direction, which is generally kept in spare loop to take care of maintenance needs. All conduit and duct shall have a diameter of

at least 100 mm. Bends are not recommended; however, if required, there should be not more than two 90° bends.

In a multi-building and multi-operator campus, an underground cable vault (UCV) may be considered close to entrance facility. Manholes/pull holes may also be provided for O&M activity in the outside plant pathway system for the pulling, placing, and splicing of cables. Size of UCV, manholes and pull holes should be decided based on the pathway capacity, number and types of cables, etc. All such points should be covered with properly marked iron covers of appropriate thickness and structural support below them keeping in view the load they are expected to bear due to pedestrian or vehicle movement.

In the planning for underground cabling, the following shall be considered:

- a) Limitations dictated by the topology (this includes land development);
- b) Grading of the underground pathway to permit proper drainage;
- c) Need to vent gaseous vapours; and
- d) Amount of vehicle traffic to determine the amount of cover over the pathway and whether or not concrete encasement is required.

3.3.2.2 Direct buried inter-building backbone pathways

In this case, the telecommunications cables are completely covered in earth. Direct burial of telecommunications cables is achieved by trenching, augering or boring (pipe-pushing). When selecting a route for the pathway, it is important to consider the landscaping, fencing, trees, paved areas and other possible services. Direct burial has the challenges of cable repair in case of faults and also in future upgradation.

3.3.2.3 Aerial inter-building backbone pathways

An aerial pathway consists of poles, cable-support strand and support system. Some considerations to make when using aerial backbone include the following:

- a) Appearance of the building and surrounding areas;
- b) Separation and clearances for electrical installations and roads;
- c) Span length, building attachments, storm loading and mechanical protection; and
- d) Initial number of cables and future growth potential.

3.3.2.4 Tunnel inter-building backbone pathways

Tunnels provide pathways for conduit, trays, wire-ways or support strand. Normally tunnels support other building services too. Hence, the location of telecom pathways within a tunnel should be planned to allow for accessibility as well as for separation from other services. In general, tunnelling inside the premises for telecom services alone may not be cost effective.

3.3.3 Backbone Building Pathways

Backbone building pathways can be built in various ways depending upon the building layout, locations of TR/TE, etc. While designing and implementing such pathways, certain requirements need to be met and precaution taken. Typical topologies and requirements/ precautions relating to these are described below:

- a) *Vertically pathway topology* — Vertically aligned TRs with connecting sleeves or slots are the most common type of backbone pathway though it is not a necessary condition. Vertical alignment is desirable because the building planner can stack them with other mechanical spaces, and it makes distribution of telecom cables more efficient because of shorter conduits and cabling runs. If the TRs/TEs are not aligned vertically above one another, the conduits interconnecting them will have horizontal offsets. There shall be not more than two 90° bends in any such conduit run. Any bend shall have an inside radius 10 times the diameter of the conduit (typically 1 m). If these conditions cannot be met, then intermediate pull boxes shall be used wherever a non-standard transition is required. This pull box shall be at least 600 mm × 600 mm in the plane of the attached conduits, and 150 mm deep. All conduits other than simple sleeves between floors shall be fitted with a continuous 1.5 mm nylon pull rope, or a 0.6 mm steel fish wire or a 0.5 mm fibre composite fish wire.

- b) *Conduits, trays, slots, sleeves, and ducts* — All rigid metallic conduit pipes shall be of steel and shall be conforming to accepted standard [D-6(1)]. The wall thickness of the pipe shall be not less than 1.6 mm (16 SWG) for conduits up to 32 mm diameter and not less than 2 mm (14 SWG) for conduits above 32 mm diameter. These shall be solid drawn or reamed by welding and finished with galvanized or stove enamelled surface.

All non-metallic conduit pipes and accessories shall be of suitable material complying with the accepted standards [D-6(2)] for rigid conduits, and accepted standards [D-6(3)] for flexible conduits. Such pipes shall not be exposed.

The vertical backbone pathway consists of telecommunications rooms located on each floor, tied together by sleeves or slots. In this context, the term ‘sleeve’ refers to a circular opening in a wall, ceiling or floor to permit the passage of cables between adjacent spaces. A ‘slot’ is the same as a ‘sleeve’, except that the shape of the opening is usually rectangular. The cable sleeves or slots are positioned adjacent to a wall on which the backbone cables can be supported. The recommendation in respect of slots and sleeves are as under:

- 1) Slots with a minimum 25 mm high curb; and
 - 2) Sleeves to extend a minimum of 25 mm above the floor level and a maximum of 77 mm above the floor level. Sleeves should be located at a minimum of 25 mm from the wall or between adjacent sleeves to provide room for bushings, but not so far from the wall that it becomes a tripping hazard or create too large a cable span from the sleeve to the backboard/tray.
- c) *Riser systems* — For connectivity to TR/TE on upper floors from the building EF, conduit or raceway should preferably be provided vertically. Vertically mounted ladder rack shall be mounted on the wall between incoming and outgoing sleeves within the TR/TE. The ladder rack should be used to provide strain-relief for cables transiting TR/TE within the riser system. If the building plan is large enough to require a second communications TR/TE on one or more floors, then the riser pathway should preferably be duplicated from the EF up to second set of TR/TE.
- d) *Vertical pathway size* — A minimum of three 100 mm diameter conduits, or equivalent raceway cross section, should be provided between the TR/TE on adjacent floors and between the EF and the first floor TR/TE.
- e) *Access to the roof* — For installations like antenna, powering the roof top/tower top equipment, access shall be provided through suitable number of conduits. This connectivity from the roof should be seepage and rodent proof. A minimum of two 75 mm and one 20 mm conduit shall extend from the topmost TR/TE to the roof terminated in a weatherproof metal enclosure. Sometimes, it may not be feasible or desirable to have direct vertical access, thus requiring horizontal extension from the rooms before vertical extension, the user may keep this in view for taking action with respect to architectural considerations.
- f) *Pathway finish details* — Conduit shall be free of burrs or sharp edges. Sheet metal sleeves, if used, shall have rolled edges. Conduits or sleeves shall protrude at least 25 mm from the surface they penetrate but not more than 75 mm. Conduits shall be fitted with a smooth bushing.
- g) *Fire protection of shafts* — All penetrations shall be filled with fire resistant material. Fire compartment plan and integrity of compartment should be maintained by either blocking permanently or by recognized systems automatically responding to temperature increase.
- h) *Open cable shafts* — Open cable shafts should be used when available and where large quantities of cables are required on a floor that is distant from the main ER. Backbone cable pathways shall not be located in elevator shafts.
- j) *Enclosed metallic raceways or conduits* — Enclosed metallic raceways or conduits are also used as vertical and horizontal cable pathways. It should be bonded to form a common bonding network.
- k) *Cable trays* — A cable tray can be used as a vertical cable pathway within shafts or as part of the pathway between vertically aligned TRs. A cable tray can be open or covered and provides a means for attaching vertical cable runs to the cable tray members. The cable tray can also accommodate ethernet LAN cables running from building management system and other low voltage automation system in the building.

3.3.4 Ethernet in the First Mile (EFM)

Ethernet in the first mile (EFM), also known as ethernet to the last mile, describes the access network from the

access point to the subscriber's premise. The first mile is the critical connection from business and residential users to the public and the public network.

Ethernet is the dominant mode of digital connectivity to user computers and other devices. Local area network (LAN) connectivity is extended to the end user computers through ethernet cable and connector. Ethernet cables generally used are Cat 5e and Cat 6. However, higher categories of cables are available (Cat 7 and Cat 8), which supports data rates of 10 Gbps and 40 Gbps. In many implementations, backbone and horizontal cabling may be on optical fibre. Such implementations address the Cat 5e/Cat 6 cable length limitations as only the last drop from TR/TE to the work area will use Cat 5e/ Cat 6 cable.

However, in many large buildings, optical to ethernet converters/switches may be required to be installed beyond TR/TE also, which may be installed in safe places in corridors false ceiling, work area, etc, where provision for supply of power should also be made. This holds true for Cat 5e/Cat 6 backbone and/or horizontal cabling also. Cabling to work area may be through conduits inside the wall or conduits on open surfaces. In cases where multiple tenants want to set up their own LAN, they may set up their system and horizontal cabling. However, they should preferably use backbone cabling of the building to reach EF for further connectivity to a telecom service provider for internet/leased bandwidth, etc.

3.4 Horizontal Cabling Media Distribution and Building Pathways

A horizontal distribution system consists of the horizontal cabling, the horizontal pathways supporting the horizontal cabling, and the telecom spaces that support the horizontal pathways. As horizontal distribution systems, cabling, and pathways often change direction, elevation, or physical orientation to accommodate obstructions, barriers, and other building systems, the use of the term horizontal in the name of the element does not require that the elements be placed or installed parallel to the ground or floor.

3.4.1 Horizontal Cabling Systems

The horizontal cabling is the portion of the telecommunications cabling system that extends from the work area telecommunications outlet/connector to the horizontal cross-connect in the telecommunications room (TR). The horizontal cabling includes horizontal cables, telecommunications outlet/connectors in the work area, mechanical terminations, and patch cords or jumpers located in the telecommunications room, and may include multi-user telecommunications outlet assemblies and consolidation points.

3.4.1.1 The following media types can be considered as options for horizontal cabling:

- a) Four-pair 100 Ω unshielded twisted-pair (UTP) or screened twisted-pair (ScTP) cables with a distance limit of 90 m taking care of 10 m patch cord/jumper cable required for work area.
- b) In case of optical fibre, OM cable normally of grade lower than or equal to the backbone cable may be selected. Such limitation is not faced if OS or appropriate ITU-T standard single-mode cables as mentioned in [3.2.4.2](#) are used. Distance limits with respect to data rate to be supported on horizontal cabling system may be seen from table under [3.2.4.2](#). Care should be taken that the equipment for corresponding fibre interface is selected.

While laying cables, maximum recommended length of cable by including provision for jumper cables/patch cords in TR and extension in work area should be kept in view.

3.4.2 The requirements for horizontal pathways, measures for avoiding electromagnetic interference in the design of cabling pathways are described in [3.4.2.1](#) and [3.4.2.2](#).

3.4.2.1 Horizontal pathways

Horizontal pathways are used for distributing, supporting, and providing access to horizontal cabling and telecom associated connecting hardware between the telecom outlets/connectors and the HC, typically located in the ER, TR or TE. Horizontal cabling is contained within horizontal pathways. Generally, the horizontal pathways are one of two types:

- a) Continuous pathways (for example, conduit, cable tray, cable matting) used for containment of telecom cabling.
- b) Non-continuous pathways (for example, the space between cable supports such as J-hooks) through which cables are placed between physical supports or containment components.

A pathway component should be designed to accommodate all standards compliant cabling and address the potential need for change and expansion during the life cycle of the cabling system and building. The emphasis should be first on the design of pathway systems and then on the cabling systems design. This approach helps to ensure a robust pathway system that supports the cabling installation over the facility's life cycle. As frequently accessing or changing the horizontal cabling leads to disruption to occupants, the choice and layout of horizontal cabling types are important to the design of the building structured cabling system. The following should be considered for the horizontal distribution system's design:

- 1) Allow for the accommodation of change over the facility's life cycle with the goal of reducing long-term maintenance and operational costs;
- 2) Utilize standardized cabling, components and systems; and
- 3) Include appropriate pathway and cabling components to accommodate ease of access and a variety of user specified technology applications.

The horizontal cabling system should be designed in order to support various telecom applications, including:

- i) Voice services;
- ii) Data services;
- iii) Audio and video services; and
- iv) Building signalling systems [for example, smart building ICT systems, building automation systems (BAS), and fire safety and security systems].

It is preferable to have provision for conduits, junction box, etc, during planning and construction stage for providing radio/wireless data equipment at the centre line of rooms and corridors to avoid unplanned provisions later on.

3.4.2.2 Avoiding electromagnetic interference (EMI)

Avoiding electromagnetic interference (EMI) is an important consideration in the design of cabling pathways. Providing physical separation from sources of EMI for these elements of the telecom infrastructure inherently provides separation of their contents (for example, cable and connecting hardware).

The telecom pathways should be located away from sources of EMI to the extent feasible, including:

- a) Electrical power cabling and transformers;
- b) RF sources;
- c) Large motors and generators;
- d) Induction heaters;
- e) Arc welders;
- f) X-ray equipment; and
- g) Photocopy equipment.

3.4.2.3 Topology

Horizontal cabling should be installed in a physical star topology. Each telecom outlet/connector should be cabled directly to an HC (FD) in the appropriate telecom space. Exceptions to this practice are possible, when:

- a) A consolidation point (CP) or multi-user telecommunications outlet assembly (MUTOA) is used to connect the open office cabling;
- b) A transition point (TP) is required to connect to under-carpet cabling; or
- c) Centralized optical fibre cabling is implemented from main cross-connect (campus distributor) [MC (CD)] to the work area(s).

Some applications may utilize a bus, ring, or tree topology, which can be implemented within a physical star topology. However, in case of large number of work areas, ring or tree topology may require intermediate distribution points like MUTOA, etc. Typical horizontal and work area cabling using a star topology is shown in Fig. 4.

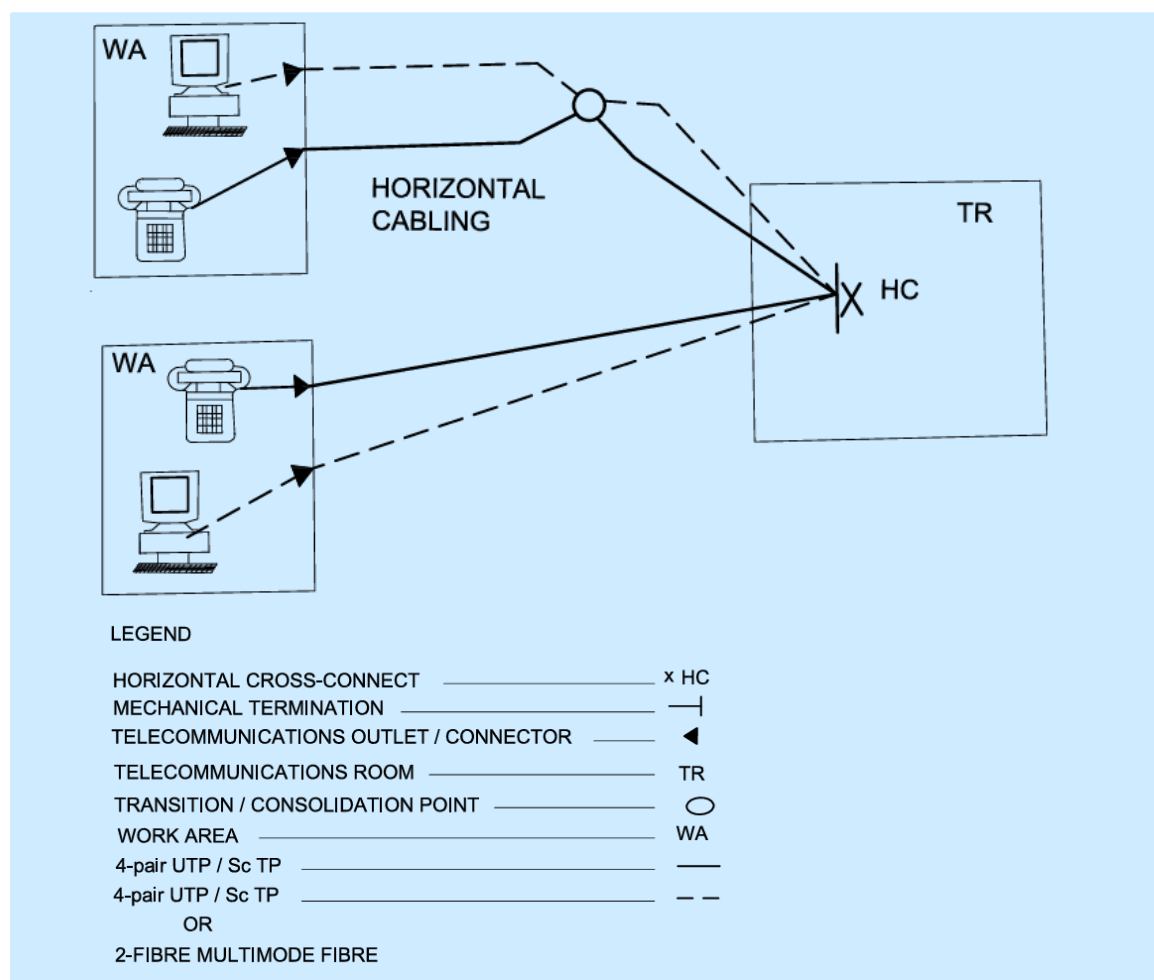


FIG. 4 TYPICAL HORIZONTAL AND WORK AREA CABLING USING A STAR TOPOLOGY

3.4.2.4 Work areas and open office cabling

3.4.2.4.1 The work area includes those spaces in a building where occupants normally work and interact with their ICT equipment. While work areas have traditionally been fixed, discrete locations, open office cabling design practices have introduced flexible layouts to support collaborative work by small teams. Such spaces are often rearranged to meet changing requirements of group work. Many other open office work situations also require frequent reconfiguration. An interconnection in the horizontal cabling allows open office spaces to be reconfigured frequently without disturbing horizontal system cabling runs. Work area equipment that may require access to the horizontal cabling includes:

- Telephones;
- Networking equipment;
- Fax machines;
- Computers;
- Network peripherals; and
- Any device plugged into a telecom outlet/connector that is located within the work area.

3.4.2.4.2 The key elements of open office cabling are the MUTOA and CP. To accommodate equipment in the work area, following components are typically used as needed:

- a) Telecom outlet/connector;
- b) Work area equipment cords;
- c) MUTOAs and CPs; and
- d) WAPs.

Some of these components and requirements or recommendations relating to them are described below:

- a) *Multi-user telecom outlet assembly (MUTOA)* — The MUTOA serves as a method of connecting more than one user (work area) to the horizontal cabling system. MUTOAs may be advantageous in open office spaces that are moved or reconfigured frequently. A MUTOA facilitates the termination of horizontal cabling system cables in a common location within a furniture cluster or similar open area. The use of MUTOAs allows the horizontal cabling to remain unchanged when the open office plan is changed. Work area equipment cords originating from the MUTOA should be routed through work area pathways (for example, furniture pathways). Each furniture cluster should have one MUTOA which may typically serve 12 work spaces. The work area equipment cords shall be connected directly to work area equipment without any additional connections. For copper cables, any combination of horizontal, work area cables and equipment cords may not exceed 100 m. The work area cables shall be connected directly to work station equipment without the use of any additional intermediate connections (see [Fig. 5](#)); and
- b) *Consolidation Point (CP)* — The consolidation point (CP) is an interconnection point within the horizontal cabling system. Like the MUTOA, a CP may be used for balanced twisted-pair cabling or optical fibre cabling. The functional difference between the CP and the MUTOA in the open office environment is that the CP introduces an additional connection for each horizontal cabling run. A CP may be useful when reconfiguration is not so frequent as to require the flexibility of the MUTOA. The CP may be located in the suspended ceilings, access floors, modular office furniture or work area. Further CPs may serve a zone consisting of number of work areas even in different rooms.

Some additional considerations and guidelines that apply specifically to the CP are as under:

- i) Cross-connections should not be used at a CP. Not more than one CP should be used within the same horizontal system cable run so as to avoid additional points of failure. Depending upon the size and number of the cables, there can be a single CP for all cables or separate CPs for different sets of cables;
- ii) For balanced twisted-pair cabling, the CP should be located at least 15 m from the HC (FD) in order to reduce the effect of near end cross talk (NEXT) and return loss from multiple connections in close proximity;
- iii) CPs shall be located in fully accessible and permanent locations. CPs shall not be located in an obstructed area; and
- iv) The CP should be sized and cabled so that it meets the telecom requirements of the zone it serves. If the floor space requirements change for an existing CP, then the CP should be reconfigured to accommodate the new requirements (see [Fig. 6](#)).

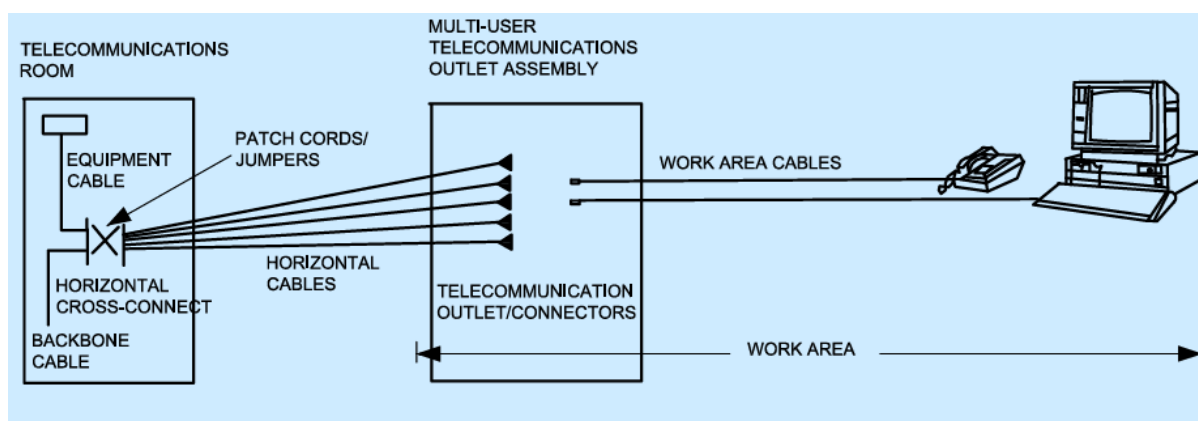


FIG. 5 APPLICATION OF MULTI-USER TELECOMMUNICATIONS OUTLET ASSEMBLY

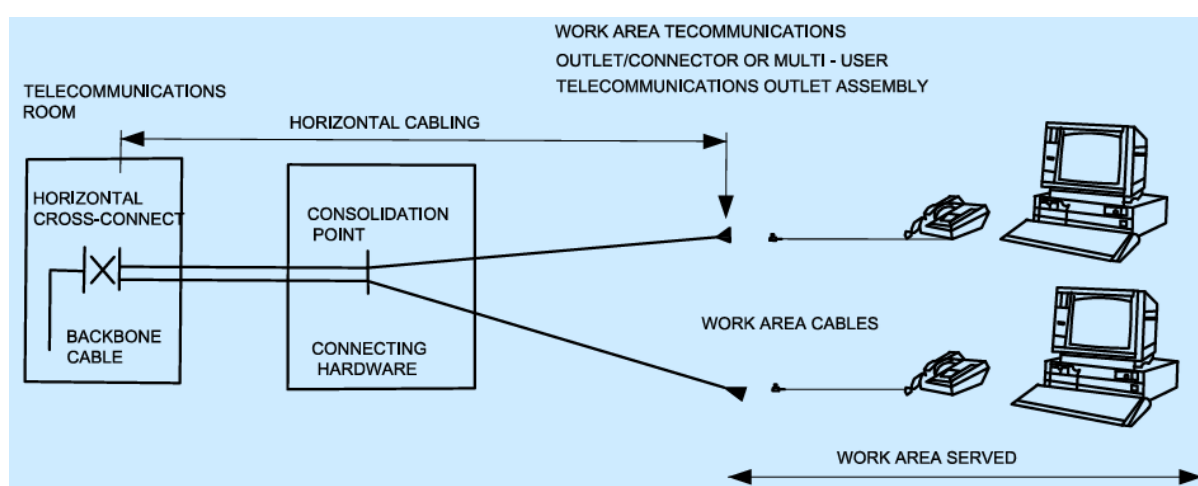


FIG. 6 APPLICATION OF CONSOLIDATION POINT

3.4.2.5 Centralized optical fibre cabling (fibre to the desk)

The HCs (FD), deployed throughout a building and located on each floor of a building, offers maximum flexibility to the user, especially in the deployment of distributed electronics or in multi-tenant buildings. In spite of the advantages of distributed cross-connections, some users may prefer data networks with centralized electronics which requires centralized cabling to provide connections from the work areas to the centralized cross-connect. Any of the following methods can be used for cabling:

- a) Pull-through cabling from the centralized cross-connection;
- b) Interconnection cabling in a floor-serving telecom space; and
- c) Spliced cabling in a floor-serving telecom space.

However, flexibility for rearrangement of work areas and need for additional work areas in future by the user should be kept in view while adopting any of the above methods.

3.5 Building Information for Telecommunication Infrastructure Planning

For facilitation of proper design or upgradation of telecommunication infrastructure, it is important to have some basic information about the building and its users. For this purpose, a template containing a typical set of data points is provided in [Annex C](#). Additional details may also be included as deemed necessary, so as to meet the objective of ensuring appropriate planning, design, and implementation of telecommunication infrastructure and services.

4 SPECIFIC REQUIREMENTS FOR TELECOM INFRASTRUCTURE CABLING

4.1 Telecommunications Bonding and Grounding

The bonding and earthing infrastructure of a telecom installation is an essential part of an information technology systems (ICT) design. Earthing systems are an integral part of the signal or telecommunications cabling system that they support. In addition to helping protect personnel and equipment from hazardous voltages, a proper earthing system may reduce electromagnetic interference (EMI) to and from the telecommunications cabling system. Improper earthing can produce induced voltages and those voltages can disrupt other telecommunications circuits. Earthing and bonding shall meet the appropriate requirements and practices of applicable standards.

The overall purpose for the ICTs bonding infrastructure is to equalize potentials between metallic surfaces (exposed and extraneous conductive parts) predominantly in the event of lightning, a.c. electrical system faults, electromagnetic induction, or electrostatic discharge.

The primary bonding busbar (PBB) serves as the dedicated extension of the building's protective earthing system for the ICTs infrastructure from the main earthing terminal (MET). It serves as the central attachment point for the telecom bonding backbone (TBB). Secondary bonding busbar (SBB) is the earthing connection point for ICTs infrastructure systems (for example, cabling, pathways) and ICTs equipment in the area served by an ER or TR. The PBB and-SBB should be a pre-drilled copper bus bar with holes for use with standard-sized lugs. A TBB is a ICTs bonding conductor to connect equipment/TRs/TEs on multiple floors of a building with an ultimate connection to the PBB. Typical telecommunication earthing and bonding arrangement is shown in [Fig. 7](#).

PBB is recommended for minimum dimensions of 6.35 mm thick, 101.6 mm wide and variable in length and the recommended minimum dimension for SBB is 6.35 mm thick and 50 mm wide. The TBB conductor sizing are as under:

<i>Sl No.</i>	<i>Maximum PBB-SBB length (l)</i> m	<i>TBB cross-sectional area</i> mm ²
(1)	(2)	(3)
i)	$l \leq 4$	16
ii)	$4 < l \leq 6$	25
iii)	$6 < l \leq 8$	35
iv)	$8 < l \leq 10$	35
v)	$10 < l \leq 13$	50
vi)	$13 < l \leq 16$	60
vii)	$16 < l \leq 20$	70
viii)	$20 < l \leq 26$	95
ix)	$26 < l \leq 32$	120
x)	$32 < l \leq 38$	150
xi)	$38 < l \leq 46$	150
xii)	$46 < l \leq 53$	185
xiii)	$53 < l \leq 76$	150
xiv)	$76 < l \leq 91$	300

NOTE — For lengths in excess of those shown above, the conductor cross-sectional area should be calculated as 3.3 mm² per meter. The figures given in parentheses are in AWG.

4.1.1 Meshed Bonded Networks

The mesh bonded networks provide enhanced immunity to EMI compared to that provided by the other bonding networks. This may be preferred for application, where high density of ICT equipment are installed, for example data centres, hospitals, airports. A merged MESH-BN and protective earthing network in buildings shall include the interconnections to the integrated lightning protection system, bonding measures of antenna installations (including satellite receiving equipment under private property) and cable networks, bonding measures of information technology cabling. The typical indicative diagram for meshed bonded network is given in [Fig 8](#).

NOTE — For further information refer special publication, ISO/IEC 30129 : 2015 'Information technology — Telecommunications bonding networks for buildings and other structures'.

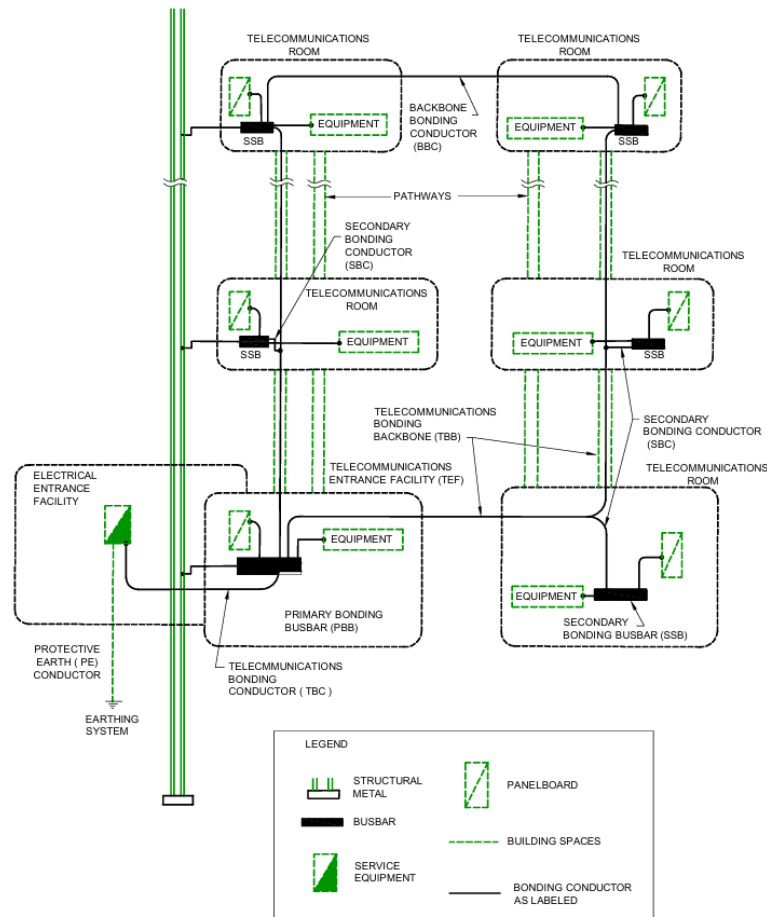


FIG. 7 TYPICAL TELECOMMUNICATION EARTHING AND BONDING ARRANGEMENT

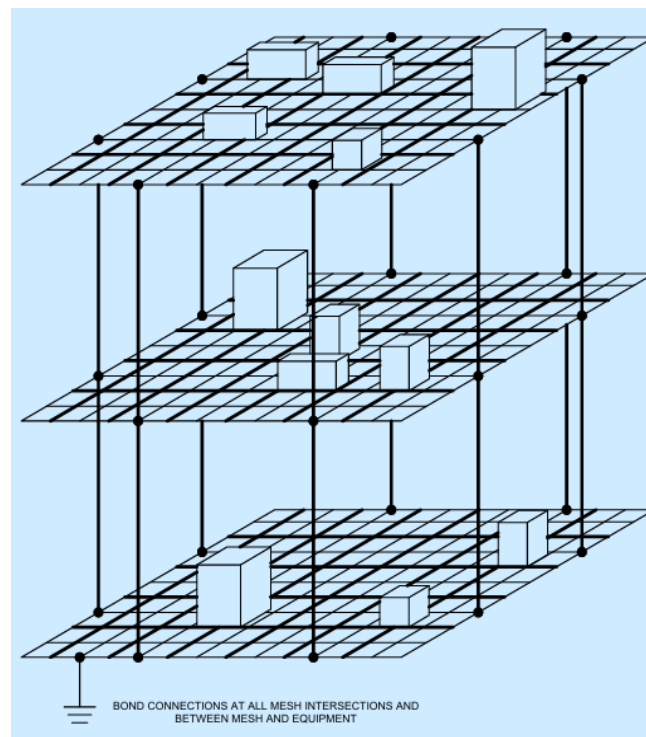


FIG. 8 MESH-BN WITH EQUIPMENT CABINETS, FRAMES, RACKS, CBN AND GLOBAL EARTHING BONDED TOGETHER

4.2 Installation and Workplace Safety

4.2.1 Cabling Installation Guidelines

The installation of system components has a tremendous effect on the final performance level of the network; therefore, it is essential to ensure that the performance of the entire network is not diminished through improper installation. The cables can be easily damaged if they are improperly handled or installed. It is imperative that certain procedures be followed in the handling of these cables to avoid damage and/or limiting their usefulness.

Care shall be taken not to stretch or abrade cables during installation, that is, the pulling tension for cables shall not be exceeded. Cables that pass through the infrastructure of the building shall be suitably protected against damage. Through walls and floors, this shall involve an appropriate type of sleeve; through any form of metalwork or stiff plastic then a rubber grommet shall be used.

To ensure cable management and also strain relief, cables shall be properly dressed using velcro cable ties. However, cable's ties should never be over tightened. On vertical runs, the cables shall be dressed and tied from the bottom up, thus putting minimum strain on the cables. In order that the system may be easily re-routed, or damaged sections quickly replaced, free access to the cable should be ensured and where possible, it is important to leave draw cords in ducting, piping, etc, for future use.

4.2.1.1 Improvement areas

- a) *Do not exceed the maximum tensile load* — On runs from 40 m to 100 m, use proper lubricants and make sure they are compatible with the cable jacket. On runs over 100 m, use proper lubricants and pull from the middle out to both ends. If possible, use an automated puller with tension control or at least a breakaway-pulling eye;
- b) *Maintaining minimum bending radius* — Sharp bends in the cable may damage the insulating material, thus causing unacceptable losses in the transmission medium. Therefore, the internal radius of every bend in a cable shall be such as not to cause damage to the cable, nor impair the characteristics of the cable;
- c) *Proper cable slack at outlet points and patch panels* — Install the system such that sufficient slack remains to enable re-termination of the outlets, a minimum of twice, and a limited scope for movement of the cabinets;
- d) *Patch panels and cable management* — Where possible, patch panels shall be installed within the communications units from the top, continuing downwards;
- e) *Electromagnetic compatibility (EMC)* — As a passive medium, structured cabling need not comply with the EMC regulations. However, telecom designers should be aware that cabling, when connected to transmission equipment, could radiate, receive and conduct electromagnetic disturbances and act accordingly. When crossing mains cables, this shall be done at right angles;
- f) *Maintaining proper cable routes* — Cable shall not be routed over pipes, conduits, other cabling, ceiling tiles, etc, but shall rest directly on the supporting surface so as to minimize the potential for sharp bends, kinks, etc. Every cable used shall be supported in such a way that it is not exposed to undue mechanical strain so that there is no appreciable mechanical strain on the terminations; and
- g) *Labelling* — The cable shall be clearly labelled at both ends, as outlined in the documentation and/or drawing. Each 8P8C connector (information outlet) shall be individually labelled. The label shall contain a unique identification, as outlined in the documentation and/or drawing, and shall be indelible and placed behind a transparent cover. At the patch panels, each socket shall be labelled according to its corresponding outlet identification.

4.2.2 Workplace Safety

Following measures shall be ensured for workplace safety:

- a) To maintain workplace safety, personal protective equipment (PPE) like protective clothing, helmets and

goggles should be used to protect the personnel from injury or infection from hazards such as physical, electrical, heat, chemical and biohazards;

- b) Lighting in telecom spaces shall be provided in an adequate amount such that continuing work operations, routine observations, and the passage of employees can be carried out in a safe and healthful manner. Certain specific tasks in centres, such as splicing cable and the maintenance and repair of equipment frame line-ups, may require a higher level of illumination and adequate provisioning should be provided for the purpose; and
- c) While working with optical fibre installation, the real issue of eye safety is getting fibre scraps into the eye. The broken ends of fibres and scraps of fibre created during the termination and splicing may also be dangerous. Therefore, protective eyewear such as safety glasses should be worn by personnel while carrying out splicing and termination operations. Further, fibre optic splicing and termination use various chemical cleaners and adhesives as part of the processes. Therefore, careful handling of fibre, cleaner and adhesive shall be ensured by the use of safety gloves and respiratory masks.

5 OTHER APPLICATIONS IN TELECOM INFRASTRUCTURE CABLING

5.1 Electronic Access Control (EAC)

EAC is important for overall personal safety and the protection of physical and intellectual property. EAC devices can include locks, integrated electronic devices controlling a single door or room, or a complex system of interconnected electronic devices controlling a zone, building, or campus. Access to private or secured spaces can be controlled in a great variety of methods. In addition, the user often has multiple levels of access required within a space. Personnel may have access to any given number of these. This access level may be required to change during the course of the day, week, or month. In contrast to a lock-and-key system, a modern computer-supported control system can meet these and many other user goals. This system employs programmable EAC. Time-of-day and day-of-the-week access levels can be applied to all personnel who have authorized entry. Additional precautions should be taken against weaknesses in the EAC system (for example, piggybacking, tailgating, unauthorized entry). A great number of possibilities exist from the passive card to biometrics and any combination in between. A user may issue smart cards.

A typical EAC cable consists of four individually shielded pairs that are used for the door strike, card reader, door contact, and request for exit. The four individual wire pairs may be supplied individually or in a group. Coloured foil shields or jacket colours may be used to designate the intended purpose of each of the wire pairs within the cables.

In a shared communication environment, the EAC data travels along with other building systems and data networks packets on the same physical network. This is often accomplished through an ethernet connection. It may also use building automation networks. Another media for shared communication relies on a private branch exchange and local analog telephone line (See [Annex A](#) for details).

5.2 Video Surveillance

Video surveillance is the extension of human vision to areas requiring surveillance. Some primary applications of this technology include investigation, prosecution, deterrence, observation and intrusion detection. In addition to traditional capture devices that operate within the visible band of the electromagnetic spectrum, other technologies provide unique viewing capabilities using IR, thermal and film cameras. (See [Annex A](#) for details.)

ANNEX A

(Clauses [3.2.4](#), [3.2.4.1](#), [3.2.6](#), [5.1](#) and [5.2](#))

ADDITIONAL GUIDELINES FOR INFORMATION AND COMMUNICATION ENABLED INSTALLATIONS

A-1 GENERAL

This Annex contains information which supplements the text of this Section. The information can be used to make decisions while exercising options from among various types of hardware, cables, etc, and also for guidance.

A-2 CONNECTORS AND CONNECTING HARDWARE

Various types of connecting hardware may be as described below.

A-2.1 Balanced Copper Twisted-Pair Connectors:

- a) *Insulation displacement contact (IDC) connectors* — The insulation displacement contact (IDC) is a gas-tight physical contact between two electrical conductors. The gas-tight contact is established by a cold weld with the elimination of the air gap between the conductor and the IDC and therefore the possibility of contact interface corrosion. Such contact creates a reliable, long-lasting connection with stable electrical properties. IDC connectors also eliminate conductor preparation (for example, insulation removal), reducing the termination time and the number of tools;
- b) *Modular plug* — Modular plugs have IDC contacts designed for either stranded or solid conductors as well as connectors having universal contacts that accept both stranded and solid conductors. Materials used to build modular plugs are typically flame retardant polycarbonate (body) and phosphor bronze with gold plating over nickel in contact area (contacts); and
- c) *Modular jack* — Modular jacks are available in various sizes and shapes (keyed and unkeyed). The number of positions indicates the connector's width, while the number of contacts installed into the available positions indicates the maximum number of conductors the connector can terminate.

A-2.2 Optical Fibre Connectors

The most common optical fibre interfaces include:

- a) *LC connector* — It is a simplex connector that can be converted to a duplex using a clip. It is keyed, low-loss, pull-proof and wiggle-proof. It can be terminated in many different ways, including using anaerobic (quick cure) adhesive, cleave and crimp, and hot melt. This is often referred to as a small form factor (SFF) connector. The LC connector provides a pull-proof design and small size perfect for high-density applications. It is available in simplex or duplex versions. The LC connector is provided with a 1.25 mm ferrule. It also incorporates a unique latching mechanism providing stability in system rack mounts.
- b) *Subscriber connector (SC)* — It is a simplex connector that can be converted to a duplex using a clip. It is keyed, low-loss, pull-proof, and wiggle-proof. It can be terminated in many different ways, including using anaerobic (quick cure) adhesive, cleave and crimp, and hot melt. The SC connector is a snap-in connector that latches with a simple push-pull motion, with a 2.5 mm ferrule and is widely used for its excellent performance. It is also available in a duplex configuration.
- c) *Straight tip (ST) compatible* — It is a simplex connector. It is a keyed, low loss connector. It can be terminated many different ways, including using anaerobic (quick cure) adhesive, cleave and crimp and hot melt.
- d) *Splices (Optical fibre connectors)* — These are used for adding optical fibre cable lengths. There are two primary splicing methods for optical fibres, fusion and mechanical. Both methods are field proven and have excellent long-term reliability when completed according to the manufacturer's instructions. Splices and stripped optical fibre cables are protected and secured by a splice enclosure. When a splice enclosure is used for splicing inside a building, it is generally secured to a rack or wall. In both cases, the splice enclosure contains the optical fibre splices in splice trays or organizers, typically in groups of 6, 12, 24, or more optical fibres per splice tray or organizer.

A-2.3 Coaxial Connectors

Connectors are installed on the end of a coaxial cable to provide electrical and mechanical connection to a system component. Either male or female connectors can be attached to coaxial cable, but most installations use male connectors on cable ends.

There are many styles of coaxial cable connectors. Three popular styles are:

- a) BNC-style connector;
- b) F-style connector; and
- c) N-style connector.

A-2.4 Connecting Hardware

Cables are terminated in outlets in work areas or intermediate equipment through connecting hardware. Basic connecting hardware styles are:

- a) Telecom outlets/connectors, including multiuser outlets;
- b) Patch panels;
- c) Connecting (wiring) blocks;
- d) Cable assemblies;
- e) Optical distribution frames; and
- f) Splices.

To facilitate their installation, servicing, administration and maintenance, connecting hardware of different styles of connectors may be considered to be grouped in one unit. However, depending upon the given situation, one or more type of connecting hardware (copper pair, coaxial or optical) may be installed on separate panels/frames.

A-3 CLASSIFICATION OF CABLES BY FIRE SAFETY PROPERTIES

A-3.1 Twisted Pair Cables

Different communication cable types and their markings on cable sheath are as given below:

<i>Sl No.</i>	<i>Marking</i>	<i>Cable Type</i>
(1)	(2)	(3)
i)	CMP	Communications plenum cable
ii)	CMR	Communications riser cable
iii)	CMG/CM	Communications general-purpose cable
iv)	CMX	Communications cable, limited use
v)	CMUC	Under carpet communications wire and cable

A-3.1.1 Type CMP (*Communications Plenum Cable*)

Type CMP communications plenum cables are suitable for use in ducts and plenums and they have adequate fire-resistant and low smoke-producing characteristics.

A-3.1.2 Type CMR (*Communications Riser Cable*)

Type CMR communications riser cables are suitable for use in a vertical run in a shaft when penetrating one or more floors as they have fire-resistant characteristics and thus be capable of preventing the carrying of fire from floor to floor.

A-3.1.3 Type CMG/CM (*Communications General-Purpose Cable*)

Type CMG general-purpose communications cables are suitable for general-purpose communications use, with the exception of risers and plenums, and these cables are resistant to the spread of fire.

A-3.1.4 Type CMX (*Communications Cable, Limited Use*)

Type CMX limited-use communications cables are suitable for use in dwellings and raceways as being resistant to flame spread.

A-3.1.5 Type CMUC (*Under Carpet Communications Wire and Cable*)

Type CMUC under carpet communications wires and cables are suitable for under carpet use and they are resistant to flame spread.

A-3.2 Optical Fibre Cables

Different optical fibre cable types and their markings on cable are as given below:

<i>Sl No.</i>	<i>Marking</i>	<i>Cable Type/Suitability</i>
(1)	(2)	(3)
i)	<i>OFNP</i>	Non-conductive optical fibre plenum cable
ii)	<i>OFCP</i>	Conductive optical fibre plenum cable
iii)	<i>OFNR</i>	Non-conductive optical fibre riser cable
iv)	<i>OFRC</i>	Conductive optical fibre riser cable
v)	<i>OFNG</i>	Non-conductive optical fibre general-purpose cable
vi)	<i>OFCC</i>	Conductive optical fibre general-purpose cable
vii)	<i>OFN</i>	Non-conductive optical fibre general-purpose cable
viii)	<i>OFC</i>	Conductive optical fibre general-purpose cable

A-3.2.1 Types *OFNP* and *OFCP*

Types OFNP and OFCP non-conductive and conductive optical fibre plenum cables are suitable for use in ducts and plenums and they have adequate fire-resistant and low smoke-producing characteristics.

A-3.2.2 Types *OFNR* and *OFRC*

Types OFNR and OFRC non-conductive and conductive optical fibre riser cables are suitable for use in a vertical run in a shaft when penetrating one or more floors and have the fire-resistant characteristics capable of preventing the carrying of fire from floor to floor.

A-3.2.3 Types *OFNG* and *OFCC*

Types OFNG and OFCC non-conductive and conductive general-purpose optical fibre cables are suitable for general-purpose use, with the exception of risers and plenums, and these cables are resistant to the spread of fire.

A-3.2.4 Types *OFN* and *OFC*

Types OFN and OFC non-conductive and conductive optical fibre cables are suitable for general-purpose use, with the exception of risers and plenums and have resistance to the spread of fire.

A-4 BALANCED TWISTED-PAIR CABLE NOMENCLATURE

Details of nomenclatures of twisted pair cables using various combinations of overall screen type and individual

pair screen type, as follows, are given in [Table 2](#):

<i>Sl No.</i>	<i>Overall Screen Type (x)</i>	<i>Individual Pair Screen Type (y)</i>
(1)	(2)	(3)
i)	Overall screen absent (<i>U</i>)	Individual screens absent (<i>U</i>)
ii)	Overall foil screen (<i>F</i>)	Individual foil screens (<i>F</i>)
iii)	Overall braid screen (<i>S</i>)	
iv)	Dual overall screen (foil + braid) (<i>SF</i>)	

Table 2 Balanced Cable Nomenclature

(Clause [A-4](#))

Sl No.	Global Abbreviations	North American	Overall Braid	Overall Foil	Individual Foil
(1)	(x/y)	Abbreviation	Screen	Screen	Screen
(1)	(2)	(3)	(4)	(5)	(6)
i)	<i>U/U</i> TP	<i>U</i> TP	No	No	No
ii)	<i>U/F</i> TP	<i>S</i> TP	No	No	Yes
iii)	<i>F/U</i> TP	<i>F</i> TP, <i>Sc</i> TP, <i>Pi</i> MF	No	Yes	No
iv)	<i>F/F</i> TP	<i>S</i> TP, <i>SS</i> TP	No	Yes	Yes
v)	<i>S/U</i> TP	<i>S</i> TP	Yes	No	No
vi)	<i>S/F</i> TP	<i>S</i> TP, <i>SS</i> TP	Yes	No	Yes
vii)	<i>SF/U</i> TP	<i>S</i> TP, <i>SS</i> TP	Yes	Yes	No
viii)	<i>SF/F</i> TP	<i>S</i> TP, <i>SS</i> TP	Yes	Yes	Yes

NOTE — The abbreviations represent as follows:

1 *F/F* TP — Foil-screened foil-screened twisted-pair (Individually foil-screened twisted-pair in overall foil screen).

2 *F/U* TP — Foil-screened unscreened twisted-pair (Unscreened twisted-pair in overall foil screen).

3 *S/F* TP — Braid-screened foil-screened twisted-pair (Individually foil-screened twisted-pair in overall braid screen).

4 *S/U* TP — Braid-screened unscreened twisted-pair (Unscreened twisted-pair in overall braid screen).

5 *Sc* TP — Screened twisted-pair.

6 *SF/F* TP — Braid-screened-foil-screened foil-screened twisted-pair (Individually foil-screened twisted-pair in overall foil and braid screen).

7 *SF/U* TP — Braid-screened-foil-screened unscreened twisted-pair (Unscreened twisted-pair in overall foil and braid screen).

8 *U/F* TP — Unscreened foil-screened twisted-pair (Individually foil-screened twisted-pair).

9 *U/U* TP — Overall unshielded twisted-pair with unshielded twisted-pair.

A-5 ELECTRONIC ACCESS CONTROL (EAC) AND VIDEO SURVEILLANCE

A-5.1 Electronic Access Control (EAC)

An EAC system allows several levels of security. Once the number and type of people are determined, a user or access level for each person is established. Access levels are incorporated into the system integration and the areas of access are determined. Factors affecting the user level include security clearances, job title, rank, short-term employee and employee type. The access level is determined by the physical layout of building and is directly related to the doors and other portals it controls.

The basic components of an EAC system include:

- A computer;

- b) One or more control panels; and
- c) One or more peripheral devices connected to the control panels (for example, card reader, siren and sensor).

The host computer runs the EAC software, manages the system parameters, maintains the system database, and controls all communication with the control panels. The host computer receives all event and alarm transactions from the control panels. It then saves them into separate history files that may be used to run reports by specific criteria.

The control panels manage the activation of peripheral devices by turning the devices on or off and controlling access to certain areas through card readers and door-locking hardware.

Media for communications may include the following:

- a) *Balanced twisted-pair* — Shielded twisted-pair, unshielded twisted-pair, and screened twisted-pair;
- b) *Optical fibre* — Multi-mode and single-mode;
- c) *Wireless* — Microwave, radio frequency (RF), and infrared (IR); and
- d) Cellular networks connection.

A-5.2 Video Surveillance

A-5.2.1 In addition to traditional capture devices that operate within the visible band of the electromagnetic spectrum, other technologies that provide unique viewing capabilities are, as follows:

- a) *IR cameras* — These refer to specially designed imagers capable of seeing into the low IR bandwidth. They are sometimes referred to as starlight cameras because of their night time viewing capabilities;
- b) *Thermal cameras* — They capture heat or temperature values of a scene rather than light values, regardless of how bright or dark the scene appears to the human eye. Although the identification of colours and details are impossible with thermal cameras (because they only view temperature), these cameras are quite useful in viewing dark scenes for activities that have heat signatures; and
- c) *Film cameras* — They have been on the market for many years. These self-sustaining units are equipped with traditional camera film, which permanently stores the scenes. The film requires replacement when exhausted.

A-5.2.2 Camera placement is based on security objectives identified through a need-assessment. Video surveillance applications include two typical types of camera mounts, as follows:

- a) *Fixed* — It refers to a camera dedicated to single view that does not change unless the camera is physically moved; and
- b) *Pan and tilt* — It refers to a camera with a pan and tilt that allows for a 180° or 360° view, which is controlled electronically from a remote viewing station.

From a practical standpoint, the trade-off between the two types is that a pan and tilt provides a more flexible area of coverage at a higher cost than a fixed camera. Pan and tilt technology allows for auto-panning where the camera moves in a predefined cycle. Camera mounting locations are typically dictated by the available physical infrastructure and are balanced against the required field of views. Connectivity of cameras to the main system in a building environment is generally through coaxial cables.

A-6 OVERVIEW OF IN-BUILDING SOLUTIONS (IBS)

A-6.1 General

The information in this annexure can be used to plan IBS to provide seamless mobile data and voice experience for indoor users including in areas like lifts, staircase and basements. IBS is a telecommunications infrastructure designed to enhance wireless network coverage. It is particularly useful in establishments such as offices, malls, hospitals, airports, sport arenas, multistorey residential complexes and other areas, where outdoor mobile signals may be weak due to distance and structural obstructions.

A-6.2 Types of In-Building Solutions

The various types of IBS technologies, to enhance indoor connectivity, are as under:

A-6.2.1 Distributed Antenna Systems (DAS) — It consist of multiple antennas distributed throughout a building or a venue. These antennas connect to a radio frequency source(s) and distribute signals effectively overcoming signal degradation in indoor environments. Typical layout of a DAS is shown in the [Fig. 9](#). Following are the main components of DAS:

- Distributed antennas* — Acts as radiating source for RF signals. These are distributed throughout the building based on radio network planning;
- Splitter* — A passive component that divides the RF signal power equally among multiple output ports;
- Coupler* — A passive device that extracts a small portion of the signal from the main feed while allowing most of it to continue;
- Radio unit* — The radio frequency or mobile signal source that connects the in-building system to the mobile network; and
- RF or fibre cable for signal distribution* — The RF is used in passive DAS to carry the radio signals up to the antennas while fibre cable is used to connect master and slave unit in case of active DAS.

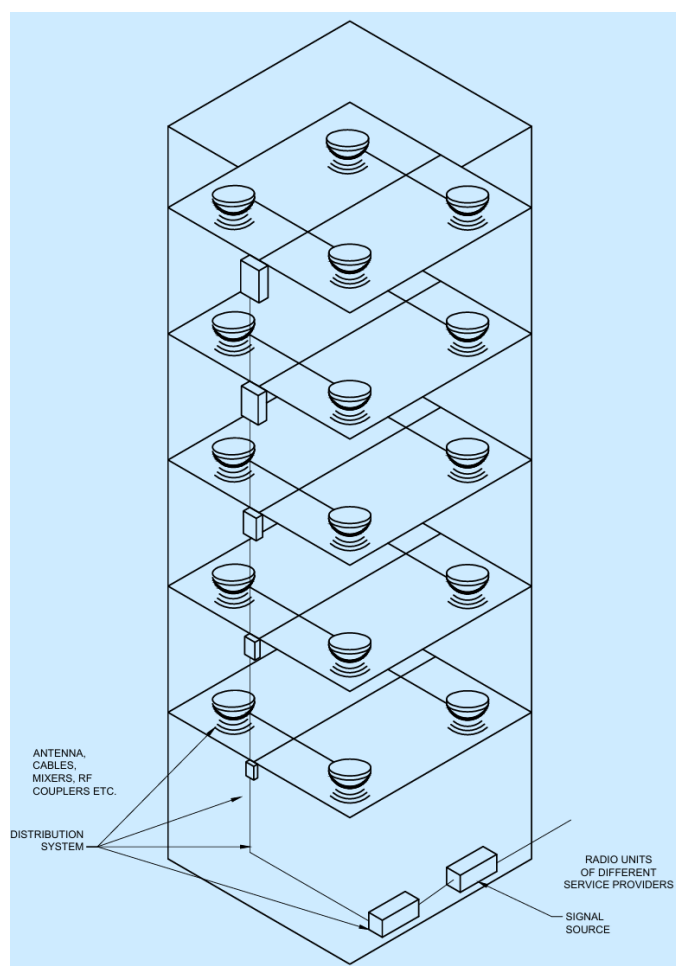


FIG. 9 TYPICAL COMPONENTS OF A DISTRIBUTED ANTENNA SYSTEM (DAS)

A DAS can support multi frequency bands to carry radio frequency signals of multiple service providers for different technologies (2G/3G/4G/5G, etc) using common DAS. This common DAS infrastructure provide considerable cost savings with improved building aesthetics while providing seamless mobile coverage including in basements, lifts and lobbies.

There are different types of distributed antenna systems suitable for different use cases. Overview of some

important DASs are covered below:

- 1) *Passive distributed antenna system (DAS)* — In passive DAS, only passive components are used to distribute the RF signals. The size of the cable depends upon the frequency in use, coverage and capacity requirement. The distribution of combined mobile signal is done using passive components in the network like RF cable, couplers and splitters. The RF signal is coupled with the antennas installed across the floor(s) for radiating in different sections of the building as indicated in the [Fig 10](#); and
- 2) *Active distributed antenna system* — This type of system is used where the coverage is required in a huge building and airport etc. where conventional passive cable system does not give adequate coverage due to high losses in the coaxial cables as shown in [Fig 11](#).

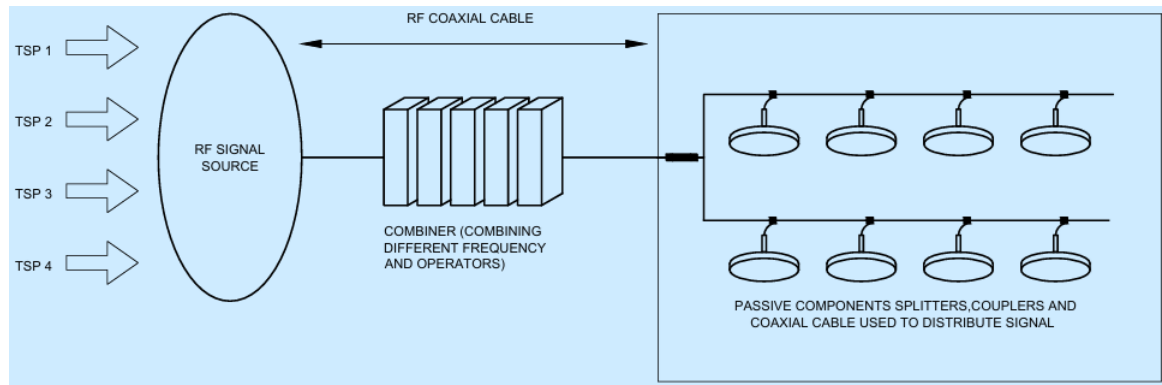


FIG. 10 KEY COMPONENTS OF TYPICAL PASSIVE DAS

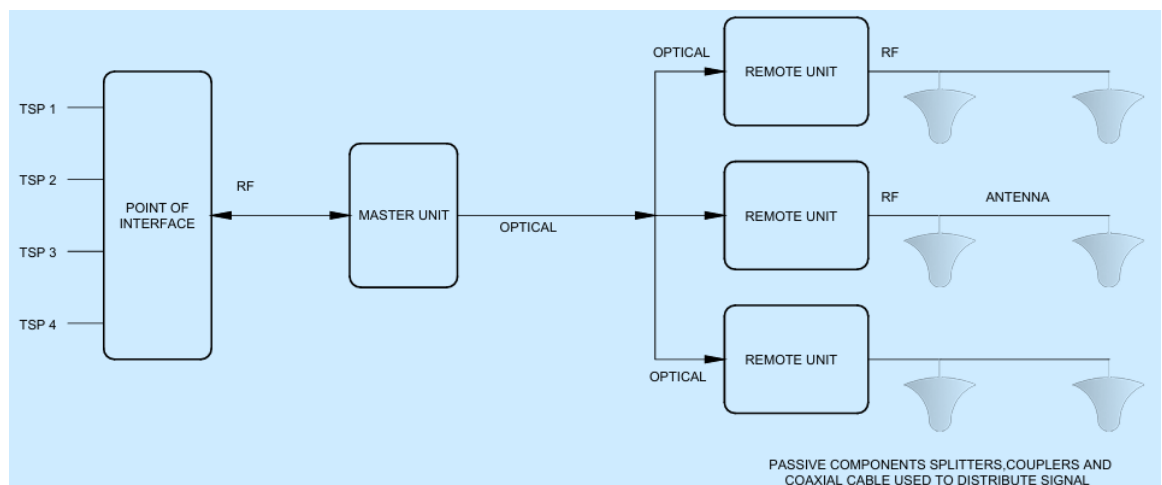


FIG. 11 KEY COMPONENTS OF TYPICAL ACTIVE DAS

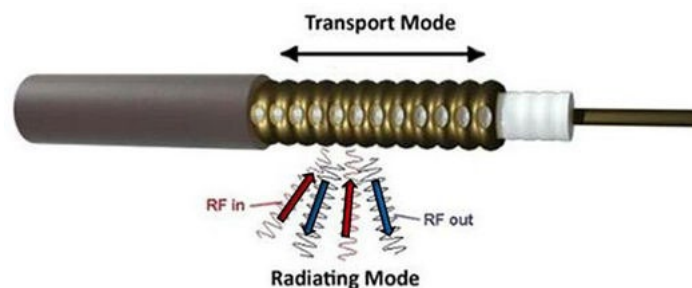


FIG. 12 TYPICAL LEAKY CABLE INSTALLATION IN A TUNNEL

A-6.2.2 Leaky Feeder Cable System — In this system this is a special type of cable known as leaky feeder cable in which the slots are cut as window in the cable itself and when the RF signal is pumped in it which get radiated through these windows which acts as an antenna as indicated in [Fig. 12](#). Such cables are generally used in the tunnels where installation of active components may be difficult.

A-6.2.3 Small Cells, Including Femtocell and Pico Cells — They are low-powered radio access nodes designed to cover small geographic areas, such as a single room or a specific area in a larger venue. They complement main cellular networks by providing targeted coverage and traffic capacity.

A-6.3 Key Considerations for Planning an In-Building Solution

- a) *Coverage requirements* — Determining the appropriate coverage levels for an in-building solution depends on user density and the types of applications being used. High-density areas require a robust network capable of handling simultaneous connections from multiple users. The design should account for variations in density across different floors and zones to ensure uniform coverage. Additionally, the nature of services being provided whether voice, video, or data — Plays a crucial role in coverage planning;
- b) *Interference management* — Effective interference management is essential to maintain signal quality within an indoor environment. Identifying potential interference sources, such as nearby wireless networks, electronic devices, and structural obstacles, helps in mitigating performance issues. To minimize interference, strategies such as frequency planning, power control, and spatial diversity should be implemented;
- c) *Capacity planning* — Proper capacity planning ensures that an in-building solution can handle both current and future network demands. By analyzing user behaviours and application requirements, network designers can forecast growth in user numbers and data consumption patterns; and
- d) *Aesthetics and integration* — Careful placement of antennas, cables, and other infrastructure components helps maintain the visual appeal of the space. Where possible, discreet installations such as ceiling-mounted antennas or concealed cabling are preferred. Engaging stakeholders including building owners, architects, and end-users during the IBS design phase ensures that the network aligns with both functional and aesthetic requirements.

NOTE — The relevant standards, prescribed by Telecom Engineering Centre (TEC), Department of Telecom, Ministry of Communications, may be referred at [\[https://www.tec.gov.in\]](https://www.tec.gov.in) for the implementation of in-building solutions.

ANNEX B

(Foreword)

STREET FURNITURE FOR ICT INSTALLATIONS

B-1 Street furniture, such as streetlights, utility poles, bus stops, and traffic signals, is expected to play a significant role in enabling ICT infrastructure, especially as India deploys next-generation technologies targeting the mobile devices used by public. As ICT installations expand, these existing structures provide an efficient and cost-effective foundation for small cell and air fibre, supporting enhanced connectivity, quicker deployments, and seamless integration into urban landscapes. The Telecom Regulatory Authority of India (TRAI) and the PM Gati Shakti National Master Plan recognize the potential of street furniture in, integrating digital connectivity into public infrastructure. Additionally, the *Telecommunications (Right of Way) Rules, 2024*, defines small cells as having a coverage range of up to 2 km, reinforcing their role in improving network reach

Further, under the PM Gati Shakti National Master Plan, 26.5 million units of street furniture have been mapped digitally to simplify planning, accelerate approvals, and reduce site survey timelines. This creates a ready-to-use foundation for 5G and future ICT infrastructure, helping embed digital connectivity into public infrastructure efficiently.

The guidelines for earmarking and use of street furniture are expected to be put in place by Department of Telecommunication (DoT)/TRAI in due course.

Some of the important aspects of use of street furniture which require consideration for effective ICT deployments are given below.

B-1.1 Structural and Positional Aspects

The aspects include, but are not limited to:

- a) Load-bearing requirements to support equipment such as small cells and associated hardware;
- b) Telecom equipment designs blend with urban aesthetics to avoid visual clutter;
- c) Proximity to existing power sources;
- d) Protection of pedestrian and vehicle traffic;
- e) Public safety; and
- f) Structural stability and safety considering the weight and required height for installation of equipment as well as sharing among service providers.

B-1.2 Administrative Aspects

The following aspects are applicable:

- a) Easy right of way (RoW) for installation and maintenance and upgrades of the equipment;
- b) Possibility of development of a GIS-mapped inventory of street furniture assets, detailing specifications like load capacity, power availability, and contact information for streamlined maintenance and coordination; and
- c) Adherence to municipal guidelines and permissions for ICT installations on public or private street furniture.

ANNEX C

(Clause 3.5)

BUILDING RELATED INFORMATION AND DIGITAL CONNECTIVITY REQUIREMENTS

C-1 GENERAL INFORMATION

<i>Sl No.</i>	<i>Item</i>	<i>Description/ Response</i>
i)	Name of building/Project and address	
ii)	Property manager detail (name, contact etc)	
iii)	Type of building	
iv)	Number of floors (including parking floors)	
v)	Total built-up area (sq. meters)	
vi)	Total carpet area (<i>see</i> NOTE)	
vii)	Year of construction completion (it will same as occupancy certificate year)	
viii)	Building/Property layout	Attachment
ix)	Ownership type (owners/tenants/developer owned/shared space)	
x)	a) Occupancy classification: 1) Group A: Residential 2) Group B: Educational 3) Group C: Institutional b) Hospitals/Healthcare: 1) Group D: Assembly c) Public buildings: 1) Airports/Railway stations/Bus stands/Metro train stations/Taxi stands, etc d) Exhibition centers: 1) Group E: Business e) Commercial retail:	
	f) Commercial office: 1) Group F: Mercantile 2) Group G: Industrial 3) Group H: Storage 4) Group I: Hazardous 5) Group J: Mixed-use buildings	
xi)	Projected date of completion	
NOTE — Digital connectivity coverage design will require all areas including open, lifts, corridors, underground parking floors, basement etc. Therefore, total area where mobile or Wi-Fi coverage is required will be important to capture.		

C-2 OCCUPANT/USER DETAILS

<i>Sl No.</i>	<i>Item</i>	<i>Description/Response</i>
(1)	(2)	(3)
i)	Total number of units (flat, office etc)	
ii)	Estimated No. of occupants/users	
iii)	Number of tenants/offices	
iv)	Occupant categories (residential, academic, business, public use etc)	

C-3 DIGITAL CONNECTIVITY REQUIREMENTS

<i>Sl No.</i>	<i>Item</i>	<i>Description/Response</i>
(1)	(2)	(3)
i)	Number of internet and mobile service providers present in the area	
ii)	Expected number of internet connections per unit	
iii)	Typical internet speed requirements	
iv)	Number of devices (for example, mobile, tablets, laptop etc) per unit	
v)	Preference for neutral FTTx infrastructure (Yes or No)	
vi)	Need for redundant paths/Backup connectivity (Yes or No)	
vii)	Use of smart devices/Home automation (Yes or No)	
viii)	Cybersecurity requirements (Yes or No)	
ix)	IoT integration requirements (if any) (Yes or No)	
x)	Fibre connectivity requirement (Yes or No)	
xi)	Requirement of secure Wifi (Yes or No)	

C-4 INFRASTRUCTURE SUPPORT (APPLICABLE IN CASE OF EXISTING BUILDINGS)

<i>Sl No.</i>	<i>Item</i>	<i>Description/Response</i>
(1)	(2)	(3)
i)	Centralized cable pathways availability	
ii)	Duct space and shaft accessibility and maintainability	
iii)	Existing server or telecommunication rooms	
iv)	Power backup with provision of uninterrupted power supply for ICT	

- Equipment in telecom room
- v) Proposed upgrades (if any)
- vi) Provision of future expansion (telecom Room, cable pathways, IoT etc)
- vii) Provision of elevated structure for TI/Telecom room to address flood situations
- viii) Redundant entry paths/Ducts for extending backhaul telecom connectivity by the service providers to the property

Declaration

This information is collected for planning and designing telecommunication infrastructure. In case of unavailability of user-specific data, data from similar building typologies shall be used.

Signature of Property Manager/Owner: _____

Date: _____

NOTE — It may be noted that in many cases, buildings or complexes are of mixed-use nature. The template provided for collecting building information may be used only as a reference document specifically for telecommunication infrastructure and services planning. The classification used in this template applies only to this section and has no relation to the occupancy classification given in any other part of this NBCS.

LIST OF STANDARDS

The following list records those standards which are acceptable as ‘good practice’ and ‘accepted standards’ in the fulfilment of the requirements of this NBCS. The latest version of a standard shall be adopted at the time of enforcement of this NBCS. The standards listed may be used for conformance with the requirements of the referred clauses in this NBCS.

In the following list, the number appearing in the first column within parentheses indicates the number of the reference in this Part/Section.

	<i>IS No.</i>	<i>Title</i>
(1)	IS 9537 (Part 2) : 1981	Specification for conduits for electrical installations: Rigid steel conduits
(2)	IS 9537 (Part 3) : 1983	Specification for conduits for electrical installations: Rigid plain conduits for insulating materials
	IS 3419 : 1988	Specification for fittings for rigid non-metallic conduits (<i>second revision</i>)
(3)	IS 9537 (Part 5) : 2000	Conduits for electrical installations — Specification Pliable conduits of insulating materials